

Baskotian Template

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388 templates · 12 sections · 24,247 lines · August 13, 2026

HOW TO FIND SOMETHING (in order of speed). **1.** You know the routine's name → the alphabetical *Index of every routine* at the very back gives the page. **2.** You know the topic → the running head on every page names the section (left) and the sub-topic (right), and the strip along the bottom edge shows which of the 12 sections you are in; thumb until it matches. **3.** You know neither → section **12 Problem Solving** maps a problem's *shape* and its *constraints* onto the technique, and docs/PLAYBOOK.md does the same at more length.

BEFORE YOU PASTE. Line 1 of every file says what it does; a `// Needs:` line names the templates it depends on; a `// NOTE:` line warns when another file declares the same symbol. Vertices and array positions are **0-based** unless the header says otherwise. Geometry: `P<ll>` for every decision, `P<ld>` only when the answer is a real number.

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1 Miscellaneous

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1 | Miscellaneous

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1.1 Template

```
#include <bits/stdc++.h>
using namespace std;

// #define int long long
#define ll long long
#define lll __int128 // used unnamed by CHT,
↳ BigInt, PrimeCounting
#define ld long double
#define fi first
#define se second
#define pb push_back
#define eb emplace_back
#define all(x) x.begin(), x.end()
#define rall(x) x.rbegin(), x.rend()
#define sz(x) (int)(x).size()
#define ones(n) __builtin_popcountll(n)
#define msb(n) (63 - __builtin_clzll(n))
#define lsb(n) __builtin_ctzll(n)
typedef vector<int> vi;
typedef pair<int, int> pii;

const int N = 2e5 + 5, M = 1e3 + 5, LOG = 31;
const int inf = 0x3f3f3f3f;
const ll llinf = 0x3f3f3f3f3f3f3f3f;
const int MOD = 1e9 + 7;
const ld eps = 1e-9, PI = acos(-1.0L); // ld, not double:
↳ geometry needs the digits

#ifdef LOCAL // compile with -DLOCAL
template <class T> void _p(const T& x);
void _p(const string& s) { cerr << "'" << s << "'"; }
void _p(char c) { cerr << '\\' << c << '\\'; }
template <class A, class B> void _p(const pair<A, B>& p) {
    cerr << '('; _p(p.fi); cerr << ", "; _p(p.se); cerr << ')';
}
template <class T> void _p(const T& x) {
    if constexpr (is_arithmetic_v<T>) cerr << x;
    else { cerr << '{'; int f = 1;
        for (auto& e : x) { if (!f) cerr << ", "; f = 0; _p(e); }
        ↪ cerr << '}'; }
}
void _pr() { cerr << "\n"; }
template <class H, class... T> void _pr(const H& h, const T&... t) {
    _p(h); if (sizeof...(t)) cerr << " | "; _pr(t...);
}
#define dbg(...) (cerr << "[" << #__VA_ARGS__ << "] = ",
    ↪ _pr(__VA_ARGS__))
#else
#define dbg(...)
#endif

void testCase() {
}

void preCompute() {
}
```

```
int32_t main() {
    ios_base::sync_with_stdio(false), cin.tie(nullptr),
    ↪ cout.tie(nullptr);
    cout << fixed << setprecision(10);
    preCompute();

    int tc = 1;
    cin >> tc; // DELETE THIS LINE for
    ↪ single-test problems
    for (int TC = 1; TC <= tc; TC++) {
        // cout << "Case #" << TC << ": ";
        testCase();
    }
}

/*

*/
```

1.2 fastIO

HAND-ROLLED FAST IO (fread/fwrite buffered). Use only when cin with sync off is measurably too

```
// slow: >1e6 numbers to read, or heavy output. Otherwise prefer the
↳ template's
// ios::sync_with_stdio.
const int MOD = 1e9 + 7;
const int BUF_SZ = 1 << 15;

inline namespace Input {
    char buf[BUF_SZ];
    int pos;
    int len;
    char next_char() {
        if (pos == len) {
            pos = 0;
            len = (int)fread(buf, 1, BUF_SZ, stdin);
            if (!len) { return EOF; }
        }
        return buf[pos++];
    }

    int read_int() {
        int x;
        char ch;
        int sgn = 1;
        while (!isdigit(ch = next_char())) {
            if (ch == '-') { sgn *= -1; }
        }
        x = ch - '0';
        while (isdigit(ch = next_char())) { x = x * 10 + (ch - '0'); }
        return x * sgn;
    }
} // namespace Input

inline namespace Output {
    char buf[BUF_SZ];
    int pos;

    void flush_out() {
        fwrite(buf, 1, pos, stdout);
        pos = 0;
    }

    void write_char(char c) {
        if (pos == BUF_SZ) { flush_out(); }
        buf[pos++] = c;
    }

    void write_int(int x) {
        static char num_buf[100];
        if (x < 0) {
            write_char('-');
            x *= -1;
        }
        int len = 0;
        for (; x >= 10; x /= 10) { num_buf[len++] = (char)('0' + (x %
        ↪ 10)); }
        write_char((char)('0' + x));
        while (len) { write_char(num_buf[--len]); }
        write_char('\n');
    }

    // auto-flush output when program exits
    void init_output() { assert(atexit(flush_out) == 0); }
} // namespace Output
```

1.3 grid

GRID DIRECTION VECTORS. First 4 entries = 4-neighbourhood (L,R,U,D), all 8 = king moves.

```
// Knight moves: dx {1,1,-1,-1,2,2,-2,-2}, dy {2,-2,2,-2,1,-1,1,-1}.
    ↳ Guard with 0<=x<n && 0<=y<m.
int dx[] = {0, 0, -1, 1, -1, -1, 1, 1};
int dy[] = {-1, 1, 0, 0, -1, 1, -1, 1};
char di[] = {'L', 'R', 'U', 'D'};

// KNIGHT DISTANCE ON AN INFINITE BOARD, O(1). BFS is fine up to a
    ↳ 1000x1000 board; past that, or
// when the coordinates are 1e9, you need this closed form. Reduce by
    ↳ symmetry to x, y >= 0, then
// the answer is driven by three lower bounds - you cover at most 2 in
    ↳ either axis per move and at
// most 3 of (x + y) per move - and parity, since every move changes x
    ↳ + y by an odd amount. Three
// positions beat the formula and must be special-cased; they are the
    ↳ classic reason a hand-rolled
// version WAS. Exact integer arithmetic, any magnitude that fits in ll.
ll knightDist(ll x, ll y) {
    x = abs(x), y = abs(y);
    if (x < y) swap(x, y);                // now x >= y
    ↳ >= 0
    if (x == 1 && y == 0) return 3;        // the three
    ↳ exceptions, all near the
    if (x == 2 && y == 2) return 4;        // origin,
    ↳ where the board "runs out"
    ll d = max({(x + 1) / 2, (y + 1) / 2, (x + y + 2) / 3});
    while ((d & 1) != ((x + y) & 1)) d++; // parity: x +
    ↳ y flips every move
    return d;
}

// ON A BOUNDED BOARD the formula is wrong near the edges and corners -
    ↳ BFS, or use the formula
// and repair the O(1) corner cases by hand. On a HALF-PLANE (x, y >=
    ↳ 0) the same three exceptions
// are exactly the corrections needed, which is what the code above
    ↳ assumes.
// KING distance is max(|dx|, |dy|); ROOK is 0/1/2; BISHOP is 0/1/2
    ↳ when the colours match and
// unreachable otherwise. Those need no code, only the reminder that
    ↳ they are not Manhattan.
```

1.4 random

RANDOMNESS. Seed from the clock so tests cannot be prepared against you.

```
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
ll rnd(ll l, ll r) { return uniform_int_distribution<ll>(l, r)(rng); }
    ↳ // inclusive

vector<int> randPerm(int n) {                // random
    ↳ permutation of 0..n-1
    vector<int> v(n);
    iota(all(v), 0);
    shuffle(all(v), rng);
    return v;
}

// Shuffling the INPUT is a real algorithm, not just a utility:
// - minimum enclosing circle is O(n) only after a shuffle
// - quickselect / random pivots
// - defeats anti-hash and anti-quicksort tests
// - random rotation of points removes degenerate ties in a geometry
    ↳ sweep
```

1.5 coordinateCompression

Coordinate Compression - O(N log N)

```
template<typename T>
struct Compress {
    vector<T> v;

    Compress(vector<T>& a) : v(a) {
        sort(all(v));
        v.erase(unique(all(v)), v.end());
        for (auto &x : a) {
            x = lower_bound(all(v), x) - v.begin();
        }
    }

    T operator[](int i) const {
        return v[i];
    }
};
```

1.6 grayCode

GRAY CODE - orders $0..2^n-1$ so consecutive values differ in exactly ONE bit. Use it to walk all

```
// subsets while changing one element at a time (incremental DP,
    ↳ XOR-basis sweeps, Tower of Hanoi).
// Gray Code Generator - O(2^N)
// g(n) = n ^ (n >> 1)
vector<int> grayCode(int n) {
    vector<int> res(1 << n);
    for (int i = 0; i < (1 << n); i++) {
        res[i] = i ^ (i >> 1);
    }
    return res;
}

int rev_g (int g) {
    int n = 0;
    for (; g; g >>= 1)
        n ^= g;
    return n;
}
```

1.7 customHashHashTable

UNORDERED_MAP THAT CANNOT BE HACKED. The default std::hash for integers is the identity, so an

```
// adversary can force every key into one bucket and turn O(1) into
    ↳ O(n). splitmix64 + a clock seed
// fixes that. gp_hash_table is ~3x faster than unordered_map;
    ↳ reserve() and max_load_factor
// help more.
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

// Custom Hash to prevent anti-hash tests in unordered_map /
    ↳ gp_hash_table
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }

    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

// Usage: gp_hash_table<ll, int, custom_hash> table;
//         unordered_map<ll, int, custom_hash> safe_map;
```

1.8 pragmas

COMPILER PRAGMAS. O3 + unrolling, and AVX2 for auto-vectorised loops (bitset, simple array math).

```
// Put them at the very top, before any include. Risk: 'target' crashes
    ↳ judges without those CPU
// features - never use avx2 on an unknown judge, and never on
    ↳ Codeforces problems with old
// machines.
// GCC Optimizations
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```

1.9 int128

__int128 Stream Operators Helper

```
typedef __int128_t int128;
typedef __uint128_t uint128;

ostream &operator<<(ostream &os, int128 n) {
    if (n == 0) return os << "0";
    if (n < 0) { os << "-"; n = -n; }
    string s;
    while (n > 0) { s += (char)('0' + (n % 10)); n /= 10; }
    reverse(all(s));
    return os << s;
}

istream &operator>>(istream &is, int128 &n) {
    string s; is >> s;
    n = 0; int i = 0, sign = 1;
```

```

if (!s.empty() && s[0] == '-') { sign = -1; i = 1; }
for (; i < (int)s.size(); i++) n = n * 10 + (s[i] - '0');
n *= sign;
return is;
}

```

1.10 SearchIdioms

The four searches. Getting the boundary right here saves more contest time than any template.

```

// BINARY SEARCH ON A PREDICATE. pred must be monotone: F F F T T T.
// firstTrue returns the first x in [lo, hi] with pred(x), or hi+1 if
↳ none.
template <class F> ll firstTrue(ll lo, ll hi, F pred) {
    hi++;
    while (lo < hi) { ll m = lo + (hi - lo) / 2; pred(m) ? hi = m : lo
↳ = m + 1; }
    return lo;
}

// lastTrue returns the last x in [lo, hi] with pred(x), or lo-1 if
↳ none. T T T F F F
template <class F> ll lastTrue(ll lo, ll hi, F pred) {
    lo--;
    while (lo < hi) { ll m = lo + (hi - lo + 1) / 2; pred(m) ? lo = m :
↳ hi = m - 1; }
    return lo;
}

// Reals: fixed iteration count beats an epsilon loop (no infinite
↳ loops, predictable time).
// 100 iterations halve the range 2^100 times - always enough.
// Returns the boundary: pred is false on [lo, x) and true on [x, hi].
↳ ('lo' and 'hi' converge.)
template <class F> ld binReal(ld lo, ld hi, F pred, int it = 100) {
    while (it--) { ld m = (lo + hi) / 2; pred(m) ? hi = m : lo = m; }
    return hi;
↳ TRUE, matching firstTrue
}

// TERNARY SEARCH on a UNIMODAL function (strictly decreasing then
↳ increasing => minimum).
// Integer version: stop when the window is tiny, then scan. Never use
↳ 'l < r' alone.
template <class F> ll ternaryInt(ll lo, ll hi, F f) {
↳ returns argmin
    while (hi - lo > 2) {
        ll m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        f(m1) <= f(m2) ? hi = m2 : lo = m1;
↳ keeps the LEFTmost min
    }
    ll best = lo;
    for (ll x = lo; x <= hi; x++) if (f(x) < f(best)) best = x;
    return best;
}

template <class F> ld ternaryReal(ld lo, ld hi, F f, int it = 200) {
    while (it--) { ld m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        f(m1) < f(m2) ? hi = m2 : lo = m1; }
    return (lo + hi) / 2;
}

// If f is only WEAKLY unimodal (has flat regions), ternary search is
↳ WRONG: one comparison
// f(m1) == f(m2) inside a plateau discards the side holding the
↳ optimum. For any DISCRETELY
// CONVEX f (non-decreasing difference sequence, plateaus allowed) use
↳ the derivative instead -
// this is strictly safer AND uses half the evaluations:
template <class F> ll convexArgmin(ll lo, ll hi, F f) {
    return firstTrue(lo, hi - 1, [&](ll x) { return f(x + 1) >= f(x);
↳ });
}

// NEWTON'S METHOD for roots of a smooth f (quadratic convergence;
↳ needs a decent start).
template <class F, class DF> ld newton(ld x, F f, DF df, int it = 60) {
    while (it--) { ld d = f(x) / df(x); x -= d;
        if (fabsl(d) < 1e-15L * max((ld)1, fabsl(x))) break;
↳ }
    return x;
}

// Integer sqrt / cbrt with correction - NEVER trust sqrt() alone near
↳ 1e18.
ll isqrt(ll n) {
    ll r = (ll)sqrtl((ld)n);
    while (r > 0 && r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    return r;
}

```

```

ll icbrt(ll n) {
    ll r = (ll)cbrtl((ld)n);
    while (r > 0 && r * r * r > n) r--;
    while ((r + 1) * (r + 1) * (r + 1) <= n) r++;
    return r;
}

```

```

// BINARY SEARCH ON THE ANSWER - the pattern, not the code:
// "minimise the maximum X" / "maximise the minimum X" / "can we
↳ finish within T"
// -> guess the answer, write a GREEDY feasibility check, binary
↳ search on it.
// PARALLEL BINARY SEARCH - when you need this for MANY queries at
↳ once: bucket the queries by
// their current midpoint, sweep the shared structure once per level,
↳ O(log) sweeps total.

```

1.11 BitTricks

BIT MANIPULATION CHEAT SHEET

```

/* ===== BIT MANIPULATION CHEAT SHEET
↳ =====
BUILTINS (add ll suffix for 64-bit: __builtin_popcountll, ...)
__builtin_popcount(x) #set bits      __builtin_ctz(x) #trailing
↳ zeros (x != 0)
__builtin_clz(x)      #leading zeros  __builtin_parity(x) popcount
↳ & 1
__lg(x) = 63 - clzll(x) = floor(log2 x)      (GCC only; write it
↳ yourself elsewhere)
lowest set bit: x & -x      clear it: x &= x - 1      set/flip:
↳ x |= b, x ^= b
is power of two: x && !(x & (x - 1))
round UP to a power of two (x >= 1): x == 1 ? 1ULL : 2ULL << (63 -
↳ __builtin_clzll(x - 1))
- clzll(0) is UNDEFINED, so x = 1 must be special-cased, and the
↳ shift must be 64-bit

```

SUBMASK ENUMERATION - iterate every subset of m:
for (int s = m; ; s = (s - 1) & m) { ...; if (!s) break; }
Over ALL masks this is $O(3^n)$ total (each element is in/out/neither).

SUPERMASK enumeration (every s with m subset of s, within n bits):
for (int s = m; s < (1 << n); s = (s + 1) | m) { ... }

K-SUBSETS in increasing order (Gosper's hack):
for (int s = (1 << k) - 1; s < (1 << n);) { ...;
int c = s & -s, r = s + c; s = (((r ^ s) >> 2) / c) | r; }

GRAY CODE g(i) = i ^ (i >> 1) inverse: x ^= x >> 1 repeatedly
↳ (see 06 - grayCode.cpp)
consecutive gray codes differ in exactly one bit - use for "change
↳ one thing at a time".

XOR FACTS
 $x \wedge x = 0$, $x \wedge 0 = x$, xor is its own inverse -> prefix xor answers
↳ range xor
xor of $0..n = [n, 1, n+1, 0][n \% 4]$
 $a + b = (a \wedge b) + 2*(a \& b)$ $a | b = (a \wedge b) + (a \& b)$ $a + b$
↳ $= (a | b) + (a \& b)$
"OR of a range equals its SUM" <=> the values are pairwise
↳ BIT-DISJOINT
(so at most 30 non-zero values fit in any such window -> two
↳ pointers)

COUNTING PER BIT - the standard decomposition: handle each of the
↳ 30/60 bits independently.
sum over pairs of $(a_i \wedge a_j) = \text{sum over bits } b \text{ of } (\# \text{with bit } b) *$
↳ $(\# \text{without bit } b) * 2^b$

```

BITSET
std::bitset<N> b; b._Find_first(), b._Find_next(i) (GCC) - iterate
↳ set bits fast
b <<= k, b |= other are  $O(N/64)$  - this is what makes bitset DP and
↳ reachability work.
=====
↳ */

```

```

// Iterate all submasks of m (including m and 0).
template <class F> void forEachSubmask(int m, F f) {
    for (int s = m; s = (s - 1) & m) { f(s); if (!s) break; }
}

// All n-bit masks with exactly k bits set, in increasing order
↳ (Gosper's hack).
vector<int> kSubsets(int n, int k) {
    vector<int> r;
    if (k == 0) return {0};
    for (int s = (1 << k) - 1; s < (1 << n); ) {

```

```

    r.push_back(s);
    int c = s & -s, x = s + c;
    s = (((x ^ s) >> 2) / c) | x;
}
return r;
}
// Sum over all pairs (i<j) of (a_i XOR a_j), by handling each bit
// independently.
ll pairXorSum(const vector<ll>& a, int B = 62) {
    ll n = a.size(), r = 0;
    for (int b = 0; b < B; b++) {
        ll c = 0;
        for (ll x : a) c += (x >> b) & 1;
        r += c * (n - c) % MOD * ((1LL << b) % MOD) % MOD;
    }
    return r % MOD;
}

```

1.12 StressTest

STRESS TESTING

```

/* ===== STRESS TESTING =====
The single highest-value 20 lines in this notebook. Use it the moment
  ↳ a solution WAS on a
test you cannot see. Write gen.cpp + brute.cpp, then loop until they
  ↳ disagree.

---- run.sh
  ↳ -----
g++ -O2 -o sol sol.cpp && g++ -O2 -o brute brute.cpp && g++ -O2 -o gen
  ↳ gen.cpp
for i in $(seq 1 100000); do
    ./gen $i > in.txt
    ./sol < in.txt > out1.txt
    ./brute < in.txt > out2.txt
    if ! diff -qb out1.txt out2.txt > /dev/null; then echo "WA on test
  ↳ $i"; cat in.txt; break; fi
done

  ↳ -----
Windows cmd: replace the loop with 'for /L %i in (1,1,100000) do (
  ↳ ... fc out1.txt out2.txt )'

```

RULES THAT MAKE IT ACTUALLY WORK

1. Generate SMALL cases ($n \leq 8$, values ≤ 5). Bugs hide in tiny
 - ↳ corner cases, not big ones.
2. Seed the generator from `argv[1]` so every iteration differs and is
 - ↳ reproducible.
3. If the answer is not unique, do not diff - write a CHECKER that
 - ↳ validates sol's output.
4. If there is no brute force, stress against yourself: compare two
 - ↳ different solutions, or
 - compare an $O(n^2)$ version of your own idea against the $O(n \log n)$
 - ↳ one.
5. For "find any valid X" problems, verify the constraints instead of
 - ↳ comparing.

```

===== */

```

```

// ---- gen.cpp ----
// #include <bits/stdc++.h>
// using namespace std;
// int main(int argc, char** argv) {
//     mt19937 rng(atoi(argv[1]));
//     auto rnd = [&](int l, int r) { return
  ↳ uniform_int_distribution<int>(l, r)(rng); };
//     int n = rnd(1, 8);
//     printf("%d\n", n);
//     for (int i = 0; i < n; i++) printf("%d ", rnd(1, 5));
//     puts("");
// }

// ---- in-process version: no files, no scripts. Put brute + fast in
  ↳ one binary. ----
// Fastest way to start; switch to the shell loop only when you need
  ↳ the failing input on disk.
template <class Gen, class Fast, class Brute>
void stress(Gen gen, Fast fast, Brute brute, int iters = 100000) {
    for (int it = 1; it <= iters; it++) {
        auto in = gen(it);
        auto a = fast(in), b = brute(in);
        if (!(a == b)) {
            printf("MISMATCH on iteration %d: fast=", it);
            cout << a << " brute=" << b << "\n";
            return;
            // print 'in' here
        }
        // too
    }
}

```

```

    puts("all tests passed");
}

```

1.13 Fraction

Exact rational. Always normalised (gcd reduced, denominator > 0).

```

// Comparison uses __int128 so it is exact for |num|, |den| up to ~9e18.
// Use whenever comparing ratios: slopes, "maximum average", scheduling
  ↳ exchange arguments.
struct Frac {
    ll n = 0, d = 1;
    Frac(ll n = 0, ll d = 1) : n(n), d(d) { norm(); }
    void norm() {
        if (d < 0) n = -n, d = -d;
        ll g = gcd(n < 0 ? -n : n, d);
        if (g) n /= g, d /= g;
        if (!d) d = 1;
        // 1/0 -> treat
        ↳ as 0/1; guard your input
    }
    Frac operator+(Frac o) const { return Frac(n * o.d + o.n * d, d *
  ↳ o.d); }
    Frac operator-(Frac o) const { return Frac(n * o.d - o.n * d, d *
  ↳ o.d); }
    Frac operator*(Frac o) const { return Frac(n * o.n, d * o.d); }
    Frac operator/(Frac o) const { return Frac(n * o.d, d * o.n); }
    bool operator<(Frac o) const { return (__int128)n * o.d <
  ↳ (__int128)o.n * d; }
    bool operator==(Frac o) const { return n == o.n && d == o.d; }
    bool operator<=(Frac o) const { return !o < *this; }
    ld val() const { return (ld)n / d; }
    friend ostream& operator<<(ostream& os, Frac f) { return os << f.n
  ↳ << '/' << f.d; }
};
// OVERFLOW: + and * multiply denominators, so chains of operations
  ↳ blow up fast. If you only
// need COMPARISONS, keep the raw pair and cross-multiply in __int128 -
  ↳ do not normalise.
// For unbounded exactness use Python/BigInt, or the Stern-Brocot
  ↳ search (ContinuedFractions).

```

1.14 Inversions

INVERSIONS: #pairs $i < j$ with $a[i] > a[j]$. $O(n \log n)$ by merge sort (also sorts a).

```

ll inversions(vector<int>& a) {
    int n = a.size();
    if (n < 2) return 0;
    vector<int> buf(n);
    function<ll(int, int)> rec = [&](int l, int r) -> ll {
        ↳ [l, r)
        if (r - l < 2) return 0;
        int m = (l + r) / 2;
        ll res = rec(l, m) + rec(m, r);
        int i = l, j = m, k = l;
        while (i < m && j < r) {
            if (a[i] <= a[j]) buf[k++] = a[i++];
            else buf[k++] = a[j++], res += m - i;
            ↳ a[i..m] all beat a[j]
        }
        while (i < m) buf[k++] = a[i++];
        while (j < r) buf[k++] = a[j++];
        copy(buf.begin() + l, buf.begin() + r, a.begin() + l);
        return res;
    };
    return rec(0, n);
}
// WHY IT KEEPS APPEARING
// - minimum number of ADJACENT SWAPS to sort = inversion count
// - minimum adjacent swaps to turn a into b: relabel b's positions,
  ↳ then count inversions
// - "how far is this permutation from sorted" / bubble-sort distance
// - inversions modulo 2 = permutation PARITY (= sign, used in
  ↳ determinants and 15-puzzle)
// BIT VERSION (needed when you must also support updates, or count
  ↳ online):
// compress values, sweep left to right, ans += (#inserted so far) -
  ↳ query(a[i]), then add a[i].

```

1.15 ExpressionParser

EXPRESSION PARSER (shunting-yard). Handles $+ - * / \% \wedge$, unary minus, parentheses.

```

// Left-associative except ^. Extend by editing prec() / applyOp().
struct Parser {
    static int prec(char op) {
        switch (op) {
            case '+': case '-': return 1;
            case '*': case '/': case '%': return 2;

```

```

    case '^': return 3;
    case '~': return 4; // unary minus
    (internal symbol)
    default: return -1;
}
static bool rightAssoc(char op) { return op == '^' || op == '~'; }

static ll apply(char op, ll a, ll b = 0) {
    switch (op) {
        case '+': return a + b;
        case '-': return a - b;
        case '*': return a * b;
        case '/': return a / b;
        case '%': return a % b;
        case '~': return ~a;
        case '^': { ll r = 1; while (b > 0) { if (b & 1) r *= a; a
    ← *= a, b >>= 1; } return r; }
    }
    return 0;
}
static ll eval(const string& s) {
    vector<ll> val;
    vector<char> ops;
    bool wantOperand = true; // distinguishes
    ← unary from binary minus
    auto pop = [&]() {
        char op = ops.back(); ops.pop_back();
        if (op == '~') { ll a = val.back(); val.pop_back();
    ← val.push_back(apply(op, a)); }
        else { ll b = val.back(); val.pop_back(); ll a =
    ← val.back(); val.pop_back();
            val.push_back(apply(op, a, b)); }
    };
    for (size_t i = 0; i < s.size(); i++) {
        char c = s[i];
        if (isspace((unsigned char)c)) continue;
        if (isdigit((unsigned char)c)) {
            ll x = 0;
            while (i < s.size() && isdigit((unsigned char)s[i])) x
    ← = x * 10 + (s[i++] - '0');
            i--;
            val.push_back(x), wantOperand = false;
        } else if (c == '(') ops.push_back(c), wantOperand = true;
        else if (c == ')') {
            while (!ops.empty() && ops.back() != '(') pop();
            if (!ops.empty()) ops.pop_back();
            wantOperand = false;
        } else {
            if (c == '-' && wantOperand) c = '~'; // unary
            while (!ops.empty() && ops.back() != '(' &&
                (prec(ops.back()) > prec(c) ||
                 (prec(ops.back()) == prec(c) &&
    ← !rightAssoc(c))))
                pop();
            ops.push_back(c), wantOperand = (c != '~') || true;
        }
    }
    while (!ops.empty()) pop();
    return val.empty() ? 0 : val.back();
}
// To build an AST instead of evaluating, push node indices instead of
// values.
// For boolean/comparison operators just add them to prec() with lower
// precedence.

```

1.16 CycleFinding

CYCLE FINDING in a FUNCTIONAL GRAPH $x \rightarrow f(x)$. Every component is a "rho": a tail feeding

into a cycle. Answers: where does x end up after k steps, cycle
 ← length, distance to cycle.

```

// Floyd (tortoise & hare), O(1) memory. Returns {start index of the
    ← cycle, cycle length}.
template <class F> pair<ll, ll> floydCycle(ll x0, F f) {
    ll t = f(x0), h = f(f(x0));
    while (t != h) t = f(t), h = f(f(h));
    ll mu = 0; // steps from x0
    ← to the cycle entry
    t = x0;
    while (t != h) t = f(t), h = f(h), mu++;
    ll lam = 1; // cycle length
    h = f(t);
    while (t != h) h = f(h), lam++;
    return {mu, lam};
}

```

```

}
// f applied k times, using the rho structure (k can be 1e18).
template <class F> ll iterate(ll x0, ll k, F f) {
    auto [mu, lam] = floydCycle(x0, f);
    if (k < mu) { while (k--) x0 = f(x0); return x0; }
    for (ll i = 0; i < mu; i++) x0 = f(x0);
    ll rem = (k - mu) % lam;
    while (rem--) x0 = f(x0);
    return x0;
}
// WHOLE-GRAPH version when f is given as an array over 0..n-1: colour
    ← DFS, O(n).
// state: 0 unvisited, 1 on the current path, 2 done. Fills
    ← cycId/cycPos/distToCycle.
struct FuncGraph {
    int n;
    vector<int> f, state, cyc, pos, dist, cycLen; // cyc[v] = id of
    ← the cycle v reaches
    FuncGraph(vector<int> f) : n(f.size()), f(f), state(n, 0), cyc(n,
    ← -1), pos(n, -1), dist(n, 0) {
        for (int i = 0; i < n; i++) if (!state[i]) {
            vector<int> path;
            int v = i;
            while (!state[v]) state[v] = 1, path.push_back(v), v = f[v];
            if (state[v] == 1) { // found a NEW
    ← cycle, starting at v
                int id = cycLen.size(), len = 0;
                for (int u = v; u = f[u]) { cyc[u] = id, pos[u] =
    ← len++; if (f[u] == v) break; }
                cycLen.push_back(len);
            }
            for (int j = path.size() - 1; j >= 0; j--) {
                int u = path[j];
                state[u] = 2;
                if (cyc[u] == -1) cyc[u] = cyc[f[u]], dist[u] =
    ← dist[f[u]] + 1;
            }
        }
    }
    int kth(int v, ll k) { // f^k(v)
        while (k > 0 && dist[v] > 0) v = f[v], k--;
        if (!k) return v;
        int L = cycLen[cyc[v]], p = (pos[v] + k % L) % L, u = v;
        for (int i = 0; i < (p - pos[v] + L) % L; i++) u = f[u];
        return u;
    }
};
// USES: "apply the permutation k times", Pollard's rho, "where does
    ← the ball stop",
// detecting a repeat in a modular sequence, linked-list loop detection.

```

1.17 ParallelBinarySearch

PARALLEL BINARY SEARCH - binary search the answer for MANY queries at once, sweeping the

```

// shared structure once per level instead of once per query.
// PRECONDITION: test(i) must be MONOTONE in time - once it is true it
    ← stays true. If updates
// can also undo things, parallel binary search does not apply; use
    ← offline divide & conquer over
// the timeline with a rollback DSU (02 - Data Structures/14) instead.
// Total: O(log(maxT) * (cost of one full sweep)).
// Shape: "after which update does query q first become satisfiable?"
// lo[q], hi[q] track each query's remaining range; every round,
    ← bucket the queries by their
// midpoint, replay the updates in order, and test each query when
    ← the sweep reaches its mid.

```

```

template <class Apply, class Reset, class Test>
vector<int> parallelBinarySearch(int q, int T, Apply applyUpdate, Reset
    ← reset, Test test) {
    // applyUpdate(t): apply update t to the shared structure
    // reset(): clear the structure
    // test(i): is query i satisfied by the current structure?
    vector<int> lo(q, 0), hi(q, T); // answer in
    ← [lo, hi]; hi == T means "never"
    while (true) {
        vector<vector<int>> at(T + 1);
        bool active = false;
        for (int i = 0; i < q; i++)
            if (lo[i] < hi[i]) at[lo[i] + hi[i] / 2].push_back(i),
    ← active = true;
        if (!active) break;
        reset();
        for (int t = 0; t <= T; t++) {
            if (t) applyUpdate(t - 1); // now updates
    ← 0..t-1 are applied
            for (int i : at[t]) { if (test(i)) hi[i] = t; else lo[i] =

```

```

    ↪ t + 1; }
    }
    }
    return lo; // lo[i] == T
    ↪ => never satisfied
}
// CONCRETE USES
// - "for each query, the first moment a path/edge set connects u and
    ↪ v" (structure = DSU)
// - k-th smallest in a range, offline (structure = BIT over values,
    ↪ updates = insert a value)
// - "smallest capacity so that a flow/matching of size k exists"
// - anything where a single query's binary search would need its own
    ↪ O(T) replay.
// If the structure supports ROLLBACK, prefer divide-and-conquer over
    ↪ the timeline instead
// (see OfflineDynamicConnectivity) - it is one pass instead of log
    ↪ passes.

```

1.18 MeetInTheMiddle

MEET IN THE MIDDLE - $n \leq 40$. Split into halves, enumerate $2^{n/2}$ subsets each side,

```

// then recombine by sorting + binary search / two pointers.  $O(2^{n/2})$ 
    ↪  $\times n$ ).

// All subset sums of a half, as a sorted vector.
vector<ll> subsetSums(const vector<ll>& a) {
    int n = a.size();
    vector<ll> s(1 << n, 0);
    for (int m = 1; m < (1 << n); m++) {
        int b = __builtin_ctz(m);
        s[m] = s[m ^ (1 << b)] + a[b];
    }
    return s;
}

// #subsets with sum <= S. Works with NEGATIVE values too (no
    ↪ pre-filter on the left half).
ll countAtMost(vector<ll> a, ll S) {
    int n = a.size(), h = n / 2;
    vector<ll> L(a.begin(), a.begin() + h), R(a.begin() + h, a.end());
    auto ls = subsetSums(L), rs = subsetSums(R);
    sort(all(rs));
    ll c = 0;
    for (ll x : ls) c += upper_bound(all(rs), S - x) - rs.begin();
    return c;
}

// Largest achievable sum <= S (the classic "closest subset sum from
    ↪ below").
ll bestAtMost(vector<ll> a, ll S) {
    int n = a.size(), h = n / 2;
    vector<ll> L(a.begin(), a.begin() + h), R(a.begin() + h, a.end());
    auto ls = subsetSums(L), rs = subsetSums(R);
    sort(all(rs));
    ll best = LLONG_MIN;
    for (ll x : ls) { // no 'x <= S'
        ↪ filter: x may be > S
        auto it = upper_bound(all(rs), S - x); // and still
        ↪ pair with a negative right sum
        if (it != rs.begin()) best = max(best, x + *prev(it));
    }
    return best;
}

// OTHER SHAPES
// - EXACT target: put one half in a hash map, look up (target - x)
    ↪ from the other.
// - 4-SUM: split the four arrays 2+2, hash all pairwise sums.
// - "choose exactly k items": bucket each half's subsets BY POPCOUNT,
    ↪ then pair bucket i with
// bucket k-i (sort each bucket once).
// - Optimising two objectives: sort one half by objective A and take
    ↪ a prefix maximum of
// objective B, so each query is one binary search.
// If  $n \leq 20$  a plain  $2^n$  loop is simpler; meet in the middle earns its
    ↪ keep from  $n = 30$  upward.

```

1.19 Scheduling

SCHEDULING / EXCHANGE-ARGUMENT THEOREM CARD

```

/* ===== SCHEDULING / EXCHANGE-ARGUMENT THEOREM CARD
    ↪ =====
Each of these turns a hard-looking problem into a sort. Prove any new
    ↪ one by the EXCHANGE
ARGUMENT: assume an optimal schedule, swap two ADJACENT jobs, and
    ↪ compare the two costs.

```

SMITH'S RULE (1 || sum $w_j C_j$: minimise the weighted sum of
 ↪ completion times)
 Sort by p_j / w_j ASCENDING. Adjacent swap of h, k improves iff
 ↪ $w_k p_h - w_h p_k > 0$,
 so ALWAYS compare by cross-multiplication ($w_k p_h$ vs $w_h p_k$),
 ↪ never by float division.

MOORE-HODGSON (1 || sum U_j : minimise the NUMBER of late jobs), $O(n \log n)$
 ↪ log n)
 Sort by deadline ascending; keep a max-heap of the processing times
 ↪ taken so far.
 For each job: take it, push p_j , add p_j to the running time; if the
 ↪ running time exceeds
 this job's deadline, pop the largest p and subtract it. The heap
 ↪ size is the answer
 (the rejected jobs can go last in any order).

EARLIEST DEADLINE FIRST (preemptive, one machine)
 If ANY schedule meets all deadlines, EDF does. Feasibility check for
 ↪ non-preemptive
 unit-machine: sort by deadline and verify prefix sums of p_j never
 ↪ exceed d_j .

JOHNSON'S RULE ($F2 || C_{\max}$: two-machine flow shop, minimise makespan)
 Set A = jobs with $p_1 < p_2$, sorted by p_1 ASCENDING
 Set B = jobs with $p_1 \geq p_2$, sorted by p_2 DESCENDING
 Optimal order = A then B.

LAWLER (1 | prec | $f_{\max}$: minimise the maximum penalty, with
 ↪ precedence, $O(n^2)$)
 Build the schedule BACK TO FRONT. With t = total remaining time,
 ↪ among the jobs that have
 no unscheduled successor pick the one minimising $f_j(t)$, and place
 ↪ it last.

CLASSIC GREEDY COMPANIONS
 Interval scheduling (max #non-overlapping): sort by RIGHT endpoint,
 ↪ take greedily.
 Interval point cover (stab all intervals): sort by right endpoint,
 ↪ place a point at it.
 Minimum #machines for overlapping intervals: max overlap = sweep
 ↪ +1/-1 over endpoints.
 Deadline scheduling with profits (each job unit length): sort by
 ↪ profit DESC, place each
 job as late as possible before its deadline (DSU "next free slot")
 ↪ makes it near-linear).
 Fractional knapsack: sort by value/weight DESC.
 Minimise sum of $|x - a_i|$: x = the MEDIAN. Minimise sum of $(x -$
 ↪ $a_i)^2$: x = the MEAN.

```

=====
    ↪ */

// Moore-Hodgson: maximum number of jobs finished on time. jobs =
    ↪ {processing time, deadline}.
int maxOnTime(vector<pair<ll, ll>> jobs) {
    sort(all(jobs), [](auto& a, auto& b) { return a.second < b.second;
    ↪ });
    priority_queue<ll> pq;
    ll t = 0;
    for (auto [p, d] : jobs) {
        t += p, pq.push(p);
        if (t > d) t -= pq.top(), pq.pop();
    }
    return pq.size();
}

// Johnson's rule: optimal two-machine flow-shop order (returns the job
    ↪ indices).
vector<int> johnson(const vector<pair<ll, ll>>& jobs) { // {p on
    ↪ machine 1, p on machine 2}
    vector<int> A, B;
    for (int i = 0; i < (int)jobs.size(); i++)
        (jobs[i].first < jobs[i].second ? A : B).push_back(i);
    sort(all(A), [&](int x, int y) { return jobs[x].first <
    ↪ jobs[y].first; });
    sort(all(B), [&](int x, int y) { return jobs[x].second >
    ↪ jobs[y].second; });
    A.insert(A.end(), all(B));
    return A;
}

```

1.20 Josephus

JOSEPHUS PROBLEM. n people in a circle $0..n-1$, every k -th is removed; who survives?

```
// Recurrence: J(1) = 0, J(n) = (J(n-1) + k) mod n. O(n).
ll josephus(ll n, ll k) { ll r = 0; for (ll i = 2; i <= n; i++) r = (r
    ↪ + k) % i; return r; }
// k = 2 closed form: write  $n = 2^m + l$  with  $0 \leq l < 2^m$ , the survivor
    ↪ is  $2 * l$  (0-indexed).
ll josephus2(ll n) { return 2 * (n - (1LL << (63 -
    ↪ __builtin_clzll(n)))); }
// LARGE n, SMALL k: one full lap kills floor(n/k) people, so recurse
    ↪ on  $n - \text{floor}(n/k)$  and map the
// index back. O(k log n).
ll josephusFast(ll n, ll k) {
    if (n == 1) return 0;
    if (k == 1) return n - 1;
    if (k > n) return (josephusFast(n - 1, k) + k) % n;
    ll r = josephusFast(n - n / k, k) - n % k;
    return r < 0 ? r + n : r + r / (k - 1);
}
// ORDER OF ELIMINATION / the j-th person to die: keep a BIT over alive
    ↪ flags and binary-search the
// ((pos + k - 1) mod alive + 1)-th alive person each step. O(n log n)
    ↪ and handles "k changes".
// LAST TWO SURVIVORS: run the recurrence starting from J(2) = {0, 1}
    ↪ instead of J(1) = 0.

// OTHER ONE-LINE CLASSICS worth having on paper:
// GRAY CODE: g(i) = i ^ (i >> 1); inverse: x ^ = x >> 1, x >> 2,
    ↪ ... until 0.
// n-th PERMUTATION (factorial number system): digit i = k / (n-1-i)!,
    ↪ then k %= (n-1-i)!.
// 15-PUZZLE SOLVABLE iff (#inversions of the tiles, blank excluded) +
    ↪ (row of the blank counted
// from the BOTTOM, 1-based) is EVEN, for a 4x4 board. For an odd
    ↪ width w, solvable iff the
// inversion count is even. General w x h: inversions + (blank row
    ↪ from bottom) parity rule above
// when w is even, plain inversion parity when w is odd.
// TOWER OF HANOI:  $2^n - 1$  moves; on move t (1-based) move disk (t &
    ↪ -t) counted by trailing zeros,
// and for odd n the smallest disk cycles A->C->B->A, for even n
    ↪ A->B->C->A.
// BULLS AND COWS / stable-marriage style pairings: Gale-Shapley is
    ↪  $O(n^2)$ , always terminates, and
// is optimal for the proposing side.
// CATALAN-STYLE BALLOT COUNTING: paths from (0,0) to (n,m) staying
    ↪ strictly above  $y = x$  are
//  $(n-m)/(n+m) * C(n+m, m)$  (Bertrand's ballot theorem).
```

1.21 BigInt

BIG INTEGERS, non-negative, base $1e9$. Enough for the usual "the answer does not fit in 64 bits"

```
// problems: exact factorials / Catalan numbers / determinants, huge
    ↪ Fibonacci, exact comparisons.
// Add/sub O(n), schoolbook multiply O(n*m) (fine to  $\sim 1e4$  digits),
    ↪ divide by a small int O(n).
// If you need big*big beyond that, convolve the limbs with FFT and
    ↪ carry.
struct Big {
    static const ll B = 1000000000;
    vector<ll> d;
    ↪ little-endian limbs, base  $1e9$ 
    Big(ll v = 0) { for (; v > 0; v /= B) d.push_back(v % B); }
    Big(const string& s) {
        for (int i = s.size(); i > 0; i -= 9)
            d.push_back(stoll(s.substr(max(0, i - 9), min(9, i))));
        trim();
    }
    void trim() { while (!d.empty() && d.back() == 0) d.pop_back(); }
    bool operator<(const Big& o) const {
        if (d.size() != o.d.size()) return d.size() < o.d.size();
        for (int i = d.size() - 1; i >= 0; i--) if (d[i] != o.d[i])
            ↪ return d[i] < o.d[i];
        return false;
    }
    Big operator+(const Big& o) const {
        Big r; ll c = 0;
        for (size_t i = 0; i < max(d.size(), o.d.size()) || c; i++) {
            if (i < d.size()) c += d[i];
            if (i < o.d.size()) c += o.d[i];
            r.d.push_back(c % B), c /= B;
        }
        return r;
    }
    Big operator-(const Big& o) const {
```

```
    ↪ requires *this >= 0
        Big r = *this; ll c = 0;
        for (size_t i = 0; i < r.d.size(); i++) {
            r.d[i] -= c + (i < o.d.size() ? o.d[i] : 0);
            if ((c = r.d[i] < 0)) r.d[i] += B;
        }
        r.trim();
        return r;
    }
    Big operator*(const Big& o) const {
        if (d.empty() || o.d.empty()) return Big();
        vector<ll> t(d.size() + o.d.size(), 0);
        for (size_t i = 0; i < d.size(); i++)
            for (size_t j = 0; j < o.d.size(); j++) t[i + j] +=
                ↪ (ll)d[i] * o.d[j];
        Big r; ll c = 0;
        for (size_t i = 0; i < t.size(); i++) c += t[i],
            ↪ r.d.push_back((ll)(c % B)), c /= B;
        for (; c; c /= B) r.d.push_back((ll)(c % B));
        r.trim();
        return r;
    }
    Big operator/(ll v) const { return *this * Big(v); }
    Big operator/(ll v) const { // small
        ↪ divisor only
        Big r = *this; ll c = 0;
        for (int i = r.d.size() - 1; i >= 0; i--) c = c * B + r.d[i],
            ↪ r.d[i] = (ll)(c / v), c %= v;
        r.trim();
        return r;
    }
    ll operator%(ll v) const {
        ll c = 0;
        for (int i = d.size() - 1; i >= 0; i--) c = (c * B + d[i]) % v;
        return (ll)c;
    }
    string str() const {
        if (d.empty()) return "0";
        string s = to_string(d.back());
        for (int i = d.size() - 2; i >= 0; i--) {
            string t = to_string(d[i]);
            s += string(9 - t.size(), '0') + t;
        }
        return s;
    }
};
// SIGNED: keep a separate sign flag and dispatch +/- to add/sub on
    ↪ magnitudes; comparison first.
// BASE  $1e9$  packs 9 decimal digits per limb, so printing is trivial -
    ↪ that is the whole point.
// FAST DIVISION of two Bigs is long and almost never needed; if a
    ↪ problem really wants it, binary
// search the quotient limb by limb, or move to Python-style thinking
    ↪ and avoid it.
// SQRT: binary search with the multiply above, or Newton on doubles
    ↪ then fix up by a few steps.
```

1.22 RankingAndDates

RANKING / UNRANKING COMBINATORIAL OBJECTS, and CALENDAR ARITHMETIC. Small, and each one costs

```
// 20 minutes to re-derive under pressure.

// --- FACTORIAL NUMBER SYSTEM: rank and unrank a PERMUTATION.  $O(n^2)$ 
    ↪ as written; use an
// order-statistic tree or a BIT for  $O(n \log n)$  when  $n$  is large.
// order(p) = how many permutations of the same multiset are
    ↪ lexicographically smaller.
ll permRank(vector<int> p) { // p is
    ↪ a permutation of  $0..n-1$ 
    int n = p.size();
    ll r = 0, f = 1;
    for (int i = n - 1; i >= 0; i--) {
        int c = 0;
        for (int j = i + 1; j < n; j++) if (p[j] < p[i]) c++; // the
            ↪ Lehmer code digit
        r += c * f, f *= (n - i);
    }
    return r;
}
vector<int> permUnrank(int n, ll k) { // k-th
    ↪ permutation, 0-indexed
    vector<int> avail(n), p(n);
    iota(all(avail), 0);
    vector<ll> f(n + 1, 1);
    for (int i = 1; i <= n; i++) f[i] = f[i - 1] * i;
    for (int i = 0; i < n; i++) {
```

```

    ll d = k / f[n - 1 - i];
    k %= f[n - 1 - i];
    p[i] = avail[d], avail.erase(avail.begin() + d);
}
return p;
}
// NEXT PERMUTATION in place is std::next_permutation, O(n) - use it
// unless you need the rank.
// COMBINATIONS: rank of a k-subset {c_1<...<c_k} of [n] is sum C(c_i,
// i); unrank greedily by
// finding the largest c with C(c, i) <= remaining. Same shape, C
// instead of factorials.
// GRAY CODE is g = i ^ (i >> 1) (06 - grayCode.cpp).

// --- BALANCED BRACKET SEQUENCES: rank, unrank, and the next one in
// lexicographic order, with
// '( < ' ). The counting table is the BALLOT NUMBERS: f[i][j] =
// number of ways to finish when i
// characters remain and j brackets are still open. f[n][0] is the
// Catalan number C_{n/2}.
// USES: "the k-th valid bracket sequence", "how many valid sequences
// are smaller than this one",
// iterating all sequences of length 2m without generating and
// filtering. Same skeleton works for
// any object counted by a DP: the rank is a walk that adds the subtree
// sizes you skip over.
// O(n^2) to build the table, O(n) per rank / unrank / next. Overflows
// for n > 60 - use __int128
// or a modulus if you only need the count.
vector<vector<ll>> ballot(int n) { // n =
    // total length, must be even
    vector<vector<ll>> f(n + 1, vector<ll>(n + 2, 0));
    f[0][0] = 1;
    for (int i = 1; i <= n; i++)
        for (int j = 0; j <= i; j++)
            f[i][j] = f[i - 1][j + 1] + (j ? f[i - 1][j - 1] : 0); //
    // place '( then ')'
    return f;
}
ll bracketRank(const string& s, const vector<vector<ll>>& f) { //
    // 0-indexed rank, s must be valid
    int n = s.size(), j = 0;
    ll r = 0;
    for (int i = 0; i < n; i++) {
        if (s[i] == ')') r += f[n - i - 1][j + 1]; // skip
        // every sequence with '(' here
        j += s[i] == '(' ? 1 : -1;
    }
    return r;
}
string bracketUnrank(int n, ll k, const vector<vector<ll>>& f) {
    string s;
    int j = 0;
    for (int i = 0; i < n; i++) {
        ll c = f[n - i - 1][j + 1]; //
        // completions starting with '('
        if (k < c) s += '(', j++;
        else k -= c, s += ')', j--;
    }
    return s;
}
bool nextBracket(string& s) { // in
    // place; false if s is the last
    int n = s.size(), d = 0;
    for (int i = n - 1; i >= 0; i--) {
        d += s[i] == '(' ? -1 : 1;
        if (s[i] == '(' && d > 0) { // flip
            // it to ')' and reset the rest
            d--;
            int open = (n - i - 1 - d) / 2;
            s = s.substr(0, i) + ')' + string(open, '(') + string(n - i
            // - 1 - open, '(');
            return true;
        }
    }
    return false;
}

// --- JULIAN DAY NUMBER: date <-> integer, closed form, no loops,
// correct for the Gregorian
// calendar including every leap rule. Date differences, "what
// weekday", and "add 1000 days" all
// become integer arithmetic once you are here.
ll dateToInt(int y, int m, int d) { // y-m-d
    // Gregorian -> day number
    return 1461 * (y + 4800 + (m - 14) / 12) / 4
        + 367 * (m - 2 - (m - 14) / 12 * 12) / 12

```

```

        - 3 * ((y + 4900 + (m - 14) / 12) / 100) / 4 + d - 32075;
}
array<int, 3> intToDate(ll jd) { //
    // inverse: {y, m, d}
    ll x, n, i, j, dd;
    x = jd + 68569, n = 4 * x / 146097, x -= (146097 * n + 3) / 4;
    i = 4000 * (x + 1) / 1461001, x -= 1461 * i / 4 - 31;
    j = 80 * x / 2447, dd = x - 2447 * j / 80, x = j / 11;
    return {(int)(100 * (n - 49) + i + x), (int)(j + 2 - 12 * x),
    // (int)dd};
}
int weekday(ll jd) { return (int)((jd + 1) % 7); } // 0 =
// Sunday
// LEAP YEAR: y % 4 == 0 && (y % 100 != 0 || y % 400 == 0).
// ZELLER'S CONGRUENCE gives the weekday directly, but the day number
// is strictly more useful.

// --- STERN-BROCOT SEARCH: the simplest fraction satisfying a MONOTONE
// predicate. ---
// Descend the median tree, but take many steps in one direction at
// once (double the step size)
// so a 1e18 denominator costs O(log^2) instead of O(answer).
// USES: "smallest denominator q with property P", recovering p/q from
// a decimal prefix, the
// simplest fraction strictly inside an interval, and best rational
// approximation with a bound.
// ok() must be MONOTONE in the value p/q: false on small fractions,
// true from some point on.
// Returns the SMALLEST such fraction with denominator <= N. Each
// iteration takes as many steps
// as possible in one direction (found by doubling, then binary
// search), so it is O(log^2 N)
// rather than O(answer) - the difference between instant and hopeless
// at N = 1e18.
template <class F>
pair<ll, ll> fracSearch(ll N, F ok) {
    ll lp = 0, lq = 1, rp = 1, rq = 0; // 0/1
    // (not ok) .. 1/0 (ok)
    for (;;) {
        ll moved = 0;
        auto walk = [&](ll& ap, ll& aq, ll& bp, ll& bq, bool want) { //
            // steps from a toward b
            ll k = 0, s = 1;
            auto test = [&](ll t) {
                return aq + t * bq <= N && ok(ap + t * bp, aq + t * bq)
            };
            while (test(k + s)) k += s, s *= 2;
            for (s /= 2; s; s /= 2) if (test(k + s)) k += s;
            ap += k * bp, aq += k * bq, moved += k;
        };
        walk(lp, lq, rp, rq, false); // push
        // the FALSE end up
        if (lq + rq > N) return {rp, rq};
        walk(rp, rq, lp, lq, true); // pull
        // the TRUE end down
        if (lq + rq > N || !moved) return {rp, rq}; //
        // !moved: nothing left to refine
    }
}
// The Stern-Brocot tree IS the continued fraction expansion: going
// left/right k times is one
// partial quotient. See 04 - Number Theory/05 - Rational for the CF
// machinery and semiconvergents.

```

1.23 MatroidIntersection

MATROID INTERSECTION - the largest set that is independent in TWO matroids at once, and the

```

// max-weight version. Ground set is {0..n-1}; each matroid arrives as
// an ORACLE object exposing
// build(S) prepare for the current independent set S (a sorted
// vector of element ids)
// canAdd(e) is S + e independent? (e not in S)
// canSwap(e, x) is S - x + e independent? (e not in S, x in S)
// build is called once per augmenting phase, so it may rebuild
// everything from scratch - that is
// exactly why the concrete oracles below can get away with |S| + 1
// rebuild structures per phase.
// matroidIssect returns a maximum-CARDINALITY common independent set,
// sorted, as element ids.
// matroidIssectW returns a maximum-WEIGHT one over all cardinalities;
// force maximum cardinality
// first by adding C > sum|w| to every weight.
// WHEN: one selection problem carrying TWO independent structural
// constraints. Colourful spanning
// tree / spanning tree with at most c_i edges of colour i, and rainbow

```

```

    ↪ spanning tree (c_i = 1):
// graphic + partition. Arborescence: graphic on the underlying
    ↪ undirected graph + partition by
// head vertex with cap 1, so "one incoming edge per vertex, no cycle".
    ↪ Spanning tree with bounded
// degrees: partition the edges by an endpoint - legal ONLY if no edge
    ↪ joins two constrained
// vertices, otherwise an edge lands in two classes and the reduction
    ↪ is silently wrong. Bipartite
// matching is two partition matroids (but use Hopcroft-Karp). THREE
    ↪ matroids is NP-hard.
// MIN-MAX: max |I| = min over U of (rank1(U) + rank2(complement of
    ↪ U)), handy for proving a bound
// or for a "why is the answer stuck at k" argument.
// COMPLEXITY in oracle calls: one phase pops each element once and
    ↪ spends O(|S|) or O(n) calls on
// it, so O(r n) calls plus 2 builds per phase, and there are r + 1
    ↪ phases: O(r^2 n) calls total.
// O(r^1.5 n) is the best known bound for this scheme and needs
    ↪ Cunningham's phase batching
// (augment along many shortest paths per BFS, Hopcroft-Karp style) -
    ↪ rarely worth typing. Budget
// r * n * (cost of one oracle call) per phase and multiply by the
    ↪ number of phases. Weighted:
// Bellman-Ford over n nodes and O(r n) arcs, so O(r n^2) per phase on
    ↪ top of the same oracle work.
// Memory is the oracles': O(r V) ints for graphic, O(63 r) for xor,
    ↪ O(colours) for partition.
// DEGENERATE: n = 0 or every element a LOOP (self-loop edge, zero
    ↪ vector, colour with cap 0)
// returns {}; loops fail canAdd on an empty S and are never picked.
    ↪ Duplicate / parallel elements
// are fine. Ties are broken arbitrarily - no lexicographic guarantee.
    ↪ Weighted with all weights
// negative returns {}. Both routines start from S = {} and cost
    ↪ nothing extra if the answer is 0.
// EXACT: pure combinatorics, no floating point; weights are ll and one
    ↪ path cost sums at most n of
// them, so |w| <= 1e17 is safe. Two traps that fail SILENTLY: the
    ↪ unweighted search must use BFS
// (a shortest augmenting path, fewest arcs) and the weighted one must
    ↪ break cost ties by fewest
// arcs - swap in a DFS or drop the tiebreak and you get a set that is
    ↪ not independent.
template <class M1, class M2>
vector<int> matroidIsect(int n, M1& m1, M2& m2) {
    vector<char> in(n, 0);
    vector<int> S;
    for (;;) {
        m1.build(S), m2.build(S);
        vector<int> par(n, -2), q; // -2 unvisited,
        ↪ -1 source; q is the BFS order
        int t = -1;
        for (int e = 0; e < n; e++) if (!in[e] && m1.canAdd(e)) par[e] =
        ↪ -1, q.pb(e);
        for (int h = 0; h < sz(q) && t < 0; h++) {
            int u = q[h];
            if (in[u]) { // leaving S: u
                ↪ out, some v in, keep M1 happy
                for (int v = 0; v < n; v++)
                    if (par[v] == -2 && !in[v] && m1.canSwap(v, u))
                        ↪ par[v] = u, q.pb(v);
                } else if (m2.canAdd(u)) t = u; // reached the
                ↪ sink side: augment
                else for (int v : S) if (par[v] == -2 && m2.canSwap(u, v))
                    ↪ par[v] = u, q.pb(v);
            }
            if (t < 0) return S; // no augmenting
            ↪ path: S is maximum
            for (int v = t; v != -1; v = par[v]) in[v] ^= 1;
            S.clear();
            for (int e = 0; e < n; e++) if (in[e]) S.pb(e);
        }
    }
}
template <class M1, class M2>
vector<int> matroidIsectW(int n, vector<ll> w, M1& m1, M2& m2) {
    vector<char> in(n, 0);
    vector<int> S, best;
    ll cur = 0, bw = 0;
    for (;;) {
        m1.build(S), m2.build(S);
        vector<pair<ll, int>> d(n, {llinf, inf}); // (path cost,
        ↪ arcs) - lexicographic
        vector<int> par(n, -1);
        vector<pii> ed;
        vector<char> snk(n, 0);
        for (int e = 0; e < n; e++) if (!in[e]) {

```

```

            if (m1.canAdd(e)) d[e] = {-w[e], 1}; // sources; a
            ↪ node costs -w in, +w out
            if (m2.canAdd(e)) snk[e] = 1;
            for (int x : S) {
                if (m1.canSwap(e, x)) ed.pb({x, e});
                if (m2.canSwap(e, x)) ed.pb({e, x});
            }
            for (int it = 0; it < n; it++) { // Bellman-Ford:
                ↪ weights are negative on entry
                bool ch = false;
                for (auto& [u, v] : ed) if (d[u].fi < llinf) {
                    pair<ll, int> nd = {d[u].fi + (in[v] ? w[v] : -w[v]),
                    ↪ d[u].se + 1};
                    if (nd < d[v]) d[v] = nd, par[v] = u, ch = true;
                }
                if (!ch) break;
            }
            int t = -1;
            for (int e = 0; e < n; e++) if (snk[e] && d[e].fi < llinf && (t
            ↪ < 0 || d[e] < d[t])) t = e;
            if (t < 0) break;
            cur -= d[t].fi; // the path cost
            ↪ is minus the weight gained
            for (int v = t; v != -1; v = par[v]) in[v] ^= 1;
            S.clear();
            for (int e = 0; e < n; e++) if (in[e]) S.pb(e);
            if (cur > bw) bw = cur, best = S; // S is now
            ↪ optimal among sets of size |S|
        }
        return best;
    }
}
// --- ORACLE 1: COLOURFUL / PARTITION matroid. Independent iff colour
    ↪ c is used <= cap[c] times.
struct ColourM {
    vector<int> col, cap, cnt; // col[e] in [0,
    ↪ |cap|)
    ColourM(vector<int> col, vector<int> cap) : col(col), cap(cap),
    ↪ cnt(cap.size()) {}
    void build(const vector<int>& S) {
        fill(all(cnt), 0);
        for (int e : S) cnt[col[e]]++;
    }
    bool canAdd(int e) { return cnt[col[e]] < cap[col[e]]; }
    bool canSwap(int e, int x) { return canAdd(e) || col[e] == col[x]; }
};
// --- ORACLE 2: GRAPHIC matroid. Ground set = edges over vertices
    ↪ 0..V-1, independent iff acyclic.
// One DSU for S and one for each S - x, all rebuilt per phase: O(r^2)
    ↪ per build, O(r V) memory.
struct GraphicM {
    int V; vector<pii> e; vector<int> pos; vector<vi> d; // d[0] = S,
    ↪ d[i + 1] = S without S[i]
    GraphicM(int V, vector<pii> e) : V(V), e(e) {}
    int f(vi& p, int x) { return p[x] == x ? x : p[x] = f(p, p[x]); }
    bool ok(vi& p, int i) { return f(p, e[i].fi) != f(p, e[i].se); }
    void build(const vector<int>& S) {
        int m = sz(S);
        pos.assign(sz(e), 0), d.assign(m + 1, vi(V));
        for (auto& p : d) iota(all(p), 0);
        for (int i = 0; i < m; i++) pos[S[i]] = i;
        for (int i = 0; i < m; i++) for (int j = 0; j <= m; j++) if (j
        ↪ != i + 1) {
            vi& p = d[j];
            p[f(p, e[S[i]].fi)] = f(p, e[S[i]].se); // union; a no-op
            ↪ when already joined
        }
    }
    bool canAdd(int x) { return ok(d[0], x); }
    bool canSwap(int x, int y) { return ok(d[pos[y] + 1], x); }
};
// --- ORACLE 3: LINEAR matroid over GF(2). Ground set = masks,
    ↪ independent iff xor-independent.
// b[j][i] is the basis vector whose top bit is i. Masks must be in [0,
    ↪ 2^62); 0 is a loop.
struct XorM {
    vector<ll> v; vector<int> pos; vector<vector<ll>> b; // b[0] = S,
    ↪ b[i + 1] = S without S[i]
    XorM(vector<ll> v) : v(v) {}
    bool ok(vector<ll>& b, ll x, bool ins = false) {
        for (int i = 62; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { if (ins) b[i] = x; return true; }
            x ^= b[i];
        }
        return false;
    }
    void build(const vector<int>& S) {

```

```

    int m = sz(S);
    pos.assign(sz(v), 0), b.assign(m + 1, vector<ll>(63, 0));
    for (int i = 0; i < m; i++) pos[S[i]] = i;
    for (int i = 0; i < m; i++)
        for (int j = 0; j <= m; j++) if (j != i + 1) ok(b[j],
    ↪ v[S[i]], true);
}
bool canAdd(int x) { return ok(b[0], v[x]); }
bool canSwap(int x, int y) { return ok(b[pos[y] + 1], v[x]); }
};
// MORE ORACLES, all trivial: UNIFORM U(k, n) - canAdd = sz(S) < k,
    ↪ canSwap = true. FREE matroid
// (everything independent) - both true; intersecting with it
    ↪ degenerates to "greedy on one
// matroid". TRANSVERSAL matroid (elements matchable into a bipartite
    ↪ graph) - one Kuhn augmenting
// path per query, legal but slow. COGRAPHIC (complement stays
    ↪ connected).
// ONE matroid only, max weight: sort by weight descending and add
    ↪ while canAdd - that is Kruskal,
// and matroid intersection is the price you pay for the second
    ↪ constraint.
// MATROID UNION and "partition the ground set into k independent sets"
    ↪ both reduce to intersection.

```

1.24 SchreierSims

SCHREIER-SIMS - a strong generating set (a stabiliser chain) for the permutation group generated

```

// by the given permutations of {0..n-1}. Gives |G| exactly, membership
    ↪ testing in O(n^2), and an
// explicit word in the generators for any member. Permutations are
    ↪ 0-indexed images: p[i] is where
// i goes, and mul(a, b) means "apply a, THEN b", so (a*b)[i] =
    ↪ b[a[i]]. Level i of the chain holds
// coset representatives of the pointwise stabiliser of 0..i-1, indexed
    ↪ by the image of i, so
// |G| = product of the transversal sizes and every element factors
    ↪ uniquely as one rep per level.
// WHEN: "you may apply these shuffles to the array any number of
    ↪ times, in any order - how many
// distinct arrangements are reachable, and is arrangement X
    ↪ reachable". Model a shuffle as the
// permutation it applies, then the reachable arrangements are exactly
    ↪ the group elements: order()
// counts them and has() decides reachability. word() prints the moves
    ↪ that get you there. Also:
// the group generated by a puzzle's legal moves, the size of a
    ↪ symmetry group before a Burnside
// count, and "do these generators generate all of S_n" (compare
    ↪ order() with n!).
// CONVENTION TRAP: an arrangement is the permutation taking the START
    ↪ to it, so ask has(q) with
// q[i] = the index the item now at position i started at (build it
    ↪ once, consistently, then stay
// consistent). If the array has DUPLICATE values, the number of
    ↪ distinct arrangements is |G| over
// the size of the stabiliser of that multiset, NOT |G| - this file
    ↪ does not give you that.
// COMPLEXITY: level i holds at most n - i reps, so O(n^2) stored
    ↪ permutations and O(n^3) ints of
// memory (under 1 MB even at n = 60). The constructor sifts every rep
    ↪ at level i against every rep
// at a level >= i: O(n^4) sifts costing O(n^2) each, so O(n^6) when
    ↪ the group is all of S_n.
// MEASURED on three random generators of S_n: n = 40 takes 0.5 s, n =
    ↪ 50 takes 2 s, n = 60 takes
// 6 s, and that is the wall - past it you need a real Schreier-Sims
    ↪ (Schreier generators kept per
// level, plus Jerrum's filter) or the randomised variant. has() and
    ↪ word() are O(n^2) per query.
// DEGENERATE: no generators or only identities gives order() == 1 and
    ↪ has(id) == true; n = 1 the
// same. Duplicate generators are free. word(identity) is the empty
    ↪ word.
// EXACT: integers only. OVERFLOW: |G| <= n! exceeds ll at n = 21 (20!
    ↪ = 2.4e18 fits), so switch
// order() to __int128 (good to 33!) or accumulate modulo the answer's
    ↪ modulus for larger n.
struct PermGroup {
    int n; bool keepW;
    vector<vi> lk; // lk[i][j] =
    ↪ index of the level i rep i->j
    vector<vector<vi>> T, Ti, W; // reps, their
    ↪ inverses, and their words
    static vi mul(const vi& a, const vi& b) {
        vi c(sz(a));

```

```

        for (int i = 0; i < sz(a); i++) c[i] = b[a[i]];
        return c;
    }
    static vi inv(const vi& a) {
        vi c(sz(a));
        for (int i = 0; i < sz(a); i++) c[a[i]] = i;
        return c;
    }
    // Strip p through the chain. Returns the level where p is NEW (and
    ↪ inserts it there if add),
    // or -1 when p reduces to the identity, i.e. p is in the group. w
    ↪ accumulates the word of the
    // running residue, so it ends as the word of p^-1 when p sifts
    ↪ through.
    int sift(vi p, vi& w, bool add) {
        for (int i = 0; i < n; i++) {
            int k = lk[i][p[i]];
            if (k < 0) {
                if (!add) return i;
                lk[i][p[i]] = sz(T[i]), T[i].pb(p), Ti[i].pb(inv(p));
                if (keepW) W[i].pb(w);
                return i;
            }
            if (keepW) for (int j = sz(W[i][k]) - 1; j >= 0; j--)
                ↪ w.pb(~W[i][k][j]); // w *= rep^-1
                bool one = true; // p *= rep^-1 in
                ↪ place; now p fixes 0..i, so
                for (int j = i; j < n; j++) // only j >= i
                ↪ can move and only they matter
                    one &= (p[j] == T[i][k][p[j]]);
                if (one) return -1; // already the
                ↪ identity: skip the rest
            }
            return -1;
        }
        PermGroup(int n, const vector<vi>& gen, bool keepW = false) : n(n),
        ↪ keepW(keepW) {
            lk.assign(n, vi(n, -1)), T.resize(n), Ti.resize(n), W.resize(n);
            vi id(n);
            iota(all(id), 0);
            for (int i = 0; i < n; i++) lk[i][i] = 0, T[i].pb(id),
            ↪ Ti[i].pb(id), W[i].pb(vi());
            vector<pii> el; // every
            ↪ non-identity rep, as (level, index)
            auto push = [&](vi p, vi w) {
                int i = sift(p, w, true);
                if (i >= 0) el.pb({i, sz(T[i]) - 1});
            };
            for (int g = 0; g < sz(gen); g++) push(gen[g], vi{g});
            for (int i = 0; i < sz(el); i++) for (int j = 0; j <= i; j++) {
                ↪ // el grows as we go
                auto prod = [&](pii a, pii b) {
                    vi w;
                    if (keepW) w = W[a.fi][a.se], w.insert(w.end(),
                    ↪ all(W[b.fi][b.se]));
                    push(mul(T[a.fi][a.se], T[b.fi][b.se]), w);
                };
                if (el[i].fi <= el[j].fi) prod(el[i], el[j]); // a
                ↪ level i rep times a
                if (el[j].fi <= el[i].fi && i != j) prod(el[j], el[i]);
                ↪ // level >= i element is
            }
            // a Schreier generator
        }
        ll order() {
            ll r = 1;
            for (int i = 0; i < n; i++) r *= sz(T[i]);
            return r;
        }
        bool has(vi p) { vi w; return sift(p, w, false) < 0; }
        vi word(vi p) { // PRECONDITION:
            ↪ has(p), and keepW was set.
            vi w; // token g >= 0
            ↪ is gen[g], ~g is its inverse
            sift(p, w, false);
            reverse(all(w));
            for (int& x : w) x = ~x; // w held p^-1;
            ↪ invert it
            return w;
        }
    };
    // Rebuild p from word(p) with res = mul(res, x >= 0 ? gen[x] :
    ↪ inv(gen[~x])) over the tokens in
    // order. Word LENGTH IS NOT BOUNDED - it is a product of stored words
    ↪ and can blow up; use it only
    // where the checker accepts any move sequence. If you need short

```

↪ words, use only for small groups.
 // ORBIT of one point (which positions a single item can reach) is a
 ↪ plain BFS on the generators
 // in $O(n \cdot t)$ - do not build the whole chain for that. The level-0
 ↪ transversal IS the orbit of 0.
 // The pointwise stabiliser of $0..i-1$ is generated by the reps at
 ↪ levels $\geq i$, which is how you
 // answer "how many arrangements fix these positions" (= product of
 ↪ $|T[j]|$ for $j \geq i$).
 // Related: Burnside / Polya for counting colourings up to the group,
 ↪ and the parity argument that
 // solves the 15-puzzle style problems in one line when the group is
 ↪ the full alternating group.

1.25 RangeBitset

RANGE BITSET - a bitset with SET / RESET / FLIP of a whole range and POPCOUNT of a range, all

// in $O((r - l) / 64 + 1)$. `std::bitset` cannot do range updates at all
 ↪ (its `set()/reset()/flip()`)
 // are either one bit or the whole thing), so anything that wants "turn
 ↪ on bits $[l, r]$ " needs
 // this. Bits are $0..n-1$, ranges are INCLUSIVE on both ends, and $l > r$
 ↪ is a silent no-op.
 // WHEN: a sweep that paints intervals into one bitset per colour
 ↪ ("recolour $[l, r]$ to c " over
 // 100 colours: AND out the range from the old bitset, OR it into the
 ↪ new one); interval DP
 // reachability; "is any position in $[l, r]$ still free" answered by
 ↪ `findNext`; knapsack-style DP
 // where the transition is $x \mid= x \ll k$.
 // COMPLEXITY: $n/64$ words. $O((r-l)/64 + 1)$ per range op, $O(1)$ test/set
 ↪ of one bit, $O(n/64)$ for
 // the word-parallel operators and for a shift. The point is the 64x
 ↪ constant, not the 0.
 // EXACT: pure unsigned integer, nothing can overflow.
 // THE WORD-BOUNDARY TRAP - the only thing that ever goes wrong in this
 ↪ file:
 // - a range touches THREE pieces: the tail of word $l \gg 6$ (bits
 ↪ $l \& 63..63$), the whole words
 // strictly between, and the head of word $r \gg 6$ (bits $0..r \& 63$).
 // - if $l \gg 6 == r \gg 6$ you must do ONE masked write and return. Falling
 ↪ through to the three-piece
 // path writes bits $l \& 63..63$ AND bits $0..r \& 63$ of that same word,
 ↪ i.e. the entire word - the
 // classic off-by-one that makes this silently wrong only when l and
 ↪ r are 64-aligned near
 // each other. Test $n = 64, 65, 128$ and l, r in the same word.
 // - never shift by 64: $\sim 0ULL \ll 64$ is UB (x86 shifts by 0 and hands
 ↪ you all-ones). `mask(a, b)`
 // below shifts by a and by $63-b$ with $0 \leq a \leq b \leq 63$, so both
 ↪ counts are in $[0, 63]$.
 // - keep the bits $\geq n$ zero (`trim()`), or `count()` and `findNext()` over
 ↪ the whole bitset lie.

`typedef unsigned long long ull;`

```

struct RangeBitset {
    int n; vector<ull> w; // w[i]
    ↪ holds bits 64i .. 64i + 63
    RangeBitset(int n = 0) : n(n), w((n + 63) / 64, 0) {}
    static ull mask(int a, int b) { // bits
    ↪ a..b of ONE word, 0<=a<=b<=63
        return (~0ULL << a) & (~0ULL >> (63 - b));
    }
    template <class F> void each(int l, int r, F f) { // f(word,
    ↪ mask) over the range
        if (l > r) return;
        int a = l >> 6, b = r >> 6;
        if (a == b) { f(a, mask(l & 63, r & 63)); return; } // SAME
    ↪ word: one write, then STOP
        f(a, mask(l & 63, 63)), f(b, mask(0, r & 63)); // the two
    ↪ partial words
        for (int i = a + 1; i < b; i++) f(i, ~0ULL); // the
    ↪ full words between them
    }
    void set(int l, int r) { each(l, r, [&](int i, ull m) { w[i] |=
    ↪ m; }); }
    void reset(int l, int r) { each(l, r, [&](int i, ull m) { w[i] &=
    ↪ ~m; }); }
    void flip(int l, int r) { each(l, r, [&](int i, ull m) { w[i] ^=
    ↪ m; }); }
    int count(int l, int r) {
        int c = 0;
        each(l, r, [&](int i, ull m) { c += ones(w[i] & m); });
        return c;
    }
    int count() { return count(0, n - 1); } // n == 0
    ↪ gives l > r, so 0
  
```

```

bool test(int i) const { return w[i >> 6] >> (i & 63) & 1; }
void set(int i) { w[i >> 6] |= 1ULL << (i & 63); }
void reset(int i) { w[i >> 6] &= ~(1ULL << (i & 63)); }
void flip(int i) { w[i >> 6] ^= 1ULL << (i & 63); }
int findNext(int i) { // first
    ↪ set bit >= i, or n if none
        if (i < 0) i = 0;
        if (i >= n) return n;
        int b = i >> 6;
        ull c = w[b] & (~0ULL << (i & 63)); // drop
    ↪ the bits before i
        while (!c) { if (++b == sz(w)) return n; c = w[b]; }
        return b * 64 + lsb(c);
    }
    void trim() { if (n & 63) w.back() &= mask(0, (n - 1) & 63); } //
    ↪ bits >= n must stay 0
    void operator|=(const RangeBitset& o) { for (int i = 0; i < sz(w);
    ↪ i++) w[i] |= o.w[i]; }
    void operator&=(const RangeBitset& o) { for (int i = 0; i < sz(w);
    ↪ i++) w[i] &= o.w[i]; }
    void operator^=(const RangeBitset& o) { for (int i = 0; i < sz(w);
    ↪ i++) w[i] ^= o.w[i]; }
    void orShl(int k) { // x |= x
    ↪ << k, the subset-sum step
        int q = k >> 6, s = k & 63;
        for (int i = sz(w) - 1; i >= q; i--) { //
    ↪ DESCENDING: every source word is
            ull v = w[i - q] << s; // below
    ↪ i, so still unmodified
            if (s && i - q) v |= w[i - q - 1] >> (64 - s); // s == 0
    ↪ would be a shift by 64
            w[i] |= v;
        }
        trim(); // bits
    }
    ↪ that slid past n
}
// VARIANTS / RELATED
// - Whole-bitset AND/OR/XOR plus _Find_first/_Find_next only: use
    ↪ std::bitset, it is faster and
// already written. Come here the moment a RANGE update appears.
// - Range ASSIGN with  $O(\log)$  amortized per op and no  $n/64$  term: the
    ↪ interval set ("Chtholly",
// a std::set of maximal constant runs). It does assign, not flip,
    ↪ and needs random data.
// - Range assign + range count with true polylog: a lazy segment tree.
    ↪ This struct wins up to
// n around  $1e7$  purely on constant factor, and is a third of the code.
// - Static "count ones in  $[l, r]$ ": a prefix popcount array, no
    ↪ structure needed.
// - findPrev: mask with mask(0, i & 63) and scan words downward,
    ↪ taking msb().
  
```

1.26 TrygubNum

TRYGUB NUMBER - a big number that eats $1e5+$ updates of the form "add $x * b^e$ " in $O(1)$

// amortized carries, and answers "what is digit k " at any time.
 // WHAT: maintains $V = \sum e \text{ of } d[e] * \text{base}^e$ exactly, where base
 ↪ $= b^K$ packs K base- b digits
 // into one machine digit. `add(x, e)` does $V += x * b^e$ with x AND e of
 ↪ either sign. `kthDigit(k)`
 // returns `floor(V / b^k) mod b`, i.e. the k -th digit of V in base b
 ↪ counting from the units digit
 // at $k = 0$. `sign()` returns $+1 / 0 / -1$.
 // THE TRICK: digits are NOT normalised. `d[e]` may be anywhere in
 ↪ $(-base, base)$, the carry is
 // pushed lazily one position at a time and stops as soon as it dies,
 ↪ and only NONZERO digits are
 // stored, in a map - so a negative e (a fractional position) costs
 ↪ nothing and the number needs
 // no length. Amortized it is $O(1)$ carry steps per add (binary-counter
 ↪ argument), each step one
 // map operation, so $O(\log D)$ with $D = \text{number of stored digits}$. Memory
 ↪ $O(D)$.
 // WHEN: $1e5$ updates "add $5 * 10^k$ " to a huge number with digit queries
 ↪ interleaved. A plain
 // vector is $O(\text{len})$ per update; a segment tree over digits drowns in
 ↪ carry logic. Also the clean
 // way to compare two such numbers: subtract, look at `sign()`.
 // NEGATIVE e : split $e = q * K + j$ with FLOOR division, not C++
 ↪ truncation. For $e = -1$, $K = 30$ the
 // language gives $e/K == 0$ and $e \% K == -1$, which would multiply x by
 ↪ $b^{(-1)}$. Take
 // $j = ((e \% K) + K) \% K$ first, then $q = (e - j) / K$, which is exact in
 ↪ both signs.
 // NEGATIVE x : nothing special in add - integer division truncates

```

    → toward zero, so after
// d[q] -= t * base the residue KEEPS the sign of d[q] and |d[q]| <
    → base. That is exactly how
// negative digits appear, and it is why kthDigit needs the borrow
    → below.
// THE BORROW RULE in kthDigit (the whole difficulty): the digits below
    → q may sum to a negative
// number. Their total is bounded by base*q in absolute value (sum of
    → (base-1)*base^e for e < q
// is base*q - 1), so the top stored digit below q decides the SIGN of
    → the whole low part, and a
// negative low part eats exactly 1 from position q: 'if (prev(it)->se
    → < 0) a--'. This is why
// zeros MUST be erased from the map - a stored 0 sitting just under q
    → would answer "not
// negative" and lose the borrow.
// OVERFLOW BUDGET: a digit lives in (-base, base) and add pushes x *
    → b^(K-1) into it, so you
// need base + |x| * b^(K-1) < 9.2e18. b = 2, K = 30 (base ~ 1.07e9)
    → and b = 10, K = 9
// (base = 1e9) both leave room for |x| <= 1e9. Bigger K = shorter
    → carry chains, so take the
// largest one the budget allows.
struct TrygubNum {
    ll b, K, base; vector<ll> pw; // pw[i] = b^i,
    → base = b^K = one machine
    map<ll, ll> d; // digit. e ->
    → digit, ZEROS ARE ERASED
    TrygubNum(ll b, ll K) : b(b), K(K), pw(K+1, 1) {
        for (ll i = 1; i <= K; i++) pw[i] = pw[i-1] * b;
        base = pw[K];
    }
    void add(ll x, ll e) { // V += x * b^e,
    → both signs allowed
        ll j = ((e % K) + K) % K, q = (e - j) / K; // FLOOR split e
    → = q*K + j, 0 <= j < K
        d[q] += x * pw[j];
        ll t;
        do {
            t = d[q] / base; // truncates
    → toward 0: residue keeps sign
            d[q] -= t * base, d[q+1] += t;
            if (!d[q]) d.erase(q); // erasing zeros
    → is load-bearing, see above
            q++;
        } while (t); // carry died ->
    → stop, O(1) amortized
        if (!d[q]) d.erase(q); // the digit
    → above may have been left at 0
    }
    int kthDigit(ll k) { // floor(V /
    → b^k) mod b, k >= 0
        ll q = k / K, a = 0;
        auto it = d.lower_bound(q);
        if (it != d.end() && it->fi == q) a = it->se;
        if (it != d.begin() && prev(it)->se < 0) a--; // low part is
    → negative: it borrows a 1
        a = ((a % base) + base) % base;
        return (int)(a / pw[k % K] % b); // the (k %
    → K)-th base-b digit of a
    }
    int sign() { // top stored
    → digit dominates all the rest
        return d.empty() ? 0 : (prev(d.end())->se > 0 ? 1 : -1);
    }
};
// DEGENERATE: empty number -> sign() 0 and kthDigit() 0 everywhere.
    → add(0, e) stores nothing.
// Adding and then subtracting the same term returns the map to empty.
    → k above the top digit
// gives 0. kthDigit is meant for V >= 0; for V < 0 it still returns
    → floor(V / b^k) mod b, which
// is the b-complement expansion (...bbb1 for -1), not the digits of
    → |V| - negate first.
// VARIANTS / RELATED
// - Reading the whole number: from prev(d.end())->fi * K + K - 1
    → downward through kthDigit.
// - Only nonnegative e and no queries until the end: just use a vector
    → and normalise once.
// - Full arithmetic (multiply, divide, print): 21 - BigInt.cpp. This
    → struct only does += x*b^e.
// - Same lazy-carry idea works for "add x at position e" in any
    → positional structure; the map is
// the only reason it stays sparse.

```

1.27 PermutationRoots

CYCLE DECOMPOSITION AND WHAT IT ANSWERS: the k-th root of a permutation, the minimum number of

```

// transpositions (with the swaps), and the order.
// Convention: p is 0-indexed, p[i] is the image of i, (p . q)[i] =
    → p[q[i]], p^k is p applied k
// times. All three routines are O(n) after the O(n) decomposition (the
    → order pays a small
// factorisation on top).
vector<vector<int>> cycles(const vector<int>& p) { // each
    → cycle as i, p[i], p[p[i]], ...
    int n = sz(p); vector<vector<int>> c; vector<char> vis(n, 0);
    for (int i = 0; i < n; i++) if (!vis[i]) {
        c.pb({});
        for (int j = i; !vis[j]; j = p[j]) vis[j] = 1, c.back().pb(j);
    }
    return c; // fixed
    → points are cycles of length 1
}
vector<int> fromCycles(int n, const vector<vector<int>>& c) {
    vector<int> p(n);
    for (auto& v : c) for (int i = 0; i < sz(v); i++) p[v[i]] = v[(i +
    → 1) % sz(v)];
    return p;
}
// --- K-TH ROOT: q with q^k = p, or {} if none exists. k >= 1 (k = 0
    → is not a question).
// DERIVATION. Raising to the k-th power splits a q-cycle of length m
    → into gcd(m, k) cycles of
// length m / gcd(m, k) - so the q-cycle behind a group of p-cycles of
    → length L must have length
// m = g*L where g is the group size, and it must satisfy g = gcd(gL,
    → k). Divide that out: g = k
// is not required, the condition is exactly
// g | k AND gcd(L, k / g) = 1.
// Let t = product over primes r dividing L of r^(exponent of r in k) -
    → computed below by
// stripping gcd(L, .) off k until it is 1. Then k/t is coprime to L,
    → every legal g is a multiple
// of t, and every g = t*z with z | (k/t) is legal. So the p-cycles of
    → length L can be packed iff
// t divides their COUNT, and greedily taking groups of exactly t
    → always works. Groups of size t
// also maximise the number of cycles of q, so this returns the root
    → with the FEWEST
// transpositions. Existence therefore = "for every L, t(L, k) divides
    → the number of cycles of
// length L". A cycle length that does not appear is vacuously fine.
// CONSTRUCTION: interleave the g cycles v_0..v_{g-1} into one array c
    → of length m = g*L by
// c[(x + y*k) mod m] = v_x[y]. That is a bijection precisely because
    → gcd(m, k) = g, and then
// stepping one place in c is q, so q^k moves v_x[y] to v_x[y+1] =
    → p(v_x[y]).
vector<int> permRoot(const vector<int>& p, ll k) {
    int n = sz(p);
    vector<vector<vector<int>>> by(n+1); // p-cycles
    → bucketed by their length
    for (auto& v : cycles(p)) by[sz(v)].pb(v);
    vector<vector<int>> out;
    for (int L = 1; L <= n; L++) {
        if (by[L].empty()) continue;
        ll t = 1, rem = k, g;
        while ((g = gcd((ll)L, rem)) != 1) t *= g, rem /= g; //
    → t = minimal legal group
        if (sz(by[L]) % t) return {}; // count
    → not divisible: NO ROOT
        for (ll i = 0; i < sz(by[L]); i += t) {
            ll m = t * L, kk = k % m; vector<int> c(m); // reduce k
    → first: y*k can overflow
            for (ll x = 0; x < t; x++) for (ll y = 0; y < L; y++)
                c[(x + y * kk) % m] = by[L][i + x][y]; // the
    → interleave
            out.pb(c);
        }
    }
    return fromCycles(n, out); // n == 0
    → returns {}, which is right
}
// --- MINIMUM TRANSPOSITIONS to sort p is n - (number of cycles), and
    → no fewer: one swap changes
// the cycle count by exactly +-1 (inside a cycle it splits, across two
    → it merges), and sorted
// means n cycles. To turn a into b it is n - cycles(b^-1 . a). Both
    → are the SAME loop: drag the
// wanted value into place, which kills one cycle per swap.

```

```
vector<pii> swapsToMatch(vector<int> a, const vector<int>& b) { //
    ↪ b = 0,1,...,n-1 means "sort"
    int n = sz(a); vector<int> pos(n); vector<pii> s;
    for (int i = 0; i < n; i++) pos[a[i]] = i; // pos[v] =
    ↪ where value v sits now
    for (int i = 0; i < n; i++) if (a[i] != b[i]) {
        int j = pos[b[i]]; // j > i
    ↪ always, so one pass suffices
        s.pb({i, j}), pos[a[i]] = j, pos[a[j]] = i, swap(a[i], a[j]);
    }
    return s; // sz(s) ==
    ↪ n - cycles(b^-1 . a)
}

// --- ORDER of p = lcm of the cycle lengths. It OVERFLOWS EARLY: the
    ↪ max over permutations of n
// (Landau's g(n) ~ e^sqrt(n ln n)) passes 1e18 around n = 130, so
    ↪ never store it in an ll. Keep
// it factored, or reduce it mod M. Returns prime -> exponent; order
    ↪ mod M is then the product of
// q^e mod M over the map, and the exponents sum to at most log2(order).
map<ll, int> permOrder(const vector<int>& p) {
    map<ll, int> f;
    for (auto& v : cycles(p)) {
        int m = sz(v);
        for (ll q = 2; q * q <= m; q++) if (m % q == 0) {
            int e = 0; while (m % q == 0) m /= q, e++;
            f[q] = max(f[q], e);
        }
        if (m > 1) f[m] = max(f[m], 1); // the
    ↪ leftover prime factor
    }
    return f; // identity
    ↪ / n == 0 gives {} = order 1
}

// VARIANTS / RELATED
// - p^k in O(n): inside each cycle of length L step by k, giving
    ↪ gcd(L, k) cycles of length
// L / gcd(L, k). That is the root loop read backwards, and the cheap
    ↪ way to check a root.
// - p^k for huge k: reduce k modulo the cycle length of each element,
    ↪ not modulo the order.
// - SQUARE ROOT (k = 2): t = 2 for even L and 1 for odd L, so a root
    ↪ exists iff the cycles of
// every even length come in pairs. Odd-length cycles always have a
    ↪ root alone.
// - Need a root with a prescribed PARITY of transposition count: once,
    ↪ merge 2t cycles of some
// length instead of t (legal iff k/t is even and that length has >=
    ↪ 2t cycles). It changes the
// cycle count by one, hence the parity.
// - MINIMUM ADJACENT swaps is a completely different quantity - the
    ↪ inversion count. See
// 14 - Inversions.cpp. Do not confuse it with n - cycles.
// - The permutation is even iff n - cycles is even; that is the sign,
    ↪ and it is what tells you
// whether the 15-puzzle position is solvable.
```

1.28 SubarrayAggregator

LOGARITHMIC SUBARRAY AGGREGATOR - every value of $op(a[l..r])$ over every subarray, in $O(n \log A)$.

```
// Fix r. As l walks from r down to 0 the aggregate  $op(a[l..r])$  only
    ↪ ever moves one way, and every
// strict move is a big jump, so only  $O(\log A)$  DISTINCT values occur
    ↪ and equal ones form contiguous
// runs of l. subarrayAgg calls cb(r, runs) for every r (0-indexed)
    ↪ with runs = {v, lo, hi} meaning
//  $op(a[l..r]) = v$  for exactly the l in [lo, hi]; the runs partition
    ↪ [0, r] and are listed from
// l = r downwards, so runs[0].hi == r and runs.back().lo == 0.
// WHEN: "count / find the subarrays whose gcd (or / and / min) is X",
    ↪ "how many subarrays have
// gcd > 1", "longest subarray with or <= k". Replaces a sparse table
    ↪ plus a binary search per l,
// and it is shorter and exact.
// PRECONDITION for correctness: op associative, and  $op(x, y)$  monotone
    ↪ in the same direction for
// every x (so equal values are adjacent and the merge below is legal).
// PRECONDITION for the  $O(\log A)$  run count: every STRICT change must
    ↪ shrink a potential of size A -
// gcd at least halves, and / or turn at least one bit off / on, lcm
    ↪ capped at LIM at least doubles.
// gcd, and, or, capped lcm all qualify. min and max are monotone so
    ↪ the code is CORRECT for them,
// but their run count is  $O(n)$ , not  $O(\log A)$  ( $a[i] = i$  gives n runs at
    ↪  $r = n-1$ ) - use a monotonic
// stack there. sum, xor and product do not qualify at all: nothing
```

```
    ↪ forces values to repeat.
// COMPLEXITY:  $O(n \log A)$  time,  $O(\log A)$  live memory plus whatever cb
    ↪ stores.
// DEGENERATE: empty a calls cb zero times. n == 1 gives one run {a[0],
    ↪ 0, 0}. All-equal gives one
// run per r. Zeros are fine for gcd ( $gcd(0, x) = x$ ), so an all-zero
    ↪ array reports v == 0.
// EXACTNESS: exact on integers, the aggregate never leaves the value
    ↪ type. Counts of subarrays
// need ll ( $n(n+1)/2$  overflows int at  $n = 65536$ ), the run endpoints
    ↪ stay int.
template <class T> struct Run { T v; int lo, hi; }; //  $op(a[l..r])$ 
    ↪ == v for every l in [lo, hi]
template <class T, class Op, class F>
void subarrayAgg(const vector<T>& a, Op op, F cb) {
    vector<Run<T>> cur, nxt; // runs for
    ↪ r-1, then the ones for r
    for (int r = 0; r < sz(a); r++) {
        nxt.clear();
        nxt.pb({a[r], r, r}); // l == r:
    ↪ the element on its own
        for (auto& e : cur) { //
    ↪ associativity:  $op(a[l..r-1], a[r])$ 
            T v = op(e.v, a[r]);
            if (v == nxt.back().v) nxt.back().lo = e.lo; //
    ↪ unchanged: swallow the whole old run
            else nxt.pb({v, e.lo, e.hi}); // changed:
    ↪ at most  $O(\log A)$  times
        }
        cur.swap(nxt);
        cb(r, cur);
    }
}

// --- CONSUMER 1: cnt[x] = number of subarrays whose gcd is exactly x.
    ↪  $O(n \log A (\log A + \log n))$ .
// std::gcd (C++17) and not __gcd: __gcd is a libstdc++ internal and
    ↪ rejects signed types on clang.
map<ll, ll> gcdHistogram(const vector<ll>& a) {
    map<ll, ll> cnt;
    subarrayAgg(a, [](ll x, ll y) { return gcd(x, y); }, [&](int, auto& v) {
    ↪ v) {
        for (auto& e : v) cnt[e.v] += e.hi - e.lo + 1;
    });
    return cnt;
}

// --- CONSUMER 2: number of subarrays with  $or(a[l..r]) \geq k$ , and the
    ↪ set of distinct subarray ORs
// (which has size  $O(n \log A)$ , not  $O(n^2)$  - that bound is the reason
    ↪ the question gets asked).
ll orAtLeast(const vector<ll>& a, ll k, set<ll>* distinct = nullptr) {
    ll c = 0;
    subarrayAgg(a, [](ll x, ll y) { return x | y; }, [&](int, auto& v) {
        for (auto& e : v) { // v is
    ↪ sorted by value, so you could
            if (e.v >= k) c += e.hi - e.lo + 1; // binary
    ↪ search - but it has  $O(\log A)$ 
            if (distinct) distinct->insert(e.v); // entries,
    ↪ a scan is already free
        }
    });
    return c;
}

// VARIANTS
// - LONGEST / SHORTEST subarray with a target property: the runs are
    ↪ sorted by l, so per r the
// best l is an endpoint of the first / last run that satisfies it.
    ↪  $O(n \log A)$  overall.
// - "gcd of  $a[l..r] = 1$ " for all pairs: gcdHistogram(a)[1].
// - LCM needs a cap or it overflows instantly:  $g = gcd(x, y)$ , then  $x$ 
    ↪ /  $g > LIM$  /  $y ? LIM$  :
//  $x / g * y$ . Capping keeps monotonicity, so the run structure
    ↪ survives.
// - Two-pointer twin: for or / and / gcd the aggregate is monotone in
    ↪ l, so "count subarrays with
//  $op == k$ " also falls to two pointers plus a sliding structure, but
    ↪ that needs undo; this does
// not and generalises to any associative monotone op in the same 10
    ↪ lines.
// - The same list carried from the LEFT (fix l, grow r) works
    ↪ identically; pick the end your
// query is indexed by.
```

1.29 MexOfAllSubarrays

MEX OF EVERY SUBARRAY - which values occur as $\text{mex}(a[l..r])$ over all $O(n^2)$ subarrays, $O(n \log n)$.

```
// Input a[0..n-1] with a[i] >= 0. Output ok[0..n], ok[m] == 1 iff some
//   ↳ NON-EMPTY subarray has mex
// exactly m. Nothing above n is reachable: mex m needs m distinct
//   ↳ values, so m <= n.
// WHY IT IS ONLY  $O(n \log n)$ : m is reachable iff some MAXIMAL m-free
//   ↳ block contains all of  $0..m-1$ .
// If a block avoids m and holds  $0..m-1$  its mex is exactly m; if the
//   ↳ maximal block misses some
// value < m then so does every sub-block of it, so testing the maximal
//   ↳ blocks alone is complete.
// The blocks for m are the gaps between consecutive occurrences of m,
//   ↳ so over all m there are only
//  $\text{sum}(\text{cnt}[m] + 1) \leq 2n + 1$  blocks to test. Each test is one range-mex
//   ↳ query, answered offline
// sorted by right end: keep last[v] = the last position of v seen so
//   ↳ far in a MIN segment tree over
// v, and  $\text{mex}(a[l..R])$  is the leftmost v with  $\text{last}[v] < l - \text{one}$ 
//   ↳ descend, go left iff the left
// child's min is < l. That descend is the reusable half of this file.
// WHEN: Codeforces 1436E ("mex of the set of all subarray mexes"), "is
//   ↳ m achievable", and any
// range-mex problem - the tree below answers those directly.
// COMPLEXITY:  $O(n \log n)$  time,  $O(n)$  memory (4n tree nodes plus <= 2n +
//   ↳ 1 buffered queries).
// DEGENERATE: n == 0 returns {0} (no subarray, nothing reachable).
//   ↳ All-zero returns only ok[1].
// All-distinct  $0..n-1$  returns every ok[0..n]. Duplicates are fine.
//   ↳ Values > n are ignored - they
// can neither BE a mex (too big) nor block one - but they still occupy
//   ↳ a position and split
// nothing, which is correct. Negative values are undefined input, mex
//   ↳ assumes non-negative.
// EXACTNESS: pure integer index arithmetic, no overflow anywhere
//   ↳ (positions fit int).
struct MexTree { // min, over a range
    ↳ of VALUES, of "last seen at"
    int V; vector<int> t; // values 0..V; -1 =
    ↳ never seen
    MexTree(int V) : V(V), t(4 * (V + 1), -1) {}
    void see(int v, int p, int x, int l, int r) {
        if (l == r) { t[x] = p; return; }
        int m = (l + r) / 2;
        v <= m ? see(v, p, 2 * x, l, m) : see(v, p, 2 * x + 1, m + 1,
        ↳ r);
        t[x] = min(t[2 * x], t[2 * x + 1]);
    }
    void see(int v, int p) { if (v <= V) see(v, p, 1, 0, V); } // v
    ↳ > V cannot affect a mex <= V
    int mex(int l, int x, int l, int r) { //
    ↳ leftmost v with last[v] < l
        if (l == r) return l; // a
        ↳ leaf always exists: [l..R] has
        int m = (l + r) / 2; // <=
        ↳ V+1 elements so some value
        return t[2 * x] < l ? mex(L, 2 * x, l, m) // in
        ↳ 0..V must be missing
            : mex(L, 2 * x + 1, m + 1, r);
    }
    int mex(int l) { return mex(l, 1, 0, V); } // == mex(a[l..R]),
    ↳ R = last position fed to see
};
vector<char> allSubarrayMex(const vector<int>& a) {
    int n = sz(a);
    vector<char> ok(n + 1, 0);
    if (!n) return ok;
    vector<vi> pos(n + 1); //
    ↳ pos[m] = positions holding m
    for (int i = 0; i < n; i++) if (a[i] <= n) pos[a[i]].pb(i);
    vector<vector<pii>> qs(n); // at
    ↳ R: (L, m), "is mex[l..R] m?"
    for (int m = 0; m <= n; m++) { // the
    ↳ maximal m-free blocks
        int lo = 0;
        for (int p : pos[m]) { if (lo < p) qs[p - 1].pb({lo, m}); lo =
        ↳ p + 1; }
        if (lo < n) qs[n - 1].pb({lo, m});
    }
    MexTree T(n);
    for (int R = 0; R < n; R++) {
        T.see(a[R], R);
        for (auto& [L, m] : qs[R]) if (T.mex(L) == m) ok[m] = 1; //
        ↳ block avoids m, so mex <= m
    }
}
```

return ok;

```
}
// CF1436E wants the mex OF THAT SET: int r = 0; while (r <= n &&
//   ↳ ok[r]) r++;
// VARIANTS
// - RANGE MEX, offline: drop the block machinery, bucket the real
//   ↳ queries by R and run the same
// - sweep - T.mex(L) after feeding a[0..R] is mex(a[l..R]).  $O((n + q) \log n)$ .
// - RANGE MEX, online: make the same "last occurrence" tree
//   ↳ PERSISTENT, one version per R, and
// - descend in version R.  $O(\log n)$  per query,  $O(n \log n)$  memory.
//   ↳ Merge-sort tree also works.
// - "smallest missing value >= x in a range": identical descend
//   ↳ restricted to values [x, V].
// - MEX OF A MULTISSET under insert/erase: segment tree over counts,
//   ↳ descend to the leftmost zero.
// - MEX OF THE WHOLE ARRAY is  $O(n)$  with a bucket array; never build a
//   ↳ tree for that.
// - mex(a[l..r]) is non-increasing in l for fixed r, and
//   ↳ non-decreasing in r for fixed l - that
//   ↳ monotonicity is what lets a two-pointer answer "shortest subarray
//   ↳ with mex >= k".
```

1.30 XorEquationRange

COUNT THE SOLUTIONS OF $x_0 \text{ xor } x_1 \text{ xor } \dots \text{ xor } x_{n-1} == X$ with $0 \leq x_i \leq a[i]$, modulo M.

```
// Returns the count mod M. a[i] >= 0, X >= 0, M >= 1.
// WHEN: "how many tuples bounded componentwise xor to X" -
//   ↳ Nim-position counting, POJ 3986, the
//   ↳ stone-game family. The bounds are ARBITRARY; if every a[i] were  $2^k$ 
//   ↳ - 1 the answer would just be
//  $2^k(k(n-1))$  [ $X < 2^k$ ], and this file is what you type when they are
//   ↳ not.
// HOW: process bit k from the top. Split on whether every  $x_i$  copies
//   ↳ a[i]'s bit k ("all tight") or
// at least one  $x_i$  drops a bit that a[i] has ("relaxed"). Relaxed is
//   ↳ countable in one pass: the
// dropper is then free over ALL k lower bits, so designate the
//   ↳ LOWEST-indexed dropper as the
// absorber - its lower bits are forced to whatever makes the xor come
//   ↳ out right, factor 1 - and
// every other variable contributes its own independent count: a[i] -
//   ↳  $2^k + 1$  if it keeps bit k,
//  $2^k$  if it drops it too, a[i] + 1 if a[i] has no bit k at all. A
//   ↳ 4-state DP over
// (parity of chosen k-bits, has anyone dropped yet) does that in  $O(n)$ 
//   ↳ per bit. The all-tight case
// forces the k-th bit of the xor to the parity of  $\# \{i : a[i] \text{ has bit}$ 
//   ↳ k}; if that misses X's bit k
// nothing survives, otherwise strip bit k everywhere and recurse to
//   ↳ bit k-1. Bottoming out leaves
// exactly the single tuple  $x_i == a[i]$ , hence the final +1.
// COMPLEXITY:  $O(n \log(\max a))$  time,  $O(1)$  extra memory. No recursion,
//   ↳ no globals.
// DEGENERATE: n == 0 returns [X == 0]. All a[i] == 0 returns [X == 0].
//   ↳ X with a bit above every
// a[i] returns 0. X == 0 always returns >= 1 (the all-zero tuple). M
//   ↳ == 1 returns 0.
// EXACTNESS: exact modulo M for any M up to  $9.2e18$  - every product
//   ↳ goes through __int128. Bounds
// a[i] up to  $9.2e18$  are fine (a[i] + 1 must not overflow, so keep a[i]
//   ↳ <  $2^{63} - 1$ ).
ll xorEqCount(vector<ll> a, ll X, ll M) {
    auto mul = [&](ll p, ll q) { return (ll)((__int128)p * q % M); };
    int K = 0;
    for (ll v : a) while (v >> K) K++; // every
    ↳ a[i] <  $2^K$ 
    if (X >> K) return 0; // xor of
    ↳ values <  $2^K$  stays <  $2^K$ 
    ll res = 0;
    for (int k = K - 1; k >= 0; k--) {
        ll two = (1LL << k) % M, dp[2][2] = {{1 % M, 0}, {0, 0}}; //
        ↳ dp[parity of bit k][dropped?]
        for (ll v : a) {
            ll nd[2][2] = {}, cnt = ((v & ~(1LL << k)) + 1) % M; // v
            ↳ -  $2^k + 1$  if bit k is set,
            for (int p = 0; p < 2; p++) for (int f = 0; f < 2; f++) {
                ↳ // v + 1 if it is not: the
                ll c = dp[p][f];
                ↳ // SAME expression
                if (!c) continue;
                if (v >> k & 1) {
                    nd[p ^ 1][f] = (nd[p ^ 1][f] + mul(c, cnt)) % M;
                }
                ↳ // keep it:  $x_i$  in  $[2^k, v]$ 
                nd[p][1] = (nd[p][1] + (f ? mul(c, two) : c)) % M;
            }
        }
    }
}
```

```

    ↪ // drop it: x_i in [0, 2^k),
        } else nd[p][f] = (nd[p][f] + mul(c, cnt)) % M;
    ↪ // and the FIRST dropper is
        }
    ↪ // the absorber: factor 1
        memcpy(dp, nd, sizeof dp);
    }
    res = (res + dp[X >> k & 1][1]) % M; // [1]:
    ↪ at least one variable dropped
    int par = 0;
    for (ll& v : a) if (v >> k & 1) par ^= 1, v ^= 1LL << k;
    ↪ // strip, stay all-tight
    if (par != (int)(X >> k & 1)) return res;
    ↪ // all-tight branch dies
    }
    return (res + 1) % M; // the
    ↪ surviving tuple is x_i == a[i]
}
// VARIANTS
// - LOWER BOUNDS TOO (lo_i <= x_i <= hi_i): you cannot shift, xor is
    ↪ not linear under +. Split
// - each range into O(log) aligned blocks "fixed prefix + j free
    ↪ bits"; a block with j >= 1 free
// - bits plays the role of a dropper (one absorber, the rest
    ↪ contribute 2^j), so the same count
// - works over the O(n log A) blocks - at the cost of a product over
    ↪ per-variable block choices.
// - AND / OR instead of xor: per bit the constraint decouples
    ↪ completely, so it is a plain
// - per-bit product, no absorber trick needed.
// - Sum over all X of the answer is prod(a[i] + 1) - a free sanity
    ↪ check on the implementation.
// - CONSTRUCTING one solution: run the same DP keeping the counts,
    ↪ then walk it back greedily.
// - If all a[i] are equal to 2^k - 1 the answer is 2^k(n-1) for X <
    ↪ 2^k, 0 otherwise.

```

1.31 SubsetUnionOfBitsets

SUBSET UNION SIZES - given $m \leq \sim 20$ sets over the ground set $\{0..n-1\}$, get $|\text{union of the chosen}$

```

// sets| for EVERY one of the 2^m subsets, in O(n + 2^m * m). Output
    ↪ u[S] for every mask S over the
// m sets, u[S] = number of ground elements covered by at least one set
    ↪ of S; u[0] == 0.
// THE TRICK IS THE TRANSPOSE, everything else is bookkeeping. You are
    ↪ handed "set j contains these
// elements"; flip it to own[e] = the mask of sets that contain element
    ↪ e. Then element e is MISSED
// by S iff S is disjoint from own[e] iff S is a SUBSET of ~own[e]. So
    ↪ drop one token at ~own[e] for
// each e and run a SUPERSET-sum (SOS) transform: cnt[S] = #{e :
    ↪ ~own[e] contains S} = number of
// elements S misses, and the union size is n - cnt[S]. Counting the
    ↪ complements is what turns a
// union (no closed form under subsets) into a plain subset-lattice sum.
// WHEN: "cheapest set of towers covering >= k houses", "for each mask,
    ↪ how much does it cover",
// any 2^m enumeration whose inner cost is a union - it removes the O(n
    ↪ / 64) bitset OR per mask.
// COMPLEXITY: O(n + 2^m * m) time, 2^m ints = 4 MB at m = 20 (plus 8
    ↪ MB if you also build costs).
// DEGENERATE: m == 0 gives the single answer u[0] == 0. n == 0 gives
    ↪ all zeros. Duplicate or empty
// sets are fine. An element in no set has own[e] == 0, is missed by
    ↪ every S, and is never counted.
// EXACTNESS: pure integer counting, no overflow (u fits int as long as
    ↪ n does). Weighted version:
// replace the ++ by += w[e] and use ll - then the budget is sum|w| <=
    ↪ 9e18.
vi transposeSets(int n, const vector<vi>& sets) { // sets[j] =
    ↪ the elements of set j
    vi own(n, 0); // own[e] =
    ↪ mask of sets holding e
    for (int j = 0; j < sz(sets); j++) for (int e : sets[j]) own[e] |=
    ↪ 1 << j;
    return own;
}
vi subsetUnionSize(int n, int m, const vi& own) {
    int full = (1 << m) - 1;
    vi u(1 << m, 0);
    for (int e = 0; e < n; e++) u[own[e] ^ full]++; // bucket e
    ↪ at "the sets that MISS e"
    for (int b = 0; b < m; b++) // SOS over
    ↪ SUPERSETS: pull from S | bit
        for (int S = 0; S <= full; S++) // into S.
    ↪ Entries with bit b set are not

```

```

        if (! (S >> b & 1)) u[S] += u[S | 1 << b]; // touched
    ↪ in this pass, so order is free
    for (int S = 0; S <= full; S++) u[S] = n - u[S]; // missed ->
    ↪ covered
    return u;
}
// --- THE DUAL CONSUMER: cheapest subset covering AT LEAST k elements,
    ↪ for every k. O(2^m * m).
vector<ll> minCostToCover(int n, int m, const vi& own, const
    ↪ vector<ll>& cost) {
    vi u = subsetUnionSize(n, m, own);
    int full = (1 << m) - 1;
    vector<ll> c(1 << m, 0), best(n + 1, llinf);
    for (int S = 1; S <= full; S++) c[S] = c[S ^ (1 << lsb(S))] +
    ↪ cost[lsb(S)]; // cost by lowest
    for (int S = 0; S <= full; S++) best[u[S]] = min(best[u[S]], c[S]);
    ↪ // set bit
    for (int k = n - 1; k >= 0; k--) best[k] = min(best[k], best[k +
    ↪ 1]); // exactly -> at least
    return best; // best[k]
    ↪ == llinf: k is unreachable
}
// VARIANTS
// - INTERSECTION SIZES for every S: bucket at own[e] instead of
    ↪ ~own[e] and run the SAME superset
// - SOS - #{e : own[e] contains S} is exactly |intersection of S|.
    ↪ One character of difference.
// - "covered by EXACTLY the sets in S": the raw histogram of own[e],
    ↪ no transform at all.
// - MAX / MIN over the covered set instead of a count: SOS with max /
    ↪ min in place of +.
// - SUBSET sum (Zeta over subsets) is the mirror: if (S >> b & 1)
    ↪ f[S] += f[S ^ 1 << b]. Mobius
// - (the inverse) is the same loop with -=, which recovers "exactly"
    ↪ counts from "at least".
// - m > 20: meet in the middle over two halves of the sets, or drop
    ↪ to a greedy / randomised
// - cover - 2^m unions is the whole budget and it is 1e6 masks at m =
    ↪ 20.
// - The ground set may be huge: only n and 2^m matter, never n * m or
    ↪ n / 64 words per mask.

```

1.32 NegativeBase

NEGATIVE-BASE ARITHMETIC. toNegBase(n, b) writes ANY integer n, either sign, in base -b with

```

// digits in [0, b): n == sum d[i] * (-b)^i, digits LEAST SIGNIFICANT
    ↪ FIRST, no leading zero, and
// zero is the single digit {0}. Every integer has exactly one such
    ↪ representation and there is no
// sign anywhere - that is the entire point of the base. fromNegBase
    ↪ inverts it.
// WHEN: "print n in base -2", "add two negabinary strings", puzzle
    ↪ problems where a value must be
// encoded without a sign, and the folklore fact that negabinary
    ↪ addition needs a NEGATIVE carry.
// THE DIVISION RULE: d = n mod b forced into [0, b), then n = (n - d)
    ↪ / (-b), an EXACT division
// since b divides n - d. Do not write n /= -b and patch d afterwards:
    ↪ C++ truncates toward zero,
// so after d += b the quotient also needs n += 1, and that is the step
    ↪ everyone forgets.
// WHY IT TERMINATES: |n - d| <= |n| + b - 1, so the new |n| is at most
    ↪ |n|/b + (b-1)/b, which is
// strictly smaller than |n| whenever |n| >= 2; from |n| == 1 it
    ↪ reaches 0 in at most two steps.
// The digit count is O(log_b |n|) + 1, and the sign of n alternates as
    ↪ you descend.
// ADDITION digit by digit: s = x[i] + y[i] + carry lands in [-b, 2b -
    ↪ 1]; force the digit into
// [0, b) by moving b, and since the place value of the next digit is
    ↪ -b, moving +b DOWN here means
// carrying -1 UP. A leftover carry of +1 appends the digit 1; a
    ↪ leftover of -1 appends TWO digits,
// b-1 then 1, because -1 == (b-1) + 1 * (-b). The same loop does
    ↪ subtraction (pass sgn = -1): the
// range of s only widens to [-b, b], still inside what the one carry
    ↪ line handles.
// COMPLEXITY: O(log_b |n|) per conversion, O(len) per add. DEGENERATE:
    ↪ n == 0 gives {0}; adding to
// or from {0} works; results are stripped of leading zeros so equality
    ↪ is plain vector ==.
// EXACTNESS: exact integers. Budget: |n| <= 9.2e18 minus b for
    ↪ toNegBase (n - d must not overflow,
// so LLONG_MIN is out), and fromNegBase must be told digits whose
    ↪ value fits in ll.
vi toNegBase(ll n, int b) { // b >= 2; the base

```

```

    ↪ used is -b
    vi d;
    while (n) {
        int r = (int)(n % b);
        if (r < 0) r += b; // C++ % keeps the
    ↪ sign of n; digits must be >= 0
        d.pb(r);
        n = (n - r) / -b; // exact division,
    ↪ never a truncation
    }
    return d.empty() ? vi{0} : d;
}
ll fromNegBase(const vi& d, int b) {
    ll v = 0;
    for (int i = sz(d) - 1; i >= 0; i--) v = v * -b + d[i]; //
    ↪ Horner, top digit down
    return v;
}
vi negAdd(vi x, const vi& y, int b, int sgn = 1) { // x +
    ↪ sgn * y, sgn is 1 or -1
    x.resize(max(sz(x), sz(y)), 0);
    int c = 0;
    for (int i = 0; i < sz(x); i++) {
        int s = x[i] + (i < sz(y) ? sgn * y[i] : 0) + c;
        c = s < 0 ? 1 : (s >= b ? -1 : 0); // next place is
    ↪ worth -b, so shifting the digit
        x[i] = s + b * c; // by -b here is a
    ↪ carry of +1, and vice versa
    }
    if (c == 1) x.pb(1);
    else if (c == -1) x.pb(b - 1), x.pb(1); // -1 == (b-1) * 1 +
    ↪ 1 * (-b)
    while (sz(x) > 1 && !x.back()) x.pop_back(); // normalise, so ==
    ↪ is equality of values
    return x;
}
// VARIANTS
// - NEGATE: negAdd({0}, x, b, -1). Do not hunt for a bit pattern,
    ↪ subtraction already does it.
// - SIGN: a normalised nonzero number is NEGATIVE iff its digit count
    ↪ is even, i.e. the leading
// - digit sits on an odd power of -b. Compare x and y by the sign of
    ↪ negAdd(x, y, b, -1).
// - MULTIPLY BY -b: prepend a zero digit (shift). Divide by -b: drop
    ↪ d[0].
// - MULTIPLY: schoolbook into a wide accumulator, then normalise from
    ↪ the bottom with the same
// - rule - while (d[i] < 0) d[i] += b, d[i+1] += 1; while (d[i] >= b)
    ↪ d[i] -= b, d[i+1] -= 1.
// - NEGABINARY (b == 2) IN ONE LINE, no loop: the digits of n as a
    ↪ bit pattern are
// - (n + 0xAAAAAAAAAAAAAAAAu11) ^ 0xAAAAAAAAAAAAAAAAu11, bit i being
    ↪ d[i] - the mask is the whole
// - carry cascade done in parallel. All 64 digits must fit, which
    ↪ covers every negative ll but
// - only n <= (4^32 - 1) / 3 = 6148914691236517205 upward; past that
    ↪ it silently truncates.
// - BASE -1+i represents every Gaussian integer with digits {0, 1}:
    ↪ take d = (re + im) & 1, then
// - (re, im) -> ((im - re + d) / 2, -(re + im - d) / 2). Same loop,
    ↪ same termination argument.

```

1.33 TestGenerators

RANDOM TEST GENERATORS for stress testing. 12 - StressTest.cpp is the harness; this is the

```

// library it feeds on. Writing a generator from scratch under pressure
    ↪ is where stress testing
// usually dies, so keep these on the page.
// SEED FROM argv SO A FAILING CASE IS REPRODUCIBLE:
// int main(int argc, char** argv) { if (argc > 1)
    ↪ rng.seed(stoull(argv[1])); ... }
// then the driver loop passes the iteration number and you can replay
    ↪ any failure exactly.
ll rnd(ll l, ll r) { return uniform_int_distribution<ll>(l, r)(rng); }
    ↪ // inclusive
bool coin(int pct = 50) { return rnd(1, 100) <= pct; }
vector<ll> randArray(int n, ll lo, ll hi) {
    vector<ll> a(n);
    for (ll& x : a) x = rnd(lo, hi);
    return a;
}
vector<int> randPermutation(int n) { //
    ↪ 0-indexed
    vector<int> p(n);
    iota(all(p), 0);
    shuffle(all(p), rng);

```

```

    return p;
}
string randString(int n, char lo = 'a', char hi = 'z') {
    string s;
    while (n--) s += (char)rnd(lo, hi);
    return s;
}
vector<string> randGrid(int n, int m, const string& alpha = ".#") {
    vector<string> g(n, string(m, '.'));
    for (auto& row : g) for (char& c : row) c = alpha[rnd(0,
    ↪ alpha.size() - 1)];
    return g;
}
// --- TREES. All 1-indexed, n-1 edges, always connected. ---
vector<pair<int, int>> randTree(int n) { //
    ↪ uniform-ish, shuffled labels
    vector<int> lab = randPermutation(n);
    vector<pair<int, int>> e;
    for (int i = 1; i < n; i++) e.push_back({lab[rnd(0, i - 1)] + 1,
    ↪ lab[i] + 1});
    return e;
}
vector<pair<int, int>> pathTree(int n) { // the
    ↪ DEEP case: recursion, HLD
    vector<pair<int, int>> e;
    for (int i = 2; i <= n; i++) e.push_back({i - 1, i});
    return e;
}
vector<pair<int, int>> starTree(int n) { // the
    ↪ WIDE case: heavy-child logic
    vector<pair<int, int>> e;
    int c = rnd(1, n);
    for (int i = 1; i <= n; i++) if (i != c) e.push_back({c, i});
    return e;
}
vector<pair<int, int>> caterpillar(int n) { // a
    ↪ spine with legs: breaks naive
    vector<pair<int, int>> e; //
    ↪ centroid and diameter code
    int spine = max(1LL, (ll)rnd(1, n));
    for (int i = 2; i <= spine; i++) e.push_back({i - 1, i});
    for (int i = spine + 1; i <= n; i++) e.push_back({(int)rnd(1,
    ↪ spine), i});
    return e;
}
// --- GRAPHS. m edges, no self loops, no duplicates unless you ask for
    ↪ them. ---
vector<pair<int, int>> randGraph(int n, int m) {
    vector<pair<int, int>> all;
    for (int i = 1; i <= n; i++) for (int j = i + 1; j <= n; j++)
    ↪ all.push_back({i, j});
    shuffle(all(all), rng);
    all.resize(min((size_t)m, all.size()));
    return all;
}
vector<pair<int, int>> connectedGraph(int n, int m) { // a
    ↪ random tree plus extra edges
    auto e = randTree(n);
    set<pair<int, int>> used;
    for (auto [u, v] : e) used.insert({min(u, v), max(u, v)});
    m = min<ll>(m, (ll)n * (n - 1) / 2);
    while ((int)e.size() < m) {
        int u = rnd(1, n), v = rnd(1, n);
        if (u == v) continue;
        if (u > v) swap(u, v);
        if (used.insert({u, v}).second) e.push_back({u, v});
    }
    shuffle(all(e), rng);
    return e;
}
vector<pair<int, int>> randDAG(int n, int m) { //
    ↪ acyclic by a random topo order
    vector<int> p = randPermutation(n);
    vector<pair<int, int>> all;
    for (int i = 0; i < n; i++) for (int j = i + 1; j < n; j++)
    ↪ all.push_back({p[i] + 1, p[j] + 1});
    shuffle(all(all), rng);
    all.resize(min((size_t)m, all.size()));
    return all;
}
vector<pair<int, int>> randQueries(int q, int n) { //
    ↪ random inclusive [l, r]
    vector<pair<int, int>> v;
    while (q--) {
        int l = rnd(1, n), r = rnd(1, n);
        if (l > r) swap(l, r);
        v.push_back({l, r});
    }

```

2 Data Structures (DS)

```
    }
    return v;
}
/* THE CASES A UNIFORM GENERATOR NEVER PRODUCES - and which is exactly
   ↳ where the bug is
   n = 1, n = 2, and an EMPTY structure
   ↳ (off-by-one, .back() on empty)
   ALL VALUES EQUAL, all zero, all negative
   ↳ (tie-breaks, strict vs non-strict)
   values at the EXTREMES of the stated range (overflow)
   a SORTED or REVERSE-SORTED array (quicksort,
   ↳ two pointers, hulls)
   a PATH tree and a STAR tree (recursion
   ↳ depth, heavy-child logic)
   a DISCONNECTED graph, an isolated vertex, a self loop, a DOUBLED
   ↳ edge (bridges, DSU, flows)
   COLLINEAR / DUPLICATE points, a degenerate polygon (all of
   ↳ geometry)
   Generate with a SMALL value range (say [1,3]) most of the time:
   ↳ collisions and ties are what
   break code, and a uniform draw from [1, 1e9] never creates one.
   SHRINK a failure before reading it: re-run the generator with
   ↳ progressively smaller bounds
   until the counterexample fits on one line, then stare at that. */
```

2 | Data Structures (DS)

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2.1 Segment Tree

2.1.1 SegmentTree

SEGMENT TREE, point update + range query, $O(\log n)$ each. Swap the merge in SEG to change the

```
// operation (sum/min/max/gcd/matrix); any associative monoid works.
↳ Size is rounded up to a power
// of two so the tree is perfect and indices are simple.
```

```
struct SEG {
    ll sum = 0;
    SEG() {}
    SEG(ll x){
        sum = x;
    }
};

struct segTree {
    vector<SEG> seg;
    int sz = 1,n;
    segTree(int nn){
        n = nn;
        while (sz < nn)
            sz *= 2;
        seg.assign(2 * sz , SEG());
    }

    SEG merge(SEG seg1, SEG seg2) {
        SEG ret;
        ret.sum = seg1.sum + seg2.sum;
        return ret;
    }

    void build(vector<int> &v,int x, int lx, int rx) {
        if (lx == rx) {
            seg[x] = SEG(v[lx]);
            return;
        }
        int mid = (lx + rx) / 2;
        int lf = 2 * x + 1, rt = 2 * x + 2;

        build(v,lf, lx, mid);
        build(v,rt, mid + 1, rx);
        seg[x] = merge(seg[lf], seg[rt]);
    }

    void build(vector<int> &v){
        build(v,0, 0, n-1);
    }

    void update(int i, ll val, int x, int lx, int rx) {
```

```
        if (lx == rx) {
            seg[x] = SEG(val);
            return;
        }
        int mid = (lx + rx) / 2,lf = 2*x+1,rt= 2*x+2;
        if(i <= mid)
            update(i, val, lf, lx, mid);
        else
            update(i, val, rt, mid + 1, rx);
        seg[x] = merge(seg[lf] , seg[rt]);
    }

    void update(int i, ll val) {
        update(i, val, 0, 0, n-1);
    }

    SEG query(int l, int r, int x, int lx, int rx) {
        if (l <= lx && r >= rx)
            return seg[x];
        if (l > rx || r < lx)
            return SEG();

        int mid = (lx + rx) / 2,lf = 2 * x + 1,rt = 2*x+2;

        return merge(query(l, r, lf, lx, mid) , query(l, r, rt, mid +
↳ 1, rx));
    }

    SEG query(int l, int r) {
        return query(l, r, 0, 0, n-1);
    }
};
```

2.1.2 LazySegmentTree

Lazy Segment Tree (range add, range sum) - build $O(n)$, update/query $O(\log n)$

```
// 1-based node indices, 0-based positions, ALL recursions on [0, n-1].
// To change the op: edit merge(), apply() and the identity in qry().
```

```
struct LazySeg {
    int n;
    vector<ll> t, lz;

    LazySeg(int n = 0) : n(n), t(4 * n, 0), lz(4 * n, 0) {}

    ll merge(ll a, ll b) { return a + b; }
    void apply(int v, int l, int r, ll x) { t[v] += x * (r - l + 1);
↳ lz[v] += x; }

    void push(int v, int l, int r) {
        if (!lz[v]) return;
        int m = (l + r) / 2;
        apply(2 * v, l, m, lz[v]);
        apply(2 * v + 1, m + 1, r, lz[v]);
        lz[v] = 0;
    }

    void build(const vector<ll>& a, int v, int l, int r) {
        if (l == r) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(a, 2 * v, l, m), build(a, 2 * v + 1, m + 1, r);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }

    void build(const vector<ll>& a) { if (n) build(a, 1, 0, n - 1); }

    void upd(int ql, int qr, ll x, int v, int l, int r) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) return apply(v, l, r, x);
        push(v, l, r);
        int m = (l + r) / 2;
        upd(ql, qr, x, 2 * v, l, m), upd(ql, qr, x, 2 * v + 1, m + 1,
↳ r);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }

    void upd(int l, int r, ll x) { upd(l, r, x, 1, 0, n - 1); }

    ll qry(int ql, int qr, int v, int l, int r) {
        if (qr < l || r < ql) return 0; // identity
        if (ql <= l && r <= qr) return t[v];
        push(v, l, r);
        int m = (l + r) / 2;
        return merge(qry(ql, qr, 2 * v, l, m), qry(ql, qr, 2 * v + 1, m
↳ + 1, r));
    }

    ll qry(int l, int r) { return qry(l, r, 1, 0, n - 1); }
};
```

2.1.3 PersistentSegmentTree

PERSISTENT SEGMENT TREE - every update returns a NEW root and old versions stay queryable.

// $O(\log n)$ time and $O(\log n)$ new nodes per update. The standard use is
 ↳ offline k-th smallest in a
 // range (build a version per prefix, then query version[r] minus
 ↳ version[l-1]).

```
struct Node {
    Node *l, *r;
    ll val = 0;

    Node(ll val) : l(NULL), r(NULL), val(val) {}

    Node() : l(NULL), r(NULL) {}

    Node(Node* l, Node* r) : l(l), r(r) {
        if (l != NULL) val += l->val;
        if (r != NULL) val += r->val;
    }

    void addChild() {
        l = new Node();
        r = new Node();
    }
};

struct PST {
    int n;

    PST(int n) : n(n + 1) {}

    Node merge(Node x, Node y) {
        Node ret;
        ret.val = x.val + y.val;
        return ret;
    }

    Node* update(Node* v, int i, ll val, int lx, int rx) {
        if (lx == rx) return new Node(v->val + val);
        int mid = (lx + rx) / 2;
        if (!v->l) v->addChild();
        if (i <= mid) return new Node(update(v->l, i, val, lx, mid),
        ↳ v->r);
        return new Node(v->l, update(v->r, i, val, mid + 1, rx));
    }

    Node* update(Node* v, int i, ll val) { return update(v, i, val, 0,
    ↳ n - 1); }

    Node query(Node* v, int l, int r, int lx, int rx) {
        if (l > rx || r < lx) return {};
        if (l <= lx && r >= rx) return *v;
        if (!v->l) v->addChild();
        int mid = (lx + rx) / 2;
        return merge(query(v->l, l, r, lx, mid), query(v->r, l, r, mid
        ↳ + 1, rx));
    }

    Node query(Node* v, int l, int r) { return query(v, l, r, 0, n -
    ↳ 1); }

    int getKth(Node* a, Node* b, int k, int lx, int rx) {
        if (lx == rx) return lx;
        if (!a->l) a->addChild();
        if (!b->l) b->addChild();
        int rem = b->l->val - a->l->val;
        int mid = (lx + rx) / 2;
        if (rem >= k) return getKth(a->l, b->l, k, lx, mid);
        return getKth(a->r, b->r, k - rem, mid + 1, rx);
    }

    int getKth(Node* a, Node* b, int k) { return getKth(a, b, k, 0, n -
    ↳ 1); }
};
```

2.1.4 SegmentTreeBeats

Rename one if you ever paste two of them together.

```
// NOTE: several segment-tree files here declare their own 'struct
↳ Node'.
// Rename one if you ever paste two of them together.
struct Node {
    ll mn;
    int mnCnt;
    ll secondMn;

    ll mx;
    int mxCnt;
    ll secondMx;

    ll sum;

    ll lazyAdd;
    ll lazyEqual;
    bool isLazyEqual;

    Node() {
        mx = -inf;
        secondMx = -inf;
        mxCnt = 0;

        mn = inf;
        secondMn = inf;
        mnCnt = 0;

        sum = 0;

        lazyAdd = 0;
        lazyEqual = 0;
        isLazyEqual = 0;
    };

    Node(int x) {
        mx = x;
        secondMx = -inf;
        mxCnt = 1;

        mn = x;
        secondMn = inf;
        mnCnt = 1;

        sum = x;

        lazyAdd = 0;
        lazyEqual = 0;
        isLazyEqual = 0;
    }
};

struct SegTree {
    int tree_size;
    vector<Node> seg_data;

    SegTree(int n) {
        tree_size = 1;
        while (tree_size < n) tree_size *= 2;
        seg_data.resize(2 * tree_size, Node());
    }

    Node merge(Node& lf, Node& ri) {
        Node ans = Node();

        ans.sum = lf.sum + ri.sum;

        ans.mx = max(lf.mx, ri.mx);
        ans.secondMx = max(lf.secondMx, ri.secondMx);
        ans.mxCnt = 0;
        if (lf.mx == ans.mx) ans.mxCnt += lf.mxCnt;
        else ans.secondMx = max(ans.secondMx, lf.mx);
        if (ri.mx == ans.mx) ans.mxCnt += ri.mxCnt;
        else ans.secondMx = max(ans.secondMx, ri.mx);

        ans.mn = min(lf.mn, ri.mn);
        ans.secondMn = min(lf.secondMn, ri.secondMn);
        ans.mnCnt = 0;
        if (lf.mn == ans.mn) ans.mnCnt += lf.mnCnt;
        else ans.secondMn = min(ans.secondMn, lf.mn);
        if (ri.mn == ans.mn) ans.mnCnt += ri.mnCnt;
        else ans.secondMn = min(ans.secondMn, ri.mn);

        return ans;
    }
};
```

```

void init(vector<int>& nums, int ni, int lx, int rx) {
    if (rx - lx == 1) {
        if (lx < nums.size())
            seg_data[ni] = Node(nums[lx]);
        return;
    }

    int mid = lx + (rx - lx) / 2;
    init(nums, 2 * ni + 1, lx, mid);
    init(nums, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void init(vector<int>& nums) {
    init(nums, 0, 0, tree_size);
}

void doPushEq(int ni, int lx, int rx, int x) {
    seg_data[ni].mn = seg_data[ni].mx = seg_data[ni].lazyEqual = x;
    seg_data[ni].isLazyEqual = 1;
    seg_data[ni].mnCnt = seg_data[ni].mxCnt = rx - lx;
    seg_data[ni].secondMx = -inf;
    seg_data[ni].secondMn = inf;

    seg_data[ni].sum = (ll)x * (rx - lx);
    seg_data[ni].lazyAdd = 0;
}

void doPushMinEq(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mn >= x) doPushEq(ni, lx, rx, x);
    else if (seg_data[ni].mx > x) {
        if (seg_data[ni].secondMn == seg_data[ni].mx)
            seg_data[ni].secondMn = x;
        seg_data[ni].sum -= (ll)(seg_data[ni].mx - x) *
↪ seg_data[ni].mxCnt;
        seg_data[ni].mx = x;
    }
}

void doPushMaxEq(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mx <= x) doPushEq(ni, lx, rx, x);
    else if (seg_data[ni].mn < x) {
        if (seg_data[ni].secondMx == seg_data[ni].mn)
            seg_data[ni].secondMx = x;

        seg_data[ni].sum += (ll)(x - seg_data[ni].mn) *
↪ seg_data[ni].mnCnt;
        seg_data[ni].mn = x;
    }
}

void doPushSum(int ni, int lx, int rx, int x) {
    if (seg_data[ni].mn == seg_data[ni].mx)
        doPushEq(ni, lx, rx, seg_data[ni].mn + x);
    else {
        seg_data[ni].mx += x;
        if (seg_data[ni].secondMx != -inf)
            seg_data[ni].secondMx += x;

        seg_data[ni].mn += x;
        if (seg_data[ni].secondMn != inf)
            seg_data[ni].secondMn += x;

        seg_data[ni].sum += (ll)(rx - lx) * x;
        seg_data[ni].lazyAdd += x;
    }
}

void propagete(int ni, int lx, int rx) {
    if (rx - lx == 1) return;

    int mid = lx + (rx - lx) / 2;
    if (seg_data[ni].isLazyEqual) {
        doPushEq(2 * ni + 1, lx, mid, seg_data[ni].lazyEqual);
        doPushEq(2 * ni + 2, mid, rx, seg_data[ni].lazyEqual);
        seg_data[ni].lazyEqual = seg_data[ni].isLazyEqual = 0;
    }
    else {
        doPushSum(2 * ni + 1, lx, mid, seg_data[ni].lazyAdd);
        doPushSum(2 * ni + 2, mid, rx, seg_data[ni].lazyAdd);
        seg_data[ni].lazyAdd = 0;

        doPushMinEq(2 * ni + 1, lx, mid, seg_data[ni].mx);
        doPushMinEq(2 * ni + 2, mid, rx, seg_data[ni].mx);
    }
}

```

```

        doPushMaxEq(2 * ni + 1, lx, mid, seg_data[ni].mn);
        doPushMaxEq(2 * ni + 2, mid, rx, seg_data[ni].mn);
    }
}

void minimize(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r || seg_data[ni].mx <= x)
        return;
    if (lx >= l && rx <= r && seg_data[ni].secondMx < x) {
        doPushMinEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    minimize(l, r, x, 2 * ni + 1, lx, mid);
    minimize(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void minimize(int l, int r, int x) {
    minimize(l, r, x, 0, 0, tree_size);
}

void maximize(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r || seg_data[ni].mn >= x)
        return;
    if (lx >= l && rx <= r && seg_data[ni].secondMn > x) {
        doPushMaxEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    maximize(l, r, x, 2 * ni + 1, lx, mid);
    maximize(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void maximize(int l, int r, int x) {
    maximize(l, r, x, 0, 0, tree_size);
}

void add(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r)
        return;
    if (lx >= l && rx <= r) {
        doPushSum(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    add(l, r, x, 2 * ni + 1, lx, mid);
    add(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

void add(int l, int r, int x) {
    add(l, r, x, 0, 0, tree_size);
}

void set(int l, int r, int x, int ni, int lx, int rx) {
    if (rx <= l || lx >= r)
        return;
    if (lx >= l && rx <= r) {
        doPushEq(ni, lx, rx, x);
        return;
    }

    propagete(ni, lx, rx);

    int mid = lx + (rx - lx) / 2;
    set(l, r, x, 2 * ni + 1, lx, mid);
    set(l, r, x, 2 * ni + 2, mid, rx);

    seg_data[ni] = merge(seg_data[2 * ni + 1], seg_data[2 * ni +
↪ 2]);
}

```

```

}

void set(int l, int r, int x) {
    set(l, r, x, 0, 0, tree_size);
}

Node get_range(int l, int r, int ni, int lx, int rx) {
    propagete(ni, lx, rx);

    if (lx >= l && rx <= r)
        return seg_data[ni];
    if (rx <= l || lx >= r)
        return Node();

    int mid = (lx + rx) / 2;

    Node lf = get_range(l, r, 2 * ni + 1, lx, mid);
    Node ri = get_range(l, r, 2 * ni + 2, mid, rx);
    return merge(lf, ri);
}

ll get_sum(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).sum;
}

int get_mx(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).mx;
}

int get_mn(int l, int r) {
    return get_range(l, r, 0, 0, tree_size).mn;
}
};

```

2.1.5 DynamicSegTree

Dynamic (Implicit) Segment Tree - Space: $O(Q \log N)$ per query

// Allocates nodes on demand for ranges up to $1e9$ or $1e18$

```

template<typename T = ll>
struct DynamicSegTree {
    struct Node {
        T val = 0;
        int lc = -1, rc = -1;
    };

    vector<Node> tree;
    ll L, R;

    DynamicSegTree(ll L = 0, ll R = 1e9) : L(L), R(R) {
        add_node();
    }

    int add_node() {
        tree.pb(Node());
        return tree.size() - 1;
    }

    void update(int node, ll l, ll r, ll pos, T val) {
        tree[node].val += val;
        if (l == r) return;
        ll mid = l + (r - l) / 2;
        if (pos <= mid) {
            if (tree[node].lc == -1) {
                int nxt = add_node();
                tree[node].lc = nxt;
            }
            update(tree[node].lc, l, mid, pos, val);
        } else {
            if (tree[node].rc == -1) {
                int nxt = add_node();
                tree[node].rc = nxt;
            }
            update(tree[node].rc, mid + 1, r, pos, val);
        }
    }

    void update(ll pos, T val) {
        update(0, L, R, pos, val);
    }

    T query(int node, ll l, ll r, ll ql, ll qr) {
        if (node == -1 || ql > r || qr < l) return 0;
        if (ql <= l && r <= qr) return tree[node].val;
        ll mid = l + (r - l) / 2;
        return query(tree[node].lc, l, mid, ql, qr) +
            query(tree[node].rc, mid + 1, r, ql, qr);
    }
};

```

```

T query(ll ql, ll qr) {
    return query(0, L, R, ql, qr);
}
};

```

2.1.6 MergeSortTree

Merge Sort Tree - Space: $O(N \log N)$, Build: $O(N \log N)$, Query: $O(\log^2 N)$

// Stores sorted elements in each node to answer 2D range count queries

```

template<typename T = int>
struct MergeSortTree {
    int n;
    vector<vector<T>> tree;

    MergeSortTree(const vector<T>& a) : n(a.size()), tree(4 * n) {
        build(1, 0, n - 1, a);
    }

    void build(int node, int l, int r, const vector<T>& a) {
        if (l == r) {
            tree[node] = {a[l]};
            return;
        }
        int mid = (l + r) / 2;
        build(2 * node, l, mid, a);
        build(2 * node + 1, mid + 1, r, a);
        merge(all(tree[2 * node]), all(tree[2 * node + 1]),
            back_inserter(tree[node]));
    }

    // Returns number of elements <= k in index range [ql, qr]
    int query_count_le(int node, int l, int r, int ql, int qr, T k) {
        if (ql > r || qr < l) return 0;
        if (ql <= l && r <= qr) {
            return upper_bound(all(tree[node]), k) - tree[node].begin();
        }
        int mid = (l + r) / 2;
        return query_count_le(2 * node, l, mid, ql, qr, k) +
            query_count_le(2 * node + 1, mid + 1, r, ql, qr, k);
    }

    int query_count_le(int ql, int qr, T k) {
        return query_count_le(1, 0, n - 1, ql, qr, k);
    }
};

```

2.1.7 IterativeSegTree

Iterative (Non-Recursive) Segment Tree - Space: $O(N)$, Update: $O(\log N)$, Query: $O(\log N)$

// Fast, cache-friendly implementation for point update and range query

```

template<typename T = ll>
struct IterativeSegTree {
    int n;
    vector<T> tree;
    T neutral;
    function<T(T, T)> combine;

    IterativeSegTree(int n = 0, T neutral = 0, function<T(T, T)>
        combine = [](T a, T b) { return a + b; })
        : n(n), tree(2 * n, neutral), neutral(neutral),
        combine(combine) {}

    IterativeSegTree(const vector<T>& a, T neutral = 0, function<T(T,
        T)> combine = [](T a, T b) { return a + b; })
        : n(a.size()), tree(2 * a.size(), neutral), neutral(neutral),
        combine(combine) {
        for (int i = 0; i < n; i++) tree[n + i] = a[i];
        for (int i = n - 1; i > 0; i--) tree[i] = combine(tree[i << 1],
            tree[i << 1 | 1]);
    }

    void update(int p, T val) {
        // combine(LEFT, RIGHT) - do NOT write combine(tree[p], tree[p
        ^ 1]): when p is the right
        // child that reverses the arguments, which is wrong for any
        non-commutative combine
        // (matrix product, affine composition, "merge two structs").
        for (tree[p += n] = val; p > 1; p >>= 1)
            tree[p >> 1] = combine(tree[p & ~1], tree[p | 1]);
    }

    // Range query on [l, r] inclusive
    T query(int l, int r) {
        T resl = neutral, resr = neutral;
        for (l += n, r += n + 1; l < r; l >>= 1, r >>= 1) {
            if (l & 1) resl = combine(resl, tree[l++]);
            if (r & 1) resr = combine(tree[--r], resr);
        }
    }
};

```

```

    }
    return combine(resl, resr);
}
};

```

2.1.8 SegTree2D

2D Segment Tree - Space: $O(N * M)$, Update: $O(\log N * \log M)$, Query: $O(\log N * \log M)$

```

// Supports point updates and 2D subgrid range sum queries
template<typename T = ll>
struct SegTree2D {
    int n, m;
    vector<vector<T>> tree;

    SegTree2D(int n = 0, int m = 0) : n(n), m(m), tree(4 * n,
    ↪ vector<T>(4 * m, 0)) {}

    void update_y(int vx, int lx, int rx, int vy, int ly, int ry, int
    ↪ x, int y, T val) {
        if (ly == ry) {
            if (lx == rx) tree[vx][vy] += val;
            else tree[vx][vy] = tree[2 * vx][vy] + tree[2 * vx + 1][vy];
            return;
        }
        int my = (ly + ry) / 2;
        if (y <= my) update_y(vx, lx, rx, 2 * vy, ly, my, x, y, val);
        else update_y(vx, lx, rx, 2 * vy + 1, my + 1, ry, x, y, val);
        tree[vx][vy] = tree[vx][2 * vy] + tree[vx][2 * vy + 1];
    }

    void update_x(int vx, int lx, int rx, int x, int y, T val) {
        if (lx != rx) {
            int mx = (lx + rx) / 2;
            if (x <= mx) update_x(2 * vx, lx, mx, x, y, val);
            else update_x(2 * vx + 1, mx + 1, rx, x, y, val);
        }
        update_y(vx, lx, rx, 1, 0, m - 1, x, y, val);
    }

    void add(int x, int y, T val) {
        update_x(1, 0, n - 1, x, y, val);
    }

    T query_y(int vx, int vy, int ly, int ry, int qy1, int qy2) {
        if (qy1 > ry || qy2 < ly) return 0;
        if (qy1 <= ly && ry <= qy2) return tree[vx][vy];
        int my = (ly + ry) / 2;
        return query_y(vx, 2 * vy, ly, my, qy1, qy2) + query_y(vx, 2 *
    ↪ vy + 1, my + 1, ry, qy1, qy2);
    }

    T query_x(int vx, int lx, int rx, int qx1, int qx2, int qy1, int
    ↪ qy2) {
        if (qx1 > rx || qx2 < lx) return 0;
        if (qx1 <= lx && rx <= qx2) return query_y(vx, 1, 0, m - 1,
    ↪ qy1, qy2);
        int mx = (lx + rx) / 2;
        return query_x(2 * vx, lx, mx, qx1, qx2, qy1, qy2) + query_x(2
    ↪ * vx + 1, mx + 1, rx, qx1, qx2, qy1, qy2);
    }

    T query(int x1, int y1, int x2, int y2) {
        return query_x(1, 0, n - 1, x1, x2, y1, y2);
    }
};

```

2.1.9 MergeSortTreeFractional

MERGE SORT TREE WITH FRACTIONAL CASCADING - static "how many values $\leq x$ in $[l, r]$ " in $O(\log n)$

```

// instead of  $O(\log^2 n)$ : each node stores its sorted values plus
    ↪ pointers into the children, so the
// binary search happens ONCE at the root and is then followed down for
    ↪ free.
const int MAXN = 30005;
int a[MAXN];

struct Node {
    vector<int> vals;
    vector<int> to_left;
    vector<int> to_right;
} tree[4 * MAXN];

void build(int node, int start, int end) {
    if (start == end) {
        tree[node].vals.push_back(a[start]);
        tree[node].to_left.assign(2, 0);
    }
}

```

```

        tree[node].to_right.assign(2, 0);
    return;
}

int mid = (start + end) / 2;
int left_child = 2 * node;
int right_child = 2 * node + 1;

build(left_child, start, mid);
build(right_child, mid + 1, end);

int left_sz = tree[left_child].vals.size();
int right_sz = tree[right_child].vals.size();
int total_sz = left_sz + right_sz;

tree[node].vals.resize(total_sz);
tree[node].to_left.resize(total_sz + 1);
tree[node].to_right.resize(total_sz + 1);

int p1 = 0, p2 = 0;

for (int i = 0; i < total_sz; i++) {
    tree[node].to_left[i] = p1;
    tree[node].to_right[i] = p2;

    if (p1 < left_sz && (p2 == right_sz ||
    ↪ tree[left_child].vals[p1] <= tree[right_child].vals[p2])) {
        tree[node].vals[i] = tree[left_child].vals[p1++];
    } else {
        tree[node].vals[i] = tree[right_child].vals[p2++];
    }
}

tree[node].to_left[total_sz] = p1;
tree[node].to_right[total_sz] = p2;
}

int query(int node, int start, int end, int l, int r, int idx) {
    if (l > end || r < start) {
        return 0;
    }

    if (l <= start && end <= r) {
        return idx;
    }

    int mid = (start + end) / 2;
    int next_idx_left = tree[node].to_left[idx];
    int next_idx_right = tree[node].to_right[idx];

    return query(2 * node, start, mid, l, r, next_idx_left) +
        query(2 * node + 1, mid + 1, end, l, r, next_idx_right);
}

```

2.1.10 RangeMajorityCandidates

```

const int MAXN = 200005; // array bound
    ↪ (Template.cpp calls this N)
int a[MAXN];
vector<int> pos[MAXN];
int K_LIMIT = 2;

struct Node {
    vector<pair<int, int>> cand;
};

Node tree_seg[4 * MAXN];

Node merge_nodes(const Node& left, const Node& right) {
    vector<pair<int, int>> combined;
    combined.reserve(left.cand.size() + right.cand.size());

    int i = 0, j = 0;
    while (i < left.cand.size() && j < right.cand.size()) {
        if (left.cand[i].first == right.cand[j].first) {
            combined.push_back({left.cand[i].first,
    ↪ left.cand[i].second + right.cand[j].second});
            i++; j++;
        } else if (left.cand[i].first < right.cand[j].first) {
            combined.push_back(left.cand[i++]);
        } else {
            combined.push_back(right.cand[j++]);
        }
    }
    while (i < left.cand.size()) combined.push_back(left.cand[i++]);
    while (j < right.cand.size()) combined.push_back(right.cand[j++]);
}

```

```

    if (combined.size() <= K_LIMIT - 1) {
        return {combined};
    }

    vector<int> counts;
    counts.reserve(combined.size());
    for (const auto& p : combined) {
        counts.push_back(p.second);
    }

    nth_element(counts.begin(), counts.begin() + K_LIMIT - 1,
        counts.end(), greater<int>());
    int penalty = counts[K_LIMIT - 1];

    vector<pair<int, int>> survived;
    survived.reserve(K_LIMIT - 1);
    for (const auto& p : combined) {
        if (p.second > penalty) {
            survived.push_back({p.first, p.second - penalty});
        }
    }

    return {survived};
}

void build(int node, int start, int end) {
    if (start == end) {
        tree_seg[node].cands.push_back({a[start], 1});
        return;
    }
    int mid = start + (end - start) / 2;
    build(2 * node, start, mid);
    build(2 * node + 1, mid + 1, end);
    tree_seg[node] = merge_nodes(tree_seg[2 * node], tree_seg[2 * node
        + 1]);
}

Node query_candidates(int node, int start, int end, int l, int r) {
    if (r < start || end < l) return {};
    if (l <= start && end <= r) return tree_seg[node];

    int mid = start + (end - start) / 2;
    Node left_res = query_candidates(2 * node, start, mid, l, r);
    Node right_res = query_candidates(2 * node + 1, mid + 1, end, l, r);

    if (left_res.cands.empty()) return right_res;
    if (right_res.cands.empty()) return left_res;

    return merge_nodes(left_res, right_res);
}

// O(log N) fast static frequency count
int get_frequency(int val, int l, int r) {
    auto& v = pos[val];
    auto right_it = upper_bound(v.begin(), v.end(), r);
    auto left_it = lower_bound(v.begin(), v.end(), l);
    return distance(left_it, right_it);
}

void solve() {
    int n, q;
    cin >> n >> q;

    for (int i = 1; i <= n; i++) {
        cin >> a[i];
        pos[a[i]].push_back(i);
    }

    build(1, 1, n);

    while (q--) {
        int l, r;
        cin >> l >> r;

        int len = r - l + 1;
        Node res = query_candidates(1, 1, n, l, r);

        int max_freq = 0;
        for (const auto& p : res.cands) {
            int cand = p.first;
            int freq = get_frequency(cand, l, r);
            max_freq = max(max_freq, freq);
        }

        if (max_freq > (len + 1) / 2) {
            cout << 2 * max_freq - len << "\n";
        } else {

```

```

            cout << 1 << "\n";
        }
    }
}

2.1.11 SegTreeDescent
Segment Tree DESCENT - answer "first / last index satisfying P" in O(log n), not O(log2 n).

// Here: max-segtree. first_ge(l, x) = smallest i >= l with a[i] >= x,
// or n if none.
// Swap the stored aggregate + the two 'good()' tests to get: first
// prefix-sum >= S,
// first zero, k-th set bit, "how far right can I extend". This is the
// workhorse behind
// greedy "extend as far as possible" and interval-cover problems.
struct SegMax {
    int n;
    vector<ll> t;

    SegMax(int n = 0) : n(n), t(4 * n, LLONG_MIN) {}

    void build(const vector<ll>& a, int v, int l, int r) {
        if (l == r) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(a, 2 * v, l, m), build(a, 2 * v + 1, m + 1, r);
        t[v] = max(t[2 * v], t[2 * v + 1]);
    }
    void build(const vector<ll>& a) { if (n) build(a, 1, 0, n - 1); }

    void set(int p, ll x, int v, int l, int r) {
        if (l == r) { t[v] = x; return; }
        int m = (l + r) / 2;
        p <= m ? set(p, x, 2 * v, l, m) : set(p, x, 2 * v + 1, m + 1,
            r);
        t[v] = max(t[2 * v], t[2 * v + 1]);
    }
    void set(int p, ll x) { set(p, x, 1, 0, n - 1); }

    ll qry(int ql, int qr, int v, int l, int r) {
        if (qr < l || r < ql) return LLONG_MIN;
        if (ql <= l && r <= qr) return t[v];
        int m = (l + r) / 2;
        return max(qry(ql, qr, 2 * v, l, m), qry(ql, qr, 2 * v + 1, m + 1,
            r));
    }
    ll qry(int l, int r) { return qry(l, r, 1, 0, n - 1); }

    // smallest i in [ql, n) with a[i] >= x (returns n if none)
    int first_ge(int ql, ll x, int v, int l, int r) {
        if (r < ql || t[v] < x) return n;
        if (l == r) return l;
        int m = (l + r) / 2, res = first_ge(ql, x, 2 * v, l, m);
        return res < n ? res : first_ge(ql, x, 2 * v + 1, m + 1, r);
    }
    int first_ge(int ql, ll x) { return first_ge(ql, x, 1, 0, n - 1); }

    // largest i in [0, qr] with a[i] >= x (returns -1 if none)
    int last_ge(int qr, ll x, int v, int l, int r) {
        if (l > qr || t[v] < x) return -1;
        if (l == r) return l;
        int m = (l + r) / 2, res = last_ge(qr, x, 2 * v + 1, m + 1, r);
        return res >= 0 ? res : last_ge(qr, x, 2 * v, l, m);
    }
    int last_ge(int qr, ll x) { return last_ge(qr, x, 1, 0, n - 1); }
};

```

2.1.12 PersistentLazySegTree

PERSISTENT LAZY SEGMENT TREE - range add and range sum on ANY past version, O(log n) time and

```

// O(log n) new nodes per update.
// WHEN: you need RANGE updates and old versions. 03 -
// PersistentSegmentTree only does point
// updates, which is all the "k-th smallest in a range" trick needs -
// prefer it when that suffices,
// it is half the code and half the memory. If you only ever query the
// LATEST version, use
// 02 - LazySegmentTree.
// KEY INVARIANT: a tag is NEVER pushed down - that would mutate a node
// an old version still points
// at. Instead t[u].sum is always the CORRECT sum of u's whole range
// (tag included) and t[u].add is
// the amount still owed to the two children. So a query carries the
// accumulated ancestor tags down
// with it, and an update fixes sum on the way back up as sum(l) +
// sum(r) + add * len.
struct PLazySeg {

```

```

struct Node { int l = 0, r = 0; ll sum = 0, add = 0; };
vector<Node> t;
int n;
LazySeg(int n) : t(1), n(n) {} // node
→ 0 = the shared all-zero node
int build(const vector<ll>& a, int lo, int hi) {
    int u = t.size();
    t.push_back({});
    if (lo == hi) { t[u].sum = a[lo]; return u; }
    int m = (lo + hi) / 2, L = build(a, lo, m), R = build(a, m + 1,
→ hi);
    t[u].l = L, t[u].r = R, t[u].sum = t[L].sum + t[R].sum;
    return u;
}
int build(const vector<ll>& a) { return build(a, 0, n - 1); }
int update(int u, int ql, int qr, ll v, int lo, int hi) { // add v
→ to [ql, qr], returns new root
    int c = t.size();
    t.push_back(t[u]);
    if (ql <= lo && hi <= qr) { t[c].sum += v * (hi - lo + 1),
→ t[c].add += v; return c; }
    int m = (lo + hi) / 2;
    if (ql <= m) t[c].l = update(t[c].l, ql, qr, v, lo, m);
    if (qr > m) t[c].r = update(t[c].r, ql, qr, v, m + 1, hi);
    t[c].sum = t[t[c].l].sum + t[t[c].r].sum + t[c].add * (hi - lo
→ + 1);
    return c;
}
int update(int u, int l, int r, ll v) { return update(u, l, r, v,
→ 0, n - 1); }
ll query(int u, int ql, int qr, ll acc, int lo, int hi) { // acc =
→ sum of the ancestors' tags
    if (ql <= lo && hi <= qr) return t[u].sum + acc * (hi - lo + 1);
    int m = (lo + hi) / 2;
    acc += t[u].add;
    ll res = 0;
    if (ql <= m) res += query(t[u].l, ql, qr, acc, lo, m);
    if (qr > m) res += query(t[u].r, ql, qr, acc, m + 1, hi);
    return res;
}
ll query(int u, int l, int r) { return query(u, l, r, 0, 0, n - 1);
→ }
};
// MEMORY is the real constraint: O((n + q log n)) nodes, ~32 bytes
→ each. reserve() up front.
// RANGE ASSIGN instead of range add: the same no-push-down trick works
→ for any tag that composes,
// but the accumulated 'acc' must then be "the last assign seen on the
→ way down", so pass the tag
// itself rather than a sum, and let a deeper tag override a shallower
→ one.
// MIN/MAX aggregate: replace v * len by v and the pull by min(sum(l),
→ sum(r)) + add.
// A COMMON ALTERNATIVE that avoids all of this: make the update
→ offline, sort by version, and use
// an ordinary lazy segment tree with rollback (14 -
→ OfflineDynamicConnectivity's timeline trick).

```

2.1.13 ArithmeticProgressionSegTree

ARITHMETIC PROGRESSION SEGMENT TREE - "add a, a+d, a+2d, ... to a[l..r]", range sum, O(log n).

// WHEN: updates whose value DEPENDS ON THE POSITION inside the range -
→ "add i-l+1 to a[l..r]",
// "add w to every node on a path, weighted by depth", any linear ramp.
→ A plain lazy segment tree
// cannot do it because its tag must be one number.
// KEY INVARIANT: a progression restricted to a sub-range is still a
→ progression, so the tag stays
// two numbers (a, d) and stays CLOSED under composition: two tags add
→ componentwise. The tag is
// ANCHORED at the node's left endpoint, so pushing to the right child
→ skips (m+1-b) terms.

```

struct APSeg {
    struct Tag { ll a = 0, d = 0; }; // a,
→ a+d, a+2d, ... from node's left
    int n;
    vector<ll> s;
    vector<Tag> lz;
    APSeg(int n) : n(n), s(4 * n, 0), lz(4 * n) {}
    static ll tot(Tag t, ll len) { return t.a * len + t.d * (len * (len
→ - 1) / 2); }
    static Tag skip(Tag t, ll k) { return {t.a + k * t.d, t.d}; }
    → // drop the first k terms
    void apply(int x, Tag t, int len) { s[x] += tot(t, len), lz[x].a +=
→ t.a, lz[x].d += t.d; }
    void push(int x, int b, int e) {
        if (!lz[x].a && !lz[x].d) return;

```

```

        int m = (b + e) / 2;
        apply(2 * x, lz[x], m - b + 1), apply(2 * x + 1, skip(lz[x], m
→ + 1 - b), e - m);
        lz[x] = {};
    }
    void build(const vector<ll>& a, int x, int b, int e) {
        lz[x] = {};
        if (b == e) { s[x] = a[b]; return; }
        int m = (b + e) / 2;
        build(a, 2 * x, b, m), build(a, 2 * x + 1, m + 1, e);
        s[x] = s[2 * x] + s[2 * x + 1];
    }
    void build(const vector<ll>& a) { build(a, 1, 0, n - 1); }
    void upd(int x, int b, int e, int l, int r, Tag t) { // t is
→ anchored at index l
        if (r < b || e < l) return;
        if (l <= b && e <= r) { apply(x, skip(t, b - l), e - b + 1);
→ return; }
        push(x, b, e);
        int m = (b + e) / 2;
        upd(2 * x, b, m, l, r, t), upd(2 * x + 1, m + 1, e, l, r, t);
        s[x] = s[2 * x] + s[2 * x + 1];
    }
    void upd(int l, int r, ll a, ll d) { upd(1, 0, n - 1, l, r, {a,
→ d}); } // a[l]+=a, a[l+1]+=a+d..
    ll qry(int x, int b, int e, int l, int r) {
        if (r < b || e < l) return 0;
        if (l <= b && e <= r) return s[x];
        push(x, b, e);
        int m = (b + e) / 2;
        return qry(2 * x, b, m, l, r) + qry(2 * x + 1, m + 1, e, l, r);
    }
    ll qry(int l, int r) { return qry(1, 0, n - 1, l, r); }
};
// OVERFLOW: tot() is O(len^2 * d) - with n = 2e5 and d = 1e9 that is
→ 2e19, past ll. Work mod p
// (replace /2 by * inv2) or bound d.
// DECREASING ramps: pass a negative d; "add a, a-d, a-2d" to [l,r]
→ read right-to-left is the same
// tag anchored at r, so mirror the array or negate d and anchor at l
→ with a' = a + (r-l)*d.
// LIGHTER ALTERNATIVES, use these when they fit:
// Range AP add + POINT query only: one BIT over the second
→ difference b[i] = a[i]-2a[i-1]+a[i-2];
// an AP add touches 4 cells, and a[i] is a double prefix sum. 5
→ lines (02 - BIT).
// OFFLINE with all updates before all queries: apply the
→ second-difference trick to a plain
// array and prefix-sum twice at the end. O(n) total.
// GENERAL RULE: any tag family closed under composition and
→ restriction works. Polynomials of
// degree k need k+1 coefficients; affine "a[i] = p*a[i] + q" needs (p,
→ q) composed as function
// composition (that one is NOT commutative - compose in the right
→ order).

```

2.1.14 XORSegTree

XOR SEGMENT TREE - sum of $a[p \wedge x]$ over $l \leq p \leq r$, for an x given per query, plus point

```

// updates. O(log n) both, and NO extra memory over an ordinary segment
→ tree.
// WHEN: "the array is permuted by xor-ing every index with x" -
→ subset/bit problems where a query
// asks about a[l..r] of a xor-shifted array, e.g. CF 1654F. Also the
→ fast way to combine two
// arrays under an xor convolution restricted to a range.
// KEY INVARIANT: an ALIGNED block [b, b + 2^k) has its low k bits
→ free, so xor-ing every element
// of it with x maps the block ONTO the aligned block starting at b ^
→ (x >> k << k) - a block goes
// to a block, never to a straddling range. The canonical decomposition
→ of [l, r] is O(log n)
// aligned blocks, so the answer is O(log n) node lookups; the only
→ work is finding each partner
// node, which at level k is just xor-ing the node's
→ position-within-level by x >> k.
struct XorSeg { // n MUST be a power
→ of two; queries use x < n
    int n;
    vector<ll> t;
    XorSeg(const vector<ll>& a) : n(a.size()), t(2 * n) {
        copy(all(a), t.begin() + n);
        for (int i = n - 1; i; i--) t[i] = t[2 * i] + t[2 * i + 1];
    }
    void add(int i, ll v) { for (t[i += n] += v; i >= 1; i) t[i] = t[2 *
→ i] + t[2 * i + 1]; }
    ll query(int l, int r, int x) { // sum of a[p ^ x]

```

```

    ↪ over l <= p <= r
        ll res = 0;
        for (int lo = l + n, hi = r + n + 1, b = n; lo < hi; lo >= 1,
    ↪ hi >= 1, b >= 1, x >= 1) {
            if (lo & 1) res += t[(lo - b) ^ x] + b];, lo++; // b =
    ↪ number of nodes on this level
            if (hi & 1) res += t[(--hi - b) ^ x] + b]; // and
    ↪ also the level's index base
        }
        return res;
    }
};
// THE AGGREGATE must be COMMUTATIVE (sum, xor, min, gcd): xor-ing
    ↪ reorders the elements inside
// each block, so anything order-sensitive (max prefix sum, matrix
    ↪ product) breaks.
// PAD n up to a power of two with identity elements - the alignment
    ↪ argument needs it.
// THE SAME TRICK, no data structure: if you only need it once,
    ↪ enumerate the O(log n) blocks of
// [l, r] yourself and answer each with a prefix sum of the ORIGINAL
    ↪ array. That is the version to
// write when there are no updates.
// RANGE UPDATES do not survive this: a lazy tag would have to travel
    ↪ to the xor-partner subtree.
// If you need them, keep the tag OUTSIDE (a separate BIT of "add v to
    ↪ a[l..r]" offsets) and add
// the contribution of the covered blocks analytically.

```

2.1.15 KineticSegTree

KINETIC SEGMENT TREE (kinetic tournament) - each index i holds a LINE $y = a[i]*T + b[i]$; the

```

// tree answers "max over i in [l, r] of a[i]*T + b[i]" at a global
    ↪ time T that only MOVES FORWARD,
// and supports adding a constant to b over a range.
// WHEN: DP where the candidate values are linear in a parameter that
    ↪ sweeps monotonically -
// "max over j of dp[j] + (i-j)*c", scheduling with a moving deadline,
    ↪ CHT where the queries are
// sorted but the LINES also change over time (a Li Chao tree cannot
    ↪ delete or shift lines).
// If T is arbitrary (not monotone) use Li Chao / convex hull trick
    ↪ instead - this degenerates.
// COMPLEXITY: O(log n) per query and per range-add; advancing T is
    ↪ O(log^2 n) AMORTISED, and the
// total over a whole run is O((n + q) log^2 n). A single advance can
    ↪ cost O(n) in the worst case.
// KEY INVARIANT (the "certificate"): every node stores the winning
    ↪ line of its subtree AND melt =
// the earliest future time at which the winner of some node below it
    ↪ changes. Advancing to T
// recurses ONLY into nodes with melt <= T; everything else is provably
    ↪ still correct. Each
// recursion strictly changes a winner, and a winner can only change
    ↪ O(log n) times amortised per
// element because two lines cross at most once.

```

```

struct KineticSeg {
    int n;
    ll T = 0;
    vector<ll> a, b, lz, melt; // a, b
    ↪ = the node's WINNING line
    KineticSeg(const vector<ll>& A, const vector<ll>& B)
        : n(A.size()), a(4 * n), b(4 * n), lz(4 * n, 0), melt(4 * n,
    ↪ llinf) {
        build(A, B, 1, 0, n - 1);
    }
    static ll fdiv(ll p, ll q) { return p / q - ((p % q != 0) && ((p <
    ↪ 0) != (q < 0))); }
    ll cross(int w, int o) { // first
    ↪ time o strictly beats w
        if (a[o] <= a[w]) return llinf; //
    ↪ parallel or flatter: never
        return fdiv(b[w] - b[o], a[o] - a[w]) + 1;
    }
    void pull(int x) {
        int L = 2 * x, R = L + 1;
        int w = a[L] * T + b[L] >= a[R] * T + b[R] ? L : R, o = L + R -
    ↪ w;
        a[x] = a[w], b[x] = b[w];
        melt[x] = min({melt[L], melt[R], cross(w, o)});
    }
    void apply(int x, ll v) { b[x] += v, lz[x] += v; } //
    ↪ uniform shift keeps every melt
    void push(int x) { if (lz[x]) apply(2 * x, lz[x]), apply(2 * x + 1,
    ↪ lz[x]), lz[x] = 0; }
    void build(const vector<ll>& A, const vector<ll>& B, int x, int lo,
    ↪ int hi) {

```

```

        lz[x] = 0, melt[x] = llinf;
        if (lo == hi) { a[x] = A[lo], b[x] = B[lo]; return; }
        int m = (lo + hi) / 2;
        build(A, B, 2 * x, lo, m), build(A, B, 2 * x + 1, m + 1, hi);
        pull(x);
    }
    void heaten(int x, int lo, int hi) {
        if (melt[x] > T || lo == hi) return; //
    ↪ certificate still valid
        int m = (lo + hi) / 2;
        push(x);
        heaten(2 * x, lo, m), heaten(2 * x + 1, m + 1, hi);
        pull(x);
    }
    void advance(ll t) { T = t; heaten(1, 0, n - 1); } // t
    ↪ must be >= the current T
    void add(int x, int lo, int hi, int l, int r, ll v) { // b[i]
    ↪ += v for i in [l, r]
        if (r < lo || hi < l) return;
        if (l <= lo && hi <= r) { apply(x, v); return; }
        int m = (lo + hi) / 2;
        push(x);
        add(2 * x, lo, m, l, r, v), add(2 * x + 1, m + 1, hi, l, r, v);
        pull(x);
    }
    void add(int l, int r, ll v) { add(1, 0, n - 1, l, r, v); }
    void set(int x, int lo, int hi, int i, ll na, ll nb) { //
    ↪ replace the line at index i
        if (lo == hi) { a[x] = na, b[x] = nb; return; }
        int m = (lo + hi) / 2;
        push(x);
        i <= m ? set(2 * x, lo, m, i, na, nb) : set(2 * x + 1, m + 1,
    ↪ hi, i, na, nb);
        pull(x);
    }
    void set(int i, ll na, ll nb) { set(1, 0, n - 1, i, na, nb); }
    ll qry(int x, int lo, int hi, int l, int r) { // max
    ↪ at the CURRENT T
        if (r < lo || hi < l) return -llinf;
        if (l <= lo && hi <= r) return a[x] * T + b[x];
        int m = (lo + hi) / 2;
        push(x);
        return max(qry(2 * x, lo, m, l, r), qry(2 * x + 1, m + 1, hi,
    ↪ l, r));
    }
    ll qry(int l, int r) { return qry(1, 0, n - 1, l, r); }
};
// MIN instead of max: negate both a and b, or flip the comparison in
    ↪ pull() and cross().
// ADDING TO THE SLOPES over a range destroys every certificate below
    ↪ and is NOT amortised - do it
// only at a single index (set()) or rebuild.
// SEGMENT TREE BEATS (04) is the same idea with a different
    ↪ certificate: recurse only where the
// tag cannot be applied wholesale, and bound the recursions by a
    ↪ potential. If you can phrase your
// update as "the answer only changes when two stored candidates swap
    ↪ order", this is the shape.
// A CHEAPER SPECIAL CASE: if the lines are only ever ADDED (never
    ↪ removed) and queries are sorted,
// a monotone CHT is O(1) amortised per op - reach for that first.

```

2.1.16 HistoricSegTree

HISTORIC-MAXIMUM SEGMENT TREE - range add, range max of the CURRENT values, and range max of

```

// the HISTORIC value hmax[i] = max over every past moment of a[i].
    ↪ O(log n) per operation.
// WHEN: "what is the largest this cell ever reached", "peak stock
    ↪ price / peak queue length /
// high-water mark over a range", and as the engine behind offline
    ↪ problems that sweep an index
// while range-adding. If you only need the current max, use 02 -
    ↪ LazySegmentTree.
// KEY INVARIANT: the tag is a PAIR (add, mx) = "the total of this
    ↪ batch of adds" and "the largest
// PARTIAL SUM (prefix) any moment inside the batch reached". That pair
    ↪ is closed under
// composition: applying t and then u gives {t.add + u.add, max(t.mx,
    ↪ t.add + u.mx)}. So a whole
// run of pending updates can sit on a node and still be replayed
    ↪ exactly for the historic max.
// Careful: mx is a max over prefixes INCLUDING the empty one, so it is
    ↪ always >= 0.
struct HistoricSeg {
    struct Tag { ll add = 0, mx = 0; };
    int n;
    vector<ll> cur, his; //

```

```

↪ current max, historic max
vector<Tag> lz;
HistoricSeg(int n) : n(n), cur(4 * n, 0), his(4 * n, 0), lz(4 * n)
↪ {}
void apply(int x, Tag t) {
    his[x] = max(his[x], cur[x] + t.mx); // peak
↪ inside this batch
    cur[x] += t.add;
    lz[x] = {lz[x].add + t.add, max(lz[x].mx, lz[x].add + t.mx)};
}
void push(int x) {
    if (!lz[x].add && !lz[x].mx) return;
    apply(2 * x, lz[x]), apply(2 * x + 1, lz[x]);
    lz[x] = {};
}
void pull(int x) {
    cur[x] = max(cur[2 * x], cur[2 * x + 1]);
    his[x] = max(his[2 * x], his[2 * x + 1]);
}
void build(const vector<ll>& a, int x, int lo, int hi) {
    lz[x] = {};
    if (lo == hi) { cur[x] = his[x] = a[lo]; return; }
    int m = (lo + hi) / 2;
    build(a, 2 * x, lo, m), build(a, 2 * x + 1, m + 1, hi);
    pull(x);
}
void build(const vector<ll>& a) { build(a, 1, 0, n - 1); }
void upd(int x, int lo, int hi, int l, int r, ll v) {
    if (r < lo || hi < l) return;
    if (l <= lo && hi <= r) { apply(x, {v, max(v, 0LL)}); return; }
    int m = (lo + hi) / 2;
    push(x);
    upd(2 * x, lo, m, l, r, v), upd(2 * x + 1, m + 1, hi, l, r, v);
    pull(x);
}
void upd(int l, int r, ll v) { upd(1, 0, n - 1, l, r, v); }
// hist = false -> max of current values; hist = true -&; max of the
↪ all-time values
ll qry(int x, int lo, int hi, int l, int r, bool hist) {
    if (r < lo || hi < l) return -llinf;
    if (l <= lo && hi <= r) return hist ? his[x] : cur[x];
    int m = (lo + hi) / 2;
    push(x);
    return max(qry(2 * x, lo, m, l, r, hist), qry(2 * x + 1, m + 1,
↪ hi, l, r, hist));
}
ll qry(int l, int r, bool hist) { return qry(1, 0, n - 1, l, r,
↪ hist); }
};
// HISTORIC MIN: mirror everything (track the smallest prefix of the
↪ batch).
// HISTORIC SUM (sum over time of every past value, i.e. integrate the
↪ array) needs a bigger tag:
// keep sum, historic sum, and a count of elapsed time steps, and let
↪ the tag carry (add, time)
// with hsum += sum * dt. Same idea, one more field - only write it if
↪ the problem forces it.
// RANGE ASSIGN + historic max is the same tag with "assign" replacing
↪ "add"; compose so the later
// assign wins and mx tracks the largest value ever assigned.
// THE OFFLINE ALTERNATIVE, almost always shorter: if the queries are
↪ known in advance, sweep time
// and answer with an ordinary max segment tree plus a BIT for "was
↪ this cell touched after t".

```

2.1.17 SegTree2DSparse

SPARSE 2D SEGMENT TREE (segment tree of TREAPS) - point assign $a[i][j] = v$ and rectangle sum on

```

// a grid whose columns are huge or unknown in advance.  $O(\log n * \log
↪ q)$  per operation and only
//  $O(q \log n)$  memory - one node per touched cell per level.
// WHEN: 08 - SegTree2D is a dense  $n \times m$  tree and needs  $O(nm)$  memory; a
↪ merge-sort tree (06) is
// static; an offline problem should use 02 - BIT/03 -
↪ Offline2DFenwickTree, which is smaller and
// faster because it can compress the coordinates first. Reach for THIS
↪ one only when the updates
// are ONLINE and the grid is too big to allocate.
// KEY INVARIANT: the outer tree is indexed by ROW and every outer node
↪ owns a treap keyed by
// COLUMN holding, for each column  $j$  that some row in its range
↪ touches, the aggregate of that
// column over the node's rows. So an outer node is only ever asked
↪ about columns that exist, and
// after a point update the parents repair a SINGLE column by
↪ recombining their two children's
// value at that one column.

```

```

struct Seg2D {
    struct T { int l = 0, r = 0, key = 0; unsigned pr = 0; ll val = 0,
↪ agg = 0; };
    vector<T> tr; //
↪ shared pool for every treap
    vector<int> rt; // treap
↪ root of each outer node
    int n;
    Seg2D(int n) : tr(1), rt(4 * n, 0), n(n) {}
    unsigned rnd() { //
↪ xorshift; any prng will do
        static unsigned s = 2463534242u;
        return s ^= s << 13, s ^= s >> 17, s ^= s << 5;
    }
    void pull(int x) { tr[x].agg = tr[tr[x].l].agg + tr[x].val +
↪ tr[tr[x].r].agg; }
    void split(int x, int k, int& a, int& b) { // a
↪ keeps keys < k
        if (!x) { a = b = 0; return; }
        if (tr[x].key < k) a = x, split(tr[x].r, k, tr[a].r, b),
↪ pull(a);
        else b = x, split(tr[x].l, k, a, tr[b].l), pull(b);
    }
    int merge(int a, int b) {
        if (!a || !b) return a | b;
        if (tr[a].pr > tr[b].pr) { tr[a].r = merge(tr[a].r, b);
↪ pull(a); return a; }
        tr[b].l = merge(a, tr[b].l);
        pull(b);
        return b;
    }
    void set(int& root, int k, ll v) { // a[k]
↪ = v inside one treap
        int a, b, c;
        split(root, k, a, b), split(b, k + 1, b, c);
        if (b) tr[b].val = v, pull(b);
        else tr.push_back({0, 0, k, rnd(), v, v}), b = tr.size() - 1;
        root = merge(merge(a, b), c);
    }
    ll get(int& root, int lo, int hi) { // sum
↪ of keys in [lo, hi]
        int a, b, c;
        split(root, lo, a, b), split(b, hi + 1, b, c);
        ll res = tr[b].agg;
        root = merge(merge(a, b), c);
        return res;
    }
    void upd(int x, int lo, int hi, int i, int j, ll v) { //
↪ a[i][j] = v
        if (lo == hi) { set(rt[x], j, v); return; }
        int m = (lo + hi) / 2;
        i <= m ? upd(2 * x, lo, m, i, j, v) : upd(2 * x + 1, m + 1, hi,
↪ i, j, v);
        set(rt[x], j, get(rt[2 * x], j, j) + get(rt[2 * x + 1], j, j));
    }
    void upd(int i, int j, ll v) { upd(1, 0, n - 1, i, j, v); }
    ll qry(int x, int lo, int hi, int r1, int r2, int c1, int c2) {
        if (r2 < lo || hi < r1) return 0;
        if (r1 <= lo && hi <= r2) return get(rt[x], c1, c2);
        int m = (lo + hi) / 2;
        return qry(2 * x, lo, m, r1, r2, c1, c2) + qry(2 * x + 1, m +
↪ 1, hi, r1, r2, c1, c2);
    }
    ll qry(int r1, int r2, int c1, int c2) { return qry(1, 0, n - 1,
↪ r1, r2, c1, c2); }
};
// ANY IDEMPOTENT OR GROUP AGGREGATE works (sum, xor, gcd, min): upd()
↪ rebuilds a parent's single
// column from its two children, so the operation only has to be
↪ associative and commutative.
// ROWS TOO BIG TOO: make the outer tree dynamic as well (05 -
↪ DynamicSegTree) - allocate the outer
// node on first touch. Memory stays  $O(q \log n)$ .
// BIT OF TREAPS is the same shape with a shorter outer layer, but it
↪ cannot do "descend to find
// the  $k$ -th" and needs a group (no min/max).
// IF THE PROBLEM IS OFFLINE, do not write this. Sort the queries by
↪ row, sweep, and keep a single
// 1D BIT over columns:  $O((n + q) \log n)$ , five lines, and no memory
↪ pressure.

```

2.1.18 MexSegTree

MEX UNDER UPDATES - the smallest non-negative integer NOT present in the multiset, maintained

```
// under insert and erase in O(log V), read in O(log V) by ONE descent
// (not a scan).
// WHEN: "after each operation print the mex", Grundy/nim-value
// maintenance (10 - Game Theory),
// sliding-window mex. If the multiset only ever GROWS, a pointer that
// walks forward is O(1)
// amortised and this file is overkill. If you need the mex of a RANGE,
// see the recipe below.
// KEY INVARIANT: a segment tree over the VALUE axis [0, V] storing the
// MINIMUM count in each value
// range. The mex is the leftmost value with count 0, so the descent is
// "if the left child's
// minimum is 0 go left, else go right" - which is why the tree stores
// a MIN and not a sum.
// V only needs to be n + 1: with n elements the mex can never exceed n.
struct MexSet {
    int V;
    vector<int> cnt, mn;
    MexSet(int V) : V(V), cnt(V + 1, 0), mn(4 * (V + 1), 0) {}
    void upd(int x, int lo, int hi, int p) {
        if (lo == hi) { mn[x] = cnt[lo]; return; }
        int m = (lo + hi) / 2;
        p <= m ? upd(2 * x, lo, m, p) : upd(2 * x + 1, m + 1, hi, p);
        mn[x] = min(mn[2 * x], mn[2 * x + 1]);
    }
    void add(int v) { if (v <= V) cnt[v]++, upd(1, 0, V, v); }
    void del(int v) { if (v <= V) cnt[v]--, upd(1, 0, V, v); }
    int mex(int x, int lo, int hi) {
        if (lo == hi) return lo;
        int m = (lo + hi) / 2;
        return mn[2 * x] == 0 ? mex(2 * x, lo, m) : mex(2 * x + 1, m +
        1, hi);
    }
    int mex() { return mex(1, 0, V); }
};
// MEX OF A RANGE [l, r], offline: sort the queries by r and keep a
// segment tree over VALUES whose
// leaf v holds the LAST POSITION where v occurs in the prefix ending
// at r (-1 if never). The mex
// of [l, r] is the leftmost v whose last position is < l - the same
// descent, comparing against l
// instead of against 0. Online: make that tree persistent (03 -
// PersistentSegmentTree), version r.
// MEX OF A RANGE with array updates: no clean polylog solution - use
// Mo's plus the O(1)-add/del
// counter in 06 - Square Root Decomposition/07, scanning for the first
// empty count bucket.
// MEX OF ALL SUBARRAYS / sum of mex over all subarrays: divide and
// conquer on the position of the
// value 0, or the "for each mex value, count the subarrays achieving
// it" sweep - not this file.
// FIRST MISSING POSITIVE (mex over 1..) is the same with the axis
// shifted by one.
```

2.2 Binary Indexed Tree (BIT)

2.2.1 FenwickTree

Fenwick Tree (Binary Indexed Tree) - 0-based indexing

// Space: O(N), Time: O(log N) per update/query

```
template<typename T> T = ll>
struct FenwickTree {
    int n;
    vector<T> tree;

    FenwickTree(int n = 0) : n(n), tree(n + 1, 0) {}

    void add(int i, T delta) {
        for (i++; i <= n; i += i & -i) tree[i] += delta;
    }

    void set(int i, T val) {
        add(i, val - query(i, i));
    }

    T query(int i) {
        T sum = 0;
        for (i++; i > 0; i -= i & -i) sum += tree[i];
        return sum;
    }

    T query(int l, int r) {
        if (l > r) return 0;
        return query(r) - query(l - 1);
    }
};
```

```
}

// Returns first 0-based index where prefix sum >= val
int lower_bound(T val) {
    int pos = 0;
    for (int sz = 1 << (n ? __lg(n) : 0); sz > 0; sz >= 1) {
        if (pos + sz <= n && tree[pos + sz] < val) {
            val -= tree[pos + sz];
            pos += sz;
        }
    }
    return pos;
};
```

2.2.2 FenwickTree2D

2D Fenwick Tree - 0-based indexing

```
// Space: O(N * M), Time: O(log N * log M) per update/query
template<typename T> T = ll>
struct FenwickTree2D {
    int n, m;
    vector<vector<T>>> tree;

    FenwickTree2D(int n = 0, int m = 0) : n(n), m(m), tree(n + 1,
    vector<T>(m + 1, 0)) {}

    void add(int x, int y, T val) {
        for (int i = x + 1; i <= n; i += i & -i) {
            for (int j = y + 1; j <= m; j += j & -j) {
                tree[i][j] += val;
            }
        }
    }

    void set(int x, int y, T val) {
        add(x, y, val - query(x, y, x, y));
    }

    T query(int x, int y) {
        T sum = 0;
        for (int i = min(x + 1, n); i > 0; i -= i & -i) {
            for (int j = min(y + 1, m); j > 0; j -= j & -j) {
                sum += tree[i][j];
            }
        }
        return sum;
    }

    // 2D Subgrid Range Sum from (x1, y1) to (x2, y2) inclusive
    T query(int x1, int y1, int x2, int y2) {
        if (x1 > x2 || y1 > y2) return 0;
        return query(x2, y2) - query(x1 - 1, y2) - query(x2, y1 - 1) +
        query(x1 - 1, y1 - 1);
    }
};
```

2.2.3 Offline2DFenwickTree

Offline 2D Fenwick Tree (Compressed 2D BIT)

// Space: O(N + Q log N), Time: O(log² N) per query

```
template<typename T> T = ll>
struct BIT2D {
    int n;
    vector<vector<int>>> vals;
    vector<vector<T>>> bit;

    int ind(const vector<int>& v, int x) {
        return upper_bound(all(v), x) - v.begin() - 1;
    }

    // mx: max row index, todo: all (r, c) update coordinates that will
    // be called
    BIT2D(int mx, vector<array<int, 2>> todo) : n(mx), vals(mx + 1),
    bit(mx + 1) {
        sort(all(todo), [](const auto& a, const auto& b) { return a[1]
        < b[1]; });

        for (int i = 0; i <= n; i++) vals[i].pb(INT_MIN);
        for (auto [r, c] : todo) {
            for (int x = r; x <= n; x |= x + 1) {
                if (vals[x].back() != c) vals[x].pb(c);
            }
        }
        for (int i = 0; i <= n; i++) bit[i].resize(vals[i].size(), 0);
    }

    void add(int r, int c, T val) {
```

```

    for (; r <= n; r |= r + 1) {
        for (int i = ind(vals[r], c); i < (int)bit[r].size(); i |=
        ↪ i + 1) {
            bit[r][i] += val;
        }
    }

    T query(int r, int c) {
        T res = 0;
        for (; r >= 0; r = (r & (r + 1)) - 1) {
            for (int i = ind(vals[r], c); i >= 0; i = (i & (i + 1)) -
        ↪ 1) {
                res += bit[r][i];
            }
        }
        return res;
    }

    T query(int r1, int c1, int r2, int c2) {
        return query(r2, c2) - query(r2, c1 - 1) - query(r1 - 1, c2) +
        ↪ query(r1 - 1, c1 - 1);
    }

    void reset() {
        for (auto& v : bit) fill(all(v), 0);
    }
};

```

2.2.4 BITRangeUpdateRangeQuery

BIT with RANGE UPDATE + RANGE QUERY (two BITs). 0-based, $O(\log n)$ per op.

```

// pre(i) = sum_{j<=i} a[j] = S1(i)*(i+1) - S2(i), where a range-add of
↪ v on [l,r] does
// S1: +v at l, -v at r+1      S2: +v*l at l, -v*(r+1) at r+1
template <typename T = ll>
struct BIT2 {
    int n;
    vector<T> b1, b2;

    BIT2(int n = 0) : n(n), b1(n + 1, 0), b2(n + 1, 0) {}
    void add(vector<T>& b, int i, T v) { for (i++; i <= n; i += i & -i)
    ↪ b[i] += v; }
    T sum(const vector<T>& b, int i) const { T s = 0; for (i++; i > 0;
    ↪ i -= i & -i) s += b[i]; return s; }

    void upd(int l, int r, T v) { // add v to [l, r]
        add(b1, l, v), add(b1, r + 1, -v);
        add(b2, l, v * l), add(b2, r + 1, -v * (r + 1));
    }
    T pre(int i) const { return i < 0 ? 0 : sum(b1, i) * (i + 1) -
    ↪ sum(b2, i); }
    T qry(int l, int r) const { return pre(r) - pre(l - 1); }
};

```

2.2.5 BITOfSegTrees

BIT OF SEGMENT TREES ("tree in tree") - k-th SMALLEST in a[l..r] WITH point assignments, in

```

//  $O(\log^2 n)$  per operation and  $O(n \log^2 n)$  nodes.
// WHEN: the array changes. Without updates the answer is a persistent
↪ segment tree over prefixes
// (01 - Segment Tree/03) at  $O(\log n)$  per query and a third of the
↪ memory, or a wavelet tree (13);
// offline, Mo's or parallel binary search is usually simpler. Reach
↪ for this only for ONLINE
// "k-th smallest / count values <= x in a range" mixed with
↪ assignments.
// KEY INVARIANT: the BIT is over POSITIONS and every BIT node owns a
↪ dynamic segment tree over the
// VALUE axis counting the values of the positions that node covers. A
↪ prefix [0, r] is a sum of
//  $O(\log n)$  BIT nodes, so [l, r] is  $O(\log n)$  value-trees to add and
↪  $O(\log n)$  to subtract; the k-th
// smallest is then ONE simultaneous top-down descent through all of
↪ them at once, adding up the
// left-child counts to decide which way to go.

```

```

struct BITSeg {
    int n, V; //
    ↪ values live in [0, V)
    vector<int> rt, lc, rc, cnt, a; // rt[i]
    ↪ = value tree of BIT node i
    BITSeg(int n, int V) : n(n), V(V), rt(n + 1, 0), lc(1, 0), rc(1,
    ↪ 0), cnt(1, 0), a(n, -1) {}
    int upd(int x, int lo, int hi, int p, int d) { // += d
    ↪ at value p, returns node id
        if (!x) x = cnt.size(), lc.push_back(0), rc.push_back(0),
    ↪ cnt.push_back(0);

```

```

        cnt[x] += d;
        if (lo < hi) {
            int m = (lo + hi) / 2, c;
            if (p <= m) c = upd(lc[x], lo, m, p, d), lc[x] = c;
            else c = upd(rc[x], m + 1, hi, p, d), rc[x] = c;
        }
        return x;
    }

    void add(int i, int v, int d) {
        for (i++; i <= n; i += i & -i) rt[i] = upd(rt[i], 0, V - 1, v,
    ↪ d);
    }

    void set(int i, int v) { // a[i]
    ↪ = v (first call = insert)
        if (a[i] >= 0) add(i, a[i], -1);
        add(i, a[i] = v, 1);
    }

    int kth(int l, int r, int k) { // k-th
    ↪ smallest in [l, r], k >= 1
        vector<int> A, B;
        for (int i = r + 1; i > 0; i -= i & -i) A.push_back(rt[i]);
        for (int i = l; i > 0; i -= i & -i) B.push_back(rt[i]);
        int lo = 0, hi = V - 1;
        while (lo < hi) {
            int m = (lo + hi) / 2, c = 0;
            for (int x : A) c += cnt[lc[x]];
            for (int x : B) c -= cnt[lc[x]];
            bool left = k <= c;
            if (!left) k -= c, lo = m + 1;
            else hi = m;
            for (int& x : A) x = left ? lc[x] : rc[x];
            for (int& x : B) x = left ? lc[x] : rc[x];
        }
        return lo;
    }

    int less(int x, int lo, int hi, int ql, int qr) { //
    ↪ count of values in [ql, qr]
        if (!x || qr < lo || hi < ql) return 0;
        if (ql <= lo && hi <= qr) return cnt[x];
        int m = (lo + hi) / 2;
        return less(lc[x], lo, m, ql, qr) + less(rc[x], m + 1, hi, ql,
    ↪ qr);
    }

    int count(int l, int r, int vl, int vr) { // how
    ↪ many a[i] in [vl, vr]
        int res = 0;
        for (int i = r + 1; i > 0; i -= i & -i) res += less(rt[i], 0, V
    ↪ - 1, vl, vr);
        for (int i = l; i > 0; i -= i & -i) res -= less(rt[i], 0, V -
    ↪ 1, vl, vr);
        return res;
    }
};

// COORDINATE COMPRESS first (01 - Miscellaneous/05) - and if updates
↪ introduce NEW values, compress
// the update values too, offline, before starting.
// MEMORY is the trap:  $n \log^2 n$  nodes at 12 bytes is ~100 MB at  $n =$ 
↪  $2e5$ . If that is too much,
// replace the inner dynamic tree by a MERGE SORT TREE rebuilt in
↪ blocks, or go offline.
// THE SAME SHAPE solves: "sum of the k smallest in a range", "rank of
↪ x in a range", "predecessor
// / successor of x in a range" - all are the same descent with a
↪ different accumulator.

```

2.2.6 OfflineDistinct

COUNT DISTINCT VALUES IN A RANGE, OFFLINE - the single most reused sweep in the whole notebook.

```

// Needs: 02 - Binary Indexed Tree (BIT)/01 - FenwickTree.cpp
//  $O((n + q) \log n)$ ,  $O(n)$  memory, ~20 lines. (SPOJ DQUERY and a hundred
↪ clones.)
// WHEN: any "how many different / how many values appear exactly once
↪ / sum of distinct values"
// question with all queries known up front. Mo's (06 - Square Root
↪ Decomposition/01) also does it
// but is  $O(n \sqrt{n})$ ; a persistent segment tree does it ONLINE at
↪  $O(\log n)$  per query with the very
// same invariant, so start here and upgrade only if the queries are
↪ forced online.
// KEY INVARIANT: sweep r left to right and keep the BIT holding a 1 at
↪ the LAST OCCURRENCE SO FAR
// of every distinct value and 0 everywhere else. Then the number of
↪ ones in [l, r] is exactly the
// number of distinct values in [l, r]: each value is counted at its
↪ rightmost occurrence, which
// lies in [l, r] if and only if the value occurs there at all. Moving
↪ r to a new position means

```

```
// clearing the previous occurrence of a[r] and setting the new one -
// → two BIT updates.
struct OfflineDistinct {
    int n;
    vector<int> a; //
    // values already compressed to [0,V)
    vector<array<int, 3>> qs; // {r,
    // l, id}
    OfflineDistinct(const vector<int>& a) : n(a.size()), a(a) {}
    void ask(int l, int r, int id) { qs.push_back({r, l, id}); }
    vector<int> solve(int V) {
        sort(all(qs));
        FenwickTree<int> bit(n);
        vector<int> last(V, -1), res(qs.size());
        int at = 0;
        for (int r = 0; r < n; r++) {
            if (last[a[r]] >= 0) bit.add(last[a[r]], -1);
            bit.add(last[a[r]] = r, 1);
            while (at < (int)qs.size() && qs[at][0] == r)
                res[qs[at][2]] = bit.query(qs[at][1], r), at++;
        }
        return res;
    }
};
// VALUES APPEARING EXACTLY ONCE in [l, r]: keep TWO BITS, one with a 1
// → at the last occurrence and
// one with a 1 at the second-to-last; the answer is ones(last) -
// → ones(secondLast) over [l, r].
// SUM of the distinct values: put a[r] instead of 1 at the last
// → occurrence. Any per-value weight
// works the same way, including "count values whose multiplicity is >=
// → k" (mark the k-th last).
// ONLINE version: build a persistent segment tree where version r is
// → "the array of last
// occurrences after processing r", then the answer is a range count on
// → version r. Same invariant,
// O(n log n) memory (01 - Segment Tree/03).
// UPDATES to the array as well: this sweep dies; use Mo's with updates
// → (06/05) or a BIT of segment
// trees (05) keyed by "previous occurrence" - both are much heavier,
// → so check first whether the
// problem really needs them.
```

2.2.7 BIT2DRangeUpdate

2D BIT WITH RANGE UPDATE AND RANGE QUERY - add v to a RECTANGLE, ask the sum of a RECTANGLE, in

```
// O(log n log m) with four BITS and no lazy propagation anywhere.
// WHEN: grid problems with rectangle add / rectangle sum. 02 -
// → FenwickTree2D only does point
// update; 04 - BITRangeUpdateRangeQuery is the 1D version of exactly
// → this trick; a 2D segment tree
// with lazy propagation does NOT exist in any usable form (you cannot
// → push a tag through two
// dimensions), which is why this identity is the standard answer.
// KEY INVARIANT: work on the 2D difference array d, where a[i][j] =
// → sum of d over the rectangle
// [0..i] x [0..j]. A rectangle add touches exactly FOUR cells of d.
// → Expanding the double prefix
// sum gives
// pre(x, y) = (x+1)(y+1) * S(d) - (y+1) * S(d*i) - (x+1) * S(d*j) +
// → S(d*i*j)
// over i <= x, j <= y - four independent BIT-able quantities, hence
// → four BITS.
template <typename T = ll>
struct BIT2DRange {
    int n, m;
    vector<vector<T>> b[4];
    BIT2DRange(int n, int m) : n(n), m(m) {
        for (int k = 0; k < 4; k++) b[k].assign(n + 2, vector<T>(m + 2,
        // → 0));
    }
    void add(int x, int y, T v) { //
        // → d[x][y] += v (internal)
        if (x > n || y > m) return;
        T c[4] = {v, v * x, v * y, v * x * y};
        for (int i = x + 1; i <= n + 1; i += i & -i)
            for (int j = y + 1; j <= m + 1; j += j & -j)
                for (int k = 0; k < 4; k++) b[k][i][j] += c[k];
    }
    void upd(int x1, int y1, int x2, int y2, T v) { // add v
        // → to the rectangle
        add(x1, y1, v), add(x1, y2 + 1, -v), add(x2 + 1, y1, -v),
        // → add(x2 + 1, y2 + 1, v);
    }
    T pre(int x, int y) { // sum
        // → over [0..x] x [0..y]
        if (x < 0 || y < 0) return 0;
    }
```

```
T s[4] = {0, 0, 0, 0};
for (int i = x + 1; i > 0; i -= i & -i)
    for (int j = y + 1; j > 0; j -= j & -j)
        for (int k = 0; k < 4; k++) s[k] += b[k][i][j];
return (T)(x + 1) * (y + 1) * s[0] - (T)(y + 1) * s[1] - (T)(x
// → + 1) * s[2] + s[3];
}
T qry(int x1, int y1, int x2, int y2) {
    return pre(x2, y2) - pre(x1 - 1, y2) - pre(x2, y1 - 1) + pre(x1
// → - 1, y1 - 1);
}
};
// MEMORY is 4 * n * m of your value type - 12.8 GB at 2000x2000 with
// → ll. If that hurts, use int
// where safe, or compress one axis and go offline (03 -
// → Offline2DFenwickTree).
// THE SAME EXPANSION generalises: range-add of a polynomial of degree
// → k in each axis needs
// (k+1)^2 BITS. In 1D the arithmetic-progression case is 01 - Segment
// → Tree/13.
// XOR instead of sum: the (x+1)(y+1) coefficients become parities, so
// → keep four BITS and multiply
// by the parity of (x+1)*(y+1) etc. Usually simpler to just use a 2D
// → segment tree of counts.
// SPARSE GRIDS: replace each inner vector by a hash map, or compress
// → coordinates first - the outer
// structure is unchanged.
```

2.3 Sparse Table

2.3.1 SparseTable

SPARSE TABLE - O(1) range query after O(N log N) build, NO updates.

```
// The operation must be IDEMPOTENT (min, max, gcd, and, or) because
// → the two covering blocks
// OVERLAP. For sum/product/xor - anything not idempotent - use 12 -
// → DisjointSparseTable.cpp
// (still O(1) query) or a Fenwick tree. Guard l <= r: __lg(0) is
// → undefined.
template<typename T, typename F = function<T(T, T)>>
struct SparseTable {
    int n;
    vector<vector<T>> mat;
    F func;

    SparseTable() = default;
    SparseTable(const vector<T>& a, F f = [](T x, T y) { return max(x,
    // → y); }) : n(a.size()), func(f) {
        int k = n ? __lg(n) + 1 : 0;
        mat.assign(k, a);
        for (int j = 1; j < k; j++) {
            for (int i = 0; i + (1 << j) <= n; i++) {
                mat[j][i] = func(mat[j - 1][i], mat[j - 1][i + (1 << j)
                // → - 1));
            }
        }

        T query(int l, int r) const {
            int j = __lg(r - l + 1);
            return func(mat[j][l], mat[j][r - (1 << j) + 1]);
        }
    }
};
```

2.3.2 SparseTable2D

2D Sparse Table - O(N * M * log N * log M) build, O(1) query

```
template<typename T = ll>
struct SparseTable2D {
    int n, m;
    // jmp[kr][kc][i][j] = max in 2^kr x 2^kc subgrid starting at (i, j)
    vector<vector<vector<vector<T>>>> jmp;

    SparseTable2D(const vector<vector<T>>& grid) {
        n = grid.size();
        if (!n) return;
        m = grid[0].size();
        if (!m) return;

        int log_n = __lg(n) + 1, log_m = __lg(m) + 1;
        jmp.assign(log_n, vector<vector<vector<T>>>>(log_m,
        // → vector<vector<T>>>(n, vector<T>(m)))));

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                jmp[0][0][i][j] = grid[i][j];
            }
        }
    }
```

```

    for (int kc = 1; kc < log_m; kc++) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j + (1 << kc) <= m; j++) {
                jmp[0][kc][i][j] = max(jmp[0][kc - 1][i][j],
                ↪ jmp[0][kc - 1][i][j + (1 << (kc - 1))]);
            }
        }

        for (int kr = 1; kr < log_n; kr++) {
            for (int kc = 0; kc < log_m; kc++) {
                for (int i = 0; i + (1 << kr) <= n; i++) {
                    for (int j = 0; j + (1 << kc) <= m; j++) {
                        jmp[kr][kc][i][j] = max(jmp[kr - 1][kc][i][j],
                ↪ jmp[kr - 1][kc][i + (1 << (kr - 1))][j]);
                    }
                }
            }
        }

        T query(int r1, int c1, int r2, int c2) const {
            int kr = __lg(r2 - r1 + 1), kc = __lg(c2 - c1 + 1);
            return max({jmp[kr][kc][r1][c1],
                jmp[kr][kc][r1][c2 - (1 << kc) + 1],
                jmp[kr][kc][r2 - (1 << kr) + 1][c1],
                jmp[kr][kc][r2 - (1 << kr) + 1][c2 - (1 << kc) +
                ↪ 1]});
        }
    };
}

```

2.3.3 SquareSparseTable

SQUARE SPARSE TABLE - $O(1)$ query for any SQUARE sub-grid, $O(N \cdot M \cdot \log(\min(N, M)))$ build.

```

// The 2D sparse table (02 - SparseTable2D.cpp) stores one table per
    ↪ (row-power, col-power) pair,
// so it costs  $\log(N) \cdot \log(M)$  memory. If every query is a SQUARE - "best
    ↪ k x k block", "does some
// k x k window contain only ones", template matching on a grid - you
    ↪ only ever need the diagonal
// of that table, which is a factor of log cheaper in both time and
    ↪ memory.
// Query a k x k square: cover it with four overlapping  $2^t \times 2^t$ 
    ↪ squares,  $t = \text{floor}(\log_2 k)$ .
// The operation must be IDEMPOTENT (min/max/gcd/and/or) because those
    ↪ four squares overlap.
template<typename T>
struct SquareSparseTable {
    int n, m;
    // jmp[k][i][j] = max of  $2^k \times 2^k$  square starting at (i, j)
    vector<vector<vector<T>>> jmp;

    SquareSparseTable(const vector<vector<T>>& grid) {
        n = grid.size();
        if (!n) return;
        m = grid[0].size();
        if (!m) return;

        int log_max = __lg(min(n, m)) + 1;
        jmp.assign(log_max, vector<vector<T>>(n, vector<T>(m)));

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) {
                jmp[0][i][j] = grid[i][j];
            }
        }

        for (int k = 1; k < log_max; k++) {
            int pw = 1 << (k - 1);
            for (int i = 0; i + (pw * 2) <= n; i++) {
                for (int j = 0; j + (pw * 2) <= m; j++) {
                    jmp[k][i][j] = max({jmp[k - 1][i][j],
                        jmp[k - 1][i][j + pw],
                        jmp[k - 1][i + pw][j],
                        jmp[k - 1][i + pw][j + pw]});
                }
            }
        }
    }

    // Queries max in square subgrid from (r1, c1) to (r2, c2)
    T query(int r1, int c1, int r2, int c2) const {
        int k = __lg(r2 - r1 + 1), pw = 1 << k;
        return max({jmp[k][r1][c1],
            jmp[k][r1][c2 - pw + 1],
            jmp[k][r2 - pw + 1][c1],

```

```

        jmp[k][r2 - pw + 1][c2 - pw + 1]});
    };
}

```

2.3.4 SqrtTree

SQRT TREE - $O(1)$ range query for ANY associative operation, WITH point updates in $O(\sqrt{n})$.

```

// Build  $O(n \log \log n)$ , memory  $O(n \log \log n)$ .
// WHEN: the operation is associative but NOT idempotent and NOT
    ↪ invertible (matrix product, "max
// suffix sum", min-plus, string hash concat), so a sparse table (01)
    ↪ is wrong and a BIT is
// impossible - and you need  $O(1)$  queries or updates on top. If the op
    ↪ is idempotent use
// 01 - SparseTable; if there are no updates use 12 -
    ↪ DisjointSparseTable ( $O(n \log n)$  build,  $O(1)$ 
// query, 20 lines); if  $O(\log n)$  queries are fine just use a segment
    ↪ tree.
// KEY INVARIANT: cut the array into  $\sqrt{n}$  blocks of  $\sqrt{n}$ . Store
    ↪ prefix and suffix aggregates
// inside every block and a table 'between[i][j]' for whole blocks
    ↪ i..j. A query then costs one
// suffix + one between + one prefix. Blocks are handled RECURSIVELY by
    ↪ the same structure, which
// is why the depth is  $\log \log n$ :  $n \rightarrow \sqrt{n} \rightarrow n^{1/4} \rightarrow \dots$ . The
    ↪ top-level 'between' table is
// itself a Sqrt Tree over the  $\sqrt{n}$  block aggregates, so an update
    ↪ rebuilds only one block per
// layer plus one entry of the index structure.

```

```

struct SqrtTree {
    typedef ll T;
    static T op(T a, T b) { return a + b; } // ANY
    ↪ associative op
    int n, lg, isz;
    vector<T> v;
    vector<int> clz, layers, onLayer;
    vector<vector<T>> pref, suf, betw;
    void buildBlock(int k, int l, int r) {
        pref[k][l] = v[l];
        for (int i = l + 1; i < r; i++) pref[k][i] = op(pref[k][i - 1],
        ↪ v[i]);
        suf[k][r - 1] = v[r - 1];
        for (int i = r - 2; i >= l; i--) suf[k][i] = op(v[i], suf[k][i
        ↪ + 1]);
    }
    void buildBetween(int k, int lb, int rb, int off) {
        int sl = (layers[k] + 1) >> 1, cl = layers[k] >> 1, bs = 1 <<
        ↪ sl;
        int bc = (rb - lb + bs - 1) >> sl;
        for (int i = 0; i < bc; i++) {
            T cur = T();
            for (int j = i; j < bc; j++) {
                T add = suf[k][lb + (j << sl)];
                cur = i == j ? add : op(cur, add);
                betw[k - 1][off + lb + (i << cl) + j] = cur;
            }
        }
        void build(int k, int lb, int rb, int off) {
            if (k >= (int)layers.size()) return;
            int bs = 1 << ((layers[k] + 1) >> 1);
            for (int l = lb; l < rb; l += bs) {
                int r = min(l + bs, rb);
                buildBlock(k, l, r), build(k + 1, l, r, off);
            }
            if (k) buildBetween(k, lb, rb, off);
            else { // layer
                ↪ 0: rebuild the index array
                int sl = (lg + 1) >> 1;
                for (int i = 0; i < isz; i++) v[n + i] = suf[0][i << sl];
                build(1, n, n + isz, (1 << lg) - n);
            }
        }
        void update(int k, int lb, int rb, int off, int x) {
            if (k >= (int)layers.size()) return;
            int sl = (layers[k] + 1) >> 1, bs = 1 << sl, bi = (x - lb) >>
            ↪ sl;
            int l = lb + (bi << sl), r = min(l + bs, rb);
            buildBlock(k, l, r);
            if (k) buildBetween(k, lb, rb, off);
            else {
                v[n + bi] = suf[0][bi << ((lg + 1) >> 1)];
                update(1, n, n + isz, (1 << lg) - n, n + bi);
            }
            update(k + 1, l, r, off, x);
        }
        void update(int x, T val) { v[x] = val, update(0, 0, n, 0, x); }
    };
}

```

```

T query(int l, int r, int off, int base) {
    ↪ inclusive [l, r]
    if (l == r) return v[l];
    if (l + 1 == r) return op(v[l], v[r]);
    int k = onLayer[clz[(l - base) ^ (r - base)]];
    int sl = (layers[k] + 1) >> 1, cl = layers[k] >> 1;
    int lb = (((l - base) >> layers[k]) << layers[k]) + base;
    int lblk = ((l - lb) >> sl) + 1, rblk = ((r - lb) >> sl) - 1;
    T res = suf[k][l];
    if (lblk <= rblk)
        res = op(res, k ? betw[k - 1][off + lb + (lblk << cl) +
    ↪ rblk]
        : query(n + lblk, n + rblk, (1 << lg) - n,
    ↪ n));
    return op(res, pref[k][r]);
}

T query(int l, int r) { return query(l, r, 0, 0); }

SqrtTree(const vector<T>& a) : n(a.size()), lg(0), v(a) {
    while ((1 << lg) < n) lg++;
    clz.assign(1 << lg, 0), onLayer.assign(lg + 1, 0);
    for (int i = 1; i < (1 << lg); i++) clz[i] = clz[i >> 1] + 1;
    for (int t = lg; t > 1; t = (t + 1) >> 1) onLayer[t] =
    ↪ layers.size(), layers.push_back(t);
    for (int i = lg - 1; i >= 0; i--) onLayer[i] = max(onLayer[i],
    ↪ onLayer[i + 1]);
    int sl = (lg + 1) >> 1, bs = 1 << sl;
    isz = (n + bs - 1) >> sl;
    v.resize(n + isz);
    pref.assign(layers.size(), vector<T>(n + isz));
    suf.assign(layers.size(), vector<T>(n + isz));
    betw.assign(max(0, (int)layers.size() - 1), vector<T>((1 << lg)
    ↪ + bs));
    build(0, 0, n, 0);
}

// NON-COMMUTATIVE ops are fine and are the whole point - op(a, b) is
    ↪ always applied left to right.
// T MUST have a sensible default constructor; it is never used as an
    ↪ identity, only as a filler.
// THE O(1)-QUERY / O(sqrt n)-UPDATE trade is unusual. Its real niche
    ↪ is "answer 1e6 queries of a
// non-invertible op with occasional updates". If updates outnumber
    ↪ queries, a segment tree wins.
// THE TWO-LEVEL VERSION (one layer of blocks, no recursion) is 20
    ↪ lines and gives O(1) query with
// O(n) build only if you can afford a sqrt(n) x sqrt(n) between-table;
    ↪ recursion is what makes the
// update cheap. Write the two-level one first if the array is static
    ↪ and small.

```

2.4 Trie

2.4.1 StringTrie

PREFIX TRIE over an alphabet. O(L) per insert/query. Reach for it for prefix counting, autocomplete,

// "is any inserted word a prefix of this one", and as the skeleton
 ↪ Aho-Corasick is built on.
 // Prefix Trie for Strings
 // Time: O(L) per insert/query, Space: O(N * L * Alphabet)

```

struct Trie {
    vector<vector<int>> nxt;
    vector<int> cnt, end_cnt;
    int alpha;

    Trie(int alpha = 26) : alpha(alpha) {
        add_node();
    }

    int add_node() {
        nxt.emplace_back(alpha, -1);
        cnt.emplace_back(0);
        end_cnt.emplace_back(0);
        return nxt.size() - 1;
    }

    void insert(const string& s, int val = 1) {
        int u = 0;
        cnt[u] += val;
        for (char c : s) {
            int ch = c - 'a';
            if (nxt[u][ch] == -1) nxt[u][ch] = add_node();
            u = nxt[u][ch];
            cnt[u] += val;
        }
        end_cnt[u] += val;
    }
}

```

```

int count_prefix(const string& s) {
    int u = 0;
    for (char c : s) {
        int ch = c - 'a';
        if (nxt[u][ch] == -1) return 0;
        u = nxt[u][ch];
    }
    return cnt[u];
}

```

2.4.2 BinaryTrie

Binary trie over $[0, 2^{(B+1)})$. insert / ERASE / query all O(B).

```

// erase() is what lets you keep the trie equal to the current
    ↪ root->node path during a DFS
// ("min XOR with any ancestor"): add on entry, add(-1) on exit.
struct BTrie {
    static const int B = 30;
    vector<array<int, 2>> ch = {{-1, -1}};
    vector<int> cnt = {0};

    int node() { ch.push_back({-1, -1}); cnt.push_back(0); return
    ↪ ch.size() - 1; }
    void add(ll x, int d = 1) { // d = +1
    ↪ insert, -1 erase
        int u = 0; cnt[0] += d;
        for (int i = B; i >= 0; i--) {
            int b = x >> i & 1;
            if (ch[u][b] < 0) ch[u][b] = node();
            u = ch[u][b], cnt[u] += d;
        }
    }
    int cn(int u, int b) const { return u < 0 || ch[u][b] < 0 ? 0 :
    ↪ cnt[ch[u][b]]; }
    int size() const { return cnt[0]; }

    ll bestXor(ll x, bool mx) const { // mx=1 -> max
    ↪ x^y, mx=0 -> min x^y
        if (!cnt[0]) return -1;
        int u = 0; ll r = 0;
        for (int i = B; i >= 0; i--) {
            int b = (x >> i & 1) ^ mx; // bit we'd
            ↪ like y to have
            if (!cn(u, b)) b ^= 1;
            r |= (1LL) << (x >> i & 1) ^ b;
            u = ch[u][b];
        }
        return r;
    }
    ll maxXor(ll x) const { return bestXor(x, 1); }
    ll minXor(ll x) const { return bestXor(x, 0); }

    int cntLess(ll x, ll k) const { // # {y in trie
    ↪ : (x ^ y) < k}
        int u = 0, r = 0;
        for (int i = B; i >= 0 && u >= 0; i--) {
            int xb = x >> i & 1;
            if (k >> i & 1) r += cn(u, xb), u = ch[u][xb ^ 1]; //
            ↪ xor-bit 0 branch is all < k
            else u = ch[u][xb];
        }
        return r;
    }
}

```

2.4.3 PersistentTrie

Persistent Binary Trie - Space: O(N * BITS), Time: O(BITS) per insert/query

// Supports versioning and range max XOR queries between versions
 ↪ [l_ver, r_ver]

```

struct PersistentTrie {
    struct Node {
        int cnt = 0;
        array<int, 2> nxt = {-1, -1};
    };

    vector<Node> tree;
    vector<int> roots;
    int max_bit;

    PersistentTrie(int max_bit = 30) : max_bit(max_bit) {
        roots.pb(add_node());
    }

    int add_node(int old_node = -1) {
        if (old_node != -1) {
            tree.pb(tree[old_node]);

```

```

    } else {
        tree.pb(Node());
    }
    return tree.size() - 1;
}

// Inserts x starting from prev_root and returns the new version
↪ root
int insert(int prev_root, ll x) {
    int new_root = add_node(prev_root);
    int u = new_root, prev_u = prev_root;
    tree[u].cnt++;

    for (int i = max_bit; i >= 0; i--) {
        int b = (x >> i) & 1;
        int prev_child = (prev_u != -1) ? tree[prev_u].nxt[b] : -1;
        tree[u].nxt[b] = add_node(prev_child);
        u = tree[u].nxt[b];
        if (prev_u != -1) prev_u = tree[prev_u].nxt[b];
        tree[u].cnt++;
    }
    roots.pb(new_root);
    return new_root;
}

// Returns max (x ^ val) over elements inserted in version range
↪ [l_ver, r_ver]
ll max_xor(int l_ver, int r_ver, ll x) const {
    int u_r = roots[r_ver], u_l = (l_ver > 0) ? roots[l_ver - 1] :
↪ -1;
    ll res = 0;

    for (int i = max_bit; i >= 0; i--) {
        int b = (x >> i) & 1;
        int want = b ^ 1;

        int count_r = (u_r != -1 && tree[u_r].nxt[want] != -1) ?
↪ tree[tree[u_r].nxt[want]].cnt : 0;
        int count_l = (u_l != -1 && tree[u_l].nxt[want] != -1) ?
↪ tree[tree[u_l].nxt[want]].cnt : 0;

        if (count_r - count_l > 0) {
            res |= (1LL << i);
            u_r = tree[u_r].nxt[want];
            u_l = (u_l != -1 && tree[u_l].nxt[want] != -1) ?
↪ tree[u_l].nxt[want] : -1;
        } else {
            u_r = (u_r != -1 && tree[u_r].nxt[b] != -1) ?
↪ tree[u_r].nxt[b] : -1;
            u_l = (u_l != -1 && tree[u_l].nxt[b] != -1) ?
↪ tree[u_l].nxt[b] : -1;
        }
    }
    return res;
}
};

```

2.5 Disjoint Set Union (DSU)

2.5.1 DSU

Disjoint Set Union (DSU) - Path Compression & Union by Size

// Time: $O(\alpha(N))$ per operation

```

struct DSU {
    vector<int> par, sz;

    DSU(int n = 0) : par(n), sz(n, 1) {
        iota(all(par), 0);
    }

    int find(int x) {
        return par[x] == x ? x : par[x] = find(par[x]);
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool join(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) return false;
        if (sz[x] < sz[y]) swap(x, y);
        sz[x] += sz[y];
        par[y] = x;
        return true;
    }
}

```

```

int size(int x) {
    return sz[find(x)];
}
};

```

2.5.2 DSURollback

DSU with Rollback - No path compression to allow backtracking

// Time: $O(\log N)$ per operation, $O(1)$ rollback

```

struct DSURollback {
    vector<int> par, siz;
    vector<pair<int, int>> history; // {child_node, old_parent_size}

    DSURollback(int n = 0) : par(n), siz(n, 1) {
        iota(all(par), 0);
    }

    int find(int x) {
        return par[x] == x ? x : find(par[x]);
    }

    bool same(int x, int y) {
        return find(x) == find(y);
    }

    bool join(int x, int y) {
        x = find(x);
        y = find(y);
        if (x == y) {
            history.eb(-1, -1);
            return false;
        }
        if (siz[x] < siz[y]) swap(x, y);
        history.eb(y, siz[x]);
        siz[x] += siz[y];
        par[y] = x;
        return true;
    }

    int size(int x) {
        return siz[find(x)];
    }

    void rollback() {
        if (history.empty()) return;
        auto [y, old_sz_x] = history.back();
        history.pop_back();
        if (y != -1) {
            siz[par[y]] = old_sz_x;
            par[y] = y;
        }
    }

    int get_state() const {
        return history.size();
    }

    void rollback_to(int state) {
        while ((int)history.size() > state) rollback();
    }
};

```

2.5.3 WeightedDSU

DSU with POTENTIALS: maintains $\text{pot}[v] = \text{value}(v) - \text{value}(\text{root})$, so you can answer

// "what is $\text{value}(u) - \text{value}(v)$ " and detect contradictions.

// Group is $(Z, +)$ here; for parity/bipartiteness use XOR: change ALL

↪ FOUR of $\text{pot}[x] += p \rightarrow ^=$,

// $pu - pv == w \rightarrow (pu \wedge pv) == w$, $\text{pot}[rv] = pu - pv - w \rightarrow pu \wedge pv \wedge$

↪ w , and DELETE the $w = -w$ line

// (negation is not the inverse in $\text{GF}(2)$), or Z_k for mod- k relations.

```

struct WDSU {
    vector<int> par;
    vector<ll> pot; // potential relative
    ↪ to the parent
    vector<int> sz;

    WDSU(int n = 0) : par(n), pot(n, 0), sz(n, 1) { iota(all(par), 0); }
    pair<int, ll> find(int x) { // {root, value(x) -
↪ value(root)}
        if (par[x] == x) return {x, 0};
        auto [r, p] = find(par[x]);
        par[x] = r, pot[x] += p; // path compression
↪ accumulates
        return {r, pot[x]};
    }

    // assert value(u) - value(v) == w. false = CONTRADICTION with what

```

↪ we already know.

```
bool join(int u, int v, ll w) {
    auto [ru, pu] = find(u);
    auto [rv, pv] = find(v);
    if (ru == rv) return pu - pv == w;
    if (sz[ru] < sz[rv]) swap(ru, rv), swap(pu, pv), w = -w;
    par[rv] = ru, pot[rv] = pu - pv - w, sz[ru] += sz[rv];
    return true;
}
// value(u) - value(v), or nullopt if they are not related yet
bool diff(int u, int v, ll& out) {
    auto [ru, pu] = find(u);
    auto [rv, pv] = find(v);
    if (ru != rv) return false;
    out = pu - pv;
    return true;
}
};
// BIPARTITENESS under edge additions: use XOR potentials with w = 1
// ↪ per edge; join() returning
// false means an odd cycle appeared. Same structure answers "are u and
// ↪ v the same colour".
```

2.5.4 RangeParallelDSU

RANGE-PARALLEL DSU ("DSU on a sparse table") - merge two whole RANGES elementwise in $O(\log n)$.

```
// The operation is: "a[u..u+len-1] must equal a[v..v+len-1]
// ↪ elementwise", i.e. union u+i with v+i
// for every i < len. Doing that naively is  $O(\text{len})$  per constraint; this
// ↪ does it in  $O(\log n)$ .
// IDEA: keep LOGN separate DSUs. par[k] answers "these two blocks of
// ↪ length  $2^k$  are identical".
// A range constraint of length len is covered by TWO blocks of length
// ↪  $2^k$  ( $k = \text{floor}(\log_2 \text{len})$ ),
// exactly like a sparse-table query - so one union at level k, twice.
// ↪ Afterwards, PUSH DOWN:
// if two blocks of length  $2^k$  are equal then their two halves of
// ↪ length  $2^{k-1}$  are equal too.
// AFTER the push-down, level 0 holds the true elementwise DSU.
// USES: "count the strings/arrays satisfying these equal-substring
// ↪ constraints" (the answer is
// alphabet^(#components at level 0)), grid/string pattern unification,
// ↪ and any problem that says
// "these two intervals must match".
struct RangeDSU {
    int n, LOG;
    vector<vector<int>> par;
    RangeDSU(int n) : n(n), LOG(_lg(max(n, 1)) + 1), par(LOG,
    // ↪ vector<int>(n)) {
        for (int k = 0; k < LOG; k++) iota(all(par[k]), 0);
    }
    int find(int k, int u) { return par[k][u] == u ? u : par[k][u] =
    // ↪ find(k, par[k][u]); }
    void join(int k, int u, int v) {
        u = find(k, u), v = find(k, v);
        if (u != v) par[k][u] = v;
    }
    // a[u .. u+len-1] equals a[v .. v+len-1], elementwise.  $O(\log n)$ 
    // ↪ amortised.
    void equalRanges(int u, int v, int len) {
        if (len <= 0) return;
        int k = _lg(len);
        join(k, u, v);
        join(k, u + len - (1 << k), v + len - (1 << k));
    }
    // Call ONCE after all the constraints. Then find(0, i) is the real
    // ↪ elementwise component.
    void pushDown() {
        for (int k = LOG - 1; k > 0; k--) {
            int half = 1 << (k - 1);
            for (int i = 0; i + (1 << k) <= n; i++) {
                int j = find(k, i);
                if (j == i) continue;
                join(k - 1, i, j);
                // ↪ both halves inherit it
                join(k - 1, i + half, j + half);
            }
        }
    }
    int components() {
        // ↪ at level 0, after pushDown
        int c = 0;
        for (int i = 0; i < n; i++) if (find(0, i) == i) c++;
        return c;
    }
};
// The same "cover a range with two power-of-two blocks, then push
```

↪ down" trick works for any

```
// idempotent relation, not just equality - it is the DSU analogue of a
// ↪ sparse table.
// If instead you need to union a range of CONSECUTIVE elements to each
// ↪ other ("a[l..r] all equal"),
// that is much simpler: a "next unvisited pointer" DSU where find(i)
// ↪ jumps to the next index not
// yet merged, giving  $O(n \alpha)$  total over all range-unions.
```

2.5.5 PartiallyPersistentDSU

PARTIALLY PERSISTENT DSU - a DSU you may QUERY at any past moment, while only the present can be

```
// modified.  $O(\log n)$  per query,  $O(\log n)$  per union, and only  $O(n)$ 
// ↪ memory - far cheaper than the
// fully persistent version in 20 - PersistentAndUndo, which costs
// ↪  $O(\log^2 n)$  and  $O(n \log n)$ .
// WHEN: "what is the earliest time at which u and v are connected",
// ↪ "how big was u's component
// after the first t edges", offline MST / bottleneck-path questions.
// ↪ Partial persistence is enough
// whenever the timeline is a LINE and not a tree of branching versions
// ↪ - which is nearly always.
// If you also want to UNDO, use 02 - DSURollback; if you want to
// ↪ delete arbitrary edges, use
// 14 - OfflineDynamicConnectivity.
// KEY INVARIANT: NO path compression and union by size, so once a
// ↪ vertex becomes a child it keeps
// that parent FOREVER. Every vertex therefore has a monotone timeline:
// ↪ a list of component sizes
// while it was a root, then (at most) one final "I became a child of p
// ↪ at time t" entry. Walking
// up while the stamp is  $\leq t$  reconstructs the forest at time t
// ↪ exactly, and the height stays
//  $O(\log n)$  because of the union by size.
struct PartialDSU {
    vector<vector<pair<int, int>>> par;
    // ↪ {-size or parent, timestamp}
    int T = 0;
    // ↪ number of unions performed
    PartialDSU(int n) : par(n, {{-1, 0}}) {}
    int root(int u, int t) {
        // ↪ representative of u at time t
        auto b = par[u].back();
        return b.fi >= 0 && b.se <= t ? root(b.fi, t) : u;
    }
    bool same(int u, int v, int t) { return root(u, t) == root(v, t); }
    int size(int u, int t) {
        // ↪ |component of u| at time t
        u = root(u, t);
        int lo = 0, hi = par[u].size() - 1, at = 0;
        while (lo <= hi) {
            // ↪ last
            int m = (lo + hi) / 2;
            par[u][m].se <= t ? at = m, lo = m + 1 : hi = m - 1;
        }
        return -par[u][at].fi;
    }
    bool join(int u, int v) {
        // ↪ happens at time ++T
        ++T;
        if ((u = root(u, T)) == (v = root(v, T))) return false;
        if (par[u].back().fi > par[v].back().fi) swap(u, v); // u is
        // ↪ the bigger (more negative)
        par[u].push_back({par[v].back().fi + par[v].back().fi, T});
        par[v].push_back({u, T});
        return true;
    }
    int firstConnected(int u, int v) {
        // ↪ earliest time, or -1 if never
        int lo = 1, hi = T, ans = -1;
        while (lo <= hi) {
            int m = (lo + hi) / 2;
            same(u, v, m) ? ans = m, hi = m - 1 : lo = m + 1;
        }
        return ans;
    }
};
// TIME IS THE UNION COUNTER, not the query index: a query "after the
// ↪ first k edges" must first map
// k to the number of SUCCESSFUL unions among those k edges (keep an
// ↪ array while you insert).
// BOTTLENECK PATH / minimax edge: sort the edges by weight, union them
// ↪ in that order, then
// firstConnected(u, v) gives the index of the edge that answers
// ↪ "minimise the maximum edge on a
// path". That is exactly the Kruskal reconstruction tree (03 -
// ↪ Graph/08 - MST) in 25 lines less
```

```
// code, at  $O(\log^2 n)$  per query instead of  $O(1)$  - use the KRT if you
//   ↳ have many queries.
// OFFLINE ALTERNATIVE: if every query is known up front, PARALLEL
//   ↳ BINARY SEARCH ( $O(1)$  - Misc/17)
// answers all of them with one plain DSU and no persistence at all.
```

2.6 Square Root Decomposition

2.6.1 MoAlgorithm

MO'S ALGORITHM - answers m OFFLINE range queries in $O((n + m) \sqrt{n})$ when you can move an endpoint

```
// by one in  $O(1)$  (add/remove an element and maintain the answer). Sort
//   ↳ queries by block of  $l$ , then by
//   ↳  $r$  (alternating direction per block). Use it for distinct counts,
//   ↳ frequency-of-frequency, XOR pairs.
```

```
class MoArray {
private:
    struct Query {
        int l, r, idx;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd)
                return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    int n;
    int block_size;
    vector<int> a;

    // --- PROBLEM SPECIFIC STATE ---
    long long ans;
    vector<int> freq;

    inline void add(int idx) {
        if (freq[a[idx]] == 0) ans++;
        freq[a[idx]]++;
    }

    inline void remove(int idx) {
        freq[a[idx]]--;
        if (freq[a[idx]] == 0) ans--;
    }

    inline long long get_answer() const {
        return ans;
    }
    // -----

public:
    MoArray(const vector<int>& arr) : n(arr.size()), a(arr) {
        int max_val = 0;
        for (int x : a) {
            max_val = max(max_val, x);
        }
        freq.assign(max_val + 1, 0);
        ans = 0;
    }

    vector<long long> solve(const vector<pair<int, int>>& raw_queries) {
        int q = raw_queries.size();
        if (q == 0) return {};

        block_size = max(1, (int)(n / sqrt(q)));
        vector<Query> queries(q);

        for (int i = 0; i < q; ++i) {
            queries[i].l = raw_queries[i].first;
            queries[i].r = raw_queries[i].second;
            queries[i].idx = i;
            queries[i].bnd = queries[i].l / block_size;
        }

        sort(queries.begin(), queries.end());

        vector<long long> answers(q);
        int cur_l = 0, cur_r = -1;

        for (const Query& qry : queries) {
            while (cur_l > qry.l) add(--cur_l);
            while (cur_r < qry.r) add(++cur_r);
            while (cur_l < qry.l) remove(cur_l++);
            while (cur_r > qry.r) remove(cur_r--);
            answers[qry.idx] = get_answer();
        }
    }
};
```

```
        return answers;
    }
};
```

2.6.2 MoOnSubtrees

MO'S ON SUBTREES - flatten the tree with an Euler tour so every subtree is a contiguous range,

```
// then run plain Mo's on it. Same  $O((n + q) \sqrt{n})$ . Use for "query
//   ↳ over the subtree of  $v$ " when no
// online structure fits (distinct colours in a subtree,  $k$ -th most
//   ↳ frequent, etc.).
```

```
class MoSubtree {
private:
    struct Query {
        int l, r, idx;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd) return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    int n;
    int block_size;
    vector<vector<int>> adj;
    vector<int> in, out, flat, clr;

    // --- PROBLEM SPECIFIC STATE ---
    int mx_freq;
    vector<int> freq;
    vector<long long> sum_freq;

    inline void add(int node_idx) {
        int c = clr[node_idx];
        sum_freq[freq[c]] -= c;
        freq[c]++;
        sum_freq[freq[c]] += c;
        if (freq[c] > mx_freq) mx_freq = freq[c];
    }

    inline void remove(int node_idx) {
        int c = clr[node_idx];
        sum_freq[freq[c]] -= c;
        freq[c]--;
        sum_freq[freq[c]] += c;
        while (mx_freq > 0 && sum_freq[mx_freq] == 0) mx_freq--;
    }

    inline long long get_answer() const {
        return sum_freq[mx_freq];
    }
    // -----

    void dfs(int u, int p) {
        in[u] = flat.size();
        flat.push_back(u);
        for (int v : adj[u]) {
            if (v == p) continue;
            dfs(v, u);
        }
        out[u] = flat.size() - 1;
    }

public:
    MoSubtree(int n, int max_val, const vector<int>& colors)
        : n(n), adj(n + 1), in(n + 1), out(n + 1), clr(colors) {
        mx_freq = 0;
        freq.assign(max_val + 1, 0);
        sum_freq.assign(n + 1, 0);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    vector<long long> solve(int root, const vector<int>& query_nodes) {
        flat.reserve(n);
        dfs(root, 0);

        int q = query_nodes.size();
        if (q == 0) return {};

        block_size = max(1, (int)(n / sqrt(q)));
        vector<Query> queries(q);
```

```

for (int i = 0; i < q; ++i) {
    int u = query_nodes[i];
    queries[i].l = in[u];
    queries[i].r = out[u];
    queries[i].idx = i;
    queries[i].bnd = queries[i].l / block_size;
}

sort(queries.begin(), queries.end());

vector<long long> answers(q);
int cur_l = 0, cur_r = -1;

for (const Query& qry : queries) {
    while (cur_l > qry.l) add(flat[--cur_l]);
    while (cur_r < qry.r) add(flat[++cur_r]);
    while (cur_l < qry.l) remove(flat[cur_l++]);
    while (cur_r > qry.r) remove(flat[cur_r--]);
    answers[qry.idx] = get_answer();
}

return answers;
}
};

```

2.6.3 MoOnPaths

MO'S ON TREE PATHS - the Euler tour trick where a vertex appearing TWICE in the range cancels out,

// plus the LCA added back by hand. Answers path queries offline in $\hookrightarrow O((n + q) \sqrt{n})$ with no HLD.

```

class MoTreePath {
private:
    int n, LOG, block_size;
    vector<vector<int>> adj, anc;
    vector<int> depth, in, out, flat;
    vector<long long> up, a;
    vector<bool> exist;

    // --- PROBLEM SPECIFIC STATE ---
    long long ans;
    vector<deque<int>> dep; // Tracks nodes by depth

    void add(int idx) {
        if (!dep[depth[idx]].empty()) {
            ans += 1ll * a[dep[depth[idx]].back()] * a[idx];
        }
        dep[depth[idx]].push_back(idx);
    }

    void remove(int idx) {
        if (dep[depth[idx]].size() == 2) {
            ans -= a[*dep[depth[idx]].begin()] * a[idx];
        }
        if (dep[depth[idx]].front() == idx) {
            dep[depth[idx]].pop_front();
        } else {
            dep[depth[idx]].pop_back();
        }
    }

    void update(int idx) {
        int node = flat[idx];
        if (exist[node]) {
            remove(node);
        } else {
            add(node);
        }
        exist[node] = !exist[node];
    }

    inline long long get_answer() const {
        return ans;
    }

    // -----

    void dfs(int u, int p = -1) {
        if (p != -1) {
            anc[u][0] = p;
            up[u] = a[u] * a[u] + up[p];
        } else {
            depth[u] = 0;
            up[u] = a[u] * a[u];
        }
        in[u] = flat.size();
        flat.push_back(u);
    }

```

```

for (int v : adj[u]) {
    if (v == p) continue;
    depth[v] = depth[u] + 1;
    dfs(v, u);
}

out[u] = flat.size();
flat.push_back(u); // Push again for Eulerian tour
}

void build_lca() {
    for (int k = 1; k < LOG; k++) {
        for (int i = 1; i <= n; i++) {
            anc[i][k] = anc[anc[i][k - 1]][k - 1];
        }
    }
}

int lift(int x, int k) {
    for (int bit = 0; bit < LOG; bit++) {
        if ((1 << bit) & k) x = anc[x][bit];
    }
    return x;
}

int lca(int u, int v) {
    if (depth[u] < depth[v]) swap(u, v);
    u = lift(u, depth[u] - depth[v]);
    if (u == v) return u;
    for (int bit = LOG - 1; bit >= 0; bit--) {
        if (anc[u][bit] != anc[v][bit]) {
            u = anc[u][bit];
            v = anc[v][bit];
        }
    }
    return anc[u][0];
}

public:
    struct Query {
        int l, r, idx, x = -1;
        int bnd;

        bool operator<(const Query& other) const {
            if (bnd != other.bnd) return bnd < other.bnd;
            return (bnd & 1) ? (r < other.r) : (r > other.r);
        }
    };

    MoTreePath(int n_nodes, const vector<long long>& node_values)
        : n(n_nodes), a(node_values), LOG(log2(n_nodes) + 2) {

        adj.assign(n + 1, vector<int>());
        anc.assign(n + 1, vector<int>(LOG, 0));
        depth.assign(n + 1, 0);
        in.assign(n + 1, 0);
        out.assign(n + 1, 0);
        up.assign(n + 1, 0);
        exist.assign(n + 1, false);
        dep.assign(n + 1, deque<int>());
        ans = 0;
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    // PRIMARY ENTRY POINT: give it the paths as vertex pairs and it
    // builds the queries itself.
    // (in[], out[] and lca() are only valid AFTER the dfs, so you
    // cannot build them yourself.)
    vector<long long> solve(int root, const vector<pair<int, int>>&
    paths) {
        flat.clear();
        dfs(root);
        build_lca();
        vector<Query> queries;
        for (int i = 0; i < (int)paths.size(); i++) {
            int u = paths[i].first, v = paths[i].second;
            if (in[u] > in[v]) swap(u, v);
            int w = lca(u, v);
            // u inside v's subtree -> plain range; otherwise use
            // out[u]..in[v] and add back the LCA
            if (w == u) queries.push_back({in[u], in[v], i, -1, 0});
            else queries.push_back({out[u], in[v], i, w, 0});
        }
    }

```

```

    }
    return solveRanges(queries);
}

vector<long long> solveRanges(vector<Query>& queries) {
    int q = queries.size();
    if (q == 0) return {};

    block_size = max(1, (int)(flat.size() / sqrt(q)));

    for (auto& qry : queries) {
        qry.bnd = qry.l / block_size;
    }

    sort(queries.begin(), queries.end());

    vector<long long> answers(q);
    int cur_l = 0, cur_r = -1;

    for (const Query& qry : queries) {
        while (cur_l > qry.l) update(--cur_l); // GROW first,
        // then shrink - the other
        while (cur_r < qry.r) update(++cur_r); // order can
        // pass through an invalid window
        while (cur_l < qry.l) update(cur_l++);
        while (cur_r > qry.r) update(cur_r--);

        answers[qry.idx] = get_answer() + (qry.x != -1 ? up[qry.x]
        // : 0);
    }
    return answers;
}
};

```

2.6.4 BlockDecomposition

SQRT / BLOCK DECOMPOSITION. Split $[0, n]$ into blocks of $B \sim \sqrt{n}$ and keep one aggregate per

```

// block. Point update  $O(1 \cdot B)$ , range query  $O(\sqrt{n})$ . A BIT/segtree
// beats it for sum, so reach for
// this when the aggregate is NOT a group / has no lazy tag:
// block-sorted arrays, block frequency
// tables, "count values > x in [l, r]", range-assign + weird query, or
// a per-block rebuild that is
// cheap. General shape below with sum; swap 'agg' and the two combine
// lines for anything else.
struct Blocks {
    int n, B;
    vector<ll> a, agg; //
    // agg[b] = aggregate of block b
    Blocks(const vector<ll>& v) : n(v.size()), B(max(1, (int)sqrt(n) +
    // 1)), a(v),
        agg((n + B - 1) / B, 0) {
        for (int i = 0; i < n; i++) agg[i / B] += a[i];
    }
    void set(int i, ll v) { agg[i / B] += v - a[i], a[i] = v; } //
    //  $O(1)$ ; for non-group aggregates
    //
    // rebuild the block instead,  $O(B)$ 
    ll query(int l, int r) { //
    // inclusive,  $O(n/B + B)$ 
        ll s = 0;
        int bl = l / B, br = r / B;
        if (bl == br) { for (int i = l; i <= r; i++) s += a[i]; return
        // s; }
        for (int i = l; i < (bl + 1) * B; i++) s += a[i];
        for (int b = bl + 1; b < br; b++) s += agg[b];
        for (int i = br * B; i <= r; i++) s += a[i];
        return s;
    }
};

// RANGE UPDATE, RANGE QUERY without lazy propagation: keep add[b] per
// block. A partial block is
// updated element-wise and its aggregate rebuilt; a full block only
// bumps add[b]. Query adds
// add[b] * (elements taken from b).
// BLOCK-SORTED ARRAY ("sqrt tree"): store each block also as a sorted
// copy. Then
// count of values  $\leq x$  in  $[l, r]$  = partial blocks scanned + binary
// search inside full blocks,
//  $O(\sqrt{n} \log n)$  per query,  $O(B \log B)$  per point update (re-sort
// one block). This is the
// offline-free alternative to a wavelet tree / merge-sort tree when
// updates are present.
// REBUILD TRICK: when an operation is expensive but rare (e.g.
// inserting into the middle), buffer
// changes and rebuild the whole decomposition every  $\sqrt{q}$  operations
//  $\rightarrow O((n+q) \sqrt{q})$  total.
// CHOOSING B: query cost  $n/B + B \cdot c$  is minimised at  $B = \sqrt{n \cdot c}$  where

```

$\hookrightarrow c$ is the per-element cost;
 // if updates are much more frequent than queries, take bigger blocks,
 $\hookrightarrow$ and vice versa.

2.6.5 MoWithUpdates

MO'S WITH UPDATES (3D Mo). Queries $\{l, r, t\}$ where $t = \# \text{updates applied before this query}$.

// Block size $n^{2/3}$ gives $O(n^{5/3})$ total. Use when the array

$\hookrightarrow$ changes between queries and no

// online structure fits.

```

struct MoUpd {
    struct Q { int l, r, t, idx; };
    struct U { int pos, oldV, newV; };

```

```

    int n, B;

```

```

    vector<int> a, rec;

```

// a = state at

$\hookrightarrow$ time 0; rec = while recording

```

    vector<Q> qs;

```

```

    vector<U> us;

```

// ---- problem state: #distinct in the window ----

```

    vector<int> cnt;

```

```

    int cur = 0;

```

```

    void add(int v) { if (cnt[v]++ == 0) cur++; }

```

```

    void rem(int v) { if (--cnt[v] == 0) cur--; }

```

```

    int answer() const { return cur; }

```

// -----

```

    MoUpd(vector<int> arr, int maxVal) : n(arr.size()), a(arr),

```

$\hookrightarrow$ rec(arr), cnt(maxVal + 1, 0) {

```

        B = max(1, (int)pow((double)max(n, 1), 2.0 / 3));
    }

```

```

    void addQuery(int l, int r) { qs.push_back({l, r, (int)us.size(),

```

$\hookrightarrow$ (int)qs.size()); };

```

    void addUpdate(int pos, int newV) { us.push_back({pos, rec[pos],

```

$\hookrightarrow$ newV}); rec[pos] = newV; }

```

    void applyU(int i, int l, int r) { // step over

```

// step over

$\hookrightarrow$ update i, forwards or backwards

```

        auto& u = us[i];

```

```

        if (l <= u.pos && u.pos <= r) rem(a[u.pos]), add(u.newV);

```

```

        a[u.pos] = u.newV;

```

```

        swap(u.oldV, u.newV);

```

// makes the

$\hookrightarrow$ operation its own inverse

```

    }

```

```

    vector<int> solve() {

```

```

        sort(all(qs), [&](const Q& x, const Q& y) {

```

```

            int bx = x.l / B, by = y.l / B;

```

```

            if (bx != by) return bx < by;

```

```

            int cx = x.r / B, cy = y.r / B;

```

```

            if (cx != cy) return (bx & 1) ? cx > cy : cx < cy;

```

```

            return (cx & 1) ? x.t > y.t : x.t < y.t;

```

```

        });

```

```

        vector<int> res(qs.size());

```

```

        int L = 0, R = -1, T = 0;

```

```

        for (auto& q : qs) {

```

```

            while (T < q.t) applyU(T++, L, R);

```

```

            while (T > q.t) applyU(--T, L, R);

```

```

            while (R < q.r) add(a[+R]);

```

```

            while (L > q.l) add(a[--L]);

```

```

            while (R > q.r) rem(a[R--]);

```

```

            while (L < q.l) rem(a[L++]);

```

```

            res[q.idx] = answer();

```

```

        }

```

```

        return res;
    }

```

```

};

```

// NOTE: 'a' stays at time 0; 'rec' tracks values while you record

$\hookrightarrow$ updates, so applyU() can

// step forwards AND backwards through time using the same swap-based

$\hookrightarrow$ inverse.

2.6.6 MoWithRollback

MO'S WITH ROLLBACK - for structures where ADD is easy but REMOVE is hard or impossible

// (running maximum, DSU, "best pair so far", knapsack-ish state).

// Per block of left endpoints: sort by r, keep a persistent right part

$\hookrightarrow$ that only grows, and

// rebuild the short left part from scratch for every query, rolling

$\hookrightarrow$ back afterwards.

// $O((n + q) \sqrt{n})$ adds, ZERO removes.

```

struct MoRollback {
    struct Q { int l, r, idx; };

```

```

    int n, B;

```

```

    vector<ll> a;

```

```

    vector<Q> qs;

```

```

// ---- problem state: max (value * count) seen; only add() and a
↳ snapshot/rollback ----
vector<int> cnt;
ll best = 0;
vector<pair<int, ll>> hist; // {value index,
↳ previous best}
void add(int v, bool rec) {
    if (rec) hist.push_back({v, best});
    cnt[v]++;
    best = max(best, (ll)cnt[v] * v);
}
void rollback(size_t to) {
    while (hist.size() > to) {
        auto [v, b] = hist.back(); hist.pop_back();
        cnt[v]--; best = b;
    }
}
void clearAll() { fill(all(cnt), 0), best = 0, hist.clear(); }
ll answer() const { return best; }
//
↳ -----
MoRollback(vector<ll> arr, int maxVal) : n(arr.size()), a(arr),
↳ cnt(maxVal + 1, 0) {
    B = max(1, (int)sqrtl((ld)max(1, n)));
}
void addQuery(int l, int r) { qs.push_back({l, r, (int)qs.size()});
↳ }

vector<ll> solve() {
    sort(all(qs), [&](const Q& x, const Q& y) {
        int bx = x.l / B, by = y.l / B;
        return bx != by ? bx < by : x.r < y.r;
    });
    vector<ll> res(qs.size());
    size_t i = 0;
    while (i < qs.size()) {
        int b = qs[i].l / B, rEnd = min(n - 1, (b + 1) * B - 1);
        size_t j = i;
        while (j < qs.size() && qs[j].l / B == b) j++;
        // pass 1: queries contained in this block - answer each
↳ from a CLEAN state
        for (size_t k2 = i; k2 < j; k2++) if (qs[k2].r <= rEnd) {
            clearAll();
            for (int k = qs[k2].l; k <= qs[k2].r; k++) add(a[k],
↳ false);
            res[qs[k2].idx] = answer();
        }
        // pass 2: long queries - right part grows monotonically,
↳ left part is rolled back
        clearAll();
        int R = rEnd;
        for (size_t k2 = i; k2 < j; k2++) if (qs[k2].r > rEnd) {
            while (R < qs[k2].r) add(a[++R], false); //
↳ grow-only, not recorded
            size_t snap = hist.size();
            for (int k = qs[k2].l; k <= rEnd; k++) add(a[k], true);
↳ // short left part
            res[qs[k2].idx] = answer();
            rollback(snap);
        }
        i = j;
    }
    return res;
}
};

```

2.6.7 RangeMode

RANGE MODE - the most frequent value in a range, and its frequency. Both structures below have

```

// O(1) ADD and O(1) DELETE, so dropped into Mo's (01 - MoAlgorithm)
↳ they solve the problem in
// O(n sqrt n) total.
// WHEN: mode / "maximum multiplicity" queries. There is no known
↳ linear-space polylog-per-query
// solution, so sqrt IS the intended answer here, not a fallback. The
↳ mode is NOT decomposable -
// the mode of a union is not a function of the two modes - which is
↳ why a segment tree cannot do
// it and why the counter has to be maintained incrementally.
// KEY INVARIANT (both): adding one element raises some count by
↳ exactly one, so the running
// maximum can only go UP by one; deleting lowers one count by one, so
↳ the maximum can only go DOWN
// by one, and only when the deleted value was the SOLE holder of it. A
↳ count-of-counts array turns

```

// "is it the sole holder" into an O(1) test.

```

// --- max FREQUENCY only. O(1) everything, O(V) memory. Use this
↳ unless you need the value. ---
struct FreqCounter {
    vector<int> cnt, f; // f[c]
    int best = 0;
    FreqCounter(int V, int n) : cnt(V, 0), f(n + 2, 0) {}
    void add(int v) { f[cnt[v]]--, f[++cnt[v]]++, best = max(best,
↳ cnt[v]); }
    void del(int v) {
        f[cnt[v]]--;
        if (cnt[v] == best && !f[cnt[v]]) best--;
        f[--cnt[v]]++;
    }
    int query() { return best; } // the
↳ frequency of the mode
};
// --- the VALUE too: bucket the value axis and keep a max per bucket.
↳ O(sqrt V) query. ---
struct ModeCounter {
    int V, B, W;
    vector<int> cnt, bmx, bc; // bc[b
↳ * W + c] flattened
    ModeCounter(int V, int n) : V(V), B(max(1, (int)sqrt((double)V) +
↳ 1)), W(n + 2),
        cnt(V, 0), bmx(V / B + 2, 0), bc((V / B
↳ + 2) * W, 0) {}
    void add(int v) {
        int b = v / B;
        bc[b * W + cnt[v]]--, bc[b * W + ++cnt[v]]++;
        bmx[b] = max(bmx[b], cnt[v]);
    }
    void del(int v) {
        int b = v / B;
        if (bmx[b] == cnt[v] && bc[b * W + cnt[v]] == 1) bmx[b]--;
        bc[b * W + cnt[v]]--, bc[b * W + --cnt[v]]++;
    }
    pii query() { //
↳ {value, count}, O(sqrt V)
        int best = 0;
        for (int x : bmx) best = max(best, x);
        for (int b = 0; b < (int)bmx.size(); b++)
            if (bmx[b] == best)
                for (int v = b * B; v < min(V, (b + 1) * B); v++)
                    if (cnt[v] == best) return {v, best};
        return {-1, 0};
    }
};
// MEMORY WARNING: bc is O(sqrt(V) * n) ints - about 160 MB at V = n =
↳ 2e5. Shrink it by passing
// the largest range length as n, by widening B (query cost O(V/B + B),
↳ memory O(V/B * n)), or by
// dropping to FreqCounter when the value itself is not needed.
// COORDINATE COMPRESS the values first so V is the number of DISTINCT
↳ values (01 - Misc/05).
// BECAUSE DELETE IS O(1) you do not need Mo's with rollback (06) -
↳ plain Mo's is enough.
// THE ONLINE VERSION, O(sqrt n) per query and no Mo's: cut the array
↳ into sqrt(n) blocks,
// precompute the mode of every block RANGE [i..j] and the prefix count
↳ of every value per block.
// A query starts from the precomputed mode of its maximal inner block
↳ range and folds in the
// O(sqrt n) leftover elements one at a time; each can only raise the
↳ answer, and the count of any
// value in the current range is (prefix count) + (leftovers seen so
↳ far), i.e. O(1). The same
// scaffolding answers any statistic that is monotone under adding one
↳ element.
// MEX WITH UPDATES is the same shape: keep cnt[] plus f[], and scan
↳ f[1], f[2], ... for the first
// zero. O(sqrt n) per query inside Mo's, O(1) amortised if elements
↳ are only ever added.

```

2.6.8 MoOnTwoIntervals

MO'S ON TWO INTERVALS ("4D Mo") - each query is a PAIR of ranges [l1, r1] and [l2, r2] and the

```

// answer depends on the multiset union of both. Four pointers instead
↳ of two.
// COMPLEXITY: with block size B the cost is O(q * B) for the three
↳ bucketed pointers plus
// O(n^4 / B^3) for the monotone one; B ~ n^(3/4) balances them at
↳ O(n^(7/4)) when q ~ n.
// WHEN: "compare two subtrees / two subarrays" - e.g. the most common
↳ value in the union of two

```

```
// subtrees (CF 1767F). Both ranges must be flattened to intervals
// → first: for subtrees use the
// Euler tour (09 - Trees/06), which is what makes a tree query a pair
// → of intervals.
// Do NOT reach for this before checking whether the two ranges can be
// → merged into one, or whether
// the answer is decomposable (then a segment tree merge, 16 -
// → SegTreeMerging, is far faster).
// KEY INVARIANT: the same amortisation as ordinary Mo's, one level
// → deeper. Sort by
// (l1/B, r1/B, l2/B, r2): the first three keys change O(B) per query
// → each because they are
// bucketed, and the last one is monotone inside every fixed triple of
// → buckets, so it sweeps the
// array once per triple. The state must support ADD and DELETE in O(1)
// → - it is shared by both
// intervals, so an element can be inserted twice and that must be
// → meaningful.
struct Mo2 {
    struct Q { int l1, r1, l2, r2, id; };
    int n, B;
    vector<Q> qs;
    Mo2(int n, int q) : n(n), B(max(1, (int)pow((double)n, 0.75))) {
        qs.reserve(q);
    }
    void addQuery(int l1, int r1, int l2, int r2, int id) {
        qs.push_back({l1, r1, l2, r2, id});
    }
    // add(i) / del(i) update the shared state with position i; get()
    // → reads the answer.
    template <class A, class D, class G>
    void run(A add, D del, G get) {
        int Bl = B;
        sort(all(qs), [&](const Q& a, const Q& b) {
            array<int, 7> x{a.l1 / Bl, a.r1 / Bl, a.l2 / Bl, a.r2,
        }
        // [a,
        // → b] and [c, d], both empty
        for (auto& q : qs) {
            while (b < q.r1) add(++b);
            while (a > q.l1) add(--a);
            while (b > q.r1) del(b--);
            while (a < q.l1) del(a--);
            while (d < q.r2) add(++d);
            while (c > q.l2) add(--c);
            while (d > q.r2) del(d--);
            while (c < q.l2) del(c--);
            get(q.id);
        }
    }
};
// THE ORDER OF THE FOUR LOOPS MATTERS: always GROW before you SHRINK,
// → exactly as in 1D Mo's,
// or the two ranges can transiently cross into an invalid state
// → (negative counts).
// OVERLAPPING RANGES: if the two intervals may intersect, an element
// → gets added twice. Decide up
// front whether the state should count it twice (multiset union) or
// → once (set union) - for set
// union keep a per-position "times present" counter and only touch the
// → state on 0 <=> 1.
// TREE PAIRS: l1 = tin[u], r1 = tout[u], l2 = tin[v], r2 = tout[v]. If
// → u is an ancestor of v the
// intervals nest, so the "counted twice" question above is not
// → hypothetical.
// 3+ INTERVALS: the same sort with more keys works but the exponent
// → gets ugly fast - at that point
// look for a completely different decomposition.
```

2.6.9 BlockList

BLOCK LIST (chunked / unrolled list, a "sqrt rope") - an array that supports insert and erase in

```
// the middle, RANGE ASSIGN, RANGE REVERSE, and range statistics, all
// → in O(sqrt n).
// WHEN: an implicit treap (07 - BBST/02) does the same in O(log n) and
// → is usually the right answer.
// Reach for this instead when the per-block statistic is something a
// → treap node cannot merge in
// O(1) - "how many elements >= x in this range" (each block keeps a
// → SORTED copy), "the k-th
// smallest", "count of distinct" - or when you want a structure you
// → can debug by printing.
// COMPLEXITY: O(sqrt n) per operation amortised, with a full rebuild
// → every ~sqrt n operations to
// stop the blocks from fragmenting. Rebuild is O(n).
```

```
// KEY INVARIANT: the array is a LIST OF BLOCKS, each holding at most
// → ~2B elements plus two lazy
// tags (assigned-value, reversed). Every range operation starts by
// → SPLITTING at both endpoints, so
// afterwards the range is an exact run of whole blocks and each one
// → can be tagged in O(1). A
// reverse then only has to flip each block's tag and reverse the ORDER
// → OF THE BLOCK LIST itself.
struct BlockList {
    struct Block {
        vector<int> v, s; // s =
        // sorted copy of v
        int len = 0, asg = -1; // asg
        // >= 0: whole block is asg
        bool rev = false;
        void resort() { s = v, sort(all(s)); }
        void solid() { // drop
        // both tags
        if (asg >= 0) v.assign(len, asg), asg = -1, resort();
        if (rev) std::reverse(all(v)), rev = false;
        }
        int atleast(int x) { // how
        // many elements >= x
        if (asg >= 0) return asg >= x ? len : 0;
        return s.end() - lower_bound(all(s), x);
        }
    };
    vector<Block> b;
    int n, B, ops = 0;
    BlockList(const vector<int>& a) : n(a.size()), B(max(1,
    // → (int)sqrt((double)n + 1))) { chunk(a); }
    void chunk(const vector<int>& a) {
        b.clear();
        for (int i = 0; i < (int)a.size(); i += B) {
            Block x;
            x.v.assign(a.begin() + i, a.begin() + min((int)a.size(), i
        // → + B));
            x.len = x.v.size(), x.resort();
            b.push_back(x);
        }
    }
    void rebuild() { // O(n),
    // → every ~B operations
    vector<int> a;
    a.reserve(n);
    for (auto& x : b) {
        x.solid();
        for (int y : x.v) a.push_back(y);
    }
    chunk(a);
    }
    int split(int i) { // index
    // → of the block starting at i
    if (i >= n) return b.size();
    int cur = 0;
    for (int k = 0; k < (int)b.size(); k++) {
        if (i < cur + b[k].len) {
            if (i == cur) return k;
            int off = i - cur;
            b[k].solid();
            Block nb;
            nb.v.assign(b[k].v.begin() + off, b[k].v.end());
            nb.len = nb.v.size(), nb.resort();
            b[k].v.resize(off), b[k].len = off, b[k].resort();
            b.insert(b.begin() + k + 1, nb);
            return k + 1;
        }
        cur += b[k].len;
    }
    return b.size();
    }
    void tick() { if (++ops % B == 0) rebuild(); } // call
    // → after every modification
    void assign(int l, int r, int x) {
        int p = split(l), q = split(r + 1);
        for (int k = p; k < q; k++) b[k].asg = x, b[k].rev = false;
        tick();
    }
    void reverse(int l, int r) {
        int p = split(l), q = split(r + 1);
        for (int k = p; k < q; k++) b[k].rev ^= 1;
        std::reverse(b.begin() + p, b.begin() + q);
        tick();
    }
    void insert(int i, int x) {
        int p = split(i);
        Block nb;
        nb.v.assign(1, x);
        nb.len = 1, nb.resort();
        b.insert(b.begin() + p, nb);
    }
    void erase(int i) {
        int p = split(i), q = split(i + 1);
        b.erase(b.begin() + p, b.begin() + q);
    }
    int kth(int k) {
        int cur = 0;
        for (int i = 0; i < (int)b.size(); i++) {
            cur += b[i].len;
            if (k < cur) return b[i].v[k - cur];
        }
        return b.back().v.back();
    }
    int range(int l, int r) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k];
        }
        return ans;
    }
    int range(int l, int r, int x) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x;
        }
        return ans;
    }
    int range(int l, int r, int x, int y) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h, int g) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h & b[k].v[k - b[k].len, k] < g;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h, int g, int f) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h & b[k].v[k - b[k].len, k] < g & b[k].v[k - b[k].len, k] < f;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h, int g, int f, int e) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h & b[k].v[k - b[k].len, k] < g & b[k].v[k - b[k].len, k] < f & b[k].v[k - b[k].len, k] < e;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h, int g, int f, int e, int d) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h & b[k].v[k - b[k].len, k] < g & b[k].v[k - b[k].len, k] < f & b[k].v[k - b[k].len, k] < e & b[k].v[k - b[k].len, k] < d;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h, int g, int f, int e, int d, int c) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h & b[k].v[k - b[k].len, k] < g & b[k].v[k - b[k].len, k] < f & b[k].v[k - b[k].len, k] < e & b[k].v[k - b[k].len, k] < d & b[k].v[k - b[k].len, k] < c;
        }
        return ans;
    }
    int range(int l, int r, int x, int y, int z, int w, int v, int u, int t, int s, int r, int q, int p, int o, int n, int m, int l, int k, int j, int i, int h, int g, int f, int e, int d, int c, int b) {
        int p = split(l), q = split(r + 1);
        int ans = 0;
        for (int k = p; k < q; k++) {
            if (b[k].rev) std::reverse(b[k].v.begin(), b[k].v.end());
            ans += b[k].v[k - b[k].len, k] >= x & b[k].v[k - b[k].len, k] < y & b[k].v[k - b[k].len, k] < z & b[k].v[k - b[k].len, k] < w & b[k].v[k - b[k].len, k] < v & b[k].v[k - b[k].len, k] < u & b[k].v[k - b[k].len, k] < t & b[k].v[k - b[k].len, k] < s & b[k].v[k - b[k].len, k] < r & b[k].v[k - b[k].len, k] < q & b[k].v[k - b[k].len, k] < p & b[k].v[k - b[k].len, k] < o & b[k].v[k - b[k].len, k] < n & b[k].v[k - b[k].len, k] < m & b[k].v[k - b[k].len, k] < l & b[k].v[k - b[k].len, k] < k & b[k].v[k - b[k].len, k] < j & b[k].v[k - b[k].len, k] < i & b[k].v[k - b[k].len, k] < h & b[k].v[k - b[k].len, k] < g & b[k].v[k - b[k].len, k] < f & b[k].v[k - b[k].len, k] < e & b[k].v[k - b[k].len, k] < d & b[k].v[k - b[k].len, k] < c & b[k].v[k - b[k].len, k] < b;
        }
        return ans;
    }
    int range(int l, int r, int x, int y,
```

```

    nb.v = {x}, nb.len = 1, nb.resort();
    b.insert(b.begin() + p, nb), n++;
    tick();
}
void erase(int l, int r) {
    int p = split(l), q = split(r + 1);
    b.erase(b.begin() + p, b.begin() + q), n -= r - l + 1;
    tick();
}
int count(int l, int r, int x) { // how
    ↪ many a[i] >= x in [l, r]
    int p = split(l), q = split(r + 1), res = 0;
    for (int k = p; k < q; k++) res += b[k].atLeast(x);
    return res;
}
};
// THE TWO TAGS INTERACT: assigning kills a pending reverse (all values
    ↪ are equal), reversing an
// assigned block is a no-op. solid() is the only place that has to get
    ↪ the order right.
// OTHER PER-BLOCK STATISTICS drop straight in: a frequency map (count
    ↪ of a value), a prefix sum
// (range sum with range assign), a max, a gcd. Anything you can
    ↪ rebuild in O(B) works.
// WITHOUT insert / erase / reverse, do not write this: 04 -
    ↪ BlockDecomposition is half the code.
// WITHOUT the sorted-copy statistic, an implicit treap is faster and
    ↪ shorter.

```

2.7 Balanced Binary Search Tree (BBST)

2.7.1 Treap

TREAP (keyed) - a BST by key and a heap by random priority, so it is balanced in expectation.

```

// NOTE: 01 - Miscellaneous/04 - random.cpp declares the same 'rng'.
    ↪ Keep one copy.
// O(log n) insert/erase/find, plus split and merge, which is what
    ↪ makes it better than std::set:
// you can cut the tree at a key, operate on a whole range, and glue it
    ↪ back.
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

```

```

template <typename T>
struct Treap {
    struct Node {
        T key, val, sum;
        T lazy = 0; // 1. Added lazy tag
        int sz = 1;
        uint64_t p = rng();
        Node *l = nullptr, *r = nullptr;

        Node(T k, T v = T()) : key(k), val(v), sum(v) {}
    } * root = nullptr;

    ~Treap() { destroy(root); }

    void destroy(Node* t) {
        if (!t) return;
        destroy(t->l);
        destroy(t->r);
        delete t;
    }

    int sz(Node* t) { return t ? t->sz : 0; }
    T sub_val(Node* t) { return t ? t->sum : 0; }

    // 2. The Push Function: Propagates pending updates to children
    void push(Node* t) {
        if (!t || t->lazy == 0) return;

        if (t->l) {
            t->l->key += t->lazy; // Shift the key (critical for the DP
            ↪ solution)
            t->l->val += t->lazy; // Shift the value
            t->l->sum += t->lazy * sz(t->l); // Update subtree sum
            t->l->lazy += t->lazy; // Pass the tag down
        }
        if (t->r) {
            t->r->key += t->lazy;
            t->r->val += t->lazy;
            t->r->sum += t->lazy * sz(t->r);
            t->r->lazy += t->lazy;
        }
        t->lazy = 0; // Clear the tag on the current node
    }

    Node* update(Node* t) {

```

```

        if (t) {
            t->sz = 1 + sz(t->l) + sz(t->r);
            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
        }
        return t;
    }

    Node* merge(Node* l, Node* r) {
        push(l); // 3. Push before making routing decisions
        push(r);
        if (!l || !r) return l ? l : r;
        if (l->p > r->p) {
            l->r = merge(l->r, r);
            return update(l);
        }
        r->l = merge(l, r->l);
        return update(r);
    }
}

```

```

pair<Node*, Node*> split(Node* t, T k) {
    if (!t) return {nullptr, nullptr};
    push(t); // 3. Push before evaluating keys for splits
    if (t->key <= k) {
        auto [l, r] = split(t->r, k);
        t->r = l;
        return {update(t), r};
    }
    auto [l, r] = split(t->l, k);
    t->l = r;
    return {l, update(t)};
}

```

// --- Core Operations ---

// Range Update Method

```

void add_range(T l_val, T r_val, T add_val) {
    if (l_val > r_val) return;

```

```

    auto [left_mid, right] = split(root, r_val);
    auto [left, mid] = split(left_mid, l_val - 1);

```

```

    // Apply the lazy tag to the entire isolated segment
    if (mid) {
        mid->key += add_val;
        mid->val += add_val;
        mid->sum += add_val * sz(mid);
        mid->lazy += add_val;
    }

```

```

    root = merge(merge(left, mid), right);
}

```

```

bool search(T x) {
    Node* curr = root;
    while (curr) {
        push(curr); // Push during traversal
        if (curr->key == x) return true;
        curr = (x < curr->key) ? curr->l : curr->r;
    }
    return false;
}

```

```

void insert(T x, T val = T()) {
    auto [l, r] = split(root, x);
    auto [l2, r2] = split(l, x - 1);
    if (!r2) {
        r2 = new Node(x, val);
    } else {
        push(r2); // Push before modifying
        r2->val += val;
        update(r2);
    }
    root = merge(merge(l2, r2), r);
}

```

```

void erase(T x) {
    auto [l, r] = split(root, x);
    auto [l2, r2] = split(l, x - 1);
    destroy(r2);
    root = merge(l2, r);
}

```

// --- Utility Operations ---

```

Node* find_by_order(Node* t, int k) {
    if (!t) return nullptr;
    push(t); // Push before accessing children sizes

```

```

    int left_sz = sz(t->l);
    if (k < left_sz) return find_by_order(t->l, k);
    if (k == left_sz) return t;
    return find_by_order(t->r, k - left_sz - 1);
}
Node* find_by_order(int k) { return find_by_order(root, k); }

int order_of_key(Node* t, T k) {
    if (!t) return 0;
    push(t); // Push before comparing keys
    if (t->key >= k) return order_of_key(t->l, k);
    return sz(t->l) + 1 + order_of_key(t->r, k);
}
int order_of_key(T k) { return order_of_key(root, k); }

void print(Node* t) {
    if (!t) return;
    push(t); // Push before printing to get true values
    print(t->l);
    cout << t->key << " ";
    print(t->r);
}
void print() {
    print(root);
    cout << '\n';
}
};

```

2.7.2 ImplicitTrep

IMPLICIT TREAP - a treap keyed by POSITION (subtree size) instead of value, i.e. an array that

// NOTE: 01 - Miscellaneous/04 - random.cpp declares the same 'rng'.
 ↳ Keep one copy.
 // supports insert/erase in the middle, reverse a range, and move a
 ↳ block, all in $O(\log n)$.
 // This is the structure for "cut [l,r] out and paste it elsewhere"
 ↳ problems.
 mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

```

template <typename T>
struct ImplicitTrep {
    struct Node {
        T val, sum;
        T lazy = 0; // Lazy tag for range additions
        bool rev = false; // Lazy tag for range reversals
        int sz = 1;
        uint64_t p = rng();
        Node *l = nullptr, *r = nullptr;

        Node(T v = T()) : val(v), sum(v) {}
    } *root = nullptr;

    // Default Constructor
    ImplicitTrep() = default;

    // Vector Constructor - Builds treap in  $O(N)$  time
    ImplicitTrep(const vector<T>& a) {
        build(a);
    }

    ~ImplicitTrep() { destroy(root); }

    void destroy(Node* t) {
        if (!t) return;
        destroy(t->l);
        destroy(t->r);
        delete t;
    }

    int sz(Node* t) { return t ? t->sz : 0; }
    T sub_val(Node* t) { return t ? t->sum : 0; }

    // Propagates pending lazy updates to children
    void push(Node* t) {
        if (!t) return;

        // 1. Process pending reversal
        if (t->rev) {
            swap(t->l, t->r);
            if (t->l) t->l->rev ^= true;
            if (t->r) t->r->rev ^= true;
            t->rev = false;
        }

        // 2. Process pending value addition
        if (t->lazy != 0) {
            if (t->l) {

```

```

                t->l->val += t->lazy;
                t->l->sum += t->lazy * sz(t->l);
                t->l->lazy += t->lazy;
            }
            if (t->r) {
                t->r->val += t->lazy;
                t->r->sum += t->lazy * sz(t->r);
                t->r->lazy += t->lazy;
            }
            t->lazy = 0; // Clear the tag
        }
    }

    Node* update(Node* t) {
        if (t) {
            t->sz = 1 + sz(t->l) + sz(t->r);
            t->sum = t->val + sub_val(t->l) + sub_val(t->r);
        }
        return t;
    }

    // ---  $O(N)$  Linear Time Build ---
    void build(const vector<T>& a) {
        destroy(root);
        root = nullptr;
        if (a.empty()) return;

        vector<Node*> st;
        for (const T& val : a) {
            Node* curr = new Node(val);
            Node* last = nullptr;

            // Maintain max-heap property on random priority 'p'
            while (!st.empty() && st.back()->p < curr->p) {
                last = st.back();
                st.pop_back();
            }

            curr->l = last;
            if (!st.empty()) {
                st.back()->r = curr;
            }
            st.push_back(curr);
        }

        // Post-order pass to compute subtree sizes and sums in  $O(N)$ 
        auto recompute = [&](auto& self, Node* t) -> void {
            if (!t) return;
            self(self, t->l);
            self(self, t->r);
            update(t);
        };

        root = st.front(); // Bottom of stack is the root
        recompute(recompute, root);
    }

    // --- Dynamic Array Operations ---

    Node* merge(Node* l, Node* r) {
        push(l);
        push(r);
        if (!l || !r) return l ? l : r;
        if (l->p > r->p) {
            l->r = merge(l->r, r);
            return update(l);
        }
        r->l = merge(l, r->l);
        return update(r);
    }

    pair<Node*, Node*> split(Node* t, int k) {
        if (!t) return {nullptr, nullptr};
        push(t);

        int left_sz = sz(t->l);
        if (left_sz < k) {
            auto [l, r] = split(t->r, k - left_sz - 1);
            t->r = l;
            return {update(t), r};
        } else {
            auto [l, r] = split(t->l, k);
            t->l = r;
            return {l, update(t)};
        }
    }
}

```

```

int size() { return sz(root); }

void insert(int idx, T val) {
    auto [l, r] = split(root, idx);
    Node* node = new Node(val);
    root = merge(merge(l, node), r);
}

void erase(int idx) {
    if (idx < 0 || idx >= sz(root)) return;
    auto [left_mid, right] = split(root, idx + 1);
    auto [left, mid] = split(left_mid, idx);
    destroy(mid);
    root = merge(left, right);
}

void reverse_range(int l, int r) {
    if (l >= r || l < 0 || r >= sz(root)) return;

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    if (mid) mid->rev ^= true;

    root = merge(merge(left, mid), right);
}

void add_range(int l, int r, T add_val) {
    if (l > r || l < 0 || r >= sz(root)) return;

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    if (mid) {
        mid->val += add_val;
        mid->sum += add_val * sz(mid);
        mid->lazy += add_val;
    }

    root = merge(merge(left, mid), right);
}

T query_range(int l, int r) {
    if (l > r || l < 0 || r >= sz(root)) return T();

    auto [left_mid, right] = split(root, r + 1);
    auto [left, mid] = split(left_mid, l);

    T ans = sub_val(mid);

    root = merge(merge(left, mid), right);
    return ans;
}

T get(int idx) {
    return query_range(idx, idx);
}

void print(Node* t) {
    if (!t) return;
    push(t);
    print(t->l);
    cout << t->val << " ";
    print(t->r);
}

void print() {
    print(root);
    cout << '\n';
}
};

```

2.7.3 Persistent Treap

PERSISTENT IMPLICIT TREAP - a persistent ARRAY that supports insert / erase / concatenate / cut

// a block in the middle, with every version staying queryable. $O(\log n)$ time and $O(\log n)$ new nodes per operation.
 // WHEN: "version v of the text, then insert this string at position $\rightarrow k$ ", copy-paste editors, and
 // the killer application: COPY A RANGE OF AN OLD VERSION AND PASTE IT $\rightarrow$ ANYWHERE - two splits of an
 // old root share all their nodes with it, so the copy is free. A
 $\rightarrow$ persistent segment tree
 // (01 - Segment Tree/03) cannot do this because its index set is
 $\rightarrow$ fixed; 02 - ImplicitTreap can,
 // but destroys the old version.

```

// KEY INVARIANT: COPY ON WRITE. split and merge only ever touch one
 $\rightarrow$  root-to-leaf path, so cloning
// each node before modifying it costs  $O(\log n)$  extra nodes and leaves
 $\rightarrow$  every earlier root pointing
// at a consistent tree. Priorities are copied too, so the shape (and
 $\rightarrow$  hence the depth bound) is
// unchanged by cloning.
struct PTreap {
    struct Node { int l = 0, r = 0, sz = 0; unsigned pr = 0; ll val =
 $\rightarrow$  0, sum = 0; };
    vector<Node> t;
    PTreap() : t(1) {} // node
 $\rightarrow$  0 = the empty tree
    unsigned rnd() {
        static unsigned s = 88172645u;
        return s ^= s << 13, s ^= s >> 17, s ^= s << 5;
    }
    int cp(int x) { t.push_back(t[x]); return t.size() - 1; }
    void pull(int x) {
        t[x].sz = t[t[x].l].sz + 1 + t[t[x].r].sz;
        t[x].sum = t[t[x].l].sum + t[x].val + t[t[x].r].sum;
    }
    pair<int, int> split(int x, int k) { // first
 $\rightarrow$  k elements go left
        if (!x) return {0, 0};
        int c = cp(x);
        if (t[t[c].l].sz < k) {
            auto [a, b] = split(t[c].r, k - t[t[c].l].sz - 1);
            t[c].r = a, pull(c);
            return {c, b};
        }
        auto [a, b] = split(t[c].l, k);
        t[c].l = b, pull(c);
        return {a, c};
    }
    int merge(int a, int b) {
        if (!a || !b) return a | b;
        if (t[a].pr > t[b].pr) {
            int c = cp(a), r = merge(t[c].r, b);
            t[c].r = r, pull(c);
            return c;
        }
        int c = cp(b), l = merge(a, t[c].l);
        t[c].l = l, pull(c);
        return c;
    }
    int mk(ll v) { t.push_back({0, 0, 1, rnd(), v, v}); return t.size()
 $\rightarrow$  - 1; }
    int build(const vector<ll>& a) { //  $O(n \log n)$ , fine for a one-off
        int r = 0;
        for (ll v : a) r = merge(r, mk(v));
        return r;
    }
    int insert(int root, int pos, ll v) { //
 $\rightarrow$  returns the new root
        auto [a, b] = split(root, pos);
        return merge(merge(a, mk(v)), b);
    }
    int erase(int root, int l, int r) { // drop
 $\rightarrow$  positions [l, r]
        auto [a, b] = split(root, l);
        auto [m, c] = split(b, r - l + 1);
        return merge(a, c);
    }
    int cut(int root, int l, int r) { // the
 $\rightarrow$  block [l, r] as its own tree
        auto [a, b] = split(root, l);
        return split(b, r - l + 1).first;
    }
    int paste(int root, int pos, int piece) { //
 $\rightarrow$  splice another tree in
        auto [a, b] = split(root, pos);
        return merge(merge(a, piece), b);
    }
    ll sum(int x, int l, int r) { //
 $\rightarrow$  read-only: allocates nothing
        if (!x || l > r || r < 0 || l >= t[x].sz) return 0;
        if (l <= 0 && r >= t[x].sz - 1) return t[x].sum;
        int L = t[t[x].l].sz;
        return sum(t[t[x].l].l, l, min(r, L - 1)) + (l <= L && L <= r ?
 $\rightarrow$  t[x].val : 0) +
            sum(t[t[x].r, max(0, l - L - 1), r - L - 1);
    }
    ll get(int root, int i) { return sum(root, i, i); }
};
// LAZY TAGS (range add, range reverse) DO work, but push() must CLONE

```

```

    ↪ both children before
// pushing into them - otherwise you mutate a node an old version still
    ↪ owns. That is one extra
// line in push and doubles the node count; write it only if the
    ↪ problem needs it.
// MEMORY is the binding constraint:  $O(q \log n)$  nodes at ~32 bytes.
    ↪ reserve() and, if a version is
// provably dead, do not bother freeing - contests end first.
// IF YOU ONLY NEED THE LATEST VERSION use 02 - ImplicitTreap (no
    ↪ allocation per split).
// IF THE OPERATIONS ARE POSITION-STABLE (no insert / erase, only
    ↪ updates) a persistent segment
// tree is 3x faster and half the memory.

```

2.8 Policy Based Data Structures

2.8.1 OrderedSet

Policy-Based Data Structure: ordered_set

```

// Time:  $O(\log N)$  per operation
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

using namespace __gnu_pbds;
template<typename T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag,
    ↪ tree_order_statistics_node_update>;

// Usage:
// st.find_by_order(k): Iterator to k-th element (0-based)
// st.order_of_key(x): Number of elements strictly smaller than x

```

2.9 Monotonic Data Structures

2.9.1 MonotonicStack

Monotonic Stack - Time: $O(N)$, Space: $O(N)$

```

// Computes Next/Previous Smaller and Greater element indices for all
    ↪ positions

```

```
template<typename T = ll>
```

```
struct MonotonicStack {
```

```

    int n;
    vector<T> a;
    vector<int> prev_less, next_less;
    vector<int> prev_greater, next_greater;

```

```

MonotonicStack(const vector<T>& arr) : n(arr.size()), a(arr),
    prev_less(n, -1), next_less(n, n),
    prev_greater(n, -1), next_greater(n, n) {

```

```

    vector<int> st;
    // Next Less & Previous Less
    for (int i = 0; i < n; i++) {
        while (!st.empty() && a[st.back()] >= a[i]) {
            next_less[st.back()] = i;
            st.pop_back();
        }
        if (!st.empty()) prev_less[i] = st.back();
        st.pb(i);
    }

```

```

    st.clear();
    // Next Greater & Previous Greater
    for (int i = 0; i < n; i++) {
        while (!st.empty() && a[st.back()] <= a[i]) {
            next_greater[st.back()] = i;
            st.pop_back();
        }
        if (!st.empty()) prev_greater[i] = st.back();
        st.pb(i);
    }
}

```

```
};
```

2.9.2 TwoStackQueue

Monotonic Queue via Two Stacks (Sliding Window Min/Max)

// Space: $O(N)$, Time: $O(1)$ amortized per push/pop/get

```

struct TwoStackQ {
    struct Node {
        ll mx = -llinf, mn = llinf, val;

        Node() : val(0) {}
        Node(ll x) : mx(x), mn(x), val(x) {}
    };

    stack<Node> a, b;

```

```

    int size() {
        return a.size() + b.size();
    }

```

```

    void mrg(Node &a, Node &b) {
        a.mn = min(a.mn, b.mn);
        a.mx = max(a.mx, b.mx);
    }

```

// Pushes value into the queue

```

    void push(ll val) {
        auto nd = Node(val);
        if (!a.empty()) mrg(nd, a.top());
        a.push(nd);
    }

```

// Transfers elements from stack a to stack b when b is empty

```

    void move() {
        while (!a.empty()) {
            auto nd = Node(a.top().val);
            if (!b.empty()) mrg(nd, b.top());
            b.push(nd);
            a.pop();
        }
    }

```

// Returns min/max of all current elements in the queue in $O(1)$

```

    Node get() {
        Node res;
        if (!b.empty()) mrg(res, b.top());
        if (!a.empty()) mrg(res, a.top());
        return res;
    }

```

// Removes the oldest element from the queue

```

    void pop() {
        if (b.empty()) move();
        if (!b.empty()) b.pop();
    }

```

```
};
```

2.9.3 MonotonicQueue2D

2D Monotonic Queue (Sliding Window 2D Min/Max) - Time: $O(N * M)$, Space: $O(N * M)$

// Computes min in every $K \times L$ subgrid of an $N \times M$ matrix

```
template<typename T = ll>
```

```
struct MonotonicQueue2D {
```

```

    int n, m;

```

```

    // Returns a matrix res where res[i][j] is the min in subgrid
    ↪ [i..i+K-1][j..j+L-1]
    vector<vector<T>> sliding_window_2d(const vector<vector<T>>& grid,
    ↪ int K, int L) {
        n = grid.size();
        m = grid[0].size();

        // 1. Sliding window min horizontally along each row
        vector<vector<T>> row_min(n, vector<T>(m - L + 1));
        for (int i = 0; i < n; i++) {
            deque<int> dq;
            for (int j = 0; j < m; j++) {
                while (!dq.empty() && grid[i][dq.back()] >= grid[i][j])
                    ↪ dq.pop_back();
                dq.pb(j);
                if (dq.front() <= j - L) dq.pop_front();
                if (j >= L - 1) row_min[i][j - L + 1] =
                    ↪ grid[i][dq.front()];
            }
        }

```

```

        // 2. Sliding window min vertically down each column of row_min
        vector<vector<T>> res(n - K + 1, vector<T>(m - L + 1));
        for (int j = 0; j < m - L + 1; j++) {
            deque<int> dq;
            for (int i = 0; i < n; i++) {
                while (!dq.empty() && row_min[dq.back()][j] >=
                    ↪ row_min[i][j]) dq.pop_back();
                dq.pb(i);
                if (dq.front() <= i - K) dq.pop_front();
                if (i >= K - 1) res[i - K + 1][j] =
                    ↪ row_min[dq.front()][j];
            }
        }

```

```

        return res;
    }

```

```
};
```

2.9.4 LazyDeleteAndTopK

TWO SMALL STRUCTURES THAT KEEP SHOWING UP, and are always retyped badly under pressure.

```
// --- LAZY-DELETION HEAP: a priority_queue that supports erase(x),
//   ↳ O(log n) amortised. ---
// Keep a second heap of "things already deleted"; before reading the
//   ↳ top, pop matching pairs.
// This is why you almost never need decrease-key: push the new value
//   ↳ and lazily drop the old one.
template <class T, class Cmp = less<T>>
struct LazyHeap {
    priority_queue<T, vector<T>, Cmp> in, del;
    void push(T x) { in.push(x); }
    void erase(T x) { del.push(x); } // x
    ↳ must actually be present
    void clean() { while (!del.empty() && in.top() == del.top())
    ↳ in.pop(), del.pop(); }
    bool empty() { clean(); return in.empty(); }
    int size() { return (int)in.size() - (int)del.size(); }
    T top() { clean(); return in.top(); }
    void pop() { clean(); in.pop(); }
};

// A multiset does the same thing with erase(find(x)) and is shorter -
//   ↳ reach for THIS when you
// need heap performance (no red-black overhead) or when the values are
//   ↳ not comparable-unique.
// TRAP: erase(x) must be matched by an earlier push(x), or 'size()'
//   ↳ and 'clean()' drift.

// --- TOP-K SUM: maintain the sum of the k largest elements under
//   ↳ insert and erase. ---
// Two multisets with a fixed frontier: 'hi' holds the k largest (and
//   ↳ its sum), 'lo' the rest.
// After every operation, move elements across the frontier until |hi|
//   ↳ == min(k, total).
// USES: "sum of the k best so far" while sweeping, sliding-window k-th
//   ↳ order statistics, and
// scheduling/greedy problems where you keep the k cheapest and pay for
//   ↳ the rest.
struct TopK {
    int k;
    ll sum = 0; // sum
    ↳ of the k largest
    multiset<ll> hi, lo; // hi =
    ↳ the chosen k, lo = the rest
    TopK(int k) : k(k) {}
    void fix() {
        while ((int)hi.size() < k && !lo.empty()) {
            auto it = prev(lo.end());
            sum += *it, hi.insert(*it), lo.erase(it);
        }
        while ((int)hi.size() > k) {
            auto it = hi.begin();
            sum -= *it, lo.insert(*it), hi.erase(it);
        }
        while (!hi.empty() && !lo.empty() && *hi.begin() <
    ↳ *prev(lo.end())) {
            auto a = hi.begin(), b = prev(lo.end());
            sum += *b - *a;
            hi.insert(*b), lo.insert(*a), hi.erase(a), lo.erase(b);
        }
    }
    void insert(ll x) { lo.insert(x), fix(); }
    void erase(ll x) {
        auto it = hi.find(x);
        if (it != hi.end()) sum -= x, hi.erase(it);
        else lo.erase(lo.find(x));
        fix();
    }
    int size() const { return hi.size() + lo.size(); }
};

// SMALLEST k instead: negate on the way in and out, or mirror the two
//   ↳ sets.
// If k CHANGES too, just set k and call fix() - the frontier moves in
//   ↳ O(|dk| log n).
// If you only ever ADD (no erase), a single size-k min-heap is
//   ↳ shorter: push, and pop while the
// size exceeds k, keeping a running sum.
```

2.10 DynamicMedian

RUNNING MEDIAN / k-th SMALLEST under insertions, via two multisets kept balanced around the split.

```
// O(log n) per insert, O(1) to read the median and either side's sum.
//   ↳ The pattern generalises to
// "maintain the sum of the k smallest" - which is what
//   ↳ cost-to-move-to-median problems need.
multiset<ll> L, R;
ll sum_L = 0, sum_R = 0;
int k;

// Rebalances the multisets so L always has the lower half/median of
//   ↳ CURRENT elements
void balance() {
    int total_size = sz(L) + sz(R);
    int k_L = (total_size + 1) / 2; // Fixed: Target size based on
    ↳ actual current elements

    // If L is too big, move its largest element to R
    while (sz(L) > k_L) {
        auto it = prev(L.end());
        ll val = *it;

        sum_L -= val;
        sum_R += val;

        R.insert(val);
        L.erase(it);
    }

    // If L is too small, move the smallest element of R to L
    while (sz(L) < k_L) {
        auto it = R.begin();
        ll val = *it;

        sum_R -= val;
        sum_L += val;

        L.insert(val);
        R.erase(it);
    }
}

void add_val(ll val) {
    if (L.empty() || val <= *L.rbegin()) {
        L.insert(val);
        sum_L += val;
    } else {
        R.insert(val);
        sum_R += val;
    }
    balance();
}

void remove_val(ll val) {
    auto it = L.find(val);
    if (it != L.end()) {
        sum_L -= val;
        L.erase(it);
    } else {
        it = R.find(val);
        if (it == R.end()) return; // not present in
        ↳ either half: do nothing
        sum_R -= val;
        R.erase(it);
    }
    balance();
}

ll get_cost() { // UB if empty - guard
    ↳ the caller
    ll median = *L.rbegin();
    return sum_R - sum_L + median * (sz(L) - sz(R));
}
```

2.11 IntervalSetODT

INTERVAL SET / "Chtholly tree" (ODT). Keeps the array as maximal runs of equal value.

```
// assign(l, r, v) is amortised near-O(log n) WHEN the data has many
//   ↳ range-assigns (random or
//   ↳ adversarial-but-assign-heavy); each assign destroys the intervals it
//   ↳ covers, so the total
//   ↳ number of intervals created is O(q).
// THIS is the standard way to support range bitwise-AND / OR / MOD /
//   ↳ assign together with
// range sum, because those operations rapidly make long stretches
//   ↳ constant.
struct ODT {
    map<int, pair<int, ll>> mp; // l -> {r,
    // value}, covering [l, r]

    ODT(int n, ll v = 0) { mp[0] = {n - 1, v}; }

    // split so that an interval starts exactly at p; returns the
    //   ↳ iterator to it
    auto split(int p) {
        auto it = prev(mp.upper_bound(p));
        if (it->first == p) return it;
        auto [r, v] = it->second;
        it->second.first = p - 1;
        return mp.emplace(p, make_pair(r, v)).first;
    }
    void assign(int l, int r, ll v) {
        auto itr = split(r + 1), itl = split(l);
        mp.erase(itl, itr);
        mp[l] = {r, v};
    }
    // Generic scan over [l, r]: f(l, r, value) is called once per run.
    //   ↳ O(#runs touched).
    template <class F> void forEach(int l, int r, F f) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) f(it->first,
        //   ↳ it->second.first, it->second.second);
    }
    ll sum(int l, int r) {
        ll s = 0;
        forEach(l, r, [&](int a, int b, ll v) { s += v * (b - a + 1);
        //   ↳ });
        return s;
    }
    // RANGE ADD: touch each run once. Runs are not merged, so adds
    //   ↳ alone do NOT keep the
    //   ↳ amortised bound - this is only cheap when assigns keep
    //   ↳ collapsing the structure.
    void add(int l, int r, ll x) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) it->second.second += x;
    }
    // RANGE AND / OR / MOD follow the same loop, applying the op to
    //   ↳ each run's value.
    //   ↳ They are fast because every value can only decrease O(log V)
    //   ↳ times before it stops
    //   ↳ changing, at which point neighbouring runs become equal and can
    //   ↳ be merged.
    template <class Op> void apply(int l, int r, Op op) {
        auto itr = split(r + 1), itl = split(l);
        for (auto it = itl; it != itr; ++it) it->second.second =
        //   ↳ op(it->second.second);
        merge(l, r);
    }
    void merge(int l, int r) { // glue
        //   ↳ neighbouring runs of equal value
        auto it = split(l);
        while (it != mp.end() && it->first <= r) {
            auto nx = next(it);
            if (nx != mp.end() && nx->second.second ==
            //   ↳ it->second.second &&
            //   ↳ it->second.first + 1 == nx->first) {
            //   ↳ it->second.first = nx->second.first;
            //   ↳ mp.erase(nx);
            //   ↳ } else it = nx;
        }
    }
};
// Careful: worst case WITHOUT assigns is O(n) per operation. If the
//   ↳ problem has only adds and
//   ↳ queries, use a segment tree instead.
```

2.12 DisjointSparseTable

DISJOINT SPARSE TABLE - O(1) query for ANY associative operation (not just idempotent ones).

```
// Build O(n log n) time and memory. Use when a plain sparse table
//   ↳ would double-count:
// sum, product, matrix product, hashing, "merge two structs",
//   ↳ min-count pairs, ...
// (For min/max/gcd/or/and a normal sparse table is smaller - use that
//   ↳ instead.)
template <typename T, class F = function<T(T, T)>>
struct DisjointST {
    int n, LOG;
    vector<vector<T>> t;
    F op;

    DisjointST(const vector<T>& a, F op) : n(a.size()), op(op) {
        LOG = 1;
        while ((1 << LOG) < n) LOG++;
        t.assign(LOG + 1, a);
        for (int lvl = 1; lvl <= LOG; lvl++) {
            int half = 1 << lvl;
            for (int mid = half; mid < n + half; mid += 2 * half) {
                if (mid - 1 < n) { // build
                    //   ↳ leftwards from the split
                    t[lvl][mid - 1] = a[mid - 1];
                    for (int i = mid - 2; i >= mid - half && i >= 0;
                    //   ↳ i--)
                        t[lvl][i] = op(a[i], t[lvl][i + 1]);
                }
                if (mid < n) { // and
                    //   ↳ rightwards
                    t[lvl][mid] = a[mid];
                    for (int i = mid + 1; i < min(n, mid + half); i++)
                        t[lvl][i] = op(t[lvl][i - 1], a[i]);
                }
            }
        }
        T query(int l, int r) const { //
            //   ↳ inclusive [l, r]
            if (l == r) return t[0][l];
            int lvl = 63 - __builtin_clzll((unsigned long long)(l ^ r));
            //   ↳ // highest differing bit
            return op(t[lvl][l], t[lvl][r]);
        }
    };
    // Every query splits at the unique power-of-two boundary between l and
    //   ↳ r, and both halves are
    //   ↳ already prefix/suffix-accumulated at that level - hence exactly one
    //   ↳ op() per query.
```

2.13 WaveletTree

WAVELET TREE over values in [lo, hi]. Build O(n log V).

```
// kth(l, r, k) : k-th SMALLEST in a[l..r]
//   ↳ O(log V)
// countLeq(l, r, x) : how many a[i] <= x in a[l..r]
//   ↳ O(log V)
// countEq(l, r, x) : how many a[i] == x in a[l..r]
//   ↳ O(log V)
// Online (no offline sorting, no persistence) - the practical
//   ↳ alternative to a merge-sort tree.
struct Wavelet {
    int lo, hi;
    Wavelet *l = nullptr, *r = nullptr;
    vector<int> b; // b[i] =
    //   ↳ #elements going left among first i

    // build on a[from, to) - the array IS reordered in place (stable
    //   ↳ partition by midpoint)
    Wavelet(vector<int>::iterator from, vector<int>::iterator to, int
    //   ↳ x, int y) : lo(x), hi(y) {
        if (from >= to) return;
        if (lo == hi) { b.assign(to - from + 1, 0); return; }
        int mid = lo + (hi - lo) / 2;
        auto f = [mid](int v) { return v <= mid; };
        b.reserve(to - from + 1);
        b.push_back(0);
        for (auto it = from; it != to; ++it) b.push_back(b.back() +
        //   ↳ f(*it));
        auto pivot = stable_partition(from, to, f);
        l = new Wavelet(from, pivot, lo, mid);
        r = new Wavelet(pivot, to, mid + 1, hi);
    }
    int kth(int L, int R, int k) { // 1-indexed k,
    //   ↳ inclusive [L, R] (1-based)
```

```

    if (L > R) return 0;
    if (lo == hi) return lo;
    int inLeft = b[R] - b[L - 1], lb = b[L - 1], rb = b[R];
    if (k <= inLeft) return l->kth(lb + 1, rb, k);
    return r->kth(L - lb, R - rb, k - inLeft);
}
int countLeq(int L, int R, int k) { // # {i in [L,R] :
    ↪ a[i] <= k}
    if (L > R || k < lo) return 0;
    if (hi <= k) return R - L + 1;
    int lb = b[L - 1], rb = b[R];
    return l->countLeq(lb + 1, rb, k) + r->countLeq(L - lb, R - rb,
    ↪ k);
}
int countEq(int L, int R, int k) {
    if (L > R || k < lo || k > hi) return 0;
    if (lo == hi) return R - L + 1;
    int lb = b[L - 1], rb = b[R], mid = lo + (hi - lo) / 2;
    if (k <= mid) return l->countEq(lb + 1, rb, k);
    return r->countEq(L - lb, R - rb, k);
}
~Wavelet() { delete l; delete r; }
};
// Usage: compress values to [0, V), copy the array, then Wavelet
    ↪ w(a.begin(), a.end(), 0, V-1);
// queries are 1-BASED on the original positions.

```

2.14 OfflineDynamicConnectivity

OFFLINE DYNAMIC CONNECTIVITY - edges appear and disappear over time; answer queries at each

```

// Needs: DSURollback (02 - DSU/02 - DSURollback.cpp).
// moment. Segment tree over the TIMELINE + DSU with rollback. O(q log
    ↪ q * log n) - DSURollback has no path compression, so find is log
    ↪ n, not alpha.
// The same skeleton answers ANY "structure supports add + rollback,
    ↪ but not delete" problem:
// put each element on the time interval where it is alive, DFS the
    ↪ segment tree, add on entry,
// roll back on exit. ("Deleting from a data structure in O(T log n)")
struct DynConn {
    int n, T;
    vector<vector<pair<int, int>>> seg; // edges alive on
    ↪ each timeline node
    DSURollback d;
    vector<int> comps; // comps[t] =
    ↪ #components at time t

    DynConn(int n, int T) : n(n), T(T), seg(4 * max(1, T)), d(n),
    ↪ comps(T, 0) {}

    void addSeg(int v, int l, int r, int ql, int qr, pair<int, int> e) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) { seg[v].push_back(e); return; }
        int m = (l + r) / 2;
        addSeg(2 * v, l, m, ql, qr, e), addSeg(2 * v + 1, m + 1, r, ql,
        ↪ qr, e);
    }
    // edge (u,v) exists during times [tl, tr] (inclusive)
    void addEdge(int u, int v, int tl, int tr) { if (tl <= tr)
    ↪ addSeg(1, 0, T - 1, tl, tr, {u, v}); }

    void dfs(int v, int l, int r, int cur) {
        int snap = d.get_state();
        for (auto [a, b] : seg[v]) cur = d.join(a, b);
        if (l == r) comps[l] = cur;
        else {
            int m = (l + r) / 2;
            dfs(2 * v, l, m, cur), dfs(2 * v + 1, m + 1, r, cur);
        }
        d.rollback_to(snap);
    }
    void solve() { if (T) dfs(1, 0, T - 1, n); }
};
// USAGE: collect every edge's [birth, death] interval first (an edge
    ↪ that is never removed dies
// at T-1), then one solve() fills comps[]. To answer "are u and v
    ↪ connected at time t", instead
// record the queries per leaf and test d.same(u, v) inside the l == r
    ↪ branch.

```

2.15 CartesianTree

CARTESIAN TREE in O(n): heap-ordered by VALUE, BST-ordered by INDEX.

```

// The root of any range is its minimum, so "range minimum" structure
    ↪ becomes a TREE -
// which turns range problems into subtree problems (and enables divide
    ↪ & conquer on the min).
// Built with a monotonic stack; ties go left so the tree is unique.
struct CartTree {
    int n, root = -1;
    vector<int> l, r, par;

    CartTree(const vector<ll>& a) : n(a.size()), l(n, -1), r(n, -1),
    ↪ par(n, -1) {
        vector<int> st;
        for (int i = 0; i < n; i++) {
            int last = -1;
            while (!st.empty() && a[st.back()] > a[i]) last =
            ↪ st.back(), st.pop_back();
            l[i] = last;
            if (last != -1) par[last] = i;
            if (!st.empty()) r[st.back()] = i, par[i] = st.back();
            st.push_back(i);
        }
        root = st[0];
    }
};
// USES
// - "for every element, the maximal range where it is the minimum" =
    ↪ its subtree range,
// - which is the contribution technique (sum over subarrays of the
    ↪ min) in tree form
// - largest rectangle in a histogram = best over nodes of value *
    ↪ subtreeSize
// - divide & conquer "solve(l, r) splits at the minimum" without
    ↪ recomputing RMQ
// - counting subarrays whose minimum is a given element
// - a Treap built on (index, random priority) is exactly a Cartesian
    ↪ tree - same code shape

```

2.16 SegTreeMerging

SEGMENT TREE MERGING - one dynamic segment tree per vertex over the VALUE domain, merged

```

// bottom-up along a tree. Total O(n log n) nodes touched over ALL
    ↪ merges (amortised), because
// each merge destroys as many nodes as it visits.
// Solves, per subtree and in one DFS: most frequent value, k-th value,
    ↪ count of a value,
// "sum of values in a range" - i.e. everything a merge-sort tree
    ↪ gives, but online per subtree.
struct SegMerge {
    struct Node { int l = 0, r = 0; ll sum = 0; int best = 0; ll
    ↪ bestCnt = 0; };
    vector<Node> t;
    int V; // value domain
    ↪ is [0, V)

    SegMerge(int V, int reserve = 0) : V(V) { t.reserve(reserve),
    ↪ t.push_back(Node()); } // 0 = null
    int newNode() { t.push_back(Node()); return t.size() - 1; }

    void pull(int v) {
        int L = t[v].l, R = t[v].r;
        t[v].sum = t[L].sum + t[R].sum;
        if (t[L].bestCnt >= t[R].bestCnt) t[v].best = t[L].best,
        ↪ t[v].bestCnt = t[L].bestCnt;
        else t[v].best = t[R].best, t[v].bestCnt = t[R].bestCnt;
    }
    // Returns the (possibly new) node index. NEVER take an 'int&' into
    ↪ t here: newNode() can
    // reallocate the vector and invalidate the reference mid-recursion.
    int update(int v, int lo, int hi, int p, ll d) {
        if (!v) v = newNode();
        if (hi - lo == 1) { t[v].sum += d, t[v].best = lo, t[v].bestCnt
        ↪ = t[v].sum; return v; }
        int m = (lo + hi) / 2;
        if (p < m) { int c = update(t[v].l, lo, m, p, d); t[v].l = c; }
        else { int c = update(t[v].r, m, hi, p, d); t[v].r = c; }
        pull(v);
        return v;
    }
    int merge(int a, int b, int lo, int hi) { // destroys b;
    ↪ returns the merged root
        if (!a || !b) return a | b;
        if (hi - lo == 1) { t[a].sum += t[b].sum, t[a].bestCnt =
        ↪ t[a].sum; return a; }

```

```

    int m = (lo + hi) / 2;
    int L = merge(t[a].l, t[b].l, lo, m), R = merge(t[a].r, t[b].r,
    ↪ m, hi);
    t[a].l = L, t[a].r = R;
    pull(a);
    return a;
}
ll query(int v, int lo, int hi, int ql, int qr) { // sum of counts
    ↪ in [ql, qr)
    if (!v || qr <= lo || hi <= ql) return 0;
    if (ql <= lo && hi <= qr) return t[v].sum;
    int m = (lo + hi) / 2;
    return query(t[v].l, lo, m, ql, qr) + query(t[v].r, m, hi, ql,
    ↪ qr);
}
int kth(int v, int lo, int hi, ll k) { // k-th smallest
    ↪ (1-indexed), -1 if k too big
    if (!v || k > t[v].sum) return -1;
    if (hi - lo == 1) return lo;
    int m = (lo + hi) / 2;
    return k <= t[t[v].l].sum ? kth(t[v].l, lo, m, k) : kth(t[v].r,
    ↪ m, hi, k - t[t[v].l].sum);
}
// wrappers
void update(int& root, int p, ll d) { root = update(root, 0, V, p,
    ↪ d); }
int merge(int a, int b) { return merge(a, b, 0, V); }
ll query(int root, int ql, int qr) { return query(root, 0, V, ql,
    ↪ qr); }
int kth(int root, ll k) { return kth(root, 0, V, k); }
};
// TREE USAGE: root[v] starts with just v's own value; after recursing,
    ↪ merge every child's root
// into v's. Answer v's query right there. Never reuse a root after
    ↪ merging it away.

```

2.17 BitsetTricks

BITSET SPEEDUPS - the /64 family. All of these are "obviously too slow" DPs that fit anyway.

```

// TRANSITIVE CLOSURE (reachability) in O(n^3 / 64). n up to ~5000.
template <int N>
void transitiveClosure(vector<bitset<N>> &g, int n) {
    for (int k = 0; k < n; k++)
        for (int i = 0; i < n; i++) if (g[i][k]) g[i] |= g[k];
}
// LCS in O(n*m/64) - bit-parallel (Allison-Dix). 'a' is the string
    ↪ whose length bounds the
// bitset width; row is a bitmask over positions of 'a'.
// x = row | match[b_i]
// row' = x & ((x - (row << 1 | 1)) ^ x)
// The "| 1" is the carry-in that makes the shift borrow correctly.
typedef unsigned long long u64;
int bitLCS(const string& a, const string& b) {
    int n = a.size(), W = (n + 63) / 64;
    if (!n || b.empty()) return 0;
    vector<vector<u64>> match(256, vector<u64>(W, 0));
    for (int i = 0; i < n; i++) match[(unsigned char)a[i]][i >> 6] |=
    ↪ 1ULL << (i & 63);
    vector<u64> row(W, 0), x(W), sh(W);
    for (char c : b) {
        const auto& M = match[(unsigned char)c];
        for (int i = 0; i < W; i++) x[i] = row[i] | M[i];
        u64 carry = 1; // sh = (row <<
    ↪ 1) | 1
        for (int i = 0; i < W; i++) { u64 nc = row[i] >> 63; sh[i] =
    ↪ (row[i] << 1) | carry; carry = nc; }
        u64 borrow = 0; // row = x & ((x
    ↪ - sh) ^ x)
        for (int i = 0; i < W; i++) {
            u64 t = sh[i] + borrow;
            borrow = (t < sh[i]) || (x[i] < t); // t < sh[i]
    ↪ catches the carry overflow
            u64 d = x[i] - t;
            row[i] = x[i] & (d ^ x[i]);
        }
    }
    int r = 0;
    for (u64 v : row) r += __builtin_popcountll(v);
    return r;
}
// SUBSET SUM / KNAPSACK FEASIBILITY: dp |= dp << w (see
    ↪ KnapsackFamily.cpp)
// BIPARTITE MATCHING on dense graphs: keep each left vertex's
    ↪ neighbourhood as a bitset and

```

```

// pick the next free right vertex with (adj[u] & free)._Find_first().
// COUNTING PAIRS with a property: bitset per value class, then
    ↪ popcount of ANDs.
// GCC EXTRAS: b._Find_first(), b._Find_next(i) iterate set bits in
    ↪ O(N/64) total.
// RULE OF THUMB: bitset turns 1e9 "simple" operations into ~1.5e7 word
    ↪ operations - budget
// about n*m/64 <= 1e8 for a comfortable pass.

```

2.18 VeniceAndSWAG

VENICE TECHNIQUE - a multiset with an O(1) "add x to EVERY element".

```

// Keep a global offset; store values shifted by it. 15 lines, and it
    ↪ turns a whole family of
// "everyone loses X health, remove the dead" problems into a one-liner.
struct Venice {
    multiset<ll> s; // global offset
    ll water = 0;

    void add(ll v) { s.insert(v + water); } // insert the
    ↪ true value v
    void erase(ll v) { auto it = s.find(v + water); if (it != s.end())
    ↪ s.erase(it); }
    void addAll(ll v) { water += v; } // O(1): add v
    ↪ to every element
    ll getMin() const { return *s.begin() - water; }
    ll getMax() const { return *s.rbegin() - water; }
    int size() const { return s.size(); }
    // Pop everything whose TRUE value is <= t (e.g. everyone who died).
    template <class F> void popLE(ll t, F onPop) {
        while (!s.empty() && *s.begin() - water <= t) {
    ↪ onPop(*s.begin() - water); s.erase(s.begin()); }
    }
};
// SAME IDEA elsewhere: a binary trie with a global XOR mask ("xor
    ↪ every element by x"),
// a BIT with a global additive offset, a lazy "add to all" on a heap.

// SWAG (Sliding Window Aggregation) for ANY monoid - non-commutative
    ↪ and non-invertible ops
// included: matrix product, affine composition, gcd, "merge two
    ↪ structs".
// Amortised O(1) push/pop/fold. This generalises 09/02 -
    ↪ TwoStackQueue.cpp, which is hardwired
// to min/max. Prefix-product-and-divide does NOT work without an
    ↪ inverse - this does.
template <class T, class Op>
struct SWAG {
    vector<pair<T, T>> front_, back_; // {value,
    ↪ aggregate from here to the bottom}
    T id;
    Op op;
    SWAG(T id, Op op) : id(id), op(op) {}

    T foldFront() const { return front_.empty() ? id :
    ↪ front_.back().second; }
    T foldBack() const { return back_.empty() ? id :
    ↪ back_.back().second; }
    T fold() const { return op(foldFront(), foldBack()); } //
    ↪ front first: order matters
    void push(T v) { back_.push_back({v, op(foldBack(), v)}); }
    void pop() {
        if (front_.empty())
            while (!back_.empty()) {
                T v = back_.back().first; back_.pop_back();
                front_.push_back({v, op(v, foldFront())}); //
    ↪ reversed accumulation
            }
        if (!front_.empty()) front_.pop_back();
    }
    int size() const { return front_.size() + back_.size(); }
};
// USAGE: SWAG<ll, function<ll(ll,ll)>> q(0, [](ll a, ll b){ return a +
    ↪ b; });
// For non-commutative ops keep the argument order exactly as written
    ↪ above, or the window
// product comes out reversed.

```

2.19 MergeableHeap

MERGEABLE (MELDABLE) HEAP - a priority queue that also supports MERGING two heaps in $O(\log n)$,

// which std::priority_queue cannot do. This is a skew heap: merge by
 ↳ walking the two right spines
 // and swapping children, which keeps the amortised depth logarithmic
 ↳ with no stored rank.
 // Supports a lazy "add v to every key in this heap" tag, because that
 ↳ is what the two algorithms
 // that need a mergeable heap actually want.
 // USES: Chu-Liu/Edmonds directed MST in $O(E \log V)$ (each contracted
 ↳ cycle merges its in-edge
 // heaps and subtracts the chosen edge's weight from the whole heap);
 ↳ Dijkstra-style k-shortest
 // paths (Eppstein); "merge the two children's candidate sets" DP on
 ↳ trees; leftist-heap solutions
 // to scheduling problems (repeatedly merge and pop the largest, e.g.
 ↳ to minimise total lateness).

```
struct MHeap {
    struct Node { ll key, lz = 0; Node *l = 0, *r = 0; };
    static void down(Node* t) { // apply
        ↳ the lazy tag downwards
        if (!t || !t->l) return;
        t->key += t->lz;
        if (t->l) t->l->lz += t->lz;
        if (t->r) t->r->lz += t->lz;
        t->lz = 0;
    }
    static Node* merge(Node* a, Node* b) { // O(log
        ↳ n) amortised, MIN-heap
        if (!a || !b) return a ? a : b;
        down(a), down(b);
        if (a->key > b->key) swap(a, b);
        a->r = merge(a->r, b);
        swap(a->l, a->r); // the
        ↳ "skew" step
        return a;
    }
    static Node* push(Node* t, ll key) { return merge(t, new
        ↳ Node(key)); }
    static ll top(Node* t) { return down(t), t->key; }
    static Node* pop(Node* t) { down(t); return merge(t->l, t->r); }
    static void addAll(Node* t, ll v) { if (t) t->lz += v; } // O(1)
        ↳ add to every key
};
// TO GET A MAX-HEAP: negate the keys, or flip the comparison in merge.
// LEFTIST HEAP is the same interface with a stored 'dist' (null-path
  ↳ length) and the rule "keep the
// larger dist on the left"; it gives WORST-CASE  $O(\log n)$  instead of
  ↳ amortised. Skew is shorter and
// fine unless you need the worst-case bound inside another amortised
  ↳ argument. 24 - PersistentHeap
// is that leftist version, made persistent (meld without destroying
  ↳ either input).
// PAIRING HEAP has  $O(1)$  merge and decrease-key in practice and is what
  ↳ you want if you need
// decrease-key (Dijkstra on very dense graphs); in contests a binary
  ↳ heap with lazy deletion wins.
// LAZY DELETION is usually enough: push (key, id), and pop until the
  ↳ top is still valid. That
// removes the need for decrease-key entirely and is 5 lines.
// SMALL-TO-LARGE with std::priority_queue also "merges" heaps:
  ↳ repeatedly pop from the smaller and
// push into the larger,  $O(n \log^2 n)$  total. Shorter than this file -
  ↳ use it unless the  $\log^2$  hurts.
```

2.20 PersistentAndUndo

PERSISTENCE AND UNDO - three ways to travel in a structure's history, and when each applies.

```
// ROLLBACK (05 - DSU/02): undo the last op. Cheapest. Needs a stack
  ↳ discipline (LIFO).
// OFFLINE DELETION (14 - OfflineDynamicConnectivity): segment tree
  ↳ over time + rollback.
// Handles arbitrary insert/delete, but you must know every
  ↳ operation in advance.
// PERSISTENCE (this file): branch the history. The only one that
  ↳ works ONLINE with deletions,
// and the only one that lets you query a past version after moving
  ↳ on.
```

// --- PERSISTENT ARRAY: a segment tree over indices. get/set in $O(\log$
 ↳ $n)$, $O(\log n)$ new nodes. ---

```
struct PArray {
    struct Node { int l = 0, r = 0; ll v = 0; };
    vector<Node> t;
```

```
int n;
PArray(int n) : t(1, n(n)) {} // node
↳ 0 is the shared all-zero
int build(const vector<ll>& a, int lo, int hi) {
    int id = t.size();
    t.push_back({});
    if (lo == hi) { t[id].v = a[lo]; return id; }
    int m = (lo + hi) / 2, L = build(a, lo, m), R = build(a, m + 1,
↳ hi);
    t[id].l = L, t[id].r = R;
    return id;
}
int build(const vector<ll>& a) { return build(a, 0, n - 1); }
ll get(int u, int i, int lo, int hi) {
    if (lo == hi) return t[u].v;
    int m = (lo + hi) / 2;
    return i <= m ? get(t[u].l, i, lo, m) : get(t[u].r, i, m + 1,
↳ hi);
}
ll get(int u, int i) { return get(u, i, 0, n - 1); }
int set(int u, int i, ll v, int lo, int hi) {
    int id = t.size();
    t.push_back(t[u]);
    if (lo == hi) { t[id].v = v; return id; }
    int m = (lo + hi) / 2;
    if (i <= m) t[id].l = set(t[u].l, i, v, lo, m);
    else t[id].r = set(t[u].r, i, v, m + 1, hi);
    return id;
}
int set(int u, int i, ll v) { return set(u, i, v, 0, n - 1); }
};
// --- PERSISTENT DSU on top of it.  $O(\log^2 n)$  per operation. ---
// The catch that makes it work: NO PATH COMPRESSION (that would mutate
  ↳ an old version), so you
// MUST union by size/rank to keep the height  $O(\log n)$ .
struct PDSU {
    PArray par, siz;
    PDSU(int n) : par(n), siz(n) {}
    pair<int, int> init(int n) { //
        ↳ returns {parRoot, szRoot}
        vector<ll> p(n), s(n, 1);
        iota(all(p), 0);
        return {par.build(p), siz.build(s)};
    }
    int find(int pr, int x) { ll p = par.get(pr, x); return p == x ? x
↳ : find(pr, p); }
    pair<int, int> join(int pr, int sr, int a, int b) {
        a = find(pr, a), b = find(pr, b);
        if (a == b) return {pr, sr};
        if (siz.get(sr, a) < siz.get(sr, b)) swap(a, b);
        return {par.set(pr, b, a), siz.set(sr, a, siz.get(sr, a) +
↳ siz.get(sr, b))};
    }
};
/* THE OTHER TWO TIME-TRAVEL TRICKS, as recipes (code: 21 -
  ↳ QueueUndoTrick, 25 - StaticToDynamic)
QUEUE UNDO. You have a structure that supports only "undo the last op"
  ↳ (rollback DSU), but the
  ↳ deletions arrive in FIFO order (a sliding window). Keep the undo
  ↳ stack partitioned so the
  ↳ OLDEST element is near the top; when it is not, pop a prefix and
  ↳ re-push it in the other order,
  ↳ always rebuilding the SMALLER half.  $O(\log n)$  amortised extra. This
  ↳ is the online alternative to
  ↳ the offline segment-tree-on-time trick, and it is what "sliding
  ↳ window connectivity" needs.
STATIC TO DYNAMIC (the logarithmic method). Any STATIC structure with
  ↳  $O(B(n))$  build and  $O(Q(n))$ 
  ↳ query becomes insert-only: keep  $O(\log n)$  instances of sizes  $2^0,$ 
  ↳  $2^1, 2^2, \dots$ ; inserting is
  ↳ binary-counter carrying, so you rebuild an instance of size  $2^k$  only
  ↳ every  $2^k$  insertions.
  ↳ Insert becomes  $O(B(n)/n * \log n)$  amortised and query  $O(Q(n) \log n)$ 
  ↳ (ask all instances).
  ↳ Use it to make a convex hull / a wavelet tree / an Aho-Corasick
  ↳ automaton accept insertions.
PARTIALLY vs FULLY PERSISTENT: partially = you may query any past
  ↳ version but only modify the
  ↳ latest - for a DSU that is 05 - DSU/05 - PartiallyPersistentDSU,
  ↳  $O(n)$  memory instead of this
  ↳ file's  $O(n \log n)$ ; fully = you may modify any version, branching the
  ↳ history into a tree.
  ↳ The code above is fully persistent.
WHEN NOT TO BOTHER: if the problem is offline and only needs "delete",
  ↳ the segment tree over the
  ↳ timeline with a rollback DSU is shorter, faster, and has a smaller
  ↳ constant than any of this. */
```

2.21 QueueUndoTrick

QUEUE UNDO TRICK - turns a structure that can only "undo the LAST operation" into one that can

```
// Needs: 05 - Disjoint Set Union (DSU)/02 - DSURollback.cpp
// also "undo the OLDEST operation", i.e. a QUEUE of operations instead
//   of a stack.
// 0(Log n) amortised extra factor on top of the underlying structure.
// WHEN: a sliding window over operations - "for each l, the largest r
//   such that edges l..r keep
// the graph bipartite / acyclic / with <= k components" (two pointers
//   over edges), or any online
// problem where insertions and deletions are FIFO. Deletions in
//   arbitrary order need
// 14 - OfflineDynamicConnectivity (offline) or a link-cut tree
//   (forests only).
// KEY INVARIANT: the undo stack is kept partitioned into a "to be
//   deleted soon" prefix and the
// rest. To pop the front, undo a suffix of the stack, then re-push it
//   with the OLDEST elements
// nearest the top; the rebuild always processes the SMALLER of the two
//   groups, which is what makes
// the potential argument work - each operation is re-pushed O(log n)
//   times over its lifetime.
// The underlying structure must support: apply(op) and undo-last, both
//   cheap; nothing else.
struct QueueUndo {
    struct Op { int u, v; bool old; }; // old =
    // "already re-pushed once"
    DSURollback d;
    vector<Op> st; //
    // mirrors the undo stack exactly
    int bot = 0; // first
    // index that is not yet 'old'
    QueueUndo(int n) : d(n) {}
    void push(Op o) { d.join(o.u, o.v), st.push_back(o); }
    void push(int u, int v) { push({u, v, false}); }
    void pop() { //
        // remove the OLDEST operation
        vector<Op> a[2];
        while (bot < (int)st.size() && st[bot].old) bot++;
        if (bot == (int)st.size()) { // no
            // old ops left: flip everything
            while (!st.empty()) a[0].push_back(st.back());
            // st.pop_back(), d.rollback();
            for (auto& o : a[0]) o.old = true, push(o);
            a[0].clear(), bot = 0;
        }
        do { // pull
            // back until the front is up
            a[st.back().old].push_back(st.back());
            st.pop_back(), d.rollback();
        } while (a[1].size() < a[0].size() && (int)st.size() > bot);
        for (int i : {0, 1}) {
            reverse(all(a[i]));
            for (auto& o : a[i]) push(o);
        }
        st.pop_back(), d.rollback(); // the
    } // front is now on top
};
// THE STRUCTURE UNDERNEATH can be anything with rollback: a DSU
// (connectivity, bipartiteness by
// storing the parity to the parent, number of components), a XOR
// basis, a max-stack, a knapsack
// with rollback. Replace push()/pop() bodies and keep the stack
// discipline identical.
// SLIDING WINDOW RECIPE: for l = 1..m, push edges while the invariant
// holds, record the answer,
// then pop() the edge at l. Total O(m log m * alpha).
// CHEAPER OPTIONS FIRST: if the problem is offline, the segment tree
// over the timeline
// (14 - OfflineDynamicConnectivity) is shorter and has a better
// constant. If the deletions are
// LIFO, plain rollback is enough. This file is only for FIFO + online.
```

2.22 PermutationTree

PERMUTATION TREE (PQ-tree / substitution decomposition) - represents ALL "common intervals" of a

```
// permutation: the ranges [l, r] whose VALUES also form a contiguous
//   range. There can be O(n^2) of
// them; this tree stores every one of them in O(n) space, in O(n log
//   n) time.
// EVERY node has a contiguous span of POSITIONS and a contiguous range
//   of VALUES, and is one of:
// LEAF - a single position.
```

```
// JOIN node - its children's value ranges are consecutive and
//   monotone (increasing = type 1,
//   decreasing = type 2). ANY consecutive run of >= 2 of
//   its children is itself a
//   common interval. This is where the O(n^2) of them
//   hide.
// CUT (prime) - no proper consecutive run of its children is a
//   common interval; only the whole
//   node is. A cut node always has >= 4 children.
// WHEN: "count subarrays whose values are consecutive integers", "the
//   number of ways to split a
// permutation into monotone blocks", simple-permutation questions, and
//   the smallest common
// interval containing a given [l, r]. If you only need to COUNT common
//   intervals, the divide and
// conquer with min/max stacks is shorter - reach for the tree when you
//   need the STRUCTURE.
// KEY INVARIANT: sweep r and keep, for every l, the quantity (max -
//   min) - (r - l) over a[l..r],
// which is >= 0 and equals 0 exactly when [l, r] is a common interval.
// Monotone stacks maintain it
// under range add in O(1) amortised each, and a min segment tree finds
// the zeros. Whenever the new
// element extends an existing node's value range by one the node
// absorbs it (join), and whenever a
// zero appears further left a cut node is created over everything
// between.
struct PermTree {
    int n, id, root = 0;
    vector<int> p, ty, par, mn, lz; // ty: 0
    // cut, 1 up-join, 2 down-join
    vector<pii> val, span; // value
    // range, position range
    vector<vector<int>> ch;
    PermTree(const vector<int>& perm) //
    // perm[1..n], a permutation of 1..n
    // : n(perm.size() - 1), id(0), p(perm), ty(2 * perm.size(), 0),
    // par(2 * perm.size(), -1),
    // mn(8 * perm.size(), 0), lz(8 * perm.size(), 0), val(2 *
    // perm.size()),
    // span(2 * perm.size(), ch(2 * perm.size())) { build(); }
    void apply(int x, int v) { mn[x] += v, lz[x] += v; }
    void push(int x) { if (lz[x]) apply(2 * x, lz[x]), apply(2 * x + 1,
    // lz[x]), lz[x] = 0; }
    void upd(int x, int b, int e, int l, int r, int v) {
        if (r < b || e < l) return;
        if (l <= b && e <= r) { apply(x, v); return; }
        int m = (b + e) / 2;
        push(x);
        upd(2 * x, b, m, l, r, v), upd(2 * x + 1, m + 1, e, l, r, v);
        mn[x] = min(mn[2 * x], mn[2 * x + 1]);
    }
    int qry(int x, int b, int e, int l, int r) {
        if (r < b || e < l || l > r) return inf;
        if (l <= b && e <= r) return mn[x];
        int m = (b + e) / 2;
        push(x);
        return min(qry(2 * x, b, m, l, r), qry(2 * x + 1, m + 1, e, l,
    // r));
    }
    void link(int u, int v) { par[v] = u, ch[u].push_back(v); }
    bool adj(int i, int j) { return val[i].se == val[j].fi - 1; }
    // j starts right after i
    int len(int i) { return val[i].se - val[i].fi + 1; }
    static pii un(pii a, pii b) { return {min(a.fi, b.fi), max(a.se,
    // b.se)}; }
    void build() {
        id = n + 1;
        vector<int> mx{0}, mi{0}, st; //
    // monotone stacks + node stack
        for (int i = 1; i <= n; i++) {
            while (mx.back() && p[mx.back()] < p[i]) {
                int r = mx.back();
                mx.pop_back();
                upd(1, 1, n, mx.back() + 1, r, p[i] - p[r]);
            }
            mx.push_back(i);
            while (mi.back() && p[mi.back()] > p[i]) {
                int r = mi.back();
                mi.pop_back();
                upd(1, 1, n, mi.back() + 1, r, p[r] - p[i]);
            }
            mi.push_back(i);
            val[i] = {p[i], p[i]}, span[i] = {i, i};
            int cur = i;
            while (true) {
                if (!st.empty() && (adj(st.back(), cur) || adj(cur,
```

```

    ↪ st.back())) {
        int b = st.back(), want = adj(b, cur) ? 1 : 2;
        if (ty[b] == want) {
            ↪ absorb into the existing join
                link(b, cur);
                val[b] = un(val[b], val[cur]), span[b] =
            ↪ un(span[b], span[cur]);
                cur = b;
        } else {
            ↪ a new join node
                ty[id] = want, link(id, b), link(id, cur);
                val[id] = un(val[b], val[cur]), span[id] =
            ↪ un(span[b], span[cur]);
                cur = id++;
        }
        st.pop_back();
    } else if (qry(1, 1, n, 1, i - len(cur)) == 0) {
        ↪ a cut node closes here
            int L = len(cur);
            pii v = val[cur], s = span[cur];
            link(id, cur);
            do {
                int b = st.back();
                L += len(b), v = un(v, val[b]), s = un(s,
            ↪ span[b]);
                link(id, b), st.pop_back();
            } while (v.se - v.fi + 1 != L);
            reverse(all(ch[id]));
            val[id] = v, span[id] = s, cur = id++;
        } else break;
    }
    st.push_back(cur);
    upd(1, 1, n, 1, i, -1);
}
root = st.back();
};
// COUNTING COMMON INTERVALS: sum over join nodes with k children of
    ↪ k*(k-1)/2, plus n (the leaves),
// plus the number of cut nodes. That is the classic "count subarrays
    ↪ forming a value range".
// SMALLEST COMMON INTERVAL CONTAINING [l, r]: binary-lift from leaf l
    ↪ to the highest ancestor whose
// span still ends before r, take its parent u. If u is a cut node the
    ↪ answer is span[u]; if u is a
// join node the answer is the run of u's children from the one
    ↪ containing l to the one containing
// r (binary search over the children, whose spans are sorted).
// NODE COUNT is < 2n, so size every array 2n. Leaves are 1..n,
    ↪ internal nodes n+1..id-1.
// SIMPLE PERMUTATIONS are exactly those whose tree is a single cut
    ↪ node over n leaves.
// THIS IS MODULAR DECOMPOSITION specialised to permutation graphs -
    ↪ the same "join / cut" shape
// appears for cographs and for general graphs, where it is much harder
    ↪ to compute.

```

2.23 DynamicTreeDiameter

DYNAMIC TREE DIAMETER - the diameter of a weighted tree under ONLINE edge-weight changes,

```

// O(log n) per update, O(1) to read the answer. The tree SHAPE is
    ↪ fixed; only the weights move.
// WHEN: "after each update print the diameter" (CF 1192B and every
    ↪ problem shaped like it). If the
// weights are static, two DFS or the tree DP in 09 - Trees/05 is O(n).
    ↪ If edges are ADDED or
// REMOVED you need a link-cut tree (09 - Trees/11) or offline divide
    ↪ and conquer instead.
// KEY INVARIANT: put the tree on its Euler tour and let f[p] = depth
    ↪ of the vertex visited at
// position p. For any x <= y <= z in the tour, f[x] - 2f[y] + f[z] <=
    ↪ dist between the vertices at
// x and z (with equality when y is their LCA, which is always visited
    ↪ between them), so the
// diameter is exactly max over x <= y <= z of f[x] - 2f[y] + f[z].
    ↪ Changing an edge's weight by
// delta adds delta to f over the CONTIGUOUS tour range of the child's
    ↪ subtree - a range add. The
// segment tree node keeps the five best partial expressions so they
    ↪ merge in O(1):
// a = max f[x], b = max -2f[y], c = max f[x]-2f[y], d = max
    ↪ -2f[y]+f[z], e = the answer.

```

```

struct DynDiameter {
    struct Node { ll a = 0, b = 0, c = 0, d = 0, e = 0; };
    struct Node mrg(const Node& L, const Node& R) {
        Node x;

```

```

        x.a = max(L.a, R.a), x.b = max(L.b, R.b);
        x.c = max({L.c, R.c, L.a + R.b});
        x.d = max({L.d, R.d, L.b + R.a});
        x.e = max({L.e, R.e, L.c + R.a, L.a + R.d});
        return x;
    }
    int n, T = 0;
    vector<Node> t;
    vector<ll> lz, w;
    vector<int> st, en, low;
    ↪ low[e] = deeper endpoint of edge e
    vector<vector<pii>> g;
    ↪ edge id
    DynDiameter(int n) : n(n), t(16 * n), lz(16 * n, 0), st(n), en(n),
    ↪ g(n) {}
    void addEdge(int u, int v, ll c) {
        int e = w.size();
        w.push_back(c), low.push_back(-1);
        g[u].push_back({v, e}), g[v].push_back({u, e});
    }
    void dfs(int u, int p) {
        st[u] = ++T;
        for (auto [v, e] : g[u])
            if (v != p) low[e] = v, dfs(v, u), ++T;
        en[u] = T;
    }
    void apply(int x, ll v) {
        t[x].a += v, t[x].b -= 2 * v, t[x].c -= v, t[x].d -= v, lz[x]
    ↪ += v;
    }
    void push(int x) { if (lz[x]) apply(2 * x, lz[x]), apply(2 * x + 1,
    ↪ lz[x]), lz[x] = 0; }
    void upd(int x, int lo, int hi, int l, int r, ll v) {
        if (r < lo || hi < l) return;
        if (l <= lo && hi <= r) { apply(x, v); return; }
        int m = (lo + hi) / 2;
        push(x);
        upd(2 * x, lo, m, l, r, v), upd(2 * x + 1, m + 1, hi, l, r, v);
        t[x] = mrg(t[2 * x], t[2 * x + 1]);
    }
    void build(int root = 0) {
        ↪ after every addEdge
        dfs(root, -1);
        for (int e = 0; e < (int)w.size(); e++) {
            ll c = w[e];
            w[e] = 0;
            setW(e, c);
        }
        ↪ start at 0, so just add
    }
    void setW(int e, ll c) {
        ↪ edge e's weight to c
        upd(1, 1, T, st[low[e]], en[low[e]], c - w[e]);
        w[e] = c;
    }
    ll diameter() { return t[1].e; }
};

```

```

// NEGATIVE WEIGHTS break it: the argument needs f[x] - 2f[LCA] + f[z]
    ↪ to be maximised at the true
// deepest pair, which assumes non-negative edges. Shift all weights up
    ↪ if you must.
// THE SAME TOUR TRICK gives dist(u, v) queries under updates: dist =
    ↪ f[st[u]] + f[st[v]] -
// 2 * min f over the tour between them, so keep a min segment tree
    ↪ alongside.
// DIAMETER ENDPOINTS: store argmaxes next to each of the five maxima;
    ↪ the merge picks them up
// unchanged. Only do this if the problem asks - it triples the node.
// SIMPLER SPECIAL CASE: if updates only ever INCREASE weights, the
    ↪ diameter is monotone and a
// small-to-large "merge the two deepest per subtree" DP is enough.

```

2.24 PersistentHeap

PERSISTENT MELDABLE HEAP - push / pop / meld that RETURN a new heap and leave the old ones

```

// intact. O(log n) time and O(log n) new nodes per operation.
// WHEN: an algorithm needs many heaps that share most of their
    ↪ contents. The canonical one is
// EPPSTEIN's k shortest walks: every vertex's candidate heap is its
    ↪ parent's heap plus its own
// out-edges, so you meld into a COPY and keep the parent's version
    ↪ alive. Also "k best paths in a
// DAG", "k-th best subset", and persistent priority queues in offline
    ↪ simulations.
// If nothing needs the old version, use 19 - MergeableHeap
    ↪ (destructive, no allocation per op).

```

```
// KEY INVARIANT: leftist heap - every node stores d = its null-path
// → length and keeps the LONGER
// path on the left, so the right spine is  $O(\log n)$  and meld only ever
// → walks right spines. Making
// it persistent then costs nothing extra: COPY each node you touch on
// → that spine instead of
// mutating it, which is exactly  $O(\log n)$  copies.
struct PHeap {
    struct Node { int l = 0, r = 0, d = 0; ll key = 0; };
    vector<Node> t;
    PHeap() : t(1) {} // node
    // 0 is the empty heap, d = 0
    int meld(int a, int b) { //
        // returns a NEW root, min-heap
        if (!a || !b) return a | b;
        if (t[a].key > t[b].key) swap(a, b);
        int c = t.size();
        t.push_back(t[a]); // copy,
        // never mutate a
        int r = meld(t[c].r, b);
        t[c].r = r;
        if (t[t[c].l].d < t[r].d) swap(t[c].l, t[c].r);
        t[c].d = t[t[c].r].d + 1;
        return c;
    }
    int push(int a, ll key) {
        t.push_back({0, 0, 1, key});
        return meld(a, (int)t.size() - 1);
    }
    ll top(int a) { return t[a].key; } // a != 0
    int pop(int a) { return meld(t[a].l, t[a].r); }
    bool empty(int a) { return !a; }
};
// MEMORY:  $O((n + q) \log n)$  nodes. If a heap is only ever pushed into
// → (never melded), a persistent
// BINARY heap over a persistent array (20 - PersistentAndUndo) is
// → smaller but has no meld.
// A LAZY ADD-TO-ALL tag does NOT combine with persistence for free: a
// → tag is a mutation, so store
// the offset OUTSIDE the heap (carry "this version's keys are all
// → shifted by s" in the caller) -
// that is what Eppstein and the directed-MST contraction actually do.
// PERSISTENT QUEUE, if that is what you came for, is much cheaper:
// → keep a persistent ARRAY plus
// two indices (head, tail) per version. push = write at tail and store
// → {head, tail+1}; pop = read
// at head and store {head+1, tail}.  $O(\log n)$  per op, no heap needed.
// PERSISTENT STACK is a linked list: every version is just a node
// → pointer.  $O(1)$  per op.
```

2.25 StaticToDynamic

STATIC TO DYNAMIC (the LOGARITHMIC METHOD) - makes ANY static structure accept insertions.

```
// If the static one builds in  $O(B(n))$  and answers in  $O(Q(n))$ , you get
// → insert in  $O(B(n)/n * \log n)$ 
// AMORTISED and query in  $O(Q(n) * \log n)$  - you ask every bucket and
// → combine.
// WHEN: you have a build-once structure (convex hull, sorted array,
// → wavelet tree, suffix
// automaton, Aho-Corasick, KD-tree, sparse table) and the problem
// → inserts items online. Do NOT
// use it when a natively dynamic structure exists and is fast enough -
// → a BIT beats "buckets of
// sorted arrays" for prefix counting, and a balanced BST beats
// → "buckets of sorted arrays" for
// order statistics.
// KEY INVARIANT: keep at most one instance of each size  $2^0, 2^1, 2^2,$ 
// → ... - a BINARY COUNTER.
// Inserting sets the lowest empty bit and merges every full bucket
// → below it, so a bucket of size
//  $2^k$  is rebuilt only once per  $2^k$  insertions; summed over k that is
// →  $O(\log n)$  rebuilds of
// amortised cost  $B(n)/n$  each. Because the buckets partition the items,
// → ANY decomposable query
// (one whose answer over a union is a cheap combination of the answers
// → over the parts: count, sum,
// min, max, "does there exist") can be answered by asking all  $O(\log n)$ 
// → of them.
```

```
template <class Static, class Item>
```

```
struct StaticToDynamic {
    vector<vector<Item>> buf; //
    // buf[k] is empty or has  $2^k$  items
    vector<Static> ds;
    void insert(Item x) {
        vector<Item> cur{x};
        int k = 0;
```

```
        while (k < (int)buf.size() && !buf[k].empty()) {
            for (auto& y : buf[k]) cur.push_back(y);
            buf[k].clear(), ds[k] = Static();
            k++;
        }
        while ((int)buf.size() <= k) buf.emplace_back(),
        // ds.emplace_back();
        // buf[k] = cur, ds[k] = Static(cur); // one
        // rebuild, size  $2^k$ 
    }
    template <class F> void each(F f) { //
        // f(instance) over every bucket
        for (int k = 0; k < (int)ds.size(); k++) if (!buf[k].empty())
        // f(ds[k]);
    }
};
// EXAMPLE static structure - a sorted array answering "how many items
// → <= x".
struct SortedBucket {
    vector<ll> v;
    SortedBucket() {}
    SortedBucket(vector<ll> a) : v(a) { sort(all(v)); }
    int count(ll x) { return upper_bound(all(v), x) - v.begin(); }
};
// USAGE: StaticToDynamic<SortedBucket, ll> d; d.insert(5);
// int c = 0; d.each([&](SortedBucket& b) { c += b.count(x); });
// DELETIONS, if the aggregate is INVERTIBLE (counts, sums): keep a
// → SECOND instance holding the
// deleted items and answer query(all) - query(deleted). This is
// → "deletion by counting" and it is
// how you get a fully dynamic convex-hull-trick or a deletable "count
// → <= x".
// DELETIONS otherwise (min, max, "exists"): you cannot subtract - use
// → offline dynamic connectivity
// style divide and conquer over the timeline (14) instead.
// WORST CASE: a single insertion can cost  $O(B(n))$  (rebuilding the
// → whole thing). If the problem has
// a per-query time limit rather than a total one, spread the rebuild
// → over the next  $2^k$  insertions
// (the "global rebuilding" / de-amortisation trick) - rarely needed in
// → contests.
```

2.26 TreeBinarization

TREE BINARIZATION - rewrite a tree so that EVERY vertex has degree ≤ 3 , by inserting dummy

```
// vertices joined with ZERO-weight edges.  $O(n)$  time, at most  $n - 2$  new
// → vertices, and every
// distance between original vertices is unchanged.
// WHEN: it is a PRECONDITION, not an algorithm. Reach for it whenever
// → a per-vertex cost is
// proportional to the DEGREE and you need it to be  $O(1)$ :
// - PERSISTENT centroid decomposition (27): copying a centroid node
// → must be  $O(1)$ , so a centroid
// may have at most 3 children in the centroid tree.
// - "merge the children's DP tables" small-to-large /
// → knapsack-on-tree DPs: after binarization
// each merge has exactly two operands, which is what makes the
// → classic  $O(n^2)$  tree knapsack
// analysis (and the  $O(n \log n)$  small-to-large bound) actually hold.
// - dynamic-tree structures whose node stores a fixed number of
// → child pointers (top trees).
// KEY INVARIANT: replace the star  $u \rightarrow c_1, c_2, \dots, c_k$  by a
// → CATERPILLAR: u keeps  $c_1$  and a chain of
// dummies  $d_1 - d_2 - \dots$  carries the rest, each dummy holding one real
// → child. Zero-weight edges mean
// distances are preserved exactly; original vertex ids are preserved,
// → so any answer indexed by
// vertex still makes sense. Depth grows by at most  $O(\deg)$ , so a DFS on
// → the result is still  $O(n)$ .
struct Binarize {
    int n, cnt; // 1..n
    // original,  $n+1..cnt$  dummies
    vector<vector<pii>> g; //
    // result: {neighbour, weight}
    Binarize(const vector<vector<pii>>& G, int root = 1)
        : n(G.size() - 1), cnt(G.size() - 1), g(2 * G.size()) { rec(G,
    // root, 0); }
    void link(int u, int v, int w) { g[u].push_back({v, w}),
    // g[v].push_back({u, w}); }
    void rec(const vector<vector<pii>>& G, int u, int p) {
        int last = u, k = 0, deg = G[u].size() - (p != 0);
        for (auto [v, w] : G[u]) {
            if (v == p) continue;
            if (++k == 1 || k == deg) link(last, v, w); //
            // first and last hang off 'last'
            else {
```

```

    int d = ++cnt; // dummy
    ↪ carrying one middle child
    link(last, d, 0), link(d, v, w), last = d;
}
for (auto [v, w] : G[u]) if (v != p) rec(G, v, u);
}
};
// SIZE EVERY ARRAY 2n - the vertex count can nearly double.
// WEIGHTS: dummy edges are 0, so shortest paths, diameters and
    ↪ centroid distances are identical.
// If your problem counts EDGES rather than weights, give the real
    ↪ edges weight 1 and the dummies 0.
// VERTEX VALUES: dummies must be neutral for whatever you aggregate (0
    ↪ for sums, +inf for minima).
// BINARY (degree ≤ 3) IS THE RIGHT TARGET, not "binary rooted tree":
    ↪ a rooted binarization that
// keeps exactly two children per node needs the same construction plus
    ↪ a root of degree ≤ 2.

```

2.27 PersistentCentroid

PERSISTENT CENTROID DECOMPOSITION - "sum of $\text{dist}(u, x)$ over all marked vertices x ", where the

```

// Needs: 26 - TreeBinarization.cpp
// set of marked vertices is versioned: version i = the first i marks.
    ↪ Insert  $O(\log n)$ , query
//  $O(\log n)$ , memory  $O(n \log n)$ .
// WHEN: queries of the form "consider only the marks with index in  $[l, r]$ , answer for vertex  $u$ " -
// subtract two versions (CF 757G). Also "the  $k$ -th mark", online, when
    ↪ marks arrive in a fixed
// order. If the mark set only ever grows and you never look back,
    ↪ plain centroid decomposition
// (09 - Trees/04) with the same accumulators is enough and far cheaper.
// COMPLEXITY REQUIRES BINARIZATION FIRST: a centroid node must have
    ↪  $O(1)$  children so that copying
// it is  $O(1)$ . Feed the tree through 26 - TreeBinarization.
// KEY INVARIANT: the centroid tree has depth  $O(\log n)$  and every vertex
    ↪  $u$  lies in exactly one
// component per level, so  $\text{dist}(u, x) = d[\text{level}][u] + d[\text{level}][x]$  where
    ↪ the level is that of the
// LOWEST common centroid of  $u$  and  $x$ . Each centroid node stores cnt
    ↪ (marks in its component), sum
// (total distance from the centroid to them), and par (their total
    ↪ distance to the centroid's
// PARENT centroid). Walking from the root of the centroid tree down to
    ↪  $u$  then adds, at each step,
// the contribution of exactly the marks that "branch off" there - the
    ↪ standard inclusion-exclusion
// that stops double counting. Marking one vertex touches only the
    ↪  $O(\log n)$  centroid nodes on that
// root-to- $u$  path, so persistence costs  $O(\log n)$  copies.

```

```

struct PCentroid {
    struct Node { array<int, 3> ct{{0, 0, 0}}; int id = 0, lvl = 0, cnt
    ↪ = 0; ll sum = 0, par = 0; };
    vector<Node> t;
    vector<vector<pii>> g; //
    ↪ BINARISED tree, 1-indexed
    vector<vector<ll>> d; //
    ↪ d[level][v]
    vector<int> st, en, sub, done;
    int n, DT = 0, tot = 0, root = 0;
    PCentroid(const vector<vector<pii>>& g, int nv, int LG = 20)
        : t(nv + 1), g(g), d(LG, vector<ll>(nv + 1, 0)), st(nv + 1),
    ↪ en(nv + 1), sub(nv + 1),
        done(nv + 1, 0), n(nv) { root = decompose(1, 0, 0); }
    void calcSz(int u, int p) {
        tot++, sub[u] = 1;
        for (auto [v, w] : g[u])
            if (v != p && !done[v]) calcSz(v, u), sub[u] += sub[v];
    }
    int findCen(int u, int p) {
        for (auto [v, w] : g[u])
            if (v != p && !done[v] && sub[v] > tot / 2) return
    ↪ findCen(v, u);
        return u;
    }
    void dist(int u, int p, ll cur, int l) {
        d[l][u] = cur;
        for (auto [v, w] : g[u])
            if (v != p && !done[v]) dist(v, u, cur + w, l);
    }
    int decompose(int u, int p, int l) {
        tot = 0, calcSz(u, p);
        int c = findCen(u, p);
        done[c] = 1, st[c] = ++DT, t[c].id = c, t[c].lvl = l;
    }
};

```

```

dist(c, 0, 0, l);
int k = 0;
for (auto [v, w] : g[c])
    if (!done[v]) t[c].ct[k++] = decompose(v, c, l + 1);
en[c] = DT;
return c;
}
bool insub(int x, int u) { return st[t[x].id] <= st[u] && en[u] <=
    ↪ en[t[x].id]; }
int mark(int cur, int u) { //
    ↪ returns the NEW version root
    t.push_back(t[cur]);
    int c = t.size() - 1;
    ll dv = d[t[c].lvl][u];
    t[c].cnt++, t[c].sum += dv;
    for (int k = 0; k < 3; k++) {
        int v = t[c].ct[k];
        if (v && insub(v, u)) {
            int nv = mark(v, u);
            t[nv].par += dv, t[c].ct[k] = nv;
        }
    }
    return c;
}
ll query(int cur, int u) { // sum
    ↪ of dist(u, x) over marks
    ll ans = 0;
    while (t[cur].id != u) {
        int v = 0;
        for (int k = 0; k < 3; k++)
            if (t[cur].ct[k] && insub(t[cur].ct[k], u)) v =
    ↪ t[cur].ct[k];
        ans += d[t[cur].lvl][u] * (t[cur].cnt - t[v].cnt) +
    ↪ t[cur].sum - t[v].par;
        cur = v;
    }
    return ans + t[cur].sum;
}
};
// USAGE: Binarize b(G); PCentroid pc(b.g, b.cnt); then ver[0] =
    ↪ pc.root and
// ver[i] = pc.mark(ver[i-1], x_i). The answer for marks  $l..r$  is
// pc.query(ver[r], u) - pc.query(ver[l-1], u).
// RESERVE t:  $n + (\text{marks}) * \log n$  nodes; each is ~40 bytes.
// OTHER ACCUMULATORS: replace cnt/sum by whatever is additive over the
    ↪ marks - count within a
// distance bound (keep a BIT per centroid instead, which kills
    ↪ persistence), sum of weights,
// number of pairs. Anything that needs a per-centroid ARRAY does not
    ↪ persist cheaply.
// DUMMY VERTICES from binarization are never marked and never queried,
    ↪ but they DO appear as
// centroids - which is fine, since query() walks the centroid tree by
    ↪ vertex id and only stops at
// the id it was asked for.

```

3 Graph Theory

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3.1 Graph Traversal

3.1.1 TopoSort

Topological Sort (Kahn's Algorithm) - $O(V + E)$

```

struct TopoSort {
    int n;
    vector<vector<int>> adj;
    vector<int> deg, order;

    TopoSort(int n = 0) : n(n), adj(n + 1), deg(n + 1, 0) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        deg[v]++;
    }

    bool solve() {
        queue<int> q;
        for (int i = 1; i <= n; i++) {
            if (!deg[i]) q.push(i);
        }
        while (!q.empty()) {
            int u = q.front();
            q.pop();

```

```

        order.pb(u);
        for (int v : adj[u]) {
            if (!--deg[v]) q.push(v);
        }
    }
    return (int)order.size() == n; // false if graph has cycles
};

```

3.1.2 BFS_DFS

BFS / DFS / 0-1 BFS / multi-source. 0-based. The bread and butter - keep them short.

```

// Unweighted shortest path + parent for reconstruction.
pair<vector<int>, vector<int>> bfs(int n, const vector<vector<int>>&
    ↪ adj, vector<int> src) {
    vector<int> d(n, -1), par(n, -1);
    queue<int> q;
    for (int s : src) d[s] = 0, q.push(s); // MULTI-SOURCE: pass
    ↪ every source at once
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int v : adj[u]) if (d[v] < 0) d[v] = d[u] + 1, par[v] = u,
    ↪ q.push(v);
    }
    return {d, par};
}

vector<int> pathTo(int t, const vector<int>& par) {
    vector<int> p;
    for (int v = t; v >= 0; v = par[v]) p.push_back(v);
    reverse(all(p));
    return p;
}

// 0-1 BFS: edge weights are only 0 or 1 -> deque instead of a heap,
    ↪ O(V + E).
// Use for "turning costs 1", "breaking one wall is free", "flip at
    ↪ most k edges".
vector<ll> bfs01(int n, const vector<vector<pair<int, int>>& adj, int
    ↪ s) {
    vector<ll> d(n, llinf);
    deque<int> q;
    d[s] = 0, q.push_back(s);
    while (!q.empty()) {
        int u = q.front(); q.pop_front();
        for (auto [v, w] : adj[u]) if (d[u] + w < d[v]) {
            d[v] = d[u] + w;
            w ? q.push_back(v) : q.push_front(v);
        }
    }
    return d;
}

// Iterative DFS with entry/exit times (avoids stack overflow at n =
    ↪ 2e5).
void dfsIter(int n, const vector<vector<int>>& adj, int root,
    ↪ vector<int>& tin, vector<int>& tout) {
    tin.assign(n, -1), tout.assign(n, -1);
    vector<int> it(n, 0), st = {root};
    int timer = 0;
    tin[root] = timer++;
    while (!st.empty()) {
        int u = st.back();
        if (it[u] < (int)adj[u].size()) {
            int v = adj[u][it[u]++];
            if (tin[v] < 0) tin[v] = timer++, st.push_back(v);
        } else tout[u] = timer++, st.pop_back();
    }
}

// GRID: 4- and 8-neighbour offsets (see 03 - grid.cpp). Encode a cell
    ↪ as r*c and reuse the
// graph routines above; add extra state dimensions (mask, parity,
    ↪ keys) by widening the index.
// CYCLE DETECTION, directed: colour DFS (0 white, 1 grey, 2 black); a
    ↪ grey->grey edge closes a
// cycle - walk the recursion stack back to extract it. Undirected: any
    ↪ edge to an already-visited
// non-parent vertex closes one.

```

3.2 Shortest Paths

3.2.1 Dijkstra

Dijkstra's Shortest Path Algorithm - $O((V + E) \log V)$

```

struct Dijkstra {
    int n;
    vector<vector<pair<int, ll>>> adj;
    vector<ll> dist;
    vector<int> par;

    Dijkstra(int n = 0) : n(n), adj(n + 1), dist(n + 1, llinf), par(n +
    ↪ 1, -1) {}

    void add_edge(int u, int v, ll w, bool directed = false) {
        adj[u].eb(v, w);
        if (!directed) adj[v].eb(u, w);
    }

    void solve(int src) {
        fill(all(dist), llinf);
        fill(all(par), -1);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
    ↪ pq;

        dist[src] = 0;
        pq.emplace(0, src);

        while (!pq.empty()) {
            auto [d, u] = pq.top();
            pq.pop();
            if (d > dist[u]) continue;

            for (auto& [v, w] : adj[u]) {
                if (dist[u] + w < dist[v]) {
                    dist[v] = dist[u] + w;
                    par[v] = u;
                    pq.emplace(dist[v], v);
                }
            }
        }
    }
};

```

3.2.2 FloydWarshall

Floyd-Warshall All-Pairs Shortest Path - $O(V^3)$

```

struct FloydWarshall {
    int n;
    vector<vector<ll>> dist;

    FloydWarshall(int n = 0) : n(n), dist(n + 1, vector<ll>(n + 1,
    ↪ llinf)) {}

    void add_edge(int u, int v, ll w, bool directed = false) {
        dist[u][v] = min(dist[u][v], w);
        if (!directed) dist[v][u] = min(dist[v][u], w);
    }

    void solve() {
        for (int k = 1; k <= n; k++) {
            for (int i = 1; i <= n; i++) {
                for (int j = 1; j <= n; j++) {
                    if (dist[i][k] < llinf && dist[k][j] < llinf) {
                        dist[i][j] = min(dist[i][j], dist[i][k] +
    ↪ dist[k][j]);
                    }
                }
            }
        }
    }
};

```

3.2.3 BellmanFord

If you paste both, keep only ONE copy of it.

```

// NOTE: 08 - Spanning Trees/01 - MST.cpp declares an identical 'struct
    ↪ Edge'.
// If you paste both, keep only ONE copy of it.
// Bellman-Ford Shortest Path with Negative Cycle Detection -  $O(V * E)$ 
struct Edge {
    int u, v;
    ll w;
};

struct BellmanFord {
    int n;
    vector<Edge> edges;
    vector<ll> dist;

```

```

vector<int> par;

BellmanFord(int n = 0) : n(n), dist(n + 1, llinf), par(n + 1, -1) {}

void add_edge(int u, int v, ll w) {
    edges.pb({u, v, w});
}

// Returns false if a negative cycle is reachable from src
bool solve(int src) {
    fill(all(dist), llinf);
    fill(all(par), -1);
    dist[src] = 0;

    for (int i = 0; i < n - 1; i++) {
        bool relaxed = false;
        for (const auto& e : edges) {
            if (dist[e.u] < llinf && dist[e.u] + e.w < dist[e.v]) {
                dist[e.v] = dist[e.u] + e.w;
                par[e.v] = e.u;
                relaxed = true;
            }
        }
        if (!relaxed) break;
    }

    for (const auto& e : edges) {
        if (dist[e.u] < llinf && dist[e.u] + e.w < dist[e.v]) {
            return false; // Negative cycle detected
        }
    }
    return true;
}
};

```

3.2.4 DominatorTree

```

/*
=====
CATALOGUE: Lengauer-Tarjan Dominator Tree (O((N + M) log N))
=====
DESCRIPTION:
Finds the "bottlenecks" in any directed graph. A node 'u' dominates
↳ 'v' if
EVERY path from the root to 'v' is forced to go through 'u'.

USAGE:
1. Build graph: Create a 2D vector 'adj' (1-based) containing the
↳ directed edges.
2. Initialize: DominatorTree dt(n, root, adj);
(Automatically builds the tree on instantiation)

EXTRACTING ANSWERS:
- dt.dom[u]: The Immediate Dominator of 'u' (the closest
↳ bottleneck to 'u').
- dt.id[u]: If dt.id[u] == 0 after building, 'u' is
↳ UNREACHABLE from the root.
=====
*/

```

```

struct DominatorTree {
    int T = 0, n;
    vector<vector<int>> rg, bucket, adj;
    vector<int> dsu, par, sdom, idom, dom, label, id, rev;

    DominatorTree(int _n, int r, vector<vector<int>>& adj):
        n(_n + 1), rg(n), bucket(n), adj(adj), dsu(n),
        par(n), sdom(n), idom(n), dom(n), label(n), id(n), rev(n) {
        dfs(r); build(r); }

    int find(int u, int x = 0) {
        if (u == dsu[u]) return x ? -1 : u;
        int v = find(dsu[u], x + 1);
        if (v < 0) return u;
        if (sdom[label[dsu[u]]] < sdom[label[u]]) label[u] =
        ↳ label[dsu[u]];
        dsu[u] = v;
        return x ? v : label[u];
    }

    void dfs(int u) {
        id[u] = ++T, rev[T] = u;
        label[T] = sdom[T] = dsu[T] = T;
        for (int& w : adj[u]) {
            if (!id[w]) {
                dfs(w);
                par[id[w]] = id[u];
            }
        }
    }
}

```

```

        rg[id[w]].push_back(id[u]);
    }

    void build(int r) {
        // FIXED: Loop MUST go down to i >= 1 to process the root's
        ↳ bucket!
        for (int i = T; i >= 1; i--) {
            for (int& u : rg[i]) sdom[i] = min(sdom[i], sdom[find(u)]);
            if (i > 1) bucket[sdom[i]].push_back(i);

            for (int& w : bucket[i]) {
                int v = find(w);
                idom[w] = sdom[v] == sdom[w] ? sdom[w] : v;
            }

            if (i > 1) dsu[i] = par[i];
        }

        for (int i = 2; i <= T; i++) {
            if (idom[i] != sdom[i]) idom[i] = idom[idom[i]];
        }

        for (int u = 1; u <= T; ++u) {
            dom[rev[u]] = rev[idom[u]];
        }
    }

    void testCase() {
        int n, m, s;
        cin >> n >> m >> s;
        s++; // Convert to 1-based indexing

        vector<vector<int>> adj(n + 1);
        for (int i = 0; i < m; i++) {
            int u, v;
            cin >> u >> v;
            adj[u + 1].push_back(v + 1);
        }

        // Automatically builds upon instantiation
        DominatorTree dt(n, s, adj);

        for (int i = 1; i <= n; i++) {
            if (i == s) {
                // pS is S
                cout << s - 1 << (i == n ? "" : " ");
            } else if (!dt.id[i]) {
                // If we can not reach i from S, print -1
                cout << -1 << (i == n ? "" : " ");
            } else {
                // Print the 0-based parent (which is now guaranteed to be
                ↳ calculated)
                cout << dt.dom[i] - 1 << (i == n ? "" : " ");
            }
        }
        cout << "\n";
    }
}

```

3.2.5 ShortestPathEdgeModify

SHORTEST PATH WITH ONE EDGE CHANGED, answered OFFLINE in O(1) per query after O((n+m) log n).

```

// Query: "if edge id had weight x instead, what is dist(s,t)?" - each
↳ query is independent.
// IDEA. Let ds/dt be the distance arrays from s and to t, P the
↳ shortest path, base = ds[t].
// * Edge NOT on P: the new path either ignores it (base) or crosses
↳ it once:
// min(base, ds[u] + x + dt[v], ds[v] + x + dt[u]).
// * Edge ON P at position i: either still use it (base - w + x), or
↳ avoid it entirely, and
// "the best s->t path avoiding path-edge i" is precomputed for
↳ every i at once:
// for a non-path edge (u,v,w), the path ds[u] -> u -> v -> dt[v]
↳ skips exactly the path edges
// in the window [L[u], R[v]], where L[x] = last path vertex on
↳ some shortest s->x path and
// R[x] = first path vertex on some shortest x->t path. Sweep i
↳ from 0 to k-1 with a heap.
struct SPEdgeModify {
    struct Edge { int u, v; ll w; };
    int n, k; // k =
    ↳ #edges on the shortest path
    ll base;
    vector<Edge> E;
    vector<ll> ds, dt, best; //
}

```

```

    best[i] = s->t avoiding path edge i
    vector<int> pos; //
    pos[e] = index on P, or -1
    SPEdgeModify(int n, vector<Edge> E, int s, int t) : n(n), E(E) {
        int m = E.size();
        vector<vector<pair<int, int>>> adj(n);
        for (int i = 0; i < m; i++)
            adj[E[i].u].push_back({E[i].v, i});
    }
    adj[E[i].v].push_back({E[i].u, i});
    vector<int> par(n, -1), pare(n, -1), ordS, ordT;
    auto dij = [&](int src, vector<ll>& d, vector<int>& ord, bool
    tree) {
        d.assign(n, llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
        greater<>> pq;
        d[src] = 0, pq.push({0, src});
        while (!pq.empty()) {
            auto [dd, u] = pq.top(); pq.pop();
            if (dd > d[u]) continue;
            ord.push_back(u);
            for (auto [v, id] : adj[u]) if (d[u] + E[id].w < d[v]) {
                d[v] = d[u] + E[id].w;
                if (tree) par[v] = u, pare[v] = id;
                pq.push({d[v], v});
            }
        }
        dij(s, ds, ordS, true), dij(t, dt, ordT, false);
        base = ds[t];
        vector<int> pv, pe; // the
        path, s ... t
        for (int c = t; c != s; c = par[c]) pv.push_back(c),
        pe.push_back(pare[c]);
        pv.push_back(s), reverse(all(pv)), reverse(all(pe));
        k = pe.size();
        pos.assign(m, -1);
        vector<char> onP(n, 0);
        for (int x : pv) onP[x] = 1;
        for (int i = 0; i < k; i++) pos[pe[i]] = i;
        vector<int> L(n, 0), R(n, k);
        for (int i = 0; i <= k; i++) L[pv[i]] = R[pv[i]] = i;
        for (int u : ordS) for (auto [v, id] : adj[u]) //
        relaxed in shortest-path order
            if (!onP[v] && ds[u] == ds[v] + E[id].w) L[v] = max(L[v],
        L[u]);
        for (int u : ordT) for (auto [v, id] : adj[u])
            if (!onP[v] && dt[u] == dt[v] + E[id].w) R[v] = min(R[v],
        R[u]);
        vector<vector<pair<ll, int>>> ev(max(k, 1)); //
        ev[start] = {cost, end}
        for (int i = 0; i < m; i++) {
            if (pos[i] >= 0) continue;
            auto [u, v, w] = E[i];
            if (L[u] < R[v]) ev[L[u]].push_back({ds[u] + w + dt[v],
        R[v]});
            if (L[v] < R[u]) ev[L[v]].push_back({ds[v] + w + dt[u],
        R[u]});
        }
        best.assign(max(k, 1), llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
        sw;
        for (int i = 0; i < k; i++) {
            for (auto& e : ev[i]) sw.push(e);
            while (!sw.empty() && sw.top().second <= i) sw.pop();
            if (!sw.empty()) best[i] = sw.top().first;
        }
        ll query(int id, ll x) const { // edge
            id gets weight x
            auto [u, v, w] = E[id];
            if (pos[id] < 0) return min({base, ds[u] + x + dt[v], ds[v] + x
            + dt[u]});
            return min(base - w + x, best[pos[id]]);
        }
    };
    // RELATED, same precomputation:
    // * "Does edge e lie on SOME shortest path?" <=> ds[u] + w + dt[v]
    // <= base (either direction).
    // * "Does e lie on EVERY shortest path?" <=> removing it
    // <= increases the distance; the sweep
    // above answers this for path edges (best[i] > base means path
    // edge i is essential).
    // * REPLACEMENT PATHS (delete one edge, ask the new distance) =
    // query(id, +infinity).
    // * SECOND SHORTEST WALK (may reuse edges): min over edges of ds[u]
    // + w + dt[v] strictly > base.
    // * If instead the whole graph changes per query, you need a fresh

```

↪ Dijkstra - this trick only
 // works because s, t and every other weight stay fixed.

3.2.6 CyclesAndConstraints

MINIMUM MEAN CYCLE, and SYSTEMS OF DIFFERENCE CONSTRAINTS - the two
 shortest-path tools that

// are not about shortest paths.

```

// --- KARP'S MINIMUM MEAN CYCLE, O(V*E). Returns the smallest possible
    (cycle weight)/(length).
// dp[k][v] = min weight of a walk of EXACTLY k edges ending at v (from
    a virtual source that
// reaches everything). Karp: the answer is min over v of max over
    k<n of (dp[n][v]-dp[k][v])/(n-k).
// Reach for it whenever the objective is an AVERAGE around a cycle:
    "minimum cost per unit time",
// "is there a cycle with negative average", profit-per-day loops.
ld minMeanCycle(int n, const vector<array<ll, 3>&& e) { // e =
    {u, v, w}, directed
    vector<vector<ll>> dp(n + 1, vector<ll>(n, llinf));
    fill(all(dp[0]), 0); //
    virtual source reaches every v
    for (int k = 1; k <= n; k++)
        for (auto [u, v, w] : e)
            if (dp[k - 1][u] < llinf) dp[k][v] = min(dp[k][v], dp[k -
        1][u] + w);
    ld best = 1e18;
    for (int v = 0; v < n; v++) {
        if (dp[n][v] == llinf) continue;
        ld cur = -1e18;
        for (int k = 0; k < n; k++)
            if (dp[k][v] < llinf) cur = max(cur, (ld)(dp[n][v] -
        dp[k][v]) / (n - k));
        best = min(best, cur);
    }
    return best; // 1e18
    // if the graph is acyclic
}
// MINIMUM COST-TO-TIME RATIO CYCLE (minimise sum(cost)/sum(time), time
    > 0): binary search x and
// test for a negative cycle with weights cost - x*time. Same skeleton,
    one Bellman-Ford per step.
// MAXIMUM mean cycle: negate the weights.

```

```

// --- SYSTEM OF DIFFERENCE CONSTRAINTS. Every constraint  $x_j - x_i \leq$ 
    w becomes an edge  $i \rightarrow j$  of
// weight w; add a super-source with a 0-edge to every vertex and run
    Bellman-Ford.
// FEASIBLE iff there is no negative cycle. The distances are then a
    solution, and they are the
// POINTWISE MAXIMUM one (every other solution is  $\leq$  it componentwise,
    after fixing  $x_{\text{source}} = 0$ ).
// TRANSLATIONS you will need:  $x_j - x_i \geq w$  is  $x_i - x_j \leq -w$ .
     $x_j - x_i = w$  is both.
//  $x_i \leq c$  is  $x_i - x_0 \leq c$  against the source. Strict " $<$ " on
    integers is " $\leq w - 1$ ".
// USES: scheduling with precedence and deadlines, "arrange these
    values so all the gaps hold",
// consistency of a set of relative measurements, and the dual of a
    min-cost-flow.

```

```

bool differenceConstraints(int n, const vector<array<ll, 3>&& c,
    vector<ll>& x) {
    vector<array<ll, 3>> e = c; // c =
    {i, j, w} meaning  $x_j - x_i \leq w$ 
    for (int v = 0; v < n; v++) e.push_back({(ll)n, (ll)v, 0}); //
    super-source n
    x.assign(n + 1, 0);
    for (int it = 0; it <= n; it++) {
        bool any = false;
        for (auto [u, v, w] : e)
            if (x[u] + w < x[v]) x[v] = x[u] + w, any = true;
        if (!any) { x.resize(n); return true; }
    }
    return false; //
    negative cycle: infeasible
}
// --- BELLMAN-FORD EXTRAS worth remembering
// NEGATIVE CYCLE RECOVERY: after n rounds, take a vertex that still
    relaxed, walk n parent
// pointers back (this lands you ON the cycle - the relaxed vertex may
    only be reachable from it),
// then follow parents until a vertex repeats.
// SPFA is Bellman-Ford with a queue; it is  $O(VE)$  worst case and
    killable, but the SLF heuristic
// (push to the front if the new distance beats the front) makes it
    fast in practice.
// ONLY SHORTEST PATHS WITH AT MOST K EDGES: run exactly k rounds,

```

↪ updating from a SNAPSHOT of the
 // previous round (otherwise one round can propagate along many edges).
 // JOHNSON'S APSP for sparse graphs with negative edges: Bellman-Ford
 ↪ from a super-source gives
 // potentials $h[]$, reweight $w'(u,v) = w + h[u] - h[v] \geq 0$, then run n
 ↪ Dijkstras. $O(VE + V E \log V)$.

3.2.7 SPFA

SPFA (Bellman-Ford with a queue) - shortest paths with NEGATIVE edges, plus negative-cycle

// detection. $O(V \cdot E)$ worst case and constructible-against, but usually
 ↪ far faster than plain
 // Bellman-Ford because only vertices whose distance actually improved
 ↪ get re-examined.
 // Returns false iff a negative cycle is reachable from s: a vertex
 ↪ relaxed n times must be on
 // (or reachable from) one, since a shortest walk uses at most n-1
 ↪ edges.
 // WHEN: negative weights and you also want the cycle check. With
 ↪ NON-negative weights always use
 // Dijkstra - SPFA is asymptotically worse and hackable. For a
 ↪ guaranteed bound use Bellman-Ford.
bool spfa(int n, int s, const vector<vector<pair<int, ll>>> g,
 ↪ vector<ll>& d) {
 d.assign(n, llinf);
 vector<int> cnt(n, 0);
 vector<char> inq(n, 0);
 deque<int> q;
 d[s] = 0, q.push_back(s), inq[s] = 1;
 while (!q.empty()) {
 int v = q.front(); q.pop_front();
 inq[v] = 0;
 for (auto [to, w] : g[v]) if (d[v] + w < d[to]) {
 d[to] = d[v] + w;
 if (!inq[to]) {
 if (++cnt[to] >= n) return false; //
 ↪ negative cycle
 inq[to] = 1;
 // SLF: a smaller tentative distance goes to the FRONT.
 ↪ Cheap, and it is what
 // makes SPFA fast in practice; without it the queue
 ↪ degenerates.
 if (!q.empty() && d[to] < d[q.front()])
 ↪ q.push_front(to);
 else q.push_back(to);
 }
 }
 return true;
 }
 // RECOVERING THE CYCLE: keep par[] and, on detection, walk n parent
 ↪ pointers back (that lands you
 // ON the cycle - the detected vertex may only be reachable from it),
 ↪ then follow parents until a
 // vertex repeats. See 06 - CyclesAndConstraints.cpp, which uses
 ↪ exactly this for difference
 // constraints and for the minimum mean cycle.
 // KILLER TESTS EXIST: grid-like graphs make SPFA quadratic. If the
 ↪ problem looks adversarial,
 // use Bellman-Ford (predictable $O(VE)$) or reweight with Johnson
 ↪ potentials and run Dijkstra.

3.2.8 SegmentTreeGraph

SEGMENT-TREE GRAPH - add "u -> every vertex in [l, r]" (or the reverse) with $O(\log n)$ edges

// instead of $O(n)$. Then run Dijkstra/BFS as usual. Total $O((n + q \log$
 ↪ $n) \log n)$.
 // Two trees: OUT (parent -> children, cost 0) to broadcast INTO a
 ↪ range,
 // IN (children -> parent, cost 0) to collect FROM a range.
 // Leaves of both trees are the original vertices.
struct SegGraph {
 int n, cnt;
 vector<int> outId, inId; // node ids of
 ↪ the two trees
 vector<vector<pair<int, ll>>> g;

 SegGraph(int n) : n(n) {
 cnt = n;
 outId.assign(4 * n, -1), inId.assign(4 * n, -1);
 g.assign(n + 8 * n, {});
 buildOut(1, 0, n - 1), buildIn(1, 0, n - 1);
 }
 int node() { return cnt++; }
 void buildOut(int v, int l, int r) {
 if (l == r) { outId[v] = l; return; }

outId[v] = node();
 int m = (l + r) / 2;
 buildOut(2 * v, l, m), buildOut(2 * v + 1, m + 1, r);
 g[outId[v]].push_back({outId[2 * v], 0});
 g[outId[v]].push_back({outId[2 * v + 1], 0});
 }
 void buildIn(int v, int l, int r) {
 if (l == r) { inId[v] = l; return; }
 inId[v] = node();
 int m = (l + r) / 2;
 buildIn(2 * v, l, m), buildIn(2 * v + 1, m + 1, r);
 g[inId[2 * v]].push_back({inId[v], 0});
 g[inId[2 * v + 1]].push_back({inId[v], 0});
 }
 void addRange(int v, int l, int r, int ql, int qr, int u, ll w,
 ↪ bool toRange) {
 if (qr < l || r < ql) return;
 if (ql <= l && r <= qr) {
 if (toRange) g[u].push_back({outId[v], w}); // u
 ↪ -> everything in [ql,qr]
 else g[inId[v]].push_back({u, w}); //
 ↪ everything in [ql,qr] -> u
 return;
 }
 int m = (l + r) / 2;
 addRange(2 * v, l, m, ql, qr, u, w, toRange);
 addRange(2 * v + 1, m + 1, r, ql, qr, u, w, toRange);
 }
 void edgeToRange(int u, int l, int r, ll w) { addRange(1, 0, n - 1,
 ↪ l, r, u, w, true); }
 void edgeFromRange(int l, int r, int u, ll w) { addRange(1, 0, n -
 ↪ 1, l, r, u, w, false); }
 void edge(int u, int v, ll w) { g[u].push_back({v, w}); }

 vector<ll> dijkstra(int s) {
 vector<ll> d(cnt, llinf);
 priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
 ↪ pq;
 d[s] = 0, pq.push({0, s});
 while (!pq.empty()) {
 auto [dd, u] = pq.top(); pq.pop();
 if (dd > d[u]) continue;
 for (auto [v, w] : g[u]) if (dd + w < d[v]) d[v] = dd + w,
 ↪ pq.push({d[v], v});
 }
 d.resize(n); // only the
 ↪ original vertices matter
 return d;
 }
 }
 // The same construction gives 2-SAT range clauses and "range -> range"
 ↪ edges (route both
 // through one fresh auxiliary node).

3.2.9 KShortestWalks

EPSTEIN'S ALGORITHM - the k SHORTEST s-t WALKS (repeated vertices and edges allowed), in

// $O(m \log m + k \log k)$. Directed, 0-indexed, non-negative weights.
 ↪ Returns the k lengths in
 // non-decreasing order, padding with -1 when fewer than k walks exist.
 // Reach for it for "the k cheapest routes", "the k-th smallest path
 ↪ length", "is there a path of
 // length exactly L" (enumerate until you pass L), and for problems
 ↪ that sum over the k best plans.
 // THE IDEA. Fix the shortest path TREE towards t, with $d[v] = \text{dist}(v,$
 ↪ t). Any walk is described by
 // the SIDETRACK edges it takes - the edges it uses that are not tree
 ↪ edges - and its length is
 // $d[s] + (\text{sum of the sidetracks' reduced costs } w + d[\text{to}] - d[\text{from}])$,
 ↪ each of which is ≥ 0 . So the
 // k shortest walks = the k cheapest ANTICHAINS in a heap-ordered
 ↪ structure, and a best-first search
 // over that structure emits them one at a time.
 // H[v] is a PERSISTENT leftist heap holding the sidetracks available
 ↪ at v and at all its ancestors
 // in the tree; persistence is what makes "inherit the parent's heap"
 ↪ $O(\log n)$ instead of $O(n)$.
struct KShortestWalks {
 struct Node { Node *l = 0, *r = 0; int sz = 1, to; ll val; };
 Node* meld(Node* a, Node* b) {
 if (!a) return b;
 if (!b) return a;
 if (a->val > b->val) swap(a, b);
 Node* c = new Node(*a);
 ↪ copy-on-write: keeps old versions
 c->r = meld(c->r, b);
 if (!c->l || c->l->sz < c->r->sz) swap(c->l, c->r);

```

c->sz = (c->r ? c->r->sz : 0) + 1;
return c;
}
Node* push(Node* h, ll val, int to) {
    Node* c = new Node();
    c->val = val, c->to = to;
    return meld(h, c);
}
// e = {u, v, w} directed arcs. Returns the k shortest s -> t walk
// lengths.
vector<ll> solve(int n, const vector<array<ll, 3>>& e, int s, int
    t, int k) {
    vector<vector<array<ll, 3>>> fwd(n), rev(n); //
    // {other endpoint, weight, edge id}
    for (int i = 0; i < (int)e.size(); i++) {
        fwd[e[i][0]].push_back({e[i][1], e[i][2], (ll)i});
        rev[e[i][1]].push_back({e[i][0], e[i][2], (ll)i});
    }
    vector<ll> d(n, llinf);
    vector<int> par(n, -1), tid(n, -1); //
    // tid[v] = v's tree edge id
    priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
    > pq;
    d[t] = 0, pq.push({0, t});
    while (!pq.empty()) { //
        // Dijkstra on the REVERSED graph
        auto [dd, u] = pq.top(); pq.pop();
        if (dd > d[u]) continue;
        for (auto [v, w, id] : rev[u]) if (dd + w < d[v])
            d[v] = dd + w, par[v] = u, tid[v] = id, pq.push({d[v],
            (int)v});
        vector<ll> ans;
        if (d[s] == llinf) return vector<ll>(k, -1);
        vector<vector<int>> kids(n);
        for (int v = 0; v < n; v++) if (par[v] >= 0)
            kids[par[v]].push_back(v);
        vector<Node*> H(n, nullptr);
        vector<int> bfs{t}; // build
        // H down the tree from t
        for (int qi = 0; qi < (int)bfs.size(); qi++) {
            int u = bfs[qi];
            if (par[u] >= 0) H[u] = meld(H[u], H[par[u]]);
            for (auto [v, w, id] : fwd[u])
                if ((int)id != tid[u] && d[v] < llinf) H[u] =
                push(H[u], w + d[v] - d[u], (int)v);
            for (int v : kids[u]) bfs.push_back(v);
        }
        auto cmp = [] (const pair<ll, Node*> & a, const pair<ll, Node*> &
        b) {
            return a.first > b.first;
        };
        priority_queue<pair<ll, Node*>, vector<pair<ll, Node*>>,
        decltype(cmp)> Q(cmp);
        Node* root = new Node();
        root->val = d[s], root->to = s;
        Q.push({d[s], root});
        while (!Q.empty() && (int)ans.size() < k) {
            auto [w, cur] = Q.top(); Q.pop();
            ans.push_back(w);
            if (cur->l) Q.push({w + cur->l->val - cur->val, cur->l});
            // swap the last sidetrack
            if (cur->r) Q.push({w + cur->r->val - cur->val, cur->r});
            if (H[cur->to]) Q.push({w + H[cur->to]->val, H[cur->to]});
            // append a new sidetrack
        }
        while ((int)ans.size() < k) ans.push_back(-1);
        return ans;
    }
};
// WALKS, NOT PATHS. The second-shortest walk may repeat a vertex. If
// the problem insists on SIMPLE
// paths you need YEN'S ALGORITHM: keep the best path, then for each
// prefix of it (the "spur"), ban
// the edges that would recreate an already-found path, run one
// Dijkstra from the spur vertex, and
// keep the best candidate. O(k * n * (m + n log n)) - much slower, but
// it is the only correct
// answer for "k shortest SIMPLE paths". On a DAG walks and paths
// coincide, so use this file.
// K-TH SHORTEST PATH WITH A COUNT LIMIT PER VERTEX: a plain Dijkstra
// that pops each vertex up to k
// times is O(k m log) and far simpler - reach for that first if k is
// small (<= a few hundred).
// SECOND SHORTEST PATH ONLY: just take ans[1] here, or do the
// two-array trick (best and
// second-best distance per vertex) in one Dijkstra.

```

3.3 Graph Connectivity

3.3.1 LowLink

LowLink: BRIDGES + ARTICULATION POINTS + 2-EDGE-CONNECTED COMPONENTS - $O(V + E)$

```

// Skips the parent by EDGE ID, so MULTI-EDGES are handled correctly
// (a doubled edge is never a bridge). 0-based vertices.
// comp[v] = id of v's 2-edge-connected component; contracting those
// gives the BRIDGE TREE,
// on which "number of bridges between u and v" is just a tree path
// length.
struct LowLink {
    int n, timer = 0, nc = 0;
    vector<vector<pair<int, int>>> adj; // {to, edge id}
    vector<int> tin, low, comp;
    vector<bool> isCut;
    vector<pair<int, int>> bridges;
    vector<int> st;

    LowLink(int n = 0) : n(n), adj(n), tin(n, -1), low(n), comp(n, -1),
    isCut(n, false) {}
    void add_edge(int u, int v, int id) { adj[u].push_back({v, id}),
    adj[v].push_back({u, id}); }

    void dfs(int u, int pe) {
        tin[u] = low[u] = timer++;
        st.push_back(u);
        int kids = 0;
        for (auto [v, id] : adj[u]) {
            if (id == pe) continue; // skip the
            // edge we came in on, not the vertex
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {
                dfs(v, id), kids++;
                low[u] = min(low[u], low[v]);
                if (low[v] > tin[u]) bridges.push_back({u, v});
                if (low[v] >= tin[u] && pe != -1) isCut[u] = true;
            }
        }
        if (pe == -1 && kids > 1) isCut[u] = true;
        if (low[u] == tin[u]) { // u is the top
            // of a 2-edge-connected comp
            while (true) {
                int v = st.back(); st.pop_back();
                comp[v] = nc;
                if (v == u) break;
            }
            nc++;
        }
    }
    void solve() { for (int i = 0; i < n; i++) if (tin[i] == -1) dfs(i,
    -1); }

    // Bridge tree: nc nodes, one edge per bridge.
    vector<vector<int>> bridgeTree() {
        vector<vector<int>> t(nc);
        for (auto [u, v] : bridges) t[comp[u]].push_back(comp[v]),
        t[comp[v]].push_back(comp[u]);
        return t;
    }
};

```

3.3.2 SCC

Strongly Connected Components (Tarjan's Algorithm) - $O(V + E)$

```

// IMPORTANT: component ids come out in REVERSE topological order - for
// every edge u->v with
// comp[u] != comp[v] we have comp[u] > comp[v]. Iterate ids DOWNWARD
// to walk the condensation
// forwards. build_dag() may emit duplicate arcs (fine for
// reachability, wrong for counting).
struct SCC {
    int n, timer = 0, num_comps = 0;
    vector<vector<int>> adj;
    vector<int> tin, low, comp;
    vector<bool> in_st;
    stack<int> st;
    vector<vector<int>> comps;

    SCC(int n = 0) : n(n), adj(n + 1), tin(n + 1, 0), low(n + 1, 0),
    comp(n + 1, -1), in_st(n + 1, false) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
    }

    void dfs(int u) {

```

```

tin[u] = low[u] = ++timer;
st.push(u);
in_st[u] = true;

for (int v : adj[u]) {
    if (!tin[v]) {
        dfs(v);
        low[u] = min(low[u], low[v]);
    } else if (in_st[v]) {
        low[u] = min(low[u], tin[v]);
    }
}

if (low[u] == tin[u]) {
    comps.pb({});
    while (true) {
        int v = st.top();
        st.pop();
        in_st[v] = false;
        comp[v] = num_comps;
        comps.back().pb(v);
        if (u == v) break;
    }
    num_comps++;
}

}

void solve() {
    for (int i = 1; i <= n; i++) {
        if (!tin[i]) dfs(i);
    }
}

vector<vector<int>> build_dag() {
    vector<vector<int>> dag(num_comps);
    for (int u = 1; u <= n; u++) {
        for (int v : adj[u]) {
            if (comp[u] != comp[v]) dag[comp[u]].pb(comp[v]);
        }
    }
    return dag;
}
};

```

3.3.3 BlockCutTree

BLOCK-CUT TREE (vertex biconnected components). $O(V + E)$.

```

// Builds a forest whose nodes are: the original CUT VERTICES, plus one
// node per BICONNECTED
// COMPONENT ("block"). Every cut vertex is joined to each block
// containing it.
// Answers "is there a path u -> v avoiding vertex w", "which blocks
// does a path cross",
// "articulation structure of the graph" - all become tree queries.
struct BlockCut {
    int n, timer = 0;
    vector<vector<int>> adj, comps, tree; // comps[i] =
// vertices of block i
    vector<int> tin, low, stk, id; // id[v] = tree
// node of original vertex v
    vector<bool> isCut;

    BlockCut(int n = 0) : n(n), adj(n), tin(n, -1), low(n), id(n, -1),
// isCut(n, false) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
// adj[v].push_back(u); }

    void dfs(int u, int p) {
        tin[u] = low[u] = timer++;
        stk.push_back(u);
        int kids = 0;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; } // skip ONE
// parent edge (multi-edge safe)
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {
                dfs(v, u), kids++;
                low[u] = min(low[u], low[v]);
                if (low[v] >= tin[u]) { // u closes a
// block
                    isCut[u] = (tin[u] > 0 || kids > 1);
                    comps.push_back({u});
                    while (true) {
                        int w = stk.back(); stk.pop_back();
                        comps.back().push_back(w);
                        if (w == v) break;
                    }
                }
            }
        }
    }
};

```

```

}
}

void build() {
    for (int v = 0; v < n; v++) if (adj[v].empty())
// comps.push_back({v}); // isolated vertex
    for (int i = 0; i < n; i++) if (tin[i] == -1) { timer = 0;
// dfs(i, -1); stk.clear(); }
    tree.assign(comps.size(), {});
    for (int v = 0; v < n; v++) if (isCut[v]) id[v] = tree.size(),
// tree.push_back({});
    for (int i = 0; i < (int)comps.size(); i++)
        for (int v : comps[i]) {
            if (!isCut[v]) id[v] = i;
            else tree[i].push_back(id[v]), tree[id[v]].push_back(i);
        }
}

};
// A vertex that is NOT a cut vertex lies in exactly one block -> id[v]
// is that block.
// An edge (u,v) always lies in exactly one block. Leaves of the tree
// are blocks; a graph is
// biconnected iff the tree has a single node.

```

3.3.4 TriangleAndCycleCounting

COUNTING SMALL SUBGRAPHS in $O(m \sqrt{m})$ - the "degree ordering" trick. Orient every edge from

```

// the lower (degree, index) endpoint to the higher one. In that DAG
// every out-degree is  $O(\sqrt{m})$ ,
// because a vertex with out-degree d has d neighbours of degree  $\geq d$ ,
// so  $d^2 \leq 2m$ .
// Undirected simple graph, 0-indexed, edges given as pairs.

// --- TRIANGLES: enumerate each once.  $O(m \sqrt{m})$ . ---
ll countTriangles(int n, const vector<pair<int, int>>& es) {
    vector<int> deg(n, 0);
    for (auto [u, v] : es) deg[u]++, deg[v]++;
    auto better = [&](int a, int b) { return deg[a] != deg[b] ? deg[a]
// < deg[b] : a < b; };
    vector<vector<int>> out(n);
    for (auto [u, v] : es) better(u, v) ? out[u].push_back(v) :
// out[v].push_back(u);
    vector<char> mark(n, 0);
    ll cnt = 0;
    for (int u = 0; u < n; u++) {
        for (int v : out[u]) mark[v] = 1;
        for (int v : out[u]) for (int w : out[v]) cnt += mark[w];
        for (int v : out[u]) mark[v] = 0;
    }
    return cnt;
// triangle counted exactly once

// LISTING them: inside the inner loop, (u,v,w) with mark[w] IS a
// triangle.
// PER-VERTEX / PER-EDGE triangle counts: add 1 to each of u,v,w (or to
// each of the three edges)
// at that moment - this is what "local clustering coefficient" and
// many CF problems want.

// --- 4-CYCLES:  $O(m \sqrt{m})$  with the same orientation. ---
// For each u, walk two steps (u -> v -> w) in the oriented graph and
// count how many times each w
// is reached; a pair of distinct paths u->w gives a 4-cycle. Sum
//  $C(cnt[w], 2)$ .
ll countFourCycles(int n, const vector<pair<int, int>>& es) {
    vector<int> deg(n, 0);
    for (auto [u, v] : es) deg[u]++, deg[v]++;
    auto better = [&](int a, int b) { return deg[a] != deg[b] ? deg[a]
// < deg[b] : a < b; };
    vector<vector<int>> adj(n), out(n);
    for (auto [u, v] : es) {
        adj[u].push_back(v), adj[v].push_back(u);
        better(u, v) ? out[u].push_back(v) : out[v].push_back(u);
    }
    vector<ll> cnt(n, 0);
    ll r = 0;
    for (int u = 0; u < n; u++) {
        for (int v : adj[u]) for (int w : out[v]) if (better(u, w)) r
// += cnt[w]++;
        for (int v : adj[u]) for (int w : out[v]) if (better(u, w))
// cnt[w] = 0;
    }
    return r;
}

// WHY IT IS  $O(m \sqrt{m})$ : the inner loop runs sum over v of  $deg(v) *
// outdeg(v) \leq O(m \sqrt{m})$ .
//

```

```
// OTHER COUNTING FACTS WORTH HAVING:
// * #triangles = trace(A^3) / 6 for the adjacency matrix A; with
//   ↳ bitsets that is  $O(n^3 / 64)$  and
//   ↳ is often the shorter answer when  $n \leq 2000$  (also gives per-pair
//     ↳ common-neighbour counts).
// * #paths of length 2 = sum over  $v$  of  $C(\deg v, 2)$ . Subtract  $3 * \#triangles$  for "paths that are
//   ↳ not part of a triangle".
// * A graph is TRIANGLE-FREE  $\Rightarrow m \leq n^2/4$  (Mantel / Turan for
//   ↳  $K_3$ ). More generally Turan:
//   ↳ a  $K_{r+1}$ -free graph has at most  $(1 - 1/r) n^2 / 2$  edges.
// * #closed walks of length  $k = \text{trace}(A^k)$  - use matrix power for  $k$ 
//   ↳ up to  $1e18$  on small  $n$ .
// * COUNTING CYCLES OF LENGTH  $\leq 5$  is polynomial by similar tricks;
//   ↳ length  $\geq 6$  is as hard as
//   ↳ matrix multiplication in general. For counting ALL simple cycles
//     ↳ use inclusion-exclusion
//   ↳ over subsets ( $2^n$ ) or DP over bitmask start-vertex.
// * GIRTH (shortest cycle) of an unweighted graph: BFS from every
//   ↳ vertex,  $O(nm)$ . For the
//   ↳ shortest cycle THROUGH a given edge, delete it and run BFS.
```

3.3.5 ThreeEdgeConnectivity

THREE-EDGE-CONNECTED COMPONENTS in ONE DFS, $O(n + m)$. Undirected, 0-indexed, may be

```
// disconnected; self loops and PARALLEL EDGES are handled (parallel
//   ↳ edges matter here - two
//   ↳ vertices joined by three parallel edges are 3-edge-connected).
// u and v land in the same component iff no set of at most TWO edges
//   ↳ separates them, i.e. iff
// there are 3 edge-disjoint u-v paths (Menger). This is the next step
//   ↳ after LowLink (01), which
// gives the 2-edge-connected components (= no single edge separates
//   ↳ them).
// Reach for it on "the network survives any two cable failures",
//   ↳ "contract every pair of parallel
//   ↳ bridges", and as the preprocessing for 3-edge-connectivity
//   ↳ augmentation problems.
// Tsin's algorithm: a DFS that maintains, for each vertex, a PATH of
//   ↳ still-undecided descendants;
// whenever a back edge proves two of them 3-edge-connected the path is
//   ↳ absorbed into one class.
// The classes live in a circular linked list, so "merge" is a single
//   ↳ swap.
vector<vector<int>> threeEdgeCC(const vector<vector<int>>& g) {
    int n = g.size(), cnt = 0;
    vector<int> in(n), out(n), low(n, n), deg(n, 0), path(n, n),
    ↳ nxt(n), it(n, 0), stk;
    vector<char> vis(n, 0);
    iota(all(nxt), 0);
    ↳ circular list = the components
    auto absorb = [&](int v, int w) { swap(nxt[v], nxt[w]), deg[v] +=
    ↳ deg[w]; };
    for (int s = 0; s < n; s++) {
        ↳ iterative DFS over the forest
        if (vis[s]) continue;
        stk = {s}, vis[s] = 1, in[s] = cnt++;
        vector<int> par(1, n), pe(1, -1);
        ↳ "parent edge already used"
        while (!stk.empty()) {
            int v = stk.back(), p = par.back();
            if (it[v] < (int)g[v].size()) {
                int w = g[v][it[v]++];
                if (w == v) continue;
                if (w == p && !pe.back()) { pe.back() = 1; continue; }
                ↳ // skip ONE parent edge
                if (vis[w]) {
                    if (in[w] < in[v]) deg[v]++, low[v] = min(low[v],
                    ↳ in[w]);
                } else {
                    ↳ edge to a descendant
                    deg[v]--;
                    int u = path[v];
                    while (u != n && in[u] <= in[w] && in[w] <
                    ↳ out[u])
                        absorb(v, u), u = path[u];
                    path[v] = u;
                }
                continue;
            }
            vis[w] = 1, in[w] = cnt++, stk.push_back(w),
            ↳ par.push_back(v), pe.push_back(0);
        } else {
            ↳ leaving v: fold it into its parent
            out[v] = cnt, stk.pop_back(), par.pop_back(),
            ↳ pe.pop_back();
            if (p == n) continue;
```

```
int w = v;
if (path[w] == n && deg[w] <= 1) { deg[p] += deg[w],
↳ low[p] = min(low[p], low[w]); }
else {
    if (deg[w] == 0) w = path[w];
    if (low[w] < low[p]) low[p] = low[w], swap(w,
    ↳ path[p]);
    while (w != n) absorb(p, w), w = path[w];
}
}
}
vector<vector<int>> comp;
↳ each circular list once
vector<char> seen(n, 0);
for (int i = 0; i < n; i++) if (!seen[i]) {
    comp.push_back({});
    for (int j = i; !seen[j]; j = nxt[j]) seen[j] = 1,
    ↳ comp.back().push_back(j);
}
return comp;
}
// CONTRACTING each component gives a graph in which every 3-edge cut
↳ is visible; the 2-edge-cuts
// are exactly the bridges plus the "cactus" pairs from LowLink.
// 3-VERTEX-CONNECTIVITY (SPQR / triconnected components) is a
↳ genuinely harder, ~400-line
// algorithm. If you only need to test 3-vertex-connectivity for small
↳ n, delete every pair of
// vertices and check connectivity in  $O(n^2 (n + m))$ .
```

3.3.6 OnlineBridges

ONLINE (INCREMENTAL) BRIDGES - maintain the number of bridges while edges are ADDED, in

```
//  $O(\alpha)$  amortised per edge. Undirected, 0-indexed, edges may repeat
↳ (a doubled edge kills the
// bridge, which is exactly right).
// LowLink (01) recomputes bridges from scratch in  $O(n + m)$ ; use this
↳ when the graph grows and you
// must answer after every insertion: "after each new road, how many
↳ roads are still critical",
// "the first moment u and v are 2-edge-connected".
// STRUCTURE. Contract every 2-edge-connected component to a node; what
↳ is left is the BRIDGE
// FOREST, kept as a rooted forest with parent pointers. Adding (u, v):
// * same 2ecc  $\rightarrow$  nothing changes;
// * different trees  $\rightarrow$  a new bridge; re-root the smaller
↳ tree and hang it under v;
// * same tree, different 2ecc  $\rightarrow$  the tree path u..v becomes a cycle:
↳ contract it, losing
// (path length) bridges.
// Two DSUs: dsu2ecc = the contracted components, dsuCC = which tree of
↳ the forest you are in.
// DELETIONS are NOT supported - that needs offline divide and conquer
↳ or a link-cut tree.
struct OnlineBridges {
    vector<int> par, e2, cc, ccSize, stamp;
    ↳ is in the BRIDGE FOREST
    int bridges = 0, iter = 0;
    OnlineBridges(int n) : par(n, -1), e2(n), cc(n), ccSize(n, 1),
    ↳ stamp(n, 0) {
        iota(all(e2), 0), iota(all(cc), 0);
    }
    int find2(int u) { return u < 0 || e2[u] == u ? u : e2[u] =
    ↳ find2(e2[u]); }
    int findCC(int u) { u = find2(u); return cc[u] == u ? u : cc[u] =
    ↳ findCC(cc[u]); }
    void makeRoot(int u) {
        ↳ reverse the parent chain above u
        u = find2(u);
        int root = u, child = -1;
        while (u != -1) { int p = find2(par[u]); par[u] = child, cc[u]
        ↳ = root, child = u, u = p; }
        ccSize[root] = ccSize[child];
    }
    void mergePath(int u, int v) {
        ↳ contract the u..v tree path
        ++iter;
        vector<int> pu, pv;
        int lca = -1;
        while (lca < 0) {
            if (u != -1) {
                u = find2(u), pu.push_back(u);
                if (stamp[u] == iter) { lca = u; break; }
                stamp[u] = iter, u = par[u];
            }
            if (v != -1) {
```

```

    v = find2(v), pv.push_back(v);
    if (stamp[v] == iter) { lca = v; break; }
    stamp[v] = iter, v = par[v];
}
for (auto& p : {pu, pv})
    for (int x : p) { if (x == lca) break; e2[x] = lca,
    ↪ bridges--; }
}
void addEdge(int u, int v) {
    u = find2(u), v = find2(v);
    if (u == v) return; //
    ↪ already 2-edge-connected
    int a = findCC(u), b = findCC(v);
    if (a != b) {
        bridges++;
        if (ccSize[a] > ccSize[b]) swap(u, v), swap(a, b); //
    ↪ re-root the SMALLER tree
        makeRoot(u);
        par[u] = cc[u] = v, ccSize[b] += ccSize[u];
    } else mergePath(u, v);
}
bool sameComponent(int u, int v) { return find2(u) == find2(v); }
    ↪ // 2-edge-connected?
};
// WHY IT IS AMORTISED O(alpha): makeRoot walks the smaller tree, which
    ↪ is the union-by-size
// argument (O(log n) total per vertex), and mergePath destroys one
    ↪ bridge per step.
// ONLINE ARTICULATION POINTS / biconnected components can be
    ↪ maintained the same way, but the
// bookkeeping is much heavier - use offline divide and conquer over
    ↪ time instead.
// OFFLINE DELETIONS: process the queries in reverse so deletions
    ↪ become insertions, or run
// divide-and-conquer on the time axis with a rollback DSU (see 02 -
    ↪ DS/05 - DSU).

```

3.3.7 STNumbering

ST-NUMBERING (bipolar orientation) - order the vertices $v_1 = s, \dots, v_n = t$ so that EVERY

```

// other vertex has at least one neighbour BEFORE it and one AFTER it.
    ↪  $O(n + m)$ , 0-indexed,
// undirected. Returns the order, or an EMPTY vector when none exists.
// EXISTS iff  $G + \text{edge}(s, t)$  is 2-VERTEX-CONNECTED (that is exactly
    ↪ what the first DFS checks).
// Why you want it: orienting every edge from the smaller number to the
    ↪ larger gives a DAG with a
// UNIQUE source  $s$  and a UNIQUE sink  $t$ , and every vertex lies on some
    ↪  $s$ - $t$  path. That is the tool
// for "orient the edges so the graph stays connected / has one source
    ↪ and one sink", for planarity
// testing and st-planar embeddings, and for "peel the graph one vertex
    ↪ at a time keeping both
// halves connected" (any prefix AND its complement induce connected
    ↪ subgraphs).
// Even-Tarjan: DFS from  $t$ , then insert the vertices into a list in
    ↪ preorder, each one just before
// or just after its parent according to a sign flipped at every
    ↪ insertion.
vector<int> stNumbering(int n, vector<vector<int>> adj, int s, int t) {
    if (n == 1) return {0};
    bool has = false; // add
    ↪ the s-t edge if missing
    for (int v : adj[s]) has |= (v == t);
    if (!has) adj[s].push_back(t), adj[t].push_back(s);
    vector<int> dis(n, 0), low(n, 0), par(n, -1), sg(n + 2, 0), pre;
    int T = 0;
    bool bicon = true;
    function<void(int, int)> dfs1 = [&](int u, int p) { //
    ↪ articulation-point check
        low[u] = dis[u] = ++T;
        int kids = 0;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; } // skip
    ↪ ONE parent edge
            if (!dis[v]) {
                kids++, dfs1(v, u), low[u] = min(low[u], low[v]);
                if (par[u] != -2 && low[v] >= dis[u]) bicon = false;
    ↪ // par[u] == -2 marks the root
            } else low[u] = min(low[u], dis[v]);
        }
        if (par[u] == -2 && kids > 1) bicon = false;
    };
    par[t] = -2, dfs1(t, -1);
    for (int i = 0; i < n; i++) if (!dis[i]) bicon = false;
    if (!bicon) return {};
}

```

```

fill(all(dis), 0), fill(all(low), 0), fill(all(par), -1), T = 0;
function<void(int, int)> dfs2 = [&](int u, int p) { //
    ↪ tree rooted at t; s is fixed
        low[u] = dis[u] = ++T;
        for (int v : adj[u]) {
            if (v == p) { p = -1; continue; }
            if (!dis[v]) pre.push_back(v), par[v] = u, dfs2(v, u),
    ↪ low[u] = min(low[u], low[v]);
            else low[u] = min(low[u], dis[v]);
        }
    };
    dis[s] = low[s] = ++T, sg[dis[s]] = -1; // s
    ↪ takes number 1, never entered
    dfs2(t, -1);
    list<int> L{s, t};
    vector<list<int>::iterator> it(n + 2);
    it[dis[s]] = L.begin(), it[dis[t]] = next(L.begin());
    for (int v : pre) { //
    ↪ insert before/after the parent
        if (sg[low[v]] == -1) it[dis[v]] = L.insert(it[dis[par[v]]], v);
        else it[dis[v]] = L.insert(next(it[dis[par[v]]]), v);
        sg[dis[par[v]]] = -sg[low[v]];
    }
    return vector<int>(all(L)); //
    ↪ order[0] = s, order[n-1] = t
}
// THE DAG. With pos[order[i]] = i, orient every edge from the smaller
    ↪ pos to the larger. Then
// every vertex is on an s-t path, so "assign directions keeping
    ↪ everything reachable from s and
    ↪ able to reach t" is solved. Reversing all edges swaps the roles of s
    ↪ and t.
// EAR DECOMPOSITION is the other standard certificate of
    ↪ 2-connectivity and is often what a
// problem really wants: an open ear decomposition exists iff the graph
    ↪ is 2-vertex-connected, and
// the ears are found from the same DFS (each back edge starts one ear).
// If the graph is only 2-EDGE-connected, use the bridge tree instead;
    ↪ st-numbering needs vertex
// connectivity.

```

3.3.8 CactusTree

CACTUS -> TREE. A CACTUS is a connected graph in which every EDGE lies on at most one simple

```

// cycle (equivalently  $m \leq 3(n-1)/2$  and the cycles are edge-disjoint).
    ↪  $O(n + m)$ , 0-indexed, no
// self loops and no parallel edges.
// Build a tree on  $n + (\text{number of cycles})$  nodes: node  $n + i$  represents
    ↪ the  $i$ -th cycle and is joined
// to every vertex of that cycle; every BRIDGE stays a tree edge. Then
    ↪ path/subtree queries on the
// cactus become ordinary tree queries (LCA, HLD, rerooting), which is
    ↪ the whole point.
// Reach for whenever the input says "each edge belongs to at most
    ↪ one cycle" or "n vertices and
    ↪ n edges" (a single cycle with trees hanging off - a "functional
    ↪ graph" shape) and the question
// is about paths, distances, or counting: "number of simple paths
    ↪ between u and v" (2 per cycle
// the path enters and leaves through two different vertices),
    ↪ "shortest path in a cactus", "how
    ↪ many vertices can a path from u to v visit".
// This is really the BLOCK-CUT TREE (03) specialised: on a cactus
    ↪ every block is a single edge or
    ↪ a single cycle, so the block nodes ARE the cycle nodes and the
    ↪ construction is much shorter.
struct CactusTree {
    int n, cycles = 0;
    vector<vector<int>> tree; // size
    ↪ n + cycles after build()
    vector<char> inCycle; // per
    ↪ EDGE id
    vector<int> par, parE, state;
    vector<vector<pair<int, int>>> g; // g[u]
    ↪ = {neighbour, edge id}
    CactusTree(int n) : n(n), par(n, -1), parE(n, -1), state(n, 0),
    ↪ g(n) {}
    void addEdge(int u, int v) {
        int id = inCycle.size();
        inCycle.push_back(0), g[u].push_back({v, id}),
    ↪ g[v].push_back({u, id});
    }
    void dfs(int u, int p) {
        state[u] = 1, par[u] = p;
        for (auto [v, id] : g[u]) {
            if (v == p) continue;
            if (!state[v]) parE[v] = id, dfs(v, u);
        }
    }
}

```

```

    else if (state[v] == 1) { // back
    ↪ edge: close a cycle
        int c = n + cycles++;
        tree.push_back({});
        for (int x = u, e = parE[u]; x != v; x = par[x], e =
    ↪ parE[x])
            tree[x].push_back(c), tree[c].push_back(x),
    ↪ inCycle[e] = 1;
            tree[v].push_back(c), tree[c].push_back(v), inCycle[id]
    ↪ = 1;
        }
        state[u] = 2;
    }
    void build() {
        tree.assign(n, {});
        for (int i = 0; i < n; i++) if (!state[i]) dfs(i, -1);
        for (int u = 0; u < n; u++) for (auto [v, id] : g[u]) // the
    ↪ bridges stay as tree edges
            if (!inCycle[id]) tree[u].push_back(v);
    }
};
// The tree is BIPARTITE between real vertices and cycle nodes only in
    ↪ the cycle parts; a bridge
// joins two real vertices directly, so do not assume alternation.
// NUMBER OF SIMPLE u-v PATHS = 2^(number of CYCLE NODES on the u-v
    ↪ tree path), because inside a
// cycle you may go either way round. That is the standard use of this
    ↪ decomposition.
// SHORTEST PATH IN A CACTUS: on the tree path, a bridge contributes
    ↪ its weight and a cycle node
// contributes min(clockwise, anticlockwise) between its two attachment
    ↪ points - store each
// vertex's position and prefix sum along its cycle to get that in O(1).
// VERTEX CACTUS (every VERTEX in at most one cycle) is a strictly
    ↪ smaller class and the same code
// builds its tree.
// IF THE GRAPH IS NOT A CACTUS this silently produces nonsense -
    ↪ verify m and the block sizes, or
// just use the block-cut tree, which is correct on any graph.

```

3.4 Satisfiability

3.4.1 TwoSat

2-SAT via Kosaraju - $O(n + m)$. Self-contained, 0-based variables.

```

// Literal encoding:  $2*i = (x_i \text{ false}), 2*i+1 = (x_i \text{ true})$ .
// Satisfiable iff x and !x are never in the same SCC; then take the
    ↪ literal whose SCC is LATER
// in topological order.
struct TwoSat {
    int n;
    vector<vector<int>> adj, radj;
    vector<int> ord, comp;
    vector<bool> used, val;

    TwoSat(int n = 0) : n(n), adj(2 * n), radj(2 * n) {}
    int lit(int i, bool t) { return 2 * i + t; }
    int add_var() { n++; adj.resize(2 * n), radj.resize(2 * n); return
    ↪ n - 1; }

    void imply(int a, int b) { // literal a
    ↪ => literal b (and contrapositive)
        adj[a].push_back(b), radj[b].push_back(a);
        adj[b ^ 1].push_back(a ^ 1), radj[a ^ 1].push_back(b ^ 1);
    }
    void either(int a, int b) { imply(a ^ 1, b); } // (a OR b)
    void force(int a) { imply(a ^ 1, a); } // a must be
    ↪ true
    void nand(int a, int b) { either(a ^ 1, b ^ 1); } // not both
    void eq(int a, int b) { imply(a, b), imply(b, a); } // a <=> b
    void xor(int a, int b) { eq(a, b ^ 1); }

    // AT MOST ONE of 'lits' is true, with O(k) edges instead of O(k^2)
    ↪ (prefix/"lightbulb" trick):
    // aux_i means "someone at index <= i was chosen".
    void at_most_one(const vector<int>& lits) {
        int prev = -1;
        for (int c : lits) {
            int cur = lit(add_var(), true);
            imply(c, cur);
            if (prev != -1) imply(prev, cur), imply(c, prev ^ 1);
            prev = cur;
        }
    }

    void dfs1(int u) { used[u] = true; for (int v : adj[u]) if

```

```

    ↪ (!used[v]) dfs1(v); ord.push_back(u); }
    void dfs2(int u, int c) { comp[u] = c; for (int v : radj[u]) if
    ↪ (comp[v] < 0) dfs2(v, c); }

    bool solve() {
        used.assign(2 * n, false), comp.assign(2 * n, -1),
    ↪ val.assign(n, false), ord.clear();
        for (int i = 0; i < 2 * n; i++) if (!used[i]) dfs1(i);
        int c = 0;
        for (int i = 2 * n - 1; i >= 0; i--) if (comp[ord[i]] < 0)
    ↪ dfs2(ord[i], c++);
        for (int i = 0; i < n; i++) {
            if (comp[2 * i] == comp[2 * i + 1]) return false;
            val[i] = comp[2 * i + 1] > comp[2 * i];
        }
        return true; // read
    ↪ val[0..base_n-1]; aux vars are padding
    }
};
// RANGE CLAUSE ("if x then ALL of [l,r]") in O(log n) edges: build a
    ↪ segment tree over the
// variables, give every internal node an aux variable, add parent =>
    ↪ both children, then
// imply(x, node) for the O(log n) covering nodes. Mirror the tree
    ↪ (child => parent) for
// "if any of [l,r] then x".

```

3.4.2 DPLL

GENERAL SAT (3-SAT and beyond) by DPLL - Davis-Putnam-Logemann-Loveland.
0-indexed variables.

```

// Exponential in the worst case, but with unit propagation and
    ↪ pure-literal elimination it decides
// most contest-sized instances (n in the hundreds, even thousands)
    ↪ instantly.
// 2-SAT is polynomial (01 - TwoSat) - always check whether the clauses
    ↪ really need three literals.
// Reach for this when a problem reduces to "assign booleans so that
    ↪ all these constraints hold" and
// some clause has three or more literals: exact cover with few
    ↪ options, graph colouring encoded as
// clauses, puzzle solvers, "is this configuration reachable" searches
    ↪ with heavy pruning.
// LITERAL ENCODING: the positive literal of variable v is v, the
    ↪ negative one is ~v (= -v-1).
// A clause is a vector of literals meaning their OR. Internally
    ↪ literal u lives at index u + n.
// THE THREE MOVES.
// UNIT PROPAGATION: a clause with all but one literal already false
    ↪ forces that last literal.
// PURE LITERAL: a variable that only ever appears with one sign can
    ↪ be set that way for free.
// BRANCH: pick the unassigned variable with the MOST remaining
    ↪ occurrences, try it, backtrack.
struct SAT {
    int n;
    vector<int> occ, pos, neg, unitS, pureS;
    vector<vector<int>> g, cls, level; //
    ↪ g[lit] = clauses containing lit
    vector<char> val; // val[u
    ↪ + n] = "literal u is true"
    SAT(int n) : n(n), occ(2 * n, 0), g(2 * n) {}
    void addClause(const vector<int>& c) {
        for (int u : c) g[u + n].push_back(cls.size()), occ[u + n]++;
        cls.push_back(c);
    }
    void set(int u) { //
    ↪ assign literal u = true
        val[u + n] = 1, level.back().push_back(u);
        for (int i : g[u + n]) if (pos[i]++ == 0) for (int w : cls[i])
    ↪ --occ[w + n];
        for (int i : g[~u + n]) { ++neg[i]; if (pos[i] == 0)
    ↪ unitS.push_back(i); }
    }
    void undo() {
        int u = level.back().back();
        level.back().pop_back(), val[u + n] = 0;
        for (int i : g[u + n]) if (--pos[i] == 0) for (int w : cls[i])
    ↪ ++occ[w + n];
        for (int i : g[~u + n]) --neg[i];
    }
    bool reduce() { //
    ↪ propagate until nothing is forced
        while (!unitS.empty() || !pureS.empty()) {
            if (!pureS.empty()) {
                int u = pureS.back(); pureS.pop_back();
                if (occ[u + n] == 1 && occ[~u + n] == 0) set(u);
            } else {

```

```

    int i = unitS.back(); unitS.pop_back();
    if (pos[i] > 0) continue; //
    ↪ clause already satisfied
    if (neg[i] == (int)cls[i].size()) return false; //
    ↪ clause already violated
    if (neg[i] + 1 == (int)cls[i].size()) {
        int w = 0;
        bool found = false;
        for (int u : cls[i]) if (!val[u + n] && !val[~u +
    ↪ n]) w = u, found = true;
        if (!found) return false;
        set(w);
    }
}
return true;
}
bool solve() { // val[v
    ↪ + n] = the model afterwards
    val.assign(2 * n, 0);
    pos.assign(cls.size(), 0), neg.assign(cls.size(), 0);
    level.assign(1, {}), unitS.clear(), pureS.clear();
    for (;;) {
        if (reduce()) {
            int s = -1, bc = 0; //
            ↪ branch on the most constrained
            for (int u = 0; u < n; u++) { //
            ↪ UNASSIGNED variable
                if (val[u + n] || val[~u + n]) continue;
                int c = occ[u + n] + occ[~u + n];
                if (c > bc) bc = c, s = u;
            }
            if (s < 0) return true; // every
            ↪ clause is satisfied
            level.push_back({}), set(s);
        } else {
            if (level.back().empty()) return false;
            int s = level.back()[0];
            while (!level.back().empty()) undo();
            level.pop_back();
            if (level.empty()) return false;
            set(~s); // the
            ↪ other branch
        }
    }
}
bool value(int v) const { return val[v + n]; }
};
// EXAMPLE. (x0 OR NOT x1) AND (NOT x1) AND (NOT x1 OR x3 OR x5) is
// addClause({0, ~1}); addClause({~1}); addClause({~1, 3, 5});
// WATCHED LITERALS + CLAUSE LEARNING (CDCL) is what real solvers add
    ↪ on top; it is far more code
// and rarely needed in a contest - if DPLL is too slow, the intended
    ↪ solution is not SAT.
// EXACTLY-ONE OF k VARIABLES: one clause (v1 OR ... OR vk) plus C(k,2)
    ↪ clauses (NOT vi OR NOT vj);
// for large k use the commander/sequential encoding to keep the clause
    ↪ count linear.
// AT MOST ONE with 2-SAT structure? Only for k = 2. Beyond that you
    ↪ are in this file.
// MAX-SAT and #SAT are harder still; for #SAT on n <= 20 variables,
    ↪ use inclusion-exclusion or a
// subset DP over the clause set instead.

```

3.5 Eulerian Path

3.5.1 EulerianPath

Hierholzer's Algorithm for Eulerian Path / Circuit - $O(V + E)$

// Works for both Directed and Undirected graphs

```

struct EulerianPath {
    int n, m;
    vector<vector<pair<int, int>>> adj; // {neighbor, edge_id}
    vector<bool> used_edge;
    vector<int> in_deg, out_deg, path;

    EulerianPath(int n = 0, int m = 0) : n(n), m(m), adj(n + 1),
    ↪ used_edge(m, false), in_deg(n + 1, 0), out_deg(n + 1, 0) {}

    void add_edge(int u, int v, int edge_id, bool directed = true) {
        adj[u].pb({v, edge_id});
        out_deg[u]++;
        in_deg[v]++;
        if (!directed) {
            adj[v].pb({u, edge_id});
            out_deg[v]++;
            in_deg[u]++;
        }
    }
}

```

```

    }
}

// Finds Eulerian path starting at start_node. Returns false if
    ↪ graph is not Eulerian
bool solve(int start_node) {
    vector<int> ptr(n + 1, 0);
    stack<int> st;
    st.push(start_node);

    while (!st.empty()) {
        int u = st.top();
        if (ptr[u] < adj[u].size()) {
            auto [v, id] = adj[u][ptr[u]++];
            if (!used_edge[id]) {
                used_edge[id] = true;
                st.push(v);
            }
        } else {
            path.pb(u);
            st.pop();
        }
    }
    reverse(all(path));

    // Verify that every edge was traversed
    for (bool used : used_edge) {
        if (!used) return false;
    }
    return true;
}
};

```

3.6 Flow and Min Cut

3.6.1 Dinic

DINIC MAX FLOW. $O(V^2 E)$ in general, $O(E \sqrt{V})$ on unit-capacity graphs, $O(V^2 E)$ worst case but

// fast in practice. After maxflow, vertices reachable from s in the
 ↪ residual graph form the min cut.
 // Scaling (start with big capacities) helps on dense graphs with large
 ↪ capacities.
 // DINIC MAX FLOW. $O(V^2 E)$ general, $O(E \sqrt{E})$ with unit capacities,
 ↪ $O(E \sqrt{V})$ for bipartite
 // matching. This is the SCALING variant (big capacities first), so it
 ↪ stays fast on wide ranges.
 // addEdge(a, b, cap, rcap): pass rcap = cap for an UNDIRECTED edge.
 ↪ Edge::flow() recovers the flow
 // on each arc. After calc(s, t), leftOfMinCut(v) is true exactly for
 ↪ the SOURCE side of a minimum
 // cut - the cut edges are those from a left vertex to a right vertex.
 ↪ 0-indexed.

```

struct Dinic {
    struct Edge {
        int to, rev;
        ll c, oc;
        ll flow() { return max(oc - c, 0LL); } // if you need flows
    };

    vector<int> lvl, ptr, q;
    vector<vector<Edge>> > adj;

    Dinic(int n) : lvl(n), ptr(n), q(n), adj(n) {}

    void addEdge(int a, int b, ll c, ll rcap = 0) {
        adj[a].push_back({b, (int) adj[b].size(), c, c});
        adj[b].push_back({a, (int) adj[a].size() - 1, rcap, rcap});
    }

    ll dfs(int v, int t, ll f) {
        if (v == t || !f) return f;
        for (int &i = ptr[v]; i < (int) adj[v].size(); i++) {
            Edge &e = adj[v][i];
            if (lvl[e.to] == lvl[v] + 1)
                if (ll p = dfs(e.to, t, min(f, e.c))) {
                    e.c -= p, adj[e.to][e.rev].c += p;
                    return p;
                }
        }
        return 0;
    }

    ll calc(int s, int t) {
        ll flow = 0;
        q[0] = s;
        for (int L = 0; L < 31; L++)

```

```

do {
    // 'int L=30' maybe faster for random data
    lvl = ptr = vector<int>(int) q.size();
    int qi = 0, qe = lvl[s] + 1;
    while (qi < qe && !lvl[t]) {
        int v = q[qi++];
        for (Edge e: adj[v])
            if (!lvl[e.to] && e.c >> (30 - L))
                q[qi++] = e.to, lvl[e.to] = lvl[v] + 1;
    }
    while (ll p = dfs(s, t, LLONG_MAX)) flow += p;
} while (lvl[t]);
return flow;
}

bool leftOfMinCut(int a) { return lvl[a] != 0; }
};

```

3.6.2 MCMF

MIN-COST MAX-FLOW with JOHNSON POTENTIALS (Dijkstra instead of SPFA on every augmentation).

```

// O(F * E log V) where F = number of augmentations. Much faster than
// the SPFA version and the
// one to use by default. Negative edge costs are allowed at
// construction PROVIDED there is no negative-cost CYCLE:
// run one Bellman-Ford
// to initialise the potentials (setPotentialsBF), after that every
// reduced cost is >= 0.
struct MCMF {
    struct E { int to, rev; ll cap, cost, flow = 0; };
    int n;
    vector<vector<E>> g;
    vector<ll> pot, dist;
    vector<int> pv, pe;

    MCMF(int n) : n(n), g(n), pot(n, 0), dist(n), pv(n), pe(n) {}
    void addEdge(int u, int v, ll cap, ll cost) {
        g[u].push_back({v, (int)g[v].size(), cap, cost});
        g[v].push_back({u, (int)g[u].size() - 1, 0, -cost});
    }
    void setPotentialsBF(int s) { // only needed if
        // some cost < 0
        fill(all(pot), llinf);
        pot[s] = 0;
        for (int it = 0; it < n; it++) {
            bool ch = false;
            for (int u = 0; u < n; u++) if (pot[u] < llinf)
                for (auto& e : g[u]) if (e.cap - e.flow > 0 && pot[u] +
                // e.cost < pot[e.to])
                    pot[e.to] = pot[u] + e.cost, ch = true;
            if (!ch) break;
        }
        for (auto& x : pot) if (x == llinf) x = 0;
    }
    bool dijkstra(int s, int t) {
        fill(all(dist), llinf);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
        // pq;
        dist[s] = 0, pq.push({0, s});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int i = 0; i < (int)g[u].size(); i++) {
                auto& e = g[u][i];
                if (e.cap - e.flow <= 0) continue;
                ll nd = d + e.cost + pot[u] - pot[e.to]; //
                // reduced cost, always >= 0
                if (nd < dist[e.to]) dist[e.to] = nd, pv[e.to] = u,
                // pe[e.to] = i, pq.push({nd, e.to});
            }
        }
        return dist[t] < llinf;
    }
    // Returns {flow, cost}. Pass maxf to cap the flow (e.g. for
    // min-cost k-flow).
    pair<ll, ll> run(int s, int t, ll maxf = llinf) {
        ll flow = 0, cost = 0;
        while (flow < maxf && dijkstra(s, t)) {
            for (int i = 0; i < n; i++) if (dist[i] < llinf) pot[i] +=
            // dist[i];
            ll aug = maxf - flow;
            for (int v = t; v != s; v = pv[v]) aug = min(aug,
            // g[pv[v]][pe[v]].cap - g[pv[v]][pe[v]].flow);
            for (int v = t; v != s; v = pv[v]) {
                auto& e = g[pv[v]][pe[v]];
                e.flow += aug, g[v][e.rev].flow -= aug;
                cost += aug * e.cost;
            }
            flow += aug;
        }
    }
};

```

```

    }
    flow += aug;
}
return {flow, cost};
}
// MIN-COST flow of exactly K units: run(s, t, K) and check flow ==
// K.
// MIN-COST (not max) flow: stop as soon as dist[t] + pot[t] -
// pot[s] > 0 (no profitable path).
};
// MODELLING: assignment on a SPARSE graph; for a dense/complete one
// use 07 - Matching/03 - Hungarian * transportation * "k disjoint
// paths of minimum total cost" *
// min-cost bipartite matching * scheduling with penalties * MCMF is
// also how you solve
// "choose k items with a matroid-ish constraint minimising cost".

```

3.6.3 FlowWithLowerBounds

FLOW WITH LOWER BOUNDS (circulation with demands). Every edge needs $d(e) \leq f(e) \leq c(e)$.

```

// Build an auxiliary network, run one max flow; feasible iff every
// edge out of s' saturates.
// excess(v) = (sum of d over edges INTO v) - (sum of d over edges
// OUT of v)
// excess(v) > 0 -> edge s' -> v of capacity excess(v); < 0 -> edge
// v -> t' of capacity -excess
// every original edge (u,v) keeps capacity c - d
// For an s-t flow, add t->s with capacity INF first, which turns it
// into a circulation.
// Needs Dinic (01 - Dinic.cpp) with addEdge(u, v, cap) and calc(s, t).
struct LowerBoundFlow {
    int n, S, T; // S, T are the
    // auxiliary super source/sink
    Dinic din;
    vector<ll> excess, low;
    vector<array<int, 2>> eid; // per original
    // edge: {u, index in din.adj[u]}

    LowerBoundFlow(int n) : n(n), S(n), T(n + 1), din(n + 2), excess(n
    // + 2, 0) {}
    void addEdge(int u, int v, ll lo, ll hi) {
        low.push_back(lo);
        eid.push_back({u, (int)din.adj[u].size()});
        din.addEdge(u, v, hi - lo);
        excess[v] += lo, excess[u] -= lo;
    }
    bool feasible() { // circulation
        // only (no s, t)
        ll need = 0;
        for (int v = 0; v < n; v++) {
            if (excess[v] > 0) din.addEdge(S, v, excess[v]), need +=
            // excess[v];
            else if (excess[v] < 0) din.addEdge(v, T, -excess[v]);
        }
        return din.calc(S, T) == need;
    }
    // Min / max s-t flow subject to the bounds - see the RECIPES block
    // below for the ONLY
    // correct order of operations. You must DELETE the t->s edge (zero
    // both residual directions)
    // before the second max flow, otherwise Dinic's first augmenting
    // path runs s -> t backwards
    // along that arc and CANCELS the feasible flow instead of
    // extending it.
    // feasible() appends the S/T edges every time it is called - call
    // it exactly once.
    ll flowOn(int i) { return low[i] +
    // din.adj[eid[i][0]][eid[i][1]].flow(); }
};
/* RECIPES
"each edge must be used at least once" -> d(e) = 1
"each vertex must be visited between L and R" -> split v into v_in
-> v_out with [L, R]
"exact degree subgraph" -> d = c = required
-> degree on the split edge
MAX s-t FLOW WITH BOUNDS: add t->s with capacity INF, run
-> feasible(); then delete that edge
and run one more max flow from s to t on the residual graph. The
-> answer is the flow that
was on t->s plus the extra augmentation.
MIN s-t FLOW WITH BOUNDS: same, but augment from t to s to cancel as
-> much as possible. */

```

3.6.4 StoerWagner

STOER-WAGNER GLOBAL MIN CUT - minimum cut over ALL vertex pairs, undirected weighted, $O(V^3)$.

```
// No source/sink. Returns {cut weight, one side of the cut}.
// Key lemma: in a maximum-adjacency order, if s is the second-to-last
//   ↳ and t the last vertex
// added, then the minimum s-t cut is exactly {t}. Merge s and t and
//   ↳ repeat n-1 times.
pair<ll, vector<int>> globalMinCut(vector<vector<ll>> w) {
    int n = w.size();
    if (n < 2) return {0, {}};
    vector<char> gone(n, 0);
    vector<vector<int>> grp(n);
    for (int i = 0; i < n; i++) grp[i] = {i};
    ll bestW = llinf;
    vector<int> best;
    for (int phase = 0; phase + 1 < n; phase++) {
        vector<ll> ww(n, 0);
        vector<char> added(n, 0);
        int prev = -1, last = -1;
        for (int it = 0; it < n - phase; it++) { //
            ↳ maximum-adjacency order
            int sel = -1;
            for (int j = 0; j < n; j++)
                if (!gone[j] && !added[j] && (sel < 0 || ww[j] >
                    ↳ ww[sel])) sel = j;
            added[sel] = 1, prev = last, last = sel;
            for (int j = 0; j < n; j++) if (!gone[j] && !added[j])
                ↳ ww[j] += w[sel][j];
        }
        ll cut = 0; //
        ↳ cut-of-the-phase separates {last}
        for (int j = 0; j < n; j++) if (!gone[j] && j != last) cut +=
            ↳ w[last][j];
        if (cut < bestW) bestW = cut, best = grp[last];
        gone[last] = 1; // merge last
        ↳ into prev
        grp[prev].insert(grp[prev].end(), all(grp[last]));
        for (int j = 0; j < n; j++) if (!gone[j] && j != prev)
            w[prev][j] += w[last][j], w[j][prev] = w[prev][j];
    }
    return {bestW, best};
}
// USES: "split the network into two non-empty parts as cheaply as
//   ↳ possible", edge connectivity
// of a weighted undirected graph, "minimum number of edges to delete
//   ↳ to disconnect the graph"
// (unit weights). Doing this with n-1 max-flows is both slower and
//   ↳ longer to write.
// KARGER (randomized contraction) finds a min cut with probability >=
//   ↳ 1/C(n,2) per run; the
// useful corollary is that a graph has AT MOST C(n,2) distinct minimum
//   ↳ cuts.
```

3.6.5 MinCutModelling

MIN-CUT MODELLING - the reason flow appears at ICPC. The code is never the hard part; choosing

```
// Needs: Dinic (01 - Dinic.cpp).
// what the vertices and the infinite capacities mean is. Rule of
//   ↳ thumb: if the objective contains
// an "infinity", you are CUTTING, not flowing.

// --- MAXIMUM WEIGHT CLOSURE / PROJECT SELECTION ---
// Choose a set S closed under the dependency arcs (if v in S and v ->
//   ↳ u is a dependency, u in S)
// maximising the total weight. Build: s -> v with capacity w(v) for
//   ↳ every w(v) > 0;
// v -> t with capacity -w(v) for every w(v) < 0; u -> v with INF for
//   ↳ every dependency u needs v.
// Answer = (sum of positive weights) - maxflow. S = the source side
//   ↳ of the min cut.
// USES: "each project earns p but needs these machines that cost c";
//   ↳ "pick cells maximising
// profit but a chosen cell forces its neighbours"; densest-subgraph
//   ↳ style selections.
ll maxClosure(int n, const vector<ll>& w, const vector<pair<int, int>>&
    ↳ need, vector<int>* chosen = nullptr) {
    Dinic din(n + 2);
    int s = n, t = n + 1;
    ll pos = 0;
    for (int v = 0; v < n; v++)
        if (w[v] > 0) pos += w[v], din.addEdge(s, v, w[v]);
        else if (w[v] < 0) din.addEdge(v, t, -w[v]);
    for (auto [u, v] : need) din.addEdge(u, v, llinf); //
    ↳ choosing u forces choosing v
    ll cut = din.calc(s, t);
```

```
if (chosen) for (int v = 0; v < n; v++) if (din.leftOfMinCut(v))
    ↳ chosen->push_back(v);
return pos - cut;
}
/* THE OTHER STANDARD REDUCTIONS
TWO LABELS WITH PAIR PENALTIES ("cats or dogs", image segmentation)
Every vertex gets label A or B. Cost a_v for A, b_v for B, and c_uv
    ↳ extra if u and v differ.
Build s -> v with b_v, v -> t with a_v, and u <-> v with c_uv in
    ↳ both directions.
The min cut is the minimum total cost; the source side takes label A.
PRECONDITION: the pair penalty must be SUBMODULAR (cost(A,A) +
    ↳ cost(B,B) <= cost(A,B) +
    ↳ cost(B,A)). A penalty that rewards disagreement cannot be modelled
    ↳ by a cut - it is NP-hard.
```

VERTEX CAPACITIES / VERTEX-DISJOINT PATHS

Split v into v_in -> v_out with the vertex capacity. Vertex-disjoint
↳ s-t paths use capacity 1.

BIPARTITE FAMILY (see 07 - Matching and 10 - Theorems)

max matching = max flow with unit capacities
min vertex cover = max matching (Konig) -> recover via
↳ the alternating-BFS set
max independent set = n - max matching
min path cover of a DAG (vertex-disjoint) = n - max matching of the
↳ split graph

"MUST BE ON THE SOURCE SIDE" -> add s -> v with INF. "MUST BE ON THE
↳ SINK SIDE" -> v -> t INF.

That is how you force decisions inside an otherwise free min cut.

EDGE / VERTEX DISJOINT PATHS = Menger. Recover the paths by
↳ decomposing the integral flow.

LOWER BOUNDS ON EDGES -> 03 - FlowWithLowerBounds.cpp.

MAXIMUM DENSITY SUBGRAPH (maximise $|E(S)| / |S|$): binary search

↳ lambda, then min cut with
s -> v capacity m, v -> t capacity m + 2*lambda - deg(v), u <-> v
↳ capacity 1. Feasible iff
the min cut is < n*m. $O(\log(n^2) * \text{maxflow})$ since the density is a
↳ ratio of integers <= n^2 .

MINIMUM CUT WITHOUT s AND t (global) -> 04 - StoerWagner.cpp, $O(V^3)$,
↳ or n-1 max flows.

ALL-PAIRS MIN CUT -> the Gomory-Hu tree: n-1 max flows build a

↳ weighted tree in which the min
u-v cut equals the MINIMUM EDGE on the u-v tree path. Use it when
↳ many pairs are asked.

COMPLEXITY TABLE - read this at minute 3, not minute 90

Dinic	$O(V^2 E)$ general $O(E \sqrt{E})$ unit caps	
↳	$O(E \sqrt{V})$ bipartite	
Edmonds-Karp	$O(V E^2)$	-- do not
Push-relabel / HLPP	$O(V^2 \sqrt{E})$	-- dense graphs,
↳	$n \sim 1000, m \sim n^2/2$	
MCMF (SSP + potentials)	$O(F * E \log V)$	-- F = total
↳	flow, so bound F first	
Hungarian	$O(n^2 m)$	-- dense
↳	assignment, $n \leq \sim 1000$	
Hopcroft-Karp	$O(E \sqrt{V})$	-- bipartite
↳	matching, $n \geq$ few thousand	
Kuhn	$O(V E)$	-- shorter, fine
↳	to ~2000 vertices	
Blossom (general)	$O(V^3)$	
Stoer-Wagner	$O(V^3)$	
Gomory-Hu	$O(V * \text{maxflow})$	

If the flow value itself can be huge (capacities up to $1e9$), an $O(F$
↳ ...) bound is not a bound -
scale the capacities or use a strongly polynomial algorithm. */

3.6.6 GomoryHu

GOMORY-HU TREE - ALL-PAIRS MINIMUM CUT from only n-1 max-flow runs.

```
// Needs: Dinic (01 - Dinic.cpp).
// The result is a weighted tree on the same vertex set with the
//   ↳ property that, for EVERY pair
// (u, v), the minimum u-v cut in the original graph equals the MINIMUM
//   ↳ EDGE WEIGHT on the u-v
// path in the tree. (There are at most n-1 distinct min-cut values in
//   ↳ any graph - that is the
// whole content of the theorem.)
// Reach for it when a problem asks about min cuts between many pairs:
//   ↳ "sum/max/k-th of all
// pairwise min cuts", "order the vertices to maximise the sum of
//   ↳ consecutive cuts", "how many
```

```
// pairs have min cut >= k". Undirected graphs only.
// Complexity: (n-1) x Dinic. The tree is returned as parent + weight
↳ arrays (par[0] = -1).
struct GomoryHu {
    int n;
    vector<array<ll, 3>> edges; // {u,
    ↳ v, capacity}, undirected
    vector<int> par;
    vector<ll> w;
    GomoryHu(int n) : n(n), par(n, 0), w(n, 0) {}
    void addEdge(int u, int v, ll c) { edges.push_back({(ll)u, (ll)v,
    ↳ c}); }
    void build() {
        par.assign(n, 0), w.assign(n, 0), par[0] = -1;
        for (int i = 1; i < n; i++) {
            Dinic din(n);
            for (auto [u, v, c] : edges) din.addEdge(u, v, c, c); //
    ↳ undirected: both directions
            w[i] = din.calc(i, par[i]);
            for (int j = i + 1; j < n; j++) //
    ↳ re-parent the same-side vertices
                if (par[j] == par[i] && din.leftOfMinCut(j)) par[j] = i;
        }
        ll minCut(int u, int v) const { // min
    ↳ edge on the tree path
            ll best = llinf;
            vector<int> du(n, 0);
            for (int x = u; x != -1; x = par[x]) du[x] = 1;
            vector<int> pv;
            int m = v;
            while (!du[m]) pv.push_back(m), m = par[m]; // m =
    ↳ the meeting vertex
            for (int x = u; x != m; x = par[x]) best = min(best, w[x]);
            for (int x : pv) best = min(best, w[x]);
            return best;
        }
    };
// EQUIVALENT-FLOW TREE (Gusfield's variant) is the same n-1 flows
↳ without the re-parenting step;
// it gets the min-cut VALUES right but not the actual cuts. The
↳ version above is the real
// Gomory-Hu, so the tree edges are genuine cuts.
// GLOBAL MIN CUT is the minimum edge of the whole tree - but
↳ Stoer-Wagner (04) is  $O(V^3)$  and
// needs no flow at all, so prefer it when that is all you want.
// DIRECTED graphs have no Gomory-Hu tree; you genuinely need  $O(n^2)$ 
↳ flows there.
```

3.6.7 PushRelabel

HIGHEST-LABEL PUSH-RELABEL max flow, $O(V^2 \sqrt{E})$, 0-indexed. Same interface as Dinic (01).

```
// WHEN TO PREFER IT OVER DINIC: dense graphs ( $V \sim 1000-5000$ ,  $E \sim$ 
↳  $V^2/2$ ) and graphs with large
// capacities, where Dinic's  $O(V^2 E)$  bound bites - usually 3-10x
↳ faster there. On unit-capacity
// or bipartite-matching graphs Dinic is  $O(E \sqrt{V})$  and wins, so keep
↳ both.
// It does NOT maintain a flow at intermediate times - only at the end
↳ - so it cannot be used for
// the incremental "add capacity and continue" tricks; Dinic can.
// IDEA: give every vertex a HEIGHT and let excess flow downhill
↳ (push); a vertex with excess and
// nowhere to go is lifted (relabel). Source starts at height V, so
↳ nothing can return to it.
// The GAP HEURISTIC (if no vertex has height h any more, lift
↳ everything above h out of range) is
// what makes it fast in practice - it is the 'co[]' bookkeeping below,
↳ do not drop it.
// addEdge(a, b, cap, rcap): rcap = cap gives an UNDIRECTED edge. After
↳ calc, leftOfMinCut(v) marks
// the SOURCE side of a minimum cut, and e.f is the flow on that arc.
struct PushRelabel {
    struct Edge { int to, rev; ll f, c; };
    int n;
    vector<vector<Edge>> g;
    vector<ll> ec;
    vector<int> H, cur, co;
    vector<vector<int>> hs;
    PushRelabel(int n) : n(n), g(n), ec(n), H(n), cur(n), co(2 * n +
    ↳ 1), hs(2 * n + 1) {}
    void addEdge(int a, int b, ll cap, ll rcap = 0) {
        if (a == b) return;
        g[a].push_back({b, (int)g[b].size(), 0, cap});
        g[b].push_back({a, (int)g[a].size() - 1, 0, rcap});
    }
    void addFlow(int u, Edge& e, ll f) {
```

```
Edge& back = g[e.to][e.rev];
    if (!ec[e.to] && f) hs[H[e.to]].push_back(e.to);
    e.f += f, e.c -= f, ec[e.to] += f;
    back.f -= f, back.c += f, ec[u] -= f;
}
ll calc(int s, int t) {
    fill(all(H), 0), fill(all(ec), 0), fill(all(co), 0),
    ↳ fill(all(cur), 0);
    for (auto& v : hs) v.clear();
    H[s] = n, ec[t] = 1, co[0] = n - 1;
    for (auto& e : g[s]) addFlow(s, e, e.c);
    for (int hi = 0;;) {
        while (hs[hi].empty()) if (!hi--) return -ec[s];
        int u = hs[hi].back(); hs[hi].pop_back();
        while (ec[u] > 0) { //
    ↳ discharge u
            if (cur[u] == (int)g[u].size()) { //
    ↳ relabel + gap heuristic
                H[u] = 2 * n;
                for (int i = 0; i < (int)g[u].size(); i++)
                    if (g[u][i].c && H[u] > H[g[u][i].to] + 1)
                        H[u] = H[g[u][i].to] + 1, cur[u] = i;
                if (++co[H[u]], !--co[hi] && hi < n)
                    for (int i = 0; i < n; i++)
                        if (hi < H[i] && H[i] < n) --co[H[i]], H[i]
    ↳ = n + 1;
                hi = H[u];
            } else if (g[u][cur[u]].c && H[u] == H[g[u][cur[u]].to]
    ↳ + 1)
                addFlow(u, g[u][cur[u]], min(ec[u],
    ↳ g[u][cur[u]].c));
                else ++cur[u];
            }
        }
        bool leftOfMinCut(int a) { return H[a] >= n; }
    };
// COMPLEXITY NOTES. FIFO push-relabel is  $O(V^3)$ ; the highest-label
↳ rule above is  $O(V^2 \sqrt{E})$ .
// The gap heuristic and the global relabel (BFS from t every so often)
↳ are the two practical
// speedups; the version here has the gap only, which is already enough
↳ for contests.
// TIE-BREAKING WITH DINIC: if you also need the actual min cut EDGES,
↳ both give them the same way
// (an edge from a leftOfMinCut vertex to a non-leftOfMinCut one). If
↳ you need the flow DECOMPOSED
// into paths, repeatedly DFS from s along positive-flow arcs and
↳ subtract the bottleneck.
```

3.6.8 CostScalingMCMF

MIN COST MAX FLOW BY COST SCALING (push-relabel), $O(V^2 E \log(V \cdot C))$ but very fast in practice -

```
// 3e4 edges are comfortable. 0-indexed, INTEGER costs and capacities.
// WHY THIS EXISTS ALONGSIDE 02 - MCMF. The successive-shortest-path
↳ version needs the residual
// graph to have NO NEGATIVE CYCLE, so it forbids negative-cost edges
↳ (Johnson potentials only fix
// negative EDGES on a DAG-like start, not negative CYCLES). This one
↳ accepts ARBITRARY costs,
// including negative cycles: it first pushes a max flow ignoring cost,
↳ then repeatedly "refines"
// the eps-optimality of the flow, halving eps each round. It also does
↳ not depend on the flow
// VALUE, so a huge max flow is fine, whereas SSP is  $O(F * E \log V)$ .
// Reach for it when: costs can be negative, or the flow value is
↳ large, or the graph is dense.
// The source and sink are fixed at construction. Costs are internally
↳ multiplied by V, so keep
// |cost| * V * V inside the cost type - use long long unless you have
↳ checked.
struct CostScalingMCMF {
    struct Edge { ll c, f; int to, rev; };
    ll eps = 0;
    int n, S, T;
    vector<vector<Edge>> g;
    vector<vector<int>> hs;
    vector<int> cur, isq, co;
    vector<ll> ex, h;
    CostScalingMCMF(int n, int s, int t) : n(n), S(s), T(t), g(n) {}
    void addEdge(int a, int b, ll cost, ll cap) { // cost
    ↳ per unit, cap >= 0
        if (a == b) return;
        cost *= n, eps = max(eps, abs(cost));
        g[a].push_back({cost, cap, b, (int)g[b].size()});
        g[b].push_back({-cost, 0, a, (int)g[a].size() - 1});
    }
```

```

void addFlow(int u, Edge& e, ll f) {
    Edge& bk = g[e.to][e.rev];
    if (!ex[e.to] && f) hs[h[e.to]].push_back(e.to);
    e.f -= f, ex[e.to] += f, bk.f += f, ex[u] -= f;
}

ll maxFlow() { // plain
    highest-label push-relabel
    ex.assign(n, 0), h.assign(n, 0), hs.assign(2 * n + 1, {}),
    co.assign(2 * n + 1, 0);
    cur.assign(n, 0), h[S] = n, ex[T] = 1, co[0] = n - 1;
    for (auto& e : g[S]) addFlow(S, e, e.f);
    if (hs[0].size()) for (int hi = 0; hi >= 0;) {
        int u = hs[hi].back(); hs[hi].pop_back();
        while (ex[u] > 0) {
            if (cur[u] == (int)g[u].size()) {
                h[u] = 2 * n;
                for (int i = 0; i < (int)g[u].size(); i++)
                    if (g[u][i].f && h[u] > h[g[u][i].to] + 1)
                        h[u] = h[g[u][i].to] + 1, cur[u] = i;
                if (++co[h[u]], !--co[hi] && hi < n)
                    for (int i = 0; i < n; i++)
                        if (hi < h[i] && h[i] < n) --co[h[i]], h[i]
            }
            hi = h[u];
        } else if (g[u][cur[u]].f && h[u] == h[g[u][cur[u]].to])
            addFlow(u, g[u][cur[u]], min(ex[u],
            g[u][cur[u]].f));
        else ++cur[u];
    }
    while (hi >= 0 && hs[hi].empty()) --hi;
    return -ex[S];
}

void push(int u, Edge& e, ll amt) {
    amt = min(amt, e.f);
    e.f -= amt, ex[e.to] += amt, g[e.to][e.rev].f += amt, ex[u] -=
    amt;
}

void relabel(int v) {
    ll nh = -llinf;
    for (int i = 0; i < (int)g[v].size(); i++)
        if (g[v][i].f && nh < h[g[v][i].to] - g[v][i].c)
            nh = h[g[v][i].to] - g[v][i].c, cur[v] = i;
    h[v] = nh - eps;
}

pair<ll, ll> solve() { // {max
    flow, its minimum cost}
    ll cost = 0;
    for (int i = 0; i < n; i++) for (Edge& e : g[i]) cost += e.c *
    e.f;
    ll flow = maxFlow();
    h.assign(n, 0), ex.assign(n, 0), isq.assign(n, 0),
    cur.assign(n, 0);
    queue<int> q;
    for (; eps; eps >= 2) { //
        refine: halve eps twice per round
        fill(all(cur), 0);
        for (int i = 0; i < n; i++) for (auto& e : g[i])
            if (h[i] + e.c - h[e.to] < 0 && e.f) push(i, e, e.f);
        for (int i = 0; i < n; i++) if (ex[i] > 0) q.push(i),
        isq[i] = 1;
        while (!q.empty()) { //
            restore feasibility
            int u = q.front(); q.pop(), isq[u] = 0;
            while (ex[u] > 0) {
                if (cur[u] == (int)g[u].size()) relabel(u);
                for (int& i = cur[u]; i < (int)g[u].size(); i++) {
                    Edge& e = g[u][i];
                    if (h[u] + e.c - h[e.to] < 0) {
                        push(u, e, ex[u]);
                        if (ex[e.to] > 0 && !isq[e.to])
                            q.push(e.to), isq[e.to] = 1;
                        if (!ex[u]) break;
                    }
                }
            }
            if (eps > 1 && (eps >= 2) == 0) eps = 4;
        }
        for (int i = 0; i < n; i++) for (Edge& e : g[i]) cost -= e.c *
        e.f;
        return {flow, cost / 2 / n};
    }
    ll flowOn(const Edge& e) const { return g[e.to][e.rev].f; }
};
// MIN COST FLOW OF A GIVEN VALUE k (not necessarily maximum): cap the

```

→ source arc at k, or add an
 // arc T → S of capacity INF and cost -BIG and take a min cost
 → CIRCULATION.
 // NEGATIVE COST CYCLES in the input are handled: the answer is the
 → true minimum, which may be
 // negative. With 02 - MCMF you would have to cancel the cycles by hand
 → first (saturate every
 // negative edge, then fix the resulting excesses with a circulation) -
 → that is what this replaces.
 // ASSIGNMENT PROBLEMS with $n \leq 500$ are still better served by
 → Hungarian (07 - Matching/03).

3.6.9 MinCostFlowLowerBounds

MIN COST FLOW WITH LOWER BOUNDS (min cost circulation with demands).
 0-indexed, costs ≥ 0 .

```

// Needs: MCMF (02 - MCMF.cpp).
// This is 03 - FlowWithLowerBounds with a price tag on every unit:
    → each edge needs
//  $lo \leq f(e) \leq hi$  and costs  $c$  per unit, and among all FEASIBLE flows
    → you want the cheapest.
// Reach for it for "every task must be done at least once, minimise
    → total cost", "each shift needs
// between L and R workers", "cover every row of the matrix at least k
    → times as cheaply as
// possible", and generally any min-cut/flow model where a constraint
    → reads "at least".
// CONSTRUCTION. Send the mandatory  $lo$  units for free by hand, then
    → repair the resulting imbalance:
// base cost +=  $lo * c$ ; the arc keeps capacity  $hi - lo$  at cost  $c$ 
// excess(v) = (sum of  $lo$  entering v) - (sum of  $lo$  leaving v)
// excess(v) > 0 → SS → v with capacity excess(v), cost 0; < 0 →
    → v → TT, cost -excess(v)
// One min cost max flow from SS to TT. FEASIBLE iff it saturates every
    → SS arc; the answer is
// base + that cost. For an s-t flow, add t → s with capacity INF and
    → cost 0 FIRST.

struct MinCostLowerBounds {
    int n, S, T;
    MCMF f;
    ll base = 0, need = 0;
    vector<ll> low, exc;
    vector<array<int, 2>> eid; // {u,
    → index in f.g[u]} per edge
    MinCostLowerBounds(int n) : n(n), S(n), T(n + 1), f(n + 2), exc(n +
    → 2, 0) {}
    void addEdge(int u, int v, ll lo, ll hi, ll cost) {
        low.push_back(lo), eid.push_back({u, (int)f.g[u].size()});
        f.addEdge(u, v, hi - lo, cost);
        base += lo * cost, exc[v] += lo, exc[u] -= lo;
    }
    pair<bool, ll> solve() { //
        → {feasible, minimum total cost}
        for (int v = 0; v < n; v++) {
            if (exc[v] > 0) f.addEdge(S, v, exc[v], 0), need += exc[v];
            else if (exc[v] < 0) f.addEdge(v, T, -exc[v], 0);
        }
        auto [fl, c] = f.run(S, T);
        return {fl == need, base + c};
    }
    ll flowOn(int i) { return low[i] + f.g[eid[i][0]][eid[i][1]].flow; }
};
// MIN COST *MAX* s-t FLOW WITH BOUNDS: add t → s with capacity INF
    → and cost 0, run solve(); the
// flow that ended up on that arc is the feasible s-t flow value. Then
    → delete the arc and run one
// more MCMF from s to t on the residual to push the flow up to maximum.
// MIN COST *MIN* s-t FLOW: same, but afterwards cancel flow by running
    → MCMF from t to s.
// NEGATIVE COSTS: this construction is still correct, but MCMF's
    → Dijkstra is not - either call
// f.setPotentialsBF(S) first (no negative cycle allowed) or switch to
    → 08 - CostScalingMCMF.
// b-FLOW (each vertex has a supply/demand b(v) instead of a single
    → source): put b(v) directly into
// exc[v] before solving - the SS/TT construction is exactly the same
    → machinery.
// SANITY CHECK: if  $lo > hi$  on any edge the instance is infeasible; the
    → code will not notice.

```

3.6.10 MaxDensitySubgraph

MAXIMUM DENSITY SUBGRAPH - choose a NON-EMPTY vertex set S maximising $|E(S)| / |S|$, the number

```
// Needs: Dinic (01 - Dinic.cpp).
// of induced edges per vertex. Undirected, 0-indexed,  $O(\log(n^2) * \maxflow)$  via parametric min cut.
// (Note this is NOT the maximum CLIQUE relaxation - it is polynomial,
//   unlike everything nearby.)
// Reach for it for "find the most tightly knit group", "the subgraph
//   with the best edges-to-people
//   ratio", spam/fraud ring detection, and as the LP-relaxation
//   certificate for densest-subgraph
// style scoring problems.
// FRACTIONAL PROGRAMMING. Ask "is there  $S$  with  $|E(S)| - g|S| > 0$ ?" for
//   a guess  $g$ . Build
//    $s \rightarrow v$  (capacity  $m$ ),  $v \rightarrow t$  (capacity  $m + 2g - \deg(v)$ ),  $u \leftrightarrow v$ 
//   (capacity 1 each way)
// then min cut =  $n * m - 2 * \max \text{ over } S \text{ of } (|E(S)| - g|S|)$ , so the
//   answer is yes iff the min cut is
// strictly below  $n * m$ . Binary search  $g$ . Two distinct densities differ
//   by at least  $1/(n(n-1))$ , so
//  $O(\log(n^2))$  iterations of the binary search are enough for an EXACT
//   answer with rationals.
// Below every capacity is scaled by  $K = n^2$  so they stay INTEGRAL and
//   the guess is  $g = g_2 / K$ .
pair<ld, vector<int>> maxDensitySubgraph(int n, const vector<pair<int,
    int>>& e) {
    int m = e.size();
    if (n == 1 || m == 0) return {0, {0}};
    ll K = (ll)n * n;
    vector<int> deg(n, 0), best{0};
    for (auto [u, v] : e) deg[u]++, deg[v]++;
    ll lo = 0, hi = (ll)m * K;
    // density * K, density <= m
    while (lo < hi) {
        ll g2 = (lo + hi + 1) / 2;
        // test
        // the density  $g = g_2 / K$ 
        Dinic din(n + 2);
        int s = n, t = n + 1;
        for (auto [u, v] : e) din.addEdge(u, v, K, K);
        // undirected, capacity 1 scaled
        for (int v = 0; v < n; v++) {
            din.addEdge(s, v, (ll)m * K);
            din.addEdge(v, t, (ll)m * K + 2 * g2 - (ll)deg[v] * K);
        }
        if (din.calc(s, t) < (ll)n * m * K) {
            // some
            // S beats density g
            lo = g2, best.clear();
            for (int v = 0; v < n; v++) if (din.leftOfMinCut(v))
                best.push_back(v);
            else hi = g2 - 1;
        }
        return {(ld)lo / K, best};
    }
    // {density, the vertex set}
}
// The source side of the min cut IS the densest set - no separate
//   extraction pass is needed.
// MAXIMUM DENSITY with VERTEX WEIGHTS or edge weights: replace  $\deg(v)$ 
//   by the weighted degree and
// the capacity-1 edges by the edge weights; the same cut works.
// AT LEAST k VERTICES / AT MOST k VERTICES turns the problem NP-hard -
//   do not try to patch this.
// MINIMUM density is trivial (a single vertex, density 0), so the
//   question is always maximum.
// RELATED: "maximum (sum of vertex profits) - (sum of edge penalties)"
//   is plain project selection
// (05 - MinCutModelling); the RATIO is what forces the binary search.
// GOLDBERG'S ORIGINAL formulation uses capacities  $m + 2g - \deg(v)$ 
//   exactly as above; if you prefer
// floating point, binary search  $g$  as a double for ~40 iterations and
//   skip the scaling.
```

3.6.11 UniqueMinCut

IS THE MINIMUM s-t CUT UNIQUE? One extra pair of DFS over the residual graph, $O(V + E)$ after the

```
// Needs: Dinic (01 - Dinic.cpp).
// max flow. 0-indexed, directed or undirected.
// After a max flow, let  $A$  = the vertices REACHABLE FROM  $s$  in the
//   residual graph (the smallest
// minimum cut) and  $B$  = the vertices that can REACH  $t$  in the residual
//   graph (the largest minimum
// cut's source side is  $V \setminus B$ ). Every minimum cut lies "between" these
//   two, so:
// the minimum cut is UNIQUE  $\Leftrightarrow A$  and  $B$  together cover every
//   vertex.
// A vertex in neither set can be put on either side without changing
//   the cut value, and moving it
// gives a genuinely different cut of the same weight.
// Reach for it for "is the answer unique / is the bottleneck forced",
//   "which edges are in EVERY
// minimum cut", and for problems that ask you to output AMBIGUOUS when
//   a tie exists.
bool uniqueMinCut(Dinic& din, int s, int t) {
    // call
    // AFTER din.calc(s, t)
    int n = din.adj.size();
    vector<char> a(n, 0), b(n, 0);
    vector<int> st[s];
    a[s] = 1;
    while (!st.empty()) {
        // forward residual reachability
        int u = st.back(); st.pop_back();
        for (auto& e : din.adj[u]) if (e.c > 0 && !a[e.to]) a[e.to] =
            1, st.push_back(e.to);
    }
    st = {t}, b[t] = 1;
    while (!st.empty()) {
        // backward: residual arcs INTO u
        int u = st.back(); st.pop_back();
        for (auto& e : din.adj[u]) if (din.adj[e.to][e.rev].c > 0 &&
            !b[e.to])
            b[e.to] = 1, st.push_back(e.to);
    }
    for (int v = 0; v < n; v++) if (!a[v] && !b[v]) return false;
    return true;
}
// THE TWO EXTREME CUTS come out of the same two searches:  $A$  is the
//   source side of the minimum cut
// with the FEWEST source-side vertices, and  $V \setminus B$  is the one with the
//   MOST.
// AN EDGE IS IN SOME MINIMUM CUT iff it is saturated and its endpoints
//   are in different strongly
// connected components of the residual graph (contract the residual
//   SCCs and the min cuts are
// exactly the closed sets of the resulting DAG).
// AN EDGE IS IN EVERY MINIMUM CUT iff it is saturated and removing its
//   capacity lowers the max
// flow, i.e. iff  $u$  is in  $A$  and  $v$  is in  $B$  - checkable directly from
//   these two arrays.
// COUNTING MINIMUM CUTS = counting antichains/closed sets in that
//   residual-SCC DAG, which is
// #P-hard in general but easy when the DAG is a path or has few
//   vertices.
```

3.7 Matching

3.7.1 HopcroftKarp

HOPCROFT-KARP - maximum bipartite matching in $O(E \sqrt{V})$, the fast alternative to Kuhn's $O(VE)$.

```
// Use it when  $n$  is above a few thousand; below that Kuhn is shorter
//   and fast enough.
// HOPCROFT-KARP - maximum bipartite matching in  $O(E \sqrt{V})$ , the fast
//   alternative to Kuhn's
//  $O(VE)$ . Use above a few thousand vertices; below that Kuhn (02) is
//   shorter and fast enough.
// INDEXING TRAP: leftMatch has size  $m$  and is indexed by a RIGHT
//   vertex; rightMatch has size  $n$  and
// is indexed by a LEFT vertex. Reading the names the obvious way is
//   the default way to get this
// wrong. For the minimum vertex cover / maximum independent set
//   (Konig), see 02 - KuhnAndKonig.
struct HopcroftKarp {
    vector<int> leftMatch, rightMatch, dist, cur;
    vector<vector<int>> > a;
    int n, m;

    HopcroftKarp() {}

    HopcroftKarp(int n, int m) {
```

```

    this->n = n;
    this->m = m;
    a = vector<vector<int>> >(n);
    leftMatch = vector<int>(m, -1);
    rightMatch = vector<int>(n, -1);
    dist = vector<int>(n, -1);
    cur = vector<int>(n, -1);
}

void addEdge(int x, int y) {
    a[x].push_back(y);
}

int bfs() {
    int found = 0;
    queue<int> q;
    for (int i = 0; i < n; i++)
        if (rightMatch[i] < 0) dist[i] = 0, q.push(i);
        else dist[i] = -1;

    while (!q.empty()) {
        int x = q.front();
        q.pop();
        for (int i = 0; i < int(a[x].size()); i++) {
            int y = a[x][i];
            if (leftMatch[y] < 0) found = 1;
            else if (dist[leftMatch[y]] < 0)
                dist[leftMatch[y]] = dist[x] + 1,
                q.push(leftMatch[y]);
        }
    }

    return found;
}

int dfs(int x) {
    for (; cur[x] < int(a[x].size()); cur[x]++) {
        int y = a[x][cur[x]];
        if (leftMatch[y] < 0 || (dist[leftMatch[y]] == dist[x] + 1
        && dfs(leftMatch[y]))) {
            leftMatch[y] = x;
            rightMatch[x] = y;
            return 1;
        }
    }
    return 0;
}

int maxMatching() {
    int match = 0;
    while (bfs()) {
        for (int i = 0; i < n; i++) cur[i] = 0;
        for (int i = 0; i < n; i++)
            if (rightMatch[i] < 0) match += dfs(i);
    }
    return match;
}
};

```

3.7.2 KuhnAndKonig

Kuhn's bipartite matching, $O(V^2E)$ but tiny and easy to modify. Left size n , right size m .

```

// In practice fast; randomise the adjacency order + greedy pre-match
// if you need more speed.
struct Kuhn {
    int n, m;
    vector<vector<int>> adj; // left -> right
    vector<int> mt; // mt[right] = matched
    // left, or -1
    vector<bool> used;

    Kuhn(int n = 0, int m = 0) : n(n), m(m), adj(n), mt(m, -1) {}
    void add_edge(int u, int v) { adj[u].push_back(v); }

    bool try_(int u) {
        for (int v : adj[u]) if (!used[v]) {
            used[v] = true;
            if (mt[v] < 0 || try_(mt[v])) { mt[v] = u; return true; }
        }
        return false;
    }

    int maxMatching() {
        int r = 0;
        vector<int> ord(n);
        iota(all(ord), 0);
        shuffle(all(ord), rng);
        for (int u : ord) { used.assign(m, false); r += try_(u); }
        return r;
    }
};

```

```

}
// KONIG: minimum VERTEX COVER = maximum matching. Recover it:
// alternate from every UNMATCHED left vertex; let Z = reached
// set.
// cover = (left \ Z) union (right and Z); max independent set =
// complement of the cover.
pair<vector<int>, vector<int>> minVertexCover() {
    vector<int> matchL(n, -1);
    for (int v = 0; v < m; v++) if (mt[v] >= 0) matchL[mt[v]] = v;
    vector<bool> zL(n, false), zr(m, false);
    vector<int> st;
    for (int u = 0; u < n; u++) if (matchL[u] < 0) zL[u] = true,
    st.push_back(u);
    while (!st.empty()) {
        int u = st.back(); st.pop_back();
        for (int v : adj[u]) if (!zr[v]) {
            zr[v] = true;
            if (mt[v] >= 0 && !zL[mt[v]]) zL[mt[v]] = true,
            st.push_back(mt[v]);
        }
    }
    vector<int> L, R;
    for (int u = 0; u < n; u++) if (!zL[u]) L.push_back(u);
    for (int v = 0; v < m; v++) if (zr[v]) R.push_back(v);
    return {L, R}; // |L| + |R| ==
    maxMatching()
};
/* THEOREMS you actually quote (bipartite graphs):
Konig: max matching = min vertex cover
max independent set = V - min vertex cover
MIN PATH COVER of a DAG (VERTEX-DISJOINT paths covering all vertices)
= n - max matching of the split graph (u_out -> v_in for
every edge)
CAREFUL: this is the vertex-disjoint cover. To get
Dilworth (FINITE poset)'s width (minimum
number of CHAINS, i.e. paths allowed to share vertices)
run it on the
TRANSITIVE CLOSURE of the DAG first.
Hall: a perfect matching on the left exists iff |N(S)| >= |S|
for every subset S of the LEFT part X
Dilworth: min chain cover of a poset = longest antichain (build the
DAG, TRANSITIVELY CLOSE
it, then use the path cover above)
Mirsky: the dual - longest chain = minimum number of antichains
partitioning the poset
Gallai: alpha + tau = n always; nu + rho = n only if there are NO
ISOLATED VERTICES
(alpha = max independent set, tau = min vertex cover, nu =
max matching,
rho = min edge cover)
Hall defect: nu(G) = |X| - max over W subset of X of (|W| - |N(W)|)
Tutte: G has a perfect matching iff odd(G - U) <= |U| for every U
Tutte-Berge: nu(G) = (1/2) * min over U of (|V| + |U| - odd(G - U))
Menger: min edge cut(s,t) = max edge-disjoint s-t paths;
min vertex cut(s,t) = max internally-vertex-disjoint s-t
paths (s,t non-adjacent) */

```

3.7.3 Hungarian

HUNGARIAN ALGORITHM - minimum-cost perfect matching on a DENSE bipartite graph, $O(n^2 m)$.

```

// a is 1-indexed: a[1..n][1..m] with n <= m. Returns the cost; ans[j]
// = row matched to column j.
// For MAXIMUM cost, negate the matrix.
pair<ll, vector<int>> hungarian(const vector<vector<ll>>& a) {
    int n = a.size() - 1, m = a[0].size() - 1;
    vector<ll> u(n + 1, 0), v(m + 1, 0);
    vector<int> p(m + 1, 0), way(m + 1, 0);
    for (int i = 1; i <= n; i++) {
        p[0] = i;
        int j0 = 0;
        vector<ll> minv(m + 1, llinf);
        vector<char> used(m + 1, false);
        do {
            used[j0] = true;
            int i0 = p[j0], j1 = 0;
            ll delta = llinf;
            for (int j = 1; j <= m; j++) if (!used[j]) {
                ll cur = a[i0][j] - u[i0] - v[j];
                if (cur < minv[j]) minv[j] = cur, way[j] = j0;
                if (minv[j] < delta) delta = minv[j], j1 = j;
            }
            for (int j = 0; j <= m; j++)
                if (used[j]) u[p[j]] += delta, v[j] -= delta;
                else minv[j] -= delta;
            j0 = j1;
        } while (p[j0] != 0);
    }
};

```

```

    do { int j1 = way[j0]; p[j0] = p[j1], j0 = j1; } while (j0);
}
vector<int> ans(m + 1);
for (int j = 1; j <= m; j++) ans[j] = p[j];
return {-v[0], ans}; // -v[0] is the
↳ optimal total cost
}
// Use it for: assignment problems, "match every task to a worker
↳ minimising total time",
// minimum-cost perfect matching in a complete bipartite graph.
// FORBIDDEN PAIRS: use a large finite cost (1e15), NOT llinf - the
↳ reduced cost a[i][j]-u[i]-v[j]
// would overflow. ans[j] == 0 means column j is unmatched (possible
↳ only when n < m).
// WHICH ONE: on a DENSE / complete graph this beats MCMF - n^2*m here
↳ versus n augmentations of
// Dijkstra over n^2 arcs (n^3 log n) there. Use MCMF only when the
↳ graph is genuinely SPARSE.

```

3.7.4 Blossom

MAXIMUM MATCHING IN A GENERAL (NON-BIPARTITE) GRAPH - Edmonds' blossom algorithm, $O(V^3)$.

```

// Everything else in this folder assumes bipartite. Reach for this the
↳ moment the two sides are
// not fixed: "pair up people who are compatible", "tile a board where
↳ the colouring argument
// fails", "maximum set of disjoint edges" on an arbitrary graph.
// The trick it exists for: an ODD cycle can be traversed in two
↳ directions, so an alternating
// search can enter and leave it on the same side. Edmonds CONTRACTS
↳ such a cycle (a "blossom")
// into a single vertex, searches the contracted graph, and expands
↳ afterwards.
// Certificate: the answer equals the Tutte-Berge bound (see 10 -
↳ Theorems/01 - GraphTheorems).
struct Blossom {
    int n;
    vector<vector<char>> g; //
    ↳ adjacency MATRIX
    vector<int> match, p, base;
    vector<char> used, inBlossom;
    Blossom(int n) : n(n), g(n, vector<char>(n, 0)), match(n, -1),
    ↳ p(n), base(n),
        used(n, inBlossom(n)) {}
    void addEdge(int u, int v) { g[u][v] = g[v][u] = 1; }
    int lca(int a, int b) {
        vector<char> seen(n, 0);
        for (;;) { a = base[a], seen[a] = 1; if (match[a] < 0) break; a
    ↳ = p[match[a]]; }
        for (;;) { b = base[b]; if (seen[b]) return b; b = p[match[b]];
    ↳ }
    }
    void mark(int v, int b, int child) { // walk
    ↳ v up to the blossom base
        while (base[v] != b) {
            inBlossom[base[v]] = inBlossom[base[match[v]]] = 1;
            p[v] = child, child = match[v], v = p[match[v]];
        }
    }
    bool augment(int root) { // one
    ↳ BFS for an augmenting path
        fill(all(used), 0), fill(all(p), -1);
        iota(all(base), 0);
        used[root] = 1;
        queue<int> q({root});
        while (!q.empty()) {
            int v = q.front(); q.pop();
            for (int to = 0; to < n; to++) {
                if (!g[v][to] || base[v] == base[to] || match[v] == to)
    ↳ continue;
                if (to == root || (match[to] >= 0 && p[match[to]] >=
    ↳ 0)) { // odd cycle -> contract
                    int b = lca(v, to);
                    fill(all(inBlossom), 0);
                    mark(v, b, to), mark(to, b, v);
                    for (int i = 0; i < n; i++) if (inBlossom[base[i]])
    ↳ {
                        base[i] = b;
                        if (!used[i]) used[i] = 1, q.push(i);
                    }
                } else if (p[to] < 0) {
                    p[to] = v;
                    if (match[to] < 0) { // free
    ↳ vertex: flip the path
                        for (int u = to, pv, nv; u >= 0; u = nv)
                            pv = p[u], nv = match[pv], match[u] = pv,
    ↳ match[pv] = u;

```

```

        return true;
    }
    used[match[to]] = 1, q.push(match[to]);
}
}
return false;
}
int solve() {
    int r = 0;
    for (int v = 0; v < n; v++) //
    ↳ greedy start: fewer BFS rounds
        if (match[v] < 0)
            for (int u = 0; u < n; u++)
                if (g[v][u] && match[u] < 0) { match[v] = u,
    ↳ match[u] = v, r++; break; }
            for (int v = 0; v < n; v++) if (match[v] < 0) r += augment(v);
            return r; //
    ↳ match[v] = partner, or -1
    }
};
// PERFECT MATCHING EXISTS iff solve() * 2 == n. Tutte's theorem gives
↳ the certificate when it
// does not: some U with odd(G - U) > |U|.
// MINIMUM EDGE COVER (no isolated vertices) = n - maximum matching
↳ (Gallai).
// MAXIMUM INDEPENDENT SET is still NP-hard here - Konig's alpha = n -
↳ nu is BIPARTITE ONLY.
// WEIGHTED general matching needs the dual-variable version (~150
↳ lines); if the graph is small
// (n <= 20) a bitmask DP over matched vertices is far shorter and
↳ usually enough.
// CHINESE POSTMAN (undirected): make every degree even by duplicating
↳ a minimum-weight perfect
// matching on the ODD-degree vertices, using all-pairs shortest paths
↳ as the edge weights, then
// take an Eulerian circuit. The MIXED version (some edges directed) is
↳ NP-hard.
// THE MATCHING GAME (a token moves along edges to unvisited vertices;
↳ stuck loses): the first
// player wins iff the start vertex is covered by EVERY maximum
↳ matching, i.e. iff deleting it
// decreases the matching size. With this file that works on a general
↳ graph, not just bipartite.

```

3.7.5 BipartiteEdgeColouring

BIPARTITE EDGE COLOURING in exactly D colours (D = maximum degree), $O(m * D)$. 0-indexed.

```

// KONIG'S EDGE COLOURING THEOREM: a bipartite (multi)graph is always
↳ D-edge-colourable, and D is
// obviously necessary. Equivalently: the edges split into D perfect
↳ matchings of the D-regular
// completion, which is the "round robin timetable" primitive.
// Reach for it for "schedule all the games/lectures in the fewest
↳ rounds so nobody plays twice in
// one round", "decompose a bipartite multigraph into matchings",
↳ "Latin square completion",
// and for the routing step of a Clos/Benes network.
// The insertion argument: give edge (u, v) the smallest colour cu free
↳ at u and look at the
// smallest colour cv free at v. If they differ, the cu/cv alternating
↳ path starting at v cannot
// reach u (it would have to ENTER u by cu, which is free there), so
↳ flipping the two colours
// along it frees cu at v as well - then colour the edge cu.
vector<int> bipartiteEdgeColouring(int nl, int nr, const
↳ vector<pair<int, int>>& e) {
    int m = e.size(), D = 0;
    vector<int> dl(nl, 0), dr(nr, 0);
    for (auto [u, v] : e) D = max({D, ++dl[u], ++dr[v]});
    vector<vector<int>> L(nl, vector<int>(D, -1)), R(nr, vector<int>(D,
    ↳ -1)); // colour -> edge id
    vector<int> col(m, -1);
    for (int i = 0; i < m; i++) {
        auto [u, v] = e[i];
        int cu = 0, cv = 0;
        while (L[u][cu] >= 0) cu++;
        while (R[v][cv] >= 0) cv++;
        if (cu != cv) { // walk
    ↳ the cu/cv alternating path
            vector<int> path;
            for (int x = v, side = 1, c = cu;;) {
                int id = side ? R[x][c] : L[x][c];
                if (id < 0) break;
                path.push_back(id);
                x = side ? e[id].first : e[id].second, side ^= 1, c =
    ↳ (c == cu ? cv : cu);

```

```

    }
    for (int id : path) L[e[id].first][col[id]] =
    ↪ R[e[id].second][col[id]] = -1;
    for (int id : path) {
        col[id] = (col[id] == cu ? cv : cu);
        L[e[id].first][col[id]] = R[e[id].second][col[id]] = id;
    }
    col[i] = cu, L[u][cu] = R[v][cu] = i;
}
return col; //
↪ col[i] in [0, D)
}
// The number of colours actually used IS the maximum degree - no
↪ search needed.
// EQUAL-SIZE ROUNDS: to split the edges into k groups whose sizes
↪ differ by at most one and where
// each vertex has at most ceil(deg/k) edges per group, split every
↪ vertex into ceil(deg/k) copies
// of degree <= k and edge-colour that graph.
// GENERAL (non-bipartite) graphs need D or D+1 colours - Vizing (06),
↪ and deciding which is
// NP-hard. On a bipartite graph the answer is always D, which is why
↪ this file is separate.
// A REGULAR bipartite graph decomposes into perfect matchings; taking
↪ the colour classes one at a
// time is much faster than running a matching algorithm D times.

```

3.7.6 VizingEdgeColouring

VIZING EDGE COLOURING - colour the edges of a SIMPLE undirected graph with at most $D+1$ colours

```

// (D = maximum degree), constructively, in  $O(n * m)$ . 0-indexed in,
↪ colours returned 0-indexed.
// VIZING'S THEOREM says D or D+1 always suffices; deciding WHICH is
↪ NP-hard, so D+1 is the answer
// you ship. (Bipartite graphs always need only D - use 05 instead, it
↪ is much shorter.)
// Reach for it for "split the matches/tasks into as few rounds as
↪ possible so no player/machine is
// used twice per round" on a general graph, and for "orient/partition
↪ the edges into D+1 matchings"
// arguments. A D+1 colouring also proves the D+1 bound constructively,
↪ which some problems ask for.
// MULTIGRAPHS are NOT covered (Shannon's bound is  $3D/2$  there); remove
↪ parallel edges first.
// Method: insert the edges one at a time. For a new edge (u, v) build
↪ a MAXIMAL FAN around u -
// a sequence of neighbours where each one's free colour is used by the
↪ next - and either rotate the
// fan (when some colour is free at both ends) or flip an alternating
↪ two-colour path and rotate.
// C[u][c] = the neighbour joined to u by colour c (0 = none); X[u] =
↪ smallest free colour at u.
// Vertices are shifted to 1..n inside, because 0 doubles as the "no
↪ such vertex" sentinel.
vector<int> vizing(int n, const vector<pair<int, int>>& e) {
    int D = 0;
    vector<int> deg(n, 0);
    for (auto [u, v] : e) D = max({D, ++deg[u], ++deg[v]});
    int K = D + 2; //
    ↪ colours 1..D+1, 0 = uncoloured
    vector<vector<int>> C(n + 1, vector<int>(K + 1, 0)), col(n + 1,
    ↪ vector<int>(n + 1, 0));
    vector<int> X(n + 1, 1);
    auto update = [&](int u) { for (X[u] = 1; C[u][X[u]]; X[u]++); };
    auto color = [&](int u, int v, int c) { // give
    ↪ edge (u,v) colour c
        int p = col[u][v];
        col[u][v] = col[v][u] = c, C[u][c] = v, C[v][c] = u, C[u][p] =
    ↪ C[v][p] = 0;
        if (p) X[u] = X[v] = p; else update(u), update(v);
        return p;
    };
    auto flip = [&](int u, int c1, int c2) { // swap
    ↪ colours c1/c2 at u
        int p = C[u][c1];
        swap(C[u][c1], C[u][c2]);
        if (p) col[u][p] = col[p][u] = c2;
        if (!C[u][c1]) X[u] = c1;
        if (!C[u][c2]) X[u] = c2;
        return p;
    };
    for (int t = 0; t < (int)e.size(); t++) {
        int u = e[t].first + 1, v0 = e[t].second + 1, v = v0, c0 =
    ↪ X[u], c = c0, d = 0, a;
        vector<pair<int, int>> L; // the
        ↪ fan: (neighbour, its free col)

```

```

        vector<char> vis(K + 1, 0);
        while (!col[u][v0]) {
            L.push_back({v, d = X[v]});
            if (!C[v][c]) for (a = (int)L.size() - 1; a >= 0; a--) c =
    ↪ color(u, L[a].first, c);
            else if (!C[u][d])
                for (a = (int)L.size() - 1; a >= 0; a--) color(u,
    ↪ L[a].first, L[a].second);
            else if (vis[d]) break; // fan
    ↪ is maximal -> flip a path
            else vis[d] = 1, v = C[u][d];
        }
        if (!col[u][v0]) {
            for (; v; v = flip(v, c, d), swap(c, d)); //
    ↪ alternating c/d path from v
            if (C[u][c0]) {
                for (a = (int)L.size() - 2; a >= 0 && L[a].second != c;
    ↪ a--);
                for (; a >= 0; a--) color(u, L[a].first, L[a].second);
            } else t--; // retry
    ↪ this edge, now it fits
        }
    }
    vector<int> res(e.size());
    for (int i = 0; i < (int)e.size(); i++) res[i] = col[e[i].first +
    ↪ 1][e[i].second + 1] - 1;
    return res; //
    ↪ res[i] in [0, D]
}
// D IS ENOUGH (class 1) for: bipartite graphs, every graph with a
↪ vertex of degree D that is not
// "overfull", and every planar graph with  $D \geq 7$ . D+1 is forced (class
↪ 2) for odd cycles, the
// Petersen graph, and any OVERFULL graph ( $m > D * \text{floor}(n/2)$ ) - a
↪ cheap sufficient test.
// EDGE COLOURING = VERTEX COLOURING OF THE LINE GRAPH, so  $\chi'(G) =$ 
↪  $\chi(L(G))$ ; that is usually the
// wrong direction to compute, but it is the right way to see a problem
↪ stated on the line graph.
// MEMORY is  $O(n^2)$  for the col matrix; at  $n \sim 2000$  that is 16 MB of
↪ int - use short if it is tight.

```

3.7.7 WeightedBlossom

MAXIMUM WEIGHT MATCHING IN A GENERAL GRAPH - blossom with DUAL VARIABLES, $O(n^3)$, $O(n^2)$ memory.

```

// 1-INDEXED (vertices 1..n; index 0 is the "none" sentinel and 0
↪ weight means "no edge").
// This is the weighted counterpart of 04 - Blossom. It returns the
↪ maximum TOTAL WEIGHT over all
// matchings (NOT necessarily a maximum-cardinality one) together with
↪ how many edges it used.
// Reach for it when the pairs are not bipartite and the pairing has a
↪ value: "pair up the people
// to maximise total compatibility", Chinese postman (08), "minimum
↪ cost to make every degree even",
// and any problem where Hungarian would work if only the graph had two
↪ sides.
// RECIPES
// MINIMUM weight PERFECT matching: add a big constant B to every
↪ edge ( $B \geq \text{sum of } |w|$ ) and
// negate, i.e. addEdge(u, v, B - w); a perfect matching then
↪ dominates any smaller one, and the
// answer is  $B * n/2 - \text{result}$ . Verify solve().second * 2 == n before
↪ trusting it.
// MAXIMUM CARDINALITY first, weight second: use  $w * (n + 1) + 1$  per
↪ edge.
// The duals lab[] are Edmonds' y-values; a blossom b keeps lab[b] = 2
↪ * (its dual). Every step
// either tightens an edge (slack 0), contracts a blossom, or expands
↪ one with zero dual.
struct WeightedBlossom {
    struct E { int u, v; ll w; };
    int n, nx, tk = 0;
    vector<vector<E>> g;
    vector<vector<int>> ff, flower; // ff =
    ↪ which sub-blossom x came from
    vector<int> match, slack, st, pa, S, vis;
    vector<ll> lab;
    deque<int> q;
    WeightedBlossom(int n) : n(n), nx(n), g(2 * n + 1, vector<E>(2 * n
    ↪ + 1)),
        ff(2 * n + 1, vector<int>(2 * n + 1, 0)), flower(2 * n + 1,
    ↪ match(2 * n + 1, 0),
        slack(2 * n + 1, 0), st(2 * n + 1, 0), pa(2 * n + 1, 0), S(2 *
    ↪ n + 1, 0),
        vis(2 * n + 1, 0), lab(2 * n + 1, 0) {
        for (int u = 1; u <= 2 * n; u++) for (int v = 1; v <= 2 * n;

```

```

↪ v++) g[u][v] = {u, v, 0};
}
void addEdge(int u, int v, ll w) { g[u][v].w = g[v][u].w =
↪ max(g[u][v].w, w); }
ll d(const E& e) { return lab[e.u] + lab[e.v] - g[e.u][e.v].w * 2;
↪ } // slack
void upSlack(int u, int x) { if (!slack[x] || d(g[u][x]) <
↪ d(g[slack[x]][x])) slack[x] = u; }
void setSlack(int x) {
    slack[x] = 0;
    for (int u = 1; u <= n; u++)
        if (g[u][x].w > 0 && st[u] != x && S[st[u]] == 0)
↪ upSlack(u, x);
}
void qPush(int x) { if (x <= n) q.push_back(x); else for (int y :
↪ flower[x]) qPush(y); }
void setSt(int x, int b) { st[x] = b; if (x > n) for (int y :
↪ flower[x]) setSt(y, b); }
int getPr(int b, int xr) {
    int pr = find(all(flower[b]), xr) - flower[b].begin();
    if (pr & 1) {
        reverse(flower[b].begin() + 1, flower[b].end());
        return (int)flower[b].size() - pr;
    }
    return pr;
}
void setMatch(int u, int v) {
    match[u] = g[u][v].v;
    if (u <= n) return;
    E e = g[u][v];
    int xr = ff[u][e.u], pr = getPr(u, xr);
    for (int i = 0; i < pr; i++) setMatch(flower[u][i], flower[u][i
↪ 1]);
    setMatch(xr, v);
    rotate(flower[u].begin(), flower[u].begin() + pr,
↪ flower[u].end());
}
void augment(int u, int v) {
    int nv = st[match[u]];
    setMatch(u, v);
    if (!nv) return;
    setMatch(nv, st[pa[nv]]), augment(st[pa[nv]], nv);
}
int getLca(int u, int v) {
    for (++tk; u || v; swap(u, v)) {
        if (!u) continue;
        if (vis[u] == tk) return u;
        vis[u] = tk, u = st[match[u]];
        if (u) u = st[pa[u]];
    }
    return 0;
}
void addBlossom(int u, int lca, int v) { //
↪ contract the odd cycle u..lca..v
    int b = n + 1;
    while (b <= nx && st[b]) ++b;
    if (b > nx) ++nx;
    lab[b] = 0, S[b] = 0, match[b] = match[lca];
    flower[b].assign(1, lca);
    for (int x = u, y; x != lca; x = st[pa[y]])
        flower[b].push_back(x), flower[b].push_back(y =
↪ st[match[x]]), qPush(y);
    reverse(flower[b].begin() + 1, flower[b].end());
    for (int x = v, y; x != lca; x = st[pa[y]])
        flower[b].push_back(x), flower[b].push_back(y =
↪ st[match[x]]), qPush(y);
    setSt(b, b);
    for (int x = 1; x <= nx; x++) g[b][x].w = g[x][b].w = 0;
    for (int x = 1; x <= n; x++) ff[b][x] = 0;
    for (int xs : flower[b]) {
        for (int x = 1; x <= nx; x++)
            if (g[b][x].w == 0 || d(g[xs][x]) < d(g[b][x]))
                g[b][x] = g[xs][x], g[x][b] = g[x][xs];
        for (int x = 1; x <= n; x++) if (ff[xs][x]) ff[b][x] = xs;
    }
    setSlack(b);
}
void expandBlossom(int b) { // only
↪ if S[b] == 1, lab[b] == 0
    for (int x : flower[b]) setSt(x, x);
    int xr = ff[b][g[b][pa[b]].u], pr = getPr(b, xr);
    for (int i = 0; i < pr; i += 2) {
        int xs = flower[b][i], xns = flower[b][i + 1];
        pa[xs] = g[xns][xs].u, S[xs] = 1, S[xns] = 0;
        slack[xs] = 0, setSlack(xns), qPush(xns);
    }
    S[xr] = 1, pa[xr] = pa[b];
}

```

```

    for (int i = pr + 1; i < (int)flower[b].size(); i++)
↪ S[flower[b][i]] = -1,
        setSlack(flower[b][i]);
    st[b] = 0;
}
bool onFound(const E& e) {
    int u = st[e.u], v = st[e.v];
    if (S[v] == -1) {
        pa[v] = e.u, S[v] = 1;
        int nu = st[match[v]];
        slack[v] = slack[nu] = 0, S[nu] = 0, qPush(nu);
    } else if (S[v] == 0) {
        int lca = getLca(u, v);
        if (!lca) return augment(u, v), augment(v, u), true;
        addBlossom(u, lca, v);
    }
    return false;
}
bool matching() { // one
↪ augmentation, or false
    fill(S.begin(), S.begin() + nx + 1, -1);
    fill(slack.begin(), slack.begin() + nx + 1, 0);
    q.clear();
    for (int x = 1; x <= nx; x++) if (st[x] == x && !match[x])
↪ pa[x] = 0, S[x] = 0, qPush(x);
    if (q.empty()) return false;
    for (;;) {
        while (!q.empty()) {
            int u = q.front(); q.pop_front();
            if (S[st[u]] == 1) continue;
            for (int v = 1; v <= n; v++) if (g[u][v].w > 0 && st[u]
↪ != st[v]) {
                if (d(g[u][v]) == 0) { if (onFound(g[u][v])) return
↪ true; }
                else upSlack(u, st[v]);
            }
        }
        ll dt = llinf; // dual
↪ update step
        for (int b = n + 1; b <= nx; b++) if (st[b] == b && S[b] ==
↪ 1) dt = min(dt, lab[b] / 2);
        for (int x = 1; x <= nx; x++) if (st[x] == x && slack[x]) {
            if (S[x] == -1) dt = min(dt, d(g[slack[x]][x]));
            else if (S[x] == 0) dt = min(dt, d(g[slack[x]][x]) / 2);
        }
        for (int u = 1; u <= n; u++) {
            if (S[st[u]] == 0) { if (lab[u] <= dt) return false;
↪ lab[u] -= dt; }
            else if (S[st[u]] == 1) lab[u] += dt;
        }
        for (int b = n + 1; b <= nx; b++) if (st[b] == b) {
            if (S[st[b]] == 0) lab[b] += dt * 2;
            else if (S[st[b]] == 1) lab[b] -= dt * 2;
        }
        q.clear();
        for (int x = 1; x <= nx; x++)
            if (st[x] == x && slack[x] && st[slack[x]] != x &&
↪ d(g[slack[x]][x]) == 0)
                if (onFound(g[slack[x]][x])) return true;
        for (int b = n + 1; b <= nx; b++)
            if (st[b] == b && S[b] == 1 && lab[b] == 0)
↪ expandBlossom(b);
    }
}
pair<ll, int> solve() { //
↪ {total weight, #matched edges}
    fill(match.begin(), match.begin() + n + 1, 0);
    nx = n;
    ll ans = 0, wmax = 0;
    int cnt = 0;
    for (int u = 0; u <= n; u++) st[u] = u, flower[u].clear();
    for (int u = 1; u <= n; u++) for (int v = 1; v <= n; v++)
        ff[u][v] = (u == v ? u : 0), wmax = max(wmax, g[u][v].w);
    for (int u = 1; u <= n; u++) lab[u] = wmax;
    while (matching()) ++cnt;
    for (int u = 1; u <= n; u++) if (match[u] && match[u] < u) ans
↪ += g[u][match[u]].w;
    return {ans, cnt}; //
↪ match[u] = partner of u, or 0
}
};
// WEIGHTS MUST BE NON-NEGATIVE here (an edge of weight 0 is
↪ indistinguishable from a non-edge);
// shift them if needed. Everything is integral - with real weights use
↪ long double and an eps.
// FASTER ALTERNATIVES: if n <= 20, a bitmask DP over matched vertices
↪ is 5 lines and beats this.

```

```
// If the graph is bipartite, use Hungarian (03) -  $O(n^3)$  too but with
    ↳ a far smaller constant.
// TUTTE MATRIX: for the UNWEIGHTED decision "is there a perfect
    ↳ matching", a random skew-symmetric
// matrix over a large field has full rank with high probability iff
    ↳ one exists -  $O(n^3)$  and 10
// lines, and rank/2 is the maximum matching size.
```

3.7.8 ChinesePostman

CHINESE POSTMAN (route inspection) - the cheapest CLOSED WALK that uses EVERY edge at least

```
// Needs: WeightedBlossom (07 - WeightedBlossom.cpp).
// once, on an undirected connected weighted graph. 0-indexed in,
    ↳  $O(n^3)$  via general matching.
// THEORY. A closed walk using every edge exists with no repeats iff
    ↳ every degree is even (Euler).
// Otherwise you must DUPLICATE some edges, and the duplicated set must
    ↳ fix exactly the odd-degree
// vertices - i.e. it is a union of paths pairing them up. So: take
    ↳ all-pairs shortest paths, build
// the complete graph on the ODD-degree vertices (there is an even
    ↳ number of them, by handshake),
// and find a MINIMUM WEIGHT PERFECT MATCHING there. Answer = total
    ↳ edge weight + matching weight.
// Reach for it for "walk every street and come back, minimum
    ↳ distance", "traverse every corridor",
// and "make every degree even at minimum cost" (that last one IS the
    ↳ matching part alone).
ll chinesePostman(int n, const vector<array<ll, 3>>& e) { // e =
    ↳ {u, v, w}, undirected
    vector<vector<ll>> d(n, vector<ll>(n, llinf));
    vector<int> deg(n, 0);
    ll total = 0;
    for (int i = 0; i < n; i++) d[i][i] = 0;
    for (auto [u, v, w] : e) {
        d[u][v] = d[v][u] = min(d[u][v], w), total += w, deg[u]++,
    ↳ deg[v]++;
    }
    for (int k = 0; k < n; k++) for (int i = 0; i < n; i++) for (int j
    ↳ = 0; j < n; j++)
        if (d[i][k] < llinf && d[k][j] < llinf) d[i][j] = min(d[i][j],
    ↳ d[i][k] + d[k][j]);
    vector<int> odd;
    for (int i = 0; i < n; i++) if (deg[i] & 1) odd.push_back(i);
    int k = odd.size();
    if (!k) return total; //
    ↳ already Eulerian
    ll mx = 0; // shift
    ↳ so that MIN becomes MAX and
    for (int i = 0; i < k; i++) for (int j = 0; j < k; j++) // a
    ↳ PERFECT matching always wins
        if (d[odd[i]][odd[j]] < llinf) mx = max(mx, d[odd[i]][odd[j]]);
    ll B = mx * k + 1;
    WeightedBlossom M(k);
    for (int i = 0; i < k; i++) for (int j = i + 1; j < k; j++)
        if (d[odd[i]][odd[j]] < llinf) M.addEdge(i + 1, j + 1, B -
    ↳ d[odd[i]][odd[j]]);
    auto [val, cnt] = M.solve();
    if (cnt * 2 != k) return -1; // odd
    ↳ vertices cannot be paired
    return total + (ll)cnt * B - val;
}
// RECOVERING THE ROUTE: duplicate every edge on the shortest path
    ↳ between each matched pair, then
// run an Eulerian circuit (05 - Eulerian Path) on the resulting
    ↳ even-degree multigraph.
// DIRECTED CHINESE POSTMAN is a MIN COST FLOW, not a matching: give
    ↳ each vertex a supply of
// (outdeg - indeg), send flow from the surplus vertices to the deficit
    ↳ ones along the original
// arcs with cost = weight and infinite capacity, and add that cost to
    ↳ the total weight.
// MIXED (some edges directed, some not) is NP-hard.
// RURAL POSTMAN (only a SUBSET of edges must be covered) is NP-hard as
    ↳ well - it contains TSP.
// OPEN version ("start anywhere, end anywhere"): leave two odd
    ↳ vertices unmatched, i.e. run a
// maximum-weight (not perfect) matching, which is what solve() gives
    ↳ you without the count check.
```

3.8.1 MST

MINIMUM SPANNING TREE. Kruskal $O(m \log m)$ - use on sparse graphs / when you also want the

```
// Needs: DSU (02 - Data Structures/05 - DSU/01).
// KRUSKAL RECONSTRUCTION TREE. Prim  $O(n^2)$  - use on dense or IMPLICIT
    ↳ complete graphs.
struct Edge { int u, v; ll w; bool operator<(const Edge& o) const {
    ↳ return w < o.w; } };

ll kruskal(int n, vector<Edge> e, vector<Edge>* used = nullptr) {
    sort(all(e));
    DSU d(n);
    ll tot = 0;
    for (auto& x : e) if (d.join(x.u, x.v)) { tot += x.w; if (used)
    ↳ used->push_back(x); }
    return tot; // check d.size(0) ==
    ↳ n for connectivity
}
// Dense / implicit graphs: cost(i,j) given by a function.  $O(n^2)$ , no
    ↳ edge list needed.
template <class F>
ll prim(int n, F cost) {
    vector<ll> best(n, llinf);
    vector<bool> in(n, false);
    best[0] = 0;
    ll tot = 0;
    for (int it = 0; it < n; it++) {
        int u = -1;
        for (int i = 0; i < n; i++) if (!in[i] && (u < 0 || best[i] <
    ↳ best[u])) u = i;
        in[u] = true, tot += best[u];
        for (int v = 0; v < n; v++) if (!in[v]) best[v] = min(best[v],
    ↳ cost(u, v));
    }
    return tot;
}
// KRUSKAL RECONSTRUCTION TREE: binary tree with 2n-1 nodes; internal
    ↳ node = merging edge weight.
// Then min possible MAX edge on a path u->v = val[lca(u,v)], and "all
    ↳ vertices reachable using
// edges of weight <= w" is exactly a subtree. Turns "binary search the
    ↳ threshold + DSU" into  $O(1)$ .
struct KRT {
    int n, cnt;
    vector<ll> val;
    vector<array<int, 2>> ch;
    KRT(int n, vector<Edge> e) : n(n), cnt(n), val(2 * n, 0), ch(2 * n,
    ↳ {-1, -1}) {
        sort(all(e));
        DSU d(2 * n);
        vector<int> top(2 * n);
        iota(all(top), 0);
        for (auto& x : e) {
            int a = top[d.find(x.u)], b = top[d.find(x.v)];
            if (a == b) continue;
            val[cnt] = x.w, ch[cnt] = {a, b};
            d.join(x.u, x.v);
            top[d.find(x.u)] = cnt++;
        }
        // root = cnt-1 (if
    ↳ connected); then build LCA on it
};
// SECOND-BEST MST: build the MST, then for each non-tree edge (u,v,w)
    ↳ the answer candidate is
// mst - maxEdgeOnPath(u,v) + w. Get maxEdgeOnPath with binary lifting
    ↳ or the KRT.
```

3.8 Spanning Trees

3.8.2 DirectedMSTandExtras

DIRECTED MST (Chu-Liu / Edmonds) - minimum spanning ARBORESCENCE rooted at r , $O(V^3E)$.

```
// Needs: struct Edge {int u,v; ll w;} and DSU (both from 01 - MST.cpp).
// Every non-root vertex takes its cheapest incoming edge; if that
//   ↳ creates a cycle, contract it,
// subtract the chosen weights, and repeat. Returns -1 if some vertex
//   ↳ is unreachable from  $r$ .
ll directedMST(int n, int r, vector<Edge> e) { // Edge = {u, v,
    ↳ w} meaning u → v
    ll total = 0;
    vector<int> id(n), par(n);
    while (true) {
        vector<ll> best(n, LLINF);
        vector<int> from(n, -1);
        for (auto& x : e) if (x.u != x.v && x.w < best[x.v]) best[x.v]
        ↳ = x.w, from[x.v] = x.u;
        best[r] = 0;
        for (int i = 0; i < n; i++) if (i != r && from[i] < 0) return
        ↳ -1; // unreachable
        int cnt = 0;
        fill(all(id), -1), fill(all(par), -1);
        for (int i = 0; i < n; i++) {
            if (i != r) total += best[i];
            int v = i;
            while (v != r && par[v] == -1 && id[v] == -1) par[v] = i, v
        ↳ = from[v];
            if (v != r && id[v] == -1 && par[v] == i) { // found a new
        ↳ cycle
                for (int u = from[v]; u != v; u = from[u]) id[u] = cnt;
                id[v] = cnt++;
            }
            if (!cnt) break; // no cycle →
        ↳ done
        for (int i = 0; i < n; i++) if (id[i] == -1) id[i] = cnt++;
        for (auto& x : e) {
            int u = x.u, v = x.v;
            x.u = id[u], x.v = id[v];
            if (x.u != x.v) x.w -= best[v]; // reduced cost
        ↳ inside the contraction
        }
        n = cnt, r = id[r];
    }
    return total;
}

// SECOND-BEST MST,  $O(E \log V + E * \log V)$ . The second-best tree
//   ↳ differs from the MST by exactly
//   ↳ ONE edge swap. For each non-tree edge (u,v,w) the candidate is mst -
//   ↳ maxEdgeOnPath(u,v) + w.
// You must track BOTH the maximum and the strict second maximum on the
//   ↳ path, because if the
//   ↳ maximum equals w the swap is not a real change - use the second
//   ↳ maximum instead.
// With binary lifting: up[k][v], mx1[k][v], mx2[k][v] merged pairwise
//   ↳ while lifting.
// (The Kruskal Reconstruction Tree in 01 - MST.cpp gives maxEdgeOnPath
//   ↳ directly as val[lca].)

// BORUVKA - MST in  $O(E \log V)$  by phases; each phase every component
//   ↳ picks its cheapest outgoing
//   ↳ edge and they are all merged at once. Use it when the edge set is
//   ↳ IMPLICIT:
// XOR-MST : cheapest cross edge per component via a binary trie
// Manhattan MST: cheapest cross edge per octant (see Transforms.cpp)
ll boruvka(int n, const vector<Edge>& e) {
    DSU d(n);
    ll total = 0;
    int comps = n;
    while (comps > 1) {
        vector<int> best(n, -1);
        for (int i = 0; i < (int)e.size(); i++) {
            int a = d.find(e[i].u), b = d.find(e[i].v);
            if (a == b) continue;
            if (best[a] < 0 || e[i].w < e[best[a]].w) best[a] = i;
            if (best[b] < 0 || e[i].w < e[best[b]].w) best[b] = i;
        }
        bool any = false;
        for (int i = 0; i < n; i++) if (best[i] >= 0) {
            auto& x = e[best[i]];
            if (d.join(x.u, x.v)) total += x.w, comps--, any = true;
        }
        if (!any) break; // disconnected
    }
    return comps == 1 ? total : -1;
}
```

3.8.3 SecondBestMST

SECOND-BEST SPANNING TREE - cheapest spanning tree whose weight is STRICTLY greater than the MST.

```
// Needs: Edge, kruskal (01 - MST.cpp).
// It differs from the MST in exactly one edge: add a non-tree edge
//   ↳ (u,v,w), drop the heaviest edge
// on the tree path u→v whose weight is < w. By the cycle property
//   ↳ every edge on that path has
// weight ≤ w, so "heaviest edge < w" is either the path maximum (if
//   ↳ it is < w) or the second
// LARGEST DISTINCT value on the path. Binary lifting carries both.  $O(m \log n)$ .
ll secondBestMST(int n, const vector<Edge>& e) {
    vector<Edge> used;
    ll mst = kruskal(n, e, &used);
    if ((int)used.size() != n - 1) return -1; //
    ↳ disconnected
    vector<vector<pair<int, ll>>> g(n);
    for (auto& x : used) g[x.u].push_back({x.v, x.w});
    ↳ g[x.v].push_back({x.u, x.w});
    typedef array<ll, 2> T; //
    ↳ {largest, 2nd largest DISTINCT}
    const T NONE = {LLONG_MIN, LLONG_MIN};
    auto join = [&](T a, T b) {
        array<ll, 4> v = {a[0], a[1], b[0], b[1]};
        sort(v.rbegin(), v.rend());
        T r = {v[0], LLONG_MIN};
        for (ll x : v) if (x < v[0]) { r[1] = x; break; }
        return r;
    };
    int L = 1;
    while ((1 << L) < n) L++;
    vector<vector<int>>> up(L + 1, vector<int>(n, 0));
    vector<vector<T>>> mx(L + 1, vector<T>(n, NONE));
    vector<int> dep(n, 0), st{0};
    vector<char> vis(n, 0);
    vis[0] = 1;
    while (!st.empty()) {
        ↳ iterative DFS from 0
        int u = st.back(); st.pop_back();
        for (auto [v, w] : g[u]) if (!vis[v])
            vis[v] = 1, dep[v] = dep[u] + 1, up[0][v] = u, mx[0][v] =
        ↳ {w, LLONG_MIN}, st.push_back(v);
    }
    for (int k = 1; k <= L; k++)
        for (int v = 0; v < n; v++) {
            int m = up[k - 1][v];
            up[k][v] = up[k - 1][m], mx[k][v] = join(mx[k - 1][v], mx[k
        ↳ - 1][m]);
        }
    auto path = [&](int u, int v) { // top-2
        ↳ distinct weights on u→v
        T r = NONE;
        if (dep[u] < dep[v]) swap(u, v);
        for (int k = L; k >= 0; k--) if (dep[up[k][u]] >= dep[v]) r =
        ↳ join(r, mx[k][u]), u = up[k][u];
        if (u == v) return r;
        for (int k = L; k >= 0; k--) if (up[k][u] != up[k][v])
            r = join(join(r, mx[k][u]), mx[k][v]), u = up[k][u], v =
        ↳ up[k][v];
        return join(join(r, mx[0][u]), mx[0][v]);
    };
    ll best = LLONG_MAX;
    for (auto& x : e) { // tree
        ↳ edges give p = {w, MIN} and
        T p = path(x.u, x.v); // fall
        ↳ through to the MIN branch
        ll rem = p[0] < x.w ? p[0] : p[1];
        if (rem != LLONG_MIN) best = min(best, mst - rem + x.w);
    }
    return best == LLONG_MAX ? -1 : best; // -1:
    ↳ no second tree exists
}

// K-TH BEST SPANNING TREE: repeat the swap argument with a priority
//   ↳ queue over (tree, forced-in /
//   ↳ forced-out edge sets) - Gabow's algorithm,  $O(k m \log n)$ . Rare; the
//   ↳ second-best case is the one
// that actually shows up.
// MINIMUM BOTTLENECK SPANNING TREE = any MST (minimising the maximum
//   ↳ edge is implied by Kruskal).
// MST OF A GRAPH PLUS ONE NEW EDGE: add it, find the cycle, drop the
//   ↳ cycle maximum.  $O(n)$ .
// MINIMUM SPANNING TREE WITH DEGREE CONSTRAINT at one vertex  $r$ : build
//   ↳ the MST of  $G - r$ , then add
// the cheapest edge from  $r$  into each component, then improve by swaps
//   ↳ while degree <  $k$ .
```

3.8.4 MinDiameterSpanningTree

MINIMUM DIAMETER SPANNING TREE - the spanning tree whose longest path is as SHORT as possible.

```
// O(V^3) (Floyd-Warshall plus one sweep per edge). Undirected,
// ↪ non-negative weights, 0-indexed.
// This is NOT the MST: the MST minimises total weight and can have a
// ↪ terrible diameter.
// KEY FACT. An MDST is always a SHORTEST PATH TREE from the ABSOLUTE
// ↪ 1-CENTRE - a point that may
// sit in the MIDDLE OF AN EDGE, not only at a vertex. So: for every
// ↪ edge (u, v) of weight w, slide
// a point c at distance x along it and minimise its eccentricity
// ↪ ecc(x) = max over i of
// min(x + d[u][i], w - x + d[v][i]). Each i contributes a "tent"
// ↪ function of x, the upper envelope
// is piecewise linear, and it is enough to test the breakpoints -
// ↪ which is the sweep below, after
// sorting the vertices by d[u][.] descending. The answer is the
// ↪ minimum eccentricity, doubled.
// Reach for it for "build a network minimising the worst-case delay
// ↪ between any two nodes",
// "connect all cities so the farthest pair is as close as possible",
// ↪ and broadcast-tree problems.
// ALL WEIGHTS ARE DOUBLED INTERNALLY so the centre's position stays an
// ↪ integer; the returned
// diameter is already in the ORIGINAL units (it is twice the
// ↪ eccentricity, which cancels out).
```

```
struct MDST {
    ll diameter = llinf;
    int edge = -1; // the
    // edge holding the centre
    ll offset = 0; // 2 *
    // distance from e[edge][0]
    vector<ll> distFromCentre; // 2 *
    // dist(centre, i), after solve
    ll solve(int n, const vector<array<ll, 3>>& e) { // e =
    // {u, v, w}
        if (n == 1) return diameter = 0;
        vector<vector<ll>> d(n, vector<ll>(n, llinf));
        for (int i = 0; i < n; i++) d[i][i] = 0;
        for (auto [u, v, w] : e) d[u][v] = min(d[u][v], 2 * w), d[v][u]
        // = min(d[v][u], 2 * w);
        for (int k = 0; k < n; k++) for (int i = 0; i < n; i++) for
        // (int j = 0; j < n; j++)
            if (d[i][k] < llinf && d[k][j] < llinf) d[i][j] =
        // min(d[i][j], d[i][k] + d[k][j]);
        vector<int> id(n);
        iota(all(id), 0);
        for (int u = 0; u < n; u++) {
            sort(all(id), [&](int a, int b) { return d[u][a] > d[u][b];
            // });
            for (int k = 0; k < (int)e.size(); k++) {
                int a = e[k][0], b = e[k][1];
                if (a != u && b != u) continue;
                int v = a ^ b ^ u;
                ll W = 2 * e[k][2], opt = 0;
                int last = id[0];
                ll val = d[u][last]; //
                // centre at u itself
                for (int t = 1; t < n; t++) { // walk
                    // the envelope breakpoints
                    int i = id[t];
                    if (d[v][i] <= d[v][last]) continue;
                    ll cx = (W - d[u][i] + d[v][last]) / 2, cy = (W +
                    // d[u][i] + d[v][last]) / 2;
                    if (val > cy) val = cy, opt = cx;
                    last = i;
                }
                if (val > d[v][last]) val = d[v][last], opt = W; //
                // centre at v itself
                if (val < diameter) diameter = val, edge = k, offset =
                // opt;
            }
        }
        distFromCentre.assign(n, llinf); //
        // Dijkstra from the virtual centre
        int a = e[edge][0], b = e[edge][1];
        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
        // pq;
        distFromCentre[a] = offset, distFromCentre[b] = 2 * e[edge][2]
        // - offset;
        pq.push({distFromCentre[a], a}), pq.push({distFromCentre[b],
        // b});
        vector<vector<pair<int, ll>>> g(n);
        for (auto [x, y, w] : e) g[x].push_back({(int)y, 2 * w}),
        // g[y].push_back({(int)x, 2 * w});
        while (!pq.empty()) {
```

```
auto [dd, x] = pq.top(); pq.pop();
if (dd > distFromCentre[x]) continue;
for (auto [y, w] : g[x]) if (dd + w < distFromCentre[y])
    distFromCentre[y] = dd + w, pq.push({distFromCentre[y],
    // y});
    }
return diameter;
}
};
// RECOVERING THE TREE: take the shortest path tree of distFromCentre
// ↪ (each vertex keeps one
// incoming edge that is tight), plus the two halves of the centre
// ↪ edge. If offset is 0 or 2*w the
// centre is a vertex and the tree is just that vertex's shortest path
// ↪ tree.
// UNWEIGHTED graphs: the answer is the shortest path tree from the
// ↪ middle of the graph's own
// diameter path, and BFS from every vertex (O(VE)) is enough - do not
// ↪ bother with this file.
// MINIMUM RADIUS spanning tree is the same computation without the
// ↪ final doubling; the absolute
// 1-centre is the object both problems share.
// MINIMUM DIAMETER with a bound on the total weight, or "diameter <= k
// ↪ with fewest edges", is
// NP-hard - the polynomial case is exactly the unconstrained one above.
```

3.9 Exact Exponential

3.9.1 CliqueColouringMIS

EXACT ALGORITHMS FOR NP-HARD GRAPH PROBLEMS ON SMALL n. The constraint IS the hint:

```
// n <= 20-24 -> plain subset DP, O(2^n * n)
// n <= 40-50 -> meet in the middle, or Bron-Kerbosch (fast in
// ↪ practice on sparse graphs)
// n <= 100+ -> only if the graph is bipartite/perfect/a tree -
// ↪ then it is polynomial (Konig)
// adj[] is an ADJACENCY BITMASK: bit j of adj[i] is set iff i-j is an
// ↪ edge. No self loops.

// --- MAXIMUM CLIQUE, Bron-Kerbosch with PIVOTING. Worst case
// ↪ O(3^(n/3)), very fast in practice. ---
// R = clique built so far, P = candidates, X = already-explored
// ↪ vertices (avoids duplicates).
// The pivot u in P|X is chosen to maximise |P & adj[u]|; only P \
// ↪ adj[u] needs branching, because
// any maximal clique either omits u or contains a non-neighbour of u.
int bestClique;
unsigned bestSet;
void bk(unsigned R, unsigned P, unsigned X, const vector<unsigned>&
    // adj) {
    if (!P && !X) {
        if (__builtin_popcount(R) > bestClique) bestClique =
        // __builtin_popcount(R), bestSet = R;
        return;
    }
    int u = __builtin_ctz(P | X), best = -1;
    for (unsigned t = P | X; t &= t - 1) {
        int v = __builtin_ctz(t), c = __builtin_popcount(P & adj[v]);
        if (c > best) best = c, u = v;
    }
    for (unsigned cand = P & ~adj[u]; cand; cand &= cand - 1) {
        int v = __builtin_ctz(cand);
        bk(R | 1u << v, P & adj[v], X & adj[v], adj);
        P &= ~(1u << v), X |= 1u << v;
    }
}
int maxClique(const vector<unsigned>& adj) { // n <=
    // 32
    bestClique = 0, bestSet = 0;
    bk(0, (1u << adj.size()) - 1, 0, adj);
    return bestClique; // the
    // clique itself is in bestSet
}
// MAXIMUM INDEPENDENT SET = maximum clique of the COMPLEMENT graph.
// ↪ Just flip adj and mask off i.
// MINIMUM VERTEX COVER = n - MIS. On BIPARTITE graphs both are
// ↪ polynomial: Konig says
// min vertex cover = maximum matching, and MIS = n - matching (see
// ↪ 07 - Matching).

// --- CHROMATIC NUMBER, O(2^n * n) by inclusion-exclusion
// ↪ (Bjorklund-Husfeldt-Koivisto). ---
// Let ind[S] = 1 if S is independent. The number of ways to cover V
// ↪ with k ORDERED independent
// sets is sum_S (-1)^(n-|S|) * I(S)^k, where I(S) = #independent
// ↪ subsets of S. The chromatic
// number is the smallest k making that non-zero. Compute I over
```

```

    ↪ subsets with one SOS pass.
int chromatic(const vector<unsigned>& adj) { // n <=
    ↪ 20 with this modulus
    int n = adj.size(), N = 1 << n;
    if (!n) return 0;
    const ll M = (1LL << 61) - 1; // any
    ↪ large prime works
    vector<ll> I(N), pw(N, 1);
    I[0] = 1;
    for (int S = 1; S < N; S++) { // I[S]
    ↪ = #independent subsets of S
        int v = __builtin_ctz(S), rest = S ^ 1 << v; // (v in
    ↪ it, or not)
        I[S] = (I[rest] + I[rest & ~adj[v]]) % M;
    }
    for (int k = 1; k <= n; k++) {
        ll tot = 0;
        for (int S = 0; S < N; S++) {
            pw[S] = (__int128)pw[S] * I[S] % M; // I(S)^k
            ll t = (n - __builtin_popcount(S)) & 1 ? M - pw[S] : pw[S];
            tot = (tot + t) % M;
        }
        if (tot % M) return k;
    }
    return n;
}
// The count is done MODULO a prime, so a zero could in principle be a
    ↪ false negative; with a
// 61-bit modulus that never happens on contest inputs. Check
    ↪ bipartiteness first - k <= 2 is O(n+m).

// --- MINIMUM STEINER TREE (connect a set of k terminals in a weighted
    ↪ graph), O(3^k n + 2^k n^2).
// dp[S][v] = cheapest tree containing terminal set S and touching
    ↪ vertex v.
// merge: dp[S][v] = min over submasks T of dp[T][v] + dp[S\T][v]
// grow: dp[S][v] = min over edges (u,v) of dp[S][u] + w
    ↪ (run Dijkstra per S)
ll steinerTree(int n, const vector<vector<pair<int, ll>>& g, const
    ↪ vector<int>& term) {
    int k = term.size();
    if (!k) return 0;
    vector<vector<ll>> dp(1 << k, vector<ll>(n, llinf));
    for (int i = 0; i < k; i++) dp[1 << i][term[i]] = 0;
    for (int S = 1; S < 1 << k; S++) {
        for (int T = (S - 1) & S; T; T = (T - 1) & S)
            for (int v = 0; v < n; v++)
                if (dp[T][v] < llinf && dp[S ^ T][v] < llinf)
                    dp[S][v] = min(dp[S][v], dp[T][v] + dp[S ^ T][v]);
        priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>>
    ↪ pq;
        for (int v = 0; v < n; v++) if (dp[S][v] < llinf)
    ↪ pq.push({dp[S][v], v});
        while (!pq.empty()) { //
    ↪ Dijkstra to relax the tree outward
            auto [d, u] = pq.top(); pq.pop();
            if (d > dp[S][u]) continue;
            for (auto [v, w] : g[u]) if (dp[S][u] + w < dp[S][v])
                dp[S][v] = dp[S][u] + w, pq.push({dp[S][v], v});
        }
    }
    return *min_element(all(dp[(1 << k) - 1]));
}
// STEINER TREE IN A GRID / with k up to ~10 is the usual contest size.
    ↪ If k = 2 it is a shortest
// path; if k = n it is the MST; the DP interpolates between them.
// RELATED EXACT-EXPONENTIAL TOOLS:
// * HAMILTONIAN PATH / TSP: dp[mask][v], O(2^n n^2). See 08 - DP/02
    ↪ - BitmaskDP.
// * SET COVER: dp over covered elements, O(2^n * #sets), or IE like
    ↪ the chromatic number.
// * GRAPH COLOURING with k fixed: try to 3-colour by 2-SAT? No -
    ↪ 3-colouring is NP-hard; use
// the IE formula above, or branch on the highest-degree vertex
    ↪ with memoisation.
// * MAXIMUM INDEPENDENT SET for n <= 40-50: meet in the middle -
    ↪ split the vertices in half,
// enumerate independent subsets of the left half, and for each
    ↪ store the best independent
// subset of the right half compatible with its neighbourhood (SOS
    ↪ over the right mask).
// * COUNTING INDEPENDENT SETS / colourings on a graph of small
    ↪ TREewidth: DP over a tree
// decomposition, O(2^tw * n) - usually a grid problem in disguise
    ↪ (see broken-profile DP).

```

3.9.2 MaxCliqueBitset

MAXIMUM CLIQUE FOR n UP TO ~150 (about 1 s at n = 155, much faster on sparse graphs). 0-indexed.

```

// 01 - CliqueColouringMIS is capped at 32 vertices by its unsigned
    ↪ mask; this is the same problem
// with a bitset adjacency matrix and a far stronger pruning rule, so
    ↪ it is the one to paste when
// n is in the 40..200 range. MAXIMUM INDEPENDENT SET = run it on the
    ↪ COMPLEMENT graph.
// The bound: greedily colour the candidate set (vertices of the same
    ↪ colour are pairwise
// non-adjacent, so at most one of them can join the clique). If
    ↪ |current| + #colours <= |best|,
// prune. The extra trick is the ADAPTIVE re-sorting - it recomputes
    ↪ the degree order in a
// sub-branch only when that branch has been visited often enough to
    ↪ pay for it (the 'limit').
// PRECONDITION: symmetric matrix, NO self edges. Set MV to the
    ↪ smallest power-of-two-ish bound
// above n; a bitset<MV> is scanned in MV/64 words, so an over-large MV
    ↪ costs real time.
const int MV = 160; // <--
    ↪ adjust to your n
struct MaxClique {
    typedef vector<bitset<MV>> Graph;
    struct Vertex { int i, d = 0; };
    double limit = 0.025, pk = 0;
    Graph e;
    vector<Vertex> V;
    vector<vector<int>> C;
    vector<int> qmax, q, S, old;
    MaxClique(Graph g) : e(g), C(g.size() + 1), S(C.size()), old(S) {
        for (int i = 0; i < (int)e.size(); i++) V.push_back({i});
    }
    void init(vector<Vertex>& r) { // order
    ↪ by degree inside r
        for (auto& v : r) v.d = 0;
        for (auto& v : r) for (auto& j : r) v.d += e[v.i][j.i];
        sort(all(r), [](Vertex a, Vertex b) { return a.d > b.d; });
        int mxD = r[0].d;
        for (int i = 0; i < (int)r.size(); i++) r[i].d = min(i, mxD) +
    ↪ 1;
    }
    void expand(vector<Vertex>& R, int lev = 1) {
        S[lev] += S[lev - 1] - old[lev], old[lev] = S[lev - 1];
        while (R.size()) {
            if ((int)q.size() + R.back().d <= (int)qmax.size()) return;
    ↪ // colour bound
            q.push_back(R.back().i);
            vector<Vertex> T;
            for (auto v : R) if (e[R.back().i][v.i]) T.push_back({v.i});
            if (T.size()) {
                if (S[lev]++ / ++pk < limit) init(T);
                int j = 0, mxk = 1, mnk = max((int)qmax.size() -
    ↪ (int)q.size() + 1, 1);
                C[1].clear(), C[2].clear();
                for (auto v : T) { //
    ↪ greedy colouring
                    int k = 1;
                    auto f = [&](int i) { return e[v.i][i]; };
                    while (any_of(all(C[k]), f)) k++;
                    if (k > mxk) mxk = k, C[mxk + 1].clear();
                    if (k < mnk) T[j++] .i = v.i;
                    C[k].push_back(v.i);
                }
                if (j > 0) T[j - 1].d = 0;
                for (int k = mnk; k <= mxk; k++) for (int i : C[k])
    ↪ T[j].i = i, T[j++] .d = k;
                expand(T, lev + 1);
            } else if (q.size() > qmax.size()) qmax = q;
            q.pop_back(), R.pop_back();
        }
    }
    vector<int> solve() { init(V), expand(V); return qmax; } // the
    ↪ clique itself
};
// MAXIMUM INDEPENDENT SET: build h with h[i][j] = (i != j && !g[i][j])
    ↪ and run this on h.
// MINIMUM VERTEX COVER = n - MIS. MINIMUM CLIQUE COVER of G =
    ↪ chromatic number of the complement.
// WEIGHTED maximum clique needs a different bound (sum of the
    ↪ per-colour maxima); the unweighted
// colour count above is not valid there.
// MAXIMUM CLIQUE ON A PERFECT GRAPH (bipartite, chordal, interval,
    ↪ comparability) is POLYNOMIAL -
// check the structure before reaching for exponential search.
// COUNTING all maximal cliques instead: Bron-Kerbosch with pivoting

```

→ (01), $0(3^{n/3})$ of them.

3.9.3 ChromaticPolynomial

CHROMATIC POLYNOMIAL $P(x)$ = the number of proper colourings with at most x colours, for $n \leq 20$.

```
// Needs: Mint (06 - Math/01 - Mint.cpp).
//  $0(2^n \cdot n^2)$  to get all of  $P(0..n)$ , then  $0(n^2)$  per evaluation
// → elsewhere. 0-indexed; adj[] is an
// adjacency BITMASK (bit j of adj[i] set iff i-j is an edge), no self
// → loops.
// 01 - CliqueColouringMIS finds the chromatic NUMBER with the same
// → inclusion-exclusion; this gets
// the whole polynomial, which is what you need for "count the proper
// → k-colourings" (k possibly
// huge), for "count acyclic orientations", and for
// → deletion-contraction arguments.
// THE FORMULA. With  $I(S)$  = the number of independent subsets of  $S$ 
// → (including the empty one),
//  $P(k) = \sum \text{over } S \text{ of } (-1)^{n-|S|} \cdot I(S)^k$ .
//  $P$  has degree  $n$ , so evaluating it at  $0..n$  pins it down; Lagrange
// → interpolation gives any other
// point. All arithmetic is mod MOD, so a zero result is "almost
// → surely" a genuine zero.
struct ChromaticPolynomial {
    int n;
    vector<Mint> p; // p[k]
    // =  $P(k)$  for  $k = 0..n$ 
    ChromaticPolynomial(const vector<int>& adj) : n(adj.size()), p(n +
    // → 1, 0) {
        int F = 1 << n;
        vector<Mint> I(F), pw(F, 1);
        I[0] = 1;
        for (int S = 1; S < F; S++) { // v is
            // in the subset, or it is not
            int v = __builtin_ctz(S), rest = S ^ 1 << v;
            I[S] = I[rest] + I[rest & ~adj[v]];
        }
        for (int k = 1; k <= n; k++) {
            Mint tot = 0;
            for (int S = 0; S < F; S++) {
                pw[S] *= I[S]; // pw[S]
            }
            // → =  $I(S)^k$ 
            tot += ((n - __builtin_popcount(S)) & 1) ? Mint(0) -
            // → pw[S] : pw[S];
            p[k] = tot;
        }
        Mint at(ll x) const { // P(x)
            // → for any x, by interpolation
            if (x >= 0 && x <= n) return p[x];
            Mint ans = 0;
            for (int i = 0; i <= n; i++) {
                Mint num = p[i];
                for (int j = 0; j <= n; j++) if (j != i) num *= (Mint(x) -
                // → j) / (Mint(i) - j);
                ans += num;
            }
            return ans;
        }
        int chromaticNumber() const {
            for (int k = 1; k <= n; k++) if (!p[k] == Mint(0)) return k;
            return n;
        }
};
// KNOWN VALUES, useful as checks: a tree on n vertices has  $P(x) = x$ 
// →  $(x-1)^{n-1}$ ; the cycle  $C_n$  has
//  $(x-1)^n + (-1)^n (x-1)$ ;  $K_n$  has  $x(x-1) \dots (x-n+1)$ ; a forest of k
// → components on n vertices in total
// has  $P(x) = x^k (x-1)^{n-k}$  only when every component is a star...
// → just use the tree formula per
// component and multiply, since P is MULTIPLICATIVE over connected
// → components.
// ACYCLIC ORIENTATIONS of  $G = |P(-1)| = (-1)^n P(-1)$  (Stanley's
// → theorem). That also counts the
// DAGs you can make by directing the edges of  $G$ , which is how "count
// → the DAGs whose underlying
// graph is  $G$ " problems are solved.
// DELETION-CONTRACTION:  $P_{-G}(x) = P_{-G/e}(x) - P_{G/e}(x)$ . Exponential
// → in general, but it is the
// right tool when the graph is nearly a tree or you can memoise on
// → small components.
// LABELLED DAGS ON n VERTICES (a different question - no fixed graph):
// → with  $a_n$  the count,
//  $a_n = \sum_{k=1..n} (-1)^{k+1} C(n,k) 2^{k(n-k)} a_{n-k}$ , by
// → inclusion-exclusion on the sources.
```

3.9.4 Hafnian

HAFNIAN of a symmetric matrix = the NUMBER OF PERFECT MATCHINGS of a general weighted graph

```
// Needs: Mint (06 - Math/01 - Mint.cpp).
// (the sum over perfect matchings of the product of the chosen
// → entries).  $0(2^{n(n/2)} \cdot n^3)$ , which
// is about 2 s for a 38x38 matrix. 0-indexed, n must be EVEN, the
// → diagonal is ignored.
// The hafnian is to the permanent what the general graph is to the
// → bipartite one:
// permanent(A) = #perfect matchings of a BIPARTITE graph → Ryser,
// →  $0(2^n \cdot n)$ 
// hafnian(A) = #perfect matchings of a GENERAL graph → this
// → file
// determinant = the SIGNED version, polynomial, and counts nothing
// → on its own
// Reach for it for "count the perfect matchings / domino tilings /
// → pairings" when the graph is not
// bipartite and n is around 30-40. If the graph IS bipartite use the
// → permanent (2x more vertices
// here would be hopeless); if you only need EXISTENCE, use Blossom or
// → a Tutte matrix.
// METHOD (Bjorklund): keep, for every pair (i, j) of the first m
// → vertices, a POLYNOMIAL in a
// formal variable that counts partial matchings by size. Removing the
// → last two vertices splits
// into "they are matched to each other" and "they are not", and the
// → recursion has depth n/2 with
// two branches that share all the work below them.
struct Hafnian {
    int h;
    void add(vector<Mint>& ans, const vector<Mint>& a, const
    // → vector<Mint>& b) {
        for (int i = 0; i < h; i++)
            for (int j = 0; j < h - 1 - i; j++) ans[i + j + 1] += a[i]
            // → * b[j];
    }
    vector<Mint> rec(const vector<vector<vector<Mint>>>& v) {
        vector<Mint> ans(h, 0);
        if (v.empty()) { ans[0] = 1; return ans; }
        int m = (int)v.size() - 2;
        auto V = v;
        V.resize(m);
        vector<Mint> zero = rec(V); // last
        // → two vertices dropped
        for (int i = 0; i < m; i++) for (int j = 0; j < i; j++) {
            add(V[i][j], v[m][i], v[m + 1][j]); // ...
        }
        // → or routed through them
        add(V[i][j], v[m + 1][i], v[m][j]);
    }
    vector<Mint> one = rec(V);
    for (int i = 0; i < h; i++) ans[i] += one[i] - zero[i];
    add(ans, one, v[m + 1][m]);
    return ans;
}
Mint solve(const vector<vector<Mint>>& a) { // a
    // → symmetric, a[i][i] unused
    int n = a.size();
    h = n / 2 + 1;
    vector<vector<vector<Mint>>> v(n);
    for (int i = 0; i < n; i++) {
        v[i].resize(i);
        for (int j = 0; j < i; j++) v[i][j].assign(h, 0),
        // → v[i][j][0] = a[i][j];
    }
    return rec(v).back();
}
// 0/1 MATRIX → plain count of perfect matchings. General weights →
// → weighted count, so this also
// answers "sum over perfect matchings of the product of edge weights"
// → (partition functions).
// PERMANENT by Ryser, for comparison, is 6 lines and  $0(2^n \cdot n)$ :
// perm(A) =  $(-1)^n \cdot \sum \text{over subsets } S \text{ of columns of } (-1)^{|S|} \cdot$ 
// → prod_i (sum_{j in S} A[i][j]).
// COUNTING MATCHINGS OF ALL SIZES (not just perfect): pad the matrix
// → with a zero-weight vertex per
// vertex - or read off rec()'s whole coefficient vector, which is
// → exactly that generating function.
// PLANAR GRAPHS have a POLYNOMIAL count via the FKT algorithm
// → (Pfaffian orientation + determinant),
//  $0(n^3)$  - that is the one to use for domino tilings of a grid.
```

3.9.5 MeetInTheMiddle

MEET IN THE MIDDLE ON GRAPHS - the $n \leq 40..50$ band, where 2^n is hopeless but $2^{(n/2)}$ is not.

```
// 0-indexed, adj[] is an adjacency BITMASK (ll, so n <= 63), no self
// loops unless stated.
// THE PATTERN. Split the vertices into halves L (first k) and R (the
// rest). Enumerate every valid
// configuration of L; each one imposes a REQUIREMENT on R that is a
// single bitmask; precompute for
// every R-mask the best/total over all valid R-configurations
// COMPATIBLE with it, using a SOS
// (subset-sum / superset-sum) transform. Then one pass over the 2^k
// left masks answers everything.

// --- COUNT ALL CLIQUES (including the empty one),  $O(2^{(n/2)} * n)$ . ---
// A set is a clique iff every chosen vertex is adjacent to all the
// others, i.e. the intersection of
// (adj[v] | bit v) over v in the set contains the set.
ll countCliques(int n, const vector<ll>& adj) {
    int k = n / 2, r = n - k;
    ll full = n == 64 ? -1LL : (1LL << n) - 1;
    vector<ll> g(n);
    for (int i = 0; i < n; i++) g[i] = adj[i] | 1LL << i;
    vector<ll> dp(1 << k, 0);
    dp[0] = 1;
    for (int m = 1; m < (1 << k); m++) { // is
        // the left mask m a clique?
        ll nw = full;
        for (int j = 0; j < k; j++) if (m >> j & 1) nw &= g[j];
        if ((nw & m) == m) dp[m] = 1;
    }
    for (int i = 0; i < k; i++) for (int m = 0; m < (1 << k); m++)
        // SOS: dp[m] = #subsets
        if (m >> i & 1) dp[m] += dp[m ^ 1 << i];
    ll ans = dp[(1 << k) - 1]; // right
    // half empty
    for (int m = 1; m < (1 << r); m++) {
        ll nw = full, p = (1LL) << k;
        for (int j = 0; j < r; j++) if (m >> j & 1) nw &= g[k + j];
        if ((nw & p) == p) ans += dp[nw & ((1LL << k) - 1)]; // left
    }
    // must sit inside nw
    return ans;
}

// --- MINIMUM COST VERTEX COVER OF A GENERAL GRAPH,  $O(2^{(n/2)} * n)$ . ---
// Every edge needs at least one endpoint chosen; minimise the total
// cost of the chosen vertices.
// A left mask fixes the left choices; the UNCHOSEN left vertices force
// their right neighbours in.
// SELF LOOP at v (bit v of adj[v]) = "v must be taken", which is how
// "forbidden" vertices are said.
// (On a BIPARTITE graph this is polynomial - Konig / min cut. This is
// the general case.)
ll minCostVertexCover(int n, const vector<ll>& adj, const vector<ll>& cost) {
    int k = n / 2, r = n - k;
    vector<ll> cl(1 << k, 0), need(1 << k, 0), dp(1 << r, 0);
    for (int m = 0; m < (1 << k); m++) {
        for (int i = 0; i < k; i++) {
            if (m >> i & 1) { cl[m] += cost[i]; continue; }
            if (adj[i] >> i & 1) { cl[m] = llinf; break; } // self
        }
        // loop: i had to be taken
        for (int j = 0; j < n; j++) if (adj[i] >> j & 1) {
            if (j < k) { if (!(m >> j & 1)) cl[m] = llinf; } //
            // uncovered left edge
            else need[m] |= 1 << (j - k);
        }
    }
    for (int m = 0; m < (1 << r); m++) {
        for (int i = 0; i < r; i++) {
            if (m >> i & 1) { dp[m] += cost[k + i]; continue; }
            for (int j = k; j < n; j++)
                if ((adj[k + i] >> j & 1) && !(m >> (j - k) & 1)) dp[m]
        }
    }
    // = llinf;
    for (int i = 0; i < r; i++) for (int m = (1 << r) - 1; m >= 0; m--)
        // SOS over SUPERSETS
        if (!(m >> i & 1)) dp[m] = min(dp[m], dp[m ^ 1 << i]);
    ll ans = llinf;
    for (int m = 0; m < (1 << k); m++)
        if (cl[m] < llinf && dp[need[m]] < llinf) ans = min(ans, cl[m]
    // + dp[need[m]]);
    return ans;
}
```

```
// MAXIMUM INDEPENDENT SET = (total cost) - (minimum cost vertex cover)
// with unit costs, so the
// second routine solves MIS for  $n \leq 50$  too. MAXIMUM CLIQUE for that
// range: 02 - MaxCliqueBitset.
// OTHER PROBLEMS THAT SPLIT THE SAME WAY: subset sum / knapsack on n
// <= 40, "count the subsets
// with property X" when X only couples the halves through one mask,
// and 3-colouring by enumerating
// the colour classes of one half.
// THE SOS DIRECTION MATTERS: use the subset transform when the right
// half must fit INSIDE a mask
// (counting), and the superset transform when the right half must
// CONTAIN a mask (covering).
```

3.10 Theorems

3.10.1 GraphTheorems

GRAPH THEOREM SHEET

/* ===== GRAPH THEOREM SHEET =====

Every hypothesis here matters. A theorem quoted without its
precondition is a wrong answer.

MATCHING AND COVERING

KONIG (BIPARTITE only): max matching = min vertex cover.

Constructive: Z = vertices reachable by alternating paths from the

UNMATCHED left vertices;
the cover is $(X \setminus Z)$ union $(Y$ and $Z)$.

GALLAI: for EVERY graph on n vertices, $\alpha + \tau = n$

(independent set + vertex cover).
for every graph WITH NO ISOLATED VERTEX, $\nu + \rho = n$

(matching + edge cover).

Bipartite + Konig gives $\alpha = n - \nu$.

HALL (bipartite): a matching saturating every vertex of the LEFT

part X exists iff

$|N(S)| \geq |S|$ for every S contained in X . (Not every S of $V -$

the side matters.)

ORE'S DEFECT FORM: $\nu(G) = |X| - \max$ over S contained in X of $(|S|$

$- |N(S)|)$.

HALL FOR REGULAR BIPARTITE: every k -regular bipartite graph ($k \geq 1$)

has a perfect matching,

so its edges split into exactly k perfect matchings. Equivalently

KONIG'S EDGE COLOURING

THEOREM: $\chi'(G) = \Delta(G)$ for every bipartite MULTIGRAPH.

(Timetabling, "split into

Delta rounds", Latin-square completion.)

BERGE: a matching is maximum iff there is no augmenting path. (Why

Kuhn / HK / Blossom stop.)

TUTTE (general graphs): G has a PERFECT matching iff $\text{odd}(G - U) \leq$

$|U|$ for every U contained

in V , where $\text{odd}(H)$ = the number of components of H with an ODD

number of vertices.

$U = \emptyset$ forces n even.

TUTTE-BERGE: $\nu(G) = (1/2) * \min$ over U of $(|V| + |U| - \text{odd}(G - U)$

$\rightarrow)$.

DILWORTH (FINITE poset): min number of chains covering P = size of

the maximum antichain.

MIRSKY (dual): min number of antichains partitioning P = length of

the longest chain.

SPERNER: the largest antichain in the subset lattice of $[n]$ has size

$\rightarrow C(n, \text{floor}(n/2))$.

PATH COVERS - the distinction that most notebooks get wrong:

min VERTEX-DISJOINT path cover of a DAG = $n - (\text{max matching of the}$

split graph, no closure).

min chain cover for Dilworth = $n - (\text{max matching of the TRANSITIVE}$

CLOSURE's split graph).

CONNECTIVITY AND FLOW

MENGER: for NON-ADJACENT s, t : min s - t VERTEX cut = max

internally-vertex-disjoint s - t paths.

For any $s \neq t$: min s - t EDGE cut = max edge-disjoint s - t paths.

WHITNEY: $\kappa(G) \leq \lambda(G) \leq \Delta(G)$ (vertex conn $\leq$ edge

conn $\leq$ min degree).

A graph on ≥ 3 vertices is 2-connected iff every two vertices lie

on a common cycle.

MAX-FLOW MIN-CUT, with INTEGRALITY: integer capacities $\Rightarrow$ some

maximum flow is integral (and

Dinic produces one). Corollary used constantly: a 0/1-capacity max

flow decomposes into that

many edge-disjoint s - t paths.

ROBBINS: an undirected graph has a STRONG orientation iff it is

connected and BRIDGELESS.

Constructive in $O(V+E)$: DFS, orient tree edges away from the root

and back edges toward it.

ESWARAN-TARJAN (augmentation):

directed - min arcs to make a digraph strongly connected =

$\rightarrow \max(\#sources, \#sinks)$ of the

condensation ($\emptyset$ if the condensation is a single node).
 undirected - for a CONNECTED graph, min edges to make it
 $\hookrightarrow$ 2-edge-connected = $\lceil L/2 \rceil$ where
 L is the number of leaves of the bridge tree ($\emptyset$ if the bridge
 $\hookrightarrow$ tree is one node).

DEGREE SEQUENCES

ERDOS-GALLAI: $d_1 \geq \dots \geq d_n \geq 0$ is the degree sequence of a
 $\hookrightarrow$ SIMPLE graph iff $\sum d_i$ is
 even and for every k in $[1, n]$:
 $\sum_{i \leq k} d_i \leq k(k-1) + \sum_{i > k} \min(d_i, k)$.
 Only k up to the largest index with $d_k \geq k$ needs checking (the
 $\hookrightarrow$ Durfee square), so it is
 $O(n)$ after sorting with prefix sums.
 HAVEL-HAKIMI: graphical iff $d_1 \leq n-1$ and, after deleting d_1 and
 $\hookrightarrow$ subtracting 1 from the next
 d_1 entries (re-sorted), the rest is graphical. Gives a
 $\hookrightarrow$ CONSTRUCTION, which Erdos-Gallai
 does not. $O(n^2)$ naive.
 LANDAU (tournaments): $s_1 \leq \dots \leq s_n$ is a score sequence iff
 $\hookrightarrow \sum_{i \leq k} s_i \geq C(k, 2)$ for
 every k , WITH EQUALITY at $k = n$.

TOURNAMENTS

REDEI: every finite tournament has a Hamiltonian PATH (insertion
 $\hookrightarrow$ sort, $O(n \log n)$).
 CAMION: every STRONGLY CONNECTED tournament has a Hamiltonian CYCLE.
 MOON: every strongly connected tournament is vertex-pancyclic - for
 $\hookrightarrow$ every vertex v and every
 k in $[3, n]$ there is a k -cycle through v .

COUNTING

MATRIX-TREE (Kirchhoff), undirected: #spanning trees (or the sum
 $\hookrightarrow$ over trees of the product of
 weights) = ANY cofactor of $L = D - A$, loops excluded.
 MATRIX-TREE, DIRECTED (Tutte), with $A_{ij} = \# \text{arcs } i \rightarrow j$:
 $L_{\text{out}} = D_{\text{out}} - A$, delete row AND column $r \rightarrow$ arborescences
 $\hookrightarrow$ CONVERGING TO r (in-trees).
 $L_{\text{in}} = D_{\text{in}} - A$, delete row AND column $r \rightarrow$ arborescences
 $\hookrightarrow$ DIVERGING FROM r (out-trees).
 These two are the pair everyone swaps. Verify on the 2-vertex
 $\hookrightarrow$ graph $1 \rightarrow 2$.
 BEST THEOREM: for a CONNECTED digraph with $\text{indeg}(v) = \text{outdeg}(v)$
 $\hookrightarrow$ everywhere, the number of
 Eulerian circuits up to starting point is $ec(G) = t_w(G) * \prod_v$
 $\hookrightarrow (\text{outdeg}(v) - 1)!$,
 where t_w is the number of arborescences CONVERGING TO any fixed w
 $\hookrightarrow$ (the L_{out} form above);
 t_w is the same for every w in such a graph. Multiply by $\text{outdeg}(v)$
 $\hookrightarrow$ to fix a start vertex.
 CAYLEY: n^{n-2} labelled trees; with prescribed degrees, $(n-2)! /$
 $\hookrightarrow \prod (d_i - 1)!$;
 labelled forests with k trees whose roots are k specified
 $\hookrightarrow$ vertices: $k * n^{n-k-1}$.

EULERIAN CONDITIONS (the check EulerianPath.cpp needs before it

$\hookrightarrow$ starts)
 UNDIRECTED: a circuit exists iff every degree is even and all EDGES
 $\hookrightarrow$ lie in one component.
 A trail exists iff exactly $\emptyset$ or 2 vertices have odd degree (start
 $\hookrightarrow$ at an odd one), same
 connectivity condition.
 DIRECTED: a circuit exists iff $\text{indeg} = \text{outdeg}$ at every vertex and
 $\hookrightarrow$ all vertices of nonzero
 degree lie in one component of the underlying undirected graph. A
 $\hookrightarrow$ trail exists iff exactly
 one vertex has out - in = 1 (the start), exactly one has in - out
 $\hookrightarrow$ = 1 (the end), the rest
 are balanced, same connectivity condition.

COLOURING

BROOKS: for connected G , $\chi(G) \leq \Delta(G)$ UNLESS G is complete or
 $\hookrightarrow$ an odd cycle, where
 $\chi = \Delta + 1$. Greedy on any order gives $\Delta + 1$; greedy on a
 $\hookrightarrow$ degeneracy order gives $d + 1$.
 VIZING (SIMPLE graphs): $\Delta \leq \chi' \leq \Delta + 1$. Multigraphs:
 $\hookrightarrow \chi' \leq \Delta + \mu$, and
 Shannon's $\chi' \leq \lceil 3\Delta / 2 \rceil$. Deciding class 1 vs class 2
 $\hookrightarrow$ is NP-complete - do not
 look for a polynomial algorithm mid-contest. Bipartite is the
 $\hookrightarrow$ exception (Konig above).
 GALLAI-HASSE-ROY-VITAVER: $\chi(G) = 1 + \min$ over orientations of
 $\hookrightarrow$ (longest directed path length).
 Equivalently every orientation contains a directed path on $\chi(G)$
 $\hookrightarrow$ vertices.

EXTREMAL

TURAN: a K_{r+1} -free graph has $e \leq (1 - 1/r) n^2 / 2$, with
 $\hookrightarrow$ equality only for the balanced
 complete r -partite Turan graph. $r = 2$ is MANTEL: triangle-free $\Rightarrow$
 $\hookrightarrow e \leq \lfloor n^2/4 \rfloor$.
 Every graph has a bipartite subgraph with $\geq m/2$ edges (greedy
 $\hookrightarrow$ max-cut).
 A graph is bipartite iff it has no odd cycle.
 RAMSEY, the known small values: $R(3,3)=6$, $R(3,4)=9$, $R(3,5)=14$,
 $\hookrightarrow R(3,6)=18$, $R(3,7)=23$,
 $R(3,8)=28$, $R(3,9)=36$, $R(4,4)=18$, $R(4,5)=25$. $R(4,6)$ in $[36,40]$ and
 $\hookrightarrow R(5,5)$ in $[43,46]$ are
 OPEN. Bound: $R(s,t) \leq C(s+t-2, s-1)$.
 NASH-WILLIAMS ARBORICITY: $a(G) = \max$ over subgraphs S with ≥ 2
 $\hookrightarrow$ vertices of
 $\lceil m_S / (n_S - 1) \rceil$. Consequences: degeneracy d satisfies $a \leq$
 $\hookrightarrow d \leq 2a - 1$; $m \leq a(n-1)$;
 triangle enumeration is $O(m * a)$.
 TUTTE / NASH-WILLIAMS TREE PACKING: G has t edge-disjoint spanning
 $\hookrightarrow$ trees iff every partition
 of V into k non-empty parts has at least $t(k-1)$ crossing edges.
 PETERSEN: every BRIDGELESS CUBIC graph has a perfect matching. Every
 $\hookrightarrow 2k$ -regular graph splits
 into k edge-disjoint 2-factors, so it has an Eulerian circuit and
 $\hookrightarrow$ an orientation with
 $\text{indeg} = \text{outdeg} = k$.

PLANARITY

EULER: connected plane graph $V - E + F = 2$; with c components, $V - E$
 $\hookrightarrow + F = 1 + c$.
 Simple planar with $V \geq 3$: $E \leq 3V - 6$. Triangle-free (girth ≥ 4):
 $\hookrightarrow E \leq 2V - 4$.
 Every planar graph has a vertex of degree ≤ 5 , i.e. it is
 $\hookrightarrow 5$ -degenerate.
 KURATOWSKI / WAGNER: planar iff no K_5 and no $K_{3,3}$ subdivision
 $\hookrightarrow$ (minor).
 FOUR COLOUR: every planar graph is 4-colourable - but there is no
 $\hookrightarrow$ usable algorithm, so never
 plan around it. FIVE COLOUR is constructive in $O(n)$ from the
 $\hookrightarrow$ degree- ≤ 5 vertex.
 K_5 , $K_{3,3}$ and the Petersen graph are non-planar. In a planar graph
 $\hookrightarrow$ min cut = shortest path
 in the dual.

HAMILTONICITY (sufficient conditions only - the general problem is

$\hookrightarrow$ NP-complete)
 DIRAC: $n \geq 3$ and min degree $\geq n/2 \Rightarrow$ Hamiltonian.
 ORE: $\deg(u) + \deg(v) \geq n$ for every non-adjacent pair $\Rightarrow$
 $\hookrightarrow$ Hamiltonian.
 Both are constructive by rotation-extension.

TREE FACTS

JORDAN: every tree has 1 or 2 CENTRES; if 2 they are adjacent; they
 $\hookrightarrow$ are the middle of any
 diameter path.
 CENTROID: every tree has 1 or 2 centroids; c is a centroid iff every
 $\hookrightarrow$ component of $T - c$ has
 $\leq n/2$ vertices iff c minimises the sum of distances. If 2 they
 $\hookrightarrow$ are adjacent and n is even.
 That $\leq n/2$ bound is exactly what makes centroid decomposition
 $\hookrightarrow O(\log n)$ deep.
 DIAMETER: the greedy two-BFS is valid on trees and on positively
 $\hookrightarrow$ weighted trees, NOT on
 general graphs. Every longest path passes through a centre.
 $\hookrightarrow$ Joining two trees by an edge
 gives diameter $\max(d_1, d_2, r_1 + r_2 + 1)$, minimised by joining
 $\hookrightarrow$ centre to centre.
 A tree is bipartite, has ≥ 2 leaves, and sum of degrees = $2(n-1)$.

WHAT IS NP-HARD, AND WHAT ONLY LOOKS IT

NP-hard: max clique, chromatic number (≥ 3 colours), Hamiltonian
 $\hookrightarrow$ path/cycle, TSP, MIXED
 Chinese postman, vertex cover / independent set on general graphs,
 $\hookrightarrow$ max cut (general),
 Steiner tree with unbounded terminals, longest path, dominating
 $\hookrightarrow$ set, feedback vertex/arc
 set, edge-colouring class decision, bandwidth.
 Polynomial despite appearances: vertex cover and independent set on
 $\hookrightarrow$ BIPARTITE graphs (Konig)
 and on trees / interval / chordal graphs; 2-colouring; 2-SAT; max
 $\hookrightarrow$ cut on PLANAR graphs;
 min mean cycle; min-cost flow; directed MST; maximum matching in a
 $\hookrightarrow$ GENERAL graph (Blossom);
 min path cover of a DAG; Steiner tree with $k \leq \sim 15$ terminals; TSP
 $\hookrightarrow$ with $n \leq \sim 20$.

===== */

3.11 Special Structures

3.11.1 StableChordalComplement

THREE GRAPH TOOLS THAT ARE NOT ABOUT PATHS OR FLOW.

```
// --- GALE-SHAPLEY STABLE MARRIAGE,  $O(n^2)$ . ---
// n proposers, n receivers, each with a full preference ranking. A
//   ↳ matching is STABLE if no pair
// mutually prefer each other to their assigned partners. One always
//   ↳ exists, and this finds the
// PROPOSER-OPTIMAL one: every proposer gets the best partner they
//   ↳ could have in ANY stable
// matching, and simultaneously every receiver gets their worst. (Swap
//   ↳ the roles to flip that.)
// pref[i] = the proposers' preference lists; rank[j][i] = receiver j's
//   ↳ ranking of proposer i.
vector<int> stableMarriage(const vector<vector<int>>& pref, const
    ↳ vector<vector<int>>& rank) {
    int n = pref.size();
    vector<int> match(n, -1), nxt(n, 0); //
    ↳ match[receiver] = proposer
    vector<int> free_(n);
    iota(all(free_), 0);
    while (!free_.empty()) {
        int m = free_.back(); free_.pop_back();
        int w = pref[m][nxt[m]++]; //
        ↳ propose down the list
        if (match[w] < 0) match[w] = m;
        else if (rank[w][m] < rank[w][match[w]])
            ↳ free_.push_back(match[w]), match[w] = m;
        else free_.push_back(m);
    }
    return match;
}

// HOSPITALS/RESIDENTS (one side has capacity c) is the same loop with
//   ↳ a heap of accepted
// proposers per hospital. STABLE ROOMMATES (one pool, no sides) may
//   ↳ have NO stable matching -
// Irving's algorithm decides it, and it is a different, longer
//   ↳ algorithm.

// --- TRAVERSAL OF THE COMPLEMENT GRAPH in  $O(n + m)$ , never
//   ↳ materialising the  $O(n^2)$  edges. ---
// Keep the not-yet-visited vertices in a linked structure; from u,
//   ↳ walk that set and step to
// every w that is NOT a neighbour of u. Each vertex leaves the set
//   ↳ once, and each REAL edge is
// inspected once, so the total is  $O(n + m)$ .
// Reach for it whenever the input describes the NON-edges: "connected
//   ↳ components of the graph of
// pairs that do NOT conflict", complement colouring, complement BFS
//   ↳ distances.
vector<int> complementComponents(int n, const vector<vector<int>>& adj)
    ↳ {
    vector<char> isAdj(n, 0);
    set<int> unvis;
    for (int i = 0; i < n; i++) unvis.insert(i);
    vector<int> comp(n, -1);
    int c = 0;
    while (!unvis.empty()) {
        int s = *unvis.begin();
        unvis.erase(unvis.begin());
        vector<int> st{s};
        comp[s] = c;
        while (!st.empty()) {
            int u = st.back(); st.pop_back();
            for (int v : adj[u]) isAdj[v] = 1;
            vector<int> take;
            for (int w : unvis) if (!isAdj[w]) take.push_back(w);
            for (int w : take) unvis.erase(w), comp[w] = c,
                ↳ st.push_back(w);
            for (int v : adj[u]) isAdj[v] = 0;
        }
        c++;
    }
    return comp; //
    ↳ comp[v] = component id
}

// The complement of a DISCONNECTED graph is connected, so a complement
//   ↳ graph almost always has
// very few components - that is why this is fast in practice as well
//   ↳ as in theory.

// --- CHORDAL GRAPHS: maximum cardinality search gives a PERFECT
//   ↳ ELIMINATION ORDERING. ---
// A graph is CHORDAL if every cycle of length  $\geq 4$  has a chord.
//   ↳ Equivalently it has a PEO: an
```

```
// order in which every vertex's LATER neighbours form a clique.
// Why you care: on a chordal graph, MAXIMUM CLIQUE, CHROMATIC NUMBER,
//   ↳ MAXIMUM INDEPENDENT SET
// and MINIMUM CLIQUE COVER are all POLYNOMIAL - the four problems that
//   ↳ are NP-hard in general.
// Interval graphs and trees are chordal, so "these intervals conflict"
//   ↳ problems land here.
vector<int> mcsOrder(int n, const vector<vector<int>>& adj) { //
    ↳ reverse of a PEO if chordal
    vector<int> w(n, 0), ord;
    vector<char> used(n, 0);
    for (int it = 0; it < n; it++) {
        int b = -1;
        for (int v = 0; v < n; v++) if (!used[v] && (b < 0 || w[v] >
            ↳ w[b])) b = v;
        used[b] = 1, ord.push_back(b);
        for (int u : adj[b]) if (!used[u]) w[u]++;
    }
    reverse(all(ord));
    return ord; // this
    ↳ IS a PEO iff chordal
}

bool isChordal(int n, const vector<vector<int>>& adj) {
    auto ord = mcsOrder(n, adj);
    vector<int> pos(n);
    for (int i = 0; i < n; i++) pos[ord[i]] = i;
    vector<char> mark(n, 0);
    for (int i = 0; i < n; i++) { // v's
        ↳ later neighbours must be
        int v = ord[i], par = -1; // a
        ↳ clique; it is enough to check
        for (int u : adj[v]) if (pos[u] > i && (par < 0 || pos[u] <
            ↳ pos[par])) par = u;
        if (par < 0) continue;
        for (int u : adj[par]) mark[u] = 1;
        for (int u : adj[v]) if (pos[u] > i && u != par && !mark[u]) {
            for (int x : adj[par]) mark[x] = 0;
            return false;
        }
        for (int u : adj[par]) mark[u] = 0;
    }
    return true;
}

// GIVEN A PEO: greedy colouring in PEO order uses exactly omega
//   ↳ colours (so  $\chi = \omega$ ); the
// maximum clique is the largest "v plus its later neighbours"; the
//   ↳ maximum independent set is the
// greedy "take v, delete its later neighbours" over the PEO.
```

3.11.2 MinimumWeightCycle

MINIMUM WEIGHT CYCLE THROUGH EVERY VERTEX (and hence the global minimum weight cycle), by one

```
// Dijkstra per source,  $O(V^3)$  with the dense version below or  $O(V(E + V \log V))$  with a heap.
// UNDIRECTED, non-negative weights, 0-indexed, g is an adjacency
//   ↳ MATRIX with llinf for "no edge".
// Returns best[u] = the weight of the lightest simple cycle CONTAINING
//   ↳ u (llinf if there is none);
// the global answer is the minimum over u, and the GIRTH is the
//   ↳ unweighted case (all weights 1).
// WHY IT WORKS. A lightest cycle through u splits into two shortest
//   ↳ paths from u that leave along
// DIFFERENT first edges and are joined by one more edge (cur, i). So
//   ↳ run Dijkstra from u, tag every
// vertex with the first edge (the "branch") its shortest path used,
//   ↳ and combine any two settled
// vertices in different branches. The 'par[cur] != cur' guard rejects
//   ↳ the degenerate 2-cycle that
// walks one edge out and back.
vector<ll> minWeightCycles(int n, const vector<vector<ll>>& g) {
    vector<ll> best(n, llinf), d(n);
    vector<int> par(n), vis(n);
    for (int u = 0; u < n; u++) {
        fill(all(vis), 0), fill(all(d), llinf);
        iota(all(par), 0);
        d[u] = 0;
        for (;;) {
            int cur = -1; // dense
            ↳ Dijkstra:  $O(n^2)$  per source
            for (int i = 0; i < n; i++) if (!vis[i] && d[i] < llinf &&
                ↳ (cur < 0 || d[i] < d[cur]))
                cur = i;
            if (cur < 0) break;
            vis[cur] = 1;
            if (par[cur] != cur && g[cur][u] < llinf) // close
                ↳ the cycle straight to u
                best[u] = min(best[u], d[cur] + g[cur][u]);
        }
    }
}
```

4 Number Theory

```
for (int i = 0; i < n; i++) {
    if (i == u || g[cur][i] == llinf) continue;
    if (vis[i]) {
        if (par[i] != par[cur]) // two
            ↪ different branches meet
            best[u] = min(best[u], d[cur] + d[i] +
            ↪ g[cur][i]);
        } else if (d[cur] + g[cur][i] < d[i])
            d[i] = d[cur] + g[cur][i], par[i] = (cur == u ? i :
            ↪ par[cur]);
        }
    }
}
return best;
}
// GLOBAL MINIMUM WEIGHT CYCLE: min over u of best[u]. If that is all
↪ you want and the graph is
// DIRECTED, Floyd-Warshall is shorter: run it, then take min over v of
↪ d[v][v] (initialise the
// diagonal to llinf, not 0).
// GIRTH of an unweighted graph: replace Dijkstra by BFS - O(V * E),
↪ and for a cycle through a
// FIXED vertex the two-branch rule is the same.
// MINIMUM CYCLE THROUGH A GIVEN EDGE (u, v): delete the edge and take
↪ dist(u, v) + w.
// MINIMUM MEAN cycle (weight divided by length) is a different problem
↪ - Karp, see 02 - Shortest
// Paths/06 - CyclesAndConstraints.
// NEGATIVE WEIGHTS break this completely (Dijkstra); a minimum weight
↪ cycle with negative edges is
// NP-hard in general, since a Hamiltonian cycle can be encoded in it.
// LONGEST cycle is NP-hard. Shortest cycle THROUGH k GIVEN VERTICES is
↪ FPT via colour coding.
```

3.11.3 IncrementalDAGReachability

INCREMENTAL DAG REACHABILITY (Italiano) - answer "is there a path $u \rightarrow v$?" while ARCS ARE BEING

```
// ADDED, in O(1) per query and O(n) amortised per insertion. Directed,
↪ 0-indexed, O(n^2) memory.
// addEdge(s, t) refuses (and returns false) any arc that would create
↪ a CYCLE, so the structure
// doubles as an online "would this arc break acyclicity?" oracle -
↪ which is the usual reason to
// reach for it: "process the constraints in order and report the first
↪ one that is contradictory",
// online topological ordering, incremental dependency graphs.
// STRUCTURE. For every vertex s it keeps a REACHABILITY TREE of
↪ everything s can reach, stored as
// par[s][v] = v's parent in that tree. Adding s -> t only ever ADDS
↪ vertices to those trees, and a
// vertex is added to a tree at most once, so the total work over all
↪ insertions is O(n^2) per
// source - i.e. O(n) amortised per update. Nothing is ever removed,
↪ hence "incremental only".
struct IncrementalDAG {
    int n;
    vector<vector<int>> par; //
    ↪ par[root][v], -1 = not reachable
    vector<vector<vector<int>>> ch; //
    ↪ ch[root][v] = children in the tree
    IncrementalDAG(int n) : n(n), par(n, vector<int>(n, -1)), ch(n,
    ↪ vector<vector<int>>(n)) {}
    bool reachable(int s, int t) const { return s == t || par[s][t] >=
    ↪ 0; }
    void meld(int root, int sub, int u, int v) { // graft
    ↪ v (and its subtree) under u
        par[root][v] = u, ch[root][u].push_back(v);
        for (int c : ch[sub][v]) if (!reachable(root, c)) meld(root,
        ↪ sub, v, c);
    }
    bool addEdge(int s, int t) { // false
    ↪ = it would close a cycle
        if (reachable(t, s)) return false;
        if (reachable(s, t)) return true; //
    ↪ nothing changes
        for (int p = 0; p < n; p++)
            if (reachable(p, s) && !reachable(p, t)) meld(p, t, s, t);
        return true;
    }
};
// FULLY DYNAMIC reachability (with deletions) has no simple solution;
↪ if the updates are known in
// advance, run OFFLINE divide and conquer over the time axis instead.
// DENSE ALTERNATIVE: bitset transitive closure, n/64 words per vertex,
↪ rebuilt from scratch in
// O(n * m / 64) - simpler and often faster for n <= 2000 when the
↪ number of queries is small.
```

```
// TRANSITIVE CLOSURE of a static DAG: process in reverse topological
↪ order and OR the successors'
// bitsets - O(n m / 64).
// COUNTING PAIRS (u, v) with u reaching v is the same bitset sweep;
↪ there is no better general
// algorithm (it is equivalent to boolean matrix multiplication).
```

4 Number Theory

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4.1 Euclidean Algorithm

4.1.1 ExtendedEuclid

EXTENDED EUCLID - returns $g = \gcd(a, b)$ and x, y with $a*x + b*y = g$. This is the one routine that

```
// gives you modular inverses for a NON-prime modulus, solves linear
↪ Diophantine equations, and
// underpins CRT.  $|x| \leq b/(2g)$  and  $|y| \leq a/(2g)$ , so no overflow for
↪ 64-bit inputs.
long long extGCD(long long a, long long b, long long &x, long long &y) {
    if (b == 0) {
        x = 1; y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}
```

```
long long modInverse(long long a, long long m) {
    long long x, y;
    long long g = extGCD(a, m, x, y);
    if (g != 1) return -1; // Inverse doesn't exist
    return (x % m + m) % m;
}
```

4.1.2 LinearDiophantine

Linear Diophantine Equations (2 Variables & N Variables)

```
// 1. Solves  $a * x + b * y = c$  (Particular solution & GCD check)
// 2. Solves  $a_1*x_1 + a_2*x_2 + \dots + a_n*x_n = c$  for N variables
struct LinearDiophantine {
    static ll ext_gcd(ll a, ll b, ll &x, ll &y) {
        if (b == 0) {
            x = 1; y = 0;
            return a;
        }
        ll x1, y1;
        ll d = ext_gcd(b, a % b, x1, y1);
        x = y1;
        y = x1 - y1 * (a / b);
        return d;
    }

    // Solves  $a * x + b * y = c$ . Returns false if no solution exists
    static bool solve_2var(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
        if (a == 0 && b == 0) {
            if (c == 0) { x = y = g = 0; return true; }
            return false;
        }
        g = ext_gcd(abs(a), abs(b), x, y);
        if (c % g != 0) return false;
        x *= c / g;
        y *= c / g;
        if (a < 0) x = -x;
        if (b < 0) y = -y;
        return true;
    }

    // Solves  $a_1*x_1 + a_2*x_2 + \dots + a_n*x_n = c$  for N variables.
    // Returns true if a solution exists and populates x vector
    static bool solve_nvar(const vector<ll>& a, ll c, vector<ll>& x) {
        int n = a.size();
        if (n == 0) return c == 0;
        x.assign(n, 0);

        vector<ll> prefix_g(n);
        prefix_g[0] = abs(a[0]);
        for (int i = 1; i < n; i++) prefix_g[i] = std::gcd(prefix_g[i -
        ↪ 1], a[i]);

        if (prefix_g.back() == 0) return c == 0;
```

```

    if (c % prefix_g.back() != 0) return false;

    ll cur_c = c;
    for (int i = n - 1; i > 0; i--) {
        ll x_prev, y_curr, g;
        solve2var(prefix_g[i - 1], a[i], cur_c, x_prev, y_curr, g);
        x[i] = y_curr;
        cur_c = prefix_g[i - 1] * x_prev;
    }
    x[0] = cur_c / a[0];
    return true;
}
};

```

4.1.3 FloorSum

EUCLIDEAN-LIKE ALGORITHM (AtCoder Library 'floor_sum').

```

// floorSum(n, m, a, b) = sum_{i=0}^{n-1} floor((a*i + b) / m)    in
//   ↪ O(log m).
// This is THE tool for "sum over a lattice line": counting lattice
//   ↪ points under a line, sums of
// floor/mod expressions, and anything of the shape sum floor((a i + b)
//   ↪ / m) that a loop cannot do
// because n is up to 1e9 or more. Works like the Euclidean algorithm:
//   ↪ reduce a mod m, then SWAP
// the roles of the axes (count by rows instead of by columns) and
//   ↪ recurse.
// Requires n >= 0, m >= 1, a >= 0, b >= 0. Overflow: a*n + b must fit
//   ↪ in ll (use __int128 if not).
ll floorSum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    if (a >= m) ans += (n - 1) * n / 2 * (a / m), a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    ll ymax = (a * n + b) / m;
    if (ymax == 0) return ans;
    ll xmax = ymax * m - b;
    ans += ymax * (n - (xmax + a - 1) / a);
    return ans + floorSum(ymax, a, m, (a - xmax % a) % a);
}
// NEGATIVE a or b: shift them into range first.
// a < 0: let a2 = a mod m in [0,m); floorSum(n,m,a,b) =
//   ↪ floorSum(n,m,a2,b) - (a2-a)/m * (n-1)n/2
// b < 0: let b2 = b mod m in [0,m); subtract n * (b2-b)/m. (Same
//   ↪ trick, easier: use __int128.)
//
// WHAT IT SOLVES, directly:
// * #lattice points (x,y) with 1 <= x <= n and 1 <= y <= (a x + b)/m
//   ↪   → floorSum itself
// * sum_{i=0}^{n-1} ((a i + b) mod m) = (a * n(n-1)/2 + b*n) - m *
//   ↪ floorSum(n, m, a, b)
// * sum_{i=1}^n floor(n / i) in O(sqrt n) instead - use DIVISOR
//   ↪ BLOCKS (below), not this.
// * "how many multiples of q lie in [l, r]" = floor(r/q) -
//   ↪ floor((l-1)/q); the basic case.
//
// THE OTHER TWO SUMS of the same family (needed together often enough
//   ↪ to be worth knowing they
// exist): S1 = sum i * floor((a i + b)/m) and S2 = sum floor((a i +
//   ↪ b)/m)^2 obey the same
// recursion with a 3-tuple state - the "universal Euclidean algorithm"
//   ↪ generalises
// this to any monoid: you supply the two operators U ("move right")
//   ↪ and R ("move up") and it
// composes the word for the line in O(log) monoid multiplications.
//
// DIVISOR BLOCKS - the other floor trick, O(sqrt n):
// floor(n / i) is constant on maximal blocks; for a block starting
//   ↪ at l the block ends at
// r = n / (n / l). Use it for sum_{i=1}^n floor(n/i), sum of
//   ↪ tau/sigma, Du sieve, Mobius sums.
// for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); /* value
//   ↪ n/l on [l, r] */ }
// Two-argument version (blocks constant for BOTH n/i and m/i): r =
//   ↪ min(n/(n/l), m/(m/l)).

```

4.1.4 FloorSumExtended

EUCLIDEAN-LIKE ALGORITHM, THE FULL TRIPLE. With $q_i = \text{floor}((a*i + b) / c)$ it returns

```

// f = sum_{i=0..n} q_i      g = sum_{i=0..n} i * q_i      h =
//   ↪ sum_{i=0..n} q_i^2
// all three at once in O(log c). NOTE i runs to n INCLUSIVE, unlike
//   ↪ floorSum in 03, which stops
// at n-1: floorSum(n, m, a, b) = fgh(n - 1, a, b, m).f.
// Reach for it when a lattice-line sum has an i or a square in it: sum
//   ↪ of i*floor, sum of
// (a i + b mod c)^2, the second moment of a modular AP, "sum over
//   ↪ lattice points under a line of

```

```

// (x + y)" and friends. The recursion is the same as floorSum: strip
//   ↪ a/c and b/c, then swap axes.
// Requires n >= -1, a >= 0, b >= 0, c >= 1. Exact (no modulus) in
//   ↪ __int128, which is good for
// n and m = (a n + b)/c up to ~1e11 (h is of order n*m^2). Take
//   ↪ everything mod p if they are
// bigger - the recursion only ever adds and multiplies, so it is
//   ↪ modulus-friendly.
struct FGH { __int128 f, g, h; };
FGH fgh(ll n, ll a, ll b, ll c) {
    if (n < 0) return {0, 0, 0};
    __int128 M = n, s1 = M * (M + 1) / 2, s2 = M * (M + 1) * (2 * M +
//   ↪ 1) / 6;
    if (a >= c || b >= c) { // q_i = A*i +
//   ↪ B + (reduced q_i)
        __int128 A = a / c, B = b / c;
        FGH t = fgh(n, a % c, b % c, c);
        return {t.f + A * s1 + B * (M + 1), t.g + A * s2 + B * s1,
//   ↪ t.h + 2 * B * t.f + 2 * A * t.g + A * A * s2 + B * B *
//   ↪ (M + 1) + 2 * A * B * s1};
    }
    __int128 m = ((__int128)a * n + b) / c; // the largest
//   ↪ value any q_i takes
    if (m == 0) return {0, 0, 0};
    FGH t = fgh((ll)m - 1, c, c - b - 1, a); // count by
//   ↪ rows instead of by columns
    __int128 f = M * m - t.f;
    return {f, (m * M * (M + 1) - t.f - t.h) / 2, M * m * (m + 1) - 2 *
//   ↪ t.g - 2 * t.f - f};
}
// DERIVED SUMS (r_i = (a*i + b) mod c = a*i + b - c*q_i):
// sum r_i = a*s1 + b*(n+1) - c*f
// sum r_i^2 = sum (a i + b)^2 - 2c * (a*g + b*f) + c^2 * h
// sum i*r_i = a*s2 + b*s1 - c*g
// The same recursion with a bigger state gives sum i^k1 * q_i^k2 for
//   ↪ any k1 + k2 <= K in
// O(K^3 log c) - rarely needed. The clean way to state the whole
//   ↪ family is the UNIVERSAL
// EUCLIDEAN ALGORITHM: walk the line as a word in {U = "step up", R =
//   ↪ "step right"} and
// multiply the letters in ANY monoid; supply U and R and you get
//   ↪ whatever statistic you like.

```

4.1.5 LatticeUnderLine

COUNT THE NONNEGATIVE LATTICE POINTS UNDER A LINE: $\#\{(x, y) : x, y \geq 0, a*x + b*y \leq c\}$

```

// in O(log min(a, b)) - a Euclidean-style descent that peels off a
//   ↪ full rectangle of solutions
// each round, never touching a single point individually.
// This is the "how many ways to pay at most c with coins a and b"
//   ↪ primitive, and the engine
// behind 06 - ModCountInRange.cpp. Requires a >= 1, b >= 1; returns 0
//   ↪ for c < 0 (a = 0 or b = 0
// would be infinite - handle that case yourself).
// OVERFLOW: the answer is about c^2 / (2ab) and the intermediate
//   ↪ (f+1)*f/2*k is of the same
// order, so it must fit in ll; c up to ~4e18 is fine as long as the
//   ↪ count itself does.
ll latticeUnderLine(ll a, ll b, ll c) {
    if (c < 0) return 0;
    if (a > b) swap(a, b); // descend on
//   ↪ the smaller coefficient
    ll ans = 0;
    while (c >= 0) // b = k*a + l,
//   ↪ then swap roles
        ll k = b / a, l = b % a, f = c / b, e = c % b / a, g = c % b %
//   ↪ a;
        ans += (f + 1) * (e + 1) + (f + 1) * f / 2 * k; // the
//   ↪ axis-aligned part
        c = f * l - a * g; // l = 0 makes
//   ↪ c < 0: the loop ends
        b = a, a = l;
    }
    return ans;
}
// VARIANTS, all from this one call:
// a*x + b*y < c → latticeUnderLine(a, b, c - 1)
// a*x + b*y <= c, x >= 1, y >= 1 → latticeUnderLine(a, b, c - a -
//   ↪ b)
// a*x + b*y == c exactly → latticeUnderLine(a,b,c) -
//   ↪ latticeUnderLine(a,b,c-1), or solve the
// Diophantine equation and count the admissible k directly (02 -
//   ↪ LinearDiophantine.cpp).
// x in [0, X] as well → subtract the shifted count
//   ↪ latticeUnderLine(a, b, c - a*(X+1)).
// Related: sum_{x} floor((c - a x)/b) is the same quantity as a floor
//   ↪ sum, so 03 - FloorSum.cpp

```

```
// also answers it in  $O(\log)$  - use whichever shape the problem hands
    → you.
```

4.1.6 ModCountInRange

HOW MANY x IN $[xl, xr]$ HAVE $(a \cdot x \bmod m)$ INSIDE $[cl, cr]$? $O(\log m)$, any $a, m \geq 1$.

```
// Needs: LatticeUnderLine (05 - LatticeUnderLine.cpp).
// The classic use: "how many of the first n multiples of a land in a
    → window mod m" - collision
// counting, clock/lap problems, counting i with  $(a \cdot i + b) \bmod m \leq c$ 
    → (shift by b: replace the
// window  $[cl, cr]$  by  $[cl - b, cr - b]$  reduced mod m, splitting it if
    → it wraps past 0).
// modCountLE counts  $0 \leq x \leq n$  with  $(a \cdot x \bmod m) \leq c$ ; everything else
    → is inclusion-exclusion
// on that. OVERFLOW: LatticeUnderLine is called with  $c \sim 2 \cdot m \cdot n$ , so  $m \cdot n$ 
    → must stay below  $\sim 4e18$ 
// (use __int128 inside LatticeUnderLine if it does not).
ll modCountLE(ll a, ll m, ll c, ll n) { // count  $0 \leq x$ 
    →  $\leq n$  with  $a \cdot x \bmod m \leq c$ 
    if (n < 0) return 0;
    ++c, ++n; // work with
    → half-open  $[0, c)$  and  $[0, n)$ 
    a %= m; if (a < 0) a += m;
    ll ec = c / m; c %= m; if (c < 0) ec--, c += m; // c outside
    →  $[0, m)$  means whole periods
    ll ans = ec * n, en = n / m; n %= m;
    if (n < 0) en--, n += m;
    auto blk = [&](ll len) { // x in  $[0,$ 
    → len): len < m, uses  $a \cdot m + m > 0$ 
        return LatticeUnderLine(m, a + m, (a + m) * (len - 1))
            - LatticeUnderLine(m, a + m, (a + m) * (len - 1) - c);
    };
    if (en) ans += en * blk(m); // en full
    → periods of length m
    if (n) ans += blk(n);
    return ans;
}
ll modCountRange(ll a, ll m, ll cl, ll cr, ll xl, ll xr) {
    return modCountLE(a, m, cr, xr) - modCountLE(a, m, cr, xl - 1)
        - modCountLE(a, m, cl - 1, xr) + modCountLE(a, m, cl - 1, xl -
    → 1);
}
// SANITY CHECK: with  $g = \gcd(a, m)$  the values  $a \cdot x \bmod m$  are exactly
    → the multiples of g, and over
// one full period x in  $[0, m)$  each of them is hit exactly g times. So
    → modCountRange over a full
// period must equal  $g \cdot (\text{number of multiples of g inside } [cl, cr])$ .
// COMPANION ROUTINES: the smallest such x is 07 -
    → SmallestModInRange.cpp; if you need the whole
// set of x, note that it is a union of  $O(\sqrt{m})$  arithmetic
    → progressions
// (03 - Modular Arithmetic/09 - ModularAPBlocks.cpp).
```

4.1.7 SmallestModInRange

SMALLEST $x \geq 0$ WITH $l \leq (a \cdot x \bmod p) \leq r$, or -1 if no such x exists. $O(\log p)$.

```
// The "when does the ticking counter first land in the window"
    → problem: jumps of size a on a
// circle of circumference p, first time inside the arc  $[l, r]$ . Also
    → solves "smallest x with
//  $a \cdot x \bmod p \leq r$ " ( $l = 0$ ) and, by symmetry, the first time OUTSIDE a
    → window.
// PRECONDITIONS:  $0 \leq a < p$ ,  $0 \leq l \leq r < p$ ,  $p \geq 1$ . p need NOT be
    → prime.
// Idea: if  $\text{ceil}(l/a) \cdot a \leq r$  we are done in one step. Otherwise write p
    →  $= k \cdot a + b$ ; asking for
//  $l \leq a \cdot x - p \cdot y \leq r$  turns into the same question one Euclidean step
    → down, with the roles of
// the modulus and the multiplier swapped - so the recursion depth is
    →  $O(\log p)$ , like gcd.
// OVERFLOW: the internal products are up to  $p \cdot p$ , hence the __int128.
ll smallestModIn(ll p, ll a, ll l, ll r) {
    if (a == 0) return l == 0 ? 0 : -1;
    auto cdiv = [ ](__int128 x, __int128 y) { return (ll)((x + y - 1) /
    → y); };
    ll c = cdiv(l, a); // first x with
    →  $a \cdot x \geq l$ , ignoring wraps
    if ((__int128) a * c <= r) return c;
    ll b = p % a; // p = (p/a) * a
    → + b
    ll y = smallestModIn(a, b, a - r % a, a - l % a); // -r <= b * y
    → (mod a) <= -l
    return y == -1 ? -1 : cdiv(__int128)l + (__int128)b * y, a) + p /
    → a * y;
}
// COMPANIONS
// count of such x in a range → 06 - ModCountInRange.cpp
```

```
// the whole solution set → it is a union of  $O(\sqrt{p})$ 
    → APs, see
// (03 - Modular Arithmetic/09
    → - ModularAPBlocks.cpp
// smallest x with  $a \cdot x \bmod p == \text{target}$  → solve the linear congruence
    → instead
// (03 - Modular Arithmetic/04
    → - CongruencesAndTowers.cpp)
// TYPICAL USE: interactive/recovery problems where you know x mod M
    → for a huge M and want the
// smallest representative that also satisfies  $1 \leq q \leq N$ ; and
    → "minimum of  $(k \cdot a \bmod p)$  over
//  $k \leq n$ ", which the three-distance theorem says is achieved at a
    → continued-fraction denominator.
```

4.1.8 CoinReachability

WHICH TOTALS ARE PAYABLE WITH COINS $a[0..n-1]$? (all $x_i \geq 0$, unbounded supply)

```
// sum  $a_i x_i = k$  solvable? how many solvable k in  $[l, r]$ ?
    → largest unsolvable k?
// THE TRICK ("congruence shortest path" / Dijkstra on residues): fix A
    → = min  $a_i$  and build a
// graph on the residues mod A with an edge  $u \rightarrow (u + a_i) \bmod A$  of
    → weight  $a_i$ . Then
//  $d[u]$  = the SMALLEST payable value congruent to u (mod A)
// and k is payable iff  $d[k \bmod A] \leq k$ . So every residue class
    → contributes the arithmetic
// progression  $d[u], d[u] + A, d[u] + 2A, \dots$  and everything below  $d[u]$ 
    → in that class is not.
// Build  $O(A \cdot n \cdot \log A)$  time,  $O(A)$  memory. Take  $A = \min a_i$  (that is
    → what bounds the table).
// PRECONDITIONS: all  $a_i \geq 1$ . Values may be up to  $1e18$ ; A up to a few
    → million is the real cap.
struct CoinReach {
    ll A;
    vector<ll> d; // d[u] = min
    → payable value = u (mod A)
    CoinReach(vector<ll> a) {
        A = *min_element(all(a));
        d.assign(A, llinf);
        d[0] = 0;
        priority_queue<pair<ll, ll>, vector<pair<ll, ll>>, greater<>>
    → pq;
        pq.push({0, 0});
        while (!pq.empty()) {
            auto [val, u] = pq.top(); pq.pop();
            if (val != d[u]) continue;
            for (ll c : a) {
                ll nv = val + c, v = nv % A;
                if (nv < d[v]) d[v] = nv, pq.push({nv, v});
            }
        }
        bool payable(ll k) { return k >= 0 && d[k % A] <= k; }
        ll count(ll l, ll r) { // #payable k
        → in  $[l, r]$ ,  $l \geq 0$ 
            ll res = 0;
            for (ll u = 0; u < A; u++) {
                if (d[u] <= r) res += (r - d[u]) / A + 1;
                if (d[u] <= l - 1) res -= (l - 1 - d[u]) / A + 1;
            }
            return res;
        }
        ll frobenius() { // largest
        → UNpayable value, or -1
            ll mx = 0; // requires
        →  $\gcd(a_i) = 1$ , else infinite
            for (ll u = 1; u < A; u++) mx = max(mx, d[u]);
            return mx == 0 ? -1 : mx - A;
        }
    };
// FROBENIUS / CHICKEN McNUGGET closed forms exist only for two coins:
    → with  $\gcd(a, b) = 1$  the
// largest unpayable value is  $a \cdot b - a - b$  and exactly  $(a-1)(b-1)/2$ 
    → values are unpayable.
// For three or more coins there is no formula - this table IS the
    → algorithm.
// SANITY: if gcd of all  $a_i$  is  $g > 1$ , only multiples of g are ever
    → payable; divide everything
// by g first, otherwise frobenius() is meaningless (infinitely many
    → unpayable values).
// SAME TABLE ANSWERS: smallest payable value  $\geq k$  (that is  $d[u] +$ 
    →  $\text{ceil}((k - d[u])/A) \cdot A$  minimised
// over u), and "can I pay exactly k with at most/at least c coins" if
    → you make d 2-dimensional.
```

4.1.9 ThreeVarDiophantine

NUMBER OF NONNEGATIVE SOLUTIONS OF $a*x + b*y + c*z = n$, in $O(\log \max(a,b,c))$.

```
// (Popoviciu's formula for two coins, extended to three by Binner -
//   ↳ the denominator of a
//   ↳ three-coin system.) Reach for it when a, b, c and n are all up to
//   ↳ 1e18 and the DP over n or
//   ↳ the  $O(n / \max)$  loop over one variable are both hopeless. For two
//   ↳ coins use the much simpler
// count of lattice points under a line, 05 - LatticeUnderLine.cpp.
// PRECONDITIONS: a, b, c >= 1, n >= 0. Handles arbitrary gcds (the
//   ↳ wrapper strips them).
// All arithmetic is __int128 because the numerator N1 is of order  $n^2$ 
//   ↳ and the products  $a*b*c$  and
//  $c*b*b1p$  reach  $\max(a,b,c)^3$ ; keep a, b, c below  $\sim 1e12$  and n below
//   ↳  $\sim 1e18$ .
__int128 gcd128(__int128 a, __int128 b) {
    while (b) { __int128 t = a % b; a = b, b = t; }
    return a;
}
__int128 xgcd128(__int128 a, __int128 b, __int128 &x, __int128 &y) {
    if (!b) { x = 1, y = 0; return a; }
    __int128 x1, y1, d = xgcd128(b, a % b, x1, y1);
    x = y1, y = x1 - y1 * (a / b);
    return d;
}
__int128 inv128(__int128 a, __int128 m) { // m >= 1,
    ↳ gcd(a, m) = 1
    __int128 x, y; xgcd128((a % m) + m, m, x, y);
    return ((x % m) + m) % m;
}
__int128 fsum128(__int128 n, __int128 m, __int128 a, __int128 b) { //
    ↳ = floorSum, in __int128
    __int128 ans = 0;
    if (a >= m) ans += (n - 1) * n / 2 * (a / m), a %= m;
    if (b >= m) ans += n * (b / m), b %= m;
    __int128 ymax = (a * n + b) / m;
    if (ymax == 0) return ans;
    __int128 xmax = ymax * m - b;
    ans += ymax * (n - (xmax + a - 1) / a);
    return ans + fsum128(ymax, a, m, (a - xmax % a) % a);
}
__int128 denum3coprime(__int128 a, __int128 b, __int128 c, __int128 n)
    ↳ { // a,b,c PAIRWISE coprime
    auto fix = [](__int128 v, __int128 m) { return v == 0 ? m : v; };
    __int128 b1 = fix((-n % a * inv128(b, a) % a + a) % a, a);
    __int128 c1 = fix(b % a * inv128(c, a) % a, a);
    __int128 c2 = fix((-n % b * inv128(c, b) % b + b) % b, b);
    __int128 a2 = fix(c % b * inv128(a, b) % b, b);
    __int128 a3 = fix((-n % c * inv128(a, c) % c + c) % c, c);
    __int128 b3 = fix(a % c * inv128(b, c) % c, c);
    __int128 N1 = n * (n + a + b + c);
    N1 += c * b * b1 * (a + 1 - c1 * (b1 - 1));
    N1 += a * c * c2 * (b + 1 - a2 * (c2 - 1));
    N1 += b * a * a3 * (c + 1 - b3 * (a3 - 1));
    return N1 / (2 * a * b * c) + fsum128(b1, a, c1, 0) + fsum128(c2,
    ↳ b, a2, 0)
        + fsum128(a3, c, b3, 0) - 2;
}
__int128 denum3(__int128 a, __int128 b, __int128 c, __int128 n) {
    __int128 g = gcd128(gcd128(a, b), c);
    if (n % g) return 0;
    a /= g, b /= g, c /= g, n /= g;
    __int128 g1 = gcd128(b, c), g2 = gcd128(c, a), g3 = gcd128(a, b);
    __int128 n1 = n * inv128(a, g1) % g1, n2 = n * inv128(b, g2) % g2,
    ↳ n3 = n * inv128(c, g3) % g3;
    return denum3coprime(a / g2 / g3, b / g3 / g1, c / g1 / g2,
        (n - a * n1 - b * n2 - c * n3) / g1 / g2 / g3);
}
// TWO COINS, for reference: #solutions of  $a*x + b*y = n$  with  $\gcd(a,b)$ 
    ↳ = 1 and  $x, y \geq 0$  is
//  $n/(a*b) - \{n * \text{inv}(a) \bmod b / b\} - \{n * \text{inv}(b) \bmod a / a\} + 1$ 
    ↳ (Popoviciu),
// i.e.  $\text{floor}(n/(ab))$  or that plus one; the largest n with NO solution
    ↳ is  $a*b - a - b$ .
// FOUR OR MORE coins: no such formula. Use 08 - CoinReachability.cpp
    ↳ (needs min  $a_i$  small), or
// a generating-function / polynomial approach when n is small.
```

4.1.10 IntegerLattice2D

A SYSTEM OF TWO LINEAR DIOPHANTINE EQUATIONS IN ANY NUMBER OF UNKNOWNNS:

```
// Needs: extGCD (01 - ExtendedEuclid.cpp).
// sum  $x_i * a_i = p$  and  $\sum x_i * b_i = q$ ,  $x_i$  ANY integers -
    ↳ is it solvable?
// Equivalently: is (p, q) in the subgroup of  $\mathbb{Z}^2$  generated by the
    ↳ vectors (a_i, b_i)? Keep that
// lattice in Hermite normal form as two vectors (g, s) and (0, t):
    ↳ every lattice point is
//  $m*(g, s) + k*(0, t)$ , with  $g \geq 0, t \geq 0$  and  $0 \leq s < t$  when  $t > 0$ .
    ↳ Adding a generator is one
// extended-gcd step, so this is ONLINE - generators may arrive
    ↳ interleaved with queries, which
// is what makes it more than "solve a  $2 \times N$  system once".
// add() is  $O(\log)$ , contains() is  $O(1)$ . No preconditions; zero vectors
    ↳ and rank 0/1 are handled.
// OVERFLOW: add() forms  $s*(c/g)$  and  $d*(g/g')$ , so keep coordinates
    ↳ below  $\sim 2e9$  or switch to __int128.
struct Lattice2D {
    ll g = 0, s = 0, t = 0; // t = 0 means
    ↳ "no free second direction"
    void add(ll c, ll d) { // insert the
        ↳ generator (c, d)
        if (c < 0) c = -c, d = -d; // (c, d) and
        ↳ its negative are equivalent
        if (c == 0) { // pure
            ↳ vertical generator
            t = gcd(t, d < 0 ? -d : d);
            if (t) s = ((s % t) + t) % t;
            return;
        }
        if (g == 0) { g = c, s = t ? ((d % t) + t) % t : d; return; }
        ll x, y, ng = extGCD(g, c, x, y); // ng = gcd(g,
        ↳ c) =  $g*x + c*y$ 
        ll cross = s * (c / ng) - d * (g / ng); // the leftover
        ↳ pure-vertical direction
        t = gcd(t, cross < 0 ? -cross : cross);
        ll ns = s * x + d * y;
        g = ng, s = t ? ((ns % t) + t) % t : ns;
    }
    bool contains(ll p, ll q) { // is (p, q)
        ↳ representable?
        if (g == 0) return p == 0 && (t ? q % t == 0 : q == 0);
        if (p % g) return false;
        ll rem = q - (p / g) * s;
        return t ? rem % t == 0 : rem == 0;
    }
};
// WHY IT WORKS: extGCD writes  $(ng, ns) = x*(g, s) + y*(c, d)$  with  $ng =$ 
    ↳  $\gcd(g, c)$ , the smallest
// positive first coordinate achievable. What remains is the sublattice
    ↳ with first coordinate 0,
// generated by  $s*(c/ng) - d*(g/ng)$  - the determinant of the two old
    ↳ generators divided by ng.
// THE SAME OBJECT in higher dimension is the Hermite normal form of
    ↳ the generator matrix; with
// one row it degenerates to gcd, and 02 - LinearDiophantine.cpp is the
    ↳ single-equation version.
// TYPICAL USES: "can I reach (p, q) using these moves any integer
    ↳ number of times", reachability
// with two conserved quantities, and answering many (p, q) queries
    ↳ against a growing move set.
// NOTE this allows NEGATIVE  $x_i$ . Requiring  $x_i \geq 0$  is a completely
    ↳ different (much harder)
// problem - see 08 - CoinReachability.cpp for the one-equation
    ↳ nonnegative version.
```

4.2 Primes and Divisors

4.2.1 Sieve

Linear Sieve & Smallest Prime Factor (SPF) - Time: $O(N)$, Space: $O(N)$

// Computes primes list, SPF array, primality test, and $O(\log N)$ prime
 ↳ factorization

```
struct Sieve {
    int n;
    vector<int> primes;
    vector<int> spf;

    Sieve(int n = 1e7) : n(n), spf(n + 1, 0) {
        for (int i = 2; i <= n; i++) {
            if (spf[i] == 0) {
                spf[i] = i;
                primes.pb(i);
            }
            for (int p : primes) {

```

```

        if (p > spf[i] || (ll)i * p > n) break;
        spf[i * p] = p;
    }
}

bool is_prime(int x) const {
    return x >= 2 && x <= n && spf[x] == x;
}

// O(log N) prime factorization using precomputed SPF array
vector<int> get_prime_factors(int x) const {
    vector<int> factors;
    while (x > 1) {
        factors.pb(spf[x]);
        x /= spf[x];
    }
    return factors;
}
};

```

4.2.2 Factorization

Trial Division Prime Factorization - Time: $O(\sqrt{N})$

```

template <typename T = ll>
struct Factorization {
    static vector<T> get_prime_factors(T n) {
        vector<T> factors;
        for (T i = 2; i * i <= n; i++) {
            while (n % i == 0) {
                factors.pb(i);
                n /= i;
            }
        }
        if (n > 1) factors.pb(n);
        return factors;
    }
};

```

4.2.3 Divisors

Trial Division Divisor Generation - Time: $O(\sqrt{N})$

```

template <typename T = ll>
struct Divisors {
    static vector<T> get_divisors(T n) {
        vector<T> divs;
        for (T i = 1; i * i <= n; i++) {
            if (n % i == 0) {
                divs.pb(i);
                if (n / i != i) divs.pb(n / i);
            }
        }
        return divs;
    }
};

```

4.2.4 Sieve1e9

Fast Bitset Sieve up to $N = 10^9$ - Time: $\sim 0.3s$ for 10^9 , Space: $N / 16$ bytes ($\sim 60MB$)

```

// Stores odd numbers only (index i represents number 2*i + 1)
// WARNING: the bitset is  $\sim 60$  MB - the instance MUST be global/static,
//           $\hookrightarrow$  never a local.
template <int MAXPR = (int)1e9>
struct Sieve1e9 {
    bitset<MAXPR / 2> is_prime_odd;

    Sieve1e9(int n = MAXPR) {
        is_prime_odd.set();
        is_prime_odd[0] = 0; // 1 is not prime
        for (int i = 1; (2 * i + 1) * (2 * i + 1) <= n; i++) {
            if (is_prime_odd[i]) {
                int p = 2 * i + 1;
                // bound is (n-1)/2, NOT n/2: n/2 writes one index past
                 $\hookrightarrow$  the end of the bitset
                for (int j = 2 * i * (i + 1); j <= (n - 1) / 2; j += p)
                    is_prime_odd[j] = 0;
            }
        }

        bool is_prime(int x) const {
            if (x == 2) return true;
            if (x < 2 || x % 2 == 0) return false;
            return is_prime_odd[x / 2];
        }
};

```

4.2.5 SegmentedSieve

Segmented Sieve for Range $[L, R]$ - Time: $O(\sqrt{R}) + (R - L) \log \log R$, Space: $O(\sqrt{R}) + R - L$

```

// Computes all prime numbers in range  $[L, R]$  where  $R \leq 1e12$  and  $(R - L) \leq 1e7$ 
struct SegmentedSieve {
    static vector<ll> get_primes_in_range(ll L, ll R) {
        ll limit = sqrtl((ld)R);
        while (limit * limit > R) limit--;
        while ((limit + 1) * (limit + 1) <= R) limit++;
        vector<bool> is_prime_small(limit + 1, true);
        vector<ll> primes;
        for (ll i = 2; i <= limit; i++) {
            if (is_prime_small[i]) {
                primes.pb(i);
                for (ll j = i * i; j <= limit; j += i)
                    is_prime_small[j] = false;
            }
        }

        vector<bool> is_prime_range(R - L + 1, true);
        for (ll p : primes) {
            ll start = max(p * p, (L + p - 1) / p * p);
            for (ll j = start; j <= R; j += p) {
                is_prime_range[j - L] = false;
            }
        }
        for (ll x = L; x <= min(R, 1LL); x++) is_prime_range[x - L] =
             $\hookrightarrow$  false; // 0 and 1

        vector<ll> result;
        for (ll i = 0; i <= R - L; i++) {
            if (is_prime_range[i]) result.pb(L + i);
        }
        return result;
    }
};

```

4.2.6 MillerRabinPollard

Primality + factorization for n up to $1e18$. isPrime $O(\log^3 n)$, factor $O(n^{1/4})$.

```

// THE gate for any problem with  $a_i$  up to  $1e18$  - divisors, phi, tau,
//  $\hookrightarrow$  sigma all follow from factor().
typedef unsigned long long u64;
u64 mulmod(u64 a, u64 b, u64 m) { return (__int128)a * b % m; }
u64 powmod(u64 a, u64 e, u64 m) {
    u64 r = 1; a %= m;
    for (; e; e >>= 1, a = mulmod(a, a, m)) if (e & 1) r = mulmod(r, a,
         $\hookrightarrow$  m);
    return r;
}

// Deterministic for all  $n < 3.3e24$  with these 7 bases (Miller-Rabin).
bool isPrime(u64 n) {
    if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
    u64 d = n - 1; int s = __builtin_ctzll(d); d >>= s;
    for (u64 a : {2, 325, 9375, 28178, 450775, 9780504, 1795265022}) {
        u64 p = powmod(a % n, d, n); int i = s;
        while (p != 1 && p != n - 1 && a % n && i--) p = mulmod(p, p,
             $\hookrightarrow$  n);
        if (p != n - 1 && i != s) return false;
    }
    return true;
}

u64 pollard(u64 n) { // returns a
     $\hookrightarrow$  non-trivial divisor (Brent)
    auto f = [n](u64 x) { return mulmod(x, x, n) + 1; };
    u64 x = 0, y = 0, t = 30, prd = 2, i = 1, q;
    while (t++ % 40 || gcd(prd, n) == 1) {
        if (x == y) x = ++i, y = f(x);
        if ((q = mulmod(prd, x > y ? x - y : y - x, n))) prd = q;
        x = f(x), y = f(f(y));
    }
    return gcd(prd, n);
}

vector<u64> factor(u64 n) { // unsorted
     $\hookrightarrow$  multiset of primes
    if (n <= 1) return {}; // guard:
     $\hookrightarrow$  factor(0) would loop forever
    if (isPrime(n)) return {n};
    u64 x = pollard(n);
    auto l = factor(x), r = factor(n / x);
    l.insert(l.end(), all(r));
    return l;
}

// {prime, exponent} pairs, sorted
vector<pair<u64, int>> factorize(u64 n) {
    auto f = factor(n); sort(all(f));
}

```

```

vector<pair<u64, int>> r;
for (u64 p : f) (!r.empty() && r.back().first == p) ?
↳ r.back().second++ : (r.push_back({p, 1}), 0);
return r;
}

vector<u64> divisors(u64 n) { // all
↳ divisors, unsorted
vector<u64> d = {1};
for (auto [p, e] : factorize(n)) {
int sz = d.size();
for (u64 pe = 1, k = 0; k < (u64)e; k++) {
pe *= p;
for (int i = 0; i < sz; i++) d.push_back(d[i] * pe);
}
}
return d;
}

u64 phi(u64 n) { u64 r = n; for (auto [p, e] : factorize(n)) r -= r /
↳ p; return r; } // Euler totient
u64 tau(u64 n) { u64 r = 1; for (auto [p, e] : factorize(n)) r *= e +
↳ 1; return r; } // #divisors
u64 sigma(u64 n) {
↳ // sum of divisors
u64 r = 1;
for (auto [p, e] : factorize(n)) { u64 t = 1, pe = 1; for (int i =
↳ 0; i < e; i++) t += (pe *= p); r *= t; }
return r;
}

```

4.2.7 MultiplicativeSieve

Linear sieve computing ANY multiplicative function for all $i \leq n$ in $O(N)$.

```

// Shipped: spf, phi, mu, tau (#divisors), sigma (sum of divisors).
// To add your own f: you need f(p), f(p^k) and f(a*b)=f(a)f(b) for
↳ coprime a,b.
struct MultSieve {
int n;
vector<int> spf, primes, phi, mu, tau, cnt; // cnt[i] =
↳ exponent of spf[i] in i
vector<ll> sigma, pw; // pw[i] =
↳ spf[i]^cnt[i]

MultSieve(int n) : n(n), spf(n + 1), phi(n + 1), mu(n + 1), tau(n +
↳ 1), cnt(n + 1),
sigma(n + 1), pw(n + 1) {
phi[1] = mu[1] = tau[1] = sigma[1] = pw[1] = 1;
for (int i = 2; i <= n; i++) {
if (!spf[i]) {
spf[i] = i, primes.push_back(i);
phi[i] = i - 1, mu[i] = -1, tau[i] = 2, cnt[i] = 1,
↳ pw[i] = i, sigma[i] = i + 1LL;
}
for (int p : primes) {
if (p > spf[i] || (ll)i * p > n) break;
int j = i * p;
spf[j] = p;
if (p == spf[i]) { // p divides i:
↳ same prime, exponent grows
cnt[j] = cnt[i] + 1, pw[j] = pw[i] * p;
phi[j] = phi[i] * p;
mu[j] = 0;
tau[j] = tau[i] / (cnt[i] + 1) * (cnt[j] + 1);
sigma[j] = sigma[i] / ((pw[i] * p - 1) / (p - 1)) *
↳ ((pw[j] * p - 1) / (p - 1));
} else { // coprime:
↳ multiply
cnt[j] = 1, pw[j] = p;
phi[j] = phi[i] * (p - 1);
mu[j] = -mu[i];
tau[j] = tau[i] * 2;
sigma[j] = sigma[i] * (p + 1);
}
}
}
}

vector<int> factors(int x) const { // O(log x) via
↳ spf
vector<int> f;
while (x > 1) f.push_back(spf[x]), x /= spf[x];
return f;
}
};

```

4.2.8 PrimeEnumeration1e9

ENUMERATE EVERY PRIME UP TO $1e9$ IN $\sim 0.5s$, and in a 30-smooth wheel that costs 1 BIT per 30

```

// integers (so  $\sim 33$  MB at  $n = 1e9$  versus  $\sim 60$  MB for the odds-only
↳ bitset in 04 - Sieve1e9.cpp).
// Credit: min_25. Use THIS when you need the primes themselves at  $n \geq$ 
↳  $1e8$ ; use 04 when you only
// need the primality bitmap and want six lines instead of sixty.
// HOW IT IS FAST: (1) a mod-30 wheel - of the 30 residues only
↳ {1,7,11,13,17,19,23,29} can be
// prime, so one byte covers 30 integers; (2) the pattern of the small
↳ primes ( $\leq Q$ ) is computed
// ONCE into 'pre' and memcpy'd over each block; (3) the sieve runs
↳ block by block so everything
// stays in L1/L2 cache, with each prime remembering where it left off
↳ in each of the 8 residues.
vector<int> sievePrimes(const int N, const int Q = 17, const int L = 1
↳ << 15) {
static const int rs[] = {1, 7, 11, 13, 17, 19, 23, 29};
struct P { P(int p) : p(p) {} int p; int pos[8]; };
auto approxPrimeCount = [](const int N) -> int {
return N > 60184 ? N / (log(N) - 1.1) : max(1., N / (log(N) -
↳ 1.11)) + 1;
};
const int v = sqrt(N), vv = sqrt(v);
vector<bool> isp(v + 1, true);
for (int i = 2; i <= vv; ++i) if (isp[i])
for (int j = i * i; j <= v; j += i) isp[j] = false;
const int rsize = approxPrimeCount(N + 30);
vector<int> primes = {2, 3, 5};
int psize = 3;
primes.resize(rsize);
vector<P> sp;
size_t pbeg = 0;
int prod = 1;
for (int p = 7; p <= v; ++p) {
if (!isp[p]) continue;
if (p <= Q) prod *= p, ++pbeg, primes[psize++] = p;
auto pp = P(p);
for (int t = 0; t < 8; ++t) {
int j = (p <= Q) ? p : p * p;
while (j % 30 != rs[t]) j += p << 1;
pp.pos[t] = j / 30;
}
sp.push_back(pp);
}
vector<unsigned char> pre(prod, 0xFF); // the
↳ small primes' pattern, once
for (size_t pi = 0; pi < pbeg; ++pi) {
auto pp = sp[pi];
const int p = pp.p;
for (int t = 0; t < 8; ++t) {
const unsigned char m = ~(1 << t);
for (int i = pp.pos[t]; i < prod; i += p) pre[i] &= m;
}
}
const int bs = (L + prod - 1) / prod * prod;
vector<unsigned char> block(bs);
unsigned char* pb = block.data();
const int M = (N + 29) / 30;
for (int beg = 0; beg < M; beg += bs, pb += bs) {
int end = min(M, beg + bs);
for (int i = beg; i < end; i += prod) copy(pre.begin(),
↳ pre.end(), pb + i);
if (beg == 0) pb[0] &= 0xFE; // 1 is
↳ not prime
for (size_t pi = pbeg; pi < sp.size(); ++pi) {
auto& pp = sp[pi];
const int p = pp.p;
for (int t = 0; t < 8; ++t) {
int i = pp.pos[t];
const unsigned char m = ~(1 << t);
for (; i < end; i += p) pb[i] &= m;
pp.pos[t] = i; //
↳ resume here in the next block
}
}
for (int i = beg; i < end; ++i)
for (int m = pb[i]; m > 0; m &= m - 1)
primes[psize++] = i * 30 + rs[__builtin_ctz(m)];
}
while (psize > 0 && primes[psize - 1] > N) --psize;
primes.resize(psize);
return primes;
}

// pi(1e9) = 50847534, so the output vector alone is  $\sim 200$  MB of int -
↳ if you only need to ITERATE

```

```
// the primes, sieve block by block and consume them inside the loop
//   ↳ instead of storing them.
// Q = 17 means the wheel pre-pattern covers 7*11*13*17 = 17017 bytes;
//   ↳ raising Q makes 'pre'
// bigger but the inner loop shorter. L = 1<<15 keeps a block inside L1.
// FOR PRIMALITY OF ONE NUMBER up to 1e18 use Miller-Rabin (06); for
//   ↳ the FACTORISATION of one
// number use Pollard rho (06); for pi(n) without the primes use
//   ↳ Lucy_Hedgehog (04 - Theorems/03).
```

4.2.9 Min25Sieve

MIN_25 SIEVE - prefix sum of ANY multiplicative f in $O(n^{3/4} / \log n)$, n up to $\sim 1e10$ - $1e11$.

// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
 // You must supply two things:
 // c[] : $f(p)$ written as a POLYNOMIAL in p , $f(p) = c[0] + c[1]*p +$
 // ↳ $c[2]*p^2$ (degree ≤ 2 here,
 // extend $\text{pref}()$ with more power sums for higher degree),
 // fpe : $f(p^e)$ for a prime power, given (p, e, p^e) .
 // PHASE 1 sieves sum of p^k over primes $\leq v$ for every $v = \text{floor}(n/i)$
 // ↳ (this is exactly Lucy's
 // method, 04 - Theorems/03 - PrimeCounting.cpp). PHASE 2 (S) walks
 // ↳ every "smallest prime factor"
 // class and re-adds the prime-power contributions. Memory $O(\sqrt{n} * \deg)$.
 // EXAMPLES: phi $\rightarrow c = \{-1, 1\}$, $fpe = p^e - p^{e-1}$; mu $\rightarrow c = \{-1\}$,
 // ↳ $fpe = (e == 1 ? -1 : 0)$;
 // tau $\rightarrow c = \{2\}$, $fpe = e + 1$; Id $\rightarrow c = \{0,$
 // ↳ $1\}$, $fpe = p^e$;
 // $f(p^e) = p^e * (p^e - 1) \rightarrow c = \{0, -1, 1\}$ (that is $f(p) =$
 // ↳ $p^2 - p$).
 // If f is COMPLETELY multiplicative, phase 1 alone is the answer (use
 // ↳ Lucy directly).

```
struct Min25 {
    ll n, rt;
    int np;
    vector<ll> pr, w; // primes <=
    ↳ sqrt(n); values floor(n/i)
    vector<int> id1, id2; // position of
    ↳ v in w, for v <= rt / v > rt
    vector<vector<Mint>> g, sp; // g[k][j] over
    ↳ w[j]; sp[k][i] = prefix p^k
    vector<Mint> c;
    function<Mint(ll, int, ll)> fpe;

    int id(ll x) { return x <= rt ? id1[x] : id2[n / x]; }
    Mint pref(ll x, int k) { // sum_{i=2..x}
    ↳ i^k (mod MOD)
        Mint z = x % MOD;
        if (k == 0) return z - 1;
        if (k == 1) return z * (z + 1) / 2 - 1;
        return z * (z + 1) * (2 * z + 1) / 6 - 1;
    }
    Min25(ll n, vector<Mint> c, function<Mint(ll, int, ll)> fpe) :
    ↳ n(n), c(c), fpe(fpe) {
        rt = (ll)sqrtl((ll)n);
        while (rt * rt > n) rt--;
        while ((rt + 1) * (rt + 1) <= n) rt++;
        int d = c.size();
        vector<char> comp(rt + 2, 0);
        for (ll i = 2; i <= rt; i++)
            if (!comp[i]) { pr.push_back(i); for (ll j = i * i; j <=
    ↳ rt; j += i) comp[j] = 1; }
        np = pr.size();
        sp.assign(d, vector<Mint>(np + 1, 0));
        for (int i = 0; i < np; i++) {
            Mint t = 1;
            for (int k = 0; k < d; k++) sp[k][i + 1] = sp[k][i] + t, t
    ↳ *= pr[i] % MOD;
        }
        id1.assign(rt + 2, 0), id2.assign(rt + 2, 0);
        for (ll i = 1, j; i <= n; i = j + 1) { // w is
    ↳ DECREASING: w[0] = n, back = 1
            ll x = n / i; j = n / x;
            w.push_back(x);
            (x <= rt ? id1[x] : id2[n / x]) = w.size() - 1;
        }
        g.assign(d, vector<Mint>(w.size()));
        for (int k = 0; k < d; k++)
            for (int j = 0; j < (int)w.size(); j++) g[k][j] =
    ↳ pref(w[j], k);
        for (int i = 0; i < np; i++) // sieve out
    ↳ multiples of pr[i]
            for (int j = 0; j < (int)w.size() && pr[i] <= w[j] / pr[i];
    ↳ j++) {
                int u = id(w[j] / pr[i]);
                Mint pk = 1;
```

```
        for (int k = 0; k < d; k++) g[k][j] -= pk * (g[k][u] -
    ↳ sp[k][i]), pk *= pr[i] % MOD;
    }
    Mint S(ll x, int j) { // sum of f(i),
    ↳ i <= x, spf(i) > pr[j-1]
        if (j >= 1 && pr[j - 1] >= x) return 0;
        Mint res = 0;
        for (int k = 0; k < (int)c.size(); k++) res += c[k] *
    ↳ (g[k][id(x)] - sp[k][j]);
        for (int i = j; i < np && pr[i] <= x / pr[i]; i++) {
            ll pw = pr[i];
            for (int e = 1; pw <= x; e++) {
                res += fpe(pr[i], e, pw) * (S(x / pw, i + 1) + Mint(e
    ↳ != 1));
                if (pw > x / pr[i]) break;
                pw *= pr[i];
            }
        }
        return res;
    }
    Mint solve() { return n < 1 ? Mint(0) : S(n, 0) + 1; } // the +1
    ↳ is f(1)
};
// The "+ Mint(e != 1)" term is  $f(p^e)$  counted for the number  $p^e$ 
//   ↳ itself (only for  $e \geq 2$ , since
//  $e = 1$  is already inside the prime sum). No memoisation is needed
//   ↳ anywhere - that is the point.
// SIBLINGS: 10 - PowerfulNumberSieve.cpp is usually FASTER ( $O(\sqrt{n})$ )
//   ↳ when you can find a  $g$  with
//  $g(p) = f(p)$  and an easy prefix sum; 04 - Theorems/05 - DuSieve.cpp
//   ↳ is the  $O(n^{2/3})$  tool when
// a Dirichlet partner makes the convolution telescope.
```

4.2.10 PowerfulNumberSieve

POWERFUL NUMBER SIEVE - prefix sum of a multiplicative f in $O(\sqrt{n})$ calls to an easy prefix

// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
 // sum. Faster and shorter than Min_25 whenever you can find a helper g
 // ↳ with
 // (1) $g(p) = f(p)$ for every prime p , and (2) the prefix sum $G(x) =$
 // ↳ $\text{sum}_{i \leq x} g(i)$ is cheap.
 // Then $f = g * h$ (Dirichlet), and $h(p) = f(p) - g(p) = 0$, so h is
 // ↳ supported ONLY on POWERFUL
 // numbers (every exponent ≥ 2) - and there are only $O(\sqrt{n})$ of
 // ↳ those below n . Hence
 // $F(n) = \text{sum}_{d \text{ powerful}, d \leq n} h(d) * G(n / d)$,
 // which this DFS evaluates. $h(p^e)$ is derived on the fly from f and g
 // ↳ by the convolution
 // recurrence $h(p^e) = f(p^e) - \text{sum}_{i < e} h(p^i) * g(p^{e-i})$, so
 // ↳ you never derive it by hand.
 // PRECONDITION: $g(p) = f(p)$ for ALL primes - check it, the algorithm
 // ↳ is silently wrong otherwise.
 // COST: $O(\sqrt{n})$ nodes, each one call to G . With G in $O(1)$ that is
 // ↳ $O(\sqrt{n})$ total; with G in
 // $O(\sqrt{x})$ it is still about $O(n^{2/3})$. Needs primes up to $\sqrt{n}$
 // ↳ (a plain sieve).
 // EXAMPLE: $f(p^e) = p^e + [e \geq 2]$ pairs with $g = \text{Id}$ ($G(x) =$
 // ↳ $x(x+1)/2$). $f = \text{"phi times something"}$
 // style problems usually pair with $g = \text{Id}$ or $g = 1$ or $g = \text{sigma}$ ($G(x) =$
 // ↳ $\text{sum } d * \text{floor}(x/d)$).

```
struct PowerfulSieve {
    ll n;
    vector<ll> pr; // primes up to
    ↳ sqrt(n)
    function<Mint(ll, int)> f, g; // f(p^e),
    ↳ g(p^e), e >= 1
    function<Mint(ll)> G; // prefix sum
    ↳ of g
    PowerfulSieve(ll n, vector<ll> pr, function<Mint(ll, int)> f,
    ↳ function<Mint(ll, int)> g,
        function<Mint(ll)> G) : n(n), pr(pr), f(f), g(g),
    ↳ G(G) {
        Mint dfs(int i, ll m, Mint h) { // m = a
    ↳ powerful number, h = h(m)
            Mint res = h * G(n / m);
            for (ll lim = n / m; i < (int)pr.size() && pr[i] <= lim /
    ↳ pr[i]; i++) {
                ll p = pr[i], pw = p * p;
                vector<Mint> hp = {1, 0}; // h(p^0) = 1,
    ↳ h(p^1) = 0
                for (int e = 2; pw <= lim; e++) {
                    Mint v = f(p, e);
                    for (int j = 0; j < e; j++) v -= hp[j] * g(p, e - j);
                    hp.push_back(v);
                    res += dfs(i + 1, m * pw, h * v);
                    if (pw > lim / p) break;
```

```

        pw *= p;
    }
    }
    return res;
}
}
Mint solve() { return dfs(0, 1, 1); }
};
// COUNTING POWERFUL NUMBERS is itself a useful fact: every powerful
// number is uniquely  $a^2 \cdot b^3$ 
// with  $b$  squarefree, so there are about  $2.173 \cdot \sqrt{n}$  of them below  $n$ .
// WHEN  $g$  IS HARD TO GUESS: try  $g = \text{Id}^k$  for the  $k$  with  $f(p) \sim p^k$ ,
// then fix the difference; or
// fall back to 09 - Min25Sieve.cpp, which needs no partner function at
// all.

```

4.2.11 DivisorSummatory

```

DIVISOR SUMMATORY FUNCTION  $D(n) = \sum_{i=1..n} \tau(i) = \sum_{i=1..n} \text{floor}(n/i)$ 
// in  $\sim O(n^{1/3})$  instead of the usual  $O(\sqrt{n})$  divisor blocks. About
// 100 ms at  $n = 1e18$ .
// Reach for it when  $n$  is  $1e14$  or more (blocks are fine up to  $\sim 1e12$ ),
// or when many queries make
// the sqrt too slow. The result exceeds 64 bits around  $n \sim 4e17$ , hence
// __int128.
// HOW: the hyperbola  $x \cdot y = n$  is walked with a STERN-BROCOT search over
// slopes. Starting from the
// point nearest the diagonal, we repeatedly take the longest lattice
// segment of the current slope
// that stays under the hyperbola (counting the trapezoid under it in
//  $O(1)$ ), and refine the slope
// by mediants when the curvature no longer allows it. Below  $y \sim$ 
//  $n^{1/3}$  the slopes get shorter
// than one step, so the tail is finished with a plain loop.
__int128 divisorSummatory(unsigned long long n) {
    if (n == 0) return 0;
    auto out = [n](unsigned long long x, unsigned y) { return x * y >
    n; }; // above the curve?
    auto cut = [n](unsigned long long x, unsigned dx, unsigned dy) {
        return (__uint128_t)x * x * dy >= (__uint128_t)n * dx;
    }; // slope too steep
    unsigned long long sn = (unsigned long long)sqrtl((ld)n);
    while (sn * sn > n) sn--;
    while ((sn + 1) * (sn + 1) <= n) sn++;
    unsigned long long cn = (unsigned long long)powl((ld)n, 0.34L);
    //  $\sim n^{1/3}$ 
    unsigned long long x = n / sn;
    unsigned y = (unsigned)(n / x + 1);
    __int128 ret = 0;
    stack<pair<unsigned, unsigned>> st;
    st.emplace(1, 0), st.emplace(1, 1); // the
    // Stern-Brocot stack of slopes
    while (true) {
        unsigned lx, ly;
        tie(lx, ly) = st.top(); st.pop();
        while (out(x + lx, y - ly)) { // walk as long
            // as the segment fits
            ret += (__uint128_t)x * ly + (__uint128_t)(ly + 1) * (lx - 1) / 2;
            x += lx, y -= ly;
        }
        if (y <= cn) break;
        unsigned rx = lx, ry = ly;
        while (true) { // pop until a
            // usable slope is on top
            tie(lx, ly) = st.top();
            if (out(x + lx, y - ly)) break;
            rx = lx, ry = ly, st.pop();
        }
        while (true) { // mediant
            // refinement
            unsigned mx = lx + rx, my = ly + ry;
            if (out(x + mx, y - my)) st.emplace(lx = mx, ly = my);
            else { if (cut(x + mx, lx, ly)) break; rx = mx, ry = my; }
        }
    }
    for (--y; y > 0; --y) ret += n / y; // the small-y
    // tail, plain loop
    return ret * 2 - (__uint128_t)sn * sn; // Dirichlet
    // hyperbola symmetry
}
// EXACT VALUES for testing:  $D(10) = 27$ ,  $D(100) = 482$ ,  $D(1e6) =$ 
//  $13970034$ ,  $D(1e9) = 20951330018$ ,
//  $D(1e18) = 31264402477813376929$ . Asymptotically  $D(n) = n \ln n +$ 
//  $(2\gamma - 1)n + O(n^{1/3})$ .
// RELATED SUMS with the same blocks:  $\sum \sigma(i) = \sum i \cdot \text{floor}(n/i)$ ,
// and any  $\sum_{i=1}^n f(n/i)$ ;
// for those, plain  $O(\sqrt{n})$  blocks (03 - FloorSum.cpp) are the right

```

→ tool.

4.2.12 MeisselLehmer

MEISSEL-LEHMER PRIME COUNTING - $\pi(n)$ in about $O(n^{2/3} / \log n)$, well under a second at

```

//  $n = 1e12$  and a few seconds at  $1e14$ . Use it when you need  $\pi(n)$  MANY
// times, or for  $n$  past the
// comfort zone of Lucy's  $O(n^{3/4})$  method (04 - Theorems/03 -
// PrimeCounting.cpp) - but Lucy
// stays the better choice when you also want the SUM of the primes,
// which this cannot give.
//  $\phi(n, k)$  = count of  $i \leq n$  free of the first  $k$  primes; the
// recursion  $\phi(n, k) =$ 
//  $\phi(n, k-1) - \phi(n/p_k, k-1)$  is memoised in a table for small
// arguments, and Lehmer's
// identity peels off the numbers with exactly two or three prime
// factors.
// CALL init() ONCE. Memory: LIM ints for the prefix table plus  $PN \cdot PK$ 
// ints (about 30 MB with the
// constants below) - shrink  $PN/PK$  if the limit is tight, raise LIM if
//  $n$  is huge.
namespace PCF {
    const int LIM = 5000000, PN = 40000, PK = 60; // sieve bound;
    // memo table dimensions
    vector<int> primes, pref; // pref[i] =
    //  $\pi(i)$  for  $i < LIM$ 
    vector<vector<int>>> memo;
    ll isqrt(ll x) { ll r = (ll)sqrtl((ld)x); while (r * r > x) r--;
    while ((r+1)*(r+1) <= x) r++;
    return r; }
    ll icbrt(ll x) { ll r = (ll)cbrtl((ld)x); while (r * r * r > x) r--;
    while ((r+1)*(r+1)*(r+1) <= x) r++; return r; }
    void init() {
        vector<bool> comp(LIM, false);
        pref.assign(LIM, 0);
        for (int i = 2; i < LIM; i++) {
            if (!comp[i]) { primes.push_back(i);
                for (ll j = (ll)i * i; j < LIM; j += i)
                    comp[j] = 1; }
            pref[i] = primes.size();
        }
        memo.assign(PN, vector<int>(PK));
        for (int i = 0; i < PN; i++) memo[i][0] = i;
        for (int k = 1; k < PK; k++)
            for (int i = 0; i < PN; i++)
                memo[i][k] = memo[i][k-1] - memo[i / primes[k-1]][k-1];
    }
    ll phi(ll n, int k) { // count  $i \leq n$  free
    // of the first  $k$  primes
    if (n < PN && k < PK) return memo[n][k];
    if (k == 1) return (n + 1) >> 1;
    if (primes[k-1] >= n) return 1;
    return phi(n, k-1) - phi(n / primes[k-1], k-1);
    }
    ll legendre(ll n) { // simplest
    // version,  $O(n^{2/3} \log n)$ -ish
    if (n < LIM) return pref[n];
    int k = pref[isqrt(n)];
    return phi(n, k) + k - 1;
    }
    ll lehmer(ll n) { // the fast one
    if (n < LIM) return pref[n];
    ll b = isqrt(n), c = lehmer(icbrt(n)), a = lehmer(isqrt(b));
    b = lehmer(b);
    ll res = phi(n, a) + (b + a - 2) * (b - a + 1) / 2;
    for (ll i = a; i < b; i++) {
        ll w = n / primes[i];
        res -= lehmer(w);
        if (i <= c) {
            ll lim = lehmer(isqrt(w));
            for (ll j = i; j < lim; j++) res += j - lehmer(w /
            // primes[j]);
        }
    }
    return res;
    }
}
// SANITY VALUES:  $\pi(1e6) = 78498$ ,  $\pi(1e9) = 50847534$ ,  $\pi(1e12) =$ 
//  $37607912018$ ,
//  $\pi(1e13) = 346065536839$ ,  $\pi(1e15) = 29844570422669$ .
//  $\pi(n) \sim n / \ln n$ , and  $\text{li}(n)$  is a much better estimate; the  $k$ -th
// prime is  $\sim k(\ln k + \ln \ln k)$ .
// NEXT STEPS: the Lagarias-Miller-Odlyzko and Deleglise-Rivat
// refinements reach  $n^{2/3} / \log^2$  and
// are what a real prime-counting program uses - not worth writing
// during a contest.

```

4.2.13 CountKDivisors

HOW MANY $x \leq n$ HAVE EXACTLY k DIVISORS? n up to $\sim 1e12$, k up to a few thousand.

```
// Needs: PCF::lehmer, PCF::primes (12 - MeisselLehmer.cpp).
// A number with exactly k divisors has the shape  $p_1^{e_1-1} \cdot p_2^{e_2-1} \cdot \dots$ 
// where  $e_1 \cdot e_2 \cdot \dots = k$ 
// and (choosing the exponents in decreasing order) the primes can be
// taken increasing. So we
// backtrack over the FACTORISATIONS of  $k$  into factors  $> 1$ , assigning a
// prime to each factor.
// Two things make it fast: (a) the case  $k = 2$  is just "count the
// primes  $\leq n$ ", answered by
// Meissel-Lehmer, and (b) a single remaining factor  $x$  means "count
// primes  $p$  with  $p^{(x-1)} \leq n$ ",
// answered by a binary search over the prime list.
// PRECONDITIONS: call PCF::init() first, and prep() once for the  $k$ 
// range you need. Sieve limit
// LIM in PCF must exceed  $n^{1/2}$  for the binary searches to be safe
// (5e6 covers  $n \leq 2.5e13$ ).
// COMPLEXITY: hard to state - it is a search whose branching is
// bounded by the divisors of  $k$ ;
// in practice a handful of milliseconds per query for  $n = 1e12$ .
namespace KDiv {
    const int MK = 2100; // max k + 1
    vector<vector<int>> dv; // dv[k] =
    // divisors of k that are > 1, desc
    vector<int> minexp; // minexp[k] =
    // sum (p-1) over primes p | k
    void prep() {
        dv.assign(MK, {}), minexp.assign(MK, 0);
        for (int i = 2; i < MK; i++) {
            for (int j = i; j < MK; j += i) dv[j].push_back(i);
            for (int p : PCF::primes) {
                if (p > i) break;
                int cur = i;
                while (cur % p == 0) cur /= p, minexp[i] += p - 1;
            }
        }
        for (int i = 2; i < MK; i++) reverse(all(dv[i])); // try the
    // big exponents first
    }
    ll powCap(ll b, int e, ll cap) { //  $b^e$ ,
    // saturating at cap + 1
    ll r = 1;
    for (int i = 0; i < e; i++) { if (r > cap / b) return cap + 1;
    r *= b; }
    return r;
    }
    // count of  $x \leq n$  with exactly  $k$  divisors, using only primes from
    // index 'first' on
    ll go(ll n, int k, int first = 0) {
        if (n <= 0 || k >= MK) return 0;
        if (k == 1) return 1;
        if (n == 1) return 0;
        if (k == 2) return max(0LL, PCF::lehmer(n) - first);
        if (dv[k].size() == 1) { // k is prime:
            // one prime power  $p^{(k-1)}$ 
            int e = dv[k][0] - 1;
            if (powCap(PCF::primes[first], e, n) > n) return 0;
            int lo = first, hi = (int)PCF::primes.size() - 1, ans =
            // first - 1;
            while (lo <= hi) { // last prime
                int mid = (lo + hi) / 2;
                if (powCap(PCF::primes[mid], e, n) > n) hi = mid - 1;
                else lo = mid + 1, ans = mid;
            }
            return ans - first + 1;
        }
        ll res = 0;
        for (int x : dv[k]) // this prime
            // carries exponent  $x - 1$ 
            for (int i = first; i < (int)PCF::primes.size(); i++) {
                ll p = powCap(PCF::primes[i], x - 1, n);
                if (p > n) break;
                if (k / x == 1) { res++; continue; }
                if (powCap(PCF::primes[i + 1], minexp[k / x], n) > n /
                // p) break;
                res += go(n / p, k / x, i + 1);
            }
        return res;
    }
}
// RANGE QUERY: answer(l, r) = go(r, k) - go(l - 1, k).
// SMALL n: just sieve tau with a linear sieve (07 -
// MultiplicativeSieve.cpp) and count.
// SMALLEST number with exactly k divisors is a different search - 15 -
```

→ SmallestWithKDivisors.cpp.

// FACTS: $\tau(x)$ is odd iff x is a perfect square; the count of $x \leq n$
// with $\tau(x) = 2$ is $\pi(n)$,
// with $\tau(x) = 3$ it is $\pi(\sqrt{n})$, and with $\tau(x) = 4$ it is $\pi(\sqrt[4]{n})$
// plus the number of
// products $p \cdot q$ of two distinct primes below n .

4.2.14 PrimeBasis

COPRIME (PRIME) BASIS - factor a set of numbers RELATIVE TO EACH OTHER,
without ever factoring

// them. Given n values up to $1e18$ (far past Pollard-rho comfort when n
// is large), it builds a set
// of PAIRWISE COPRIME basis elements such that every input is a
// product of basis powers. Then
// each input becomes an exponent vector, and gcd / lcm / divisibility
// / "product is a square" all
// become coordinatewise min / max / $\leq$ on those vectors.
// WHEN: "is $a_i \mid \prod a_j$ ", "make all numbers equal by moving prime
// factors", "count subsets
// whose product is a perfect k -th power", any problem where only the
// RELATIVE factorisations of
// the given numbers matter and the numbers are too big to factor.
// SIZE: at most $n \cdot \log \log(\max a)$ basis elements; build is $O(n^2)$
// $\log(\max a)$ worst case, and
// far less in practice. factorize(x) is $O(|\text{basis}| \cdot \log x)$.
// NOTE the basis elements are not necessarily prime - they are only
// pairwise coprime, which is
// all that matters. They ARE prime whenever the inputs force it.

```
template <typename T>
struct PrimeBasis {
    vector<T> basis;
    void reducePair(T& x, T& y) { // strip the
    // common part greedily
        bool sw = 0;
        if (x > y) sw ^= 1, swap(x, y);
        while (x > 1 && y % x == 0) { y /= x; if (x > y) sw ^= 1,
        swap(x, y); }
        if (sw) swap(x, y);
    }
    void split(int pos, T& val) { // basis[pos]
    // and val share a factor
        while (basis[pos] % val == 0) basis[pos] /= val;
        vector<T> cur = {basis[pos], val};
        for (int ptr = 1; ptr < (int)cur.size(); ptr++)
            for (int i = 0; i < ptr; i++) {
                reducePair(cur[i], cur[ptr]);
                if (cur[ptr] == 1 || cur[i] == 1) break;
                T g = gcd(cur[ptr], cur[i]);
                if (g > 1) cur[ptr] /= g, cur[i] /= g, cur.push_back(g);
            }
        basis[pos] = cur[0], val = cur[1];
        for (int i = 2; i < (int)cur.size(); i++) if (cur[i] > 1)
        basis.push_back(cur[i]);
        if (basis[pos] == 1) swap(basis[pos], basis.back()),
        basis.pop_back();
    }
    void add(T val) {
        for (int i = 0; i < (int)basis.size(); i++) {
            reducePair(val, basis[i]);
            if (basis[i] == 1) { swap(basis[i], basis.back());
            basis.pop_back(); break; }
            if (val == 1) return;
            if (gcd(basis[i], val) > 1) split(i, val);
        }
        if (val > 1) basis.push_back(val);
    }
    vector<int> factorize(T x) { // exponent
    // vector of x over the basis
        vector<int> f(basis.size());
        for (int i = 0; i < (int)basis.size(); i++) while (x % basis[i]
        // == 0) f[i]++, x /= basis[i];
        return f;
    }
};
```

// AFTER BUILDING: add() EVERY input first, then factorize() them - the
// basis must be final, or
// old vectors have the wrong length. Verify with: every factorize(a_i)
// must multiply back to a_i
// (i.e. dividing a_i by all basis powers leaves 1), and all pairs of
// basis elements are coprime.
// The vectors support: gcd = coordinatewise min, lcm = max, $a \mid b$ iff
// $\text{vec}(a) \leq \text{vec}(b)$
// coordinatewise, $a \cdot b = \text{sum}$, $a/b = \text{difference}$, x is a perfect square
// iff all coordinates are even.

4.2.15 SmallestWithKDivisors

THE SMALLEST NUMBER WITH EXACTLY k DIVISORS. The answer explodes ($k = 1e9$ already needs ~ 100

```
// Needs: powmod (06 - MillerRabinPollard.cpp), divisors of k (03 -
//   Divisors.cpp or factorize).
// digits), so it is compared by the SUM OF LOGARITHMS and reported
//   modulo MOD.
// Structure:  $x = p_0^{e_0} * p_1^{e_1} * \dots$  over the FIRST primes in
//   order, with
//  $(e_0+1)(e_1+1)\dots = k$  and  $e_0 \geq e_1 \geq \dots$  (swapping a larger
//   exponent onto a smaller prime
// never hurts). So: walk the primes in order, and at prime  $i$  choose a
//   divisor  $x \mid k$  for the
// factor  $(e_i + 1) = x$ . Memoise on (prime index, remaining  $k$ ).
// PRECONDITIONS:  $pr$  must hold the first  $\sim 64$  primes (2, 3, 5, ...);  $dvs$ 
//    $\hookrightarrow$  = all divisors of  $k$  that
//   are  $> 1$ , ASCENDING (that order is what makes the break-early pruning
//   valid).  $k \geq 1$ .
// COMPLEXITY:  $O(\#states * \tau(k))$  with tiny constants - milliseconds
//    $\hookrightarrow$  for  $k$  up to  $1e12$ .
struct SmallestKDiv {
    vector<ll> pr, dvs;
    vector<ld> lg;
    vector<map<ll, pair<ld, ll>>> memo; // memo[i][rem]
     $\hookrightarrow$  = {log of  $x$ ,  $x \bmod MOD$ }
    SmallestKDiv(vector<ll> pr, vector<ll> dvs) : pr(pr), dvs(dvs) {
        for (ll p : pr) lg.push_back(logl((ld)p));
        memo.assign(pr.size() + 1, {});
    }
    pair<ld, ll> go(int i, ll rem) { // use primes
     $\hookrightarrow$  pr[i], pr[i+1], ... only
        if (rem == 1) return {0.0L, 1};
        auto it = memo[i].find(rem);
        if (it != memo[i].end()) return it->second;
        pair<ld, ll> best = {1e30L, 0};
        for (ll x : dvs) {
            if (rem % x) continue;
            ld z = lg[i] * (ld)(x - 1); // cost of
     $\hookrightarrow$  putting exponent  $x-1$  on pr[i]
            if (z > best.first) break; // dvs
     $\hookrightarrow$  ascending  $\Rightarrow$  all later are worse
            auto cur = go(i + 1, rem / x);
            cur.first += z;
            cur.second = cur.second * powmod(pr[i], x - 1, MOD) % MOD;
            best = min(best, cur);
        }
        return memo[i][rem] = best;
    }
    ll solve(ll k) { return k == 1 ? 1 : go(0, k).second; }
};
// EXACT SMALL ANSWERS (for a sanity check):  $k = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$ 
//    $\hookrightarrow \rightarrow 1, 2, 4, 6, 16, 12, 64,$ 
// 24, 36, 48.  $k = 100 \rightarrow 45360$ . The sequence is OEIS A005179.
// LONG DOUBLE is enough here: two candidates tie only when the numbers
//   are equal, and the log
// sums differ by more than  $1e-9$  otherwise for any  $k$  that fits in 64
//   bits. If paranoid, compare
// the exponent vectors exactly with big integers, or compare logs and
//   break ties by the vectors.
// RELATED: the MAXIMUM number of divisors below  $n$  (highly composite
//   numbers) is found by the
// same DFS with the objective flipped - keep the primes in order and
//   try  $e_i = 0..60$  with
//  $e_0 \geq e_1 \geq \dots$ ; tau maxes at 1344 for  $n \leq 1e9$ , 6720 for  $1e12$ ,
//   103680 for  $1e18$ .
```

4.3 Modular Arithmetic

4.3.1 CRT

```
// NOTE: extGCD below is the same one as in 01 - Euclidean Algorithm.
//   Paste only one copy.
long long extGCD(long long a, long long b, long long &x, long long &y) {
    if (b == 0) {
        x = 1; y = 0;
        return a;
    }
    long long x1, y1;
    long long d = extGCD(b, a % b, x1, y1);
    x = y1;
    y = x1 - y1 * (a / b);
    return d;
}
// Solves  $x = a_i \pmod{m_i}$ . Returns  $\{x, \text{lcm}(m_1, \dots, m_k)\}$ .
pair<long long, long long> CRT(const vector<long long>& a, const
     $\hookrightarrow$  vector<long long>& m) {
```

```
long long res = a[0], lcm = m[0];
for (int i = 1; i < a.size(); i++) {
    long long x, y;
    long long g = extGCD(lcm, m[i], x, y);
    if ((a[i] - res) % g != 0) return {-1, -1}; // No solution

    long long step = m[i] / g;
    long long k = ((a[i] - res) / g) % step * (x % step) % step;
    res = (long long)((__int128)k * lcm + res) % ((__int128)lcm *
     $\hookrightarrow$  step));
    lcm *= step;
    res = (res % lcm + lcm) % lcm;
}
return {res, lcm};
}
```

4.3.2 DiscreteLogAndRoots

Discrete log, modular square root, primitive root, discrete k -th root, multiplicative order.

// Needs: powmod (02 - Primes and Divisors/06 - MillerRabinPollard.cpp).
// All need mulmod/powmod from MillerRabinPollard.

```
// BSGS: smallest  $x \geq 0$  with  $a^x = b \pmod{m}$ . Works for ANY  $m$  (handles
//    $\hookrightarrow \text{gcd}(a, m) \neq 1$ ).  $O(\sqrt{m})$ .
ll discreteLog(ll a, ll b, ll m) {
    a %= m, b %= m;
    ll k = 1, add = 0, g;
    while ((g = gcd(a, m)) > 1) { // strip common
     $\hookrightarrow$  factors
        if (b == k) return add;
        if (b % g) return -1;
        b /= g, m /= g, add++, k = k * (a / g) % m;
    }
    ll n = (ll)sqrtl((ld)m) + 1, an = 1;
    for (ll i = 0; i < n; i++) an = an * a % m;
    unordered_map<ll, ll> vals;
    for (ll q = 0, cur = b; q <= n; q++) vals[cur] = q, cur = cur * a %
     $\hookrightarrow$  m;
    for (ll p = 1, cur = k; p <= n; p++) {
        cur = cur * an % m;
        if (vals.count(cur)) { ll ans = n * p - vals[cur] + add; if
     $\hookrightarrow$  (ans >= 0) return ans; }
    }
    return -1;
}

// Tonelli-Shanks:  $x$  with  $x^2 = a \pmod{p}$ ,  $p$  odd prime. Returns -1 if  $a$ 
//    $\hookrightarrow$  is not a QR.
ll sqrtMod(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a == 0) return 0;
    if (powmod(a, (p - 1) / 2, p) != 1) return -1; // Euler's
     $\hookrightarrow$  criterion
    if (p % 4 == 3) return powmod(a, (p + 1) / 4, p);
    ll s = p - 1, n = 2; int r = 0;
    while (s % 2 == 0) s /= 2, r++;
    while (powmod(n, (p - 1) / 2, p) != p - 1) n++; // find a
     $\hookrightarrow$  non-residue
    ll x = powmod(a, (s + 1) / 2, p), b = powmod(a, s, p), g =
     $\hookrightarrow$  powmod(n, s, p);
    for (int m = r; m > 0; m--) {
        ll t = b;
        for (r = 0; r < m && t != 1; r++) t = t * t % p;
        if (r == 0) return x;
        ll gs = powmod(g, 1LL << (m - r - 1), p);
        g = gs * gs % p, x = x * gs % p, b = b * g % p;
    }
}

// Smallest primitive root of a PRIME  $p$  (generator of  $\mathbb{Z}_p^*$ ).
ll primitiveRoot(ll p) {
    if (p == 2) return 1;
    auto f = factorize(p - 1);
    for (ll g = 2; g < p; g++) {
        bool ok = true;
        for (auto [q, e] : f) if (powmod(g, (p - 1) / q, p) == 1) { ok
     $\hookrightarrow$  = false; break; }
        if (ok) return g;
    }
    return -1;
}
```

```
// Multiplicative order of  $a \bmod m$  (smallest  $e > 0$  with  $a^e = 1$ ).
//   Requires  $\text{gcd}(a, m) = 1$ .
ll multOrder(ll a, ll m) {
    ll e = phi(m), r = e;
    for (auto [q, k] : factorize(e))
        while (r % q == 0 && powmod(a, r / q, m) == 1) r /= q;
```

```

    return r;
}

// All x with x^k = a (mod p), p prime. Reduce to a discrete log in
// base g.
vector<ll> discreteRoot(ll k, ll a, ll p) {
    if (a == 0) return {0};
    ll g = primitiveRoot(p), y = discreteLog(powmod(g, k, p), a, p);
    if (y < 0) return {};
    ll x0 = powmod(g, y, p), d = gcd(k, p - 1), step = powmod(g, (p - 1) / d, p);
    vector<ll> r;
    for (ll i = 0, cur = x0; i < d; i++, cur = cur * step % p)
        r.push_back(cur);
    sort(all(r)), r.erase(unique(all(r)), r.end());
    return r;
}

```

4.3.3 LucasAndBinomialMod

$C(n, r) \bmod m$ for HUGE n . Pick by what m is:

```

// Needs: Mint (06 - Math/01), and factorize (02 - Primes and
// Divisors/06) for the last section.
// m prime and small -> lucas
// m = p^e -> binomPrimePower (Andrew Granville /
// Wilson generalisation)
// m arbitrary (squarefree-ish)-> binomAnyMod (CRT over prime powers)

// LUCAS. p prime, needs fact/invFact built up to p. C(n,r) = prod
// C(n_i, r_i) over base-p digits.
Mint lucas(ll n, ll r) {
    if (r < 0 || r > n) return 0;
    Mint res = 1;
    while (n || r) {
        ll a = n % MOD, b = r % MOD;
        if (b > a) return 0;
        res *= fact[a] * invFact[b] * invFact[a - b];
        n /= MOD, r /= MOD;
    }
    return res;
}

// --- arbitrary modulus ---
ll pw(ll a, ll e, ll m) { ll r = 1; a %= m; for (; e; e >>= 1, a = a *
// a % m) if (e & 1) r = r * a % m; return r; }
ll inv_mod(ll a, ll m) { // requires
// gcd(a, m) = 1
    ll g = m, x = 0, x1 = 1, a1 = a % m;
    while (a1) { ll q = g / a1; g -= q * a1, swap(g, a1); x -= q * x1,
// swap(x, x1); }
    return (x % m + m) % m;
}

// n! with all factors of p removed, mod p^e. O(p^e + log n).
ll factNoP(ll n, ll p, ll pe) {
    if (n == 0) return 1;
    vector<ll> f(pe);
    f[0] = 1;
    for (ll i = 1; i < pe; i++) f[i] = (i % p ? f[i - 1] * i : f[i - 1] / p) % pe;
    ll r = 1;
    for (ll m = n; m; m /= p) {
        r = r * pw(f[pe - 1], m / p, pe) % pe;
        r = r * f[m % p] % pe;
    }
    return r;
}

// C(n, r) mod p^e
ll binomPrimePower(ll n, ll r, ll p, ll pe) {
    if (r < 0 || r > n) return 0;
    ll k = 0; // exponent of
// p in C(n,r) (Kummer's theorem:
    for (ll m = n / p, a = r / p, b = (n - r) / p; m; m /= p, a /= p, b
// /= p) k += m - a - b;
    if (k >= 60 || pw(p, k, pe * p) % pe == 0) { ll t = pw(p, k, pe);
// if (t == 0) return 0; }
    ll num = factNoP(n, p, pe), d1 = factNoP(r, p, pe), d2 = factNoP(n
// - r, p, pe);
    ll res = num * inv_mod(d1, pe) % pe * inv_mod(d2, pe) % pe;
    return res * pw(p, k, pe) % pe;
}

// C(n, r) mod m for arbitrary m: factor m, solve mod each p^e, CRT.
ll binomAnyMod(ll n, ll r, ll m) {
    ll res = 0, M = m;
    for (auto [p, e] : factorize(m)) {
        ll pe = 1; for (int i = 0; i < e; i++) pe *= p;
        ll c = binomPrimePower(n, r, p, pe), t = M / pe;
        res = (res + c % pe * t % M * inv_mod(t % pe, pe)) % M;
    }
}

```

```

    return res;
}

// KUMMER'S THEOREM: the exponent of prime p in C(n,r) = number of
// CARRIES when adding
// r and n-r in base p. (Used above to pull out the
// p-part.)
// WILSON: (p-1)! = -1 (mod p) for prime p.

```

4.3.4 CongruencesAndTowers

Linear congruences and power towers. Needs powmod / phi / gcd.

```

// Needs: powmod, phi (04 - NT/02 - Primes and Divisors/06 -
// MillerRabinPollard.cpp).

// Solve a*x = b (mod m). Returns all solutions in [0, m) (there are
// gcd(a,m) of them, or none).
vector<ll> linCongruence(ll a, ll b, ll m) {
    a = ((a % m) + m) % m, b = ((b % m) + m) % m;
    ll g = gcd(a, m);
    if (b % g) return {};
    ll m2 = m / g, a2 = a / g, b2 = b / g;
    // a2 is invertible mod m2
    ll x0 = 0, x1 = 1, g2 = m2, a3 = a2 % m2;
    while (a3) { ll q = g2 / a3; g2 -= q * a3, swap(g2, a3); x0 -= q *
// x1, swap(x0, x1); }
    ll inv = ((x0 % m2) + m2) % m2, x = b2 % m2 * inv % m2;
    vector<ll> r;
    for (ll i = 0; i < g; i++) r.push_back((x + i * m2) % m);
    return r;
}

// GENERALISED EULER: a^k = a^(k mod phi(m) + phi(m)) (mod m)
// whenever k >= log2(m).
// This is what makes power towers computable even when gcd(a, m) != 1.
ll capTower(const vector<ll>& a, int i, ll CAP = 64) { // min(true
// tower value, CAP)
    if (i == (int)a.size()) return 1;
    if (a[i] == 0) return capTower(a, i + 1, CAP) == 0 ? 1 : 0; //
// 0^0 = 1, 0^(k>0) = 0
    if (a[i] == 1) return 1;
    ll e = capTower(a, i + 1, CAP), r = 1;
    for (ll k = 0; k < e; k++) { r *= a[i]; if (r >= CAP) return CAP; }
    return r;
}

// a[i] ^ a[i+1] ^ ... ^ a[last] (mod m), right-associative.
ll powTower(const vector<ll>& a, int i, ll m) {
    if (m == 1) return 0;
    if (i == (int)a.size()) return 1 % m;
    ll ph = phi(m), t = capTower(a, i + 1);
    ll e = t < 64 ? t : powTower(a, i + 1, ph) + ph; // exact when
// tiny, +phi when large
    return powmod(a[i] % m, e, m);
}

// Extended CRT for NON-coprime moduli: x = a1 (mod m1), x = a2 (mod
// m2).
// Returns {x, lcm} or {-1, -1}. Chain it to merge many congruences.
pair<ll, ll> crt2(ll a1, ll m1, ll a2, ll m2) {
    ll g = gcd(m1, m2), l = m1 / g * m2;
    if ((a2 - a1) % g) return {-1, -1};
    ll m2g = m2 / g;
    ll x0 = 0, x1 = 1, gg = m2g, aa = (m1 / g) % m2g;
    while (aa) { ll q = gg / aa; gg -= q * aa, swap(gg, aa); x0 -= q *
// x1, swap(x0, x1); }
    ll inv = ((x0 % m2g) + m2g) % m2g;
    __int128 k = (__int128)((a2 - a1) / g % m2g) * inv % m2g;
    ll x = (ll)((__int128)k * m1 + a1) % l;
    return {(x % l + l) % l, l};
}

```

4.3.5 Garner

GARNER'S ALGORITHM - reconstruct x from residues modulo PAIRWISE COPRIME m_i , in mixed-radix

```

// form, so you can emit x modulo a target MOD without ever building
// prod(m_i).
// This is what CRT cannot do once the product overflows 64 bits.
// x = x1 + x2*m1 + x3*m1*m2 + ... , 0 <= xi < mi
ll garner(vector<ll> a, vector<ll> m, ll MODOUT) {
    int k = a.size();
    vector<ll> x(k);
    auto inv = [](ll a, ll M) { // requires
// gcd(a, M) = 1
        ll g = M, X = 0, x1 = 1, A = ((a % M) + M) % M;
        while (A) { ll q = g / A; g -= q * A, swap(g, A); X -= q * x1,
// swap(X, x1); }
        return (X % M + M) % M;
    };
}

```

```

for (int i = 0; i < k; i++) {
    x[i] = ((a[i] % m[i]) + m[i]) % m[i];
    ll mul = 1;
    for (int j = 0; j < i; j++) {
        x[i] = (x[i] - x[j] % m[i] * mul) % m[i];
        mul = mul * (m[j] % m[i]) % m[i];
    }
    x[i] = (x[i] % m[i] + m[i]) % m[i] * inv(mul, m[i]) % m[i];
}
ll res = 0, mul = 1;
for (int i = 0; i < k; i++) {
    res = (res + x[i] % MODOUT * mul) % MODOUT;
    mul = mul * (m[i] % MODOUT) % MODOUT;
}
return (res % MODOUT + MODOUT) % MODOUT;
}
// MAIN USE: arbitrary-modulus convolution. Run NTT under three NTT
// → primes, then Garner the
// three residues down to the target modulus.
// NTT PRIMES: 998244353 = 119*2^23 + 1 (g=3)    167772161 =
// → 5*2^25 + 1 (g=3)    469762049 = 7*2^26 + 1 (g=3)    1004535809 =
// → 479*2^21 + 1 (g=3)    754974721 = 45*2^24 + 1 (g=11)    <- note: NOT
// → 9*2^23+1, a common misprint
// The 3-prime product 5.95e25 covers n * (MOD-1)^2 for any MOD < 2^30
// → and n up to ~5e7.

```

4.3.6 Montgomery

MONTGOMERY MULTIPLICATION - modular multiplication for a FIXED ODD 64-bit modulus without any

```

// hardware division: 2-3x faster than ((__int128)a*b % m. Use it in the
// → hot loop of Miller-Rabin,
// Pollard rho, BSGS, matrix power or any modpow-heavy routine with a
// → 64-bit modulus. For a
// COMPILE-TIME modulus the compiler already turns % into a
// → multiply-shift, so this buys nothing;
// the win is for a modulus known only at run time.
// TRICK: work with the representative a*R mod m where R = 2^64.
// → reduce(x) computes x*R^-1 mod m
// with two multiplications and no division, so (aR)(bR)*R^-1 = (ab)R
// → stays in the same form.
// PRECONDITIONS: m must be ODD (the Newton iteration inverts m modulo
// → 2^64) and m < 2^63.
// Values returned by get() are in [0, m); mul/add work on the
// → Montgomery form.
typedef unsigned long long u64; // same
// → typedef as 02/06 - keep one copy
struct Mont {
    u64 mod, minv, r2; // minv = m^-1
    // → mod 2^64
    void set(u64 m) {
        mod = minv = m;
        for (int i = 0; i < 5; i++) minv *= 2 - minv * m; //
        // → Newton: doubles the bits each time
        r2 = (u64)(-((__uint128_t)m % m)); // R^2
        // → mod m
    }
    u64 reduce(__uint128_t x) const {
        u64 y = (u64)(x >> 64) - (u64)((__uint128_t)((u64)x * minv) *
        // → mod) >> 64);
        return (long long)y < 0 ? y + mod : y;
    }
    u64 in(u64 a) const { return reduce((__uint128_t)(a % mod) * r2); }
    // → // to Montgomery form
    u64 get(u64 a) const { u64 y = reduce(a); return y >= mod ? y - mod
    // → : y; } // back to normal
    u64 mul(u64 a, u64 b) const { return reduce((__uint128_t)a * b); }
    u64 add(u64 a, u64 b) const { a += b - mod; return (long long)a < 0
    // → ? a + mod : a; }
    u64 sub(u64 a, u64 b) const { a -= b; return (long long)a < 0 ? a +
    // → mod : a; }
    u64 pow(u64 a, u64 e) const { // a already
    // → in Montgomery form
        u64 r = in(1);
        for (; e >>= 1, a = mul(a, a)) if (e & 1) r = mul(r, a);
        return r;
    }
};
// USAGE: Mont M; M.set(n); u64 x = M.in(3); x = M.pow(x, d); cout
// → << M.get(x);
// Do NOT mix forms: everything inside the loop stays in Montgomery
// → form, converting only at the
// boundaries. Comparisons against 1 and m-1 (Miller-Rabin) must
// → compare against M.in(1) and
// M.in(m-1), which you compute once.
// EVEN MODULUS: split m = 2^k * odd, run Montgomery on the odd part

```

→ and the trivial bit-mask on
 // the power of two, then CRT. Usually not worth it - just use __int128.
 // ALTERNATIVE: BARRETT reduction works for even moduli too (precompute
 // → floor(2^64/m) and replace
 // the division by a multiply-shift; it is a little slower but has no
 // → parity restriction.

4.3.7 QuadraticExtension

ARITHMETIC IN $F_p[\sqrt{D}] = \{a + b\sqrt{D} : a, b \text{ in } F_p\}$ - the quadratic extension field

```

// Needs: powmod (02 - Primes and Divisors/06 - MillerRabinPollard.cpp).
// when D is a NON-RESIDUE mod p (then it is F_{p^2}), or a ring of
// → "split" numbers when it is.
// WHAT IT BUYS YOU:
// * CIPOLLA's square root mod p: always O(log p) multiplications,
// → and unlike Tonelli-Shanks it
// does not degrade when p - 1 has a huge power of two. (Tonelli is
// → in 02 - DiscreteLog...)
// * Closed forms that need an irrational: Fibonacci via
// → ((1+sqrt(5))/2)^n mod p, and ANY order-2
// linear recurrence x_{n+1} = a x_n + b x_{n-1}, even when the
// → discriminant is a non-residue
// mod p (no matrix needed - one field power).
// * Lucas sequences / Frobenius primality tests, and n-th terms of
// → (a + b sqrt(D))^n identities.
// PRECONDITIONS: QP an odd prime, all inputs reduced into [0, QP). QP
// → up to ~4.6e18 (the __int128
// products need QP^2 < 2^127).
ll QP, QD; // the modulus
// → and the radicand
ll qmul(ll x, ll y) { return (ll)((__int128)x * y % QP); }
struct Quad {
    ll a, b; // a +
    // → b*sqrt(QD)
    Quad(ll a = 0, ll b = 0) : a(a), b(b) {}
    Quad operator*(const Quad& o) const {
        return Quad((qmul(a, o.a) + qmul(QD, qmul(b, o.b))) % QP,
        // → (qmul(a, o.b) + qmul(b, o.a)) % QP);
    }
    Quad operator+(const Quad& o) const { return Quad((a + o.a) % QP,
    // → (b + o.b) % QP); }
    Quad operator-(const Quad& o) const { return Quad((a - o.a + QP) %
    // → QP, (b - o.b + QP) % QP); }
    Quad conj() const { return Quad(a, (QP - b) % QP); } // the
    // → Frobenius x -> x^p
    ll norm() const { return (qmul(a, a) - qmul(QD, qmul(b, b)) % QP +
    // → QP) % QP; }
};
Quad qpow(Quad x, ll e) { // O(log e)
    // → with 3 multiplications a step
    Quad r(1, 0);
    for (; e >>= 1, x = x * x) if (e & 1) r = r * x;
    return r;
}
// CIPOLLA: x with x^2 = a (mod p), p an odd prime. Returns -1 if a is
// → not a quadratic residue.
ll cipolla(ll a, ll p) {
    a %= p; if (a < 0) a += p;
    if (a <= 1) return a;
    if (powmod(a, (p - 1) / 2, p) != 1) return -1; // Euler's
    // → criterion
    QP = p;
    ll t = 1;
    while (true) { // find t with
    // → t^2 - a a NON-residue
        QD = ((__int128)t * t - a) % p; QD = (QD + p) % p;
        if (QD && powmod(QD, (p - 1) / 2, p) == p - 1) break;
        t++;
    }
    return qpow(Quad(t, 1), (p + 1) / 2).a; // the sqrt(QD)
    // → part is provably 0
}
// The other root is p - x. Expected O(log p) tries for t (half the
// → residues work).
// FIBONACCI mod p: set QD = 5, phi = (1 + sqrt 5)/2 = Quad((p+1)/2,
// → (p+1)/2); then
// F(n) = (phi^n - (1 - phi)^n) / sqrt(5), i.e. qpow(phi, n).b * 2
// → (mod p) since the two
// conjugate parts cancel. Works for every odd p, whether or not 5 is
// → a residue.
// GENERAL ORDER-2 RECURRENCE x_{n+1} = A x_n + B x_{n-1}: the roots of
// → z^2 = A z + B live in
// F_p[sqrt(A^2 + 4B)], so x_n is a linear combination of their n-th
// → powers - one qpow call.

```

4.3.8 SqrtModPrimePower

factorize (02 - Primes and Divisors/06 - MillerRabinPollard.cpp).

```
// Needs: sqrtMod (02 - DiscreteLogAndRoots.cpp), extGCD (01 -
//   ↳ Euclidean Algorithm), CRT (01),
//   ↳ factorize (02 - Primes and Divisors/06 -
//   ↳ MillerRabinPollard.cpp).
// SQUARE ROOTS MODULO A PRIME POWER, AND MODULO A COMPOSITE - what
//   ↳ Tonelli-Shanks alone cannot
// do. Everything here is HENSEL LIFTING: a root mod  $p^j$  lifts to a
//   ↳ root mod  $p^{j+1}$  by one
// Newton step  $x \leftarrow x - f(x)/f'(x)$ , valid whenever  $f'(x) = 2x$  is
//   ↳ invertible mod  $p$ , i.e.  $p$  odd and
//  $p \nmid x$ .  $p = 2$  is the exception (the derivative is even) and is
//   ↳ handled separately below.
// PRECONDITION for both routines:  $\gcd(a, p) = 1$ . If  $p \mid a$ , first pull
//   ↳ out the largest  $p^{2t}$ 
// dividing  $a$  (an ODD exponent of  $p$  in  $a$  means NO solution unless it is
//   ↳  $\geq k$ ), solve for  $a/p^{2t}$ 
// modulo  $p^{k-2t}$ , multiply the roots by  $p^t$ , and add every multiple
//   ↳ of  $p^{k-t}$ .
// COMPLEXITY:  $O(k \log p)$  per lift;  $O(\log^2)$  for the composite version
//   ↳ plus the factorisation.
ll invMod(ll a, ll m) { ll x, y; extGCD((a % m) + m, m, m, x, y);
//   ↳ return ((x % m) + m) % m; }
// one root of  $x^2 = a \pmod{p^k}$ ,  $p$  an ODD prime,  $\gcd(a, p) = 1$ . -1 if
//   ↳  $a$  is a non-residue mod  $p$ .
ll sqrtPrimePower(ll a, ll p, int k) {
    ll x = sqrtMod((a % p) + p, p);
    if (x < 0) return -1;
    ll pk = p;
    for (int j = 1; j < k; j++) {
        ll np = pk * p;
        ll fx = (((__int128)x * x - a) % np + np) % np;
        x = (ll)(((__int128)x - (__int128)fx * invMod(2 * x % np, np))
//   ↳ % np + np) % np;
        pk = np;
    }
    return x;
//   ↳ the other
//   ↳ root is  $p^k - x$ 
}
// one root of  $x^2 = a \pmod{2^k}$ ,  $a$  ODD. -1 if there is none.
ll sqrtPow2(ll a, int k) {
    a &= (1LL << k) - 1;
    if (k == 1) return 1;
//   ↳ a odd  $\Rightarrow x$ 
//   ↳ = 1
    if (k == 2) return (a & 3) == 1 ? 1 : -1;
    if ((a & 7) != 1) return -1;
//   ↳ a = 1 (mod
//   ↳ 8) is necessary for  $k \geq 3$ 
    ll x = 1;
    for (int j = 3; j < k; j++) {
//   ↳ lift  $2^j \rightarrow$ 
//   ↳  $2^{j+1}$ ; pick the right one
        ll nm = 1LL << (j + 1);
        if (((__int128)x * x - a) % nm + nm) % nm) x += 1LL << (j - 1);
    }
    return x;
}
// NUMBER OF SOLUTIONS of  $x^2 = a \pmod{p^k}$  with  $\gcd(a, p) = 1$ :
//   ↳  $p$  odd: 2 if  $a$  is a residue mod  $p$ , else 0. The roots are  $\pm x$ .
//   ↳  $p = 2$ :  $k = 1 \rightarrow 1$  root;  $k = 2 \rightarrow 2$  roots ( $a = 1 \pmod{4}$ );  $k \geq 3 \rightarrow$ 
//   ↳ 4 roots ( $a = 1 \pmod{8}$ ),
//   ↳ namely  $\pm x$  and  $\pm x + 2^{k-1}$ .
// COMPOSITE  $m$ : factor  $m = \prod p_i^{k_i}$ , solve each, and CRT every
//   ↳ COMBINATION of signs - the
// solution count is the product of the per-prime-power counts, so it
//   ↳ can be  $2^{\omega(m)}$  many.
// THE SPECIAL CASE  $a = 1$  is worth memorising: the number of  $x$  with  $x^2$ 
//   ↳ = 1 (mod  $m$ ) is
//   ↳  $\prod$  over odd  $p^k \mid m$  of 2, times (1 if  $2 \nmid m$  is  $2^1$ , 2 if  $4 \mid m$ , 4
//   ↳ if  $8 \mid m$ ).
ll numSqrt1(ll m) {
//   ↳ count of  $x$ 
//   ↳ in  $[0, m)$  with  $x^2 = 1$ 
    ll res = 1;
    for (auto [p, e] : factorize(m))
        res *= (p != 2 ? 2 : (e == 1 ? 1 : e == 2 ? 2 : 4));
    return res;
}
// USES: counting self-inverse elements, "how many  $x$  with  $x^2 = 1$ " in
//   ↳ group-theory counting,
// solving quadratic congruences  $ax^2 + bx + c = 0$  (complete the
//   ↳ square:  $(2ax + b)^2 = b^2 - 4ac$ ,
// needs  $\gcd(2a, m) = 1$ , and lifting solutions of ANY polynomial  $f -$ 
//   ↳ the same Newton step works
// whenever  $\gcd(f'(x_0), p) = 1$ .
```

4.3.9 ModularAPBlocks

THE SEQUENCE $(a^k + b) \pmod{m}$, FOR $k = 0..n-1$, AS $O(\sqrt{m})$ ARITHMETIC PROGRESSIONS.

```
// Each returned block {start, diff, len} lists consecutive terms
//   ↳ start, start+diff, ... that do
// NOT wrap past  $m$  - so a value-sorted / value-structured query over
//   ↳ the whole sequence becomes
//  $O(\sqrt{m})$  range operations instead of  $n$  individual ones. This is the
//   ↳ constructive form of the
// THREE-DISTANCE THEOREM: the points  $k \cdot a \pmod{m}$  take at most three
//   ↳ distinct gaps.
// USE IT FOR: sum of  $x^k((a^k + b) \pmod{m})$ , sum over  $k$  of  $f((a^k + b) \pmod{m})$ 
//   ↳ for an  $f$  with prefix
// sums, min/max of  $(a^k + b) \pmod{m}$  over a range of  $k$ , counting terms
//   ↳ in a value window when the
// generic  $O(\log)$  count (01 - Euclidean Algorithm/06) does not fit the
//   ↳ shape of the problem.
// PRECONDITIONS for decomposeModAP:  $\gcd(a, m) = 1$  and  $n \leq m$ . The
//   ↳ wrapper solveModAP below
// removes both restrictions:  $\gcd$  is peeled off by rewriting the
//   ↳ sequence, and  $n > m$  is split
// into  $\text{floor}(n/m)$  identical full cycles plus a remainder.
// COMPLEXITY:  $O(\sqrt{m} \log m)$  for the decomposition (the sort
//   ↳ dominates).
struct APBlock { ll start, diff, len; };
// start +
//   ↳  $i \cdot \text{diff}$  for  $0 \leq i < \text{len}$ 
vector<APBlock> decomposeModAP(ll a, ll b, ll m, ll n) {
    vector<APBlock> res;
    ll w = (ll)sqrtl((ld)m) + 1;
    if (n <= w) {
//   ↳ too short to
//   ↳ be worth the machinery
        for (ll i = 0; i < n; i++) res.push_back({(a * i + b) % m, 0,
//   ↳ 1});
        return res;
    }
    vector<pair<ll, ll>> v(w + 1);
//   ↳ (residue,
//   ↳ index) for the first  $w+1$  terms
    for (ll i = 0; i <= w; i++) v[i] = {a * i % m, i};
    sort(all(v));
    ll d = 1, h = m;
//   ↳  $a \cdot d = \pm h$ 
//   ↳ (mod  $m$ ) with  $h \leq m/w$ ,  $|d| \leq w$ 
    for (ll i = 0; i < w; i++) {
//   ↳ pigeonhole:
//   ↳ two residues within  $m/w$ 
        ll diff = v[i + 1].first - v[i].first;
        if (diff < h) h = diff, d = v[i + 1].second - v[i].second;
    }
    if (d > 0) {
//   ↳ stepping k
//   ↳ by  $d$  increases the value by  $h$ 
        for (ll r = 0; r < d; r++)
            for (ll j = r; j < n; j++) {
                ll val = (a * j + b) % m;
                ll step = min((m - val - 1) / h + 1, (n - j - 1) / d +
//   ↳ 1);
                res.push_back({val, h, step});
                j += step * d;
            }
    } else {
//   ↳ stepping k
//   ↳ by  $-d$  decreases it by  $h$ 
        for (ll r = n - 1; r >= n + d; r--)
            for (ll j = r; j >= 0; j--) {
                ll val = (a * j + b) % m;
                ll step = min((m - val - 1) / h + 1, j / -d + 1);
                res.push_back({val, h, step});
                j += step * d;
            }
    }
    return res;
}
// FULL WRAPPER for arbitrary  $a, b, m, n$ : reduce  $\gcd$ , then full cycles,
//   ↳ then one decomposition.
//  $g = \gcd(a, m) > 1$ : write  $b = b_1 \cdot g + b_2$  with  $b_2 < g$ ; then  $(a^k + b)$ 
//   ↳ mod  $m$ 
//   ↳ =  $b_2 + g \cdot ((a/g)^k + b_1) \pmod{(m/g)}$  -
//   ↳ recurse on the smaller modulus.
//  $n \geq m$  (after  $\gcd = 1$ ): each residue in  $[0, m)$  appears exactly
//   ↳ once per cycle, so
//   ↳  $\text{floor}(n/m)$  full cycles contribute a constant,
//   ↳ and  $n \bmod m$  is left over.
// EXAMPLE (sum of  $x^k((a^k + b) \pmod{m}) \pmod{p}$ ): per block the answer is
//   ↳ the geometric-like sum
//  $x^{\text{start}} \cdot (1 + x^{\text{diff}} + \dots + x^{\text{diff} \cdot (\text{len}-1)})$ , computable in  $O(\log$ 
//   ↳  $\text{len})$  by the standard
// "return  $(x^n, 1 + x + \dots + x^{(n-1)})$  together" doubling recursion.
```

4.3.10 RootsOfUnityFilter

SUM OF BINOMIALS OVER A CONGRUENCE CLASS: $r[j] = \text{sum over } k = j \pmod{m} \text{ of } C(n, k)$, for every

```
// Needs: multiply (06 - Math/05 - Convolution/02 - NTT.cpp), modulus
// → NMOD = 998244353.
// j at once, with n up to 1e18 and m up to ~1e5.  $O(m \log m \log n)$ .
// THE IDEA:  $(1 + x)^n$  has  $C(n, k)$  as its coefficients, so the answer
// → is that polynomial reduced
// MODULO  $x^m - 1$  (i.e. exponents folded by  $k \bmod m$ ).
// → Binary-exponentiate  $(1 + x)$  in the ring
//  $Z_p[x] / (x^m - 1)$ : one NTT product plus a fold per bit.
// PRECONDITIONS: works modulo NMOD (998244353); no condition on m at
// → all - unlike the classical
// roots-of-unity filter, this needs no primitive  $m$ -th root to exist.
vector<ll> binomByResidue(ll n, int m) { // r[j] =
    sum_{k = j mod m} C(n, k)
    auto fold = [&](vector<ll> v) { // reduce
        modulo  $x^m - 1$ 
        for (int i = (int)v.size() - 1; i >= m; i--) v[i - m] = (v[i -
        m] + v[i]) % NMOD;
        v.resize(min<int>(v.size(), m));
        return v;
    };
    vector<ll> res = {1}, base = fold({1, 1});
    for (; n; n >>= 1) {
        if (n & 1) res = fold(multiply(res, base));
        base = fold(multiply(base, base));
    }
    res.resize(m);
    return res;
}
// THE CLASSICAL ROOTS-OF-UNITY FILTER, which is what you use when m is
// → TINY (2, 3, 4, ...) and
// you want a closed form instead of a convolution. With w a primitive
// →  $m$ -th root of unity:
// sum_{k = r (mod m)} a_k = (1/m) * sum_{j=0}^{m-1} w^{(-j r)} *
// →  $A(w^j)$ ,  $A(x) = \text{sum } a_k x^k$ 
// so for  $a_k = C(n, k)$ ,  $A(x) = (1+x)^n$  and
// sum_{k = r (mod m)} C(n, k) = (1/m) * sum_{j=0}^{m-1} w^{(-j r)} *
// →  $(1 + w^j)^n$ .
// Examples: m = 2 gives  $2^{(n-1)}$  for both classes (n >= 1); m = 4 gives
// → the classic
//  $(2^n + 2^{((n+2)/2)} * \cos(n \pi / 4)) / 4$  style answers.
// WHERE TO GET w: if  $m \mid p - 1$  use  $g^{((p-1)/m)}$  for a primitive root g
// → mod p; otherwise work in
// the extension  $F_p[x]/(\text{cyclotomic})$ , or just use this polynomial
// → version, which sidesteps it.
// SAME TECHNIQUE, other sequences: replace  $(1 + x)$  by ANY generating
// → function and you filter its
// coefficients by residue - e.g. "number of subsets with sum divisible
// → by m", "number of paths of
// length n whose displacement is  $0 \bmod m$ ". Reduce modulo  $x^m - 1$  and
// → exponentiate.
```

4.3.11 BinomialPrefixQueries

PARTIAL SUMS OF A BINOMIAL ROW: $f(n, k) = C(n, 0) + C(n, 1) + \dots + C(n, k)$, for q queries.

```
// Needs: Mint, buildFact, C (06 - Math/01 - Modular Arithmetic/01 -
// → Mint.cpp).
// There is NO closed form for  $f(n, k)$ , but both coordinates move in
// →  $O(1)$ :
//  $f(n+1, k) = 2*f(n, k) - C(n, k)$   $f(n, k+1) = f(n, k) + C(n,$ 
// →  $k+1)$ 
//  $f(n-1, k) = (f(n, k) + C(n-1, k)) / 2$   $f(n, k-1) = f(n, k) - C(n,$ 
// →  $k)$ 
// so treat the queries as points (n, k) and run MO'S ALGORITHM on
// → them:  $O((n + q) \sqrt{n})$  with a
// tiny constant, versus  $O(n)$  per query naively.
// REACH FOR IT when a problem asks "probability that at most k of n
// → coins came up heads" or
// "number of subsets of size <= k" for many (n, k) pairs offline.
// PRECONDITIONS: MOD prime, buildFact() called, all  $n < N$ . Queries
// → must be answered OFFLINE.
vector<Mint> binomPrefixQueries(vector<pair<int, int>> qs) {
    int q = qs.size(), B = max(1, (int)(N / max(1.0, sqrt((double)q))));
    vector<int> ord(q);
    iota(all(ord), 0);
    sort(all(ord), [&](int x, int y) {
        int bx = qs[x].first / B, by = qs[y].first / B;
        if (bx != by) return bx < by;
        return (bx & 1) ? qs[x].second > qs[y].second : qs[x].second <
        qs[y].second;
    });
    vector<Mint> ans(q);
    Mint cur = 1, inv2 = Mint((MOD + 1) / 2); // f(1, 0) =
    C(1, 0) = 1
```

```
int n = 1, k = 0;
for (int i : ord) {
    auto [L, R] = qs[i];
    while (n < L) cur = cur * 2 - C(n, k), n++;
    while (n > L) --n, cur = (cur + C(n, k)) * inv2;
    while (k < R) ++k, cur += C(n, k);
    while (k > R) cur -= C(n, k), k--;
    ans[i] = cur;
}
return ans;
}
// EXACT VALUES you can use as identities instead, when they apply:
//  $f(n, n) = 2^n$ ;  $f(n, k) = 2^n - f(n, n-k-1)$ ;  $f(2m, m) = (2^{(2m)} +$ 
// →  $C(2m, m)) / 2$ .
// A SINGLE query with n up to 1e18 and k small is just Cbig summed k
// → times.
// The same Mo trick works for any quantity with  $O(1)$  moves in both
// → indices - partial sums of
//  $C(n, i) * x^i$ , of Catalan-like rows, or of a fixed row of any
// → Pascal-style triangle.
```

4.3.12 IntersectAP

INTERSECTION OF TWO ARITHMETIC PROGRESSIONS $\{a_1 + d_1 * i : i \geq 0\}$ and $\{a_2 + d_2 * j : j \geq 0\}$.

```
// Needs: crt2 (04 - CongruencesAndTowers.cpp) or CRT (01 - CRT.cpp).
// The intersection is itself an AP with difference  $\text{lcm}(d_1, d_2)$ , whose
// → first term is the smallest
// common value that is  $\geq \max(a_1, a_2)$ . Empty exactly when the CRT
// → system is unsolvable, i.e.
// when  $(a_1 - a_2)$  is not divisible by  $\text{gcd}(d_1, d_2)$ .
// Returns {first, step}; step = 0 means EMPTY.  $O(\log)$  via one CRT.
// WHEN: merging periodic events ("both traffic lights green"),
// → intersecting the answer sets of
// several modular constraints, and range-counting over a set defined
// → by many congruences.
// OVERFLOW:  $\text{lcm}(d_1, d_2)$  can reach  $d_1 * d_2$  - use __int128 inside CRT (the
// → notebook version does).
pair<ll, ll> intersectAP(ll a1, ll d1, ll a2, ll d2) {
    auto [r, m] = crt2(((a1 % d1) + d1) % d1, d1, ((a2 % d2) + d2) %
    d2, d2);
    if (m == -1) return {0, 0}; // no common
    term at all
    ll start = max(a1, a2);
    if (r < start) r += (start - r + m - 1) / m * m; // advance to
    the first term >= start
    return {r, m};
}
// COUNTING the common terms in  $[lo, hi]$  is then  $(hi - \text{first}) / \text{step} + 1$ 
// → when first <= hi.
// MANY progressions: fold them left to right with this function; the
// → difference multiplies up to
// the lcm of all of them, so it overflows fast - keep an eye on it, or
// → cap at "beyond the range".
// THE INFINITE-IN-BOTH-DIRECTIONS version (i and j any integer) is
// → just the CRT answer with no
// max(a1, a2) clamp - that is the subgroup  $a_1 + \text{gcd-shifted lattice}$ ,
// → see also
// 01 - Euclidean Algorithm/10 - IntegerLattice2D.cpp for the
// → two-coordinate analogue.
```

4.3.13 PohligHellman

discreteLog (02 - DiscreteLogAndRoots.cpp), CRT (01 - CRT.cpp).

```
// Needs: powmod, factorize (02 - Primes and Divisors/06 -
// → MillerRabinPollard.cpp),
// discreteLog (02 - DiscreteLogAndRoots.cpp), CRT (01 -
// → CRT.cpp).
// POHLIG-HELLMAN DISCRETE LOG - solve  $g^x = h \pmod{p}$  in  $O(\text{sum over}$ 
// →  $q^e \mid\mid n$  of  $e \cdot (\log n + \sqrt{q})$ )
// instead of the  $O(\sqrt{p})$  of plain baby-step giant-step, where  $n =$ 
// →  $\text{ord}(g)$ . It is a massive win
// exactly when n is SMOOTH (all prime factors small), which is common
// → in constructed problems:
//  $p - 1 = 2^k * \text{small stuff}$ , or a subgroup of small order.
// THE IDEA: the discrete log is determined modulo each prime power  $q^e$ 
// → dividing n. Inside the
// order-q subgroup a log costs only  $O(\sqrt{q})$ , and the q-adic digits
// → of x are recovered one at a
// time; CRT glues the residues back together.
// PRECONDITIONS: p PRIME (the inverse below uses Fermat), g of
// → multiplicative order exactly n
// (pass  $n = p - 1$  when g is a primitive root),  $0 < h < p$ . Returns x in
// →  $[0, n)$ , or -1 if h is not
// in the subgroup generated by g.
ll pohligHellman(ll g, ll h, ll p, ll n) {
    vector<ll> rs, ms;
    for (auto [q, e] : factorize(n)) {
```

```

    ll qe = 1;
    for (int i = 0; i < e; i++) qe *= q;
    ll gq = powmod(g, n / q, p), x = 0, cur = 1; // gq generates
    ↪ the subgroup of order q
    for (int k = 0; k < e; k++) { // recover the
    ↪ q-adic digits of x
        ll gx = powmod(powmod(g, x, p), p - 2, p); // g^-x
        ll hk = powmod((ll)((__int128)h * gx % p), n / (cur * q),
    ↪ p);
        ll d = discreteLog(gq, hk, p); // a log inside
    ↪ a group of size q
        if (d < 0) return -1;
        x += d * cur, cur *= q;
    }
    rs.push_back(x % qe), ms.push_back(qe);
}
return CRT(rs, ms).first;
}
// WHEN TO USE WHICH
// n smooth -> this file (can be near-instant even for p
    ↪ ~ 1e18)
// n has a large prime -> plain BSGS, O(sqrt p) time AND memory (02
    ↪ - DiscreteLogAndRoots.cpp)
// composite modulus -> the BSGS in 02 already strips gcd(a, m);
    ↪ Pohlig-Hellman needs a
// group, so factor m and CRT the
    ↪ per-prime-power answers yourself.
// SANITY: check powmod(g, n, p) == 1 and that n really is the ORDER of
    ↪ g (multOrder), otherwise
// the CRT reassembles a residue modulo the wrong number and the answer
    ↪ is silently wrong.
// SECURITY TRIVIA THAT IS ACTUALLY A HINT: this is why cryptographic
    ↪ primes are chosen so that
// p - 1 has a large prime factor. If a problem hands you a "random"
    ↪ prime whose p - 1 factors
// into small pieces, that is the intended solution.

```

4.4 Theorems

4.4.1 NumberTheoryFormulas

NUMBER THEORY - FORMULA SHEET

```

/* ===== NUMBER THEORY - FORMULA SHEET =====
MULTIPLICATIVE FUNCTIONS (f(ab)=f(a)f(b) for gcd(a,b)=1)
tau(n)=#divisors=prod(e_i+1) sigma(n)=sum of divisors=prod
    ↪ (p^i(e_i+1)-1)/(p-1)
phi(n)=n*prod(1-1/p) mu(n)= 0 if NOT SQUAREFREE (some p^2
    ↪ | n), else (-1)^#primes
sum_{d|n} phi(d) = n sum_{d|n} mu(d) = [n==1] (this
    ↪ is the key identity)
sum_{d|n} mu(d)/d = phi(n)/n sum_{d|n} mu(d)*tau(n/d) = 1

DIRICHLET CONVOLUTION (f*g)(n) = sum_{d|n} f(d) g(n/d)
1*1 = tau Id*1 = sigma phi*1 = Id mu*1 = eps([n==1])
mu is the inverse of 1, so f = g*1 <=> g = f*mu.

MOBIUS INVERSION
g(n) = sum_{d|n} f(d) <=> f(n) = sum_{d|n} mu(d) g(n/d)
g(n) = sum_{n|d, d<=N} f(d) <=> f(n) = sum_{n|d, d<=N} mu(d/n)
    ↪ g(d) (the "upward" form)
Standard use: [gcd(i,j)==1] = sum_{d | gcd(i,j)} mu(d).

GCD / LCM SUMS (all O(N log N) or better with a mu/phi sieve)
#coprime pairs (i,j)<=n = sum_{d=1..n} mu(d) * floor(n/d)^2
sum_{i,j<=n} gcd(i,j) = sum_{d=1..n} phi(d) * floor(n/d)^2
sum_{d|n} d*phi(n/d) = sum_{i=1..n} gcd(i,n)
sum_{i=1..n} lcm(i,n) = n/2 * (1 + sum_{d|n} d*phi(d))
sum_{i,j<=n} lcm(i,j) = sum_g g * sum_{a,b<=n/g,
    ↪ gcd(a,b)=1} a*b
general: sum_{i,j} f(gcd(i,j)) = sum_d (f*mu)(d) * floor(n/d)^2

DIVISOR BLOCKING - evaluates sum_{i=1..n} F(floor(n/i)) in O(sqrt n):
for (ll l = 1, r; l <= n; l = r + 1) { r = n / (n / l); } //
    ↪ floor(n/i) == n/l on [l, r]
Gives sum tau(i), sum sigma(i), sum floor(n/i), and Min_25 /
    ↪ Lucy-style prefix sums.

MODULAR
Fermat: a^(p-1) = 1 (mod p), p prime, p !| a => a^-1 = a^(p-2)
Euler: a^phi(m) = 1 (mod m) if gcd(a,m)=1
GENERALISED EULER (a and m NOT coprime), for n >= log2(m):
a^n = a^(n mod phi(m) + phi(m)) (mod m) <- power
    ↪ towers
Wilson: (p-1)! = -1 (mod p) iff p prime
CRT: x = a_i (mod m_i) has a unique solution mod lcm(m_i) iff
    ↪ a_i = a_j (mod gcd(m_i, m_j))
Legendre: exponent of p in n! = sum_{i>=1} floor(n / p^i) = (n -

```

```

    ↪ s_p(n)) / (p - 1)
Kummer: exponent of p in C(n,r) = #carries when adding r and (n-r)
    ↪ in base p
LTE (p odd, p | a-b, p !| a,b): v_p(a^n - b^n) = v_p(a-b) + v_p(n)
p=2 (a, b both ODD):
n odd : v_2(a^n - b^n) = v_2(a-b) <- the case
    ↪ everyone forgets
n even : v_2(a^n - b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1
n odd : v_2(a^n + b^n) = v_2(a+b)
n even : v_2(a^n + b^n) = 1
LTE for SUMS (p odd, p | a+b, p !| a,b):
n odd -> v_p(a^n + b^n) = v_p(a+b) + v_p(n)
n even -> p does not divide a^n + b^n at all

```

QUADRATIC RESIDUES (p odd prime)

Euler's criterion: $a^{((p-1)/2)} = +1$ if a is a QR, -1 if not
 #QRs = $(p-1)/2$. $x^2=a$ has 0 or 2 roots. Use Tonelli-Shanks.
 -1 is a QR iff $p \equiv 1 \pmod{4}$; 2 is a QR iff $p \equiv \pm 1 \pmod{8}$

PRIMITIVE ROOTS exist exactly for $m = 1, 2, 4, p^k, 2p^k$ (p odd
 ↪ prime). Count = $\phi(\phi(m))$.

REPRESENTATIONS

n is a sum of two squares iff every prime $p \equiv 3 \pmod{4}$ has an EVEN
 ↪ exponent
 n is a sum of three squares iff $n \neq 4^a(8b+7)$ every n is a
 ↪ sum of four squares
 Primitive Pythagorean triples: $(m^2-k^2, 2mk, m^2+k^2)$, $m>k>0$,
 ↪ $\gcd(m,k)=1$, $m-k$ odd

SIZE FACTS (for choosing an algorithm)

max tau(n): 1344 for $n \leq 1e9$, 6720 for $n \leq 1e12$, 103680 for $n \leq 1e18$
 $\sum_{i \leq n} \tau(i) \sim n \ln n$; $\sum_{i \leq n} \phi(i) \sim 3n^2/\pi^2$
 $\pi(n) \sim n/\ln n$; n-th prime $\sim n \ln n$; Bertrand: a prime always
 ↪ lies in $(n, 2n)$
 gaps: for $n \leq 1e18$ the max prime gap is < 1500
 Pisano period $\pi(m) \mid \dots : F(n) \bmod m$ repeats with period $\leq 6m$
 ===== */

4.4.2 MobiusInversion

Working code for the identities on the formula sheet. Needs MultSieve (mu, phi).

```

// Needs: MultSieve (04 - NT/02 - Primes and Divisors/07 -
    ↪ MultiplicativeSieve.cpp) for mu/phi.

// #pairs (i,j) with 1<=i,j<=n and gcd(i,j)=1 0(n)
ll coprimePairs(int n, const MultSieve& S) {
    ll r = 0;
    for (int d = 1; d <= n; d++) r += (ll)S.mu[d] * (n / d) * (n / d);
    return r;
}
// sum over all pairs of gcd(i,j) 0(n)
ll gcdSum(int n, const MultSieve& S) {
    ll r = 0;
    for (int d = 1; d <= n; d++) r += (ll)S.phi[d] * (n / d) * (n / d);
    return r;
}
// General: sum_{i,j<=n} f(gcd(i,j)) = sum_d (f*mu)(d) * floor(n/d)^2
// Build g = f*mu once by sieving over multiples: for d: for
    ↪ m=d, 2d, ...: g[m] += mu[m/d]*f[d]
vector<ll> dirichletMu(const vector<ll>& f, const MultSieve& S) { //
    ↪ g = f * mu 0(n log n)
    int n = f.size() - 1;
    vector<ll> g(n + 1, 0);
    for (int d = 1; d <= n; d++) if (f[d])
        for (int m = d; m <= n; m += d) g[m] += (ll)S.mu[m / d] * f[d];
    return g;
}

// DIVISOR BLOCKING: sum_{i=1..n} floor(n/i) in O(sqrt n). Same
    ↪ skeleton for sum tau / sum sigma.
ll sumFloorDiv(ll n) {
    ll r = 0;
    for (ll l = 1, rr; l <= n; l = rr + 1) { rr = n / (n / l); r += (n
    ↪ / l) * (rr - l + 1); }
    return r;
}

// sum_{i=1..n} tau(i) = sum_{i=1..n} floor(n/i) (count multiples)
// sum_{i=1..n} sigma(i) = sum_{i=1..n} i * floor(n/i)

// COUNT MULTIPLES / INCLUSION-EXCLUSION OVER DIVISORS:
// "how many i<=n are divisible by at least one of p1..pk" -> IE over
    ↪ the 2^k squarefree products,
// sign = mu of the product. For k up to ~20 enumerate submasks; beyond
    ↪ that, sieve mu.

// sum_{i=1..n} gcd(i, n) = sum_{d|n} d * phi(n/d)
ll gcdSumWithN(ll n) {

```

```

    ll r = 0;
    for (auto d : divisors(n)) r += (ll)d * phi(n / d);
    return r;
}
// sum_{i=1..n} lcm(i, n) = n/2 * (1 + sum_{d|n} d*phi(d))
ll lcmSumWithN(ll n) {
    ll s = 1;
    for (auto d : divisors(n)) s += (ll)d * phi(d);
    return n / 2 * s + (n % 2 ? s / 2 : 0);
}

```

4.4.3 PrimeCounting

LUCY_HEDGEHOG: $\pi(n) = \#\text{primes} \leq n$, and the SUM of primes $\leq n$, in $O(n^{3/4})$ time and

```

// O(sqrt n) memory. Handles n up to ~1e12 in well under a second; 1e13
// in a few seconds.
// Idea: S(v, p) = (count or sum) of the integers in [2, v] left after
// sieving by all primes < p.
// Sieving by p only touches v >= p^2, so the whole table lives on the
// O(sqrt n) distinct
// values of floor(n / i).
ll primePi(ll n) {
    if (n < 2) return 0;
    ll r = (ll)sqrtl((ld)n);
    while (r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    vector<ll> S(r + 1), L(r + 1); // S[v] for v
    for (<= r, L[i] for v = n/i)
        for (ll i = 1; i <= r; i++) S[i] = i - 1, L[i] = n / i - 1;
    for (ll p = 2; p <= r; p++) {
        if (S[p] == S[p - 1]) continue; // p is not
        // prime
        ll sp = S[p - 1], p2 = p * p;
        for (ll i = 1; i <= min(r, n / p2); i++)
            L[i] -= (i * p <= r ? L[i * p] : S[n / (i * p)]) - sp;
        for (ll v = r; v >= p2; v--) S[v] -= S[v / p] - sp;
    }
    return L[1];
}
// Sum of all primes <= n. Same recurrence with f(x) = x(x+1)/2 - 1 and
// a factor of p.
ll primeSum(ll n) {
    if (n < 2) return 0;
    ll r = (ll)sqrtl((ld)n);
    while (r * r > n) r--;
    while ((r + 1) * (r + 1) <= n) r++;
    // __int128: n(n+1)/2 wraps for n > 4.29e9, and the ANSWER (~n^2/2
    // ln n) wraps at n ~ 3e10.
    auto f = [](ll x) { return x % 2 ? (lll)(x + 1) / 2 * x - 1 :
    (lll)x / 2 * (x + 1) - 1; };
    vector<ll> S(r + 1), L(r + 1);
    for (ll i = 1; i <= r; i++) S[i] = f(i), L[i] = f(n / i);
    for (ll p = 2; p <= r; p++) {
        if (S[p] == S[p - 1]) continue;
        ll sp = S[p - 1], p2 = p * p;
        for (ll i = 1; i <= min(r, n / p2); i++)
            L[i] -= p * ((i * p <= r ? L[i * p] : S[n / (i * p)]) - sp);
        for (ll v = r; v >= p2; v--) S[v] -= p * (S[v / p] - sp);
    }
    return L[1];
}
// SAME SKELETON computes the prefix sum of ANY
// completely-multiplicative f (change the two
// initialisations and the multiplier). For general multiplicative f
// use the Min_25 sieve, which
// runs this as its first phase and then re-adds the prime-power
// contributions.
// APPROXIMATIONS for sanity checks: pi(n) ~ n/ln n, the k-th prime ~ k
// ln k.

```

4.4.4 Representations

Representing integers: sums of two squares, Pythagorean triples, Legendre symbol.

```

// Needs: powmod, factorize (02 - Primes and Divisors/06 -
// MillerRabinPollard.cpp).
// Needs factorize / sqrtMod / isPrime.

int legendre(ll a, ll p) { // 1 QR, -1
    if (a % p == 0) return 0;
    if (a < 0) a += p;
    return powmod(a, (p - 1) / 2, p) == 1 ? 1 : -1;
}
// n is a sum of two squares <=> every prime p = 3 (mod 4) appears to
// an EVEN power.
bool isSumOfTwoSquares(ll n) {
    if (n <= 0) return n == 0;

```

```

    for (auto [p, e] : factorize(n)) if (p % 4 == 3 && (e & 1)) return
    false;
    return true;
}
// One representation n = x^2 + y^2 (Cornacchia). Returns {-1,-1} if
// none exists.
pair<ll, ll> sumOfTwoSquares(ll n) {
    if (n == 0) return {0, 0};
    ll t = n, pw2 = 1;
    while (t % 4 == 0) t /= 4, pw2 *= 2; // pull out
    // squares of 2
    if (!isSumOfTwoSquares(t)) return {-1, -1};
    // Gaussian-integer product over the prime factorisation
    ll xr = 1, yr = 0; // current
    // value as xr + yr*i
    for (auto [p, e] : factorize(t)) {
        if (p == 2) { for (int i = 0; i < e; i++) { ll nx = xr - yr, ny
        = xr + yr; xr = nx, yr = ny; } continue; }
        if (p % 4 == 3) { ll k = 1; for (int i = 0; i < e / 2; i++) k
        *= p; xr *= k, yr *= k; continue; }
        ll r = sqrtMod(p - 1, p); // r^2 = -1
        (mod p)
        ll a = p, b = r;
        while (b * b > p) { ll c = a % b; a = b, b = c; } // Cornacchia
        // descent: p = b^2 + c^2
        ll c = (ll)sqrtl((ld)(p - b * b));
        while (c * c < p - b * b) c++;
        while (c * c > p - b * b) c--;
        for (int i = 0; i < e; i++) { ll nx = xr * b - yr * c, ny = xr
        * c + yr * b; xr = nx, yr = ny; }
    }
    return {llabs(xr) * pw2, llabs(yr) * pw2};
}
// #representations of n as an ORDERED pair of squares (with signs) = 4
// * (d_1(n) - d_3(n)),
// where d_i counts divisors = i (mod 4). [Jacobi's two-square
// theorem]

// PRIMITIVE Pythagorean triples with hypotenuse <= L: (m^2-k^2, 2mk,
// m^2+k^2),
// m > k > 0, gcd(m,k) = 1, m-k odd. Every triple is a multiple of
// exactly one primitive triple.
vector<array<ll, 3>> pythagorean(ll L) {
    vector<array<ll, 3>> r;
    for (ll m = 2; m * m <= L; m++)
        for (ll k = 1 + (m % 2); k < m; k += 2) // m-k must be
        // ODD => opposite parity
            if (gcd(m, k) == 1 && m * m + k * k <= L)
                r.push_back({m * m - k * k, 2 * m * k, m * m + k * k});
    return r;
}
// n is a sum of THREE squares <=> n != 4^a * (8b + 7). Every n is
// a sum of FOUR squares.

```

4.4.5 DuSieve

DU SIEVE (Dirichlet prefix sums) - prefix sums of a multiplicative f in SUBLINEAR time.

```

// Pick g with an easy prefix sum G and an easy (f*g) prefix sum; then
// g(1)*S(n) = sum_{i=1..n} (f*g)(i) - sum_{i=2..n} g(i) *
// S(floor(n/i))
// Sieve S(x) directly for x <= LIM (take LIM ~ n^(2/3)); recurse with
// memoisation above that.
// Total O(n^(2/3)) with the sieve, O(n^(3/4)) without.
// mu * 1 = eps -> M(n) = 1 - sum_{i=2..n} M(n/i)
// phi * 1 = Id -> Phi(n) = n(n+1)/2 - sum_{i=2..n} Phi(n/i)
struct DuSieve {
    ll LIM;
    vector<ll> mu, phi; // prefix sums
    // for x <= LIM
    unordered_map<ll, ll> memoM, memoP;

    DuSieve(ll n) {
        LIM = max((ll)1000, (ll)powl((ld)n, 2.0L / 3));
        vector<int> spf(LIM + 1, 0), primes, m(LIM + 1, 0), p(LIM + 1,
        0);
        m[1] = p[1] = 1;
        for (ll i = 2; i <= LIM; i++) {
            if (!spf[i]) spf[i] = i, primes.push_back(i), m[i] = -1,
            p[i] = i - 1;
            for (int q : primes) {
                if (q > spf[i] || i * q > LIM) break;
                spf[i * q] = q;
                if (q == spf[i]) m[i * q] = 0, p[i * q] = p[i] * q;
                else m[i * q] = -m[i], p[i * q] = p[i] * (q - 1);
            }
        }
        mu.assign(LIM + 1, 0), phi.assign(LIM + 1, 0);
        for (ll i = 1; i <= LIM; i++) mu[i] = mu[i - 1] + m[i], phi[i]

```

```

↪ = phi[i - 1] + p[i];
}
ll M(ll n) { // sum_{i=1..n}
↪ mu(i) (Mertens)
    if (n <= LIM) return mu[n];
    auto it = memoM.find(n);
    if (it != memoM.end()) return it->second;
    ll r = 1;
    for (ll l = 2, rr; l <= n; l = rr + 1) { rr = n / (n / l); r -=
↪ (rr - l + 1) * M(n / l); }
    return memoM[n] = r;
}
ll P(ll n) { // sum_{i=1..n}
↪ phi(i)
    if (n <= LIM) return phi[n];
    auto it = memoP.find(n);
    if (it != memoP.end()) return it->second;
    ll r = n % 2 ? (n + 1) / 2 * n : n / 2 * (n + 1);
    for (ll l = 2, rr; l <= n; l = rr + 1) { rr = n / (n / l); r -=
↪ (rr - l + 1) * P(n / l); }
    return memoP[n] = r;
}
};
// USES: squarefree count up to n = sum_{d<=sqrt n} mu(d) *
↪ floor(n/d^2);
// #coprime pairs up to n = 2*Phi(n) - 1; sum of gcd/lcm over
↪ pairs at n ~ 1e10;
// any "sum over 1..n of a multiplicative function" that a linear
↪ sieve cannot reach.

```

4.4.6 MoreFormulaCards

NUMBER THEORY, SECOND FORMULA SHEET

```

/* ===== NUMBER THEORY, SECOND FORMULA SHEET =====
Everything here was machine-verified against brute force / sympy.

QUADRATIC RECIPROCITY. p, q distinct ODD primes.
(p/q)(q/p) = (-1)^((p-1)/2 * (q-1)/2)
⇒ (p/q) = (q/p), UNLESS p = q = 3 (mod 4), where (p/q) = -(q/p)
SUPPLEMENTS: (-1/p) = 1 iff p = 1 (mod 4); (2/p) = 1 iff p = +-1
↪ (mod 8);
(3/p) = 1 iff p = +-1 (mod 12)
EULER'S CRITERION (p odd prime AND p ∤ a): a^((p-1)/2) = (a/p)
↪ (mod p).
If p ∤ a the value is 0, not +-1 - state the hypothesis.
#k-th ROOTS of a != 0 mod p: d = gcd(k, p-1); the count is d if
↪ a^((p-1)/d) = 1, else 0.
JACOBI SYMBOL (a/n) for ODD n > 0, n NOT necessarily prime - O(log n),
↪ no factoring: */
int jacobi(ll a, ll n) { // n
↪ odd, n > 0
    int r = 1;
    a = ((a % n) + n) % n; // (a/n)
↪ is periodic in a mod n
    while (a) {
        while (a % 2 == 0) { a /= 2; if (n % 8 == 3 || n % 8 == 5) r =
↪ -r; }
        swap(a, n);
        if (a % 4 == 3 && n % 4 == 3) r = -r;
        a %= n;
    }
    return n == 1 ? r : 0;
}
/* CAUTION: (a/n) = 1 does NOT prove a is a QR mod a composite n. (a/n)
↪ = -1 DOES prove it is
not. Use it as a cheap filter, and to find a non-residue fast for
↪ Tonelli-Shanks.

```

```

FIBONACCI (F_0 = 0, F_1 = 1)
FAST DOUBLING, O(log n), no matrices:
F(2k) = F(k) * (2 F(k+1) - F(k)), F(2k+1) = F(k)^2 + F(k+1)^2
gcd(F_m, F_n) = F_gcd(m,n) ⇒ F_m | F_n iff m | n (m >= 3)
Cassini: F_{n-1} F_{n+1} - F_n^2 = (-1)^n
d'Ocagne: F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}
sum_{i<=n} F_i = F_{n+2} - 1 sum_{i<=n} F_i^2 = F_n F_{n+1}
F_92 is the last Fibonacci number fitting in a signed 64-bit integer.
PISANO PERIOD pi(m) = the period of F(n) mod m
pi(1)=1, pi(2)=3, and pi(m) is EVEN for every m > 2
pi(lcm(a,b)) = lcm(pi(a), pi(b)) pi(p^k) = p^(k-1) * pi(p)
pi(m) <= 6m, with EQUALITY exactly at m = 2 * 5^k (verified: m =
↪ 10, 50, 250)
p = +-1 (mod 5) ⇒ pi(p) | p-1; p = +-2 (mod 5) ⇒ pi(p) | 2(p+1)
ZECKENDORF: every n >= 1 is a UNIQUE sum of NON-CONSECUTIVE Fibonacci
↪ numbers, found
greedily from the top. The basis of "Fibonacci base" digit DP and of
↪ Wythoff's game.
PELL: x^2 - D y^2 = 1, D > 0 non-square. Always infinitely many

```

```

↪ solutions.
sqrt(D) = [a0; a1, ..., a_L] is periodic with a_L = 2 a0 and a
↪ palindromic middle.
FUNDAMENTAL SOLUTION = the convergent h_k / k_k with
k = L - 1 if the period L is EVEN, k = 2L - 1 if L is
↪ ODD.
ALL solutions: x_m + y_m sqrt(D) = (x1 + y1 sqrt(D))^m, i.e.
x_{m+1} = x1 x_m + D y1 y_m, y_{m+1} = x1 y_m + y1 x_m
NEGATIVE PELL x^2 - D y^2 = -1 is solvable IFF the period L is ODD;
↪ then the fundamental
solution is the convergent at k = L - 1.
SIZE WARNING: solutions explode. D = 61 gives x = 1766319049, but D
↪ = 661 gives a
39-digit x. 'll' is useless past D ~ 150 - use bignum, or work
↪ modulo the answer.
CF MACHINERY (a0 = floor(sqrt D); m = 0, d = 1, a = a0), all exact
↪ integers:
m' = d*a - m, d' = (D - m'^2) / d, a' = floor((a0 + m') /
↪ d')
BEST RATIONAL APPROXIMATION: the closest p/q to x with q <= N is a
↪ convergent OR a
semiconvergent (t h1 + h2)/(t k1 + k2) with 1 <= t <= a_next. Check
↪ both.
FAREY MEDIANT: for neighbours a/b < c/d (bc - ad = 1), (a+c)/(b+d)
↪ is the unique fraction
with the SMALLEST denominator strictly between them.

```

SUMS OF SQUARES

```

TWO: n = x^2 + y^2 possible IFF every prime p = 3 (mod 4) has an
↪ EVEN exponent.
r_2(n) = 4 (d_1(n) - d_3(n)), d_i = #divisors congruent to i
↪ mod 4.
THREE: n = x^2 + y^2 + z^2 possible IFF n is NOT of the form 4^a (8b
↪ + 7).
FOUR: ALWAYS possible (Lagrange). r_4(n) = 8 * sum over d | n with
↪ 4 ∤ d of d.
PRIMITIVE PYTHAGOREAN TRIPLES: (m^2 - k^2, 2mk, m^2 + k^2), m > k >
↪ 0, gcd(m,k) = 1,
m - k odd.
GAUSSIAN INTEGERS Z[i]: the norm N(a+bi) = a^2 + b^2 is
↪ multiplicative. Primes of Z[i]:
1+i (over 2); p = 1 (mod 4) SPLITS as pi * conj(pi); p = 3 (mod 4)
↪ stays prime.
gcd by rounded division: q = round(a conj(b) / N(b)), r = a - qb,
↪ then Euclid on norms.

```

FROBENIUS / CHICKEN McNUGGET. a, b >= 2, gcd(a,b) = 1:

```

largest non-representable g(a,b) = ab - a - b; #non-representable
↪ = (a-1)(b-1)/2.
For k >= 3 coins there is NO closed form: run Dijkstra on residues
↪ mod a_1 (d[r] = the
smallest representable value congruent to r), then g = max_r d[r] -
↪ a_1. O(a_1 k log a_1).

```

BEATTY / RAYLEIGH. r, s > 1 IRRATIONAL with 1/r + 1/s = 1 ⇒

```

↪ floor(n r) and floor(n s),
n >= 1, PARTITION the positive integers exactly. (Wythoff's losing
↪ positions are exactly
(floor(n phi), floor(n phi^2)).)

```

THREE-DISTANCE (Steinhaus). The points {a}, {2a}, ..., {na} cut [0,1)

```

↪ into intervals of AT
MOST 3 distinct lengths, and when there are 3 the largest is the sum
↪ of the other two.
Use: min over k <= n of (k*a mod m) in O(log) via the continued
↪ fraction of a/m.

```

DETERMINISTIC MILLER-RABIN BASE SETS (all machine-verified)

```

n < 2^32 : {2, 7, 61} (valid
↪ to 4759123141)
n < 3.2e18 : the first 12 primes {2..37}
n < 3.317e24 > 2^64 : {2, 325, 9375, 28178, 450775, 9780504,
↪ 1795265022}
{2,3,5,7} is NOT enough even for 32 bits: 3215031751 = 151*751*28351
↪ passes all four.
Skip a base a when n divides a, or small primes are reported
↪ composite.

```

NTT PRIMES (form, primality and primitive root all machine-verified)

p	= c * 2^k + 1	g	max transform length
998244353	= 119 * 2^23 + 1	3	2^23 <- the
↪ default			
167772161	= 5 * 2^25 + 1	3	2^25
469762049	= 7 * 2^26 + 1	3	2^26
754974721	= 45 * 2^24 + 1	11	2^24 <- NOT
↪ 9*2^23+1 (common misprint)			
1004535809	= 479 * 2^21 + 1	3	2^21
2013265921	= 15 * 2^27 + 1	31	2^27
1224736769	= 73 * 2^24 + 1	3	2^24

3221225473 = $3 \cdot 2^{30} + 1$ 5 2^{30}
 9223372036737335297 = $549755813881 \cdot 2^{24} + 1$, $g = 3$ (64-bit,
 $\hookrightarrow$ needs `__int128 mulmod`)
 THE 3-PRIME TRIPLE for an arbitrary modulus: $167772161 \cdot 469762049 \cdot$
 $\hookrightarrow 754974721 = 5.95e25$.
 THE BINDING CONSTRAINT IS LENGTH, NOT MAGNITUDE: 754974721 caps the
 $\hookrightarrow$ transform at 2^{24} , so
 the input length is capped at 2^{23} . At that length the magnitude
 $\hookrightarrow$ headroom is still 6x.

64-BIT CEILINGS - memorise these

$a \cdot b$ fits in signed ll iff $a, b < 3037000499$ (= $\hookrightarrow \text{floor}(\sqrt{2^{63} - 1})$)
 $n(n+1)/2$ fits in ll iff $n \leq 4294967295$
 sum of primes $\leq n$ fits iff $n < 3e10$ (the sum is $\sim$
 $\hookrightarrow n^2 / (2 \ln n)$)
 sum of $\phi(i)$, $i \leq n$, fits iff $n \leq 5508509576$ (the sum is $\sim$
 $\hookrightarrow 0.3040 n^2$)
 max $\tau(n)$: 1344 for $n \leq 1e9$, 6720 for $1e12$, 103680 for $1e18$
 NEED `__int128` FOR: any modmul with modulus $> 3.04e9$ (Pollard, BSGS,
 $\hookrightarrow$ Tonelli, CRT, Garner);
 $\hookrightarrow$ floorSum with $n > 1e9$; Cornacchia's $b \cdot b > p$ test; prime/totient
 $\hookrightarrow$ SUMS at $n \geq 1e10$.
 FFT PRECISION, the real criterion (not "`|result| < 1e15`"):
 safe iff $(\text{sum } a_i^2 + \text{sum } b_i^2) \cdot \log_2(N) < 9e14$.
 Splitting at $\sqrt{M}$ additionally needs $M^{1.5} < 9.2e18$, i.e. $M <$
 $\hookrightarrow 4.4e12$, with the
 inputs already reduced into $[0, M)$.
 FWHT: the forward transform inflates magnitudes by up to N , so the
 $\hookrightarrow$ pointwise product is
 bounded by $(N \cdot \max |a|)^2$. Under a modulus the XOR inverse must
 $\hookrightarrow$ MULTIPLY BY $\text{inv}(N) \bmod p$ -
 an integer ' $x \neq n$ ' is exact over Z and WRONG mod p .
 ===== */

4.4.7 MultiplicativeGroupStructure

THE STRUCTURE OF Z_n^* (the units mod n) - the facts that decide whether a modular equation

```
// Needs: factorize (02 - Primes and Divisors/06 -
  // MillerRabinPollard.cpp).
// has solutions, how many, and what the true exponent of the group is.
// Z_n^* = product over p^e || n of Z_{(p^e)^*}, and
// p odd : Z_{(p^e)^*} is CYCLIC of order  $\phi(p^e) = p^{e-1}(p-1)$ 
// -> primitive root exists
// p = 2 : Z_{2^e} trivial, Z_{4^e} cyclic of order 2, and for  $e \geq 3$ 
// Z_{(2^e)^*} = Z_2 \times Z_{(2^{e-2})} -> NOT cyclic, no
  // primitive root
// A PRIMITIVE ROOT mod n exists iff  $n = 1, 2, 4, p^e$  or  $2 \cdot p^e$  for an
  // odd prime p.
// CARMICHAEL LAMBDA = the true exponent of  $Z_n^*$ : the smallest L with
  //  $a^L = 1$  for every unit a.
// It divides  $\phi(n)$  and is often much smaller - use it (not  $\phi$ ) to
  // reduce exponents, and as the
// modulus bound in power towers.
ll carmichael(ll n) {
  ll res = 1;
  for (auto [p, e] : factorize(n)) {
    ll pe = 1;
    for (int i = 0; i < e; i++) pe *= p;
    ll l = pe / p * (p - 1); //  $\phi(p^e)$ 
    if (p == 2 && e >= 3) l /= 2; // the one
    // exception
    res = res / gcd(res, l) * l; // lcm
  }
  return res;
}
// HOW MANY DISTINCT VALUES DOES  $a^k \bmod n$  TAKE, over ALL a in  $[0, n)$ ?
// (Not just the units - the
// non-units contribute the recursive term, which is what makes this
// more than  $|Z_n^*| / \gcd$ .)
// Multiplicative in n, so handle each prime power and multiply.
ll ipow(ll b, ll e) { ll r = 1; while (e-- > 0) r *= b; return r; }
ll kthPowersPrimePower(ll p, ll e, ll k) { // count of
  // distinct  $a^k \bmod p^e$ 
  if (e <= 0 || k == 0) return 1;
  if (p == 2) {
    if (e == 1) return 2;
    ll u = ipow(2, e - 2) / gcd(k, ipow(2, e - 2));
    if (k % 2) u *= 2;
    return u + kthPowersPrimePower(2, e - k, k);
  }
  ll ph = ipow(p, e) - ipow(p, e - 1);
  return ph / gcd(k, ph) + kthPowersPrimePower(p, e - k, k);
}
ll kthPowersMod(ll k, ll n) { // #distinct
  //  $a^k \bmod n$  over all a
  ll res = 1;
```

```
for (auto [p, e] : factorize(n)) res *= kthPowersPrimePower(p, e,
  // k);
  return res;
}
// COMPANION FACTS (all about  $x^k = a \bmod p$ , p prime,  $a \neq 0$ ):
// the k-th powers form the subgroup of index  $d = \gcd(k, p-1)$ , so
// there are  $(p-1)/d$  of them;
// a is a k-th power iff  $a^{(p-1)/d} = 1$ , and then it has exactly d
// roots.
// For a MODULUS n,  $x^k = a$  is solved prime-power by prime-power and
// glued with CRT.
// ORDER OF AN ELEMENT:  $\text{ord}(a) \mid \lambda(n)$ ; compute it by starting from
//  $\lambda(n)$  and dividing
// out prime factors while  $a^{(\lambda/q)} = 1$  (that is multOrder in 03 -
// Modular Arithmetic/02).
// EXPONENT REDUCTION:  $a^e = a^{(e \bmod \lambda(n) + \lambda(n)) \bmod n}$ 
// for  $e \geq \log_2(n)$ , which is
// the correct form even when  $\gcd(a, n) > 1$  - the same guard the power
// tower uses.
```

4.4.8 FourSquares

sumOfTwoSquares (04 - Representations.cpp).

```
// Needs: isPrime (02 - Primes and Divisors/06 -
  // MillerRabinPollard.cpp),
// sumOfTwoSquares (04 - Representations.cpp).
// LAGRANGE'S FOUR-SQUARE DECOMPOSITION - actually CONSTRUCT x, y, z, w
  // with
//  $n = x^2 + y^2 + z^2 + w^2$ 
// for n up to  $1e18$ , in expected  $O(\log^2 n)$  primality tests. The
// existence theorem is famous; the
// construction is the part people cannot reproduce under pressure, and
// brute force over two of
// the four variables is  $O(n)$  - hopeless past  $1e12$ .
// THE ALGORITHM (Rabin-Shallit, simplified):
// 1. strip factors of 4:  $n = 4^a \cdot m$ , solve for m and multiply every
  // answer by  $2^a$ ;
// 2. if  $m \equiv 7 \pmod 8$  then m is NOT a sum of three squares, so spend
  // one square:  $m \leftarrow 1$ ;
// 3. now m IS a sum of three squares. Pick x at random and test
  // whether  $m - x^2$  is a PRIME
  // congruent to 1 mod 4 (or a trivial  $0/1/2$ ). Such x are dense, so
  // a handful of tries suffice;
// 4. split that prime into two squares with Cornacchia (04 -
  // Representations.cpp).
// PRECONDITION:  $n \geq 0$ . Uses randomness, so it is expected-time, not
  // worst-case.
array<ll, 4> fourSquares(ll n) {
  array<ll, 4> r{0, 0, 0, 0};
  if (n <= 0) return r;
  ll mul = 1;
  while (n % 4 == 0) n /= 4, mul *= 2; //  $4^a$  factors
  // out of every square
  int idx = 0;
  if (n % 8 == 7) r[idx++] = 1, n -= 1; // 7 mod 8
  // needs a fourth nonzero square
  if (n < 2000) { // small: brute
    // force, and always succeeds
    for (ll x = 0; x * x <= n && idx < 4; x++)
      for (ll y = x; x * x + y * y <= n; y++) {
        ll t = n - x * x - y * y, z = (ll)sqrtl((ld)t);
        while (z * z > t) z--;
        while ((z + 1) * (z + 1) <= t) z++;
        if (z * z == t) { r[idx++] = x, r[idx++] = y, r[idx++] = z;
          break; }
      }
  } else {
    mt19937_64
    rng(chrono::steady_clock::now().time_since_epoch().count());
    ll lim = (ll)sqrtl((ld)n);
    while (lim * lim > n) lim--;
    while (true) {
      ll x = rng() % (lim + 1), m = n - x * x;
      if (m % 4 != 1 && m != 2) continue; // want a prime
      // = 1 (mod 4), or 2
      if (m != 2 && !isPrime(m)) continue;
      auto [u, v] = sumOfTwoSquares(m);
      if (u < 0) continue;
      r[idx++] = x, r[idx++] = u, r[idx++] = v;
      break;
    }
  }
  for (ll& v : r) v *= mul;
  return r;
}
// FACTS WORTH KNOWING
// THREE squares suffice unless  $n = 4^a \cdot (8b + 7)$  (Legendre); TWO
  // suffice iff every prime
```

```
// 3 mod 4 has an even exponent; ONE iff n is a perfect square.
// r_4(n) = 8 * sum of the divisors of n that are not multiples of 4
//   ↳ (Jacobi) - so the number
// of ORDERED signed representations is easy even though finding one
//   ↳ is not.
// Every n is a sum of at most 9 cubes and at most 19 fourth powers
//   ↳ (Waring); the "at most 4
// squares" bound is tight exactly on the numbers 4^a(8b+7).
// IF n IS ALREADY FACTORED, a deterministic route exists: multiply
//   ↳ Hurwitz quaternions the way
// sumOfTwoSquares multiplies Gaussian integers. The randomized version
//   ↳ above is far shorter.
```

4.5 Rational

4.5.1 ContinuedFractions

Continued fractions, convergents, best rational approximation, Stern-Brocot search.

```
// x = [a0; a1, a2, ...]. For a rational p/q this is just the Euclidean
//   ↳ algorithm's quotients.
vector<ll> cfrac(ll p, ll q) {
    vector<ll> a;
    // FLOOR division: C++ truncates toward zero, so cfrac(-7,2) would
    //   ↳ give [-3,-2] instead of
    //   ↳ the correct [-4,2] and every convergent would be wrong.
    while (q) { ll d = p / q - ((p % q) && ((p < 0) ^ (q < 0)));
        //   ↳ a.push_back(d); ll r = p - d * q; p = q, q = r; }
        return a;
    }
// Convergents h[i]/k[i] of a continued fraction: h = a*h1 + h2, k =
//   ↳ a*k1 + k2.
// Every convergent is a BEST approximation: no fraction with a smaller
//   ↳ denominator is closer.
vector<pair<ll, ll>> convergents(const vector<ll>& a) {
    vector<pair<ll, ll>> c;
    ll h1 = 1, h2 = 0, k1 = 0, k2 = 1;
    for (ll x : a) {
        ll h = x * h1 + h2, k = x * k1 + k2;
        c.push_back({h, k});
        h2 = h1, h1 = h, k2 = k1, k1 = k;
    }
    return c;
}
// Best rational approximation to p/q with denominator <= N.
// The answer is a convergent OR a SEMICONVERGENT (t*h1+h2)/(t*k1+k2) -
//   ↳ you must compare both.
pair<ll, ll> bestApprox(ll p, ll q, ll N) {
    ll h1 = 1, h2 = 0, k1 = 0, k2 = 1, P = p, Q = q;
    pair<ll, ll> best = {p >= 0 ? 0 : -((-p + q - 1) / q), 1};
    //   ↳ 0/1 (or floor) is legal
    auto better = [&](pair<ll, ll> a, pair<ll, ll> b) {
        //   ↳ is a closer to p/q than b?
        __int128 da = (__int128)llabs(p * a.second - a.first * q) *
        //   ↳ b.second;
        __int128 db = (__int128)llabs(p * b.second - b.first * q) *
        //   ↳ a.second;
        return da != db ? da < db : a.second < b.second;
    };
    while (Q) {
        ll a = P / Q, r = P % Q;
        if (k1 && a * k1 + k2 > N) {
            //   ↳ cannot take the full step
            ll t = (N - k2) / k1;
            if (t > 0) { pair<ll, ll> c = {t * h1 + h2, t * k1 + k2};
                //   ↳ if (better(c, best)) best = c; }
                break;
            }
            ll h = a * h1 + h2, k = a * k1 + k2;
            if (k > N) break;
            h2 = h1, h1 = h, k2 = k1, k1 = k;
            if (better({h1, k1}, best)) best = {h1, k1};
            P = Q, Q = r;
        }
        return best;
    }
}
// STERN-BROCOT SEARCH: find the simplest fraction p/q with f(p/q)
//   ↳ true, when f is monotone.
// Walk the mediant tree; take exponential jumps in each direction so
//   ↳ it is O(log^2).
// Classic uses: "smallest denominator between two reals", recovering a
//   ↳ fraction from a decimal,
// and answering "is the answer rational with small denominator" style
//   ↳ problems.
// FAREY / mediant fact: if a/b < c/d are neighbours (bc - ad = 1) then
//   ↳ their mediant
// (a+c)/(b+d) is the unique simplest fraction strictly between them.
```

4.5.2 PellEquation

PELL'S EQUATION $x^2 - D \cdot y^2 = 1$ ($D > 0$ and NOT a perfect square) - the fundamental solution,

```
// then all of them. Every solution is a CONVERGENT of the continued
//   ↳ fraction of sqrt(D), so walk
// the convergents and stop at the first one that satisfies the
//   ↳ equation: O(sqrt D) steps, since
// the period of sqrt(D) is O(sqrt D log D).
// ALL SOLUTIONS from the fundamental (x1, y1): x_{m+1} = x1*x_m +
//   ↳ D*y1*y_m,
//   ↳ y_{m+1} = x1*y_m +
//   ↳ y1*x_m,
// i.e. x_m + y_m sqrt(D) = (x1 + y1 sqrt(D))^m. So there are
//   ↳ infinitely many, growing
// EXPONENTIALLY - D = 61 already needs x = 1766319049, and D = 661
//   ↳ needs 39 digits. __int128
// carries you to about D = 200; past that use bignum, or work modulo
//   ↳ the required output.
// Returns {0, 0} when D is a square or <= 0 (only the trivial (1, 0)
//   ↳ exists).
pair<__int128, __int128> pell(ll D) {
    ll a0 = (ll)sqrtl((ld)D);
    while (a0 * a0 > D) a0--;
    while ((a0 + 1) * (a0 + 1) <= D) a0++;
    if (D <= 0 || a0 * a0 == D) return {0, 0};
    ll m = 0, d = 1, a = a0;
    // the exact CF
    // recurrence for sqrt(D)
    __int128 h1 = 1, h = a0, k1 = 0, k = 1;
    // convergents
    // h/k
    while (h * h - (__int128)D * k * k != 1) {
        m = d * a - m;
        d = (D - m * m) / d;
        a = (a0 + m) / d;
        __int128 h2 = h1; h1 = h; h = a * h1 + h2;
        __int128 k2 = k1; k1 = k; k = a * k1 + k2;
    }
    return {h, k};
}
// NEGATIVE PELL x^2 - D*y^2 = -1 is solvable IFF the period of the CF
//   ↳ of sqrt(D) is ODD; the
// same loop finds it if you test for -1 instead (and it always appears
//   ↳ before the +1 solution).
// Necessary condition: D has no prime factor congruent to 3 mod 4 (not
//   ↳ sufficient: D = 34).
// GENERAL x^2 - D*y^2 = k: find one solution by brute force /
//   ↳ Lagrange-Matthews-Mollin, then
// multiply by powers of the fundamental unit to walk the whole class;
//   ↳ there can be several
// inequivalent classes, so collect every solution with y below the
//   ↳ standard bound.
// CF MACHINERY, exact in integers (a0 = floor(sqrt D), start m = 0, d
//   ↳ = 1, a = a0):
// m' = d*a - m, d' = (D - m'^2)/d, a' = floor((a0 + m')/d') -
//   ↳ the period ends when
// a' = 2*a0, and the a-sequence between is a palindrome.
// USES: minimal solutions of "x^2 - D y^2 = 1" style Diophantine
//   ↳ puzzles, the fundamental unit of
// the ring Z[sqrt D], square-triangular numbers (D = 8), and any
//   ↳ recurrence whose growth factor
// is a quadratic irrational.
```

4.5.3 RationalReconstruction

RATIONAL RECONSTRUCTION - recover the fraction p/q from its value $r = p * q^{-1} \pmod m$.

```
// The answer is UNIQUE as long as |p| and q are both below sqrt(m/2),
//   ↳ and it is found by running
// the extended Euclidean algorithm on (m, r) and stopping halfway: the
//   ↳ remainder is p and the
// cofactor is q. O(log m).
// WHY YOU WANT IT: do a whole computation in modular arithmetic
//   ↳ (Gaussian elimination, a DP over
// fractions, an interpolation, a determinant) and then read off the
//   ↳ EXACT rational answer -
// no bignum fractions, no gcd-ing at every step. Also the standard way
//   ↳ to guess a closed form
// from modular data, and to answer "print the answer as an irreducible
//   ↳ fraction" when the
// intermediate values would explode.
// PRECONDITIONS: m >= 2, gcd(q, m) = 1 (implicit in r existing).
//   ↳ Returns {0, 0} on failure -
// which means no such small fraction exists, so take a bigger m (or
//   ↳ several and CRT them).
pair<ll, ll> ratRecon(ll r, ll m) {
    ll a = m, b = ((r % m) + m) % m, x = 0, y = 1;
    // invariant: b
    //   ↳ = y * r (mod m)
    ll lim = (ll)sqrtl((ld)m / 2);
```

```

while (b > lim) {
    ll q = a / b;
    ll t = a - q * b; a = b, b = t;
    t = x - q * y; x = y, y = t;
}
if (y < 0) b = -b, y = -y;
if (y == 0 || gcd(b < 0 ? -b : b, y) != 1) return {0, 0};
return {b, y}; // b / y = r
↪ (mod m), in lowest terms
}
// CHECK YOUR RESULT: verify b * inverse(y) mod m == r before trusting
↪ it - the reconstruction can
// return a wrong small fraction if the true one is not small enough.
↪ Verifying is O(log m).
// LARGER BOUNDS: if you know |p| ≤ P and q ≤ Q with 2*P*Q < m, stop
↪ the loop at b ≤ P instead
// and the same argument applies. Choosing m as a product of several
↪ NTT primes (Garner) is the
// usual way to make m big enough.
// RELATED: the same halfway-Euclid stopping rule is the
↪ Berlekamp-Massey / Pade approximant of a
// power series, and 01 - ContinuedFractions.cpp is the
↪ exact-arithmetic version of this walk.

```

4.6 Sequences

4.6.1 FibonacciAndPisano

FIBONACCI, FAST. The notebook had the identities on a formula sheet and no routine; this is it.

```

// F(0)=0, F(1)=1. Fast doubling is O(log n) with two multiplications
↪ per bit and NO matrices -
// shorter and ~2x faster than 2x2 matrix exponentiation.
// F(2k) = F(k) * (2*F(k+1) - F(k))
// F(2k+1) = F(k)^2 + F(k+1)^2
pair<ll, ll> fibPair(ll n, ll mod) { //
    ↪ returns {F(n), F(n+1)} mod
    if (!n) return {0, 1 % mod};
    auto [a, b] = fibPair(n >> 1, mod);
    ll c = a * ((2 * b % mod - a % mod + mod) % mod) % mod; // F(2k)
    ll d = (a * a + b * b) % mod; //
    ↪ F(2k+1)
    return (n & 1) ? make_pair(d, (c + d) % mod) : make_pair(c, d);
}
ll fib(ll n, ll mod) { return fibPair(n, mod).first; }
// LUCAS NUMBERS L(n) = F(n-1) + F(n+1) = 2F(n+1) - F(n).
// NEGATIVE index: F(-n) = (-1)^(n+1) F(n).
// The mod must be < ~3e9 for the ll products above, or use __int128 /
↪ Mint.

// --- PISANO PERIOD pi(m): the period of F(n) mod m. Needed to reduce
↪ a huge index. ---
// pi(1) = 1, pi(2) = 3, and pi(m) is EVEN for every m > 2.
// pi(lcm(a,b)) = lcm(pi(a), pi(b)) pi(p^k) = p^(k-1) * pi(p)
// pi(m) ≤ 6m, with equality exactly at m = 2 * 5^k.
// p = +1 (mod 5) => pi(p) divides p-1; p = +-2 (mod 5) => pi(p)
↪ divides 2(p+1).
// So: factor m, get pi for each prime power by testing the divisors of
↪ that bound, then lcm.
ll pisanoPrime(ll p) { // p
    ↪ prime
    if (p == 2) return 3;
    if (p == 5) return 20;
    ll cand = (p % 5 == 1 || p % 5 == 4) ? p - 1 : 2 * (p + 1);
    vector<ll> divs;
    for (ll d = 1; d * d ≤ cand; d++)
        if (cand % d == 0) divs.push_back(d), divs.push_back(cand / d);
    sort(all(divs));
    for (ll d : divs) {
        auto [a, b] = fibPair(d, p);
        if (a == 0 && b == 1 % p) return d;
    }
    return cand;
}
// Full pi(m): factorise m (Pollard rho), take pisanoPrime(p) * p^(e-1)
↪ per prime power, lcm them.
/* MORE FIBONACCI FACTS worth the page
gcd(F_m, F_n) = F_gcd(m,n) => F_m | F_n iff m | n (m >= 3)
Cassini: F_{n-1} F_{n+1} - F_n^2 = (-1)^n
Catalan: F_n^2 - F_{n-r} F_{n+r} = (-1)^{n-r} F_r^2
d'Ocagne: F_m F_{n+1} - F_{m+1} F_n = (-1)^n F_{m-n}
sum_{i<=n} F_i = F_{n+2} - 1 sum_{i<=n} F_i^2 = F_n F_{n+1}
sum_{i<=n} F_{2i-1} = F_{2n} sum_{i<=n} F_{2i} = F_{2n+1} - 1
F_n = (phi^n - psi^n)/sqrt(5); F_92 is the last one fitting in a
↪ signed 64-bit integer.
ZECKENDORF: every n >= 1 is a UNIQUE sum of NON-CONSECUTIVE
↪ Fibonacci numbers, found greedily
from the largest. This is the "Fibonacci base", and it is what

```

↪ makes Fibonacci-nim work.
 LCM of F_1..F_n and "which F are divisible by k" both follow from
 ↪ the gcd identity.
 Fibonacci mod p when 5 is a QR mod p: sqrt(5) exists, so the closed
 ↪ form works in F_p directly.
 When it is not, work in F_p[x]/(x^2-5) (a "Phi field") - two
 ↪ components, same O(log n). */

```

// --- TRIBONACCI / any linear recurrence: matrix exponentiation (06 -
↪ Math/03), or Kitamasa /
// Bostan-Mori for a long recurrence. Berlekamp-Massey RECOVERS the
↪ recurrence from ~2k terms.
// --- LUCAS SEQUENCES U_n, V_n with U_{n+1} = P U_n - Q U_{n-1}
↪ generalise all of this and have
// the same doubling identities; they are what Baillie-PSW primality is
↪ built on.

```

4.6.2 FibonacciLCM

LCM OF FIBONACCI NUMBERS lcm(F(a1), F(a2), ..., F(an)) mod p - the numbers themselves have

```

// Needs: fib (01 - FibonacciAndPisano.cpp), powmod (02 - Primes and
↪ Divisors/06).
// billions of digits, so everything is done modulo p and division is a
↪ modular inverse.
// THE KEY IDENTITY is gcd(F(i), F(j)) = F(gcd(i, j)), which turns the
↪ usual
// lcm(S) = lcm(S') * x / gcd(lcm(S'), x) (x = the last element)
// into -- all
↪ indices, no big numbers --
// lcm(F(A)) = lcm(F(A')) * F(x) / lcm(F(gcd(a_i, x)) for a_i in A')
// i.e. the denominator is again an lcm of Fibonacci numbers, on the
↪ SMALLER index set of gcds.
// So recurse on sets of indices; memoise on the sorted, deduplicated
↪ index vector.
// COMPLEXITY: O(#distinct gcd-closed subsets * n * log) - bounded by
↪ the number of divisors of
// the a_i in practice, fast for n up to a few hundred with a_i up to
↪ 1e9.
// CAUTION: this DIVIDES modulo p, so p must be prime AND no F(k)
↪ involved may be 0 mod p. With
// p = 1e9+7 that is safe for the usual constraints; if you hit a zero,
↪ use a different prime.
map<vector<int>, ll> fibLcmMemo;
ll fibLcm(vector<int> a) {
    sort(a.rbegin(), a.rend());
    while (!a.empty() && a.back() ≤ 2) a.pop_back(); // F(1) = F(2)
    ↪ = 1 contribute nothing
    a.erase(unique(all(a)), a.end());
    if (a.empty()) return 1;
    if (a.size() == 1) return fib(a[0], MOD);
    auto it = fibLcmMemo.find(a);
    if (it != fibLcmMemo.end()) return it->second;
    vector<int> b(a.begin(), a.end() - 1);
    ll res = fibLcm(b);
    for (int& x : b) x = gcd(x, a.back());
    res = res * fib(a.back(), MOD) % MOD * powmod(fibLcm(b), MOD - 2,
    ↪ MOD) % MOD;
    return fibLcmMemo[a] = res;
}
// SAME PATTERN, other sequences: any sequence with gcd(u_i, u_j) =
↪ u_gcd(i,j) (a "strong
// divisibility sequence" works verbatim - F(n), 2^n - 1 (lcm of
↪ Mersenne numbers), and the
// Lucas sequences U_n(P, Q) with gcd(P, Q) = 1 in general.
// USEFUL COMPANIONS: F(m) | F(n) iff m | n (for m >= 3); gcd(F(m),
↪ F(n)) = F(gcd(m,n));
// lcm over ALL i ≤ n of F(i) has a product formula via the primitive
↪ parts (Zsigmondy), but the
// recursion above is what you want in a contest.

```

5 Combinatorics

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5.1 StarsAndBars

Stars and Bars (Number of Solutions to $x_1 + x_2 + \dots + x_k = n$) - O(1)

// 1. Non-negative integer solutions ($x_i \geq 0$): $C(n + k - 1, k - 1)$
 // 2. Positive integer solutions ($x_i \geq 1$): $C(n - 1, k - 1)$

```

struct StarsAndBars {
    // Requires precomputed Combination C(n, k)
    static ll count_non_negative(ll n, ll k, function<ll(ll, ll)> nCr) {

```

```

    if (n < 0 || k <= 0) return 0;
    return nCr(n + k - 1, k - 1);
}

static ll count_positive(ll n, ll k, function<ll(ll, ll)> nCr) {
    if (n < k || k <= 0) return 0;
    return nCr(n - 1, k - 1);
}
};

```

5.2 BurnsidePolya

```

Burnside / Polya - count colourings up to symmetry. #orbits = (1/|G|) sum_g |fix(g)|
// Needs: Mint (06 - Math/01); phi (02 - Primes and Divisors/06) for
    ↳ the necklace formulas.
// Needs Mint and phi() (single-value Euler totient).

// Necklaces: n beads in a circle, k colours, ROTATIONS only.
// A rotation by r has gcd(r,n) cycles -> k^gcd(r,n) fixed colourings;
    ↳ group by d = gcd.
Mint necklaces(ll n, ll k) {
    Mint s = 0;
    for (ll d = 1; d * d <= n; d++) if (n % d == 0) {
        s += Mint(phi(n / d)) * Mint(k).pow(d);
        if (d != n / d) s += Mint(phi(d)) * Mint(k).pow(n / d);
    }
    return s * Mint(n).inv();
}
// Bracelets: rotations AND reflections (dihedral group, |G| = 2n).
Mint bracelets(ll n, ll k) {
    Mint s = necklaces(n, k) * Mint(n); // = sum over
    ↳ the n rotations
    if (n & 1) s += Mint(n) * Mint(k).pow((n + 1) / 2);
    else s += Mint(n / 2) * (Mint(k).pow(n / 2) + Mint(k).pow(n / 2 +
    ↳ 1));
    return s * Mint(2 * n).inv();
}
// GENERAL RECIPE: for each group element g, count its CYCLES c(g) on
    ↳ the objects being
// coloured; it fixes k^c(g) colourings. Answer = (1/|G|) * sum_g
    ↳ k^c(g).
// square grid n x n under the 4 rotations: cycles = n^2,
    ↳ ceil(n^2/4)... (work them out per problem)
// cube faces under 24 rotations: cycle counts 6,3,3,4,2,2,... ->
    ↳ (k^6+3k^4+12k^3+8k^2)/24
// POLYA (colours have weights): replace k^c by prod over cycles of
    ↳ (sum of x_i^len).
Mint burnside(const vector<ll>& cyclesPerElement, ll k) {
    Mint s = 0;
    for (ll c : cyclesPerElement) s += Mint(k).pow(c);
    return s * Mint((ll)cyclesPerElement.size()).inv();
}

```

5.3 CombinatoricsFormulas

COMBINATORICS - FORMULA SHEET

```

/* ===== COMBINATORICS - FORMULA SHEET =====
BASICS C(n,k)=n!/(k!(n-k)!) P(n,k)=n!/(n-k)! multinomial
    ↳ n!/(k1!...km!)
STARS AND BARS: x1+...+xk = n, xi>=0 -> C(n+k-1, k-1); xi>=1 ->
    ↳ C(n-1, k-1)
with upper bounds xi<=ci: IE over which variables
    ↳ overflow
Compositions of n into k parts C(n-1,k-1); into any parts 2^(n-1)

BINOMIAL IDENTITIES (memorise these - recognising one is worth 30
    ↳ minutes)
Pascal C(n,k) = C(n-1,k-1) + C(n-1,k)
Symmetry C(n,k) = C(n,n-k)
Absorption k*C(n,k) = n*C(n-1,k-1)
Hockey stick sum_{i=r..n} C(i,r) = C(n+1, r+1)
Vandermonde sum_k C(m,k) C(n,p-k) = C(m+n, p)
sum_k C(n,k) = 2^n sum_k (-1)^k C(n,k) = [n==0]
sum_k k*C(n,k) = n*2^(n-1) sum_k C(n,k)^2 = C(2n,n)
sum_k C(k,r) x^k ... upper negation: C(-n,k) = (-1)^k C(n+k-1, k)
sum_{k} C(n,k) C(k,m) = C(n,m) 2^(n-m)
Binomial inversion: f(n)=sum_k C(n,k) g(k) <=> g(n)=sum_k
    ↳ (-1)^(n-k) C(n,k) f(k)

```

INCLUSION-EXCLUSION

```

|A1 ∪ ... ∪ An| = sum |Ai| - sum |Ai ∩ Aj| + ...
#surjections [n]->[k] = sum_{i=0..k} (-1)^i C(k,i) (k-i)^n = k! *
    ↳ S2(n,k)
"exactly m" from "at least m": E(m) = sum_{j>=m} (-1)^(j-m) C(j,m)
    ↳ A(j)
derangements D(n) = n! sum_{k=0..n} (-1)^k/k! =

```

↳ (n-1)(D(n-1)+D(n-2)), D0=1,D1=0
 permutations with exactly k fixed points = C(n,k) * D(n-k)

CATALAN Cat(n) = C(2n,n)/(n+1) = C(2n,n) - C(2n,n+1), Cat(n+1) = sum
 ↳ Cat(i)Cat(n-i)
 1,1,2,5,14,42,132,429,1430,4862 - counts: balanced bracket strings
 ↳ of length 2n;
 binary trees with n nodes; triangulations of an (n+2)-gon; monotone
 ↳ lattice paths under
 the diagonal; stack-sortable permutations; non-crossing
 ↳ partitions/chords.
 BALLOT / CYCLE LEMMA: paths from 0 to k in n steps staying > 0 =
 ↳ (k/n) C(n, (n+k)/2)
 REFLECTION: #paths (0,0)->(a,b) touching the line y=x+c = #paths
 ↳ to the reflected point

STIRLING

2nd kind S2(n,k) = #partitions of n LABELLED items into k non-empty
 ↳ UNLABELLED blocks
 S2(n,k) = k*S2(n-1,k) + S2(n-1,k-1); S2(n,k) = (1/k!) sum_i
 ↳ (-1)^i C(k,i)(k-i)^n
 x^n = sum_k S2(n,k) * x_falling(k) surjections = k!
 ↳ S2(n,k)
 1st kind (unsigned) c(n,k) = #permutations of n with exactly k cycles
 c(n,k) = c(n-1,k-1) + (n-1) c(n-1,k); sum_k c(n,k) x^k =
 ↳ x(x+1)..(x+n-1)
 BELL B(n) = sum_k S2(n,k) = #set partitions. B(n+1) = sum_k C(n,k)
 ↳ B(k). Bell triangle.

PARTITIONS (unordered, into positive parts)

p(n) via pentagonal number theorem, 0(n sqrt n):
 p(n) = sum_{k>=1} (-1)^(k+1) [p(n - k(3k-1)/2) + p(n - k(3k+1)/2)
 ↳]
 partitions into <= k parts = partitions into parts <= k (conjugate)

TWELFOLD WAY (n balls into k boxes)

balls	boxes	any	injective	surjective
dist.	dist.	k^n	k!/(k-n)!	k! S2(n,k)
ident. dist.	C(n+k-1, n)	C(k,n)	C(n-1, n-k)	
dist. ident.	sum_{i<=k} S2(n,i)	[n<=k]	S2(n,k)	
ident. ident.	p_k(n)	[n<=k]	p(n into exactly ↳ k parts)	

BURNSIDE / POLYA #orbits = (1/|G|) sum_{g in G} |fix(g)|
 necklaces (rotations), n beads, k colours: (1/n) sum_{d|n} phi(n/d)
 ↳ k^d
 bracelets (+ reflections): (necklace sum + reflection sum) / (2n),
 reflection sum = n*k^((n+1)/2) if n odd, (n/2)(k^(n/2) +
 ↳ k^(n/2+1)) if n even

TREES / GRAPHS

Cayley: n^(n-2) labelled trees on n vertices; with given degrees
 ↳ d_i: (n-2)! / prod (d_i-1)!
 Prufer code <=> labelled tree, bijective
 Matrix-Tree: #spanning trees = any cofactor of the Laplacian (D - A)
 LGV: #non-intersecting path systems = det(M) with M_ij = #paths
 ↳ a_i -> b_j

GENERATING FUNCTIONS

OGF 1/(1-x) = sum x^n 1/(1-x)^k = sum C(n+k-1, k-1) x^n
 x/(1-x-x^2) = Fibonacci (1-sqrt(1-4x))/(2x) = Catalan
 prod 1/(1-x^i) = partitions prod (1+x^i) = distinct-part
 ↳ partitions
 EGF e^x = sum x^n/n! EGF of set partitions = e^(e^x -
 ↳ 1)
 EXPONENTIAL FORMULA: if C(x) is the EGF of connected structures,
 ↳ e^C(x) is the EGF of all
 Coefficient extraction: [x^n] A(x)B(x) = sum_k a_k b_{n-k} (a
 ↳ convolution -> NTT)
 Lagrange inversion, for F(0)=0, F'(0)!=0: [x^n] G(F^-1)(x) =
 ↳ (1/n)[x^(n-1)] G'(x)(x/F(x))^n

MISC

hook length: #standard Young tableaux of shape lambda = n! /
 ↳ prod(hook lengths)
 Erdos-Ko-Rado, Dilworth (min chain cover = max antichain), Sperner
 ↳ (max antichain C(n,n/2))

5.4 CatalanStirlingBell

Catalan, Stirling (both kinds), Bell, derangements, partitions. Needs Mint + buildFact().

```

Mint catalan(int n) { return C2 * n, n) * inv_n[n + 1]; }
// paths from 0 to k in n steps (+1) that stay strictly positive -
    ↳ ballot / cycle lemma
Mint ballot(int n, int k) {

```

```

    if (n == 0) return k == 0; // the empty path
    if (k <= 0 || k > n || (n + k) % 2) return 0; // k <= 0 was
    ↪ unguarded: Mint(-2) wraps
    return Mint(k) * inv_n[n] * C(n, (n + k) / 2);
}

Mint derange(int n) { // D(n) =
    ↪ (n-1)(D(n-1)+D(n-2))
    Mint a = 1, b = 0; // D0=1, D1=0
    for (int i = 2; i <= n; i++) { Mint c = Mint(i - 1) * (a + b); a =
    ↪ b, b = c; }
    return n ? b : a;
}

// permutations of n with exactly k fixed points
Mint fixedPoints(int n, int k) { return C(n, k) * derange(n - k); }

// Stirling 2nd kind, single value: S2(n,k) = (1/k!) sum_i (-1)^i
    ↪ C(k,i) (k-i)^n 0(k log n)
Mint stirling2(ll n, int k) {
    Mint r = 0;
    for (int i = 0; i <= k; i++) {
        Mint t = C(k, i) * Mint(k - i).pow(n);
        (i & 1) ? r -= t : r += t;
    }
    return r * invFact[k];
}

Mint surjections(ll n, int k) { return stirling2(n, k) * fact[k]; }

// Whole ROW of Stirling 2nd (all k for fixed n) in O(n^2), or O(n log
    ↪ n) as a convolution of
// a_i = (-1)^i/i! and b_i = i^n/i! (NTT).
vector<Mint> stirling2Row(int n) {
    vector<Mint> a(n + 1), b(n + 1), r(n + 1, 0);
    for (int i = 0; i <= n; i++) a[i] = (i & 1 ? Mint(0) - invFact[i] :
    ↪ invFact[i]), b[i] = Mint(i).pow(n) * invFact[i];
    for (int k = 0; k <= n; k++) for (int i = 0; i <= k; i++) r[k] +=
    ↪ a[i] * b[k - i];
    return r;
}

// Stirling 1st kind (unsigned), whole row: prod_{i=0..n-1} (x + i)
    ↪ O(n^2)
vector<Mint> stirling1Row(int n) {
    vector<Mint> p = {1};
    for (int i = 0; i < n; i++) {
        vector<Mint> q(p.size() + 1, 0);
        for (size_t j = 0; j < p.size(); j++) q[j + 1] += p[j], q[j] +=
    ↪ p[j] * Mint(i);
        p = q;
    }
    return p; // p[k] = c(n,
    ↪ k)
}

// Bell numbers B(0..n) via the Bell triangle, O(n^2)
vector<Mint> bell(int n) {
    vector<Mint> B(n + 1), row = {1};
    B[0] = 1;
    for (int i = 1; i <= n; i++) {
        vector<Mint> nr = {row.back()};
        for (Mint x : row) nr.push_back(nr.back() + x);
        row = nr, B[i] = row[0];
    }
    return B;
}

// Integer partitions p(0..n) via the pentagonal number theorem, O(n
    ↪ sqrt n)
vector<Mint> partitions(int n) {
    vector<Mint> p(n + 1, 0);
    p[0] = 1;
    for (int i = 1; i <= n; i++)
        for (int k = 1;; k++) {
            ll g1 = (ll)k * (3 * k - 1) / 2, g2 = (ll)k * (3 * k + 1) /
    ↪ 2;
            if (g1 > i) break;
            Mint t = p[i - g1] + (g2 <= i ? p[i - g2] : Mint(0));
            (k & 1) ? p[i] += t : p[i] -= t;
        }
    return p;
}

```

5.5 InclusionExclusion

INCLUSION-EXCLUSION recipes. Needs Mint + buildFact().

```

// x1+...+xk = n with 0 <= xi <= ub[i]. IE over which variables
    ↪ overflow their bound.
// O(2^k) - fine for k <= ~20. With equal bounds use the closed form
    ↪ below instead.

```

```

Mint boundedStarsBars(ll n, const vector<ll>& ub) {
    int k = ub.size();
    Mint r = 0;
    for (int m = 0; m < (1 << k); m++) {
        ll rem = n;
        for (int i = 0; i < k; i++) if (m >> i & 1) rem -= ub[i] + 1;
        if (rem < 0) continue;
        Mint t = C(rem + k - 1, k - 1);
        (__builtin_popcount(m) & 1) ? r -= t : r += t;
    }
    return r;
}

// Same with a COMMON bound c: sum_j (-1)^j C(k,j) C(n - j(c+1) + k-1,
    ↪ k-1). 0(k)
Mint boundedStarsBarsEqual(ll n, ll k, ll c) {
    Mint r = 0;
    for (ll j = 0; j <= k && j * (c + 1) <= n; j++) {
        Mint t = C(k, j) * C(n - j * (c + 1) + k - 1, k - 1);
        (j & 1) ? r -= t : r += t;
    }
    return r;
}

// #surjections [n] -> [k] (equivalently k! * S2(n,k))
Mint surj(ll n, ll k) {
    Mint r = 0;
    for (ll i = 0; i <= k; i++) {
        Mint t = C(k, i) * Mint(k - i).pow(n);
        (i & 1) ? r -= t : r += t;
    }
    return r;
}

// "AT LEAST" -> "EXACTLY" (binomial / Mobius inversion over the subset
    ↪ lattice):
// if A(j) = #configurations having at least a chosen set of j
    ↪ properties (summed over sets),
// then E(m) = sum_{j>=m} (-1)^(j-m) C(j,m) A(j) counts those with
    ↪ EXACTLY m.
Mint exactlyFromAtLeast(const vector<Mint>& A, int m) {
    Mint r = 0;
    for (int j = m; j < (int)A.size(); j++) {
        Mint t = C(j, m) * A[j];
        ((j - m) & 1) ? r -= t : r += t;
    }
    return r;
}

// BINOMIAL INVERSION: f(n) = sum_k C(n,k) g(k) <=> g(n) = sum_k
    ↪ (-1)^(n-k) C(n,k) f(k)
vector<Mint> binomialInvert(const vector<Mint>& f) {
    int n = f.size();
    vector<Mint> g(n, 0);
    for (int i = 0; i < n; i++)
        for (int k = 0; k <= i; k++) {
            Mint t = C(i, k) * f[k];
            ((i - k) & 1) ? g[i] -= t : g[i] += t;
        }
    return g;
}

// COUNTING COPRIME PAIRS / "gcd is exactly g": count multiples then
    ↪ Mobius-invert downwards:
// exact[g] = cntMultiples[g] - sum_{g | d, d > g} exact[d]
    ↪ (iterate g downwards)
vector<ll> exactGcdCounts(const vector<ll>& cntMultiples) {
    int n = cntMultiples.size() - 1;
    vector<ll> e(n + 1);
    for (int g = n; g >= 1; g--) {
        e[g] = cntMultiples[g];
        for (int d = 2 * g; d <= n; d += g) e[g] -= e[d];
    }
    return e;
}

```

5.6 TreeCounting

COUNTING TREES AND GRAPHS.

```

// Needs: Mint (06 - Math/01) and determinant (06 - Math/04 - Linear
    ↪ Algebra/02).

```

```

// PRUFER CODE: bijection between labelled trees on n>=2 vertices and
    ↪ sequences in [0,n)^(n-2).
// Immediate corollaries: Cayley n^(n-2) labelled trees; with
    ↪ prescribed degrees d_i the count is
// (n-2)! / prod (d_i - 1)! (because vertex i appears exactly d_i - 1
    ↪ times in the code).
vector<int> toPrufer(const vector<vector<int>>& adj) { //
    ↪ repeatedly strip the SMALLEST leaf
    int n = adj.size();
    if (n <= 2) return {};

```

```

vector<int> deg(n), code;
vector<bool> gone(n, false);
priority_queue<int, vector<int>, greater<int>> q;
for (int i = 0; i < n; i++) if ((deg[i] = adj[i].size()) == 1)
    q.push(i);
for (int it = 0; it < n - 2; it++) {
    int leaf = q.top(); q.pop();
    gone[leaf] = true;
    int nb = -1;
    for (int v : adj[leaf]) if (!gone[v]) nb = v;
    code.push_back(nb);
    if (--deg[nb] == 1) q.push(nb);
}
return code;
}

vector<pair<int, int>> fromPrufer(vector<int> code) {
    int n = code.size() + 2;
    vector<int> deg(n, 1);
    for (int x : code) deg[x]++;
    priority_queue<int, vector<int>, greater<int>> q;
    for (int i = 0; i < n; i++) if (deg[i] == 1) q.push(i);
    vector<pair<int, int>> e;
    for (int x : code) {
        int leaf = q.top(); q.pop();
        e.push_back({leaf, x});
        if (--deg[x] == 1) q.push(x);
    }
    int a = q.top(); q.pop();
    e.push_back({a, q.top()});
    return e;
}

// MATRIX-TREE THEOREM: #spanning trees of a graph = any cofactor of
// the Laplacian L = D - A.
// Multigraphs work (put the multiplicity in A). Needs determinant()
// from GaussianElimination.
Mint spanningTrees(int n, const vector<pair<int, int>>& edges) {
    if (n == 1) return 1;
    vector<vector<Mint>> L(n, vector<Mint>(n, 0));
    for (auto [u, v] : edges) {
        L[u][u] += 1, L[v][v] += 1;
        L[u][v] -= 1, L[v][u] -= 1;
    }
    vector<vector<Mint>> M(n - 1, vector<Mint>(n - 1)); // delete
    // row/col 0
    for (int i = 1; i < n; i++) for (int j = 1; j < n; j++) M[i - 1][j - 1] = L[i][j];
    return determinant(M);
}

// DIRECTED version (Tutte): #arborescences rooted at r = cofactor of
// (Dout - A) deleting row/col r
// for in-trees, or (Din - A) for out-trees.
// WEIGHTED version: put edge weights in A and the weighted degree in D
// -> sum over spanning
// trees of the product of weights (useful mod p for "sum over all
// spanning trees" problems).

// EXPONENTIAL FORMULA: if C(x) is the EGF of CONNECTED structures then
// e^C(x) is the EGF of
// all structures. Concretely, with g_n = #labelled graphs on n nodes
// = 2^C(n,2):
// c_n = g_n - sum_{k=1}^{n-1} C(n-1, k-1) * c_k * g_{n-k}
// (connected labelled graphs)
vector<Mint> connectedLabelledGraphs(int n) {
    vector<Mint> g(n + 1, c(n + 1, 0));
    for (int i = 0; i <= n; i++) g[i] = Mint(2).pow((ll)i * (i - 1) / 2);
    for (int i = 1; i <= n; i++) {
        c[i] = g[i];
        for (int k = 1; k < i; k++) c[i] -= C(i - 1, k - 1) * c[k] * g[i - k];
    }
    return c;
}

```

5.7 SubsetConvolution

TRANSFORMS OVER THE SUBSET LATTICE. n bits, arrays of size 2^n .

// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).

```

// ZETA (sum over subsets) and its inverse MOBIUS.  $O(n 2^n)$ . This is
// SOS DP.
void zetaSub(vector<Mint>& f, int n) { // f[m] :=
    // sum_{s subset of m} f[s]
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (m >> i & 1) f[m] += f[m
        // ^ (1 << i)];

```

```

}
void mobiusSub(vector<Mint>& f, int n) { // exact
    // inverse of zetaSub
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (m >> i & 1) f[m] -= f[m
        // ^ (1 << i)];
}
void zetaSuper(vector<Mint>& f, int n) { // f[m] :=
    // sum_{m subset of s} f[s]
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (!(m >> i & 1)) f[m] +=
        // f[m | (1 << i)];
}
void mobiusSuper(vector<Mint>& f, int n) {
    for (int i = 0; i < n; i++)
        for (int m = 0; m < (1 << n); m++) if (!(m >> i & 1)) f[m] -=
        // f[m | (1 << i)];
}
// OR-convolution:  $h[m] = \sum_{a|b=m} f[a] g[b]$  -> zetaSub,
// multiply, mobiusSub
// AND-convolution:  $h[m] = \sum_{a\&b=m} f[a] g[b]$  -> zetaSuper,
// multiply, mobiusSuper
vector<Mint> orConv(vector<Mint> f, vector<Mint> g, int n) {
    zetaSub(f, n), zetaSub(g, n);
    for (int m = 0; m < (1 << n); m++) f[m] *= g[m];
    return mobiusSub(f, n), f;
}
vector<Mint> andConv(vector<Mint> f, vector<Mint> g, int n) {
    zetaSuper(f, n), zetaSuper(g, n);
    for (int m = 0; m < (1 << n); m++) f[m] *= g[m];
    return mobiusSuper(f, n), f;
}

// SUBSET (disjoint) CONVOLUTION:  $h[m] = \sum_{a|b=m, a\&b=0} f[a] g[b]$ 
// ->  $g[b] \cdot 0(2^n n^2)$ .
// The ranked-zeta trick: index by popcount, convolve the ranks like
// polynomials.
vector<Mint> subsetConv(const vector<Mint>& f, const vector<Mint>& g,
    // int n) {
    int N = 1 << n;
    vector<vector<Mint>> F(n + 1, vector<Mint>(N, 0)), G = F, H = F;
    for (int m = 0; m < N; m++) F[__builtin_popcount(m)][m] = f[m],
    // G[__builtin_popcount(m)][m] = g[m];
    for (int i = 0; i <= n; i++) zetaSub(F[i], n), zetaSub(G[i], n);
    for (int i = 0; i <= n; i++)
        for (int j = 0; j <= n; j++)
            for (int m = 0; m < N; m++) H[i + j][m] += F[i][m] *
            // G[j][m];
    for (int i = 0; i <= n; i++) mobiusSub(H[i], n);
    vector<Mint> h(N);
    for (int m = 0; m < N; m++) h[m] = H[__builtin_popcount(m)][m];
    return h;
}

// USES: exact set cover / "partition the ground set into k valid
// groups" (repeated subset
// convolution, or exponentiate the EGF-like series); counting perfect
// matchings by subsets;
// "for each mask, best way to split it into two disjoint valid halves".
// SUBMASK ENUMERATION over all masks is  $O(3^n)$ : for (s = m; s; s =
// (s-1) & m)

```

5.8 SpecialSequences

Named sequences that keep showing up, with what each one counts. Needs Mint + buildFact().

```

// MOTZKIN M(n): lattice paths (0,0)->(n,0) with steps U/D/FLAT staying
// >= 0; also the number of
// ways to draw non-crossing chords on n points on a circle. M:
// 1,1,2,4,9,21,51,127
vector<Mint> motzkin(int n) {
    vector<Mint> m(n + 1, 0);
    m[0] = 1;
    if (n >= 1) m[1] = 1;
    for (int i = 2; i <= n; i++) { // M(i) =
        // M(i-1) + sum M(k)M(i-2-k)
        m[i] = m[i - 1];
        for (int k = 0; k <= i - 2; k++) m[i] += m[k] * m[i - 2 - k];
    }
    return m;
}
// NARAYANA N(n,k): Dyck paths of length 2n with exactly k peaks. sum_k
// N(n,k) = Catalan(n).
Mint narayana(int n, int k) { return k < 1 || k > n ? Mint(0) : C(n, k)
    // * C(n, k - 1) * inv_n[n]; }
// SCHRODER: large R(n) counts lattice paths (0,0)->(n,n) with steps
// E/N/NE staying below y=x.

```

```
// R: 1,2,6,22,90,394. R(n) = R(n-1) + sum_{k} R(k) R(n-1-k) (little
↳ schroeder s(n) = R(n)/2 for n >= 1 only (s(0) = 1, not 1/2))
vector<Mint> schroder(int n) {
    vector<Mint> r(n + 1, 0);
    r[0] = 1;
    for (int i = 1; i <= n; i++) { r[i] = r[i - 1]; for (int k = 0; k <
↳ i; k++) r[i] += r[k] * r[i - 1 - k]; }
    return r;
}
// EULERIAN A(n,k): permutations of n with exactly k ascents.
// A(n,k) = (k+1) A(n-1,k) + (n-k) A(n-1,k-1). sum_k A(n,k) = n!
vector<vector<Mint>> eulerian(int n) {
    vector<vector<Mint>> a(n + 1, vector<Mint>(n + 1, 0));
    a[0][0] = 1;
    for (int i = 1; i <= n; i++)
        for (int k = 0; k < i; k++)
            a[i][k] = Mint(k + 1) * a[i - 1][k] + Mint(i - k) * (k ?
↳ a[i - 1][k - 1] : Mint(0));
    return a;
}
// q-BINOMIAL (Gaussian) coefficient: #k-dim subspaces of an n-dim
↳ space over F_q; also
// #partitions fitting in a k x (n-k) box, weighted by q^area.
// PRECONDITION: ord_p(q) > k. The denominator is prod (q^i - 1), which
↳ vanishes in F_p
// whenever ord_p(q) <= k - NOT only at q == 1. (q = -1 mod 998244353
↳ breaks every k >= 2.)
// Root-of-unity-safe fallback, O(nk): Pascal's rule [n,k]_q =
↳ [n-1,k-1]_q + q^k [n-1,k]_q.
Mint qBinom(int n, int k, Mint q) {
    //
    ↳ prod_{i=1..k} (q^(n-k+i)-1)/(q^i-1)
    if (k < 0 || k > n) return 0;
    if (q == Mint(1)) return C(n, k); // q=1 is the
    ↳ 0/0 limit
    Mint num = 1, den = 1;
    for (int i = 1; i <= k; i++) num *= q.pow(n - k + i) - 1, den *=
↳ q.pow(i) - 1;
    return num / den;
}
// HOOK LENGTH FORMULA: #standard Young tableaux of shape lambda = n! /
↳ prod(hook lengths).
// hook(i,j) = (arm to the right) + (leg below) + 1.
Mint youngTableaux(const vector<int>& lam) {
    int n = 0;
    for (int x : lam) n += x;
    vector<int> conj(lam.empty() ? 0 : lam[0], 0);
    for (int i = 0; i < (int)lam.size(); i++) for (int j = 0; j <
↳ lam[i]; j++) conj[j]++;
    Mint den = 1;
    for (int i = 0; i < (int)lam.size(); i++)
        for (int j = 0; j < lam[i]; j++) den *= Mint(lam[i] - j +
↳ conj[j] - i - 1);
    return fact[n] / den;
}
// LGV LEMMA: for sources a_1..a_k and sinks b_1..b_k in a DAG, the
↳ number of NON-INTERSECTING
// path systems equals det(M) with M[i][j] = #paths a_i -> b_j (when
↳ only the identity
// permutation can be non-intersecting). Counts plane partitions,
↳ rhombus tilings, etc.
```

5.9 EulerTransformAndSymbolic

SYMBOLIC METHOD: spec -> generating function

```
// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
/* ===== SYMBOLIC METHOD: spec -> generating function
↳ =====
Decide labelled (EGF) or unlabelled (OGF), then translate the
↳ construction:
UNLABELLED (OGF), A built from atoms with GF B(z):
    SEQ (ordered list) A = 1 / (1 - B)
    MSET (unordered, rep) A = exp( sum_{k>=1} B(z^k) / k )
↳ <- Euler transform
    PSET (unordered, no rep) A = exp( sum_{k>=1} (-1)^(k-1) B(z^k)/k )
    CYC (up to rotation) A = sum_{k>=1} (phi(k)/k) * log(1/(1 -
↳ B(z^k)))
LABELLED (EGF):
    SEQ A = 1/(1-B) SET A = exp(B) CYC A =
↳ log(1/(1-B))
"SET of connected pieces = exp(connected)" is the exponential formula.
=====
↳ */

// EULER TRANSFORM: a[n] = #atoms of size n -> b[n] = #MULTISETS of
↳ atoms of total size n.
// prod_{n>=1} (1 - x^n)^{-a_n} = 1 + sum b_n x^n. O(n log n) via
↳ the harmonic sum.
```

```
vector<Mint> eulerTransform(const vector<Mint>& a) { // a[0] unused;
    ↳ returns b with b[0] = 1
    int n = a.size() - 1;
    vector<Mint> c(n + 1, 0), b(n + 1, 0);
    for (int d = 1; d <= n; d++) // c_n = sum_{d
↳ | n} d * a_d
        for (int m = d; m <= n; m += d) c[m] += Mint(d) * a[d];
    b[0] = 1;
    for (int i = 1; i <= n; i++) { // b_n = (c_n +
↳ sum_{k<n} c_k b_{n-k}) / n
        Mint s = c[i];
        for (int k = 1; k < i; k++) s += c[k] * b[i - k];
        b[i] = s * inv_[i];
    }
    return b;
}
int muSmall(int n) { // Mobius of one
    ↳ small n, by trial division
    int r = 1;
    for (int p = 2; (ll)p * p <= n; p++) if (n % p == 0) {
        n /= p;
        if (n % p == 0) return 0;
        r = -r;
    }
    return n > 1 ? -r : r;
}
// INVERSE Euler transform: b -> a (given multiset counts, recover the
↳ atom counts).
vector<Mint> eulerInverse(const vector<Mint>& b) {
    int n = b.size() - 1;
    vector<Mint> c(n + 1, 0), a(n + 1, 0);
    for (int i = 1; i <= n; i++) { // c_n = n*b_n -
↳ sum_{d<n} c_d b_{n-d}
        Mint s = Mint(i) * b[i];
        for (int d = 1; d < i; d++) s -= c[d] * b[i - d];
        c[i] = s;
    }
    for (int i = 1; i <= n; i++) { // a_n = (1/n)
↳ sum_{d|n} mu(n/d) c_d
        Mint s = 0;
        for (int d = 1; d <= i; d++) if (i % d == 0) s +=
↳ Mint(muSmall(i / d)) * c[d];
        a[i] = s * inv_[i];
    }
    return a;
}
// UNLABELLED ROOTED TREES: r_1 = 1, and r_{n+1} = (Euler transform of
↳ r_1..r_n)[n]
// (a rooted tree = a root plus a MULTISSET of rooted subtrees).
vector<Mint> rootedTrees(int n) {
    vector<Mint> r(n + 1, 0);
    if (n >= 1) r[1] = 1;
    for (int i = 1; i < n; i++) {
        vector<Mint> a(i + 1, 0);
        for (int j = 1; j <= i; j++) a[j] = r[j];
        auto b = eulerTransform(a);
        r[i + 1] = b[i];
    }
    return r; //
    ↳ 0,1,1,2,4,9,20,48,115,286,719,...
}
// PARTITIONS INTO PARTS FROM A SET S = eulerTransform with a[s] = 1
↳ for s in S.
// COMPOSITIONS (ordered) use SEQ instead: 1/(1 - sum x^s).
```

5.10 PermanentAndSymmetric

RYSER'S FORMULA - permanent of an $n \times n$ matrix in $O(2^n * n)$.

```
// perm(A) counts PERFECT MATCHINGS of a bipartite graph (weighted: sum
↳ over matchings of the
// product of weights). Use for n <= ~20: "count ways to assign n tasks
↳ to n people",
// "count systems of distinct representatives", "count permutations
↳ avoiding forbidden cells".
Mint permanent(const vector<vector<Mint>>& a) {
    int n = a.size();
    vector<Mint> row(n, 0); // row[i] =
↳ sum_{j in S} a[i][j]
    Mint total = 0;
    int prev = 0;
    for (int g = 1; g < (1 << n); g++) { // Gray code:
↳ one column toggles per step
        int cur = g ^ (g >> 1), diff = cur ^ prev, j =
↳ __builtin_ctz(diff);
        if (cur & diff) for (int i = 0; i < n; i++) row[i] += a[i][j];
↳ // column j entered S
        else for (int i = 0; i < n; i++) row[i] -= a[i][j];
    }
```

```

↪ // column j left S
    prev = cur;
    Mint p = 1;
    for (int i = 0; i < n; i++) p *= row[i];
    (__builtin_popcount(cur) & 1) ? total -= p : total += p;
}
return (n & 1) ? Mint(0) - total : total; // perm = (-1)^n
↪ * sum_S (-1)^|S| prod_i rowSum
}
// PERMANENT vs DETERMINANT: identical formula without the signs. det
↪ is polynomial-time,
// perm is #P-hard - never expect an O(n^3) permanent.

// NEWTON'S IDENTITIES - convert between power sums p_k = sum x_i^k and
↪ elementary symmetric e_k.
// e_k = sum over k-subsets of the product; this is exactly "for every
↪ k, the sum over all
// k-subsets of the product of the chosen values".
// e_0 = 1; k * e_k = sum_{i=1..k} (-1)^(i-1) * e_{k-i} * p_i
vector<Mint> powerToElementary(const vector<Mint>& p, int n) { //
    ↪ p[1..n] -> e[0..n]
    vector<Mint> e(n + 1, 0);
    e[0] = 1;
    for (int k = 1; k <= n; k++) {
        Mint s = 0;
        for (int i = 1; i <= k; i++) {
            Mint t = e[k - i] * p[i];
            (i & 1) ? s += t : s -= t;
        }
        e[k] = s * inv_[k];
    }
    return e;
}
// p_k = (-1)^(k-1) * k * e_k + sum_{i=1..k-1} (-1)^(k-1+i) * e_{k-i}
↪ * p_i
vector<Mint> elementaryToPower(const vector<Mint>& e, int n) { //
    ↪ e[0..n] -> p[1..n]
    vector<Mint> p(n + 1, 0);
    for (int k = 1; k <= n; k++) {
        Mint s = Mint(k) * e[k];
        if ((k - 1) & 1) s = Mint(0) - s; // (-1)^(k-1) *
    ↪ k * e_k
        for (int i = 1; i < k; i++) {
            Mint t = e[k - i] * p[i];
            ((k - 1 + i) & 1) ? s -= t : s += t;
        }
        p[k] = s;
    }
    return p;
}
// e_k are also the coefficients of prod (x + x_i): prod_{i} (x + x_i)
↪ = sum_k e_k * x^(n-k).
// So "sum over k-subsets of the product" = coefficient extraction from
↪ that product, which you
// can also get in O(n log^2 n) by divide-and-conquer multiplication of
↪ the linear factors.

```

5.11 BernoulliFaulhaber

BERNOULLI NUMBERS and the CLOSED-FORM FAULHABER polynomial.

```

// Use this (not Lagrange interpolation) when you need the POLYNOMIAL
↪ COEFFICIENTS of
// sum_{k=1..n} k^p, or when p is large and many n are queried.
// Convention here: B_1 = -1/2 (the "B^-" convention that the formula
↪ below expects).
// B_0 = 1; sum_{k=0}^m C(m+1, k) * B_k = [m == 0]
// => B_m = -(1/(m+1)) * sum_{k=0}^{m-1} C(m+1, k) * B_k
vector<Mint> bernoulli(int m) { // B[0..m],
    ↪ O(m^2)
    vector<Mint> B(m + 1, 0);
    B[0] = 1;
    for (int i = 1; i <= m; i++) {
        Mint s = 0;
        for (int k = 0; k < i; k++) s += C(i + 1, k) * B[k];
        B[i] = Mint(0) - s * inv_[i + 1];
    }
    return B; // 1, -1/2, 1/6,
    ↪ 0, -1/30, 0, 1/42, ...
}
// O(m log m) alternative: B_m / m! = [t^m] inv( sum_{i>=0} t^i /
↪ (i+1)! ) -- one Poly::inv.

// FAULHABER: sum_{k=1}^n k^p = (1/(p+1)) * sum_{r=0}^p (-1)^r *
↪ C(p+1, r) * B_r * n^(p+1-r)
Mint powerSumClosed(ll n, int p, const vector<Mint>& B) {
    Mint s = 0, np = Mint(n).pow(p + 1); // n^(p+1-r),
    ↪ decreasing r

```

```

vector<Mint> pw(p + 2);
pw[0] = 1;
for (int i = 1; i <= p + 1; i++) pw[i] = pw[i - 1] * Mint(n);
for (int r = 0; r <= p; r++) {
    Mint t = C(p + 1, r) * B[r] * pw[p + 1 - r];
    (r & 1) ? s -= t : s += t;
}
return s * inv_[p + 1];
}
// The COEFFICIENTS of the Faulhaber polynomial in n (degree p+1):
// [n^(p+1-r)] = (-1)^r * C(p+1, r) * B_r / (p+1)
vector<Mint> faulhaberPoly(int p, const vector<Mint>& B) {
    vector<Mint> c(p + 2, 0);
    for (int r = 0; r <= p; r++) {
        Mint t = C(p + 1, r) * B[r] * inv_[p + 1];
        c[p + 1 - r] = (r & 1) ? Mint(0) - t : t;
    }
    return c; // c[i] =
    ↪ coefficient of n^i
}
// RELATED SUMS
// sum_{k=0}^n C(k, r) = C(n+1, r+1)
↪ (hockey stick)
// sum_{k=1}^n k^p * r^k : degree-p polynomial times r^n; solve
↪ by ansatz or by
// differentiating the geometric series
↪ p times.
// Bernoulli also gives Euler-Maclaurin and the von Staudt-Clausen
↪ denominator facts.

```

5.12 CycleLemmaFussCatalan

CYCLE LEMMA (Dvoretzky-Motzkin) and the FUSS-CATALAN / rational-Catalan family.

```

// The general-k version of the ballot problem: it handles every "+1 /
↪ -k step" path problem.
//
// CYCLE LEMMA: a word with m a's and n b's, m >= k*n, has EXACTLY m -
↪ k*n cyclic shifts that
// are k-dominating (every non-empty prefix has #a > k * #b).
// k = 1 ballot: p A's and q B's with p > q -> exactly p - q
↪ dominating shifts,
// so the probability that A leads the whole count is
↪ (p - q) / (p + q).

// #paths with p up-steps and q down-steps that stay STRICTLY positive
↪ after the first step.
Mint ballotStrict(ll p, ll q) { // p > q >= 0
    if (p <= q) return 0;
    return C(p + q, q) * Mint(p - q) * inv_[(int)(p + q)];
}
// FUSS-CATALAN: A_m(p, r) = (r / (m*p + r)) * C(m*p + r, m)
Mint fussCatalan(ll m, ll p, ll r) {
    if (m < 0) return 0;
    return C(m * p + r, m) * Mint(r) * Mint(m * p + r).inv();
}
// #p-ary trees with m internal nodes = A_m(p, 1) = C(m*p + 1, m) /
↪ (m*p + 1)
Mint pAryTrees(ll m, ll p) { return fussCatalan(m, p, 1); }
// Catalan is the case p = 2, r = 1.
// #sequences of m opens and (p-1)*m closes where every prefix keeps
↪ the stack non-negative
// is also A_m(p, 1).

// RATIONAL CATALAN: for gcd(a, b) = 1, the number of monotone lattice
↪ paths from (0,0) to
// (b, a) that never rise above the line a*x = b*y is C(a + b, a) / (a
↪ + b).
Mint rationalCatalan(ll a, ll b) { return C(a + b, a) * Mint(a +
    ↪ b).inv(); }

// CHUNG-FELLER: among the C(2n,n) balanced +-1 paths of length 2n, the
↪ number having exactly
// 2r steps above the axis is Catalan(n) for EVERY r in 0..n. Hence
↪ C(2n,n) = (n+1)*Cat(n).
// Use it for "count sequences with exactly k flaws" - the answer does
↪ not depend on k.

// REFLECTION PRINCIPLE (the tool behind all of the above):
// #paths (0,0)->(x,y) touching the line y = m = #paths (0,0)->(x,
↪ 2m - y)
// so "#paths avoiding the line" = C(total, up) - C(total, up shifted
↪ by the reflection).
// PRECONDITION: the barrier must SEPARATE the start (height 0) from
↪ the end y, i.e. m != 0
// and 0, y strictly on the same side of m. Without it the reflection
↪ subtracts too much:
// (x,y,m) = (2,2,1) returns -1 where the true count is 0.

```

```

Mint pathsAvoidingLine(ll x, ll y, ll m) {
    if (m == 0 || (m > 0 && y >= m) || (m < 0 && y <= m)) return 0;
    // steps +1/-1, must never reach y = m
    if (((x + y) & 1) || y > x) return 0;
    ll up = (x + y) / 2, up2 = (x + 2 * m - y) / 2;
    return C(x, up) - ((x + 2 * m - y) % 2 || up2 < 0 || up2 > x ?
    // Mint(0) : C(x, up2));
}

```

5.13 LGVandPolya

LGV LEMMA and the POLYA CYCLE INDEX - the two theorems that were stated but never coded.

```

// Needs: Mint, determinant (06 - Math/04/02), phi (04 - NT/02/06),
// muSmall (09 - EulerTransform...).

// LINDSTROM-GESSEL-VIENNOT: with sources a_1..a_k and sinks b_1..b_k
// in a DAG,
// det(M), M[i][j] = #paths a_i -> b_j
// counts NON-INTERSECTING path systems, PROVIDED the only permutation
// that admits a
// non-intersecting system is the identity (true for "sorted"
// sources/sinks in a grid).
// Grid instance: monotone lattice paths, so M[i][j] = C(dx + dy, dx).
Mint lgvGrid(const vector<pair<ll, ll>>& src, const vector<pair<ll,
// ll>>& dst) {
    int k = src.size();
    vector<vector<Mint>> M(k, vector<Mint>(k));
    for (int i = 0; i < k; i++)
        for (int j = 0; j < k; j++) {
            ll dx = dst[j].first - src[i].first, dy = dst[j].second -
            // src[i].second;
            M[i][j] = (dx < 0 || dy < 0) ? Mint(0) : C(dx + dy, dx);
        }
    return determinant(M); // needs
    // GaussianElimination.cpp
}

// USES: k non-crossing monotone paths, rhombus/domino tilings of a
// region, plane partitions,
// "k tokens walk on a grid and must never share a cell".

// POLYA CYCLE INDEX. Z(G)(a_1..a_n) = (1/|G|) sum_{g in G} prod_k
// a_k^{(#k-cycles of g)}.
// The UNWEIGHTED orbit count is Z(G)(m, m, ..., m) for m colours -
// that is what necklaces()
// and bracelets() already give. The reason to carry the cycle index is
// the WEIGHTED case:
// "necklaces with exactly c_1 of colour 1, c_2 of colour 2, ..." which
// the numeric form cannot do.
// Z(C_n) = (1/n) * sum_{d | n} phi(d) * a_d^{(n/d)}
// Z(D_n) = (1/2) Z(C_n) + (1/2) a_1 a_2^{((n-1)/2)}
// [n odd]
// Z(D_n) = (1/2) Z(C_n) + (1/4) ( a_1^2 a_2^{((n-2)/2)} + a_2^{(n/2)} )
// [n even]
// Z(S_n) = (1/n) * sum_{l=1..n} a_l * Z(S_{n-l}), Z(S_0) = 1
// Weighted PET: substitute a_k -> (t_1^k + t_2^k + ... + t_m^k) and
// read off the coefficient
// of t_1^{c_1} * ... * t_m^{c_m}.
// Necklaces with a PRESCRIBED colour multiset, n beads, counts c[]
// summing to n (rotations only):
Mint necklacesWithCounts(int n, const vector<int>& c) {
    Mint total = 0;
    for (int d = 1; d <= n; d++) {
        if (n % d) continue; // rotations of
        // order n/d have d cycles
        bool ok = true;
        for (int x : c) if (x % (n / d)) ok = false; // each colour
        // must split evenly
        if (!ok) continue;
        Mint ways = fact[d]; // multinomial
        // over the d cycle slots
        for (int x : c) ways *= invFact[x / (n / d)];
        total += Mint(phi(n / d)) * ways;
    }
    return total * Mint(n).inv();
}

// APERIODIC necklaces (Lyndon words) with k colours: M(n,k) = (1/n)
// sum_{d|n} mu(d) k^{(n/d)}.
Mint lyndonCount(ll n, ll k) {
    Mint s = 0;
    for (ll d = 1; d <= n; d++) if (n % d == 0) s += Mint(muSmall(d)) *
    // Mint(k).pow(n / d);
    return s * Mint(n).inv();
}

// TRANSFER MATRIX: count length-L walks in a finite automaton. Build
// A[s][t] = #transitions,

```

```

// then (A^L)[s][t]. The generating function sum_L (A^L)[i][j] x^L =
// ((I - xA)^{-1})[i][j] is
// RATIONAL with denominator det(I - xA) of degree <= #states, so the
// sequence satisfies a linear
// recurrence of that order - feed a prefix to Berlekamp-Massey and
// finish in O(k^2 log L)
// instead of O(states^3 log L).

```

5.14 Hafnian

HAFNIAN - the number of PERFECT MATCHINGS of a GENERAL (non-bipartite) graph, weighted:

```

// Needs: Mint (06 - Math/01 - Modular Arithmetic/01 - Mint.cpp).
// haf(A) = sum over perfect matchings M of prod over {i,j} in M of
// A[i][j],
// for a symmetric A with a zero diagonal and n EVEN. O(2^{(n/2)} * n^2),
// n up to ~38 in a couple of
// seconds. This is the general-graph analogue of the permanent (which
// counts perfect matchings of
// a BIPARTITE graph, 10 - PermanentAndSymmetric.cpp); haf(A) is the
// "square root" of the permanent
// of the corresponding symmetric matrix.
// REACH FOR IT for "count the ways to pair up 2m items with forbidden
// or weighted pairs", domino
// tilings of an irregular board, "partition into pairs" counting, and
// any #matching problem where
// Blossom only gives you the MAXIMUM matching and you need the COUNT
// of perfect ones.
// COUNTING is much harder than finding: max matching is polynomial
// (Blossom), counting is #P-hard.
// For PLANAR graphs use the FKT algorithm (a Pfaffian orientation,
// then one determinant) - that is
// polynomial, and is how you count domino tilings of a grid.
// The recursion removes the last two rows/columns and tracks a
// truncated polynomial in a marker
// variable whose exponent counts how many pairs have been committed;
// only degree <= n/2 survives.

namespace Hafnian {
    int H; // = n/2 + 1,
    // the truncation length
    void addTo(vector<Mint>& r, const vector<Mint>& a, const vector<Mint>&
    // b) {
        for (int i = 0; i < H; i++) if (a[i].v)
            for (int j = 0; j + i + 1 < H; j++) r[i + j + 1] += a[i] * b[j];
    }
    vector<Mint> rec(const vector<vector<vector<Mint>>>& v) {
        vector<Mint> r(H, 0);
        if (v.empty()) { r[0] = 1; return r; }
        int m = (int)v.size() - 2;
        auto V = v;
        V.resize(m);
        vector<Mint> zero = rec(V); // vertex pair
        // (m, m+1) left unmatched here
        for (int i = 0; i < m; i++)
            for (int j = 0; j < i; j++) { // ... or
                // routed through i and j
                addTo(V[i][j], v[m][i], v[m + 1][j]);
                addTo(V[i][j], v[m + 1][i], v[m][j]);
            }
        vector<Mint> one = rec(V);
        for (int i = 0; i < H; i++) r[i] += one[i] - zero[i];
        addTo(r, one, v[m + 1][m]); // ... or
        // matched to each other
        return r;
    }
    Mint solve(const vector<vector<Mint>>& a) { // a
        // symmetric, zero diagonal, n even
        int n = a.size();
        H = n / 2 + 1;
        vector<vector<vector<Mint>>> v(n);
        for (int i = 0; i < n; i++) {
            v[i].resize(i);
            for (int j = 0; j < i; j++) v[i][j].assign(H, 0), v[i][j][0] =
            // a[i][j];
        }
        return rec(v).back();
    }
}

// SMALL n: a plain bitmask DP over "set of matched vertices", always
// matching the lowest unmatched
// vertex, is O(2^n * n) - simpler and faster up to n = 22. Use the
// Hafnian only past that.
// RELATED
// PFAFFIAN: for a SKEW-symmetric matrix, pf(A)^2 = det(A) and pf is
// computable in O(n^3); the
// Hafnian is its "no signs" version, which is exactly why one is
// easy and the other is not.

```

```
// NUMBER OF MATCHINGS OF EVERY SIZE (not just perfect): that is the
//   ↳ matching polynomial; the same
// recursion with a longer truncation, or  $O(2^n)$  subset DP.
// MODULUS: everything here is Mint arithmetic, so the count comes out
//   ↳ mod MOD; for exact small
// counts use unsigned long long and drop Mint.
```

5.15 SubsetSumCounts

NUMBER OF SUBSETS WITH EACH SUM, for every sum $0..m$, in $O(m \log m)$ - not $O(n * m)$.

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / exp_ / log_ / cut (05
//   ↳ Convolution/04).
// Given a multiset of n positive values, the generating function is
//   ↳  $\prod_i (1 + x^{a_i})$ , which
// naively costs n multiplications. Take logarithms: with  $\text{cnt}[v]$  = how
//   ↳ many times v occurs,
//  $\log \prod_v (1 + x^v)^{\text{cnt}[v]} = \sum_v \text{cnt}[v] * \sum_{j \geq 1} (-1)^{j+1}$ 
//   ↳  $x^{(v*j)} / j$ ,
// and the double sum is a HARMONIC sum,  $O(m \log m)$  terms in total.
//   ↳ Then one exp brings it back.
// REACH FOR IT for Yosupo "#P subset sum", "for every k, how many
//   ↳ subsets sum to k", partition
// counting with a bounded multiset of part sizes, and knapsack
//   ↳ COUNTING when m is  $1e5$  and n is
// large. A plain  $O(n*m)$  knapsack (or the bitset one) is better when n
//   ↳ is small or you only need
// feasibility rather than counts.
// PRECONDITION: all values  $\geq 1$  (handle  $a_i = 0$  separately - each zero
//   ↳ doubles every count).
Poly subsetSumCounts(const vector<int>& a, int m) { // result[k] =
//   ↳ #subsets summing to k
    vector<int> cnt(m + 1, 0);
    for (int v : a) if (v >= 1 && v <= m) cnt[v]++;
    Poly p(m + 1, 0);
    for (int v = 1; v <= m; v++) if (cnt[v])
        for (int j = 1; (ll)v * j <= m; j++) {
            ll t = (ll)cnt[v] % NMOD * npow(j, NMOD - 2) % NMOD;
            if (j & 1) p[v * j] = (p[v * j] + t) % NMOD;
            else p[v * j] = (p[v * j] - t + NMOD) % NMOD;
        }
    return exp_(p, m + 1);
}
// THE SAME MACHINE, DIFFERENT SIGN - this is the symbolic method (09 -
//   ↳ EulerTransformAndSymbolic):
// MULTISET (each value usable any number of times) = exp(  $\sum_j$ 
//   ↳  $B(x^j) / j$  )
// POWERSET (each value usable at most once) = exp(  $\sum_j$ 
//   ↳  $(-1)^{j+1} B(x^j) / j$  )
// with  $B(x) = \sum_v \text{cnt}[v] x^v$ . Swapping the sign is the only
//   ↳ difference between "partitions into
// parts from a set" and "subsets of a multiset".
// GOING BACKWARDS: given the counts, log_ recovers B and hence the
//   ↳ multiset - that is the inverse
// Euler transform, the standard "recover the item sizes from the
//   ↳ subset-sum counts" problem.
// ADD OR REMOVE ONE ITEM in  $O(m)$ : multiply or divide the answer by  $(1$ 
//   ↳  $+ x^v)$  directly (division by
//  $1 + x^v$  is the recurrence  $r[k] \leftarrow r[k - v]$  - much cheaper than
//   ↳ rebuilding.
// K ITEMS ONLY: to also track the SIZE of the subset you need a second
//   ↳ variable; do the same
// construction with  $x^v y$  and truncate in both, or run a 2-D
//   ↳ convolution (05 - Convolution/08).
```

6 Math

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6.1 Modular Arithmetic

6.1.1 Mint

Modular integer (MOD must be PRIME for inv/division) + $O(N)$ factorial tables.

```
struct Mint {
    int v;
    Mint(ll x = 0) { v = int(x % MOD); if (v < 0) v += MOD; }
    Mint& operator+=(Mint o) { if ((v += o.v) >= MOD) v -= MOD; return
    ↳ *this; }
    Mint& operator-=(Mint o) { if ((v -= o.v) < 0) v += MOD; return
    ↳ *this; }
    Mint& operator*=(Mint o) { v = int(1LL * v * o.v % MOD); return
    ↳ *this; }
    Mint& operator/=(Mint o) { return *this *= o.inv(); }
```

```
friend Mint operator+(Mint a, Mint b) { return a += b; }
friend Mint operator-(Mint a, Mint b) { return a -= b; }
friend Mint operator*(Mint a, Mint b) { return a *= b; }
friend Mint operator/(Mint a, Mint b) { return a /= b; }
friend bool operator==(Mint a, Mint b) { return a.v == b.v; }
Mint pow(ll e) const { Mint r = 1, b = *this; for (; e > 0; e >>=
    ↳ 1, b *= b) if (e & 1) r *= b; return r; }
Mint inv() const { return pow(MOD - 2); } // Fermat:
    ↳  $a^{p-2} = a^{-1} \pmod{p}$ 
friend ostream& operator<<(ostream& o, Mint m) { return o << m.v; }
friend istream& operator>>(istream& i, Mint& m) { ll x; i >> x; m =
    ↳ Mint(x); return i; }
```

```
const int N = 200005; // ALSO in 01 -
    ↳ Template.cpp: keep one
Mint fact[N], invFact[N], inv_[N];
void buildFact() { //  $O(N)$  total,
    ↳ not  $O(N \log \text{MOD})$ 
    fact[0] = invFact[0] = inv_[1] = 1;
    for (int i = 1; i < N; i++) fact[i] = fact[i - 1] * i;
    invFact[N - 1] = fact[N - 1].inv();
    for (int i = N - 1; i > 0; i--) invFact[i - 1] = invFact[i] * i;
    for (int i = 2; i < N; i++) inv_[i] = Mint(MOD - MOD / i) *
    ↳ inv_[MOD % i];
}
Mint C(ll n, ll r) { if (r < 0 || r > n) return 0; return fact[n] *
    ↳ invFact[r] * invFact[n - r]; }
Mint Perm(ll n, ll r) { if (r < 0 || r > n) return 0; return fact[n] *
    ↳ invFact[n - r]; }
//  $C(n, r)$  for HUGE n and small r (no tables):  $\prod_{i < r} (n - i) / r!$ 
Mint Cbig(ll n, ll r) {
    if (r < 0 || n < 0 || r > n) return 0;
    Mint num = 1;
    for (ll i = 0; i < r; i++) num *= Mint(n - i);
    return num * invFact[r];
}
```

6.1.2 BinaryFieldGF2k

ARITHMETIC IN THE FINITE FIELD $\text{GF}(2^K)$: elements are K-bit masks = polynomials over $\text{GF}(2)$

```
// reduced mod a fixed IRREDUCIBLE polynomial of degree K. Add = xor,
//   ↳ multiply = carry-less
// multiply then reduce. mul is  $O(K)$ , pow / inverse  $O(K \log)$  - or  $O(1)$ 
//   ↳ with the CLMUL intrinsic.
// REACH FOR IT when a problem is "linear algebra over a field of
//   ↳ characteristic 2 with more than
// two elements": random algebraic identity testing over  $\text{GF}(2^K)$ 
//   ↳ (Freivalds / Schwartz-Zippel with
// xor-shaped constraints), Reed-Solomon-flavoured coding, "assign
//   ↳ field values to symbols and test
// whether a determinant vanishes", and Wiedemann over  $\text{GF}(2)$ .
// PRECONDITION: POLY below must be irreducible over  $\text{GF}(2)$  and of
//   ↳ degree exactly K.
// K=8: 0x11B K=16: 0x1002B K=30: 0x40000003 K=31: 0x80000009
//   ↳ K=32: 0x1000000AF
// K=64:  $x^{64} + x^4 + x^3 + x + 1$  (constant 0x1B), the usual choice
//   ↳ for unsigned long long.
namespace GF2K {
    const int K = 30;
    const ll POLY = (1LL << K) | 3; //  $x^{30} + x + 1$ ,
    ↳ irreducible

    ll add(ll a, ll b) { return a ^ b; } // subtraction is the
    ↳ same operation
    ll mul(ll a, ll b) {
        ll r = 0;
        while (b) {
            if (b & 1) r ^= a;
            b >>= 1, a <<= 1;
            if (a >> K & 1) a ^= POLY; // reduce as soon as
            ↳ the degree reaches K
        }
        return r;
    }
    ll pw(ll a, ll e) {
        ll r = 1;
        for (; e; e >>= 1, a = mul(a, a)) if (e & 1) r = mul(r, a);
        return r;
    }
    ll inv(ll a) { return pw(a, (1LL << K) - 2); } //  $a^{2^K - 1} = 1$ , so
    ↳  $a^{-1} = a^{(2^K - 2)}$ 
}
// FACTS YOU NEED WHEN USING IT
//  $|\text{GF}(2^K)| = 2^K$ , the multiplicative group is CYCLIC of order  $2^K -$ 
//   ↳ 1, and every element
// satisfies  $a^{(2^K)} = a$ . Squaring is a field automorphism
```

```

    ↪ (FROBENIUS): (a + b)^2 = a^2 + b^2,
// so squaring is LINEAR over GF(2) and can be tabulated as a matrix.
// GF(2^K) contains GF(2^d) exactly when d | K.
// x^(2^K) - x factors as the product of all irreducible polynomials
    ↪ whose degree divides K.
// A random degree-K polynomial is irreducible with probability about
    ↪ 1/K; test with
// gcd(x^(2^d) - x, f) == 1 for all d < K dividing K.
// NOT THE SAME AS NIMBERS. Nim-multiplication (10 - Game Theory/05 -
    ↪ NimbersAndSums.cpp) is a
// DIFFERENT field structure on the same set: it needs no irreducible
    ↪ polynomial, it is defined by
// the mex rules, and 2^(2^k) values form subfields. Use numbers for
    ↪ game values, this for coding
// and linear-algebra style problems - they are isomorphic but NOT
    ↪ equal element by element.
// SPEED: on x86 with -mpclmul, _mm_cmlulepi64_sil28 does the
    ↪ carry-less product in one
// instruction; you then only need the Barrett-style reduction. Worth
    ↪ it only in a hot loop.

```

6.2 Multiplicative Functions

6.2.1 EulerTotientPhi

EULER TOTIENT SIEVE - $\phiArr[i] = \#\{1 \leq k \leq i : \gcd(k, i) = 1\}$ for all $i < N$ in $O(N \log \log N)$.

```

// Needed for: a*b mod m with gcd(a,m)=1 (Euler's theorem), counting
    ↪ coprime pairs, Mobius identities
// (sum_{d|n} phi(d) = n), and the order of the multiplicative group
    ↪ mod n.
// NAME: phiArr, not phi - a global 'phi' collides with the phi()
    ↪ FUNCTION in
// 04 - Number Theory/02 - Primes and Divisors/06 -
    ↪ MillerRabinPollard.cpp.
int phiArr[N];

void calculatePhi() {
    for (int i = 0; i < N; ++i) phiArr[i] = i & 1 ? i : i / 2;
    for (int i = 3; i < N; i += 2)
        if (phiArr[i] == i)
            for (int j = i; j < N; j += i) phiArr[j] -= phiArr[j] / i;
}

```

6.2.2 MobiusFunction

Mobius Function $\mu(n)$ Sieve - Time: $O(N)$, Space: $O(N)$

```

// mu(n) = 1 if n is square-free with an even number of prime factors
// mu(n) = -1 if n is square-free with an odd number of prime factors
// mu(n) = 0 if n has a squared prime factor
struct Mobius {
    int n;
    vector<int> mu, primes;
    vector<bool> is_prime;

    Mobius(int n = 1e7) : n(n), mu(n + 1, 0), is_prime(n + 1, true) {
        is_prime[0] = is_prime[1] = false;
        mu[1] = 1;

        for (int i = 2; i <= n; i++) {
            if (is_prime[i]) {
                primes.pb(i);
                mu[i] = -1;
            }
            for (int p : primes) {
                if ((ll)i * p > n) break;
                is_prime[i * p] = false;
                if (i % p == 0) {
                    mu[i * p] = 0;
                    break;
                } else {
                    mu[i * p] = -mu[i];
                }
            }
        }
    }
};

```

6.3 Matrices

6.3.1 MatrixExponentiation

Matrix over Mint. mul $O(n^3)$ (cache-friendly i-k-j order), pow $O(n^3 \log e)$.

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).
// Fix the size at compile time (array<array<Mint,K>,K>) if you need
    ↪ the last 2x speedup.
typedef vector<vector<Mint>> Mat;

Mat operator*(const Mat& a, const Mat& b) {
    int n = a.size(), k = b.size(), m = b[0].size();
    Mat c(n, vector<Mint>(m));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < k; t++) if (a[i][t].v)
            for (int j = 0; j < m; j++) c[i][j] += a[i][t] * b[t][j];
    return c;
}

Mat eye(int n) { Mat r(n, vector<Mint>(n)); for (int i = 0; i < n; i++)
    ↪ r[i][i] = 1; return r; }

Mat mpow(Mat a, ll e) {
    Mat r = eye(a.size());
    for (; e > 0; e >>= 1, a = a * a) if (e & 1) r = r * a;
    return r;
}

// RECIPES
// Linear recurrence f(n) = c1*f(n-1) + ... + ck*f(n-k):
// companion matrix, first row = (c1..ck), subdiagonal = 1. f(n) =
    ↪ (M^(n-k) * [f(k-1)..f(0)]^T)[0]
// Add a "d" constant term: enlarge by one row/col with a 1 on the
    ↪ diagonal, put d in the first row.
// Count walks of length L between u and v: (A^L)[u][v] where A is the
    ↪ adjacency matrix.
// Shortest walk with exactly L edges: same, but replace (+,*) with
    ↪ (min,+) - "min-plus product".

```

6.3.2 CharacteristicPolynomial

CHARACTERISTIC POLYNOMIAL $p(x) = \det(xI - A)$ of an $n \times n$ matrix in $O(n^3)$. MOD must be PRIME.

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp), Mat (03 -
    ↪ Matrices/01).
// REACH FOR IT when you need det(A - t*I) for MANY t (n = 500, q = 1e5
    ↪ queries: build p once,
// then each query is one Horner evaluation), or A^k for a huge k: by
    ↪ Cayley-Hamilton p(A) = 0, so
// A^k = (x^k mod p(x))(A), which costs  $O(n^2 \log k)$  polynomial work
    ↪ plus  $O(n^3)$  to expand - much
// better than  $O(n^3 \log k)$  when n is large. Also: det(A) = (-1)^n *
    ↪ p[0], trace(A) = -p[n-1].
// Gaussian elimination CANNOT do this directly because the entries are
    ↪ polynomials in x.

// STEP 1: reduce to upper HESSENBERG form (a[i][j] = 0 for i > j + 1)
    ↪ by SIMILARITY transforms.
// The row op "row i -= f * row (c+1)" is undone by "col (c+1) += f *
    ↪ col i", so A -> E A E^-1 and
// the whole spectrum (hence the characteristic polynomial) is
    ↪ unchanged.  $O(n^3)$ .
Mat hessenberg(Mat a) {
    int n = a.size();
    for (int c = 0; c + 2 < n; c++) {
        int p = -1;
        for (int i = c + 1; i < n; i++) if (a[i][c].v) { p = i; break; }
        if (p < 0) continue; // column already
        ↪ clear below the subdiagonal
        if (p != c + 1) { // swapping rows AND
        ↪ cols keeps it a similarity;
            swap(a[c + 1], a[p]); // column c is
        ↪ untouched because c < c+1 <= p
            for (int i = 0; i < n; i++) swap(a[i][c + 1], a[i][p]);
        }
        for (int i = c + 2; i < n; i++) if (a[i][c].v) {
            Mint f = a[i][c] / a[c + 1][c];
            for (int j = 0; j < n; j++) a[i][j] -= a[c + 1][j] * f;
            for (int j = 0; j < n; j++) a[j][c + 1] += a[j][i] * f;
        }
        ↪ // uses the UPDATED col i
    }
    return a;
}

// STEP 2: expand det(x*I - H) along the last row. With f[m] = char
    ↪ poly of the leading m x m block,
// f[m] = (x - h[m-1][m-1]) f[m-1] - sum_{i=1..m-1} h[m-1-i][m-1] *
    ↪ (prod_{j=m-i..m-1} h[j][j-1])
// *
    ↪ f[m-1-i]
// Only Hessenberg matrices satisfy such a short recurrence; that is
    ↪ the whole point of step 1.
vector<Mint> charPoly(const Mat& a) { // returns p[0..n],

```

```

    ↪ p[i] = coeff of x^i, p[n] = 1
    int n = a.size();
    Mat h = hessenberg(a);
    vector<vector<Mint>> f(n + 1);
    f[0] = {1};
    for (int m = 1; m <= n; m++) {
        f[m].assign(m + 1, 0);
        for (int i = 0; i < m; i++) {
            f[m][i + 1] += f[m - 1][i];
            f[m][i] -= h[m - 1][m - 1] * f[m - 1][i];
        }
        Mint prod = 1;
        for (int i = 1; i < m; i++) {
            prod *= h[m - i][m - i - 1];
            Mint c = prod * h[m - 1 - i][m - 1];
            if (!c.v) continue;
            for (int j = 0; j <= m - 1 - i; j++) f[m][j] -= c * f[m - 1
    ↪ - i][j];
        }
    }
    return f[n];
}

// A^k VIA CAYLEY-HAMILTON: r(x) = x^k mod p(x) (O(n^2 log k) with
    ↪ schoolbook, O(n log n log k)
// with NTT), then A^k = sum r[i] A^i. If you only need u^T A^k v, that
    ↪ is sum r[i] * (u^T A^i v)
// and the A^i v are n matrix-vector products - O(n * nnz * n) for a
    ↪ sparse A, no n^3 anywhere.
// EIGENVALUES over F_p are the roots of p; over R use the QR algorithm
    ↪ (not a contest routine).
// det(A + t*B) as a polynomial in t: reduce (A, B) to
    ↪ Hessenberg/triangular pencil form, or just
// evaluate at n+1 values of t with plain Gaussian determinants and
    ↪ interpolate - O(n^4), fine to
// n = 100 and much easier to get right.
// SIMILAR-LOOKING BUT DIFFERENT: the MINIMAL polynomial divides p; get
    ↪ it from the Krylov
// sequence u^T A^k v with Berlekamp-Massey (see 03 -
    ↪ SparseDeterminant.cpp).

```

6.3.3 SparseDeterminant

```

rnd (01 - Miscellaneous/04 - random.cpp).

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp), berlekampMassey
    ↪ (06 - Interpolation/02),
// rnd (01 - Miscellaneous/04 - random.cpp).
// WIEDEMANN: determinant and minimal polynomial of a SPARSE n x n
    ↪ matrix in O(n * nnz + n^2),
// using O(n) memory. MOD must be PRIME. Gaussian elimination is O(n^3)
    ↪ and fills in; this is the
// routine for n = 5e4 with 1e5 non-zeros (Yosupo "sparse matrix det"),
    ↪ for adjacency-like
// matrices, and for the Matrix-Tree determinant of a sparse graph.
// IDEA: for random vectors u, v the scalar sequence s_k = u^T A^k v
    ↪ obeys the same linear
// recurrence as A itself (Cayley-Hamilton), so Berlekamp-Massey on
    ↪ 2n+2 samples recovers the
// minimal polynomial. Each sample costs one sparse mat-vec, hence n *
    ↪ nnz total.
// PRECONDITIONER: right-multiplying by a random diagonal D makes
    ↪ minpoly(AD) = charpoly(AD) with
// probability >= 1 - O(n^2 / MOD), which is what lets us read the
    ↪ determinant off the last
// coefficient. det(AD) = det(A) * prod d_i, so divide the d_i back out.
struct Elem { int r, c; Mint v; }; // A[r][c] = v; all
    ↪ (r, c) distinct, rest zero

vector<Mint> sparseMinPoly(int n, const vector<Elem>& a) {
    vector<Mint> u(n), x(n), s;
    for (int i = 0; i < n; i++) u[i] = rnd(1, MOD - 1), x[i] = rnd(1,
    ↪ MOD - 1);
    for (int k = 0; k < 2 * n + 2; k++) {
        Mint t = 0;
        for (int i = 0; i < n; i++) t += u[i] * x[i];
        s.push_back(t);
        vector<Mint> y(n, 0); // y = A * x, one pass
    ↪ over the non-zeros
        for (const Elem& e : a) y[e.r] += e.v * x[e.c];
        x = y;
    }
    return berlekampMassey(s); // s[i] = sum_j C[j] *
    ↪ s[i-1-j]
}

Mint sparseDet(int n, vector<Elem> a) {
    vector<Mint> d(n);
    for (int i = 0; i < n; i++) d[i] = rnd(1, MOD - 1);
    for (Elem& e : a) e.v *= d[e.c]; // A -> A * diag(d)
    vector<Mint> c = sparseMinPoly(n, a);
}

```

```

    if ((int)c.size() < n) return 0; // minpoly shorter
    ↪ than n => singular (whp)
    Mint r = c.back(); // charpoly = x^n - c0
    ↪ x^(n-1) - ... - c(n-1)
    if (n % 2 == 0) r = Mint(0) - r; // det = (-1)^(n+1) *
    ↪ c[n-1]
    for (int i = 0; i < n; i++) r /= d[i];
    return r;
}

// RANK of a sparse matrix: same trick on a random projection; simpler
    ↪ in practice is to run
// blocked elimination only if the fill-in stays small.
// SOLVING A x = b sparsely: with the minimal polynomial m(x) = sum m_i
    ↪ x^i and m_0 != 0,
// x = -(1/m_0) * sum_{i>=1} m_i A^(i-1) b -- again only sparse
    ↪ mat-vecs.
// FAILURE MODE: if the returned recurrence is shorter than n, either
    ↪ the matrix is singular or
// the random draw was unlucky; rerun once before believing the 0.
// INTEGER determinant with no modulus: run this under several primes
    ↪ and CRT, sizing with
// Hadamard's bound |det| <= prod of the row 2-norms.

```

6.3.4 CirculantDeterminant

DETERMINANT OF A CIRCULANT in $O(n \log^2 n)$ instead of $O(n^3)$.

```

// Needs: resultant (05 - Convolution/11 - PolyGcdResultant.cpp), which
    ↪ needs NTT & Polynomial.
// C[i][j] = c[(j - i) mod n] (every row is the previous one rotated
    ↪ right by 1). Then
// det C = prod_{k=0}^{n-1} f(w^k) = Res(x^n - 1, f), f(x) = c0 +
    ↪ c1 x + ... + c(n-1) x^(n-1),
// where w is a primitive n-th root of unity. Eigenvalue f(w^k) has
    ↪ eigenvector (1, w^k, w^2k, ...)
// - the Fourier modes diagonalise EVERY circulant simultaneously,
    ↪ which is why circulants commute
// and why multiplying two of them is just a cyclic convolution of
    ↪ their first rows.
// The resultant form is what you use when n does NOT divide MOD - 1,
    ↪ so the roots of unity do not
// exist in the field. Everything below is mod NMOD = 998244353 (the
    ↪ Poly modulus).
ll circulantDet(const vector<ll>& c) {
    int n = c.size();
    Poly xn(n + 1, 0);
    xn[0] = NMOD - 1, xn[n] = 1; // x^n - 1
    return resultant(xn, c);
}

// If n | (NMOD - 1) you do not need the resultant at all: evaluate f
    ↪ at the n-th roots of unity
// with one length-n NTT and multiply the results, O(n log n).
// PERMUTANT MATRIX (a circulant along a permutation): M[i][j] = a[k]
    ↪ where sigma^k(i) = j,
// for a permutation sigma that is a SINGLE n-cycle. Relabel the
    ↪ indices along the cycle
// (q[0] = 0, q[t] = sigma^t(0)) and M becomes an honest circulant; the
    ↪ determinant picks up the
// SIGN of that relabelling permutation. If sigma has more than one
    ↪ cycle the matrix is not a
// permutant and the trick does not apply.
// BLOCK-CIRCULANT (blocks of size m, n blocks): det = prod_k
    ↪ det(F(w^k)) with F the matrix-valued
// symbol - same argument, m x m determinants instead of scalars.
// OTHER DETERMINANTS YOU CAN DO WITHOUT ELIMINATION
// MATRIX DETERMINANT LEMMA: det(A + u v^T) = det(A) * (1 + v^T A^-1
    ↪ u). Rank-k update:
// det(A + U V^T) = det(A) * det(I_k + V^T A^-1 U). So det(x*I + a
    ↪ b^T) = x^(n-1) * (x + a.b),
// which closes "M[i][j] = a_i b_j off the diagonal, x + a_i b_i on
    ↪ it" in O(n).
// CAUCHY-BINET: for A (n x m) and B (m x n), det(AB) = sum over
    ↪ n-subsets S of [m] of
// det(A[:,S]) * det(B[S,:]). This is the LGV lemma's algebraic twin.
// VANDERMONDE: det = prod_{i<j} (x_j - x_i). CAUCHY:
    ↪ det(1/(x_i + y_j)) =
// prod_{i<j} (x_i - x_j)(y_i - y_j) / prod_{i,j} (x_i + y_j).
// TRIDIAGONAL: d_k = b_k d_{k-1} - a_k c_{k-1} d_{k-2}, O(n) (this is
    ↪ the Hessenberg recurrence).

```

6.3.5 PolyMatrixPower

MATRIX EXPONENTIATION WHERE THE ENTRIES ARE POWER SERIES, truncated mod x^K .

```
// Needs: multiply (05 - Convolution/02 - NTT.cpp), Poly / cut (05 -
//   Convolution/04).
// M^e in  $O(n^3 * K \log K * \log e)$ . Everything is mod NMOD = 998244353.
// REACH FOR IT for a linear recurrence in n that also tracks a SECOND
//   statistic you want the full
// distribution of: put  $x^{(\text{contribution})}$  in the transition instead of
//   1, and the (i, j) entry of
// M^e becomes the generating function over that statistic. Classic:
//   "count length-n walks by
// number of times a marked edge is used", "f(n) for every k = number
// of ..." (CF 755G), any DP
// whose answer is asked for ALL values of a parameter at once rather
// than one value.
// A plain Mat over Mint gives you one number; this gives you K numbers
// for a log K factor.
typedef vector<vector<Poly>> PMat;

PMat pmul(const PMat& a, const PMat& b, int K) {
    int n = a.size(), m = b[0].size(), k = b.size();
    PMat c(n, vector<Poly>(m, Poly(1, 0)));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < k; t++) {
            if (a[i][t].size() == 1 && !a[i][t][0]) continue; //
            // skip the zero series
            for (int j = 0; j < m; j++) c[i][j] = c[i][j] +
            // cut(multiply(a[i][t], b[t][j]), K);
        }
    return c;
}

PMat peye(int n) {
    PMat r(n, vector<Poly>(n, Poly(1, 0)));
    for (int i = 0; i < n; i++) r[i][i] = Poly(1, 1);
    return r;
}

PMat ppow(PMat a, ll e, int K) { // a^e mod x^K
    PMat r = peye(a.size());
    for (; e > 0; e >= 1, a = pmul(a, a, K)) if (e & 1) r = pmul(r, a,
    // K);
    return r;
}

// EXAMPLE (Fibonacci refined by the number of "1" steps, CF 755G
//   shape):
// M = {{1 + x, x}, {1, 0}} and  $(M^{(n-1)} * v)[0]$  is the generating
//   function you want.
// CHEAPER ALTERNATIVES, try these first:
// - if K is tiny ( $\leq 3$ ), just hand-roll the truncated series as k
//   separate Mint matrices.
// - if the statistic is "how many times did we take transition t",
//   one variable suffices; if it
// is a pair of statistics you need a bivariate series - flatten it
//   to  $x^{(i*W + j)}$ .
// - if n (the matrix size) is 1, this is just power-series
//   exponentiation: use pow_ directly.
// - if you want ONE coefficient rather than all K, Bostan-Mori on the
//   rational function is
//  $O(d \log d \log e)$  and much faster (06 - Interpolation/03 -
//   BostanMori.cpp).
// COST WARNING:  $n^3$  poly multiplications per squaring. n = 8, K = 1e5,
//   e = 1e18 is already at the
// edge; keep the matrix small, and truncate (cut to K) after EVERY
//   product, never at the end.
```

6.3.6 Freivalds

FREIVALDS: verify $A * B == C$ in $O(t * n^2)$ instead of computing the product in $O(n^3)$.

```
// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp), rnd (01 -
//   Miscellaneous/04 - random.cpp).
// Error probability  $\leq 2^{-t}$  (one-sided: it never rejects a correct
//   product). t = 30..40 is plenty.
// REACH FOR IT when the problem is "is this claimed product right",
//   when A or B changes between
// queries so recomputing is too slow, or as a subroutine inside a
//   bigger randomised check.
bool freivalds(const vector<vector<Mint>>& a, const
//   vector<vector<Mint>>& b,
//   const vector<vector<Mint>>& c, int t = 40) {
    int n = a.size();
    vector<Mint> r(n), x(n), y(n);
    while (t--) {
        for (int i = 0; i < n; i++) r[i] = rnd(0, 1); //
        // random 0/1 vector is enough
        for (int i = 0; i < n; i++) { // x =
            // B r, y = C r
```

```
x[i] = y[i] = 0;
    for (int j = 0; j < n; j++) x[i] += b[i][j] * r[j], y[i] +=
    // c[i][j] * r[j];
    }
    for (int i = 0; i < n; i++) { //
        // check A x == y
        Mint s = 0;
        for (int j = 0; j < n; j++) s += a[i][j] * x[j];
        if (!s == y[i]) return false;
    }
    return true;
}

// WHY IT WORKS: D = AB - C is non-zero, so some row of D is non-zero;
//   for a uniform 0/1 vector r,
// Pr[D r == 0]  $\leq 1/2$  (fix all coordinates but one - at most one of
//   the two choices can cancel).
// Independent rounds multiply, hence  $2^{-t}$ . Do the products RIGHT to
//   LEFT: A(Br), never (AB)r.

// THE GENERAL PATTERN - RANDOMISED ALGEBRAIC IDENTITY TESTING
// SCHWARTZ-ZIPPEL: a non-zero polynomial of total degree d over a
//   field, evaluated at a uniform
// point of  $S^n$ , is zero with probability  $\leq d / |S|$ . Every check below
//   is one application.
// - "are these two expressions equal?" - evaluate both at a random
//   point mod a large prime.
// - PERFECT MATCHING in a general graph: build the TUTTE matrix T
//   with  $T[i][j] = x_{ij}$  and
//  $T[j][i] = -x_{ij}$  for each edge  $i < j$ , substitute random values,
//   and G has a perfect matching
// iff  $\det T \neq 0$  (whp).  $O(n^3)$  existence check with no Blossom.
//   rank(T) = 2 * (max matching).
// - BIPARTITE matching: same with the Edmonds matrix (random value on
//   each edge, 0 elsewhere).
// - "do these two multisets/strings/trees agree?" - hash them as
//   polynomial evaluations.
// - CHECKING A PRODUCT OF SPARSE OBJECTS: a random-vector probe
//   generalises far past matrices
// (it is exactly the idea behind Wiedemann in 03 -
//   SparseDeterminant.cpp).
// Prefer a random FIELD element over 0/1 when the identity has degree
//    $> 1$ ; the failure bound is
// then  $d / \text{MOD}$  rather than  $1/2$  per round.
```

6.3.7 MinPlusMatrixPower

TROPICAL (MIN-PLUS) MATRIX POWER: replace (+, *) by (min, +) and matrix exponentiation answers

```
// "cheapest walk using EXACTLY k edges" in  $O(n^3 \log k)$ . The min-plus
//   semiring has additive
// identity llnf and multiplicative identity 0, and it is associative
//   - which is all matrix
// exponentiation ever needed. Everything below works verbatim for
//   MAX-PLUS by flipping min to max
// and llnf to -llnf.
// REACH FOR IT when the edge count is a hard constraint and k is
//   large: "shortest path with
// exactly k edges", "cheapest k-step schedule", periodic/cyclic
//   transport problems, and DPs whose
// transition is the same min-plus matrix every layer (n  $\leq 100$ -200, k
//   up to  $1e18$ ).
// If k is small, just do k layers of DP -  $O(k * n^2)$  beats  $O(n^3 \log
//   k)$  whenever  $k < n \log k$ .
typedef vector<vector<ll>> TMat;
```

```
TMat tmul(const TMat& a, const TMat& b) {
    int n = a.size(), m = b[0].size(), p = b.size();
    TMat c(n, vector<ll>(m, llnf));
    for (int i = 0; i < n; i++)
        for (int t = 0; t < p; t++) {
            if (a[i][t] >= llnf) continue; //
            // skip, and avoid overflow
            for (int j = 0; j < m; j++)
                if (b[t][j] < llnf) c[i][j] = min(c[i][j], a[i][t] +
            // b[t][j]);
        }
    return c;
}

TMat teye(int n) { //
    // min-plus identity
    TMat r(n, vector<ll>(n, llnf));
    for (int i = 0; i < n; i++) r[i][i] = 0;
    return r;
}

TMat tpow(TMat a, ll k) { //
    // exactly k edges
    TMat r = teye(a.size());
```

```

    for (k > 0; k >= 1, a = tmul(a, a)) if (k & 1) r = tmul(r, a);
    return r;
}
// SET-UP: a[i][j] = weight of edge i->j, llinf when there is no edge.
//   ↳ Then tpow(a, k)[u][v] is the
//   ↳ minimum total weight over walks u->v with exactly k edges (llinf if
//   ↳ none exists).
// AT MOST k EDGES: add a zero-weight self-loop at every vertex
//   ↳ (a[i][i] = min(a[i][i], 0)).
// COUNTING instead of minimising: the same code over (+, *) is 03 -
//   ↳ Matrices/01, and over
// (or, and) with bitsets it answers reachability in exactly k steps in
//   ↳  $O(n^3 / 64)$ .
// LONGEST path with exactly k edges: max-plus, but ONLY if the graph
//   ↳ is a DAG or k is bounded;
//   ↳ otherwise a positive cycle makes it unbounded.
// OVERFLOW is the standard bug: guard both operands against llinf
//   ↳ before adding, as above, or
// llinf + llinf silently wraps negative and poisons the whole matrix.
// FASTER SPECIAL CASES
// - if the matrix is CONVEX in the Monge sense, min-plus products
//   ↳ drop to  $O(n^2)$  via SMAWK, so
//   ↳ the whole power costs  $O(n^2 \log k)$  (see 08 - DP/01/11 -
//   ↳ MinPlusConvolution.cpp).
// - if you only need one row (a fixed source), keep a VECTOR and
//   ↳ multiply it by the precomputed
//   ↳ powers  $a^{(2^i)}$ :  $O(n^2 \log k)$  total instead of  $n^3 \log k$ .
// - for eventually-periodic behaviour, the tropical eigenvalue (the
//   ↳ minimum mean cycle, Karp's
//   ↳ algorithm) gives the per-step growth rate and lets you skip the
//   ↳ huge exponent entirely.

```

6.4 Linear Algebra

6.4.1 LinearXorBasis

XOR basis (vector space over GF(2)). insert / query $O(B)$.

```

// span size =  $2^{sz}$ . Use MERGE to keep a basis in each segment-tree
//   ↳ node -> max-xor over a RANGE.
struct Basis {
    static const int B = 62; // 60 silently drops bits 61-62 of an
    // ll up to 9e18
    ll b[B + 1] = {}; // b[i] has
    // leading bit i
    int sz = 0;

    bool insert(ll x) { // false if x
        // already in the span
        for (int i = B; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { b[i] = x, sz++; return true; }
            x ^= b[i];
        }
        return false;
    }
    bool contains(ll x) const {
        for (int i = B; i >= 0; i--) if (x >> i & 1) { if (!b[i])
        // return false; x ^= b[i]; }
        return true;
    }
    ll maxXor(ll r = 0) const { for (int i = B; i >= 0; i--) r = max(r,
    // r ^ b[i]); return r; }
    ll minXor(ll r = 0) const { for (int i = B; i >= 0; i--) if (r >> i
    // & 1) r ^= b[i]; return r; }
    void reduce() { // reduced row
    // echelon form
        for (int i = B; i >= 0; i--) if (b[i])
            for (int j = i - 1; j >= 0; j--) if (b[i] >> j & 1) b[i] ^=
        // b[j];
    }
    ll kth(ll k) { // k-th
    // SMALLEST value in the span, 0-indexed
        reduce(); // kth(0) == 0;
    // -1 if k >=  $2^{sz}$ 
        ll r = 0;
        for (int i = 0; i <= B; i++) if (b[i]) { if (k & 1) r ^= b[i];
        // k >>= 1; }
        return k ? -1 : r;
    }
    ll spanSize() const { return sz >= 63 ? -1 : 1LL << sz; }
    friend Basis merge(Basis a, const Basis& c) {
        for (int i = 0; i <= B; i++) if (c.b[i]) a.insert(c.b[i]);
        return a;
    }
};

```

6.4.2 GaussianElimination

Gaussian elimination mod a PRIME. Returns rank; solves $Ax = b$; also det and inverse.

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp) for the modular
//   ↳ routines.
//  $O(n^2 m)$ . Pass the augmented matrix (n rows, m+1 cols) to solve().
int gaussMod(vector<vector<Mint>>& a, int m) { // reduce
    // first m columns to RREF
    int n = a.size(), rank = 0;
    for (int col = 0; col < m && rank < n; col++) {
        int piv = -1;
        for (int i = rank; i < n; i++) if (a[i][col].v) { piv = i;
        // break; }
        if (piv < 0) continue;
        swap(a[rank], a[piv]);
        Mint iv = a[rank][col].inv();
        for (auto& x : a[rank]) x *= iv;
        for (int i = 0; i < n; i++) if (i != rank && a[i][col].v) {
            Mint f = a[i][col];
            for (size_t j = 0; j < a[i].size(); j++) a[i][j] -= f *
        // a[rank][j];
        }
        rank++;
    }
    return rank;
}
// Solve  $Ax = b$ . Returns {} if inconsistent, else one solution (free
//   ↳ vars = 0).
// 'free' receives the indices of the free variables (solution space
//   ↳ has  $MOD^{|free|}$  points).
vector<Mint> solveLinear(vector<vector<Mint>> a, vector<Mint> b,
    // vector<int>* freeIdx = nullptr) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) a[i].push_back(b[i]);
    int rank = gaussMod(a, m);
    for (int i = rank; i < n; i++) if (a[i][m].v) return {}; // 0
    // = nonzero
    vector<Mint> x(m, 0);
    int r = 0;
    for (int c = 0; c < m; c++) { // must run over
    // EVERY column: stopping at
        if (r < rank && a[r][c].v) x[c] = a[r][m], r++; // r == rank
    // drops the free columns
        else if (freeIdx->push_back(c); // that come
    // after the last pivot
        }
    return x;
}
Mint determinant(vector<vector<Mint>> a) { //  $O(n^3)$ ,
    // destroys a
    int n = a.size();
    Mint det = 1;
    for (int col = 0; col < n; col++) {
        int piv = -1;
        for (int i = col; i < n; i++) if (a[i][col].v) { piv = i;
        // break; }
        if (piv < 0) return 0;
        if (piv != col) swap(a[col], a[piv]), det = Mint(0) - det;
        det *= a[col][col];
        Mint iv = a[col][col].inv();
        for (int i = col + 1; i < n; i++) if (a[i][col].v) {
            Mint f = a[i][col] * iv;
            for (int j = col; j < n; j++) a[i][j] -= f * a[col][j];
        }
    }
    return det;
}
vector<vector<Mint>> inverse(vector<vector<Mint>> a) { // {} if
    // singular
    int n = a.size();
    for (int i = 0; i < n; i++) { a[i].resize(2 * n, 0); a[i][n + i] =
    // 1; }
    if (gaussMod(a, n) < n) return {};
    vector<vector<Mint>> r(n, vector<Mint>(n));
    for (int i = 0; i < n; i++) for (int j = 0; j < n; j++) r[i][j] =
    // a[i][n + j];
    return r;
}
// MATRIX-TREE: #spanning trees = determinant of the Laplacian (deg on
//   ↳ diagonal, -1 per edge)
// with any one row AND the matching column deleted.

```

6.4.3 GaussXorAndReal

XOR-equation systems over GF(2) with bitset - $O(n \cdot m / 64)$. n equations, m variables,

```
// row[i][m] holds the RHS. Returns rank, or -1 if inconsistent.
template<int M>
int gaussXor(vector<bitset<M + 1>> &a, int m, bitset<M>* sol = nullptr)
{
    int n = a.size(), rank = 0;
    vector<int> where(m, -1);
    for (int col = 0; col < m && rank < n; col++) {
        int piv = -1;
        for (int i = rank; i < n; i++) if (a[i][col]) { piv = i; break; }
        if (piv < 0) continue;
        swap(a[rank], a[piv]);
        for (int i = 0; i < n; i++) if (i != rank && a[i][col]) a[i] ^= a[rank];
        where[col] = rank++;
    }
    for (int i = rank; i < n; i++) if (a[i][m]) return -1;
    if (sol) { sol->reset(); for (int c = 0; c < m; c++) if (where[c]
    <= 0) (*sol)[c] = a[where[c]][m]; }
    return rank; // #solutions =
    <= 2^(m - rank)
}

// USES: "light switch" puzzles, xor-equation counting, rank of a set
// of vectors over GF(2),
// linear independence of xor values (dense sibling of the XOR
// basis).

// ---- Gaussian elimination over the reals, with partial pivoting ----
// Returns rank; a is n x (m+1) augmented. Solution in x (free vars =
// 0). eps-based.
int gaussReal(vector<vector<ld>> &a, int m, vector<ld>& x) {
    int n = a.size(), rank = 0;
    vector<int> where(m, -1);
    for (int col = 0; col < m && rank < n; col++) {
        int piv = rank;
        for (int i = rank; i < n; i++) if (fabs(a[i][col]) >
        <= fabs(a[piv][col])) piv = i;
        if (fabs(a[piv][col]) < 1e-12) continue;
        swap(a[rank], a[piv]);
        for (int i = 0; i < n; i++) if (i != rank) {
            ld f = a[i][col] / a[rank][col];
            for (int j = col; j <= m; j++) a[i][j] -= f * a[rank][j];
        }
        where[col] = rank++;
    }
    x.assign(m, 0);
    for (int c = 0; c < m; c++) if (where[c] >= 0) x[c] =
    <= a[where[c]][m] / a[where[c]][c];
    return rank;
}

// The classic use is EXPECTED-VALUE DP with cyclic state dependencies:
// write one equation per
// state, E[s] = 1 + sum p(s->t) E[t], and solve. n <= ~500 fits O(n^3).
```

6.4.4 WeightedXorBasis

WEIGHTED XOR BASIS - keep the MAXIMUM-WEIGHT independent set of vectors over GF(2). $O(B)$ per

```
// insert, online, no sorting and no rollback needed.
// The vectors of a matroid's ground set with a "keep the heaviest"
// greedy: linear independence is
// greedy, so the greedy is optimal. Standard basis insertion keeps
// whichever vector arrives
// first; this one keeps the heavier and pushes the lighter down to be
// re-inserted, which is the
// same as running Kruskal's exchange argument implicitly.
// REACH FOR IT: "pick a maximum-weight subset of vectors that stays
// linearly independent",
// maximum spanning forest over GF(2) constraints", cycle-space
// problems where each edge carries
// both an xor value and a cost, and any "add items online, report the
// best independent set now".
struct WBasis {
    static const int B = 62;
    ll b[B + 1] = {}; // b[i] has
    // leading bit i
    ll w[B + 1] = {}; // weight of
    // the vector stored in b[i]
    int sz = 0;

    void insert(ll x, ll wt) {
        for (int i = B; i >= 0; i--) if (x >> i & 1) {
            if (!b[i]) { b[i] = x, w[i] = wt, sz++; return; }
            if (w[i] < wt) swap(b[i], x), swap(w[i], wt); // keep the
            // heavier one at this pivot
        }
    }
};
```

```
        x ^= b[i];
    }
    ll totalWeight() const { ll s = 0; for (int i = 0; i <= B; i++) s
    <= += w[i]; return s; }
    ll maxXor(ll r = 0) const { for (int i = B; i >= 0; i--) r = max(r,
    <= r ^ b[i]); return r; }
};

// MINIMUM-weight independent set: negate the weights (or compare with
// > instead of <).
// THE TRAP: you may NOT just sort by weight and insert greedily into a
// plain basis if the vectors
// arrive online - but if they are all known up front, sorting
// descending and using the ordinary
// Basis::insert gives exactly the same answer and is simpler. Use this
// version when insertions are
// interleaved with queries.
// MATROID INTERSECTION generalises this to "independent in TWO
// matroids at once" (e.g. linearly
// independent AND a forest); that needs augmenting paths in an
// exchange graph, not a greedy.
// LINEAR MATROID over a general field: same greedy with Gaussian
// row-reduction instead of xor.
// TYPICAL ENCODING TRICK: to force "at most one vector per group",
// append group-identifier bits
// above the value bits (a 128-bit word via __int128 gives you 62 value
// bits plus 66 tag bits).
// That turns "spanning-forest + xor" constraints into one basis insert
// per edge.
```

6.4.5 XorSystemExtremes

LEXICOGRAPHICALLY EXTREME SOLUTION of a GF(2) system, on top of 03 - GaussXorAndReal.cpp.

```
// gaussXor returns SOME solution (free variables = 0), which is the
// lexicographically SMALLEST one
// when variables are compared in index order x[0], x[1], ...
// Problems that ask for the
// lexicographically LARGEST assignment, or that weight the variables,
// need this wrapper.
// THE TRICK: reverse the variable order before eliminating and reverse
// the answer back. Reduced
// row echelon form always puts a pivot as early as possible, so "free
// variables = 0" minimises the
// reversed vector, i.e. maximises the original.  $O(n \cdot m / 64)$ , same as
// one elimination.
// Careful: "lexicographically largest" here means as a SEQUENCE x[0] >
// x[1] > ...; if the problem
// means "largest as a binary NUMBER with x[0] the most significant
// bit", that is the same thing.
```

```
template<int M>
int gaussXorLexMax(vector<bitset<M + 1>> &a, int m, bitset<M>& sol) {
    int n = a.size();
    for (int i = 0; i < n; i++) { // reverse the
    // m variable columns
        bitset<M + 1> t;
        for (int j = 0; j < m; j++) t[j] = a[i][m - 1 - j];
        t[m] = a[i][m];
        a[i] = t;
    }
    bitset<M> s;
    int rank = gaussXor<M>(a, m, &s);
    if (rank < 0) return -1; // inconsistent
    for (int j = 0; j < m; j++) sol[j] = s[m - 1 - j]; // and
    // reverse the answer back
    return rank; // #solutions
    <= 2^(m - rank)
}

// PRIORITY ORDER other than "by index": permute the columns by the
// priority you want (most
// important variable first), solve, permute back. Only the ORDER of
// the columns matters.
// MAXIMISE A LINEAR OBJECTIVE sum w_i x_i over the solution set: that
// is NOT lexicographic and is
// NP-hard in general (max-XOR-SAT). It is easy only when all w_i >= 0
// and you may take each pivot
// greedily - i.e. when the objective is itself lexicographic.
// MAXIMISE THE XOR of the solution vector viewed as a number: build a
// basis of the NULL SPACE
// (one vector per free variable: set that free variable to 1,
// back-substitute the pivots), then
// run Basis::maxXor starting from the particular solution (04 - Linear
// Algebra/01).
// COUNTING: the solution set is a coset, so every count is  $2^{(m -
// rank)}$  or 0; "number of solutions
// with x[i] = 1" is either 0, the full count, or half of it, depending
// on whether e_i is forced.
```

6.4.6 GF2MatrixInverse

INVERSE OF A MATRIX OVER GF(2) with bitsets - $O(n^3 / 64)$. $n = 2000$ runs in well under a second,

```
// where a Mint-based inverse at the same size is 8e9 operations and
// ↳ hopeless.
// Returns {} if the matrix is singular. Rows are bitset<M> with bit j
// ↳ = entry (i, j).
// REACH FOR IT for xor-system problems that need the FULL inverse
// ↳ rather than one solution:
// - "for each edge, is it in EVERY / SOME spanning structure" style
// ↳ queries, which reduce to a
// single entry of the inverse (CF 736D: after inverting the
// ↳ incidence-style matrix, entry
// inv[b][a] decides whether edge (a, b) is forced);
// - answering many right-hand sides against one fixed matrix ( $x =$ 
// ↳  $A^{-1}b$  in  $O(n^2 / 64)$  each);
// - determinant over GF(2) ( $= 1$  iff invertible), rank, and the null
// ↳ space in one pass.
template <int M>
vector<bitset<M>> inverseGF2(vector<bitset<M>> a) {
    int n = a.size();
    vector<bitset<M>> inv(n);
    for (int i = 0; i < n; i++) inv[i][i] = 1;
    for (int i = 0; i < n; i++) { // forward
        ↳ elimination
        int piv = -1;
        for (int j = i; j < n; j++) if (a[j][i]) { piv = j; break; }
        if (piv < 0) return {}; // singular
        swap(a[i], a[piv]), swap(inv[i], inv[piv]);
        for (int j = i + 1; j < n; j++) if (a[j][i]) a[j] ^= a[i],
        ↳ inv[j] ^= inv[i];
    }
    for (int i = n - 1; i >= 0; i--) // back
        ↳ substitution
        for (int j = i + 1; j < n; j++) if (a[i][j]) inv[i] ^= inv[j];
    return inv;
}
// NO PIVOT VALUES to divide by: over GF(2) every non-zero entry is 1,
// ↳ so elimination is pure xor
// and there is no numerical anything. That is why this is the cleanest
// ↳ Gaussian elimination there
// is - and why you should reduce to GF(2) whenever a problem lets you.
// DETERMINANT over GF(2) is 1 iff the forward pass finds a pivot in
// ↳ every column; equivalently
// rank == n. For the rank alone, or a solution of  $Ax = b$ , use
// ↳ gaussXor (04 - Linear Algebra/03)
// which is the same loop with an augmented column.
// NULL SPACE: run gaussXor to RREF; for each free column c, the vector
// ↳ with 1 at c and the pivot
// entries copied from column c is a basis vector of the kernel. Its
// ↳ size is  $n - \text{rank}$ , and the
// solution set of  $Ax = b$  is (any particular solution) + (that span).
// PRODUCT over GF(2) in  $O(n^3 / 64)$ : transpose b, then  $c[i][j] =$ 
// ↳  $((a[i] \& b[j]).count() \& 1)$ .
// Or, faster,  $c[i] = \text{xor of } a\text{'s rows selected by } b - \text{that is } O(n^2 /$ 
// ↳  $64)$  per row with no popcounts.
// M IS A COMPILE-TIME BOUND: bitset<M> must be at least n wide; keep M
// ↳ a multiple of 64.
```

6.5 Convolution

6.5.1 FFT

FFT over complex doubles. `conv()` is exact while $|\text{result coefficient}| < \sim 1e15$.

```
// Prefer NTT when the answer is taken modulo an NTT prime; use convMod
// ↳ for arbitrary moduli.
typedef complex<double> C;
void fft(vector<C>& a) {
    int n = a.size(), L = 31 - __builtin_clz(n);
    static vector<complex<long double>> R(2, 1);
    static vector<C> rt(2, 1);
    for (static int k = 2; k < n; k *= 2) {
        R.resize(n), rt.resize(n);
        auto x = polar(1.0/L, acos(-1.0/L) / k);
        for (int i = k; i < 2 * k; i++) rt[i] = R[i] * x * R[i / 2]
        ↳ * x : R[i / 2];
    }
    vector<int> rev(n);
    for (int i = 0; i < n; i++) rev[i] = (rev[i / 2] | (i & 1) << L) /
    ↳ 2;
    for (int i = 0; i < n; i++) if (i < rev[i]) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k)
            for (int j = 0; j < k; j++) {
                C z = rt[j + k] * a[i + j + k];
                a[i + j + k] = a[i + j] - z;
                a[i + j] += z;
            }
}
```

```
}
}
// Real convolution. Result is rounded to the nearest integer.
vector<ll> conv(const vector<ll>& a, const vector<ll>& b) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <= 1;
    vector<C> in(n), out(n);
    for (size_t i = 0; i < a.size(); i++) in[i].real((double)a[i]);
    for (size_t i = 0; i < b.size(); i++) in[i].imag((double)b[i]);
    fft(in);
    for (C& x : in) x *= x; //  $(a+bi)^2 =$ 
    ↳  $a^2 - b^2 + 2abi$ 
    for (int i = 0; i < n; i++) out[i] = in[-i & (n - 1)] - conj(in[i]);
    fft(out);
    vector<ll> res(rs);
    for (int i = 0; i < rs; i++) res[i] = llround(imag(out[i]) / (4 *
    ↳ n));
    return res;
}
// Convolution modulo an ARBITRARY M, splitting coefficients at sqrt(M)
// ↳ to keep precision.
vector<ll> convMod(const vector<ll>& a, const vector<ll>& b, ll M) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, B = 32 - __builtin_clz(rs), n = 1
    ↳ << B;
    ll cut = (ll)sqrtl((ld)M);
    vector<C> L(n), R(n), os(n), ol(n);
    for (size_t i = 0; i < a.size(); i++) L[i] = C((double)(a[i] /
    ↳ cut), (double)(a[i] % cut));
    for (size_t i = 0; i < b.size(); i++) R[i] = C((double)(b[i] /
    ↳ cut), (double)(b[i] % cut));
    fft(L), fft(R);
    for (int i = 0; i < n; i++) {
        int j = -i & (n - 1);
        ol[j] = (L[i] + conj(L[j])) * R[i] / (2.0 * n);
        os[j] = (L[i] - conj(L[j])) * R[i] / (2.0 * n) / C(0, 1);
    }
    fft(ol), fft(os);
    vector<ll> res(rs);
    for (int i = 0; i < rs; i++) {
        ll av = llround(real(ol[i])), cv = llround(imag(os[i]));
        ll bv = llround(imag(ol[i])) + llround(real(os[i]));
        res[i] = ((av % M * cut + bv) % M * cut + cv) % M;
    }
    return res;
}
}
```

6.5.2 NTT

NTT modulo 998244353 ($= 119 \cdot 2^{23} + 1$, primitive root 3). $O(n \log n)$.

```
// multiply() is the entry point. MAX LENGTH  $2^{23}$  for 998244353 (see
// ↳ the assert in ntt()).
// For an ARBITRARY modulus: run this under three of the primes below
// ↳ and Garner them together
// (06 - Garner.cpp is written for exactly that), or split the
// ↳ coefficients at sqrt(M) and use
// convMod in 01 - FFT.cpp.
const ll NMOD = 998244353, NROOT = 3;
ll npow(ll a, ll e, ll m = NMOD) { ll r = 1; a %= m; for (; e; e >>= 1,
// ↳ a = a * a % m) if (e & 1) r = r * a % m; return r; }
```

```
void ntt(vector<ll>& a, bool inv) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        assert((NMOD - 1) % len == 0); // 998244352 =  $119 \cdot 2^{23}$ : len >
        ↳  $2^{23}$  truncates the division
        ll w = npow(NROOT, (NMOD - 1) / len); // and returns GARBAGE
        ↳ with no other symptom
        if (inv) w = npow(w, NMOD - 2);
        for (int i = 0; i < n; i += len) {
            ll cur = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] * cur % NMOD;
                a[i + j] = (u + v) % NMOD;
                a[i + j + len / 2] = (u - v + NMOD) % NMOD;
                cur = cur * w % NMOD;
            }
        }
        if (inv) { ll invn = npow(n, NMOD - 2); for (ll& x : a) x = x *
        ↳ invn % NMOD; }
    }
}
```

```

}
vector<ll> multiply(vector<ll> a, vector<ll> b) {
    if (a.empty() || b.empty()) return {};
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <= 1;
    a.resize(n), b.resize(n);
    ntt(a, false), ntt(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % NMOD;
    ntt(a, true);
    a.resize(rs);
    return a;
}
// ARBITRARY MODULUS: split each coefficient as a = hi*2^15 + lo and
//   ↳ combine three products
// (or run NTT under three NTT-friendly primes and CRT). Values must
//   ↳ stay below ~1e18.
// COMMON NTT PRIMES: 998244353 (g=3), 167772161 (g=3), 469762049
//   ↳ (g=3), 1004535809 (g=3).
// USES: polynomial multiplication; "count pairs/triples with a given
//   ↳ sum"; convolving
// distributions; string matching with wildcards; sum of C(a_i, k) over
//   ↳ all i for every k.

```

6.5.3 FWHT

Fast Walsh-Hadamard Transform (FWHT) - Time: $O(N \log N)$

// Computes Bitwise Convolutions: XOR, AND, OR for array of size $N = 2^k$

```

struct FWHT {
    enum Type { XOR, AND, OR };

    static void transform(vector<ll> &a, bool invert, Type type = XOR) {
        int n = a.size();
        for (int len = 1; 2 * len <= n; len <= 1) {
            for (int i = 0; i < n; i += 2 * len) {
                for (int j = 0; j < len; j++) {
                    ll u = a[i + j], v = a[i + len + j];
                    if (type == XOR) {
                        a[i + j] = u + v;
                        a[i + len + j] = u - v;
                    } else if (type == AND) {
                        a[i + j] = invert ? u - v : u + v;
                    } else if (type == OR) {
                        a[i + len + j] = invert ? v - u : u + v;
                    }
                }
            }
        }
        if (type == XOR && invert) {
            for (ll &x : a) x /= n;
        }
    }

    // Computes  $C[k] = \sum_i \{i \text{ op } j = k\} (A[i] * B[j])$  where op in {XOR,
    //   ↳ AND, OR}
    static vector<ll> multiply(vector<ll> a, vector<ll> b, Type type =
    //   ↳ XOR) {
        int n = 1;
        while (n < max(a.size(), b.size())) n <= 1;
        a.resize(n); b.resize(n);

        transform(a, false, type);
        transform(b, false, type);
        for (int i = 0; i < n; i++) a[i] *= b[i];
        transform(a, true, type);
        return a;
    }
};

```

6.5.4 Polynomial

FORMAL POWER SERIES over $NMOD = 998244353$, on top of multiply() from 02-NTT.cpp.

// Every routine works "mod x^k ". inv/log/exp/sqrt/pow are all Newton
 // ↳ iteration: each step
 // doubles the number of correct coefficients, so the cost is the cost
 // ↳ of the last multiply.

typedef vector<ll> Poly;

```

Poly cut(Poly a, int k) { a.resize(k, 0); return a; }
Poly operator+(Poly a, const Poly& b) {
    a.resize(max(a.size(), b.size()), 0);
    for (size_t i = 0; i < b.size(); i++) a[i] = (a[i] + b[i]) % NMOD;
    return a;
}
Poly operator-(Poly a, const Poly& b) {
    a.resize(max(a.size(), b.size()), 0);
    for (size_t i = 0; i < b.size(); i++) a[i] = (a[i] - b[i] + NMOD) %
    //   ↳ NMOD;

```

```

    return a;
}
Poly operator*(const Poly& a, ll c) { Poly r = a; for (ll& x : r) x = x
    //   ↳ * (c % NMOD) % NMOD; return r; }

Poly deriv(const Poly& a) {
    if (a.size() <= 1) return {};
    Poly r(a.size() - 1);
    for (size_t i = 1; i < a.size(); i++) r[i - 1] = a[i] * i % NMOD;
    return r;
}
Poly integ(const Poly& a) {
    Poly r(a.size() + 1, 0);
    for (size_t i = 0; i < a.size(); i++) r[i + 1] = a[i] * npow(i + 1,
    //   ↳ NMOD - 2) % NMOD;
    return r;
}
// 1/a mod  $x^k$ . Requires  $a[0] \neq 0$ .  $g_{-2m} = g_m * (2 - a * g_m)$ 
Poly inv(const Poly& a, int k) {
    Poly g = {npow(a[0], NMOD - 2)};
    for (int m = 1; m < k; m <= 1) {
        Poly t = multiply(cut(a, 2 * m), g);
        t = cut(t, 2 * m);
        for (ll& x : t) x = (NMOD - x) % NMOD;
        t[0] = (t[0] + 2) % NMOD;
        g = cut(multiply(g, t), 2 * m);
    }
    return cut(g, k);
}
// log(a) mod  $x^k$ . Requires  $a[0] = 1$ .  $\log a = \text{integral}(a' / a)$ 
Poly log_(const Poly& a, int k) { return
    //   ↳ cut(integ(cut(multiply(deriv(a), inv(a, k)), k - 1)), k); }
// exp(a) mod  $x^k$ . Requires  $a[0] = 0$ .  $g_{-2m} = g_m * (1 + a - \log$ 
    //   ↳  $g_m)$ 
Poly exp_(const Poly& a, int k) {
    Poly g = {};
    for (int m = 1; m < k; m <= 1) {
        Poly t = cut(a, 2 * m) - log_(cut(g, 2 * m), 2 * m);
        t[0] = (t[0] + 1) % NMOD;
        g = cut(multiply(g, t), 2 * m);
    }
    return cut(g, k);
}
// sqrt(a) mod  $x^k$ , for  $a[0] = 1$  (general  $a[0]$  needs a modular square
    //   ↳ root of  $a[0]$ ).
Poly sqrt_(const Poly& a, int k) {
    Poly g = {};
    ll i2 = npow(2, NMOD - 2);
    for (int m = 1; m < k; m <= 1) {
        Poly t = cut(a, 2 * m) + multiply(g, g);
        t = cut(t, 2 * m);
        g = cut(multiply(t, inv(cut(g, 2 * m) * 2, 2 * m)), 2 * m);
    }
    return cut(g, k);
}
//  $a^e \text{ mod } x^k$ , for  $a[0] = 1$ :  $\exp(e * \log a)$ . For  $a[0] \neq 1$  factor
    //   ↳ out  $c * x^d$  first.
Poly pow_(const Poly& a, ll e, int k) { return exp_(log_(a, k) * (e %
    //   ↳ NMOD), k); }

// Polynomial DIVISION with remainder:  $a = q * b + r$ ,  $\deg r < \deg b$ .  $O(n$ 
    //   ↳  $\log n)$ 
pair<Poly, Poly> divmod(Poly a, Poly b) {
    while (a.size() > 1 && !a.back()) a.pop_back(); // trim BOTH, or
    //   ↳ the reversal is wrong
    while (b.size() > 1 && !b.back()) b.pop_back();
    int n = a.size(), m = b.size();
    if (n < m) return {{}, a};
    Poly ra(a.rbegin(), a.rend()), rb(b.rbegin(), b.rend());
    Poly q = cut(multiply(cut(ra, n - m + 1), inv(cut(rb, n - m + 1), n
    //   ↳ - m + 1)), n - m + 1);
    reverse(all(q));
    Poly r = cut(a - multiply(q, b), max(1, m - 1)); // cut AFTER
    //   ↳ subtracting, not before
    return {q, r};
}
// USES: counting with generating functions (exp of a "connected"
    //   ↳ series = all structures),
// partition numbers via  $\prod 1/(1-x^i) = \exp(\text{sum } \dots)$ , Catalan-style
    //   ↳ algebraic equations via
// sqrt, k-th power of a GF, linear recurrence coefficients via inv,
    //   ↳ and Lagrange inversion.

```

6.5.5 CyclicConvolution

CYCLIC CONVOLUTION and CYCLIC CORRELATION of two length- n arrays, $O(n \log n)$.

```
// Needs: multiply (02 - NTT.cpp) [or swap in conv() from 01 -
// FFT.cpp].
// Cyclic = multiplication in the ring modulo  $x^n - 1$ : indices wrap
// around instead of growing.
// REACH FOR IT whenever the index set is a CIRCLE rather than a line:
// necklaces and rotations,
// "for every rotation  $r$ , the score of aligning  $a$  against  $b$  rotated by
//  $r$ ", counting pairs  $(i, j)$ 
// with  $(i + j) \bmod n = k$ , and any DP on residues modulo  $n$ .
typedef vector<ll> Poly;
Poly cyclicConv(const Poly& a, const Poly& b, int n) { //  $c[k] =$ 
    sum_{(i+j) mod n == k} a[i]b[j]
    Poly p = multiply(a, b), c(n, 0);
    for (size_t i = 0; i < p.size(); i++) c[i % n] = (c[i % n] + p[i])
    % NMOD;
    return c;
}
//  $c[k] = \text{sum}_i a[i] * b[(i + k) \bmod n]$  -- "slide  $b$  by  $k$  over  $a$  and
// take the dot product".
// This is the one you actually want for rotations. Reverse  $a$ 
// cyclically, then convolve.
Poly cyclicCorr(const Poly& a, const Poly& b, int n) {
    Poly ar(n);
    for (int i = 0; i < n; i++) ar[i] = a[(n - i) % n];
    return cyclicConv(ar, b, n);
}
// NEGACYCLIC convolution (modulo  $x^n + 1$ , "wrap with a sign flip"):
// same fold but subtract the
// wrapped part,  $c[i \% n] += (i / n \text{ odd} ? -p[i] : p[i])$ . Shows up in
// ring-LWE style problems and in
// the Schoenhage-Strassen recursion.
// FASTER when  $n \mid (NMOD - 1)$ : a length- $n$  NTT IS the cyclic convolution
// transform, multiply
// pointwise, transform back, with no padding to a power of two and no
// fold. When  $n$  is arbitrary,
// Bluestein (06 - ChirpZ.cpp) gives you the length- $n$  DFT anyway.
// USES
// ROTATION SCORING: given two binary necklaces, the number of
// positions where both are 1 under
// rotation  $r$  is exactly cyclicCorr(a, b)[r] - one call answers all  $n$ 
// rotations.
// RESIDUE DP: "number of pairs whose sum is congruent to  $k \bmod n$ " for
// all  $k$  at once.
// CIRCULAR STRING MATCHING with wildcards: encode as sums of products
// (see the wildcard trick in
// 07 - Strings/01 - String Matching/04) and fold cyclically.
// A circulant matrix times a vector IS a cyclic convolution, so this
// is also how you apply  $a$ 
// circulant in  $O(n \log n)$  instead of  $O(n^2)$ .
```

6.5.6 ChirpZ

CHIRP-Z / BLUESTEIN: evaluate a polynomial at a GEOMETRIC progression of points,

```
// Needs: multiply, npow (02 - NTT.cpp); Poly / cut (04 -
// Polynomial.cpp).
//  $P(1), P(z), P(z^2), \dots, P(z^{(m-1)})$  in  $O((n + m) \log(n + m))$ ,
// where  $n = \deg P$ . Multipoint evaluation at arbitrary points is  $O(n$ 
//  $\log^2 n)$ ; this is a full log
// factor cheaper and much shorter, and it covers most of the cases
// that actually come up.
// PRECONDITION:  $z$  must be INVERTIBLE mod  $NMOD$  ( $z \neq 0$ ). Any  $m$  and  $n$ 
// are fine - no power-of-two.
// REACH FOR IT for: a DFT of ARBITRARY length (set  $z =$  a primitive
//  $m$ -th root of unity - this is
// how you do a length-7 or length-1000000 transform with an NTT that
// only likes powers of two);
// "evaluate  $f$  at  $1, r, r^2, \dots$ " which is what  $q$ -analogue and
// geometric-weight problems ask for;
// and cyclic convolution of a non-power-of-two length.
// IDENTITY:  $i * j = C(i + j, 2) - C(i, 2) - C(j, 2)$ , which turns the
// exponent  $z^{(i * j)}$  into a product
// of three one-index factors, so the sum becomes a convolution
// ("chirp" = the  $z^C(k, 2)$  sequence).
Poly chirpZ(const Poly& a, ll z, int m) {
    int n = a.size() - 1; // deg P
    if (m <= 0) return {};
    ll iz = npow(z, NMOD - 2);
    auto tri = [&](ll k) { return k * (k - 1) / 2 % (NMOD - 1); };
    //  $C(k, 2) \bmod (p-1)$ 
    Poly f(n + 1), g(n + m);
    for (int j = 0; j <= n; j++) f[n - j] = a[j] * npow(iz, tri(j)) %
    NMOD; // reversed
    for (int t = 0; t < n + m; t++) g[t] = npow(z, tri(t));
    Poly h = multiply(f, g);
    Poly r(m);
```

```
for (int k = 0; k < m; k++) r[k] = h[n + k] * npow(iz, tri(k)) %
NMOD;
return r;
}
// ARBITRARY-LENGTH DFT: with  $w$  a primitive  $m$ -th root of unity mod  $NMOD$ 
// (needs  $m \mid NMOD - 1$ ),
// chirpZ(a, w, m) is the length- $m$  DFT of  $a$ ; the inverse is chirpZ(.,
//  $w^{-1}, m)$  scaled by  $1/m$ .
// That gives cyclic convolution of ANY length  $m$  in  $O(m \log m)$  without
// a fold.
// THE OTHER DIRECTION: given the  $m$  values  $P(z^k)$ , recovering  $P$  is a
// chirp- $z$  with  $z^{-1}$  followed by
// a rescale - the transform is its own inverse up to the chirp
// factors, exactly like a DFT.
// EVALUATING AT AN ARITHMETIC progression instead ( $P(a), P(a+d), \dots$ ):
// that is a Taylor SHIFT plus
// a scale, see 10 - PolyShiftAndComposition.cpp, also  $O(n \log n)$ .
// OVERFLOW:  $\text{tri}(k)$  is taken mod  $(NMOD - 1)$  by Fermat, so  $k$  may be as
// large as you like; without
// that reduction  $k * (k-1) / 2$  overflows around  $k = 4e9$ .
// RELATED: the same "split the exponent" trick underlies baby-step
// giant-step for discrete log and
// the  $O(\sqrt{n})$  evaluation of  $n! \bmod p$ .
```

6.5.7 RelaxedConvolution

ONLINE (RELAXED) CONVOLUTION. You feed $a[i]$ and $b[i]$ one index at a time and immediately get

```
// Needs: multiply (02 - NTT.cpp).
//  $c[i] = \text{sum}_{j+k=i} a[j] b[k]$ , BEFORE you have to supply  $a[i+1]$  -
// which is exactly what  $a$ 
// self-referential DP needs. Total  $O(n \log^2 n)$  for all  $n$  steps;  $\sim 1s$ 
// at  $n = 2e5$ .
// REACH FOR IT whenever  $f[i]$  depends on a convolution of  $f$  with itself
// (or with something  $f$ 
// determines) up to index  $i - 1$ :  $f[n] = \text{sum } f[k] f[n-1-k]$ 
// (Catalan-shaped DPs),  $f[n] = \text{sum}$ 
//  $g[k] f[n-k]$  where you only learn  $g[n]$  after computing  $f[n-1]$ ,
// probability DPs with a feedback
// loop, ABC315-H style. Divide and conquer over the recursion is the
// alternative and is usually
// harder to write; a plain  $O(n^2)$  double loop is the thing you are
// replacing.
// The blocks are scheduled so that the product of two length- $w$  blocks
// is done exactly when both
// halves are known, and every coefficient is covered exactly once.
struct Relaxed {
    vector<ll> a, b, c;
    int k = 0;
    Relaxed(int n) : a(n), b(n), c(2 * n, 0) {}
    // supply  $a[i] = x$  and  $b[i] = y$  ( $i$  must equal the current index);
    // returns  $c[i]$ .
    ll extend(ll x, ll y) {
        a[k] = x % NMOD, b[k] = y % NMOD;
        int s = k + 2;
        for (int w = 1; s % w == 0 && w < s; w <= 1) {
            for (int t = 0; t < 2; t++) {
                if (t == 0 || w * 2 != s) { // the
                    vector<ll> f(a.begin() + w - 1, a.begin() + 2 * w -
                    1);
                    vector<ll> g(b.begin() + k - w + 1, b.begin() + k +
                    1);
                    f = multiply(f, g);
                    for (size_t i = 0; i < f.size() && k + i <
                    c.size(); i++)
                        c[k + i] = (c[k + i] + f[i]) % NMOD;
                    swap(a, b); // and again,
                    // so a/b are restored
                }
            }
            return c[k++];
        }
    }
};
// SELF-CONVOLUTION shorthand: pass the same value twice, extend(f[i],
// f[i]).
// TYPICAL SHAPE
// Relaxed R(n); f[0] = 1;
// for (i = 1; i <= n; i++) { ll t = R.extend(f[i-1], f[i-1]); f[i] =
// /* uses t */; }
// Note the OFF-BY-ONE: after the  $i$ -th extend you know coefficient
//  $i-1$  of the product, so the
// value you feed in must be the one that is already final.
// WHEN NOT TO: if the recurrence is  $f[n] = \text{sum}_{k < n} g[k] f[n-k]$  with
//  $g$  KNOWN in advance, the
// cheaper standard answer is divide and conquer over  $[l, r)$ 
// (contribute  $[l, \text{mid})$  to  $[\text{mid}, r)$  with
```

```
// one multiply), same  $O(n \log^2 n)$  but a smaller constant and no
    ↳ bookkeeping.
// If instead the whole system is polynomial ( $f = F(f)$  as power
    ↳ series), Newton iteration on the
// series (04 - Polynomial.cpp:  $\text{inv} / \log / \exp / \text{sqrt}$ ) is  $O(n \log n)$ 
    ↳ and beats both.
```

6.5.8 NTT2D

2-D CONVOLUTION of two bivariate polynomials mod $\text{NMOD} = 998244353$:

```
// Needs: ntt, npow (02 - NTT.cpp).
//  $c[i][j] = \sum_{p+r=i, q+s=j} a[p][q] * b[r][s]$ , in
    ↳  $O(R * C * \log(R * C))$  with  $R, C$  the padded
// dimensions. The transform SEPARATES: NTT every row, then every
    ↳ column, multiply pointwise,
// invert columns then rows. Same idea works for any number of
    ↳ dimensions.
// REACH FOR IT when the state has TWO additive coordinates: "count
    ↳ paths with a given (x, y)
// displacement", "polynomials in two variables" (a variable per
    ↳ statistic you must track jointly),
// 2-D pattern matching by sums of products, and grid DPs whose
    ↳ transition is a translation.
typedef vector<vector<ll>> Poly2;
void ntt2d(Poly2& a, bool inv) {
    int R = a.size(), C = a[0].size();
    for (int i = 0; i < R; i++) ntt(a[i], inv);
    vector<ll> col(R);
    for (int j = 0; j < C; j++) {
        for (int i = 0; i < R; i++) col[i] = a[i][j];
        ntt(col, inv);
        for (int i = 0; i < R; i++) a[i][j] = col[i];
    }
}
Poly2 multiply2D(const Poly2& a, const Poly2& b) {
    int rr = a.size() + b.size() - 1, cc = a[0].size() + b[0].size() -
    ↳ 1, R = 1, C = 1;
    while (R < rr) R <= 1;
    while (C < cc) C <= 1;
    Poly2 x(R, vector<ll>(C, 0)), y(R, vector<ll>(C, 0));
    for (size_t i = 0; i < a.size(); i++)
        for (size_t j = 0; j < a[0].size(); j++) x[i][j] = a[i][j];
    for (size_t i = 0; i < b.size(); i++)
        for (size_t j = 0; j < b[0].size(); j++) y[i][j] = b[i][j];
    ntt2d(x, false), ntt2d(y, false);
    for (int i = 0; i < R; i++)
        for (int j = 0; j < C; j++) x[i][j] = x[i][j] * y[i][j] % NMOD;
    ntt2d(x, true);
    x.resize(rr);
    for (auto& row : x) row.resize(cc);
    return x;
}
// CHEAPER ALTERNATIVE - FLATTENING. If one dimension is small ( $C_0$ 
    ↳ columns in a and  $C_1$  in b), pack
// the 2-D array into a 1-D one with stride  $W \geq C_0 + C_1 - 1$  (index  $i * W$ 
    ↳ + j) and do a SINGLE 1-D
// convolution: the strides are wide enough that the coordinates never
    ↳ interfere. One NTT of length
//  $R * W$  instead of  $R + C$  transforms - usually faster, and it is three
    ↳ lines. Use the 2-D routine
// when both dimensions are large and comparable.
// The same separability argument gives 2-D FWHT (xor in both
    ↳ coordinates) and 2-D subset-sum
// transforms; nothing about it is special to the NTT.
// MEMORY:  $R * C$  longs.  $4096 \times 4096$  is already 128 MB - flatten instead
    ↳ when the product is large.
```

6.5.9 FWHTAnyBase

FWHT IN AN ARBITRARY BASE k : convolution where indices combine by DIGIT-WISE ADDITION MOD k in

```
// Needs: npow (02 - NTT.cpp).
// base  $k$ . For  $k = 2$  this is exactly xor and the routine collapses to
    ↳ the usual FWHT; for  $k = 10$  it
// is "add the decimal digits without carrying"; for  $k = 3$  it is the
    ↳ ternary "no-carry" addition
// that Nim-like and ternary-state problems produce.
// Length must be  $n = k^d$ . Cost  $O(n * k * \log_k n)$  - the transform
    ↳ along each digit is a length- $k$ 
// DFT, done with the  $k \times k$  matrix  $M[p][q] = w^k(pq)$ .
// PRECONDITION: a primitive  $k$ -th root of unity must exist mod  $\text{NMOD}$ ,
    ↳ i.e.  $k \mid \text{NMOD} - 1$ .
//  $998244352 = 2^{23} * 7 * 17$ , so  $k$  in  $\{2, 4, 8, \dots, 2^{23}, 7, 14, 17, 34,$ 
    ↳  $119, \dots\}$  all work;  $k = 3$ 
// does NOT - see the note at the bottom. Use a different NTT prime if
    ↳ you need another  $k$ .
void fwhtBase(vector<ll>& a, int k, bool inv) {
    int n = a.size();
```

```
ll w = npow(NROOT, (NMOD - 1) / k); // primitive
↳  $k$ -th root of unity
if (inv) w = npow(w, NMOD - 2);
vector<ll> wp(k);
wp[0] = 1;
for (int i = 1; i < k; i++) wp[i] = wp[i - 1] * w % NMOD;
vector<ll> v(k);
for (int len = 1; len < n; len *= k)
    for (int i = 0; i < n; i += len * k)
        for (int j = 0; j < len; j++) {
            for (int p = 0; p < k; p++) v[p] = a[i + j + p * len];
            for (int p = 0; p < k; p++) {
                ll s = 0;
                for (int q = 0; q < k; q++) s = (s + v[q] * wp[p *
                ↳  $q \% k$ ]) % NMOD;
                a[i + j + p * len] = s;
            }
        }
if (inv) { ll ni = npow(n, NMOD - 2); for (ll& x : a) x = x * ni %
    ↳ NMOD; }
}
vector<ll> multiplyBase(vector<ll> a, vector<ll> b, int k) {
    int n = 1;
    while (n < (int)max(a.size(), b.size())) n *= k; // n must be a
    ↳ power of  $k$ 
    a.resize(n, 0), b.resize(n, 0);
    fwhtBase(a, k, false), fwhtBase(b, k, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % NMOD;
    fwhtBase(a, k, true);
    return a;
}
//  $k$ -TH POWER OF THE OPERATOR: to convolve one array with itself  $e$ 
    ↳ times, transform once, raise
// every entry to the  $e$ -th power, transform back -  $O(n \log n + n \log$ 
    ↳  $e$ ), not  $e$  convolutions.
// TERNARY when 3 does not divide  $\text{MOD} - 1$ : work in  $F_p[w] / (w^2 + w +$ 
    ↳  $1)$ , i.e. represent each
// value as  $a + b * w$  with  $w^2 = -1 - w$ . That is a genuine cube root of
    ↳ unity as long as  $3 \nmid p + 1$ .
// Multiplication is  $(a_1 + b_1 w)(a_2 + b_2 w) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1$ 
    ↳  $- b_1 b_2) w$ . Everything else
// in the routine above is unchanged.
// PACKING TRICK: if your indices are BITMASKS but the operation is
    ↳ digit-wise addition mod 3
// (e.g. "each element used 0, 1 or 2 times"), rewrite each mask in
    ↳ base 3 first: the value with
// binary digits  $d_i$  becomes  $\sum d_i 3^i$ .
// WHY NOT SUBSET-SUM/OR: OR-convolution is the zeta/Mobius transform
    ↳ ( $08 - \text{DP}/02 - \text{SOS\_DP.cpp}$ );
// this file is only for the group  $Z_{k^d}$ , where the transform is
    ↳ invertible pointwise.
```

6.5.10 PolyShiftAndComposition

TAYLOR SHIFT $p(x) \rightarrow p(x + c)$, and COMPOSITION $p(g(x))$. Both mod $\text{NMOD} = 998244353$.

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / cut (04 -
    ↳ Polynomial.cpp).
// TAYLOR SHIFT in  $O(n \log n)$ .  $[x^k] p(x+c) = \sum_{j \geq k} a_j * C(j, k)$ 
    ↳  $* c^{j-k}$ , which is a
// correlation once you divide by the factorials, hence ONE
    ↳ multiplication.
// REACH FOR IT to re-centre an interpolation ("I have  $f$  sampled at
    ↳  $0..n$ , I want it at  $m..m+n$ "),
// to turn "evaluate at an ARITHMETIC progression  $a, a+d, a+2d, \dots$ "
    ↳ into a shift plus a scale, and
// inside falling-factorial / Stirling conversions.
Poly taylorShift(Poly a, ll c) {
    int n = a.size();
    vector<ll> f(n), fi(n);
    f[0] = 1;
    for (int i = 1; i < n; i++) f[i] = f[i - 1] * i % NMOD;
    fi[n - 1] = npow(f[n - 1], NMOD - 2);
    for (int i = n - 1; i > 0; i--) fi[i - 1] = fi[i] * i % NMOD;
    Poly A(n), B(n);
    for (int i = 0; i < n; i++) A[n - 1 - i] = a[i] * f[i] % NMOD;
    ↳ // reversed  $a_j * j!$ 
    ll p = 1;
    for (int i = 0; i < n; i++) B[i] = p * fi[i] % NMOD, p = p * (c %
    ↳  $\text{NMOD} + \text{NMOD}$ ) % NMOD;
    Poly h = multiply(A, B);
    Poly r(n);
    for (int k = 0; k < n; k++) r[k] = h[n - 1 - k] * fi[k] % NMOD;
    return r;
}
// COMPOSITION  $p(g(x)) \bmod x^m$  in  $O(\sqrt{n} * m \log m)$  by the BLOCK
    ↳ (baby-step giant-step) method:
```

```
// write  $p(y) = \sum_{i < K} y^{i \cdot D} \cdot q_i(y)$  with  $\deg q_i < D$ , precompute
//  $g^{\cdot D}$  once, build each
//  $q_i(g)$  by linear combinations, then Horner over the  $K$  blocks with
// the single power  $g^D$ .
// REACH FOR IT for "substitute one generating function into another":
//  $\exp/\log$  of a series with  $a$ 
// non-constant argument, EGF of a labelled construction, "apply  $a$ 
// transform given as a series".
Poly composition(const Poly& p, const Poly& g, int m) {
    int n = p.size(), K = 1;
    while (K * K < n) K++; // K blocks of
    int D = (n + K - 1) / K;
    vector<Poly> pw(D + 1, Poly{1});
    for (int i = 1; i <= D; i++) pw[i] = cut(multiply(pw[i - 1], g), m);
    Poly res(m, 0), gd{1};
    for (int i = 0; i < K; i++) {
        Poly blk(m, 0);
        for (int j = 0; j < D && i * D + j < n; j++) {
            ll c = p[i * D + j];
            if (!c) continue;
            for (int t = 0; t < (int)pw[j].size() && t < m; t++)
                blk[t] = (blk[t] + c * pw[j][t]) % NMOD;
        }
        blk = cut(multiply(blk, gd), m);
        for (int t = 0; t < m && t < (int)blk.size(); t++) res[t] =
            (res[t] + blk[t]) % NMOD;
        gd = cut(multiply(gd, pw[D]), m);
    }
    return res;
}
// COMPOSITIONAL INVERSE  $f^{-1}$  (the series  $h$  with  $f(h(x)) = x$ ), for
//  $f[0] = 0, f[1] \neq 0$ :
// LAGRANGE INVERSION gives a single coefficient directly,
//  $[x^n] h = (1/n) * [x^{n-1}] (x / f(x))^n$ , and more generally
//  $[x^n] G(h(x)) = (1/n) * [x^{n-1}] G'(x) * (x / f(x))^n$ .
// One coefficient costs  $O(n \log n)$ ; all  $n$  of them by this route costs
//  $O(n^2 \log n)$ , so use it when
// you need ONE answer (which is the usual case: Catalan, Fuss-Catalan,
// tree counts).
// K-TH ROOT of a series with  $a[0] = 1$ :  $\text{pow}_k(a, \text{inv}(k) \bmod (NMOD - 1))$ 
//  $\rightarrow$  i.e.  $\exp(\log(a)/k)$ . For
//  $a[0] \neq 1$  factor out  $c * x^d$  first; a  $k$ -th root exists only if  $k \mid d$ 
// and  $c$  is a  $k$ -th power
// residue, and then it is unique up to a  $k$ -th root of unity.
// COST WARNING: composition is the expensive one. If  $g$  is a MONOMIAL
//  $\rightarrow c * x^t$ , substitute by hand; if
//  $p$  is short, plain Horner with truncated products is  $O(n * m \log m)$ 
// and simpler.
```

6.5.11 PolyGcdResultant

POLYNOMIAL GCD and RESULTANT over F_p ($p = NMOD = 998244353$, prime).

```
// Needs: multiply / npow (02 - NTT.cpp), Poly / divmod / cut (04 -
// Polynomial.cpp).
//  $\deg \gcd(a, b)$  = the number of COMMON ROOTS of  $a$  and  $b$  counted with
// multiplicity (over the
// algebraic closure), so gcd answers "do these two polynomials share a
// root" and "how many".
// Euclid with the fast divmod is  $O(n \log n)$  per step and  $O(n)$  steps;
// the half-gcd algorithm brings
// it to  $O(n \log^2 n)$  but is long - this is the version you write in a
// contest.
Poly trim(Poly a) { while (a.size() > 1 && !a.back()) a.pop_back();
    return a; }

Poly polyGcd(Poly a, Poly b) { // monic gcd; returns
    {1} for coprime inputs
    a = trim(a), b = trim(b);
    while (b.size() > 1 || b[0]) { Poly r = trim(divmod(a, b).second);
        a = b, b = r; }
    if (a.size() == 1 && !a[0]) return a; // gcd(0, 0)
    ll ic = npow(a.back(), NMOD - 2); // normalise
    // to monic
    for (ll& x : a) x = x * ic % NMOD;
    return a;
}
// RESULTANT  $\text{Res}(a, b) = \text{lc}(b)^{\deg(a)} * \text{prod over roots beta of b of}$ 
//  $a(\text{beta})$ 
//  $= (-1)^{(\deg(a) * \deg(b))} * \text{lc}(a)^{\deg(b)} * \text{prod over}$ 
//  $\alpha \text{ roots of } a \text{ of } b(\alpha)$ 
// It is 0 EXACTLY when  $a$  and  $b$  share a root, and it is the determinant
// of the Sylvester matrix -
// so it is the polynomial-time way to eliminate a variable from two
// equations.
// Euclidean recursion: with  $r = a \bmod b$ ,
//  $\text{Res}(a, b) = \text{lc}(b)^{(\deg(a) - \deg(r))} * (-1)^{(\deg(b) * \deg(r))} * \text{Res}(b,$ 
//  $r)$ .
```

```
//  $O(n \log n)$  per step,  $O(n)$  steps.
ll resultant(Poly a, Poly b) {
    a = trim(a), b = trim(b);
    if ((a.size() == 1 && !a[0]) || (b.size() == 1 && !b[0])) return 0;
    ll res = 1;
    while (b.size() > 1) {
        int da = a.size() - 1, db = b.size() - 1;
        Poly r = trim(divmod(a, b).second);
        int dr = r.size() - 1;
        if (r.size() == 1 && !r[0]) return 0; // common root
        res = res * npow(b.back(), da - dr) % NMOD;
        if ((ll)db * dr % 2) res = (NMOD - res) % NMOD;
        a = b, b = r;
    }
    return res * npow(b[0], a.size() - 1) % NMOD; // Res(a,
    // const c) = c^deg(a)
}
// USES
// DISCRIMINANT of  $f$ :  $\text{Res}(f, f') / \text{lc}(f)$  - zero iff  $f$  has a repeated
// root.  $\gcd(f, f')$  isolates the
// squarefree part, which is the first step of any polynomial
// factorisation.
// ELIMINATION: given  $P(x, y) = 0$  and  $Q(x, y) = 0$ ,  $\text{Res}_y(P, Q)$  is a
// polynomial in  $x$  alone whose
// roots are exactly the  $x$ -coordinates of the common solutions.
//  $\det(A + x \cdot B)$  style problems and "determinant of a circulant" (03 -
// Matrices/04) are resultants.
// COMMON ROOT COUNTING mod  $p$  for two polynomials given as products,
// without factoring either.
// PRECONDITION on the Euclidean chain:  $p$  must be PRIME (we invert
// leading coefficients). Over  $a$ 
// composite modulus use the Sylvester determinant with  $\detMod$  (07 -
// Numerical/02) instead.
// DEGREE GOTCHA: divmod in 04 - Polynomial.cpp keeps a padded
// remainder, so trim() before reading
// any degree - the sign in the resultant recursion depends on the TRUE
// degree.
```

6.5.12 CyclotomicPolynomial

CYCLOTOMIC POLYNOMIALS $\Phi_n(x) = \text{prod over primitive } n\text{-th roots of unity of } (x - w)$.

```
// They are the IRREDUCIBLE factors of  $x^n - 1$  over the integers:  $x^n$ 
//  $- 1 = \text{prod}_{d \mid n} \Phi_d(x)$ ,
//  $\deg \Phi_n = \phi(n)$ , and every  $\Phi_n$  has integer coefficients (tiny
// ones - the first  $n$  with  $a$ 
// coefficient outside  $\{-1, 0, 1\}$  is  $n = 105$ ).
// By Mobius inversion  $\Phi_n(x) = \text{prod}_{d \mid n} (x^d - 1)^{\mu(n/d)}$ , and
// both multiplying and
// dividing by  $(x^d - 1)$  are  $O(\deg)$  in place, so one  $\Phi_n$  costs
//  $O(\sigma(n))$  - linear, no FFT.
// REACH FOR IT when a problem factors  $x^n - 1$  (roots-of-unity filters,
// "count sequences whose sum
// is 0 mod  $n$ ", cyclic codes), when you need the minimal polynomial of
// a root of unity, or for
// determinants of circulants factored per divisor.
void mulXd(vector<ll>& a, int d) { // a *= (x^d - 1)
    int n = a.size();
    a.insert(a.begin(), d, 0);
    for (int i = 0; i < n; i++) a[i] -= a[i + d];
}
void divXd(vector<ll>& a, int d) { // a /= (x^d - 1),
    // exact division assumed
    int n = a.size();
    for (int i = n - 1 - d; i >= 0; i--) a[i] += a[i + d];
    a.erase(a.begin(), a.begin() + d);
}
int muOne(int n) { // Mobius of a single
    // n, by trial division
    int r = 1;
    for (int p = 2; (ll)p * p <= n; p++) if (n % p == 0) {
        n /= p;
        if (n % p == 0) return 0;
        r = -r;
    }
    return n > 1 ? -r : r;
}
vector<ll> cyclotomic(int n) { // coefficients, low
    // degree first; size  $\phi(n)+1$ 
    vector<ll> a{1};
    for (int d = 1; d <= n; d++) if (n % d == 0 && muOne(n / d) == 1)
        mulXd(a, d);
    for (int d = 1; d <= n; d++) if (n % d == 0 && muOne(n / d) == -1)
        divXd(a, d);
    return a; // do the
    // multiplications FIRST, then the divisions
}
// FACTS
```

```
// Phi_p(x) = 1 + x + ... + x^(p-1) for prime p; Phi_2n(x) =
//   ↳ Phi_n(-x) for odd n > 1;
// Phi_{n p}(x) = Phi_n(x^p) if p | n, and Phi_n(x^p) / Phi_n(x)
//   ↳ otherwise.
// Phi_n(1) = p if n is a prime power p^k, and 1 for any other n > 1.
// ORDER OF 2 MOD n and friends: the irreducible factors of x^n - 1
//   ↳ over F_q are the factors of
// the Phi_d, each splitting into phi(d) / ord_d(q) factors of degree
//   ↳ ord_d(q).
// x^n - 1 over F_p (p not dividing n) is squarefree; if p | n, x^n -
//   ↳ 1 = (x^(n/p) - 1)^p.
// ROOTS-OF-UNITY FILTER (the usual reason you are here):
// sum_{k=r mod m} [x^k] F(x) = (1/m) * sum_{j=0}^{m-1} w^(-jr) *
//   ↳ F(w^j), w a primitive m-th
// root of unity. Mod p you need m | p - 1 for w to exist; otherwise
//   ↳ work in F_p[x] / Phi_m(x),
// where x IS a primitive m-th root by construction - which is
//   ↳ exactly what this file gives you.
// OVERFLOW: coefficients stay small (bounded by a few hundred for n <
//   ↳ 1e5) so ll is safe; if you
// want them mod p, reduce afterwards.
```

6.5.13 ConvAnyMod

NTT, because 02 - NTT.cpp hardcodes NMOD.

```
// Needs: npow (02 - NTT.cpp) -- self-contained otherwise: it carries
//   ↳ its own modulus-parametric
// CONVOLUTION MODULO AN ARBITRARY M (composite, not NTT-friendly): run
//   ↳ the NTT under THREE
// NTT-friendly primes and recombine with Garner. O(n log n) with a
//   ↳ constant of three transforms.
// The true coefficients are bounded by len * (M-1)^2; the three primes
//   ↳ below multiply to ~9.6e25,
// so for M <= 1e9 and len <= 1e7 the CRT reconstruction is EXACT, and
//   ↳ only then reduced mod M.
// PREFER THIS over convMod in 01 - FFT.cpp when M is large (> ~1e9),
//   ↳ when the arrays are long
// (>= 1e6, where the double-based split loses precision), or when you
//   ↳ do not want to reason about
// floating point at all. It is exact by construction; the FFT version
//   ↳ is not.
const ll CP[3] = {167772161, 469762049, 1224736769}; // all
//   ↳ c=2^k + 1, primitive root 3
void nttP(vector<ll>& a, bool inv, ll p) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        ll w = npow(3, (p - 1) / len, p);
        if (inv) w = npow(w, p - 2, p);
        for (int i = 0; i < n; i += len) {
            ll cur = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] * cur % p;
                a[i + j] = (u + v) % p, a[i + j + len / 2] = (u - v +
//   ↳ p) % p;
                cur = cur * w % p;
            }
        }
        if (inv) { ll ni = npow(n, p - 2, p); for (ll& x : a) x = x * ni %
//   ↳ p; }
    }
}
vector<ll> convP(const vector<ll>& a, const vector<ll>& b, ll p) {
    int rs = a.size() + b.size() - 1, n = 1;
    while (n < rs) n <= 1;
    vector<ll> x(n, 0), y(n, 0);
    for (size_t i = 0; i < a.size(); i++) x[i] = (a[i] % p + p) % p;
    for (size_t i = 0; i < b.size(); i++) y[i] = (b[i] % p + p) % p;
    nttP(x, false, p), nttP(y, false, p);
    for (int i = 0; i < n; i++) x[i] = x[i] * y[i] % p;
    nttP(x, true, p);
    x.resize(rs);
    return x;
}
vector<ll> convAnyMod(const vector<ll>& a, const vector<ll>& b, ll M) {
    if (a.empty() || b.empty()) return {};
    vector<ll> r0 = convP(a, b, CP[0]), r1 = convP(a, b, CP[1]), r2 =
//   ↳ convP(a, b, CP[2]);
    ll i01 = npow(CP[0] % CP[1], CP[1] - 2, CP[1]);
    ll i02 = npow(CP[0] % CP[2], CP[2] - 2, CP[2]);
    ll i12 = npow(CP[1] % CP[2], CP[2] - 2, CP[2]);
    vector<ll> res(r0.size());
    for (size_t i = 0; i < r0.size(); i++) { // Garner:
```

```
//   ↳ mixed-radix digits x0,x1,x2
    ll x0 = r0[i];
    ll x1 = (r1[i] - x0 % CP[1] + CP[1]) % CP[1] * i01 % CP[1];
    ll t = (r2[i] - x0 % CP[2] + CP[2]) % CP[2] * i02 % CP[2];
    ll x2 = (t - x1 % CP[2] + CP[2]) % CP[2] * i12 % CP[2];
    __int128 v = (__int128)x2 * CP[1] % M * CP[0] % M +
//   ↳ (__int128)x1 * CP[0] % M + x0;
    res[i] = (ll)(v % M);
}
return res;
}
// PRIME MENU (all with primitive root 3; the 2-power in p-1 caps the
//   ↳ transform length):
// 167772161 = 5*2^25 + 1 469762049 = 7*2^26 + 1 754974721 =
//   ↳ 45*2^24 + 1
// 998244353 = 119*2^23 + 1 1004535809 = 479*2^21 + 1 1224736769 =
//   ↳ 73*2^24 + 1
// TWO PRIMES are enough only when len * (M-1)^2 < P0 * P1 (~7.9e16) -
//   ↳ i.e. tiny M. Use three.
// SPLIT-COEFFICIENT ALTERNATIVE: convMod in 01 - FFT.cpp writes each
//   ↳ coefficient as hi*2^15 + lo
// and does 3 complex FFTs - shorter, but its error grows with len and
//   ↳ with M.
// EXACT INTEGER convolution with no modulus: pass an M larger than any
//   ↳ possible coefficient, or
// drop the final reduction and reconstruct into __int128.
```

6.6 Interpolation

6.6.1 LagrangeAndFaulhaber

Lagrange interpolation + power sums. Needs Mint + buildFact().

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).

```
// Given y[0..n] = f(0), f(1), ..., f(n) for a polynomial f of degree
//   ↳ <= n, evaluate f(x). O(n).
// The consecutive-points case is the one you actually want; it kills a
//   ↳ whole class of problems
// whose answer is "a polynomial in n of degree d" (guess d, sample d+1
//   ↳ values, interpolate).
Mint lagrangeConsecutive(const vector<Mint>& y, ll x) {
    int n = y.size() - 1;
    if (0 <= x && x <= n) return y[x]; // x < 0 would index
//   ↳ y[negative]
    vector<Mint> pre(n + 2, 1), suf(n + 2, 1);
    for (int i = 0; i <= n; i++) pre[i + 1] = pre[i] * Mint(x - i);
    for (int i = n; i >= 0; i--) suf[i + 1] = suf[i + 1] * Mint(x - i);
    Mint r = 0;
    for (int i = 0; i <= n; i++) {
        Mint t = y[i] * pre[i] * suf[i + 1] * invFact[i] * invFact[n -
//   ↳ i];
        ((n - i) & 1) ? r -= t : r += t;
    }
    return r;
}
// Arbitrary sample points, O(n^2).
Mint lagrange(const vector<Mint>& xs, const vector<Mint>& ys, Mint x) {
    int n = xs.size();
    Mint r = 0;
    for (int i = 0; i < n; i++) {
        Mint num = ys[i], den = 1;
        for (int j = 0; j < n; j++) if (j != i) num *= x - xs[j], den
//   ↳ *= xs[i] - xs[j];
        r += num / den;
    }
    return r;
}
// FAULHABER: sum_{i=1..n} i^k is a polynomial in n of degree k+1 ->
//   ↳ sample k+2 points. O(k log k).
Mint powerSum(ll n, int k) {
    vector<Mint> y(k + 2, 0);
    for (int i = 1; i <= k + 1; i++) y[i] = y[i - 1] + Mint(i).pow(k);
    return lagrangeConsecutive(y, n);
}
// Same trick works for: sum of C(i,k), sum of i^k * r^i (degree k in n
//   ↳ after dividing by r^n),
// and any DP whose answer is eventually polynomial.
```

6.6.2 BerlekampMassey

Berlekamp-Massey: find the shortest linear recurrence fitting a sequence, O(n^2).

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp).
// Then jump to the k-th term in O(len^2 log k). THE weapon for "brute
// ↳ force small n, find the
// pattern, extrapolate" - works on any sequence that satisfies a
// ↳ constant-coefficient recurrence.
// Feed it ~2*len terms (60-100 is usually plenty).
vector<Mint> berlekampMassey(vector<Mint> s) {

```

int n = s.size(), L = 0, m = 0;
vector<Mint> C(n), B(n), T;
C[0] = B[0] = 1;
Mint b = 1;
for (int i = 0; i < n; i++) {
    m++;
    Mint d = s[i];
    for (int j = 1; j <= L; j++) d += C[j] * s[i - j];
    if (d.v == 0) continue;
    T = C;
    Mint coef = d / b;
    for (int j = m; j < n; j++) C[j] -= coef * B[j - m];
    if (2 * L > i) continue;
    L = i + 1 - L, B = T, b = d, m = 0;
}
C.resize(L + 1), C.erase(C.begin());
for (auto& x : C) x = Mint(0) - x;
return C; // s[i] = sum_j
    ↪ C[j] * s[i-1-j]
}
// k-th term (0-indexed) of the recurrence tr with initial values S.
    ↪ O(|tr|^2 log k).
Mint linearRec(const vector<Mint>& S, const vector<Mint>& tr, ll k) {
    if (tr.empty()) return 0; // e[1] would be out of
    ↪ bounds
    int n = tr.size();
    auto comb = [&](vector<Mint> a, vector<Mint> b) {
        vector<Mint> r(2 * n + 1, 0);
        for (int i = 0; i <= n; i++) if (a[i].v)
            for (int j = 0; j <= n; j++) r[i + j] += a[i] * b[j];
        for (int i = 2 * n; i > n; i--)
            for (int j = 0; j < n; j++) r[i - 1 - j] += r[i] * tr[j];
        r.resize(n + 1);
        return r;
    };
    vector<Mint> pol(n + 1, 0), e(pol);
    pol[0] = e[1] = 1;
    for (++k; k; k /= 2) {
        if (k % 2) pol = comb(pol, e);
        e = comb(e, e);
    }
    Mint r = 0;
    for (int i = 0; i < n; i++) r += pol[i + 1] * S[i];
    return r;
}

```

6.6.3 BostanMori

BOSTAN-MORI: the k-th term of a linear recurrence of order d in $O(d \log d \log k)$, mod NMOD.

```

// Needs: multiply / npow (02 - NTT.cpp).
// This is the fast replacement for Kitamasa ( $O(d^2 \log k)$ ), 06 -
    ↪ Interpolation/02 - linearRec and
// for companion-matrix exponentiation ( $O(d^3 \log k)$ ). At d = 500, k =
    ↪ 1e18 it is the only one that
// runs. Pair it with Berlekamp-Massey: guess the recurrence from
    ↪ brute-forced terms, then jump.
// INPUT: a[0..d-1] the first d terms, c[0..d-1] with a[n] = sum_i c[i]
    ↪ * a[n-1-i] for n >= d.
// IDEA: the generating function is RATIONAL,  $\sum a_n x^n = P(x) / Q(x)$ 
    ↪ with
//  $Q = 1 - c_0 x - \dots - c_{d-1} x^d$ ,  $P = (A * Q) \bmod x^d$ .
// Then  $[x^k] P/Q$ : multiply top and bottom by  $Q(-x)$ . The new
    ↪ denominator  $V(x) = Q(x)Q(-x)$  is EVEN,
// so  $V(x) = W(x^2)$ ; keeping the even or odd part of the numerator (by
    ↪ the parity of k) halves k.
ll bostanMori(vector<ll> a, const vector<ll>& c, ll k) {
    int d = c.size();
    if (k < (ll)a.size() && k < d) return a[k] % NMOD;
    vector<ll> Q(d + 1, 0);
    Q[0] = 1;
    for (int i = 0; i < d; i++) Q[i + 1] = (NMOD - c[i] % NMOD) % NMOD;
    a.resize(d, 0);
    vector<ll> P = multiply(a, Q);
    P.resize(d); // P = (A * Q)
    ↪ mod  $x^d$ 
    while (k) {
        vector<ll> Qm = Q;
        for (size_t i = 1; i < Qm.size(); i += 2) Qm[i] = (NMOD -
            ↪ Qm[i]) % NMOD;
        vector<ll> U = multiply(P, Qm), V = multiply(Q, Qm);
        P.assign((U.size() + 1) / 2, 0);
        for (size_t i = k & 1; i < U.size(); i += 2) P[i / 2] = U[i];
        Q.assign((V.size() + 1) / 2, 0);
        for (size_t i = 0; i < V.size(); i += 2) Q[i / 2] = V[i];
        P.resize(d), Q.resize(d + 1);
        k >>= 1;
    }
}

```

```

return P[0] * npow(Q[0], NMOD - 2) % NMOD;
}
// THE RATIONAL GENERATING FUNCTION IS THE POINT, not just the k-th
    ↪ term:
//  $\sum_n a_n = P(1) / Q(1)$  (formally; use it when the series really
    ↪ does terminate/converge,
// e.g. absorbing-state probabilities where  $a_n$  is the chance of
    ↪ stopping at step n).
//  $\sum_n n * a_n = (P'(1) Q(1) - P(1) Q'(1)) / Q(1)^2$  - this is the
    ↪ EXPECTED STOPPING TIME, the
// reason this trick shows up in probability problems: brute-force
    ↪ 2d terms, run Berlekamp-
// Massey, build P/Q, differentiate at  $x = 1$ . Beware  $Q(1) = 0$ , which
    ↪ means the series does not
// sum and you must handle the pole separately.
// PARTIAL FRACTIONS of P/Q give the closed form  $a_n = \sum c_i r_i^n$ 
    ↪ over the roots of Q.
// PREFIX SUMS:  $\sum_{n \leq k} a_n$  is the k-th coefficient of  $P / ((1-x)
    ↪ Q) -$  just multiply Q by
//  $(1 - x)$  and run the same routine.
// PRECONDITION: NMOD prime,  $Q[0] = 1$  (automatic here). The recurrence
    ↪ must be exact; if you got it
// from Berlekamp-Massey, feed BM at least 2*d terms or the answer is
    ↪ silently wrong.

```

6.6.4 MultipointEval

MULTIPOINT EVALUATION and FAST INTERPOLATION at ARBITRARY points, both $O(n \log^2 n)$, mod NMOD.

```

// Needs: multiply (02 - NTT.cpp), Poly / divmod / deriv / cut /
    ↪ operator+ (04 - Polynomial.cpp).
// Evaluating a degree-n polynomial at n points naively is  $O(n^2)$ ;
    ↪ interpolating from n arbitrary
// pairs naively is  $O(n^2)$ . These are the log-squared versions, built
    ↪ on the same PRODUCT TREE:
// node [l, r] stores  $\text{prod}_i \{ \text{in } [l, r] \} (x - x_i)$ .
// REACH FOR IT when n is 1e5 and the points are NOT consecutive
    ↪ integers (if they are, use
// lagrangeConsecutive,  $O(n)$ ) and NOT a geometric progression (then use
    ↪ chirp-z,  $O(n \log n)$ ).
// Typical: "evaluate this generating function at n given points",
    ↪ "recover the polynomial behind
// n scattered samples", shifting a sampled polynomial to a new set of
    ↪ arguments.
vector<Poly> ptree; // 4n nodes;
    ↪ call buildTree first
void buildTree(int v, int l, int r, const vector<ll>& x) {
    if (l == r) { ptree[v] = Poly{(NMOD - x[l] % NMOD) % NMOD, 1};
        ↪ return; }
    int m = (l + r) / 2;
    buildTree(2 * v, l, m, x), buildTree(2 * v + 1, m + 1, r, x);
    ptree[v] = multiply(ptree[2 * v], ptree[2 * v + 1]);
}
void evalRec(int v, int l, int r, Poly f, vector<ll>& out) {
    f = divmod(f, ptree[v]).second; // f mod prod
    ↪  $(x - x_i)$  keeps the values
    if (l == r) { out[l] = f.empty() ? 0 : f[0]; return; }
    int m = (l + r) / 2;
    evalRec(2 * v, l, m, f, out), evalRec(2 * v + 1, m + 1, r, f, out);
}
vector<ll> evalAll(const Poly& f, const vector<ll>& x) {
    int n = x.size();
    if (!n) return {};
    ptree.assign(4 * n, {});
    buildTree(1, 0, n - 1, x);
    vector<ll> out(n);
    evalRec(1, 0, n - 1, f, out);
    return out;
}
// INTERPOLATION: with  $M(x) = \text{prod } (x - x_i)$ , the Lagrange basis
    ↪ denominator at  $x_i$  is  $M'(x_i)$ , so
// one multipoint evaluation of  $M'$  gives every denominator; then merge
    ↪ the leaves bottom-up with
// result =  $A * (\text{right subtree product}) + B * (\text{left subtree product})$ .
Poly interpRec(int v, int l, int r, const vector<ll>& d) {
    if (l == r) return Poly{d[l]};
    int m = (l + r) / 2;
    Poly A = interpRec(2 * v, l, m, d), B = interpRec(2 * v + 1, m + 1,
        ↪ r, d);
    return multiply(A, ptree[2 * v + 1]) + multiply(B, ptree[2 * v]);
}
Poly interpolate(const vector<ll>& x, const vector<ll>& y) { //
    ↪ all  $x_i$  must be DISTINCT
    int n = x.size();
    if (!n) return {};
    ptree.assign(4 * n, {});
    buildTree(1, 0, n - 1, x);
    Poly root = ptree[1];

```

```

vector<ll> out(n);
evalRec(1, 0, n - 1, deriv(root), out);          // out[i] =
↳ M'(xi) != 0 since xi distinct
vector<ll> d(n);
for (int i = 0; i < n; i++) d[i] = y[i] % NMOD * npow(out[i], NMOD
↳ - 2) % NMOD;
return cut(interpRec(1, 0, n - 1, d), n);
}
// COMPLEXITY: each level of the tree does O(n) total polynomial work
↳ with logarithmic factors, so
// n log^2 n; the constant is big - expect ~1s at n = 1e5, and prefer
↳ the specialised routines.
// PRECONDITIONS: NMOD prime; the xi must be pairwise distinct for
↳ interpolate (duplicates make
// M'(xi) = 0 and the inverse blows up). deriv/divmod come from 04 -
↳ Polynomial.cpp.
// APPLICATIONS BEYOND THE OBVIOUS
// Composing "evaluate then re-interpolate" implements ANY linear map
↳ given by its action on
// points, e.g. shifting a sampled polynomial by a constant in O(n
↳ log^2 n) (or O(n log n) with
// the Taylor shift in 05 - Convolution/10).
// The product tree alone is worth having: prod (x - xi) in O(n log^2
↳ n) is how you build the
// characteristic polynomial of a set of roots, the elementary
↳ symmetric polynomials of n values,
// and "the number of ways to pick k of these items" for every k at
↳ once.

```

6.6.5 InterpolationCoefficients

GETTING THE COEFFICIENTS, not just one value. 01 - LagrangeAndFaulhaber.cpp evaluates an

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp); binomialBasis
↳ needs 02 - NTT.cpp instead.
// interpolating polynomial at a point; sometimes the problem wants the
↳ polynomial itself - to
// integrate it, to compose it, to read off a single coefficient, or to
↳ answer queries at points
// that are not known in advance.

// ARBITRARY sample points, explicit coefficients, O(n^2). xi must be
↳ pairwise DISTINCT.
// Build M(x) = prod (x - xi) once, then divide out each (x - xi) by
↳ synthetic division.
vector<Mint> interpCoeffs(const vector<Mint>& x, const vector<Mint>& y)
↳ {
    int n = x.size();
    vector<Mint> M(n + 1, 0), r(n, 0);
    M[0] = 1;
    for (int i = 0; i < n; i++)                // M *= (x -
↳ xi)
        for (int j = n; j >= 0; j--) { M[j] *= Mint(0) - x[i]; if (j)
↳ M[j] += M[j - 1]; }
    for (int i = 0; i < n; i++) {
        vector<Mint> q(n);                    // q = M / (x
↳ - xi), synthetic division
        Mint rem = M[n];
        for (int j = n - 1; j >= 0; j--) q[j] = rem, rem = M[j] + rem *
↳ x[i];
        Mint den = 1;
        for (int j = 0; j < n; j++) if (j != i) den *= x[i] - x[j];
        Mint sc = y[i] / den;
        for (int j = 0; j < n; j++) r[j] += q[j] * sc;
    }
    return r;                                // r[j] =
↳ coefficient of x^j
}
// THE INVERSE VANDERMONDE MATRIX is exactly this: V[i][j] = xi^j, and
↳ (V^-1)[j][i] is the
// coefficient of x^j in the i-th Lagrange basis polynomial - the
↳ vector interpCoeffs produces when
// y = ei. So "solve V a = y" IS interpolation, in O(n^2) instead of
↳ O(n^3), and
// det V = prod_{i < j} (xj - xi).
// If the first column starts at xi instead of 1 (i.e. V[i][j] =
↳ xi^(j+1)), divide row i by xi.

// BINOMIAL (Newton forward-difference) BASIS, from CONSECUTIVE samples
↳ p(0..n), in O(n log n):
// p(x) = sumi ai C(x, i), ai = i-th finite difference at 0 =
↳ sumj (-1)^(i-j) C(i, j) p(j)
// which is one convolution of p[j]/j! with (-1)^j/j!. The ai are
↳ integers whenever p maps
// integers to integers, so this is THE basis for "p has integer
↳ values" arguments, and evaluating
// p at a HUGE x is then sum ai * C(x, i) with C(x, i) built by a
↳ running product - O(n) and no

```

```

// division by anything that could vanish.
// Written mod NMOD on top of multiply() from 02 - NTT.cpp (Poly =
↳ vector<ll>); if your modulus is
// not NTT-friendly, run the same two lines with an O(n^2) convolution
↳ or with convMod.
vector<ll> binomialBasis(const vector<ll>& p) {          // p[j] =
↳ value at j, j = 0..n
    int n = p.size();
    vector<ll> f(n), fi(n), u(n), v(n);
    f[0] = 1;
    for (int i = 1; i < n; i++) f[i] = f[i - 1] * i % NMOD;
    fi[n - 1] = npow(f[n - 1], NMOD - 2);
    for (int i = n - 1; i > 0; i--) fi[i - 1] = fi[i] * i % NMOD;
    for (int i = 0; i < n; i++) {
        u[i] = p[i] % NMOD * fi[i] % NMOD;
        v[i] = (i & 1) ? (NMOD - fi[i]) % NMOD : fi[i];
    }
    vector<ll> w = multiply(u, v);
    vector<ll> a(n);
    for (int i = 0; i < n; i++) a[i] = w[i] * f[i] % NMOD;
    return a;
}
// FALLING FACTORIAL basis x^*(i) = x(x-1)...(x-i+1) is the same thing
↳ scaled: ai * i! are its
// coefficients. Converting between the monomial and falling-factorial
↳ bases is exactly the
// STIRLING transform (Stirling numbers of the first/second kind are
↳ the two change-of-basis
// matrices), which is how "sum of i^k" becomes "sum of C(i, j)" and
↳ telescopes.
// WHICH ROUTINE WHEN
// one value, consecutive samples ..... lagrangeConsecutive, O(n)
// one value, arbitrary samples ..... lagrange, O(n^2)
// coefficients, arbitrary samples ..... interpCoeffs, O(n^2) (or
↳ 04 - MultipointEval)
// coefficients in the binomial basis ... binomialBasis, O(n log n)

```

6.6.6 PRecursive

LINEAR RECURRENCE WITH POLYNOMIAL COEFFICIENTS (a "P-recursive" or holonomic sequence):

```

// Needs: Mint (01 - Modular Arithmetic/01 - Mint.cpp). MOD must be
↳ PRIME.
// sum_{i=0..order} Pi(m - i) * a[m - i] = 0 for all large m,
↳ deg Pi <= deg.
// This is the strictly stronger sibling of Berlekamp-Massey. BM only
↳ finds CONSTANT-coefficient
// recurrences, so it fails on n!, C(2n, n), Motzkin, derangements,
↳ Hertzprung, and essentially
// every sequence built from binomials or factorials. Those are all
↳ P-recursive.
// REACH FOR IT in the classic contest workflow: brute-force the first
↳ ~60-100 terms, guess the
// recurrence here, then extend it to n = 1e7 in O(n * order * deg).
↳ Also tells you the answer IS
// holonomic, which is itself a strong hint about the intended solution.
// COST: guessing is O(n^3 / deg^2)-ish Gaussian elimination on an R x
↳ C matrix; extending is
// linear. You must GUESS deg (try 0, 1, 2, ... until it succeeds) -
↳ order comes out automatically.
// You need roughly (deg + 2) * (order + 1) terms; give it more and the
↳ verification is meaningful.

// coeffs[i][d] = coefficient of m^d in Pi. Returns {} if no such
↳ recurrence fits the data.
vector<vector<Mint>> findPRec(const vector<Mint>& t, int deg) {
    int n = t.size(), B = (n + 2) / (deg + 2), C = B * (deg + 1), R = n
↳ - (B - 1);
    if (B < 2 || R < C - 1) return {};
    vector<vector<Mint>> mat(R, vector<Mint>(C));
    for (int y = 0; y < R; y++)
        for (int b = 0; b < B; b++) {
            Mint v = t[y + b];
            for (int d = 0; d <= deg; d++) mat[y][b * (deg + 1) + d] =
↳ v, v *= Mint(y + b);
        }
    int rank = 0;
    for (int x = 0; x < C && rank < R; x++) {                // RREF
        int piv = -1;
        for (int y = rank; y < R; y++) if (mat[y][x].v) { piv = y;
↳ break; }
        if (piv < 0) continue;
        swap(mat[rank], mat[piv]);
        Mint iv = Mint(1) / mat[rank][x];
        for (int j = x; j < C; j++) mat[rank][j] *= iv;
        for (int y = 0; y < R; y++) if (y != rank && mat[y][x].v) {
            Mint c = mat[y][x];
            for (int j = x; j < C; j++) mat[y][j] -= c * mat[rank][j];
        }
    }
}

```

```

    }
    rank++;
}
if (rank == C) return {}; // no
↳ non-trivial kernel vector
// column 'rank' is free: set it to 1 and read the pivot columns
↳ off (the matrix is in RREF)
int order = rank / (deg + 1);
vector<vector<Mint>> r(order + 1, vector<Mint>(deg + 1, 0));
r[0][rank % (deg + 1)] = 1;
for (int y = rank - 1; y >= 0; y--)
    r[order - y / (deg + 1)][y % (deg + 1)] = Mint(0) -
↳ mat[y][rank];
return r;
}
// Extend the sequence to index n. 'terms' must hold at least 'order'
↳ correct initial values.
// PRECONDITION: the leading polynomial P_0(m) must be non-zero mod MOD
↳ at every m you generate -
// if it vanishes the recurrence genuinely does not determine a[m] (a
↳ "singularity" of the
// recurrence); shift the start or pick a different MOD.
vector<Mint> extendPrec(int n, const vector<vector<Mint>>& c,
↳ vector<Mint> terms) {
    int order = c.size() - 1, deg = c[0].size() - 1;
    vector<Mint> a = terms;
    a.resize(max<int>(n + 1, terms.size()), 0);
    for (int m = terms.size(); m <= n; m++) {
        Mint s = 0;
        for (int i = 1; i <= order; i++) {
            Mint p = 1, k = Mint(m - i);
            for (int d = 0; d <= deg; d++) s += a[m - i] * c[i][d] * p,
↳ p *= k;
        }
        Mint den = 0, p = 1;
        for (int d = 0; d <= deg; d++) den += c[0][d] * p, p *= Mint(m);
        a[m] = (Mint(0) - s) / den;
    }
    return a;
}
// EXAMPLES that come out immediately (deg, order):
// n! -> (1, 1): a[m] - m * a[m-1] = 0. Catalan -> (1, 1).
↳ Motzkin -> (1, 2).
// derangements -> (1, 2). C(2n, n) -> (1, 1). Hertzprung -> (1,
↳ 4). sum C(n, k)^3 -> (2, 2).
// ALWAYS SANITY-CHECK: re-run extendPrec on the initial terms and
↳ compare against the input; a
// recurrence that fits by accident is the main failure mode when you
↳ feed too few terms.
// THEORY: holonomic sequences are closed under +, Cauchy product, and
↳ Hadamard product, and the
// generating function satisfies a linear ODE with polynomial
↳ coefficients. That is why "sum of
// products of binomials" is essentially always P-recursive
↳ (Zeilberger's algorithm proves it).
// If you also need the k-th term for k = 1e18, P-recursive sequences
↳ admit an O(sqrt(k) log k)
// evaluation by the factorial-style baby-step/giant-step product of
↳ matrices - far more code.

```

6.6.7 ExpPolynomialSum

SUM OF A GEOMETRIC TIMES A POLYNOMIAL: $S(n) = \sum_{i=0}^n a^i \cdot f(i)$ for n up to 1e18,

```

// Needs: Mint + buildFact (01 - Mint.cpp), lagrangeConsecutive (06 -
↳ Interpolation/01).
// where f is a polynomial of degree <= k given by its values f(0),
↳ f(1), ..., f(k). O(k log MOD).
// REACH FOR IT for "sum i^k r^i", weighted prefix sums in a linear
↳ recurrence, expected values
// where each step is discounted by a factor a, and anything of the
↳ shape sum a^i * (polynomial).
// WHY IT WORKS: for a != 1 the answer has the closed form  $S(n) = a^n \cdot$ 
↳  $g(n) + c$  with g a
// polynomial of degree exactly k and c a constant. Apply the operator
↳  $(D h)(n) = h(n+1) - a \cdot h(n)$ :
// it kills  $a^n \cdot g$  one degree at a time, so  $D^{k+1}$  annihilates the
↳  $a^n g(n)$  part completely and
// turns the constant into  $c \cdot (1 - a)^{k+1}$ . That single equation
↳ gives c; then g is sampled and
// Lagrange-interpolated. For a == 1 the sum is just a polynomial of
↳ degree k+1 in n.
Mint expPolySum(vector<Mint> y, Mint a, ll n) { // y[i] = f(i)
    for i = 0..k
        int k = y.size() - 1;
        if (n < 0) return 0;
        y.push_back(lagrangeConsecutive(y, k + 1)); // need f(k+1)
    }
    ↳ too, so k+2 samples of S

```

```

vector<Mint> S(k + 2);
Mint p = 1;
for (int i = 0; i <= k + 1; i++) {
    S[i] = (i ? S[i - 1] : Mint(0)) + p * y[i];
    p *= a;
}
if (a == Mint(1)) return lagrangeConsecutive(S, n); // S is a
↳ polynomial of degree k+1
Mint c = 0; // c *
↳  $(1-a)^{k+1} = D^{k+1} S$  at 0
for (int i = 0; i <= k + 1; i++) {
    Mint t = C(k + 1, i) * (Mint(0) - a).pow(k + 1 - i) * S[i];
    c += t;
}
c /= (Mint(1) - a).pow(k + 1);
vector<Mint> g(k + 1); // g(i) =
↳  $(S(i) - c) / a^i$ 
Mint ia = Mint(1) / a, q = 1;
for (int i = 0; i <= k; i++) g[i] = (S[i] - c) * q, q *= ia;
return lagrangeConsecutive(g, n) * a.pow(n) + c;
}
// SPECIAL CASES WORTH KNOWING
// f == 1: sum_{i<=n} a^i = (a^{n+1} - 1) / (a - 1), and n+1
↳ when a == 1. Do NOT route a
// trivial geometric series through this - and watch a
↳ == 1 mod MOD, which happens
// for a = MOD + 1 style inputs and makes the closed
↳ form divide by zero.
// a == 1: pure Faulhaber, powerSum in 06 - Interpolation/01.
// INFINITE sum with |a| < 1 formally: sum_{i>=0} a^i f(i) = (applied
↳ k+1 times) - or read it off
// the rational generating function, see 03 -
↳ BostanMori.cpp.
// PRECONDITION: MOD prime and larger than k + 2 (we invert factorials
↳ and  $(1 - a)^{k+1}$ ); a != 1
// for the second branch, which the code checks. n may be as large as
↳ you like: a.pow(n) and
// lagrangeConsecutive both take a ll argument.
// SAMPLING f: if you are handed the COEFFICIENTS of f instead of its
↳ values, evaluate it at
// 0..k+1 by Horner in O(k^2) or by one multipoint evaluation.

```

6.6.8 PowerSumsAtOnce

ALL POWER SUMS AT ONCE: $s[i] = \sum_{p=0}^i t^p \cdot p^i$ for EVERY $i = 0..k$, in $O(k \log k)$.

```

// Needs: multiply / npow (02 - NTT.cpp), Poly / inv / cut (04 -
↳ Polynomial.cpp).
// (Convention  $0^0 = 1$ , so  $s[0] = t + 1$ .) powerSum in 06 -
↳ Interpolation/01 gives one exponent in
//  $O(k)$ ; this gives all k+1 of them for the same asymptotic cost, which
↳ is what you need when the
// answer is "for each k, output sum i^k" or when a Newton-identity /
↳ symmetric-function step needs
// the whole vector of power sums.
// EGF IDENTITY: sum_i s[i] x^i / i! = sum_{p=0}^k t^p e^x = (e^n
↳ x - 1) / (e^x - 1)
// = [(e^n x - 1) / x] * [x / (e^x
↳ - 1)], n = t + 1.
// The right factor is the BERNOULLI generating function,  $B[i] = i! \cdot$ 
↳  $[x^i] x / (e^x - 1)$ , and
//  $x / (e^x - 1) = 1 / \sum_{i>=0} x^i / (i+1)! -$  one power-series
↳ inverse,  $O(k \log k)$ . That beats the
//  $O(k^2)$  Bernoulli recurrence in 05 - Combinatorics/11.
Poly bernoulliEGF(int m) { // returns B[i] / i!
    for i = 0..m-1
        Poly d(m);
        d[0] = 1;
        for (int i = 1; i < m; i++) d[i] = d[i - 1] * npow(i + 1, NMOD - 2)
↳ % NMOD; // 1 / (i+1)!
        return inv(d, m);
}
Poly powerSums(ll t, int k) { // s[i] =
    ↳ sum_{p=0}^k t^p \cdot p^i, i = 0..k
    Poly B = bernoulliEGF(k + 2), P(k + 2, 0);
    ll n = (t + 1) % NMOD, cur = 1;
    for (int i = 1; i <= k + 1; i++) { //
        ↳ P[i] = n^i / i!
        cur = cur * n % NMOD * npow(i, NMOD - 2) % NMOD;
        P[i] = cur;
    }
    Poly Q = cut(multiply(P, B), k + 2);
    Poly s(k + 1);
    ll f = 1;
    for (int i = 0; i <= k; i++) { s[i] = Q[i + 1] * f % NMOD; f = f *
↳ (i + 1) % NMOD; }
    return s;
}

```

```
// RELATED ONE-LINERS BUILT FROM THE SAME PIECES
// SUM OVER A LIST rather than an interval: sum_{j} a_j^i for all i =
//   ↪ 0..k in O((n + k) log^2 n) -
//   build the rational function sum_j 1 / (1 - a_j x) by divide and
//   ↪ conquer over the list (keep
//   numerator and denominator, merge with up = A*D + B*C, down =
//   ↪ B*D), then expand it mod x^(k+1).
//   Those coefficients ARE the power sums; feed them to Newton's
//   ↪ identities (05 - Comb/10)
//   to get the elementary symmetric polynomials.
//   sum_{p=0}^t p^i * r^p for all i: same shape, use 07 -
//   ↪ ExpPolynomialSum.cpp per i, or convolve
//   with the EGF of the geometric series.
// The Faulhaber POLYNOMIAL in n (coefficients rather than a value) is
//   ↪ faulhaberPoly in
//   05 - Combinatorics/11 - BernoulliFaulhaber.cpp.
// PRECONDITION: NMOD prime and > k + 2 (we invert 1..k+1 and take a
//   ↪ series inverse whose constant
// term is 1). t may be huge - only t + 1 mod NMOD enters.
```

6.6.9 ReedsSloane

REEDS-SLOANE: the shortest linear recurrence fitting a sequence MODULO A COMPOSITE m.

```
// Needs: modInverse / extGCD (04 - Number Theory/01), CRT (04 - Number
//   ↪ Theory/03/01 - CRT.cpp).
// Berlekamp-Massey (06 - Interpolation/02) divides by the discrepancy,
//   ↪ so it needs a FIELD and
//   silently produces garbage whenever a leading coefficient is not
//   ↪ invertible mod m. This is the
// replacement: it factors m into prime powers, runs a shift-register
//   ↪ synthesis over each Z/p^e
// (tracking, for every 2-adic-style valuation g, the best
//   ↪ approximation whose discrepancy has that
// valuation), and CRTs the coefficient vectors back together.
// REACH FOR IT when the problem's modulus is 1e9, 2^32, 10^k or any
//   ↪ other composite and you still
// want the "brute force a few terms, guess the recurrence,
//   ↪ extrapolate" attack. O(n^2 log m).
// OUTPUT: A[0..L] with A[0] = 1 and sum_j A[j] * s[i-j] = 0 mod m, i.e.
//   s[i] = -(A[1] s[i-1] + ... + A[L] s[i-L]) mod m. Feed 2L or
//   ↪ more terms.
namespace RS {
int deg(const vector<ll>& a) { // -inf encoded as a
    ↪ large negative number
    return (a.size() > 1 || (a.size() == 1 && a[0])) ? (int)a.size() -
    ↪ 1 : -40000;
}
int L(const vector<ll>& a, const vector<ll>& b) { return max(deg(a),
    ↪ deg(b) + 1); }
void ext(vector<ll>& a, size_t d) { if (d > a.size()) a.resize(d, 0); }

// shift-register synthesis modulo p^e (m = p^e); returns the
//   ↪ connection polynomial
vector<ll> primePower(const vector<ll>& s, ll m, ll p, int e) {
    vector<vector<ll>> a(e), b(e), an(e), bn(e), ao(e), bo(e);
    vector<ll> t(e), u(e), r(e), to(e, 1), uo(e), pw(e + 1, 1);
    for (int i = 1; i <= e; i++) pw[i] = pw[i - 1] * p;
    for (int i = 0; i < e; i++) {
        a[i] = an[i] = {pw[i]};
        b[i] = {0}, bn[i] = {s[0] * pw[i] % m};
        t[i] = s[0] * pw[i] % m;
        if (!t[i]) t[i] = 1, u[i] = e;
        else for (u[i] = 0; t[i] % p == 0; t[i] /= p) u[i]++;
    }
    for (size_t k = 1; k < s.size(); k++) {
        for (int g = 0; g < e; g++) if (L(an[g], bn[g]) > L(a[g],
        ↪ b[g])) {
            int h = e - 1 - u[g];
            ao[g] = a[h], bo[g] = b[h], to[g] = t[h], uo[g] = u[h],
        ↪ r[g] = k - 1;
        }
        a = an, b = bn;
        for (int o = 0; o < e; o++) {
            ll d = 0;
            for (size_t i = 0; i < a[o].size() && i <= k; i++) d = (d +
        ↪ a[o][i] * s[k - i]) % m;
            if (!d) { t[o] = 1, u[o] = e; continue; }
            for (u[o] = 0, t[o] = d; t[o] % p == 0; t[o] /= p) u[o]++;
            int g = e - 1 - u[o];
            if (L(a[g], b[g]) == 0) { ext(bn[o], k + 1); bn[o][k] =
        ↪ (bn[o][k] + d) % m; continue; }
            ll c = t[o] % m * modInverse(to[g] % m, m) % m * pw[u[o] -
        ↪ uo[g]] % m;
            int sh = k - r[g];
            ext(an[o], ao[g].size() + sh), ext(bn[o], bo[g].size() +
        ↪ sh);
            for (size_t i = 0; i < ao[g].size(); i++)
```

```
        an[o][i + sh] = (an[o][i + sh] - c * ao[g][i]) % m;
        for (size_t i = 0; i < bo[g].size(); i++)
            bn[o][i + sh] = (bn[o][i + sh] - c * bo[g][i]) % m;
        for (ll& x : an[o]) x = (x % m + m) % m;
        for (ll& x : bn[o]) x = (x % m + m) % m;
        while (an[o].size() > 1 && !an[o].back()) an[o].pop_back();
        while (bn[o].size() > 1 && !bn[o].back()) bn[o].pop_back();
    }
}
return an[0];
}
}
vector<ll> reedsSloane(vector<ll> s, ll m) {
    vector<array<ll, 3>> fac; // {p^e, p, e}
    ll t = m;
    for (ll i = 2; i * i <= t; i++) if (t % i == 0) {
        ll c = 0, q = 1;
        while (t % i == 0) t /= i, c++, q *= i;
        fac.push_back({q, i, c});
    }
    if (t > 1) fac.push_back({t, t, 1});
    vector<vector<ll>> per;
    size_t n = 0;
    for (auto& f : fac) {
        vector<ll> ss = s;
        for (ll& x : ss) x = (x % f[0] + f[0]) % f[0];
        per.push_back(RS::primePower(ss, f[0], f[1], (int)f[2]));
        n = max(n, per.back().size());
    }
    vector<ll> res(n);
    for (size_t i = 0; i < n; i++) {
        vector<ll> c(fac.size(), md(fac.size()));
        for (size_t j = 0; j < fac.size(); j++)
            md[j] = fac[j][0], c[j] = i < per[j].size() ? per[j][i] : 0;
        res[i] = CRT(c, md).first;
    }
    return res; // res[0] == 1; s[i] =
    ↪ -sum_{j>=1} res[j]*s[i-j]
}
// AFTER YOU HAVE THE RECURRENCE: extending it is trivial (the leading
//   ↪ coefficient is 1, so no
//   inversion is ever needed) - just iterate s[i] = -(sum_{j>=1} A[j]
//   ↪ s[i-j]) mod m.
// For the k-th term with k = 1e18 mod a COMPOSITE, do Kitamasa-style
//   ↪ polynomial exponentiation
// mod the connection polynomial: it only multiplies, never divides, so
//   ↪ a composite modulus is fine.
// CHEAPER FIRST TRIES, in order:
//   - if m is prime, use Berlekamp-Massey; this file is strictly
//   ↪ heavier.
//   - if m is squarefree, run BM under each prime factor separately and
//   ↪ CRT. That is 10 lines and
//   covers m = 1e9+6-style inputs; the prime-POWER case is the only
//   ↪ one that really needs this.
//   - if the sequence is not constant-coefficient at all (factorials,
//   ↪ binomials), you want
//   06 - PRecursive.cpp instead, and it also works over composite
//   ↪ moduli when the leading
//   polynomial happens to be invertible.
// PRECONDITION: m >= 1 and the factorisation loop is trial division,
//   ↪ so m up to ~1e18 needs
// Pollard rho (04 - Number Theory/02/06) instead of the loop above.
```

6.7 Numerical

6.7.1 Integration

ADAPTIVE SIMPSON - numeric integral of a black-box f on [a,b] to a requested absolute error.

```
// Simpson approximates f by a parabola through the two ends and the
//   ↪ midpoint; "adaptive" means it
// splits an interval only where the parabola does not yet agree with
//   ↪ the two half-parabolas.
// Reach for it when the answer is an integral you cannot do in closed
//   ↪ form: area of a union of
// circles/ellipses cut by a line, area swept by a moving shape,
//   ↪ probability over a continuous
// parameter, arc lengths. Cost is proportional to how wiggly f is, not
//   ↪ to (b-a).
template <class F> ld simpson(F f, ld a, ld b) {
    return (b - a) * (f(a) + 4 * f((a + b) / 2) + f(b)) / 6;
}
template <class F> ld asr(F f, ld a, ld b, ld eps, ld S, int dep) {
    ld c = (a + b) / 2, L = simpson(f, a, c), R = simpson(f, c, b);
    if (dep <= 0 || fabs(L + R - S) <= 15 * eps) return L + R + (L + R
    ↪ - S) / 15;
    return asr(f, a, c, eps / 2, L, dep - 1) + asr(f, c, b, eps / 2, R,
    ↪ dep - 1);
```

```

}
template <class F> ld integrate(F f, ld a, ld b, ld eps = 1e-9) {
    return asr(f, a, b, eps, simpson(f, a, b), 50);
}
// TRAPS. (1) If f has a KINK or a jump, split the interval at that
//    ↳ point by hand and integrate the
// pieces - adaptive Simpson will burn its depth budget trying to
//    ↳ resolve it. (2) The (L+R-S)/15
// term is Richardson extrapolation; keeping it makes the result much
//    ↳ more accurate for free.
// (3) A too-loose eps on a nearly flat region can stop early and miss
//    ↳ a spike: if f can be zero on
// most of the range and huge somewhere, integrate the pieces
//    ↳ separately.
//
// OTHER NUMERICAL TOOLS WORTH THE PAGE:
// TERNARY SEARCH on a strictly unimodal f over reals: 100 iterations of
//    m1 = l + (r-l)/3, m2 = r - (r-l)/3; f(m1) < f(m2) ? r = m2 : l =
//    ↳ m1; (minimising)
// fixes the answer to ~1e-30 relative. On INTEGERS use binary search
//    ↳ on the difference
// f(m+1) - f(m) instead - the three-way split has off-by-one traps
//    ↳ at the last two points.
// NEWTON'S METHOD for a root of g: x ← g(x) / g'(x). Doubles the
//    ↳ correct digits per step but
// needs a good start; bisect first if the sign changes, then Newton.
// BISECTION on a monotone predicate: 100 iterations of (l+r)/2 is
//    ↳ always safe and never diverges.
// GOLDEN-SECTION SEARCH: like ternary search but one function
//    ↳ evaluation per step instead of two -
// worth it only when f is expensive.
// SOLVING f(x) = 0 WHERE f IS A POLYNOMIAL: find all real roots by
//    ↳ recursing on the derivative
// (roots of f' split the line into monotone pieces; bisect inside
//    ↳ each).
// SIMULATED ANNEALING / HILL CLIMBING when the objective is not
//    ↳ unimodal: start from several
// random points, take a step of size T, accept worse solutions with
//    ↳ probability exp(-dE/T),
// cool T geometrically (T *= 0.995). Good for geometric-median /
//    ↳ facility-location style
// problems where an exact method is out of reach; always seed the
//    ↳ RNG from the clock.
// WEISZFELD for the geometric median (minimise sum of Euclidean
//    ↳ distances): iterate
// x ← (sum p_i / |x - p_i|) / (sum 1 / |x - p_i|), guarding the
//    ↳ case x == p_i. Converges
// linearly; nested ternary search is the safer contest answer (the
//    ↳ objective is convex).

```

6.7.2 LinearSolvers

SPECIAL-SHAPE LINEAR SOLVERS. Gaussian elimination is $O(n^3)$; these are the shapes where the

```

// structure buys you a factor of n or removes a precondition.

// --- THOMAS ALGORITHM: TRIDIAGONAL system in  $O(n)$ . ---
// a[i] x[i-1] + b[i] x[i] + c[i] x[i+1] = d[i], with a[0] = c[n-1]
//    ↳ = 0.
// This is THE closer for random walks on a path ("expected steps to
//    ↳ escape [0, n]"), 1-D heat /
// diffusion DPs, and any DP whose state depends on its two neighbours.
//    ↳ Gaussian elimination on
// the same system is  $O(n^3)$  and will TLE at n = 1e5; this is 8 lines.
// STABLE without pivoting when the system is diagonally dominant (|b|
//    ↳ > |a| + |c|), which every
// hitting-time system is.
vector<ld> thomas(vector<ld> a, vector<ld> b, vector<ld> c, vector<ld>
    ↳ d) {
    int n = b.size();
    for (int i = 1; i < n; i++) {
        ld m = a[i] / b[i - 1];
        b[i] -= m * c[i - 1], d[i] -= m * d[i - 1];
    }
    vector<ld> x(n);
    x[n - 1] = d[n - 1] / b[n - 1];
    for (int i = n - 2; i >= 0; i--) x[i] = (d[i] - c[i] * x[i + 1]) /
        ↳ b[i];
    return x;
}
// BANDED systems of half-width w (state depends on w neighbours)
//    ↳ generalise this at  $O(n w^2)$  -
// the shape of "robot on a grid with m columns", where w = m.
// CYCLIC tridiagonal (x[0] and x[n-1] also coupled): Sherman-Morrison
//    ↳ - solve two tridiagonal
// systems and combine, still  $O(n)$ .

// --- DETERMINANT MODULO AN ARBITRARY m (composite allowed),  $O(n^3 \log$ 
```

```

    ↳ m). ---
// Gaussian elimination inverts the pivot, so it needs a PRIME modulus.
//    ↳ This version never
// divides: it runs the Euclidean algorithm between two rows, swapping
//    ↳ them, until the lower
// entry is 0. Use it when the modulus is  $2^k$ , 1e9, or anything you
//    ↳ cannot invert in.
ll detMod(vector<vector<ll>> a, ll m) {
    int n = a.size();
    ll det = 1 % m;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++)
            while (a[j][i]) { //
                ↳ Euclid on the two rows
                ll t = a[i][i] / a[j][i];
                for (int k = i; k < n; k++) a[i][k] = (a[i][k] - t *
                    ↳ a[j][k]) % m;
                swap(a[i], a[j]), det = -det;
            }
        det = det * a[i][i] % m;
        if (!a[i][i]) return 0;
    }
    return (det % m + m) % m;
}
// MATRIX RANK over a field: the usual elimination, counting non-zero
//    ↳ pivot rows.
// RANK over GF(2): use bitsets -  $O(n^3 / 64)$ , see 04 - Linear
//    ↳ Algebra/03.
// INTEGER determinant with no modulus: compute it modulo several
//    ↳ primes and CRT. Size it with
// HADAMARD'S BOUND |det| ≤ prod of the row 2-norms ≤  $n^{(n/2)}$  *
//    ↳ max|a_ij|^n, and take enough
// primes that the product exceeds  $2x$  that.
// SPARSE determinant (n = 1e5 with few non-zeros): Wiedemann - pick
//    ↳ random u, v, feed the Krylov
// sequence  $u^T A^k v$  to Berlekamp-Massey to get the minimal
//    ↳ polynomial, whose constant term is
// +-det after a random diagonal preconditioner.  $O(n * nnz)$ .
// CHARACTERISTIC POLYNOMIAL in  $O(n^3)$ : reduce to upper HESSENBERG form
//    ↳ by similarity transforms
// (so  $\det(xI - H)$  satisfies a short recurrence in the leading
//    ↳ principal minors), then read the
// coefficients off. With Cayley-Hamilton that gives  $A^k$  in  $O(n^3 + n^2
//    ↳ \log k)$  instead of
//  $O(n^3 \log k)$  - worth it only for large k and small n.

```

6.7.3 Romberg

ROMBERG INTEGRATION - the other numerical quadrature. Repeatedly halve the step (trapezoid rule)

```

// and RICHARDSON-EXTRAPOLATE the error away: T(h) has an error
//    ↳ expansion in  $h^2$  only, so combining
// two step sizes cancels the  $h^2$  term, three cancel  $h^4$ , and so on.
//    ↳ Row k of the table is accurate
// to  $O(h^{(2k)})$ . Cost:  $2^k$  function evaluations, each sample reused
//    ↳ across the whole table.
// REACH FOR IT over adaptive Simpson (07 - Numerical/01) when the
//    ↳ integrand is SMOOTH and cheap
// and you want many correct digits: Romberg reaches 1e-12 in ~10 rows
//    ↳ where Simpson needs a deep
// adaptive recursion. Use adaptive Simpson instead when the function
//    ↳ has kinks, spikes, or a
// singular endpoint - Romberg assumes the error expansion exists and
//    ↳ quietly lies when it does not.
template <class F> ld romberg(F f, ld a, ld b, ld tol = 1e-10, int
    ↳ maxIter = 25) {
    vector<ld> t;
    ld h = b - a;
    t.push_back(h * (f(a) + f(b)) / 2); // T(h), one
    ↳ trapezoid
    for (int k = 1, m = 1; k <= maxIter; k++, m <= 1) {
        ld s = 0, x = a + h / 2;
        for (int j = 0; j < m; j++) s += f(x), x += h; // only the
        ↳ NEW midpoints
        h /= 2;
        ld cur = t[0] / 2 + h * s; // refined
        ↳ trapezoid, previous row's T
        ld prev = t.back();
        for (int j = 0; j < k; j++) { // Richardson:
            ↳ t[j] holds the old row
            ld nxt = (powl(4, j + 1) * cur - t[j]) / (powl(4, j + 1) -
                ↳ 1);
            t[j] = cur, cur = nxt;
        }
        t.push_back(cur);
        if (k > 3 && fabsl(prev - cur) < tol) break;
    }
    return t.back();
}

```

```

}
// THE TABLE: R[k][0] = trapezoid with 2^k intervals, R[k][j] = (4^j
  ↳ R[k][j-1] - R[k-1][j-1])
// / (4^j - 1). R[k][1] is exactly Simpson's rule; R[k][2] is Boole's.
  ↳ The code above keeps only
// one row, which is all you need.
// PRACTICAL NOTES
// - convergence check on |R[k][k] - R[k-1][k-1]| is a heuristic;
  ↳ require a few rows first (the
// k > 3 guard) or a flat integrand will make it stop after one step.
// - INFINITE range: substitute x = t/(1-t) or integrate to a large
  ↳ cut-off and bound the tail.
// - ENDPOINT SINGULARITY (1/sqrt(x) at 0): substitute it away, or use
  ↳ the tanh-sinh (double
// exponential) rule, which handles endpoint singularities and
  ↳ converges even faster.
// - OSCILLATORY integrands: split at the zeros of the oscillating
  ↳ factor and sum, or Romberg will
// happily converge to nonsense.
// - the same Richardson idea accelerates ANY sequence with a known
  ↳ error expansion, e.g.
// numerical derivatives (central differences) and extrapolating a
  ↳ DP in a step size.

```

6.8 Linear Programming

6.8.1 Simplex

TWO-PHASE SIMPLEX. maximise $c \cdot x$ subject to $A \cdot x \leq b$, $x \geq 0$.

```

// Returns the optimum, or +inf if unbounded, or -inf if infeasible; x
  ↳ is filled with a solution.
// Exponential in theory,  $O(n \cdot m)$  per pivot and instant at contest sizes
  ↳ (n, m up to a few hundred).
// REACH FOR IT when the problem is "optimise a linear objective under
  ↳ linear constraints" and no
// combinatorial structure presents itself - resource allocation,
  ↳ blending, fractional covering,
// zero-sum matrix games, and any min-cost-flow-with-side-constraints
  ↳ that flow cannot express.
typedef vector<ld> vd;
struct Simplex {
    int m, n;
    vector<int> B, N;
    vector<vd> D;
    Simplex(const vector<vd>& A, const vd& b, const vd& c)
        : m(b.size()), n(c.size()), B(m), N(n + 1), D(m + 2, vd(n + 2,
  ↳ 0)) {
        for (int i = 0; i < m; i++) for (int j = 0; j < n; j++) D[i][j]
  ↳ = A[i][j];
        for (int i = 0; i < m; i++) B[i] = n + i, D[i][n] = -1, D[i][n
  ↳ + 1] = b[i];
        for (int j = 0; j < n; j++) N[j] = j, D[m][j] = -c[j];
        N[n] = -1, D[m + 1][n] = 1;
    }
    void pivot(int r, int s) {
        ld inv = 1 / D[r][s];
        for (int i = 0; i < m + 2; i++) if (i != r && fabs(D[i][s]) >
  ↳ 1e-9) {
            ld f = D[i][s] * inv; // the
  ↳ pivot COLUMN is fixed up by
            for (int j = 0; j < n + 2; j++) if (j != s) D[i][j] -= f *
  ↳ D[r][j]; // the loop below
        }
        for (int j = 0; j < n + 2; j++) if (j != s) D[r][j] *= inv;
        for (int i = 0; i < m + 2; i++) if (i != r) D[i][s] *= -inv;
        D[r][s] = inv;
        swap(B[r], N[s]);
    }
    bool simplex(int phase) {
        int x = m + phase - 1;
        for (;;) {
            int s = -1;
            for (int j = 0; j < n; j++)
                if (N[j] != -phase && (s < 0 || make_pair(D[x][j],
  ↳ N[j]) < make_pair(D[x][s], N[s]))) s = j;
            if (D[x][s] >= -1e-9) return true;
            int r = -1;
            for (int i = 0; i < m; i++) {
                if (D[i][s] <= 1e-9) continue;
                if (r < 0 || make_pair(D[i][n + 1] / D[i][s], B[i]) <
  ↳ make_pair(D[r][n + 1] / D[r][s], B[r])) r = i;
            }
            if (r < 0) return false; //
  ↳ unbounded
            pivot(r, s);
        }
    }
}

```

```

ld solve(vd& x) {
    int r = 0;
    for (int i = 1; i < m; i++) if (D[i][n + 1] < D[r][n + 1]) r =
  ↳ i;
    if (D[r][n + 1] < -1e-9) { // phase
  ↳ 1: find any feasible point
        pivot(r, n);
        if (!simplex(2) || D[m + 1][n + 1] < -1e-9) return -1.0L /
  ↳ 0.0L; // infeasible
        for (int i = 0; i < m; i++) if (B[i] == -1) {
            int s = 0;
            for (int j = 1; j <= n; j++)
                if (make_pair(D[i][j], N[j]) < make_pair(D[i][s],
  ↳ N[s])) s = j;
            pivot(i, s);
        }
        bool ok = simplex(1);
        x.assign(n, 0);
        for (int i = 0; i < m; i++) if (B[i] < n) x[B[i]] = D[i][n + 1];
        return ok ? D[m][n + 1] : 1.0L / 0.0L; // +inf
    }
    = unbounded
}

/* MODELLING NOTES - the part that actually costs you time
MINIMISE instead: negate c and negate the answer.
>= CONSTRAINT: multiply the row by -1. EQUALITY: add it twice, as <=
  ↳ and as >=.
FREE VARIABLE (no x >= 0): substitute x = u - v with u, v >= 0.
  ↳ Doubles that column.
UPPER BOUND x_i <= U: just another row; simplex has no special
  ↳ handling and does not need any.
LINEAR-FRACTIONAL objective (c.x + a) / (d.x + b): CHARNES-COOPER -
  ↳ substitute y = x/(d.x+b),
  t = 1/(d.x+b) and maximise c.y + a t subject to A y <= b t, d.y + b
  ↳ t = 1, t >= 0. Linear again.
INTEGER answers are NOT guaranteed. If the constraint matrix is
  ↳ TOTALLY UNIMODULAR (network
  matrices, interval matrices, bipartite incidence) the optimum IS
  ↳ integral - that is exactly why
  flow problems have integral optima. Otherwise you need ILP, which is
  ↳ NP-hard.
ZERO-SUM MATRIX GAME: shift A so every entry is > 0, then min sum(y)
  ↳ s.t. A y >= 1, y >= 0 gives
  value v = 1/sum(y) and the column player's mixed strategy y*v; the
  ↳ DUAL gives the row player's.
  Von Neumann: max_x min_y x^T A y = min_y max_x x^T A y.
DUALITY, which is often the actual solution: max c.x, Ax <= b, x >= 0
  ↳ is dual to
  min b.y, A^T y >= c, y >= 0; the two optima are equal. COMPLEMENTARY
  ↳ SLACKNESS at the optimum:
  x_j > 0 => the j-th dual constraint is tight, and y_i > 0 => the
  ↳ i-th primal row is tight.
  The combinatorial specialisations you already know are
  ↳ max-flow/min-cut and Konig.
PRECISION: this is floating point. If the data is integral and small,
  ↳ and you need an exact
  answer, prefer a combinatorial model (flow, matching) over simplex.
  ↳ */

```

7 Strings

String Matching 122 · Hashing 124 · Aho Corasick 125 · Suffix Structures 128 · Palindromes 132 · Rotations 136 · Sequences 137 · Theorems 140

7.1 String Matching

7.1.1 KMP

KMP. $\text{pi}[i]$ = length of the longest proper border of $s[0..i]$ (prefix that is also a suffix).

// Gives $O(n \cdot m)$ substring search, the period of a prefix ($i+1 - \text{pi}[i]$),
 ↳ and the automaton for DP.

```

vector<int> compute_pi(const string &s) {
    int n = s.size();
    vector<int> pi(n);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j])
            j = pi[j - 1];
        if (s[i] == s[j])
            j++;
        pi[i] = j;
    }
    return pi;
}

vector<int> find_matches(const string &text, const string &pat) {

```

```

int n = pat.length(), m = text.length();
string s = pat + "#" + text;
vector<int> pi = compute_pi(s), ans;
for (int i = n + 1; i <= n + m; i++) { /* n + 1 is where the text
    ↳ starts */
    if (pi[i] == n) {
        ans.push_back(i - 2 * n); /* i - (n - 1) - (n + 1) */
    }
}
return ans;
}

void compute_automaton(string s, vector<vector<int>> &aut) {
    s += '#';
    int n = s.size();
    vector<int> pi = compute_pi(s);
    aut.assign(n, vector<int>(26));
    for (int i = 0; i < n; i++) {
        for (int c = 0; c < 26; c++) {
            if (i > 0 && 'a' + c != s[i])
                aut[i][c] = aut[pi[i - 1]][c];
            else
                aut[i][c] = i + ('a' + c == s[i]);
        }
    }
}

```

7.1.2 RabinKarp

RABIN-KARP - rolling-hash substring search in $O(n+m)$ expected. Prefer 05 - Hashing for anything

```

// serious (this uses a FIXED base, which is hackable); keep this for
    ↳ one-off matching.
vector<int> rabin_karp(string const &s, string const &t) {
    const int p = 31;
    const int m = 1e9 + 9;
    int S = s.size(), T = t.size();

    vector<ll> p_pow(max({S, T, 1})); // max(0,0) would
    ↳ leave it empty
    p_pow[0] = 1;
    for (int i = 1; i < (int) p_pow.size(); i++)
        p_pow[i] = (p_pow[i - 1] * p) % m;

    vector<ll> h(T + 1, 0);
    for (int i = 0; i < T; i++)
        h[i + 1] = (h[i] + (t[i] - 'a' + 1) * p_pow[i]) % m;
    ll h_s = 0;
    for (int i = 0; i < S; i++)
        h_s = (h_s + (s[i] - 'a' + 1) * p_pow[i]) % m;

    vector<int> occurrences;
    for (int i = 0; i + S - 1 < T; i++) {
        ll cur_h = (h[i + S] + m - h[i]) % m;
        if (cur_h == h_s * p_pow[i] % m)
            occurrences.push_back(i);
    }
    return occurrences;
}

```

7.1.3 ZFunction

$z[i]$ is the length of the longest substring starting from $s[i]$ which is also a prefix of s

```

vector<int> z_function(string s) {
    int n = s.length();
    vector<int> z(n);
    int l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if (i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}

```

7.1.4 WildcardMatching

STRING MATCHING WITH WILDCARDS in $O(n \log n)$. A wildcard matches any character, on EITHER side.

```

// Needs: conv /06 - Math/05 - Convolution/01 - FFT.cpp.
// Map every real character to a positive value and every wildcard to
    ↳ 0. Then position i matches iff
//      sum_j (p[j] - t[i+j])^2 * p[j] * t[i+j] == 0,
// because each term is  $\geq 0$  and vanishes exactly when the characters
    ↳ agree or one of them is 0.
// Expanding gives three correlations: sum  $p^3 t - 2 \sum p^2 t^2 + \sum$ 
    ↳  $p t^3$ , and a correlation is a
// convolution with the pattern reversed.
vector<int> wildcardMatch(const string& t, const string& p, char wild =
    ↳ '?') {
    int n = t.size(), m = p.size();
    if (m > n) return {};
    // The identity needs values  $\geq 0$  and wildcard == 0. Compress the
    ↳ characters that actually
    // occur to 1..k - with a raw 'c - 'a' + 1' any character below 'a'
    ↳ is NEGATIVE and the terms
    // can cancel, which reports matches that do not exist (e.g. text
    ↳ "b" against pattern "Z").
    ll id[256] = {}, k = 0; // a
    ↳ flat table, not a map: the
    for (char c : t) if (c != wild) id[(unsigned char)c] = 1; //
    ↳ conversion is  $O(n)$ , not  $O(n \log n)$ 
    for (char c : p) if (c != wild) id[(unsigned char)c] = 1;
    for (int c = 0; c < 256; c++) if (id[c]) id[c] = ++k;
    auto val = [&](char c) { return c == wild ? 0LL : id[(unsigned
    ↳ char)c]; };
    vector<ll> a(m), b(n);
    for (int i = 0; i < m; i++) a[i] = val(p[m - 1 - i]); //
    ↳ reversed pattern
    for (int i = 0; i < n; i++) b[i] = val(t[i]);
    auto pw = [](vector<ll> v, int k) { for (ll& x : v) { ll y = 1; for
    ↳ (int i = 0; i < k; i++) y *= x; x = y; } return v; };
    auto c1 = conv(pw(a, 3), b), c2 = conv(pw(a, 2), pw(b, 2)), c3 =
    ↳ conv(a, pw(b, 3));
    vector<int> res;
    for (int i = 0; i + m <= n; i++)
        if (c1[i + m - 1] - 2 * c2[i + m - 1] + c3[i + m - 1] == 0)
            res.push_back(i);
    return res; //
    ↳ 0-based start positions
}

// PRECISION: the largest intermediate is about  $k^4 * m$ , where k is the
    ↳ number of DISTINCT real
// characters (which is why compressing is free). With  $k = 26$  and  $m =$ 
    ↳  $1e5$  that is
//  $\sim 4.6e10$ , comfortable for double FFT. For a big alphabet, compress
    ↳ the characters that actually
// occur first, or run the three convolutions under two NTT primes and
    ↳ CRT.
// COUNT MISMATCHES at every shift (no wildcards, small alphabet): for
    ↳ each letter c convolve the
// 0/1 indicator of c in p (reversed) with the indicator of c in t;
    ↳ summing gives the number of
// MATCHES per shift, and mismatches =  $m - \text{matches}$ .  $O(|\Sigma| n \log n)$ .
// k-MISMATCH: same counts, keep shifts with matches  $\geq m - k$ .
// SUBSET / "at most one character class" matching: encode each
    ↳ position as a bitmask over the
// alphabet and convolve the complement indicators; a shift is valid
    ↳ iff the total is 0.

```

7.1.5 PrefixFunctionApplications

WHAT THE PREFIX FUNCTION ANSWERS once you have $pi[]$ - every one of these is $O(n)$ and every one

```

// Needs: 07 - Strings/01 - String Matching/01 - KMP.cpp (compute_pi)
// of them is a contest problem on its own. INDEX CONVENTION: s is
    ↳ 0-indexed and
//       $pi[i]$  = length of the longest PROPER border of the prefix
    ↳  $s[0..i]$  (so  $pi[0] = 0$ ).
// A "border" is a string that is both a proper prefix and a proper
    ↳ suffix.
// WHEN: any question about periodicity, repetition, "shortest string
    ↳ whose k-th power is s",
// "how many times does every prefix occur", "which prefixes are also
    ↳ suffixes".

// --- ALL BORDERS of the whole string, longest first. The border of a
    ↳ border is a border, so the
// chain  $pi[n-1] \rightarrow pi[pi[n-1]-1] \rightarrow \dots$  enumerates them ALL, in
    ↳  $O(\text{number of borders}) \leq O(n)$ .
vector<int> allBorders(const vector<int>& pi) {
    vector<int> b;
    for (int k = pi.empty() ? 0 : pi.back(); k > 0; k = pi[k - 1])

```

```

    ↪ b.push_back(k);
    return b; //
    ↪ lengths, decreasing
}
// --- PERIODS. p is a period of s (s[i] == s[i+p] wherever both exist)
    ↪ IFF n - p is a border.
// So the periods are exactly { n - b : b a border }, and the SMALLEST
    ↪ period is n - pi[n-1].
int smallestPeriod(int n, const vector<int>& pi) { return n - (n ? pi[n
    ↪ - 1] : 0); }
// s is a perfect power t^k IFF n % smallestPeriod == 0, and then k = n
    ↪ / smallestPeriod.
// (Careful: "aabaa" has period 3 but 5 % 3 != 0, so it is NOT a power
    ↪ - the test is required.)

// --- OCCURRENCE COUNT OF EVERY PREFIX inside s. cnt[len] = number of
    ↪ occurrences of s[0..len-1].
// Every position ends the prefix of length i+1, and an occurrence of a
    ↪ prefix implies an
// occurrence of all of ITS borders - so push the counts down the pi
    ↪ chain in reverse order.
vector<int> prefixOccurrences(const string& s, const vector<int>& pi) {
    int n = s.size();
    vector<int> cnt(n + 1, 0);
    for (int i = 0; i < n; i++) cnt[pi[i]]++;
    for (int i = n - 1; i > 0; i--) cnt[pi[i - 1]] += cnt[i];
    for (int i = 0; i <= n; i++) cnt[i]++; // the
    ↪ prefix itself
    cnt[0] = 0;
    return cnt;
}
// The same routine on p + '#' + t counts occurrences of every prefix
    ↪ of p inside t: read
// cnt[p] for the number of matches, and cnt[k] - (contribution
    ↪ inside p) for shorter prefixes.

// --- BORDER TREE: parent(i) = pi[i-1] over nodes 0..n (node i = the
    ↪ prefix of length i). Then
// "the borders of the prefix of length i" = the ancestors of i, so
// is u a border of v? -> u is an ancestor of v
    ↪ (Euler-tour interval test, O(1))
// longest common border of two -> LCA
// count borders of prefix i -> depth(i)
vector<int> borderTree(const vector<int>& pi) {
    int n = pi.size();
    vector<int> par(n + 1, 0);
    par[0] = -1;
    for (int i = 1; i <= n; i++) par[i] = pi[i - 1];
    return par;
}
/* MORE PREFIX-FUNCTION FACTS worth having on the page
pi vs Z: z[i] = length of the lcp of s and s[i..]. Either converts to
    ↪ the other in O(n), but pi
    is the one that extends ONLINE (append a character, update in O(1)
    ↪ amortised) and the one that
    turns into an automaton; z is the one that answers "match at offset
    ↪ i" directly.
AUTOMATON aut[i][c] (see 01 - KMP.cpp) makes the DP "count strings of
    ↪ length L avoiding p"
    trivial; without it that DP is O(L n |A|) with the pi chain walked
    ↪ per step.
DISTINCT SUBSTRINGS incrementally: append c to s, run the prefix
    ↪ function of the REVERSED new
    string; the new distinct substrings number (i+1) - max(z-values) -
    ↪ O(n^2) total, and the right
    answer whenever the input arrives online and a suffix automaton is
    ↪ overkill.
FINE AND WILF: if p and q are periods of s and p + q - gcd(p,q) <= n,
    ↪ then gcd(p,q) is a period.
    This is why the border lengths of any prefix form O(log n)
    ↪ ARITHMETIC PROGRESSIONS - the fact
    that makes "sum over all borders" DPs O(n log n) instead of O(n^2).
COUNTING STRINGS with a given pi array, or with a given set of
    ↪ borders, is Fine and Wilf plus
    inclusion-exclusion over the progressions; do NOT try to enumerate
    ↪ borders one by one. */

```

7.2 Hashing

7.2.1 StringHashing

Double-mod polynomial hashing. RANDOM bases => not anti-hackable.

```

// No modular inverses: h(l,r) is compared already scaled, so get() is
    ↪ 2 mults.
ll HB1, HB2;
struct Hash {
    static const ll M1 = 1000000007, M2 = 998244353;
    int n;
    vector<ll> h1, h2, p1, p2;

    Hash(const string& s) : n(s.size()), h1(n + 1, 0), h2(n + 1, 0),
    ↪ p1(n + 1, 1), p2(n + 1, 1) {
        if (!HB1) {
            mt19937_64
            ↪ rng(chrono::steady_clock::now().time_since_epoch().count());
            HB1 = 256 + rng() % (M1 - 300), HB2 = 256 + rng() % (M2 -
            ↪ 300); // FULL range: a base drawn
            // from only 1e6 values gives collision probability n/1e6 per
            ↪ modulus (0.1 at n = 1e5)
            for (int i = 0; i < n; i++) {
                h1[i + 1] = (h1[i] * HB1 + s[i]) % M1, p1[i + 1] = p1[i] *
            ↪ HB1 % M1;
                h2[i + 1] = (h2[i] * HB2 + s[i]) % M2, p2[i + 1] = p2[i] *
            ↪ HB2 % M2;
            }
            pair<ll, ll> get(int l, int r) const { // [l, r]
            ↪ inclusive, 0-based
                ll a = (h1[r + 1] - h1[l] * p1[r - l + 1]) % M1;
                ll b = (h2[r + 1] - h2[l] * p2[r - l + 1]) % M2;
                return {a < 0 ? a + M1 : a, b < 0 ? b + M2 : b};
            }
            bool eq(int l1, int r1, int l2, int r2) const {
                return r1 - l1 == r2 - l2 && get(l1, r1) == get(l2, r2);
            }
            // concat(A over lenA, B): h = hA * base^lenB + hB (works per
            ↪ modulus)
        };
        // USES: substring equality O(1) * longest common prefix of two
            ↪ suffixes by binary search
        // * dedup of substrings * 2D grid patterns (hash rows, then hash
            ↪ the row-hashes)
        // * tree isomorphism (hash the sorted multiset of child hashes)
    }

```

7.2.2 Hashing2D

2D HASHING - O(1) hash of any submatrix, for grid pattern matching.

```

// 2D prefix hash with random bases b1 (down) and b2 (right), single
    ↪ 62-bit modulus via __int128.
ll HB2D1 = 0, HB2D2 = 0; // ONE base pair
    ↪ per RUN, at file scope.
struct Hash2D { // Per-object
    ↪ bases would make two grids
    static const ll M = (ll)1e18 + 9; // incomparable -
    ↪ and "find this pattern in
    int n, m; // that grid"
    ↪ needs exactly two objects.
    ll b1, b2;
    vector<ll> P1, P2;
    vector<vector<ll>> h;

    static ll mul(ll a, ll b) { return (ll)((__int128)a * b % M); }
    Hash2D(const vector<string>& g) : n(g.size()), m(n ? g[0].size() :
    ↪ 0),
        P1(n + 1, 1), P2(m + 1, 1), h(n +
    ↪ 1, vector<ll>(m + 1, 0)) {
        if (!HB2D1) {
            mt19937_64
            ↪ rng(chrono::steady_clock::now().time_since_epoch().count());
            HB2D1 = 256 + rng() % (M - 300), HB2D2 = 256 + rng() % (M -
            ↪ 300);
            b1 = HB2D1, b2 = HB2D2;
            for (int i = 1; i <= n; i++) P1[i] = mul(P1[i - 1], b1);
            for (int j = 1; j <= m; j++) P2[j] = mul(P2[j - 1], b2);
            for (int i = 1; i <= n; i++)
                for (int j = 1; j <= m; j++) {
                    __int128 v = (__int128)mul(h[i - 1][j], b1) +
            ↪ mul(h[i][j - 1], b2)
                    - mul(mul(h[i - 1][j - 1], b1), b2) + g[i -
            ↪ 1][j - 1];
                    h[i][j] = (ll)((v % M + M) % M);
                }
        }
        // hash of the submatrix with top-left (r, c) and size dr x dc,
        ↪ 0-based
    }

```

```

ll get(int r, int c, int dr, int dc) const {
    int r2 = r + dr, c2 = c + dc;
    __int128 v = (__int128)h[r2][c2] - mul(h[r][c2], P1[dr]) -
    ↪ mul(h[r2][c], P2[dc])
    + mul(mul(h[r][c], P1[dr]), P2[dc]);
    return (ll)((v % M + M) % M);
};
// USES: count occurrences of a pattern grid * largest common square
↪ submatrix (binary search the
// side, hash all windows of both grids) * count distinct k x k
↪ submatrices.
// CHEAPER when the pattern is fixed: 1D-hash each row of the pattern,
↪ then run KMP /
// Aho-Corasick over the rows treated as single symbols.

```

7.2.3 HashingWithUpdates

HASHING UNDER POINT UPDATES, forward AND backward - the structure that answers "is s[l..r] a

```

// palindrome" while characters keep changing. Static prefix hashes
↪ cannot survive an update, and
// Manacher / eertree cannot be updated at all, so this is the only
↪ tool for that query shape.
// Two Fenwick trees over the modular values s[i]*p^i and
↪ s[i]*p^(n+1-i): point update O(log n),
// substring hash O(log n), palindrome test O(log n), longest
↪ palindrome at a centre O(log^2 n).
// MODULUS 2^61 - 1 with 128-bit multiplication: one modulus of that
↪ size is safer than two
// 32-bit ones (birthday bound ~ q^2 / 2^61) and shorter to write.
// INDEX CONVENTION: 1-INDEXED, s[1..n], and every l, r below is
↪ inclusive 1-based. get(l, r) is
// scaled so that equal substrings of EQUAL LENGTH compare equal -
↪ never compare across lengths.

```

```

struct DynHash {
    static const unsigned long long MD = (1ULL << 61) - 1;
    static unsigned long long mul(unsigned long long a, unsigned long
    ↪ long b) {
        __int128 c = (__int128)a * b;
        unsigned long long lo = (unsigned long long)(c & MD), hi =
        ↪ (unsigned long long)(c >> 61);
        lo += hi;
        return lo >= MD ? lo - MD : lo;
    }
    static unsigned long long ad(unsigned long long a, unsigned long
    ↪ long b) {
        return a + b >= MD ? a + b - MD : a + b;
    }
    static unsigned long long sb(unsigned long long a, unsigned long
    ↪ long b) {
        return a >= b ? a - b : a + MD - b;
    }
    static unsigned long long pw(unsigned long long b, unsigned long
    ↪ long e) {
        unsigned long long r = 1;
        for (; e >= 1, b = mul(b, b)) if (e & 1) r = mul(r, b);
        return r;
    }
    int n;
    string s;
    ↪ s[1..n]
    vector<unsigned long long> P, IP, ft[2];
    DynHash(const string& in) {
        n = in.size(), s = string(n + 1, 0);
        ↪ unused; filled by upd below
        unsigned long long base = 131 + (unsigned long
        ↪ long)chrono::steady_clock::now()
        .time_since_epoch().count()
        ↪ % 1000000;
        P.resize(n + 2), IP.resize(n + 2), ft[0].assign(n + 1, 0),
        ↪ ft[1].assign(n + 1, 0);
        P[0] = IP[0] = 1;
        unsigned long long ib = pw(base, MD - 2);
        for (int i = 1; i <= n + 1; i++) P[i] = mul(P[i - 1], base),
        ↪ IP[i] = mul(IP[i - 1], ib);
        for (int i = 1; i <= n; i++) upd(i, in[i - 1]);
    }
    void add(int k, int i, unsigned long long v) {
        for (; i <= n; i += i & -i) ft[k][i] = ad(ft[k][i], v);
    }
    unsigned long long qr(int k, int i) {
        unsigned long long r = 0;
        for (; i > 0; i -= i & -i) r = ad(r, ft[k][i]);
        return r;
    }
    unsigned long long rng(int k, int l, int r) { return sb(qr(k, r),
    ↪ qr(k, l - 1)); }

```

```

void upd(int i, char c) { // set
    ↪ s[i] = c
    unsigned long long d = sb(c, s[i]);
    add(0, i, mul(d, P[i])), add(1, i, mul(d, P[n + 1 - i]));
    s[i] = c;
}
unsigned long long get(int l, int r) { return mul(rng(0, l, r),
    ↪ IP[l]); }
unsigned long long rev(int l, int r) { return mul(rng(1, l, r),
    ↪ IP[n + 1 - r]); }
bool isPal(int l, int r) { return get(l, r) == rev(l, r); }
int oddRadius(int c) { //
    ↪ longest s[c-k..c+k] palindrome
    int lo = 0, hi = min(c - 1, n - c), res = 0;
    while (lo <= hi) {
        int m = (lo + hi) / 2;
        isPal(c - m, c + m) ? (res = m, lo = m + 1) : hi = m - 1;
    }
    return res; //
    ↪ length 2 * res + 1
}
int evenRadius(int c) { //
    ↪ longest s[c-k+1..c+k] palindrome
    int lo = 1, hi = min(c, n - c), res = 0;
    while (lo <= hi) {
        int m = (lo + hi) / 2;
        isPal(c - m + 1, c + m) ? (res = m, lo = m + 1) : hi = m -
    ↪ 1;
    }
    return res; //
    ↪ length 2 * res
}
}
/* NOTES

```

THE SCALING IS THE WHOLE GAME. Forward stores $s[i] * p^i$, so the raw
 ↪ range sum of $[l, r]$ is
 p^l times the hash of the substring - multiply by p^{-l} . Backward
 ↪ stores $s[i] * p^{(n+1-i)}$, whose
 range sum is $p^{(n+1-r)}$ times the reversed substring's hash. Get one
 ↪ exponent wrong and every
 answer is silently false; test `isPal` on a known palindrome first.
 A SEGMENT TREE does the same with range ASSIGN ("set $s[l..r]$ to c ")
 ↪ lazily, which a Fenwick
 cannot: keep (hash, length) per node and lazily set $hash = c * p^{len}$
 ↪ $(p^{len} - 1) / (p - 1)$.
 SUBSTRING COMPARISON / SORTING under updates: binary search the first
 ↪ differing position with
`get()`, $O(\log^2 n)$ per comparison - the standard "compare two
 ↪ suffixes with updates".
 LONGEST COMMON EXTENSION under updates is the same binary search, and
 ↪ is what makes this a
 drop-in replacement for a suffix array in dynamic problems.
 ANTI-HASH: the base is randomised at construction. Never hard-code a
 ↪ base in an open-hacking
 contest; with a fixed base and mod $2^{61}-1$ the string is still
 ↪ forgeable by a prepared test. */

7.3 Aho Corasick

7.3.1 AhoCorasick

AHO-CORASICK - a trie of all patterns plus suffix (fail) links, so one pass over the text finds

```

// every occurrence of every pattern in  $O(|text| + \text{total pattern length} + \#matches)$ .
// The goto-automaton form (nxt filled in by BFS) is also the
↪ transition table for DP over strings.

```

```

struct Mapper {
    int operator()(char c) const { return c - 'a'; }
};

```

```

template <int ALPHABET, typename Mapper>

```

```

struct Aho {
    struct Node {
        array<int, ALPHABET> nxt;
        int link = 0;
        int terminal_idx = -1; // -1 if not a terminal, else idx of
    ↪ string insertion
        int exit_link = 0; // Points to the nearest terminal suffix
        int depth = 0; // Size of the prefix the node represents
        Node() { nxt.fill(0); }
    };
    vector<Node> t;
    vector<int> bfs_order;
    Mapper get_idx; // Instance of the mapping function
    int word_count = 0; // Tracks the order of inserted strings

```

```

// Constructor optionally accepts a custom mapper instance
Aho(Mapper mapper = Mapper()) : get_idx(mapper) {
    t.emplace_back();
}

int insert(const string& p) {
    int u = 0;
    for (char c : p) {
        int v = get_idx(c);
        if (!t[u].nxt[v]) {
            t[u].nxt[v] = t.size();
            t.emplace_back();
            t.back().depth = t[u].depth + 1; // New node's depth is
            ↪ parent's depth + 1
        }
        u = t[u].nxt[v];
    }

    // If this exact string hasn't been inserted before, assign it
    ↪ the current index
    if (t[u].terminal_idx == -1) {
        t[u].terminal_idx = word_count;
    }
    word_count++; // Increment for the next insertion

    return u;
}

void build() {
    queue<int> q;
    for (int c = 0; c < ALPHABET; ++c) {
        if (t[0].nxt[c]) {
            q.push(t[0].nxt[c]);
        }
    }

    while (!q.empty()) {
        int u = q.front();
        q.pop();
        bfs_order.push_back(u);

        for (int c = 0; c < ALPHABET; ++c) {
            int v = t[u].nxt[c];
            if (v) {
                t[v].link = t[t[u].link].nxt[c];

                // Compute the exit link:
                // If the immediate suffix link is a terminal,
                ↪ point to it.
                // Otherwise, copy its exit link.
                t[v].exit_link = (t[t[v].link].terminal_idx != -1)
                ↪ ? t[v].link : t[t[v].link].exit_link;

                q.push(v);
            }
            else {
                t[u].nxt[c] = t[t[u].link].nxt[c];
            }
        }
    }

    int advance(int u, char c) {
        return t[u].nxt[get_idx(c)];
    }
};

```

7.3.2 AhoCorasickDP

DP OVER THE AHO-CORASICK AUTOMATON - the high-value use of Aho, more than matching itself.

// Needs: Mint (06 - Math/01) and AhoCorasick (01 - AhoCorasick.cpp).
 // After build(), nxt[] is a total transition function, so the
 ↪ automaton is a DFA over states
 // 0..S-1 and you can run any DP on (position, state, extra).
 // "bad" marks states that contain a forbidden pattern as a suffix;
 ↪ propagate it along exit links.

```

struct AhoDP {
    static const int A = 26; // trie fan-out
    int alpha = 26; // alphabet the DP
    ↪ counts over (set smaller!)
    vector<array<int, A>> nxt;
    vector<int> link, term;
    vector<char> bad;
    vector<int> order; // BFS order of
    ↪ states

    AhoDP() { newNode(); }
};

```

```

int newNode() { nxt.push_back({}); nxt.back().fill(0);
    ↪ link.push_back(0); term.push_back(0); bad.push_back(0); return
    ↪ nxt.size() - 1; }

void insert(const string& p, int id = 1) {
    int u = 0;
    for (char c : p) {
        int v = c - 'a';
        if (!nxt[u][v]) nxt[u][v] = newNode();
        u = nxt[u][v];
    }
    term[u] = id, bad[u] = 1;
}

void build() {
    queue<int> q;
    for (int c = 0; c < A; c++) if (nxt[0][c]) q.push(nxt[0][c]);
    while (!q.empty()) {
        int u = q.front(); q.pop();
        order.push_back(u);
        bad[u] = bad[u] || bad[link[u]]; // a state is
        ↪ bad if any SUFFIX is a pattern
        for (int c = 0; c < A; c++) {
            int v = nxt[u][c];
            if (v) link[v] = nxt[link[u]][c], q.push(v);
            else nxt[u][c] = nxt[link[u]][c]; // total
            ↪ transition function
        }
    }

    // #strings of length L over the first 'alpha' letters containing
    ↪ NO pattern. O(L*S*alpha).
    Mint countAvoiding(int L) {
        int S = nxt.size();
        vector<Mint> dp(S, 0), nd;
        dp[0] = 1;
        for (int i = 0; i < L; i++) {
            nd.assign(S, 0);
            for (int u = 0; u < S; u++) if (!bad[u] && dp[u].v)
                for (int c = 0; c < alpha; c++) {
                    int v = nxt[u][c];
                    if (!bad[v]) nd[v] += dp[u];
                }
            dp = nd;
        }
        Mint r = 0;
        for (int u = 0; u < S; u++) if (!bad[u]) r += dp[u];
        return r;
    }

    // #occurrences of every pattern in a text: walk the text, count
    ↪ visits per state, then push
    // the counts DOWN the suffix-link tree (reverse BFS order).
    vector<ll> occurrences(const string& s) {
        vector<ll> cnt(nxt.size(), 0);
        int u = 0;
        for (char c : s) u = nxt[u][c - 'a'], cnt[u]++;
        for (int i = order.size() - 1; i >= 0; i--) cnt[link[order[i]]]
        ↪ += cnt[order[i]];
        return cnt; // cnt[state of
        ↪ pattern p] = #occurrences
    }
};

// SHORTEST / LEXICOGRAPHICALLY SMALLEST string avoiding all patterns:
    ↪ BFS over (state) with
    // the alphabet in order. If a cycle among good states is reachable,
    ↪ arbitrarily long strings
    // exist - detect it with a cycle check on the good-state subgraph.

```

7.3.3 DynamicAhoCorasick

DYNAMIC (ONLINE) AHO-CORASICK - the pattern set grows while queries arrive. A plain

// Aho-Corasick must be rebuilt from scratch after every insertion,
 ↪ which is O(total length) per
 // insert; here the patterns are kept in O(log P) STATIC automata of
 ↪ sizes 1, 2, 4, 8, ... and
 // inserting merges the small ones the way binary counter increments
 ↪ carry.
 // AMORTISED O(|p| log P) per insert, O(|s| log P) per query (log P <=
 ↪ ~20 automata to feed).
 // WHEN: "add a word, then count occurrences of all words in this
 ↪ text", online dictionary
 // problems (CF 710F), and any offline-impossible interleaving of
 ↪ insert and query.
 // DELETION: keep a SECOND identical structure of deleted patterns and
 ↪ subtract its answer.
 // INDEX CONVENTION: node 0 is the root; cnt[v] is already the SUM over
 ↪ the whole suffix-link
 // chain, so a query adds cnt[cur] once per text position and never
 ↪ walks exit links.

```

struct StaticAho {
    ↪ rebuilt wholesale, never patched
    vector<array<int, 26>> nxt;
    vector<int> link, cnt;
    StaticAho() { clear(); }
    void clear() { nxt.assign(1, {}), nxt[0].fill(0), link.assign(1,
    ↪ 0), cnt.assign(1, 0); }
    void insert(const string& p) {
        int u = 0;
        for (char ch : p) {
            int c = ch - 'a';
            if (!nxt[u][c]) {
                nxt[u][c] = nxt.size();
                nxt.push_back({}), nxt.back().fill(0),
    ↪ link.push_back(0), cnt.push_back(0);
            }
            u = nxt[u][c];
        }
        cnt[u]++;
    }
    void build() {
    ↪ goto-automaton + cumulative cnt
        queue<int> q;
        for (int c = 0; c < 26; c++) if (nxt[0][c]) q.push(nxt[0][c]);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            cnt[u] += cnt[link[u]];
            for (int c = 0; c < 26; c++) {
                int v = nxt[u][c];
                if (v) link[v] = nxt[link[u]][c], q.push(v);
                else nxt[u][c] = nxt[link[u]][c];
            }
        }
        ll count(const string& s) const {
    ↪ occurrences of all patterns
            ll r = 0;
            int u = 0;
            for (char ch : s) u = nxt[u][ch - 'a'], r += cnt[u];
            return r;
        }
};

struct DynamicAho {
    static const int B = 20;
    ↪ supports 2^20 patterns
    vector<string> words[B];
    StaticAho aho[B];
    void insert(const string& p) {
        int lvl = 0;
        while (lvl < B && !words[lvl].empty()) lvl++;
    ↪ carry of a binary counter
        words[lvl].push_back(p);
        for (int i = 0; i < lvl; i++) {
            for (auto& w : words[i]) words[lvl].push_back(w);
            words[i].clear(), aho[i].clear();
        }
        aho[lvl].clear();
        for (auto& w : words[lvl]) aho[lvl].insert(w);
        aho[lvl].build();
    }
    ll count(const string& s) {
    ↪ occurrences of all patterns in s
        ll r = 0;
        for (int i = 0; i < B; i++) if (!words[i].empty()) r +=
    ↪ aho[i].count(s);
        return r;
    }
};

/* NOTES AND VARIANTS
WHY THE COST IS AMORTISED: a pattern is copied into a bigger automaton
    ↪ at most log P times, so
    the total rebuild work is O(sum |p| * log P) - exactly the same
    ↪ argument as binary lifting on
    a knapsack, or as the "logarithmic method" for making any static
    ↪ structure insert-only.
THE SAME WRAPPER makes ANY static structure dynamic (suffix arrays,
    ↪ convex hulls, kd-trees):
    keep buckets of size 2^i and merge on carry. That is the reusable
    ↪ idea here.
PER-PATTERN counts instead of a total: store an id per terminal and
    ↪ walk the exit-link chain, or
    put the query positions on the fail tree of each bucket - the total
    ↪ stays O(matches log P).
MEMORY: 26 ints per node in every bucket. With big alphabets use a
    ↪ hash map per node or the
    sorted-children trick, otherwise 20 rebuilt copies will not fit.
IF ALL PATTERNS ARE KNOWN OFFLINE, do not use this - build one static

```

↪ automaton. */

7.3.4 PatternContainmentPoset

ALL-PAIR OCCURRENCE RELATION between the patterns themselves: for every ordered pair (i, j),

↪ does pattern i occur as a SUBSTRING of pattern j? That relation is a
 ↪ partial order (a poset),
 ↪ and problems then ask for its longest chain, its maximum antichain
 ↪ (Dilworth = minimum path
 ↪ cover = k - maximum bipartite matching, see 03 - Graph Theory), or
 ↪ its transitive reduction.
 ↪ CF 590E "Birthday" is the canonical use: pick the largest subset
 ↪ with no containments.
 ↪ NAIVE is O(k^2 * L) with a matcher per pair. This is O(sum |p| * A)
 ↪ for the automaton plus
 ↪ O(k^3 / 64) for the closure, and the closure is the whole trick:
 ↪ during one pass over p_j you record ONLY the NEAREST terminal suffix
 ↪ at each position (the
 ↪ exit link). Everything else follows by TRANSITIVITY, because "occurs
 ↪ in" is transitive.
 ↪ INDEX CONVENTION: patterns are 0-indexed in insertion order and must
 ↪ be DISTINCT (dedup first,
 ↪ otherwise p_i occurs in p_j and p_j in p_i and the poset
 ↪ degenerates).
 ↪ Needs: 07 - Strings/03 - Aho Corasick/01 - AhoCorasick.cpp

```

template <int K>
    ↪ K = compile-time
vector<bitset<K>> containment(const vector<string>& p) {
    Aho<26, Mapper> ac;
    vector<int> node(p.size());
    for (size_t i = 0; i < p.size(); i++) node[i] = ac.insert(p[i]);
    ac.build();
    int k = p.size();
    vector<bitset<K>> in(k);
    ↪ in[j][i] = p_i occurs in p_j
    for (int j = 0; j < k; j++) {
        int u = 0;
        for (char c : p[j]) {
            u = ac.advance(u, c);
            ↪ the nearest terminal suffix - skipping p_j itself, which
            ↪ ends at its own last
            ↪ position and would otherwise hide the shorter terminal
            ↪ suffixes underneath it
            int tu = ac.t[u].terminal_idx;
            int v = (tu != -1 && tu != j) ? u : ac.t[u].exit_link;
            if (v && ac.t[v].terminal_idx != -1 && ac.t[v].terminal_idx
    ↪ != j)
                in[j][ac.t[v].terminal_idx] = 1;
        }
    }
    vector<int> ord(k);
    ↪ shorter patterns first: a proper
    iota(ord.begin(), ord.end(), 0);
    ↪ substring is strictly shorter
    sort(ord.begin(), ord.end(), [&](int a, int b) { return p[a].size()
    ↪ < p[b].size(); });
    for (int j : ord) {
    ↪ transitive closure
        auto cur = in[j];
        for (int i = 0; i < k; i++) if (cur[i] && in[i])
            return in;
    }
}

/* USING THE POSET
LARGEST SUBSET WITH NO CONTAINMENT (maximum antichain) = k - (minimum
    ↪ chain cover), and by
    ↪ Dilworth on the TRANSITIVELY CLOSED relation, minimum chain cover =
    ↪ k - maximum matching in
    the bipartite graph "i on the left, j on the right, edge if p_i
    ↪ occurs in p_j". Recover the
    actual antichain from the Konig vertex cover: keep the vertices NOT
    ↪ in the cover.
LONGEST CHAIN = longest path in the DAG = the usual O(k^2) DP on the
    ↪ closed relation.
COUNTING occurrences instead of deciding: replace the bitset with a
    ↪ fail-tree subtree sum -
    push +1 at every visited node of p_j and read the subtree sum of
    ↪ each terminal, O((L + k) log).
MEMORY: k * K / 8 bytes. k = 750 (the CF 590E bound) is 70 KB; k = 1e5
    ↪ is not viable, and at
    that size the question is always something the fail tree answers
    ↪ directly.
IF THE PATTERNS REPEAT, dedup and remember the multiplicity; the poset
    ↪ is on DISTINCT strings. */

```

7.4 Suffix Structures

7.4.1 SuffixArray

SUFFIX ARRAY + LCP + sparse table for $O(1)$ LCE. Build is $O(n \log n)$ (prefix doubling with a radix

// sort), Kasai gives the LCP array in $O(n)$. Everything you can ask
 ↳ about substrings reduces to a

// range on this array - see 03 - SuffixArrayApplications.cpp.

```
struct SuffixArray {
    string S;
    // sa is the suffix array with the empty suffix being sa[0]
    // lcp[i] holds the lcp between sa[i], sa[i - 1]
    vector<int> logs, sa, lcp, rank;
    vector<vector<int>> table;

    SuffixArray() {}

    SuffixArray(string &s, int lim = 256) {
        S = s;
        int n = s.size() + 1, k = 0, a, b;
        vector<int> c(s.begin(), s.end() + 1), tmp(n), frq(max(n, lim));
        c.back() = 0; // 0 is less than any character
        sa = lcp = rank = tmp, iota(sa.begin(), sa.end(), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
            p = j, iota(tmp.begin(), tmp.end(), n - j);
            for (int i = 0; i < n; i++) {
                if (sa[i] >= j)
                    tmp[p++] = sa[i] - j;
            }
            fill(frq.begin(), frq.end(), 0);

            for (int i = 0; i < n; i++) frq[c[i]]++;
            for (int i = 1; i < lim; i++) frq[i] += frq[i - 1];
            for (int i = n; i--;) sa[--frq[c[tmp[i]]]] = tmp[i];

            swap(c, tmp), p = 1, c[sa[0]] = 0;
            for (int i = 1; i < n; i++)
                a = sa[i - 1], b = sa[i], c[b] = (tmp[a] == tmp[b] &&
                ↳ tmp[a + j] == tmp[b + j]) ? p - 1 : p++;
        }

        for (int i = 1; i < n; i++) rank[sa[i]] = i;
        for (int i = 0, j; i < n - 1; lcp[rank[i++]] = k)
            for (k && k--, j = sa[rank[i] - 1];
                s[i + k] == s[j + k];
                k++);

        void preLcp() {
            int n = S.size() + 1;
            logs = vector<int>(n + 5);
            for (int i = 2; i < n + 5; ++i) {
                logs[i] = logs[i / 2] + 1;
            }
            // level-major: sized by logs[n], not a hard-coded 20 (which is
            ↳ OUT OF BOUNDS at
            // |s| >= 2^20 - and Codeforces routinely gives |s| up to 1e6),
            ↳ and no dead columns.
            table = vector<vector<int>>(logs[n] + 1, vector<int>(n));
            for (int i = 0; i < n; ++i) table[0][i] = lcp[i];
            for (int j = 1; j <= logs[n]; ++j)
                for (int i = 0; i + (1 << j) <= n; ++i)
                    table[j][i] = min(table[j - 1][i], table[j - 1][i + (1
                    ↳ << (j - 1))]);
        }

        // ARGUMENTS ARE SUFFIX-ARRAY INDICES, not string positions. For
        ↳ two string positions
        // p and q use lce(p, q) below, which is what every application
        ↳ actually wants.
        int queryLcp(int i, int j) {
            if (i == j) return (int)S.size() - sa[i];
            if (i > j) swap(i, j);
            i++;
            int len = logs[j - i + 1];
            return min(table[len][i], table[len][j - (1 << len) + 1]);
        }
        int lce(int p, int q) { return queryLcp(rank[p], rank[q]); } //
        ↳ longest common extension
    };
};
```

7.4.2 SuffixAutomaton

SUFFIX AUTOMATON - $O(n)$ states/transitions, accepts exactly the substrings of s.

```
// st[v].len = longest string in v's endpos class link = suffix
↳ link (v's parent)
// cnt[v] = |endpos(v)| = #occurrences of every string in v
// distinct substrings = sum over v != root of (len[v] - len[link[v]])
struct SAM {
    struct S { int len = 0, link = -1; array<int, 26> nxt; S() {
        ↳ nxt.fill(-1); } };
    vector<S> st;
    vector<ll> cnt; // occurrence
    ↳ counts (after countOcc)
    vector<int> firstPos; // end position of
    ↳ the first occurrence
    int last = 0;
    bool counted = false; // countOcc() is
    ↳ idempotent

    SAM(int reserve = 0) { st.reserve(2 * reserve), st.push_back(S()),
    ↳ cnt.push_back(0), firstPos.push_back(0); }

    void extend(char ch) {
        int c = ch - 'a', cur = st.size();
        st.push_back(S()), cnt.push_back(1), firstPos.push_back(0);
        st[cur].len = st[last].len + 1, firstPos[cur] = st[cur].len - 1;
        int p = last;
        while (p != -1 && st[p].nxt[c] == -1) st[p].nxt[c] = cur, p =
        ↳ st[p].link;
        if (p == -1) st[cur].link = 0;
        else {
            int q = st[p].nxt[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;
            else {
                int cl = st.size();
                st.push_back(st[q]), cnt.push_back(0),
                ↳ firstPos.push_back(firstPos[q]);
                st[cl].len = st[p].len + 1;
                while (p != -1 && st[p].nxt[c] == q) st[p].nxt[c] = cl,
                ↳ p = st[p].link;
                st[q].link = st[cur].link = cl;
            }
        }
        last = cur;
    }

    void build(const string& s) { for (char c : s) extend(c); }

    vector<int> byLen() const { // states sorted
        ↳ by len (counting sort)
        int n = st.size(), mx = st[0].len;
        for (auto& x : st) mx = max(mx, x.len);
        vector<int> c(mx + 2, 0), ord(n);
        for (auto& x : st) c[x.len]++;
        for (int i = 1; i <= mx; i++) c[i] += c[i - 1];
        for (int v = n - 1; v >= 0; v--) ord[--c[st[v].len]] = v;
        return ord; // increasing len;
    }
    ↳ reverse for link-tree order
    void countOcc() { // push counts
        ↳ down the suffix-link tree
        if (counted) return; // NOT idempotent
        counted = true;
        auto ord = byLen();
        for (int i = (int)ord.size() - 1; i > 0; i--) {
            int v = ord[i];
            if (st[v].link >= 0) cnt[st[v].link] += cnt[v];
        }
        cnt[0] = 0;
    }

    ll distinctSubstrings() const { // number of
        ↳ DISTINCT non-empty substrings
        ll r = 0;
        for (int v = 1; v < (int)st.size(); v++) r += st[v].len -
        ↳ st[st[v].link].len;
        return r;
    }

    ll totalOccurrences() { // sum over
        ↳ distinct substrings of #occurrences
        countOcc();
        ll r = 0;
        for (int v = 1; v < (int)st.size(); v++) r += cnt[v] *
        ↳ (st[v].len - st[st[v].link].len);
        return r;
    }

    int countOf(const string& p) { // #occurrences of
        ↳ p in s (call countOcc first)
        int v = 0;
```

```

    for (char ch : p) { v = st[v].nxt[ch - 'a']; if (v < 0) return
    ↪ 0; }
    return cnt[v];
}
// K-th SMALLEST distinct substring (k is 1-indexed). Needs 'paths'
↪ = #substrings from v.
vector<ll> paths;
void buildPaths() {
    paths.assign(st.size(), 1); // 1 counts the
    ↪ empty continuation
    auto ord = byLen();
    for (int i = (int)ord.size() - 1; i >= 0; i--) {
        int v = ord[i];
        paths[v] = 1;
        for (int c = 0; c < 26; c++) if (st[v].nxt[c] >= 0)
            ↪ paths[v] += paths[st[v].nxt[c]];
    }
}
string kthSubstring(ll k) { // "" if fewer
    ↪ than k distinct substrings
    string r;
    int v = 0;
    while (k > 0) {
        int c = 0;
        for (; c < 26; c++) {
            int u = st[v].nxt[c];
            if (u < 0) continue;
            if (k <= paths[u]) { r += char('a' + c), v = u, k--;
            ↪ break; }
            k -= paths[u];
        }
        if (c == 26) break;
    }
    return r;
}
};
// LONGEST COMMON SUBSTRING of s and t: build the SAM of s, then walk t
↪ keeping the current
// length; on a missing transition follow suffix links and clamp the
↪ length to st[v].len.
int longestCommonSubstring(SAM& A, const string& t) {
    int v = 0, l = 0, best = 0;
    for (char ch : t) {
        int c = ch - 'a';
        while (v && A.st[v].nxt[c] < 0) v = A.st[v].link, l =
        ↪ A.st[v].len;
        if (A.st[v].nxt[c] >= 0) v = A.st[v].nxt[c], l++;
        best = max(best, l);
    }
    return best;
}
// GENERALISED SAM (several strings): reset 'last = 0' before each
↪ string and, in extend(),
// if a transition already exists reuse/clone it instead of creating a
↪ new state.

```

7.4.3 SuffixArrayApplications

WHAT YOU ACTUALLY EXTRACT FROM A SUFFIX ARRAY.

```

// Convention of 01 - SuffixArray.cpp: sa has n+1 entries (sa[0] = the
↪ empty suffix),
// and lcp[i] = LCP(sa[i], sa[i-1]). Everything below assumes that.

// 1) NUMBER OF DISTINCT SUBSTRINGS = sum over i of (len of suffix i) -
↪ lcp with the previous.
ll distinctSubstrings(const SuffixArray& S) {
    ll n = S.S.size(), r = 0;
    for (ll i = 1; i <= n; i++) r += (n - S.sa[i]) - S.lcp[i];
    return r;
}
// 2) LONGEST REPEATED SUBSTRING = max lcp; the substring starts at
↪ sa[argmax].
pair<int, int> longestRepeated(const SuffixArray& S) { //
    ↪ {length, start position}
    int n = S.S.size(), bi = 0, best = 0;
    for (int i = 1; i <= n; i++) if (S.lcp[i] > best) best = S.lcp[i],
    ↪ bi = S.sa[i];
    return {best, bi};
}
// 3) OCCURRENCES of a pattern = the contiguous SA block whose suffixes
↪ start with it.
// Two binary searches, O(|P| log n).
pair<int, int> patternRange(const SuffixArray& S, const string& p) {
    ↪ // [lo, hi], empty if lo>hi
    int n = S.S.size();
    auto cmp = [&](int idx, const string& q) { return S.S.compare(idx,
    ↪ q.size(), q) < 0; };
    int lo = 1, hi = n, L = n + 1, R = n;

```

```

    while (lo <= hi) { int m = (lo + hi) / 2; cmp(S.sa[m], p) ? lo = m
    ↪ + 1 : (L = m, hi = m - 1); }
    lo = L, hi = n, R = L - 1; // R
    ↪ MUST be reset here:
    //
    ↪ otherwise an absent pattern
    //
    ↪ returns the range [L, n]
    while (lo <= hi) {
        int m = (lo + hi) / 2;
        S.S.compare(S.sa[m], p.size(), p) == 0 ? (R = m, lo = m + 1) :
        ↪ hi = m - 1;
    }
    return {L, R}; // count
    ↪ = R - L + 1
}
// 4) LONGEST COMMON SUBSTRING of two strings (named lcsViaSuffixArray
↪ to avoid clashing
// with the SAM version in 02 - SuffixAutomaton.cpp): build SA of a
↪ + sep + b (sep smaller than every
// real character), then take the max lcp over ADJACENT pairs coming
↪ from different sides.
int lcsViaSuffixArray(const string& a, const string& b) {
    string t = a + char(1) + b;
    SuffixArray S(t);
    int n = t.size(), best = 0, cut = a.size();
    auto side = [&](int p) { return p < cut ? 0 : (p == cut ? -1 : 1);
    ↪ };
    for (int i = 2; i <= n; i++) {
        int s1 = side(S.sa[i]), s2 = side(S.sa[i - 1]);
        if (s1 >= 0 && s2 >= 0 && s1 != s2) best = max(best, S.lcp[i]);
    }
    return best;
}
// 5) K-TH SMALLEST DISTINCT SUBSTRING: SA entry i contributes (n -
↪ sa[i]) - lcp[i] new
// substrings, all sharing the prefix of length lcp[i]. Prefix-sum
↪ those counts and binary
// search for k; inside the block the answer is a prefix of suffix
↪ sa[i].
// 6) LCS OF K STRINGS: concatenate with distinct separators, then
↪ slide a window over the SA
// that contains at least one suffix of every string; the answer is
↪ the max over windows of
// the minimum lcp inside (monotonic deque), O(n) after the SA.
// 7) COMPARE TWO SUBSTRINGS in O(1): let l = S.lcp(i, j) (STRING
↪ positions - queryLcp takes
// SUFFIX-ARRAY indices, do not mix them up); if l >= both
// lengths they are equal, otherwise compare s[i+l] with s[j+l].

```

7.4.4 GeneralizedSAM

GENERALIZED SUFFIX AUTOMATON - one automaton accepting every substring of a SET of strings.

```

// Same size bound O(total length). The only change vs a plain SAM:
↪ when extending, the transition
// may ALREADY exist (a previous string walked here), so handle that
↪ case instead of blindly adding.
// Feeding strings one after another into a plain SAM is WRONG - it
↪ creates junk states.
struct GenSAM {
    struct S { int len = 0, link = -1; array<int, 26> nxt; S() {
        ↪ nxt.fill(-1); } };
    vector<S> st;
    GenSAM() { st.push_back(S()); }
    int clone(int q, int len) { int c = st.size(); st.push_back(st[q]),
    ↪ st[c].len = len; return c; }
    int extend(int last, int c) { //
        ↪ returns the new "last"
        if (st[last].nxt[c] != -1) { // the
            ↪ edge is already there
            int q = st[last].nxt[c];
            if (st[q].len == st[last].len + 1) return q;
            int cl = clone(q, st[last].len + 1);
            for (int p = last; p != -1 && st[p].nxt[c] == q; p =
            ↪ st[p].link) st[p].nxt[c] = cl;
            st[q].link = cl;
            return cl;
        }
        int cur = st.size();
        st.push_back(S()), st[cur].len = st[last].len + 1;
        int p = last;
        while (p != -1 && st[p].nxt[c] == -1) st[p].nxt[c] = cur, p =
        ↪ st[p].link;
        if (p == -1) st[cur].link = 0;
        else {
            int q = st[p].nxt[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;

```

```

    else {
        int cl = clone(q, st[p].len + 1);
        for (; p != -1 && st[p].nxt[c] == q; p = st[p].link)
            st[p].nxt[c] = cl;
        st[q].link = st[cur].link = cl;
    }
    return cur;
}

void add(const string& s) { int last = 0; for (char ch : s) last =
    extend(last, ch - 'a'); }
vector<int> byLen() const {
    // states in increasing len
    int n = st.size(), mx = 0;
    for (auto& x : st) mx = max(mx, x.len);
    vector<int> c(mx + 2, 0), ord(n);
    for (auto& x : st) c[x.len]++;
    for (int i = 1; i <= mx; i++) c[i] += c[i - 1];
    for (int v = n - 1; v >= 0; v--) ord[--c[st[v].len]] = v;
    return ord;
}

// HOW MANY OF THE k STRINGS CONTAIN each substring: mark the
// states each string walks through,
// then push the marks up the link tree. k <= 64 here; for larger k
// use small-to-large sets, or
// a bitset<K>, or process the strings in batches of 64 and add up
// the popcounts.
vector<int> countStringsContaining(const vector<string>& v) {
    // requires v.size() <= 64
    vector<unsigned long long> mask(st.size(), 0);
    for (int i = 0; i < (int)v.size(); i++)
        for (int p = 0, j = 0; j < (int)v[i].size(); j++)
            p = st[p].nxt[v[i][j] - 'a'], mask[p] |= 1ULL << i;
    auto ord = byLen();
    for (int i = ord.size() - 1; i > 0; i--)
        mask[st[ord[i]].link] |= mask[ord[i]];
    vector<int> r(st.size());
    for (size_t i = 0; i < st.size(); i++) r[i] =
        __builtin_popcountll(mask[i]);
    return r;
}

// for every state v
};

// LONGEST COMMON SUBSTRING OF k STRINGS: build the generalized SAM,
// take the deepest state with
// count == k; its longest string is len[v] (walk suffix links for the
// actual string).
// DISTINCT SUBSTRINGS OF THE UNION = sum over v != 0 of (len[v] -
// len[link[v]]), same formula.
// SUBSTRING PRESENT IN EXACTLY ONE STRING / IN AT LEAST k: filter by
// the same counts.
// ALTERNATIVE when the strings are few: concatenate with distinct
// separators and use one SUFFIX
// ARRAY + a sliding window over lcp. Shorter to write, O(n log n), but
// no online extension.

```

7.4.5 SuffixArraySAIS

SUFFIX ARRAY IN O(n) - SA-IS. Use it instead of 01 - SuffixArray.cpp (O(n log n) doubling)

```

// when n is 1e6 or more, when the input is a vector<int> with a huge
// alphabet (doubling needs a
// radix sort per round), or when the doubling version is the
// bottleneck of a heavy solution.
// It is ~4x faster than doubling at n = 1e6 and needs no sorting of
// pairs at all.
// INDEX CONVENTION - DIFFERENT FROM 01 - SuffixArray.cpp, and mixing
// them up is a wrong answer:
// sa has EXACTLY n entries (no empty suffix); sa[i] = start of the
// i-th smallest suffix.
// lcp has n entries with lcp[0] = 0 and lcp[i] = LCP(sa[i-1], sa[i]).
// HOW IT WORKS (enough to debug it): classify every position as S-type
// (its suffix is smaller
// than the next one) or L-type. The LMS positions (an S right after an
// L) split the string into
// blocks; sort those blocks by recursing on a shorter string, then
// INDUCE the whole order from
// them with two linear scans. The recursion halves the length, so the
// total is O(n).
vector<int> saIs(const vector<int>& s, int up) {
    // values in [0, up], up >= 0
    int n = s.size();
    if (n == 0) return {};
    if (n == 1) return {0};
    if (n == 2) return s[0] < s[1] ? vector<int>{0, 1} : vector<int>{1, 0};
    vector<int> sa(n);
    vector<char> ls(n, 0);
    // 1 =

```

```

    S-type
    for (int i = n - 2; i >= 0; i--) ls[i] = s[i] == s[i + 1] ? ls[i + 1] : s[i] < s[i + 1];
    vector<int> sumL(up + 2, 0), sumS(up + 2, 0);
    for (int i = 0; i < n; i++) ls[i] ? sumL[s[i] + 1]++ : sumS[s[i]]++;
    for (int i = 0; i <= up; i++) { sumS[i] += sumL[i]; if (i < up)
        sumL[i + 1] += sumS[i]; }
    auto induce = [&](const vector<int>& lms) {
        fill(sa.begin(), sa.end(), -1);
        vector<int> buf = sumS;
        for (int d : lms) if (d != n) sa[buf[s[d]]++] = d;
        buf = sumL;
        sa[buf[s[n - 1]]++] = n - 1;
        for (int i = 0; i < n; i++) {
            int v = sa[i];
            if (v >= 1 && !ls[v - 1]) sa[buf[s[v - 1]]++] = v - 1;
        }
        buf = sumL;
        for (int i = n - 1; i >= 0; i--) {
            int v = sa[i];
            if (v >= 1 && ls[v - 1]) sa[--buf[s[v - 1] + 1]] = v - 1;
        }
    };
    vector<int> lmsMap(n + 1, -1), lms;
    int m = 0;
    for (int i = 1; i < n; i++) if (!ls[i - 1] && ls[i]) lmsMap[i] =
        m++, lms.push_back(i);
    induce(lms);
    if (m) {
        vector<int> sorted;
        for (int v : sa) if (lmsMap[v] != -1) sorted.push_back(v);
        vector<int> rec(m);
        int upRec = 0;
        rec[lmsMap[sorted[0]]] = 0;
        for (int i = 1; i < m; i++) {
            int l = sorted[i - 1], r = sorted[i];
            int el = lmsMap[l] + 1 < m ? lms[lmsMap[l] + 1] : n;
            int er = lmsMap[r] + 1 < m ? lms[lmsMap[r] + 1] : n;
            bool same = el - l == er - r;
            if (same) {
                while (l < el && s[l] == s[r]) l++, r++;
                if (l == n || s[l] != s[r]) same = false;
            }
            if (!same) upRec++;
            rec[lmsMap[sorted[i]]] = upRec;
        }
        vector<int> recSa = saIs(rec, upRec);
        for (int i = 0; i < m; i++) sorted[i] = lms[recSa[i]];
        induce(sorted);
    }
    return sa;
}

vector<int> suffixArray(const string& s) {
    // n
    // entries, no empty suffix
    vector<int> a(s.begin(), s.end());
    for (int& x : a) x = (unsigned char)x;
    return saIs(a, 255);
}

// KASAI: lcp[i] = LCP(suffix sa[i-1], suffix sa[i]), lcp[0] = 0. O(n).
vector<int> kasai(const string& s, const vector<int>& sa) {
    int n = s.size(), k = 0;
    vector<int> rnk(n), lcp(n, 0);
    for (int i = 0; i < n; i++) rnk[sa[i]] = i;
    for (int i = 0; i < n; i++) {
        if (!rnk[i]) { k = 0; continue; }
        int j = sa[rnk[i] - 1];
        while (i + k < n && j + k < n && s[i + k] == s[j + k]) k++;
        lcp[rnk[i]] = k;
        if (k) k--;
    }
    return lcp;
}

/* NOTES
GENERAL ALPHABETS: call saIs directly on a vector<int> after
    coordinate-compressing the values
    to [0, up]; every value must be >= 0 and the LAST element does NOT
    have to be a sentinel -
    this implementation handles the end of string internally.
CONCATENATING MANY STRINGS (generalised suffix array): join with
    distinct separators SMALLER
    than every real character, or - better with SA-IS - compress to ints
    and use 0, 1, 2, ... as
    separators, shifting real characters up.
EVERYTHING ELSE (distinct substrings, longest repeated, pattern range,
    LCE sparse table) is in
    03 - SuffixArrayApplications.cpp; shift its formulas by one because
    that file's sa[0] is the

```

empty suffix and this one has no empty suffix.
 MEMORY is $\sim 3n$ ints at the top level plus the recursion, which sums to
 $\hookrightarrow < 2x$ that. Doubling needs
 5 arrays of n and a radix sort - SA-IS also wins on memory.
 DC3 / SKEW is the other linear construction; it is shorter to write
 $\hookrightarrow$ but slower in practice and
 needs $3n$ extra memory, so SA-IS is the one to carry. */

7.4.6 IsomorphicSuffixArray

PARAMETERIZED / ISOMORPHIC SUFFIX ARRAY. Two strings are ISOMORPHIC (p-match) when some

```
// Needs: 07 - Strings/04 - Suffix Structures/05 - SuffixArraySAIS.cpp
// (saIs, kasai)
// BIJECTION of the alphabet turns one into the other: "abab" ~ "cdcd"
// ~ "baba", but not "aabb".
// WHEN: "how many substrings of s are isomorphic to s[l..r]",
// ~ detecting renamed patterns,
// register-allocation style matching. Plain hashing cannot do this -
// ~ the equality is not on
// characters but on the PATTERN OF REPEATS.
// THE ENCODING that makes it work: a[i] = (i - previous position of
// ~ s[i]) + 1, or 1 when s[i] is
// the first occurrence of its character. Two substrings are isomorphic
// ~ iff their encodings match
// AFTER CAPPING: at offset L inside a substring the value is a[x] if
// ~ a[x] <= L + 1, else 1 (a
// reference that points outside the window becomes "first
// ~ occurrence"). The cap is why this is
// not just an ordinary suffix array on a[].
// INDEX CONVENTION: sa has n entries, no empty suffix; sa[i] = start
// ~ of the i-th smallest suffix
// UNDER ISOMORPHISM, and lcp[i] = longest common ISOMORPHIC prefix of
// ~ sa[i] and sa[i+1] (note:
// indexed by the LEFT of the pair, unlike 01 - SuffixArray.cpp - one
// ~ entry per adjacent pair).
// COMPLEXITY:  $O(n \log n * A)$  worst case ( $A$  = alphabet), because one
// ~ comparison may restart at
// every distinct character; in practice each comparison is a couple of
// ~ jumps.
```

```
struct IsoSuffixArray {
    int n, LG;
    vector<int> a, sa, rnk, lcp, hsa, hrnk, hlcp, lg;
    vector<vector<int>>> sp; //
    ~ sparse table over hlcp
    IsoSuffixArray(const string& s, int A = 26, char base = 'a') {
        n = s.size(), a.resize(n);
        vector<int> last(A, -1);
        for (int i = 0; i < n; i++)
            a[i] = last[s[i] - base] < 0 ? 1 : i - last[s[i] - base] +
        ~ 1, last[s[i] - base] = i;
        hsa = saIs(a, n + 1); //
        ~ helper: plain SA of the encoding
        hrnk.resize(n);
        for (int i = 0; i < n; i++) hrnk[hsa[i]] = i;
        hlcp.assign(n, 0);
        for (int i = 0, k = 0; i < n; i++) { // Kasai
            ~ on the integer array
            if (!hrnk[i]) { k = 0; continue; }
            int j = hsa[hrnk[i] - 1];
            while (i + k < n && j + k < n && a[i + k] == a[j + k]) k++;
            hlcp[hrnk[i]] = k;
            if (k) k--;
        }
        lg.assign(n + 1, 0);
        for (int i = 2; i <= n; i++) lg[i] = lg[i / 2] + 1;
        LG = lg[n] + 1;
        sp.assign(LG, vector<int>(n, 0));
        sp[0] = hlcp;
        for (int k = 1; k < LG; k++)
            for (int i = 0; i + (1 << k) <= n; i++)
                sp[k][i] = min(sp[k - 1][i], sp[k - 1][i + (1 << (k -
        ~ 1))]);
        sa.resize(n, iota(sa.begin(), sa.end(), 0));
        sort(sa.begin(), sa.end(), [&](int i, int j) { return cmp(i,
        ~ j); });
        rnk.resize(n);
        for (int i = 0; i < n; i++) rnk[sa[i]] = i;
        lcp.assign(max(0, n - 1), 0);
        for (int i = 0; i + 1 < n; i++) lcp[i] = lce(sa[i], sa[i + 1]);
    }
    int rawLce(int i, int j) { // lcp
        ~ of a[i..] and a[j..]
        if (i == j) return n - i;
        int l = hrnk[i], r = hrnk[j];
        if (l > r) swap(l, r);
        int k = lg[r - l];
        return min(sp[k][l + 1], sp[k][r - (1 << k) + 1]);
    }
};
```

```
}
int lce(int i, int j) { //
    ~ longest common ISOMORPHIC prefix
    for (int L = 0;;) {
        int x = i + L, y = j + L;
        if (x >= n || y >= n) return L;
        int vx = a[x] > L + 1 ? 1 : a[x], vy = a[y] > L + 1 ? 1 :
        ~ a[y];
        if (vx != vy) return L;
        L += max(1, rawLce(x, y)); // raw
    }
    ~ equality implies iso equality
}
bool cmp(int i, int j) {
    int L = lce(i, j);
    if (i + L >= n) return true;
    if (j + L >= n) return false;
    int vx = a[i + L] > L + 1 ? 1 : a[i + L], vy = a[j + L] > L + 1
    ~ ? 1 : a[j + L];
    return vx < vy;
}
// Number of substrings isomorphic to s[p .. p+len-1]: the SA block
// ~ of suffixes whose
// isomorphic prefix of that length matches.  $O(\log^2 n)$  with the
// ~ lcp array as a min-sparse
// table; the linear scan below is fine when len is large or
// ~ queries are few.
pii occurrences(int p, int len) {
    int lo = rnk[p], hi = rnk[p];
    while (lo > 0 && lcp[lo - 1] >= len) lo--;
    while (hi + 1 < n && lcp[hi] >= len) hi++;
    return {lo, hi}; // count
    ~ = hi - lo + 1
}
};
/* NOTES
```

THE OTHER ENCODINGS you may meet: "previous occurrence index" (0 if
 $\hookrightarrow$ none) is the same idea;
 "order of first appearance" (abcbab $\rightarrow$ 01201) is NOT enough for
 $\hookrightarrow$ substrings, because a substring
 restarts the numbering - that is exactly what the cap fixes here.
 HASHING ALTERNATIVE: hash the capped encoding with a segment tree over
 $\hookrightarrow$ "distance to previous
 occurrence" and answer isomorphism of two substrings in $O(\log n)$ -
 $\hookrightarrow$ shorter if you only need
 EQUALITY tests and not an order.
 STRUCTURAL / ORDER-ISOMORPHISM (the same RELATIVE ORDER of numbers,
 $\hookrightarrow$ not equality pattern) is a
 different problem: encode each position by (rank among the previous
 $\hookrightarrow$ window) and use a similar
 suffix structure, or hash the pair (previous smaller, previous
 $\hookrightarrow$ larger) positions.
 THE COMPARATOR IS NOT TRANSITIVE-SAFE if the cap is wrong; test on
 $\hookrightarrow$ "aabb" vs "bbab" (isomorphic)
 and "aabb" vs "abab" (not) before trusting a modified version. */

7.4.7 DistinctSubstringsInRange

NUMBER OF DISTINCT SUBSTRINGS OF $s[l..r]$, for q queries, OFFLINE in $O((n + q) \log n)$.

```
// Needs: 07 - Strings/04 - Suffix Structures/02 - SuffixAutomaton.cpp
// For the whole string this is one line off a suffix automaton or a
// ~ suffix array; for an
// arbitrary WINDOW it is a genuinely hard problem, and this is the
// ~ standard solution
// (HackerRank "How Many Substrings", CF gym 102129I).
// INDEX CONVENTION: 0-based inclusive queries [l, r]; internally
// ~ everything is 1-based over
// positions 1..n, and LCT node id = SAM state id + 1 (id 0 is the null
// ~ node).
// THE INVARIANT: sweep r from left to right and keep, in a Fenwick
// ~ with range add and range sum,
// val[p] = number of distinct substrings whose LAST occurrence in
// ~ s[1..r] STARTS at p.
// Then the answer for [l, r] is sum of val[l..r] - every distinct
// ~ substring of the window is
// counted at the start position of its last occurrence, which is
// ~ inside the window exactly when
// the substring occurs in the window at all.
// MAINTAINING IT is the trick: appending s[r] creates r new
// ~ occurrences (the suffixes of the
// prefix), so add +1 to every position 1..r, and then REMOVE the
// ~ previous last-occurrence entries
// of exactly those substrings. Those substrings are the ones on the
// ~ suffix-link path from the
// state of prefix r to the root - and their old "last occurrence"
// ~ positions are constant along
// each PREFERRED PATH of a link-cut tree. So maintain the suffix-link
```

```

    ↪ tree in an LCT where
// access(state, r) PAINTS the path with the colour r: every
    ↪ preferred-child change reports one
// maximal group (a range of lengths that shared an old colour), which
    ↪ is one range update.
// That "access as painting" pattern is reusable: any "assign a colour
    ↪ to a root path, and pay
// only for the O(log n) amortised colour groups you destroy" problem
    ↪ is this code.
struct DistinctSubstringsInRange {
    int n = 0, sz = 0;
    vector<int> par, lz, col, len;
    vector<array<int, 2>> ch;
    vector<pii> mods;
    ↪ {segment top len, old colour}
    vector<ll> b1, b2;
    ↪ range-add range-sum Fenwick
    void bAdd(vector<ll>& b, int i, ll v) { for (; i <= n; i += i & -i)
    ↪ b[i] += v; }
    ll bSum(const vector<ll>& b, int i) const {
        ll r = 0;
        for (; i > 0; i -= i & -i) r += b[i];
        return r;
    }
    void rangeAdd(int l, int r, ll v) {
    ↪ to positions [l, r]
        if (l > r) return;
        bAdd(b1, l, v), bAdd(b2, l, v * (l - 1));
        if (r + 1 <= n) bAdd(b1, r + 1, -v), bAdd(b2, r + 1, -v * r);
    }
    ll pref(int i) const { return bSum(b1, i) * i - bSum(b2, i); }
    ll rangeSum(int l, int r) const { return l > r ? 0 : pref(r) -
    ↪ pref(l - 1); }
    void mark(int x, int v) { lz[x] = col[x] = v; }
    void push(int x) {
        if (!lz[x]) return;
        if (ch[x][0]) mark(ch[x][0], lz[x]);
        if (ch[x][1]) mark(ch[x][1], lz[x]);
        lz[x] = 0;
    }
    bool isRoot(int x) { return ch[par[x]][0] != x && ch[par[x]][1] !=
    ↪ x; }
    void rotate(int x) {
        int y = par[x], z = par[y], k = ch[y][1] == x, w = ch[x][!k];
        if (!isRoot(y)) ch[z][ch[z][1] == y] = x;
        ch[x][!k] = y, ch[y][k] = w;
        if (w) par[w] = y;
        par[y] = x, par[x] = z;
    }
    void splay(int x) {
        static vector<int> st;
        st.clear(), st.push_back(x);
        for (int i = x; !isRoot(i); i = par[i]) st.push_back(par[i]);
        while (!st.empty()) push(st.back()), st.pop_back();
        while (!isRoot(x)) {
            int y = par[x], z = par[y];
            if (!isRoot(y)) rotate((ch[y][1] == x) == (ch[z][1] == y) ?
    ↪ y : x);
            rotate(x);
        }
    }
    void access(int x, int v) {
    ↪ root..x with colour v
        mods.clear();
        for (int z = 0; x; z = x, x = par[x]) {
            splay(x);
            mods.push_back({len[x] - 1, col[x]});
        }
    ↪ preferred group dies now
        ch[x][1] = z;
        mark(x, v);
    }
    vector<ll> solve(const string& s, const vector<pii>& queries) {
        n = s.size();
        SAM sam(n);
        vector<int> pos(n + 1, 1);
        for (int i = 0; i < n; i++) sam.extend(s[i]), pos[i + 1] =
    ↪ sam.last + 1;
        sz = sam.st.size();
        par.assign(sz + 1, 0), lz.assign(sz + 1, 0), col.assign(sz + 1,
    ↪ 0);
        ch.assign(sz + 1, {0, 0}), len.assign(sz, 0);
        for (int i = 0; i < sz; i++) len[i] = sam.st[i].len;
        for (int i = 1; i <= sz; i++) par[i] = sam.st[i - 1].link + 1;
        b1.assign(n + 2, 0), b2.assign(n + 2, 0);
        vector<vector<pii>> byR(n + 1);
    ↪ query id, 1-based r

```

```

        for (int i = 0; i < (int)queries.size(); i++)
            byR[queries[i].second + 1].push_back({queries[i].first + 1,
    ↪ i});
        vector<ll> ans(queries.size(), 0);
        for (int r = 1; r <= n; r++) {
            rangeAdd(1, r, 1);
            ↪ suffix of s[1..r] is new
            access(pos[r], r);
            int prevLen = 0;
            for (int j = (int)mods.size() - 1; j >= 1; j--) {
            ↪ root-most group first
                auto [topLen, oldCol] = mods[j];
                if (!topLen) continue;
            ↪ root state (empty string)
                if (oldCol) rangeAdd(oldCol - topLen + 1, oldCol -
    ↪ prevLen, -1);
                prevLen = topLen;
            }
            for (auto [l, id] : byR[r]) ans[id] = rangeSum(l, r);
        }
        return ans;
    }
};
/* NOTES
WHY THE LENGTHS LINE UP: a preferred group is a set of
    ↪ suffix-link-tree nodes whose strings all
    had the same last-occurrence end position 'oldCol', and the group
    ↪ covers exactly the lengths
    (prevLen, topLen]. Those substrings therefore started at positions
    ↪ oldCol - topLen + 1 ..
    oldCol - prevLen, which is the single range you cancel. Get the
    ↪ direction of prevLen wrong and
    the answers drift slowly - test against a brute force on n <= 12
    ↪ before using it.
THE SAM IS BUILT FIRST, in full, and the LCT is built over the FINAL
    ↪ suffix-link tree; no links
    or cuts ever happen, only access(). Clones created later do not
    ↪ break the sweep because the
    path from the state of prefix r to the root always lists exactly the
    ↪ suffixes of that prefix.
COMPLEXITY: O(n log n) amortised for the accesses (the number of
    ↪ preferred-child changes is
    O(n log n)), each producing one O(log n) Fenwick range update.
ONLINE variant: replace the Fenwick with a PERSISTENT one, keeping a
    ↪ root per r; queries then
    need no sorting. Memory becomes O(n log^2 n).
RELATED, MUCH EASIER: distinct substrings of the WHOLE string (sum of
    ↪ len - len[link] over SAM
    states), and distinct VALUES in a range (the classic
    ↪ BIT-over-last-occurrence sweep, which is
    this same idea with one position per value instead of a whole path).
    ↪ */

```

7.5 Palindromes

7.5.1 Manacher

MANACHER - the radius of the longest palindrome centred at every position, in O(n).

↪ Answers "is s[l..r] a palindrome" in O(1) and counts all palindromic
 ↪ substrings in one pass.

```

struct Manacher {
    vector<int> p;

    // Returns a vector where p[i] is the radius of the longest
    ↪ palindrome around center i
    // Centers are interleaved: 0 is before string, 1 is first char, 2
    ↪ is between 1st & 2nd, etc.
    Manacher(string s) {
        string t = "#";
        for (char c : s) {
            t += c;
            t += "#";
        }
        int n = t.size();
        p.assign(n, 0);
        int c = 0, r = 0;
        for (int i = 0; i < n; i++) {
            int mirror = 2 * c - i;
            if (i < r) p[i] = min(r - i, p[mirror]);
            while (i - 1 - p[i] >= 0 && i + 1 + p[i] < n && t[i - 1 -
    ↪ p[i]] == t[i + 1 + p[i]]) {
                p[i]++;
            }
            if (i + p[i] > r) {
                c = i;
                r = i + p[i];
            }
        }
    }
}

```

```

    }
}

// Check if substring [l, r] (0-based) is a palindrome in O(1)
bool is_palindrome(int l, int r) {
    int len = r - l + 1;
    int center = l + r + 1;
    return p[center] >= len;
}
};

```

7.5.2 Palindromic Tree

PALINDROMIC TREE (eertree) - $O(n)$ nodes, one per DISTINCT palindromic substring, built online in

// $O(n)$. Gives the count of distinct palindromes, occurrence counts,
 ↳ and the longest palindromic
 // suffix at every prefix - which is what palindromic-factorization DP
 ↳ needs.

```

struct PalindromeTree {
    int n, id, cur, tot;
    vector<array<int, 26>> go;

    // Core structural links
    vector<int> suflink, len, cnt;

    // Extensions for Hard DP (Palindrome Partitioning)
    vector<int> diff; // Difference in length to the suffix link
    vector<int> slink; // Series link (skips arithmetic progressions
    ↳ of suffix links)

    // Extensions for counting
    vector<int> num; // Number of palindromic suffixes ending at
    ↳ this node
    long long total_overlapping; // Running sum of total palindromes

    PalindromeTree() {}

    PalindromeTree(const string &s) {
        build(s);
    }

    void init(int max_len) {
        n = max_len;
        go.assign(n + 2, {});
        suflink.assign(n + 2, 0);
        len.assign(n + 2, 0);
        cnt.assign(n + 2, 0);
        diff.assign(n + 2, 0);
        slink.assign(n + 2, 0);
        num.assign(n + 2, 0);

        suflink[0] = suflink[1] = 1;
        len[1] = -1;
        id = 2;
        cur = 0;
        tot = 0;
        total_overlapping = 0;
    }

    int get(const string &s, int i, int v) {
        while (i - len[v] - 1 < 0 || s[i - len[v] - 1] != s[i]) {
            v = suflink[v];
        }
        return v;
    }

    void add(const string &s, int i) {
        int ch = s[i] - 'a';
        cur = get(s, i, cur);

        if (go[cur][ch] == 0) {
            len[id] = 2 + len[cur];
            suflink[id] = go[get(s, i, suflink[cur])][ch];

            // 1. Number of palindromes ending exactly here
            num[id] = num[suflink[id]] + 1;

            // 2. Arithmetic Progression / Series Links for DP
            diff[id] = len[id] - len[suflink[id]];
            if (diff[id] == diff[suflink[id]]) {
                slink[id] = slink[suflink[id]];
            } else {
                slink[id] = suflink[id];
            }

            tot++;
            go[cur][ch] = id++;
        }
    }
};

```

```

    }
    cur = go[cur][ch];
    cnt[cur]++;
    total_overlapping += num[cur];
}

// Standard build
void build(const string &s) {
    init(s.length());
    for (int i = 0; i < n; i++) {
        add(s, i);
    }
    countAll();
}

// Call this to push frequencies down the suffix links
void countAll() {
    for (int i = id - 1; i >= 2; --i) {
        cnt[suflink[i]] += cnt[i];
    }
}

// --- Black-Box Solvers ---

// 1. Returns number of unique palindromes
int cntDistinct() {
    return tot;
}

// 2. Returns total number of palindromes (including
↳ overlaps/duplicates)
long long cntTotal() {
    return total_overlapping;
}

// 3. Solves Minimum Palindrome Partitioning in  $O(N \log N)$ 
// Rebuilds the tree internally to compute the DP concurrently.
int min_partitions(const string &s) {
    init(s.length());
    const int INF = 1e9;
    vector<int> dp(s.length() + 1, INF);
    vector<int> series_ans(s.length() + 2, INF);

    dp[0] = 0;

    for (int i = 0; i < s.length(); ++i) {
        add(s, i);

        for (int v = cur; v > 1; v = slink[v]) {
            series_ans[v] = dp[i + 1 - (len[slink[v]] + diff[v])];

            if (diff[v] == diff[suflink[v]]) {
                series_ans[v] = min(series_ans[v],
                ↳ series_ans[suflink[v]]);
            }

            dp[i + 1] = min(dp[i + 1], series_ans[v] + 1);
        }
    }
    return dp[s.length()];
}

// 4. Returns an array of frequencies for string B's palindromes
↳ inside this tree (String A)
// Note: You must 'build()' the tree with string A before calling
↳ this with B.
vector<int> count_string(const string &B) {
    vector<int> out_cnt(id, 0);
    int curr = 0;

    for (int i = 0; i < B.length(); ++i) {
        int ch = B[i] - 'a';

        while (true) {
            if (i - len[curr] - 1 >= 0 && B[i - len[curr] - 1] ==
            ↳ B[i]) {
                if (go[curr][ch]) {
                    curr = go[curr][ch];
                    break;
                }
            }
            if (curr == 1) {
                curr = 0;
                break;
            }
            curr = suflink[curr];
        }
    }
}

```

```

        out_cnt[curr]++;
    }

    for (int i = id - 1; i >= 2; --i) {
        out_cnt[suflink[i]] += out_cnt[i];
    }

    return out_cnt;
}
};

```

7.5.3 PalindromicTreeRollback

PALINDROMIC TREE WITH ROLLBACK - same structure, but every extend() can be undone, so you can DFS

```

// NOTE: 02 - PalindromicTree.cpp declares the same 'struct
↳ PalindromeTree'. Paste one.
// a trie/tree of characters and maintain the palindromic structure of
↳ the current root-to-node string.
struct PalindromeTree {
    int n, id, cur, tot;
    vector<array<int, 26>> go;

    // Core structural links
    vector<int> suflink, len, cnt;

    // Extensions for Hard DP (Palindrome Partitioning)
    vector<int> diff; // Length difference to suffix link
    vector<int> slink; // Series link

    // Extensions for counting
    vector<int> num; // Number of palindromic suffixes ending at
↳ this node
    long long total_overlapping; // Running total of all palindromic
↳ substrings

    // --- ROLLBACK EXTENSIONS ---
    string S; // Maintains current active string
    struct History {
        int old_cur;
        int added_node; // ID of created node, or 0 if edge already
↳ existed
        int parent_node; // Node where go[parent_node][ch] was updated
        int ch; // Character index (0..25)
    };
    vector<History> history;

    PalindromeTree() {}

    PalindromeTree(int max_len) {
        init(max_len);
    }

    void init(int max_len) {
        n = max_len;
        go.assign(n + 2, {});
        suflink.assign(n + 2, 0);
        len.assign(n + 2, 0);
        cnt.assign(n + 2, 0);
        diff.assign(n + 2, 0);
        slink.assign(n + 2, 0);
        num.assign(n + 2, 0);

        suflink[0] = suflink[1] = 1;
        len[1] = -1;
        id = 2;
        cur = 0;
        tot = 0;
        total_overlapping = 0;

        S.clear();
        history.clear();
    }

    int get(int i, int v) {
        while (i - len[v] - 1 < 0 || S[i - len[v] - 1] != S[i]) {
            v = suflink[v];
        }
        return v;
    }

    // 1. PUSH / ADD CHARACTER (WITH ROLLBACK HISTORY)
    void push_back(char c) {
        S.push_back(c);
        int i = (int)S.length() - 1;
        int ch = c - 'a';

        int old_cur = cur;

```

```

        cur = get(i, cur);

        int added_node = 0;
        int parent_node = 0;

        if (go[cur][ch] == 0) {
            added_node = id;
            parent_node = cur;

            len[id] = 2 + len[cur];
            suflink[id] = go[get(i, suflink[cur])][ch];

            num[id] = num[suflink[id]] + 1;

            diff[id] = len[id] - len[suflink[id]];
            if (diff[id] == diff[suflink[id]]) {
                slink[id] = slink[suflink[id]];
            } else {
                slink[id] = suflink[id];
            }

            tot++;
            go[cur][ch] = id++;
        }

        cur = go[cur][ch];
        cnt[cur]++;
        total_overlapping += num[cur];

        // Save state transition into history stack
        history.push_back({old_cur, added_node, parent_node, ch});
    }

    // 2. POP / ROLLBACK LAST CHARACTER
    void pop_back() {
        if (history.empty()) return;

        History h = history.back();
        history.pop_back();

        // Revert occurrence count & running palindrome count
        total_overlapping -= num[cur];
        cnt[cur]--;

        // If a new node was created during this push_back, delete it
        if (h.added_node != 0) {
            int u = h.added_node;
            go[h.parent_node][h.ch] = 0; // Disconnect child edge

            // Reset node u's properties
            go[u].fill(0);
            suflink[u] = len[u] = cnt[u] = diff[u] = slink[u] = num[u]
↳ = 0;

            id--;
            tot--;
        }

        // Restore previous active node and string length
        cur = h.old_cur;
        S.pop_back();
    }

    // --- Black-Box Solvers ---

    int cntDistinct() {
        return tot;
    }

    long long cntTotal() {
        return total_overlapping;
    }

    // Call this if you need to push end-occurrence counts down suffix
↳ links
    void countAll() {
        for (int i = id - 1; i >= 2; --i) {
            cnt[suflink[i]] += cnt[i];
        }
    }
};

```

7.5.4 PalindromicFactorization

COUNTING PALINDROMIC FACTORISATIONS - the number of ways to cut s into palindromic pieces,

```
// Needs: 07 - Strings/05 - Palindromes/02 - PalindromicTree.cpp
// in  $O(n \log n)$ . The tree file already has min_partitions (the MINIMUM
//   ↳ number of pieces); this is
// the same series-link machinery with (min, +1) replaced by (sum, *1),
//   ↳ plus the even-length
// variant that CF 932G needs. Doing it naively is  $O(n^2)$ : "for each  $i$ ,
//   ↳ for each palindromic
// suffix" - and a string like  $a^n$  has  $n$  palindromic suffixes at every
//   ↳ position.
// WHY IT IS  $O(n \log n)$ : the palindromic suffixes of any prefix fall
//   ↳ into  $O(\log n)$  ARITHMETIC
// PROGRESSIONS by length (Fine and Wilf). diff[v] = len[v] -
//   ↳ len[link[v]] is the common
// difference of  $v$ 's progression and slink[v] jumps to the previous
//   ↳ progression, so walking
//  $v = \text{cur}$ , slink[v], slink[slink[v]], ... visits  $O(\log n)$  nodes.
//   ↳ series[v] aggregates the whole
// progression of  $v$  in  $O(1)$  by reusing series[link[v]] from the
//   ↳ PREVIOUS position - that reuse is
// the entire trick and is why series[] must be indexed by NODE and not
//   ↳ by position.
// INDEX CONVENTION: dp is 1-indexed over prefixes, dp[i] = answer for
//   ↳  $s[0..i-1]$ , dp[0] = 1.
ll countPalindromicFactorizations(const string& s, bool
//   ↳ evenLengthPiecesOnly = false) {
    int n = s.size();
    PalindromicTree t;
    t.init(n);
    vector<ll> dp(n + 1, 0), series(n + 2, 0);
    dp[0] = 1;
    for (int i = 0; i < n; i++) {
        t.add(s, i);
        for (int v = t.cur; t.len[v] > 0; v = t.slink[v]) {
            series[v] = dp[i + 1 - (t.len[t.slink[v]] + t.diff[v])];
            int sl = t.suflink[v];
            if (t.diff[v] == t.diff[sl]) series[v] = (series[v] +
//   ↳ series[sl]) % MOD;
            dp[i + 1] = (dp[i + 1] + series[v]) % MOD;
        }
        if (evenLengthPiecesOnly && (i + 1) % 2) dp[i + 1] = 0; // an
//   ↳ odd prefix cannot be filled
    }
    return dp[n];
}
/* THE VARIANTS THIS UNLOCKS - all the same loop, only the semiring
//   ↳ changes
// MINIMUM number of pieces: (min, +1) - already in 02 -
//   ↳ PalindromicTree.cpp as min_partitions.
// NUMBER of factorisations: (sum, *1) - the code above.
// FACTORISATIONS INTO AN EVEN NUMBER OF PIECES, or with a parity
//   ↳ constraint: carry two dp arrays
//   dp[i][0], dp[i][1] and one series[] per parity, exactly as
//   ↳ min_partition does for (even, odd).
// CF 932G "PALINDROME PARTITION" (split  $s$  into  $k$  pieces with  $p_1 = p_k$ ,
//   ↳  $p_2 = p_{k-1}$ , ...):
//   build  $t = s[0] s[n-1] s[1] s[n-2] \dots$  and count factorisations of  $t$ 
//   ↳ into EVEN-LENGTH
// palindromes - that is the call with evenLengthPiecesOnly = true.  $n$ 
//   ↳ must be even.
// WEIGHTED pieces (cost per palindrome length): the series aggregate
//   ↳ must then be a min/sum over
// the progression WITH the weight, which only stays  $O(1)$  if the weight
//   ↳ is affine in the length -
// otherwise you are back to  $O(n^2)$  and should look for a different
//   ↳ decomposition.
// SANITY: countPalindromicFactorizations("aaa") = 4 (aaa | a+aa | aa+a |
//   ↳ a+a+a), and any string of
// distinct characters gives exactly 1. */
```

7.5.5 PalindromesInRange

HOW MANY PALINDROMIC SUBSTRINGS LIE ENTIRELY INSIDE $s[l..r]$? Offline, $O((n + q) \log n)$.

```
// Needs: 07 - Strings/05 - Palindromes/01 - Manacher.cpp
// The whole-string count is one Manacher pass; the RANGE version is
//   ↳ the hard part, because a
// centre's contribution is CLIPPED by both ends of the query.
// INDEX CONVENTION (0-based, inclusive): with the interleaved radii
//   ↳  $p[i]$  of 01 - Manacher.cpp,
//  $d1[i] = (p[2i + 1] + 1) / 2$  = number of odd palindromes centred AT
//   ↳  $i$  (longest =  $2 * d1 - 1$ )
//  $d2[i] = p[2i] / 2$  = number of even palindromes centred
//   ↳ BETWEEN  $i-1$  and  $i$ 
// Then the answer for  $[l, r]$  is
// sum over  $i$  in  $[l, r]$  of  $\min(d1[i], i - l + 1, r - i + 1)$ 
// + sum over  $i$  in  $[l+1, r]$  of  $\min(d2[i], i - l, r - i + 1)$ .
// SPLIT AT THE MIDPOINT so that only ONE of the two clips can bind: on
//   ↳ the left half the binding
// clip is the distance to  $l$ , on the right half it is the distance to
//   ↳  $r$ . Each half becomes
// sum over  $i$  in a range of (  $\text{key}[i] \geq T ? \text{affine}(i) : v[i]$  )
// with  $\text{key}[i] = v[i] - i$  (left) or  $v[i] + i$  (right) - a threshold
//   ↳ sweep with three Fenwicks.
struct PalinRange {
    int n;
    vector<ll> d1, d2;
    PalinRange(const string& s) {
        Manacher M(s);
        n = s.size(), d1.assign(n, 0), d2.assign(n, 0);
        for (int i = 0; i < n; i++) d1[i] = (M.p[2 * i + 1] + 1) / 2,
//   ↳ d2[i] = M.p[2 * i] / 2;
    }
    struct BIT {
        vector<ll> f;
        BIT(int n = 0) : f(n + 1, 0) {}
        void add(int i, ll v) { for (i++; i < (int)f.size(); i += i &
//   ↳ -1) f[i] += v; }
        ll qr(int i) { ll r = 0; for (i++; i > 0; i -= i & -i) r +=
//   ↳ f[i]; return r; }
        ll rng(int l, int r) { return l > r ? 0 : qr(r) - qr(l - 1); }
    };
    // sum over  $i$  in  $[ql, qr]$  of (  $\text{key}[i] \geq T ? c0 + c1 * i : v[i]$  ),
//   ↳ for many queries at once.
    vector<ll> sweep(const vector<ll>& v, const vector<ll>& key,
//   ↳ const vector<array<ll, 5>>& qs) { // {ql,
//   ↳ qr, T, c0, c1}
        int q = qs.size();
        vector<ll> res(q, 0), pre(n + 1, 0);
        for (int i = 0; i < n; i++) pre[i + 1] = pre[i] + v[i];
        vector<int> ord(n), qo(q);
        iota(ord.begin(), ord.end(), 0), iota(qo.begin(), qo.end(), 0);
        sort(ord.begin(), ord.end(), [&](int a, int b) { return key[a]
//   ↳ > key[b]; });
        sort(qo.begin(), qo.end(), [&](int a, int b) { return qs[a][2]
//   ↳ > qs[b][2]; });
        BIT cnt(n), sIdx(n), sVal(n);
        int ptr = 0;
        for (int id : qo) {
            auto [ql, qr, T, c0, c1] = qs[id];
            while (ptr < n && key[ord[ptr]] >= T) {
                int i = ord[ptr++];
                cnt.add(i, 1), sIdx.add(i, i), sVal.add(i, v[i]);
            }
            if (ql > qr) continue;
            ll c = cnt.rng(ql, qr), si = sIdx.rng(ql, qr), sv =
//   ↳ sVal.rng(ql, qr);
            res[id] = c0 * c + c1 * si + (pre[qr + 1] - pre[ql] - sv);
        }
        return res;
    }
    vector<ll> solve(const vector<pii>& queries) { // {l,
//   ↳ r} 0-based inclusive
        int q = queries.size();
        vector<ll> km(n), kp(n), res(q, 0);
        vector<array<ll, 5>> a(q), b(q), c(q), d(q);
        for (int i = 0; i < n; i++) km[i] = d1[i] - i, kp[i] = d1[i] +
//   ↳ i;
        for (int j = 0; j < q; j++) {
            ll l = queries[j].first, r = queries[j].second, m = (l + r)
//   ↳ / 2;
            a[j] = {l, m, 1 - l, 1 - l, 1}; // odd,
//   ↳ left half:  $i - l + 1$ 
            b[j] = {m + 1, r, r + 1, r + 1, -1}; // odd,
//   ↳ right half:  $r - i + 1$ 
        }
        auto ra = sweep(d1, km, a), rb = sweep(d1, kp, b);
        for (int i = 0; i < n; i++) km[i] = d2[i] - i, kp[i] = d2[i] +
```

```

    ↪ i;
    for (int j = 0; j < q; j++) {
        ll l = queries[j].first, r = queries[j].second, m = (l + r
    ↪ + 1) / 2;
        c[j] = {l + 1, m, -l, -l, 1}; //
    ↪ even, left half: i - l
        d[j] = {m + 1, r, r + 1, r + 1, -1}; //
    ↪ even, right half: r - i + 1
    }
    auto rc = sweep(d2, km, c), rd = sweep(d2, kp, d);
    for (int j = 0; j < q; j++) res[j] = ra[j] + rb[j] + rc[j] +
    ↪ rd[j];
    return res;
}
}
/* NOTES
WHOLE-STRING COUNT is just sum(d1) + sum(d2) - no sweep needed, and
    ↪ the number of DISTINCT
    palindromic substrings is the node count of an eertree (05 -
    ↪ Palindromes/02).
ONLINE queries: replace the sweep with a WAVELET TREE (02 - DS/13)
    ↪ that also stores prefix SUMS
    per node, or with a merge-sort tree - same formulas,  $O(\log^2 n)$  per
    ↪ query, no sorting of
    queries. The offline version above is shorter and faster; use it
    ↪ unless the queries are forced
    to be online.
PALINDROMES STARTING IN [l, r] (rather than contained in it) is a
    ↪ different and easier sweep:
    each centre contributes a contiguous range of start positions.
COUNTING DISTINCT palindromes inside a range is much harder (offline
    ↪ eertree + BIT over the
    "last occurrence" trick, like distinct values in a range); do not
    ↪ confuse the two.
CHECK THE SPLIT POINT: the odd half uses  $m = (l+r)/2$  and the even half
    ↪  $m = (l+r+1)/2$ . Both come
    from solving  $i - l + 1 \leq r - i + 1$  and  $i - l \leq r - i + 1$ ; an
    ↪ off-by-one there double counts
    the centre and is the only bug this code ever has. */

```

7.6 Rotations

7.6.1 LyndonAndRotations

LYNDON FACTORISATION (Duval) and MINIMAL CYCLIC ROTATION (Duval), both $O(n)$, $O(1)$ memory.

```

// A Lyndon word is strictly smaller than all of its proper suffixes.
    ↪ Every string factors
// UNIQUELY as  $w_1 \geq w_2 \geq \dots \geq w_k$  with each  $w_i$  Lyndon.
vector<string> duval(const string& s) {
    int n = s.size(), i = 0;
    vector<string> f;
    while (i < n) {
        int j = i + 1, k = i;
        while (j < n && s[k] <= s[j]) s[k] < s[j] ? k = i : k++, j++;
        while (i <= k) f.push_back(s.substr(i, j - k)), i += j - k;
    }
    return f;
}
// Smallest cyclic rotation: run Duval on s+s and take the factor
    ↪ starting before n.
int minRotation(string s) {
    int n = s.size();
    s += s;
    int i = 0, ans = 0;
    while (i < n) {
        ans = i;
        int j = i + 1, k = i;
        while (j < (int)s.size() && s[k] <= s[j]) s[k] < s[j] ? k = i :
    ↪ k++, j++;
        while (i <= k) i += j - k;
    }
    return ans; // index in the
    ↪ ORIGINAL string
}
// USES: canonical form of a cyclic string (necklace dedup -> pairs
    ↪ with Burnside);
// "is A a rotation of B" (compare minRotation forms, or search A
    ↪ in B+B);
// lexicographically smallest result of rotating; runs /
    ↪ repetitions via Lyndon roots.

```

7.6.2 RunsAndTandemRepeats

ALL RUNS (maximal repetitions) of s in $O(n \log n) - O(n)$ if the LCE structure is linear.

```

// Needs: 07 - Strings/04 - Suffix Structures/05 - SuffixArraySAIS.cpp
    ↪ (saIs, kasai)
// A RUN is a triple (l, r, p):  $s[l..r]$  has period p,  $r - l + 1 \geq 2p$ ,
    ↪ and it cannot be extended
// left or right with the same period. Every tandem repeat, every
    ↪ square, every "x repeated k
// times" substring lives inside exactly one run, so runs are THE
    ↪ compact encoding of all
// repetitions - and by the RUNS THEOREM there are FEWER THAN  $n$  of
    ↪ them, with total exponent
//  $\sum(r - l + 1)/p < 3n$ . Enumerating squares directly is quadratic;
    ↪ enumerating runs is not.
// WHEN: count/locate squares or k-th powers, "longest substring that
    ↪ is a repetition", periodic
// structure of a string, and as the input to Lempel-Ziv style
    ↪ factorisations.
// INDEX CONVENTION: 0-based, l and r INCLUSIVE, p is the SMALLEST
    ↪ period of  $s[l..r]$ .
// HOW (Bannai et al.): a run's period is a LYNDON ROOT - for one of
    ↪ the two alphabet orders, the
// run contains a Lyndon word of length p starting at some i. So take
    ↪ every i, let  $l[i]$  be the
// longest Lyndon prefix of the suffix i (that is "next smaller suffix"
    ↪ minus i), and try to
// extend the pair (i, i + l[i]) both ways with LCE. Do it for BOTH
    ↪ orders (< and its reverse).
struct RunsFinder {
    struct LceSA { //
        ↪ suffix array +  $O(1)$  LCE
        int n, LG;
        vector<int> sa, rnk, lcp, lg;
        vector<vector<int>> sp;
        void build(const string& s) {
            n = s.size(), sa = suffixArray(s), lcp = kasai(s, sa),
    ↪ rnk.assign(n, 0);
            for (int i = 0; i < n; i++) rnk[sa[i]] = i;
            lg.assign(n + 1, 0);
            for (int i = 2; i <= n; i++) lg[i] = lg[i / 2] + 1;
            LG = lg[n] + 1, sp.assign(LG, vector<int>(n, 0)), sp[0] =
    ↪ lcp;
            for (int k = 1; k < LG; k++)
                for (int i = 0; i + (1 << k) <= n; i++)
                    sp[k][i] = min(sp[k - 1][i], sp[k - 1][i + (1 << (k
    ↪ - 1))]);
        }
        int lce(int i, int j) { // lcp
    ↪ of the suffixes i and j
            if (i >= n || j >= n) return 0;
            if (i == j) return n - i;
            int l = rnk[i], r = rnk[j];
            if (l > r) swap(l, r);
            int k = lg[r - l];
            return min(sp[k][l + 1], sp[k][r - (1 << k) + 1]);
        }
    };
    int n;
    LceSA F, B; //
    ↪ forward, and on the reversal
    vector<int> crnk; //
    ↪ suffix ranks under the FLIPPED
    RunsFinder(const string& s) { //
    ↪ alphabet order
        n = s.size();
        string r(s.rbegin(), s.rend()), c = s;
        for (char& ch : c) ch = (char)(255 - (unsigned char)ch);
        F.build(s), B.build(r);
        vector<int> csa = suffixArray(c);
        crnk.assign(n, 0);
        for (int i = 0; i < n; i++) crnk[csa[i]] = i;
        // NOT the reverse of F.rnk: flipping the alphabet flips every
    ↪ comparison EXCEPT the
        // prefix rule ("abc" < "abcd" in both orders), so the second
    ↪ order needs its own array.
    }
    int lceF(int i, int j) { return F.lce(i, j); } //
    ↪ common prefix of s[i..], s[j..]
    int lceB(int i, int j) { //
    ↪ common suffix of s[i..], s[j..]
        if (i < 0 || j < 0) return 0;
        return B.lce(n - 1 - i, n - 1 - j);
    }
    vector<array<int, 3>> runs() {
        vector<array<int, 3>> res;
        for (int ord = 0; ord < 2; ord++) { // both
    ↪ alphabet orders

```

```

    const vector<int>& rk = ord ? crnk : F.rnk;
    vector<int> nxt(n, n), st; // next
    ↪ suffix smaller in this order
    for (int i = n - 1; i >= 0; i--) {
        while (!st.empty() && rk[st.back()] > rk[i])
    }
    ↪ st.pop_back();
    nxt[i] = st.empty() ? n : st.back();
    st.push_back(i);
    }
    for (int i = 0; i < n; i++) {
        int p = nxt[i] - i, j = i + p; // p =
    ↪ Lyndon root length
        if (j > n) continue;
        int b = j < n ? lceF(i, j) : 0, a = lceB(i - 1, j - 1);
        if (a + b < p || p == 0) continue;
        int l = i - a, r = j + b - 1;
        if (r - l + 1 >= 2 * p) res.push_back({l, r, p});
    }
    sort(res.begin(), res.end()); // (l,
    ↪ r, p) with p increasing
    vector<array<int, 3>> out; // keep
    ↪ the SMALLEST p per (l, r)
    for (auto& x : res)
        if (out.empty() || out.back()[0] != x[0] || out.back()[1]
    ↪ != x[1]) out.push_back(x);
        return out; // < n
    }
};
/* WHAT TO DO WITH THE RUNS
NUMBER OF SQUARES (as occurrences, x x with |x| = q): a run (l, r, p)
    ↪ with length L = r - l + 1
    contains, for every multiple q of ... no: for every q with p <= q
    ↪ and 2q <= L AND q a multiple
    of p, exactly L - 2q + 1 occurrences of a square of half-length q.
    ↪ Summing over runs is O(n)
    amortised by the runs theorem (total exponent < 3n).
NUMBER OF DISTINCT SQUARES is at most n (the "three squares lemma");
    ↪ get them by taking, per
    run, the primitively rooted squares and deduplicating with the run's
    ↪ root.
K-TH POWERS: a run (l, r, p) contains a k-th power iff L >= k p, and
    ↪ then L - k p + 1 of them.
LONGEST REPETITION = the run maximising L / p, read straight off the
    ↪ list.
MAIN-LORENTZ is the alternative: divide & conquer with z-functions,
    ↪ O(n log n), ~40 lines, and
    it reports tandem repeats as O(n log n) grouped triples instead of <
    ↪ n runs. Prefer runs.
PRIMITIVITY of s[l..r] (is it a proper power?) is a prefix-function
    ↪ question, not a runs one:
    see 01 - String Matching/05, "n % (n - pi) == 0".
TESTING: on a^n there is exactly ONE run (0, n-1, 1); on "abababa"
    ↪ exactly one, (0, 6, 2). If
    you get duplicates with different p, the Lyndon-root loop ran with
    ↪ only one order. */

```

7.7 Sequences

7.7.1 DeBruijnAndBitap

DE BRUIJN SEQUENCES and BIT-PARALLEL MATCHING.

```

// --- DE BRUIJN SEQUENCE B(k, n): a CYCLIC string over an alphabet of
    ↪ size k, of length k^n, in
// which every one of the k^n words of length n appears exactly once as
    ↪ a (cyclic) substring.
// USES: "open the safe with the fewest keypresses" (the classic),
    ↪ minimum-length probe sequences,
// and hashing tricks (a de Bruijn constant is how you find the index
    ↪ of the lowest set bit).
// CONSTRUCTION (FKM / Lyndon): concatenate, in lexicographic order,
    ↪ every Lyndon word whose
// length DIVIDES n. That is exactly what this O(k^n) recursion emits.
string deBruijn(int k, int n) {
    string s;
    vector<int> a(k * n, 0);
    function<void(int, int)> db = [&](int t, int p) {
        if (t > n) {
            if (n % p == 0) for (int i = 1; i <= p; i++) s += char('0'
    ↪ + a[i]);
        } else {
            a[t] = a[t - p];
            db(t + 1, p);
            for (int j = a[t - p] + 1; j < k; j++) a[t] = j, db(t + 1,
    ↪ t);
        }
    };
}

```

```

    }
    };
    db(1, 1);
    return s; //
    ↪ length k^n, read cyclically
}
// The same object is an EULERIAN CIRCUIT in the de Bruijn graph
    ↪ (vertices = words of length n-1,
// edges = words of length n), so Hierholzer on that graph is the
    ↪ alternative construction - and
// the BEST theorem then counts how many distinct de Bruijn sequences
    ↪ exist: (k!)^(k^(n-1)) / k^n.

// --- BITAP / SHIFT-AND: exact matching in O(n*m/64) with bitsets, and
    ↪ the natural home for
// approximate matching. Build one bitmask per character marking where
    ↪ it occurs in the pattern;
// then one pass over the text ANDs shifted copies together.
// Beats KMP only when you need the extra flexibility (k mismatches,
    ↪ wildcards, "any of these
// patterns"), or when you want ALL occurrences as a bitset to
    ↪ intersect with a range.
vector<int> bitap(const string& t, const string& p) {
    int n = t.size(), m = p.size();
    if (!m || m > n) return {};
    vector<unsigned long long> mask(256, 0);
    // Works for m <= 64 with a single word; for longer patterns use
    ↪ bitset<M> and shift it.
    if (m > 64) return {};
    for (int i = 0; i < m; i++) mask[(unsigned char)p[i]] |= 1ULL << i;
    unsigned long long st = 0, hit = 1ULL << (m - 1);
    vector<int> res;
    for (int i = 0; i < n; i++) {
        st = ((st << 1) | 1) & mask[(unsigned char)t[i]];
        if (st & hit) res.push_back(i - m + 1);
    }
    return res;
}
/* WHAT BITAP GENERALISES TO, which is the reason to carry it
k MISMATCHES: keep k+1 state words R0..Rk, where Rj = "matched a
    ↪ prefix with at most j errors".
Rj_new = ((Rj << 1) & mask[c]) | Rj-1 | (Rj-1 << 1) | (Rj-1_old <<
    ↪ 1) | 1
covering substitution, insertion and deletion respectively - that is
    ↪ Wu-Manber, O(k n m / 64).
WILDCARDS in the PATTERN: set mask[c] bits for every character the
    ↪ class accepts. A '?' just
means "every mask has that bit". This is free - no FFT needed - and
    ↪ is the right tool when the
alphabet is small and the pattern is short. Use the FFT method (01 -
    ↪ String Matching/04) when
the pattern is long or wildcards appear in the TEXT too.
MULTIPLE PATTERNS of the same length: OR their masks and check which
    ↪ one actually matched.
OCCURRENCES INSIDE [l, r]: keep the whole occurrence set as a bitset
    ↪ and popcount a range - that
is what makes "how many times does p occur in t[l..r]" O(n/64) per
    ↪ query with no suffix
structure at all.
MYERS' BIT-VECTOR ALGORITHM computes full edit distance in O(n m / 64)
    ↪ with the same idea
(carry-save arithmetic on the DP's difference vectors); it is the
    ↪ fastest practical exact edit
distance and the escape hatch when the O(nm) DP is 1e10. */

```

7.7.2 EditDistance

EDIT DISTANCE (Levenshtein): fewest single-character insertions, deletions and substitutions

```

// turning a into b. dp[i][j] = distance between the prefixes a[0..i-1]
    ↪ and b[0..j-1], so the
// table is (n+1) x (m+1) and the answer is dp[n][m]. O(n m) time,
    ↪ O(min(n,m)) memory here.
// WHEN: fuzzy matching, "minimum operations to make equal", DNA-style
    ↪ alignment. If you only need
// insert+delete (no substitution) the answer is n + m - 2*LCS(a, b) -
    ↪ use the LCS DP instead.
int editDistance(const string& a, const string& b) {
    int n = a.size(), m = b.size();
    vector<int> prv(m + 1), cur(m + 1);
    iota(prv.begin(), prv.end(), 0);
    for (int i = 1; i <= n; i++) {
        cur[0] = i;
        for (int j = 1; j <= m; j++)
            cur[j] = min({prv[j] + 1, cur[j - 1] + 1, prv[j - 1] + (a[i
    ↪ - 1] != b[j - 1])});
        swap(prv, cur);
    }
}

```

```

    return prv[m];
}
// --- BANDED: when you only care whether the distance is <= k. Cells
//   ↳ off the diagonal by more
//   ↳ than k can never be on an optimal path (each step off the diagonal
//   ↳ costs >= 1), so evaluate
//   ↳ only |i - j| <= k. O(n k). Returns k + 1 as "more than k".
int editDistanceAtMost(const string& a, const string& b, int k) {
    int n = a.size(), m = b.size(), BIG = k + 1;
    if (abs(n - m) > k) return BIG;
    vector<int> prv(m + 2, BIG), cur(m + 2, BIG);
    for (int j = 0; j <= min(m, k); j++) prv[j] = j; //
    ↳ row 0
    for (int i = 1; i <= n; i++) {
        int lo = max(0, i - k), hi = min(m, i + k);
        cur[lo] = lo ? min({prv[lo] + 1, prv[lo - 1] + (a[i - 1] !=
        ↳ b[lo - 1]), BIG}) : min(i, BIG);
        for (int j = lo + 1; j <= hi; j++) //
        ↳ cur[lo-1] is OUTSIDE the band
            cur[j] = min({prv[j] + 1, cur[j - 1] + 1, prv[j - 1] + (a[i
        ↳ - 1] != b[j - 1]), BIG});
        if (hi + 1 <= m) cur[hi + 1] = BIG; //
        ↳ the cell the next row reads
        swap(prv, cur);
    }
    return min(prv[m], BIG);
}
// --- DAMERAU-LEVENSHTEIN, RESTRICTED (optimal string alignment): also
//   ↳ allows swapping two
//   ↳ ADJACENT characters at cost 1, but forbids editing a substring
//   ↳ twice. This is the version
//   ↳ contests mean; the unrestricted one needs the "last row/column per
//   ↳ character" table.
int damerau(const string& a, const string& b) {
    int n = a.size(), m = b.size();
    vector<vector<int>> dp(n + 1, vector<int>(m + 1));
    for (int i = 0; i <= n; i++) dp[i][0] = i;
    for (int j = 0; j <= m; j++) dp[0][j] = j;
    for (int i = 1; i <= n; i++)
        for (int j = 1; j <= m; j++) {
            dp[i][j] = min({dp[i - 1][j] + 1, dp[i][j - 1] + 1,
            dp[i - 1][j - 1] + (a[i - 1] != b[j - 1])});
            if (i > 1 && j > 1 && a[i - 1] == b[j - 2] && a[i - 2] ==
            ↳ b[j - 1])
                dp[i][j] = min(dp[i][j], dp[i - 2][j - 2] + 1);
        }
    return dp[n][m];
}
// --- MYERS' BIT-VECTOR ALGORITHM: exact Levenshtein in O(n * m / 64),
//   ↳ the escape hatch when the
//   ↳ O(n m) table is 1e10. It stores the VERTICAL differences dp[i][j] -
//   ↳ dp[i-1][j] (always in
//   ↳ {-1,0,+1}) as two bitmasks Pv/Mv over the rows, and advances a whole
//   ↳ column with 12 word ops.
//   ↳ Here a is the "pattern" and must satisfy |a| <= 64; loop over blocks
//   ↳ of 64 rows for longer a.
int myersEditDistance(const string& a, const string& b) {
    int m = a.size();
    if (!m) return b.size();
    unsigned long long peq[256] = {};
    for (int i = 0; i < m; i++) peq[(unsigned char)a[i]] |= 1ULL << i;
    unsigned long long Pv = ~0ULL, Mv = 0, high = 1ULL << (m - 1);
    int score = m;
    for (char ch : b) {
        unsigned long long Eq = peq[(unsigned char)ch];
        unsigned long long Xv = Eq | Mv;
        unsigned long long Xh = (((Eq & Pv) + Pv) ^ Pv) | Eq;
        unsigned long long Ph = Mv | ~(Xh | Pv), Mh = Pv & Xh;
        if (Ph & high) score++;
        if (Mh & high) score--;
        Ph = (Ph << 1) | 1; //
        ↳ dp[0][j] = j (GLOBAL alignment)
        Mh <= 1;
        Pv = Mh | ~(Xv | Ph);
        Mv = Ph & Xv;
    }
    return score;
}
/* VARIANTS AND TRAPS
SHIFT IN 0 INSTEAD OF 1 ('Ph <= 1') and Myers computes, for every j,
↳ the distance from a to the
BEST-MATCHING substring of b ending at j - i.e. approximate pattern
↳ matching, k-differences.
RECONSTRUCTING the alignment needs the table; with O(n) memory use
↳ Hirschberg
(08 - DP/04 - Classics), which is the same divide & conquer for any
↳ such 2-string DP.

```

WEIGHTED costs (ins != del != sub) work in the plain DP but break
↳ Myers and the LCS identity.
LONGEST COMMON SUBSTRING (contiguous) is NOT this DP: use hashing +
↳ binary search, a suffix
automaton, or $dp[i][j] = dp[i-1][j-1] + 1$ with the answer read as a
↳ max over all cells.
EDIT DISTANCE TO A REGULAR LANGUAGE / to any string of a trie: same DP
↳ with the automaton state
as the second dimension, $O(n * \text{states})$.
LOWER BOUND: $|n - m| \leq \text{dist} \leq \max(n, m)$, and $\text{dist} \geq (\text{number of}$
↳ characters whose counts
differ) / 2 style bounds - cheap filters before running the real DP
↳ on many pairs. */

7.7.3 BitParallelLCS

BIT-PARALLEL LCS - the length of the longest common subsequence of a and b in $O(n m / 64)$,

```

// where n = |a|. The classic DP is O(n m) and dies at n = m = 5e4;
//   ↳ this survives n = m = 1e5.
//   ↳ IDEA: the LCS row is monotone and its DIFFERENCE vector is a bit
//   ↳ string over the positions of
//   ↳ a (bit i = "the LCS value increases at position i of a"). One row
//   ↳ update is then a handful of
//   ↳ word operations, of which the carry-propagating ADD is what does the
//   ↳ actual work.
//   ↳ PRECONDITION: none - this is exact, for arbitrary alphabets (map
//   ↳ characters to 0..A-1).
//   ↳ The answer is the POPCOUNT of the final vector; the bit positions
//   ↳ themselves are NOT an LCS.
struct BitLCS {
    int n, W; // W =
    ↳ number of 64-bit words
    vector<vector<unsigned long long>> match; //
    ↳ match[c] = positions of c in a
    BitLCS(const string& a, int A = 26, char base = 'a') {
        n = a.size(), W = n / 64 + 1;
        match.assign(A, vector<unsigned long long>(W, 0));
        for (int i = 0; i < n; i++) match[a[i] - base][i >> 6] |= 1ULL
        ↳ << (i & 63);
    }
    int lcs(const string& b, char base = 'a') {
        vector<unsigned long long> row(W, 0), x(W);
        for (char ch : b) {
            const auto& mc = match[ch - base];
            for (int i = 0; i < W; i++) x[i] = row[i] | mc[i]; // x =
            ↳ row | match[c]
            unsigned long long carry = 1; // row
            ↳ = (row << 1) | 1
            for (int i = 0; i < W; i++) {
                unsigned long long nc = row[i] >> 63;
                row[i] = (row[i] << 1) | carry;
                carry = nc;
            }
            unsigned long long borrow = 0; // row
            ↳ = x - row
            for (int i = 0; i < W; i++) {
                unsigned long long d = x[i] - row[i] - borrow;
                borrow = (x[i] < row[i] + borrow) || (borrow && row[i]
                ↳ == ~0ULL);
                row[i] = d;
            }
            for (int i = 0; i < W; i++) row[i] = x[i] & ~row[i]; // row
            ↳ = x & ~(x - row)
            int res = 0;
            for (int i = 0; i < W; i++) res += __builtin_popcountll(row[i]);
            return res;
        }
    }
};
// USAGE: build once on the SHORTER string (that is the one that costs
//   ↳ memory and words),
//   ↳ then call lcs() for every other string - the build is reused.
//   ↳ BitLCS L(a); int k = L.lcs(b);
/* NOTES
WHY POPCOUNT: after processing all of b, bit i of 'row' is set iff
↳ LCS(a[0..i], b) is strictly
greater than LCS(a[0..i-1], b) - the difference vector of a monotone
↳ row of the DP table.
EDIT DISTANCE has the same treatment (Myers, 07 - Sequences/02) with
↳ two vectors instead of one,
because its differences live in {-1, 0, +1} while the LCS row only
↳ ever steps by 0 or 1.
LCS ITSELF (the subsequence, not its length) still needs the O(n m)
↳ table or Hirschberg
(08 - DP/04 - Classics) - bit-parallelism gives you the LENGTH only.
LCS OF PREFIXES: read the popcount after each character of b to get
↳ LCS(a, b[0..j]) for all j at

```

no extra cost. That is what makes this the practical answer to
 ↳ "all-prefix" LCS queries.
 SPARSE ALTERNATIVE: when the number of matching pairs r is small, the
 ↳ Hunt-Szymanski algorithm
 is $O(r \log n)$ - better on near-distinct alphabets, worse on binary
 ↳ strings.
 ALPHABET: 'match' is $A * W$ words. For a large alphabet, map to the
 ↳ distinct characters of a
 first (everything else can never match and can be dropped from b). */

7.7.4 AllSubstringLCS

ALL-SUBSTRING LCS - after ONE $O(nm)$ pass you can answer $\text{lcs}(s, t[i..j])$ for EVERY one of the

```
// m^2 substrings of t. The classic DP gives one pair per  $O(nm)$ ; this
↳ gives all of them at once.
// INDEX CONVENTION: s and t are 1-indexed below; the returned array F
↳ is indexed 1..m and
// lcs(s, t[i..j]) = #{ k in [i, j] : F[k] < i }.
// So one array of m integers encodes all  $m^2$  answers, and a single
↳ query is a RANGE COUNT of
// values below a threshold - BIT offline, wavelet tree, or merge-sort
↳ tree,  $O(\log m)$  each.
// WHEN: "for every substring of t, how similar is it to s",
↳ edit-distance-to-a-window sweeps,
// and problems asking for the shortest/longest window of t whose LCS
↳ with s reaches a target.
// WHY IT WORKS: for a fixed left end i the row of the LCS table is
↳ monotone with steps of 0/1;
// F[k] is the largest left end for which position k still contributes
↳ a step. f/g below track
// the row and column "critical points" of the table - that is the
↳ Alves-Caceres-Song theory.
vector<int> allSubstringLCS(const string& s, const string& t) {
    int n = s.size(), m = t.size();
    vector<int> f(m + 1), g(m + 1, 0), nf(m + 1, 0);
    for (int j = 0; j <= m; j++) f[j] = j; // row 0
    ↳ of the table
    for (int i = 1; i <= n; i++) {
        nf[0] = 0;
        for (int j = 1; j <= m; j++) {
            int gp = g[j - 1], fp = f[j]; //
            ↳ g[i][j-1] and f[i-1][j]
            if (s[i - 1] == t[j - 1]) nf[j] = gp, g[j] = fp;
            else nf[j] = max(fp, gp), g[j] = min(gp, fp);
        }
        swap(f, nf);
    }
    return f; //
    ↳ F[1..m], F[0] is unused
}
// DIRECT USE - all answers for one left end i in  $O(m)$ :
// int cnt = 0; for (int j = i; j <= m; j++) { if (F[j] < i) cnt++;
↳ /* lcs(s, t[i..j]) = cnt */ }
// MANY (i, j) QUERIES: sort by i, sweep i from m down to 1 adding
↳ position k when  $F[k] < i$  to a
// BIT, then the answer is a prefix-sum query on [i, j].  $O((m + q) \log$ 
↳  $m)$  total.
/* RELATED SHAPES
ALL-SUBSTRING LCS OF s AGAINST ITS OWN SUBSTRINGS is the same call
↳ with  $t = s$ .
THE TRANSPOSE (all substrings of s against the whole t) is the same
↳ routine with the arguments
swapped - the construction is NOT symmetric, so swap deliberately.
SEMI-LOCAL LCS is the full theory this is a corner of: it also answers
↳ "s[a..b] vs t[i..j]"
using a permutation matrix and range counting, in  $O(nm \log)$ 
↳ preprocessing. Rarely needed.
EDIT DISTANCE has no such compact all-substring encoding; only the
↳ +1/-1 step structure of LCS
makes the answers a range count.
SANITY CHECK:  $\text{lcs}(s, t[i..i])$  is 1 exactly when  $t[i]$  appears in s, and
↳  $F[k] < i$  is then the test
"the first position of  $t[k]$ 's character in s is early enough" -
↳ verify on a tiny case before
trusting the direction of the inequality. */
```

7.7.5 CyclicLCS

CYCLIC LCS - max over all n cyclic rotations of s of $\text{lcs}(\text{rotation}, t)$, in $O(nm)$ TOTAL.

```
// The naive answer is n separate LCS computations,  $O(n^2m)$ . WHEN: the
↳ two sequences are
// necklaces / circular arrangements and the cut point is yours to
↳ choose.
// HOW: build the LCS table of s+s against t once, then RE-ROOT it n
↳ times. Deleting the first
// row of the table only ever cancels ONE unit of the answer, and the
↳ path that loses it can be
```

```
// followed in  $O(m)$  - so all n rotations cost  $O(nm)$  together, not
↳  $O(n^2m)$ .
// INDEX CONVENTION: rows 1..2n of dp read the character  $s[(\text{row} - 1) \% n]$ ;  $\text{dp}[i][j]$  is the LCS of
// the first i characters of the doubled s with  $t[0..j-1]$ . Rotation r
↳ reads  $\text{dp}[r + n][m]$ .
//  $\text{from}[i][j]$  records WHICH transition was taken: 0 = came from the
↳ left, 1 = diagonal match,
// 2 = came from above. The re-rooting walk needs it, so it is not
↳ optional bookkeeping.
int cyclicLCS(const string& s, const string& t) {
    int n = s.size(), m = t.size();
    if (!n || !m) return 0;
    vector<vector<int>> dp(2 * n + 1, vector<int>(m + 1, 0));
    vector<vector<char>> from(2 * n + 1, vector<char>(m + 1, 0));
    auto eq = [&](int i, int j) { return s[(i - 1) % n] == t[j - 1]; };
    for (int i = 0; i <= 2 * n; i++)
        for (int j = 0; j <= m; j++) {
            dp[i][j] = 0, from[i][j] = 0;
            if (j && dp[i][j - 1] > dp[i][j]) dp[i][j] = dp[i][j - 1],
            ↳ from[i][j] = 0;
            if (i && j && eq(i, j) && dp[i - 1][j - 1] + 1 > dp[i][j])
            ↳ dp[i][j] = dp[i - 1][j - 1] + 1, from[i][j] = 1;
            if (i && dp[i - 1][j] > dp[i][j]) dp[i][j] = dp[i - 1][j],
            ↳ from[i][j] = 2;
        }
    int res = 0;
    for (int r = 0; r < n; r++) {
        res = max(res, dp[r + n][m]); //
        ↳ rotation starting at  $s[r]$ 
        int x = r + 1, y = 0; // now
        ↳ delete row r + 1
        while (y <= m && from[x][y] == 0) y++; // find
        ↳ where that row is used
        for (; y <= m && x <= 2 * n; x++) {
            from[x][y] = 0, dp[x][m]--;
            if (x == 2 * n) break;
            for (; y <= m; y++) { //
                if (from[x + 1][y] == 2) break;
                if (y + 1 <= m && from[x + 1][y + 1] == 1) { y++;
                }
            }
        }
        return res;
    }
}
/* WHAT THIS DOES AND DOES NOT COMPUTE
IT ROTATES ONLY s. That is the standard definition of CLCS and it is
↳ enough: rotating BOTH
strings is a strictly larger maximum in general, and needs  $n * m$ 
↳ separate cut pairs - if a
problem really wants both, rotate t in an outer loop ( $O(nm^2)$ ) or
↳ think again.
MEMORY is  $O(nm)$  for two tables - at  $n = m = 2000$  that is ~32 MB for
↳ dp plus ~8 MB for from.
Make s the SHORTER string; the doubled dimension is the expensive
↳ one.
THE ANSWER IS NOT MONOTONE in the rotation, so binary searching the
↳ cut point is wrong.
CYCLIC EDIT DISTANCE is the same re-rooting idea (Maes' algorithm) but
↳ is  $O(nm \log n)$  via
divide & conquer on non-crossing optimal paths - do not expect this
↳ code to adapt.
SANITY:  $\text{cyclicLCS}(s, s) == |s|$  and  $\text{cyclicLCS}(s, t) \geq \text{lcs}(s, t)$ 
↳ always. */
```

7.7.6 SubsequenceAutomaton

SUBSEQUENCE AUTOMATON (a.k.a. the "next occurrence" table). $O(n * A)$ build, $O(1)$ transition.

```
// INDEX CONVENTION:  $\text{nxt}[i][c]$  = the SMALLEST  $j \geq i$  with  $s[j] == c$ , or
↳ n if there is none.
// There are  $n + 1$  rows,  $i = 0..n$ ; row i is the state "we have consumed
↳  $s[0..i-1]$ ".
// After matching a character at j you move to row  $j + 1$  - never to row
↳ j.
// WHEN: everything of the form "is t a subsequence of s", counting or
↳ ranking DISTINCT
// subsequences, and shortest strings that are NOT subsequences. Build
↳ it from the RIGHT.
vector<array<int, 26>> subseqAutomaton(const string& s) {
    int n = s.size();
    vector<array<int, 26>> nxt(n + 1);
    for (int c = 0; c < 26; c++)  $\text{nxt}[n][c] = n$ ;
    for (int i = n - 1; i >= 0; i--)  $\text{nxt}[i] = \text{nxt}[i + 1]$ ,  $\text{nxt}[i][s[i] -$ 
    ↳ 'a'] = i;
    return nxt;
```

```

}
// --- IS t A SUBSEQUENCE OF s? O(|t|) per query after the build (the
    ↳ reason to build at all:
// the two-pointer scan is O(|s| + |t|) per query, which loses the
    ↳ moment there are many queries).
bool isSubsequence(const string& t, const vector<array<int, 26>>& nxt) {
    int n = nxt.size() - 1, i = 0;
    for (char ch : t) {
        int j = nxt[i][ch - 'a'];
        if (j == n) return false;
        i = j + 1;
    }
    return true;
}
// --- NUMBER OF DISTINCT NON-EMPTY SUBSEQUENCES, O(n * A). Each
    ↳ distinct subsequence is read by
// exactly ONE path in the automaton (always take the earliest
    ↳ occurrence), so just count paths.
// f[i] = number of distinct non-empty subsequences of s[i..], counted
    ↳ as "first letter is c".
ll distinctSubsequences(const string& s, const vector<array<int, 26>>&
    ↳ nxt, ll mod = MOD) {
    int n = s.size();
    vector<ll> f(n + 2, 0);
    for (int i = n; i >= 0; i--) {
        f[i] = 0;
        for (int c = 0; c < 26; c++) if (nxt[i][c] < n) f[i] = (f[i] +
    ↳ f[nxt[i][c] + 1] + 1) % mod;
        }
        return f[0];
    }
}
// --- K-TH DISTINCT SUBSEQUENCE in lexicographic order (1-indexed,
    ↳ non-empty). Use the UNCAPPED
// counts f[] above (saturating at k, not modular - a modular count
    ↳ cannot be compared).
string kthSubsequence(const string& s, const vector<array<int, 26>>&
    ↳ nxt, ll k) {
    int n = s.size();
    vector<ll> f(n + 2, 0); // f[i]
    ↳ saturated at 2e18
    for (int i = n; i >= 0; i--)
        for (int c = 0; c < 26; c++) if (nxt[i][c] < n)
            f[i] = min((ll)2e18, f[i] + f[nxt[i][c] + 1] + 1);
    if (k > f[0]) return ""; // fewer
    ↳ than k subsequences
    string res;
    int i = 0;
    while (k > 0) {
        for (int c = 0; c < 26; c++) {
            int j = nxt[i][c];
            if (j == n) continue;
            ll cur = f[j + 1] + 1; //
            ↳ subsequences starting with c
            if (k <= cur) { res += char('a' + c), i = j + 1, k--;
            ↳ break; }
            k -= cur;
        }
    }
    return res;
}
// --- SHORTEST STRING OVER THE ALPHABET THAT IS NOT A SUBSEQUENCE of
    ↳ s, O(n * A).
// g[i] = that length for s[i..]; a missing letter gives 1, otherwise
    ↳ pay one letter and recurse.
int shortestNonSubsequence(const string& s, const vector<array<int,
    ↳ 26>>& nxt, int A = 26) {
    int n = s.size();
    vector<int> g(n + 2, 0);
    for (int i = n; i >= 0; i--) {
        g[i] = inf;
        for (int c = 0; c < A; c++)
            g[i] = min(g[i], nxt[i][c] == n ? 1 : 1 + g[nxt[i][c] + 1]);
    }
    return g[0]; //
    ↳ reconstruct by replaying the min
}
/* VARIANTS
DISTINCT COMMON SUBSEQUENCES of s and t: run both automata in lockstep,
    f[i][j] = sum over c of (f[nxt1[i][c] + 1][nxt2[j][c] + 1] + 1), O(n
    ↳ * m A).
COUNTING SUBSEQUENCES EQUAL TO t (occurrences, not distinct) is the
    ↳ other classic DP,
    dp[i][j] = dp[i-1][j] + (s[i] == t[j]) * dp[i-1][j-1], O(n m) - the
    ↳ automaton does not help.
DISTINCT SUBSEQUENCES OF LENGTH EXACTLY k: add the length as a
    ↳ dimension, O(n k A).
LARGE ALPHABET: store nxt as a vector of sorted position lists per

```

↳ character and binary search;
 that is $O(n + A)$ memory and $O(\log n)$ per transition, which is the
 ↳ right trade when A is $1e5$.
 DO NOT CONFUSE with the SUFFIX automaton (04 - Suffix Structures/02),
 ↳ which recognises
 SUBSTRINGS. This one recognises SUBSEQUENCES and is a DAG with no
 ↳ suffix links. */

7.8 Theorems

7.8.1 String Theorems

STRING THEOREMS - the facts that turn an $O(n^2)$ idea into an $O(n)$ one. No code: these are the

/* STRING THEOREMS - the facts that turn an $O(n^2)$ idea into an $O(n)$
 ↳ one. No code: these are the
 statements you need to REMEMBER, because none of them is derivable
 ↳ at the table.

PERIODICITY

BORDER-PERIOD DUALITY: p is a period of s ($s[i] = s[i+p]$) whenever both
 ↳ exist IFF $n - p$ is the
 length of a border. So the periods and the borders are the same set
 ↳ read backwards, and the
 prefix function gives all of them (07 - Strings/01/05).

FINE AND WILF: if p and q are both periods of s and $p + q - \gcd(p, q)$
 ↳ $\leq n$, then $\gcd(p, q)$ is
 also a period. The bound is TIGHT (Fibonacci words). Consequences:
 - the borders of any string form $O(\log n)$ ARITHMETIC PROGRESSIONS
 ↳ (each progression is one
 "period class"); this is what makes palindromic-factorisation and
 ↳ border-sum DPs $O(n \log n)$
 instead of $O(n^2)$ - see 05 - Palindromes/04.
 - two occurrences of a pattern P closer than $|P| / 2$ force P to be
 ↳ periodic; hence in any text
 window of length $2|P|$ a non-periodic pattern occurs at most twice
 ↳ (the basis of constant-space
 matching and of "the occurrences form $O(\log)$ arithmetic
 ↳ progressions").

CRITICAL FACTORISATION: every string s with period p has a position i
 ↳ $< p$ such that the local
 period at i equals the global period. Splitting there makes the two
 ↳ halves comparable
 independently - the basis of the two-way (Crochemore-Perrin) $O(n)$
 ↳ constant-space matcher.

PRIMITIVITY: s is a power t^k with $k \geq 2$ IFF $n \% (n - \text{pi}[n-1]) == 0$
 ↳ and $n - \text{pi}[n-1] < n$. A
 primitive word has exactly n distinct rotations; a non-primitive one
 ↳ has n / k .

THE CYCLE LEMMA / rotations: s and t are rotations of each other IFF t
 ↳ is a substring of $s + s$
 IFF $\text{minRotation}(s) == \text{minRotation}(t)$ (06 - Rotations/01).

REPETITIONS

RUNS THEOREM (Bannai et al. 2015): a string of length n has FEWER THAN
 ↳ n runs (maximal
 repetitions), and the sum of their exponents is $< 3n$. Both bounds
 ↳ are what make "enumerate all
 repetitions" linear rather than quadratic. Construction: 06 -
 ↳ Rotations/02.

THREE SQUARES LEMMA: if three squares start at the same position and u
 ↳ is a prefix of v is a
 prefix of w , then $|u| + |v| \leq |w|$. Hence at most $O(\log n)$ squares
 ↳ start at any position, and a
 string of length n contains fewer than n DISTINCT squares.

CHEN-FOX-LYNDON: every string factors uniquely as $w_1 \geq w_2 \geq \dots \geq$
 ↳ w_k with each w_i Lyndon, and
 Duval computes it in $O(n)$ with $O(1)$ memory. The last factor is the
 ↳ smallest suffix; the first
 is the longest Lyndon prefix. Lyndon roots are the pivot of the runs
 ↳ algorithm.

A LYNDON WORD is strictly smaller than all of its proper suffixes;
 ↳ equivalently it is the unique
 minimum among its own rotations, and it is primitive. There are
 ↳ $(1/n) * \sum_{d | n} \mu(d) * A^{(n/d)}$ Lyndon words of length exactly n over an alphabet of
 ↳ size A (Witt's formula).

SUBSTRINGS AND SUFFIX STRUCTURES

ENDPOS EQUIVALENCE (suffix automaton): two substrings are equivalent
 ↳ when they occur at exactly
 the same set of END positions. The classes form a TREE under "is a
 ↳ suffix of" (the suffix-link
 tree), there are at most $2n - 1$ states and $3n - 4$ transitions, and
 ↳ one class contains exactly
 the strings of lengths $(\text{len}[\text{link}[v]], \text{len}[v])$ - so a class
 ↳ contributes $\text{len}[v] - \text{len}[\text{link}[v]]$
 distinct substrings. Summing that is the count of distinct

8 | Dynamic Programming (DP)

→ substrings.
THE SUFFIX-LINK TREE of a SAM built on the REVERSED string IS the
→ suffix tree of the string.
A STRING OF LENGTH n HAS AT MOST n DISTINCT PALINDROMIC SUBSTRINGS
→ (eertree node count), and the
palindromic suffixes of any prefix again form $O(\log n)$ arithmetic
→ progressions.
DISTINCT SUBSTRINGS = $n(n+1)/2$ - sum of the LCP array; adding one
→ character to the end changes
that by (new length) - (longest suffix that already occurred), which
→ is the SAM's `len[last]`.

MATCHING BOUNDS

KNUTH-MORRIS-PRATT does at most $2n$ character comparisons; Boyer-Moore
→ is sublinear on average
but $O(nm)$ in the worst case without Galil's rule; Z and prefix
→ functions are exactly linear.
A PATTERN WITH WILDCARDS is not solvable by automata in linear time;
→ the FFT convolution
sum $(p_i - t_j)^2$ p_i t_j is the standard $O(n \log n)$ route (01 -
→ String Matching/04).
EDIT DISTANCE cannot be computed in $O(n^{2-\epsilon})$ unless SETH fails -
→ so the $O(nm / 64)$
bit-parallel version (07 - Sequences/02) is essentially the last
→ word.
LCS of k STRINGS is NP-hard in k ; for $k = 2$ it is $O(nm / 64)$ and no
→ truly subquadratic
algorithm is known (same SETH barrier).

COUNTING

NUMBER OF DISTINCT NECKLACES (rotation classes) of length n over A
→ letters:
 $(1/n) * \sum \text{over } d \mid n \text{ of } \phi(d) * A^{(n/d)}$
→ (Burnside)
NUMBER OF BRACELETS adds the reflections; see 05 - Combinatorics/02 -
→ BurnsidePolya.cpp.
DE BRUIJN SEQUENCES $B(A, n)$: there are $(A!)^{(A^{(n-1)})} / A^n$ distinct
→ ones (07 - Sequences/01). */

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8.1 DP Optimizations

8.1.1 DivideAndConquerDP

Divide & Conquer DP Optimization - Time: $O(K * N \log N)$, Space: $O(N)$

```
// Applies when DP transition is dp[k][i] = min_{j < i} (dp[k-1][j] +  
→ cost(j+1, i))  
// Condition: opt(i, j) <= opt(i, j + 1) (Monotonicity of optimal split  
→ points)  
const int N = 200005; // ALSO in 01 -  
→ Template.cpp: keep one copy  
int a[N];  
ll curCost;  
int curL = 0, curR = -1, freq[N];  
pair<int, ll> pre[N], cur[N];  
// Modify at will :  
int lo(int ind) { return 1; }  
int hi(int ind) { return ind; }  
  
void add(int val) {  
    curCost += freq[val];  
}  
  
void remove(int val) {  
    curCost -= --freq[val];  
}  
  
ll Cost(int l, int r) {  
    while (curR < r)  
        add(a[++curR]);  
    while (curL > l)  
        add(a[--curL]);  
    while (curR > r)  
        remove(a[curR--]);  
    while (curL < l)  
        remove(a[curL++]);  
    return curCost;  
}  
  
ll f(int ind, int k) { return pre[k - 1].second + Cost(k, ind); }  
void store(int ind, int k, ll v) { cur[ind] = pair(k, v); }  
  
void rec(int L, int R, int LO, int HI) {
```

```
if (L > R) return;  
int mid = (L + R) >> 1;  
pair best(linf, LO);  
for (int k = max(LO, lo(mid)); k <= min(HI, hi(mid)); ++k)  
    best = min(best, pair(f(mid, k), k));  
store(mid, best.second, best.first);  
rec(L, mid - 1, LO, best.second);  
rec(mid + 1, R, best.second, HI);  
}  
  
void solve(int L, int R) { rec(L, R, -inf, inf); }
```

8.1.2 CHT

DYNAMIC CONVEX HULL TRICK (UPPER hull -> MAXIMUM of $m*x + b$). Lines may
be inserted in ANY order

```
// and queried at any x,  $O(\log n)$  each. If your slopes AND queries are  
→ both monotone, the deque  
// version at the bottom of this file is  $O(1)$  amortised and much  
→ shorter - prefer it.  
// For a MINIMUM, insert  $(-m, -b)$  and negate the answer.  
// To maximise over all  $x$  in  $[l, r]$ , query only  $l$  and  $r$ : the upper  
→ envelope is convex.  
struct Line {  
    ll m, b;  
    mutable function<const Line *(> succ;  
  
    bool operator<(const Line &other) const {  
        return m < other.m;  
    }  
  
    bool operator<(const ll &x) const {  
        const Line *s = succ();  
        if (!s)  
            return 0;  
        return (lll)(b - s->b) < (lll)(s->m - m) * x; // __int128:  
→ |dm|*|x| overflows ll  
    }  
};  
  
// will maintain upper hull for maximum  
struct HullDynamic : public multiset<Line, less<>> {  
    bool bad(iterator y) {  
        auto z = next(y);  
        if (y == begin()) {  
            if (z == end())  
                return 0;  
            return y->m == z->m && y->b <= z->b;  
        }  
        auto x = prev(y);  
        if (z == end())  
            return y->m == x->m && y->b <= x->b;  
        // __int128, NOT long double: each factor reaches  $\sim 2e18$ , the  
→ product needs  $\sim 123$  bits,  
        // and this test is consulted exactly when three lines are  
→ nearly collinear.  
        return (lll)(x->b - y->b) * (z->m - y->m) >= (lll)(y->b - z->b)  
→ * (y->m - x->m);  
    }  
  
    void insert_line(ll m, ll b) {  
        // for minimum  
        // m *= -1;  
        // b *= -1;  
        auto y = insert({m, b});  
        y->succ = [=] { return next(y) == end() ? 0 : &*next(y); };  
        if (bad(y)) {  
            erase(y);  
            return;  
        }  
        while (next(y) != end() && bad(next(y)))  
            erase(next(y));  
        while (y != begin() && bad(prev(y)))  
            erase(prev(y));  
    }  
  
    ll query(ll x) { // UB if empty -  
→ insert at least one line first  
        auto l = *lower_bound(x);  
        // for minimum  
        // return -(l.m * x + l.b);  
        return l.m * x + l.b;  
    }  
};  
//  
→ -----  
// MONOTONE CHT (deque) - the version you actually want when the DP  
→ feeds lines in slope order.
```

```
// MINIMISES m*x + b. Requires slopes added in DECREASING order;
// queries in increasing x are O(1)
// amortised (move the head), arbitrary x is O(log n) by binary search
// on the hull.
// For a MAXIMUM, negate m and b on the way in and negate the answer on
// the way out.
struct MonoCHT {
    vector<ll> M, B;
    int head = 0;
    bool bad(int a, int b, int c) { // is line b
        // unnecessary between a and c?
        return (lll)(B[c] - B[a]) * (M[a] - M[b]) <= (lll)(B[b] - B[a])
            * (M[a] - M[c]);
    }
    void add(ll m, ll b) { // m must be <=
        // the last slope added
        M.push_back(m), B.push_back(b);
        while (M.size() >= 3 && bad(M.size() - 3, M.size() - 2,
            M.size() - 1))
            M.erase(M.end() - 2), B.erase(B.end() - 2);
        head = min(head, (int)M.size() - 1);
    }
    ll f(int i, ll x) { return M[i] * x + B[i]; }
    ll queryInc(ll x) { // x
        // non-decreasing across calls
        while (head + 1 < (int)M.size() && f(head + 1, x) <= f(head,
            x)) head++;
        return f(head, x);
    }
    ll query(ll x) { // any x, O(log n)
        int lo = 0, hi = M.size() - 1;
        while (lo < hi) { int m = (lo + hi) / 2; f(m + 1, x) <= f(m, x)
            // ? lo = m + 1 : hi = m; }
        return f(lo, x);
    }
};
// WHEN EACH ONE: monotone slopes + monotone queries ->
// MonoCHT::queryInc, O(n).
// monotone slopes, arbitrary queries ->
// MonoCHT::query, O(n log n).
// arbitrary slopes -> HullDynamic
// above, or Li Chao (04),
// which needs no
// ordering at all.
// lines inserted and removed -> CHT with
// rollback (03) or Li Chao merge.
```

8.1.3 CHTRollback

Rollback Convex Hull Trick - Space: O(N), Add: O(log N), Query: O(log N), Rollback: O(1)

```
// Supports inserting monotonic lines with history stack rollback for
// tree/dsu DP
// Template parameters:
// IS_MAX: Set to true for Maximum queries, false for Minimum queries.
// IS_INC: Set to true if slopes added are strictly Increasing, false
// for Decreasing.
```

```
template<bool IS_MAX = false, bool IS_INC = false>
```

```
struct RollbackCHT {
    struct Line {
        ll m, c;
        Line() : m(0), c(llinf) {}
        Line(ll m, ll c) : m(m), c(c) {}
        ll eval(ll x) const { return m * x + c; }
    };

    struct History {
        int pos;
        int sz;
        Line old_line;
    };

    vector<Line> st;
    vector<History> hist;
    int sz;

    RollbackCHT(int max_nodes) {
        st.resize(max_nodes);
        sz = 0;
    }

    bool redundant(Line l1, Line l2, Line l3) {
        __int128 dx1 = l1.m - l2.m;
        __int128 dc1 = l2.c - l1.c;
        __int128 dx2 = l2.m - l3.m;
        __int128 dc2 = l3.c - l2.c;
        return dc1 * dx2 >= dc2 * dx1;
    }
};
```

```
void add(ll m, ll c) {
    // The magic happens here: automatically formats internal state
    ll m_int = IS_INC ? -m : m;
    ll c_int = IS_MAX ? -c : c;

    Line l_new(m_int, c_int);
    int pos = sz;
    int low = 1, high = sz - 1;

    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (redundant(st[mid - 1], st[mid], l_new)) {
            pos = mid;
            high = mid - 1;
        } else {
            low = mid + 1;
        }
    }

    hist.push_back({pos, sz, st[pos]});
    st[pos] = l_new;
    sz = pos + 1;
}

void rollback() {
    if (hist.empty()) return;
    History h = hist.back();
    hist.pop_back();
    sz = h.sz;
    st[h.pos] = h.old_line;
}

ll query(ll x) {
    if (sz == 0) return IS_MAX ? -llinf : llinf;

    // The magic happens here: matches x to the internal geometry
    ll x_int = (IS_INC != IS_MAX) ? -x : x;

    int low = 0, high = sz - 2;
    int best = sz - 1;

    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (st[mid].eval(x_int) <= st[mid + 1].eval(x_int)) {
            best = mid;
            high = mid - 1;
        } else {
            low = mid + 1;
        }
    }

    ll ans = st[best].eval(x_int);
    return IS_MAX ? -ans : ans;
}
};
```

8.1.4 LiChaoTree

LI CHAO TREE - MAXIMUM of a set of lines $y = m*x + b$ at any query x , with NO order requirement

```
// on the slopes or on the queries (that is the whole point; CHT needs
// monotone slopes).
// O(log X) per line insert and per query, O(log^2 X) to insert a line
// only on a SEGMENT [lx, rx].
// Space O(n log X). Works for any family of functions where two
// members cross AT MOST ONCE -
// lines, and also  $a*x^2 + b*x + c$  with a FIXED a.
// EXACT INTEGER ARITHMETIC: never compute the crossing abscissa.
// Compare the two candidates at
// ns, mid and ne only - a division in double is off by  $\sim 1e3$  in x when
// the coefficients reach  $1e18$ ,
// which silently sends the loser down the wrong half.
// SET the domain [X0, X1] before use; it must cover every x you will
// ever query.
```

```
struct LiChao {
    struct Line { ll m = 0, b = LLONG_MIN / 4; }; //
    // neutral = -infinity
    ll X0, X1;
    vector<Line> t;
    vector<int> lf, rt;
    LiChao(ll X0, ll X1) : X0(X0), X1(X1) { newNode(); }
    int newNode() { t.push_back({}); lf.push_back(-1);
        // rt.push_back(-1); return t.size() - 1; }
    static ll f(const Line& L, ll x) { return L.m == 0 && L.b ==
        // LLONG_MIN / 4 ? L.b : L.m * x + L.b; }
    void add(Line nw, int v, ll l, ll r) { //
        // insert on the whole node range
```

```

    while (true) {
        ll m = l + (r - l) / 2; // NOT
        ↪ (l+r)/2: negative l+r truncates up
        bool lef = f(nw, l) > f(t[v], l), mid = f(nw, m) > f(t[v],
        ↪ m);
        if (mid) swap(t[v], nw);
        if (l == r) return;
        if (lef != mid) {
            if (lf[v] < 0) lf[v] = newNode();
            v = lf[v], r = m;
        } else {
            if (f(nw, r) > f(t[v], r)) {
                if (rt[v] < 0) rt[v] = newNode();
                v = rt[v], l = m + 1;
            } else return;
        }
    }
}

void addLine(ll m, ll b) { add({m, b}, 0, X0, X1); }
void addSegment(ll m, ll b, ll lx, ll rx, int v = 0, ll l =
↪ LLONG_MIN, ll r = LLONG_MIN) {
    if (l == LLONG_MIN) l = X0, r = X1;
    if (rx < l || r < lx) return;
    if (lx <= l && r <= rx) { add({m, b}, v, l, r); return; }
    ll md = l + (r - l) / 2;
    if (lf[v] < 0) lf[v] = newNode();
    if (rt[v] < 0) rt[v] = newNode();
    addSegment(m, b, lx, rx, lf[v], l, md), addSegment(m, b, lx,
↪ rx, rt[v], md + 1, r);
}

ll query(ll x) {
    ll r = LLONG_MIN, l = X0, rr = X1;
    for (int v = 0; v >= 0; v++) {
        r = max(r, f(t[v], x));
        if (l == rr) break;
        ll m = l + (rr - l) / 2;
        if (x <= m) v = lf[v], rr = m;
        else v = rt[v], l = m + 1;
    }
    return r; //
    ↪ LLONG_MIN/4 if no line covers x
}

};
// FOR A MINIMUM: insert (-m, -b) and negate the answer.
// PERSISTENT / MERGEABLE: see 05 - PersistentLiChaoTree.cpp. MERGING
    ↪ two Li Chao trees costs
// O(n log X) amortised in total (push the loser line down into the
    ↪ other subtree, exactly like
// segment-tree merging) - that is how you do convex-hull DP on a tree
    ↪ without small-to-large.

```

8.1.5 PersistentLiChaoTree

Persistent Li Chao Tree - Space: $O(N \log X)$, Insert Line: $O(\log X)$, Query: $O(\log X)$
 // Version-persistent Li Chao Segment Tree for tree / history DP queries
 const ll maxN = 1e7 + 5;

```

struct Line {
    ll m, c;

    Line() : m(0), c(LLINF) {}

    Line(ll m, ll c) : m(m), c(c) {}
};

ll sub(ll x, Line l) {
    return x * l.m + l.c;
}

```

```

// Persistent Li Chao
struct Node {
    // range I am responsible for
    Line line;
    Node *left, *right;
}

```

```

Node() {
    left = right = NULL;
}

```

```

Node(ll m, ll c) {
    line = Line(m, c);
    left = right = NULL;
}

```

```

void extend(int l, int r) {
    if (left == NULL && l != r) {
        left = new Node();
        right = new Node();
    }
}

```

```

}

Node *copy(Node *node) {
    Node *newNode = new Node;
    newNode->left = node->left;
    newNode->right = node->right;
    newNode->line = node->line;
    return newNode;
}

Node *addLine(toAdd, int l, int r) {
    int mid = l + (r - l) / 2; // NOT (l+r)/2: that truncates
    ↪ toward ZERO, so on a negative // interval the left child equals
    ↪ the parent -> infinite recursion
    Node *cur = copy(this);
    if (l == r) {
        if (sub(l, toAdd) < sub(l, cur->line))
            swap(toAdd, cur->line);
        return cur;
    }
    bool lef = sub(l, toAdd) < sub(l, cur->line);
    bool midE = sub(mid + 1, toAdd) < sub(mid + 1, cur->line);
    if (midE)
        swap(cur->line, toAdd);
    cur->extend(l, r);
    if (lef != midE) {
        cur->left = cur->left->add(toAdd, l, mid);
    } else {
        if (mid + 1 > r)
            return cur;
        cur->right = cur->right->add(toAdd, mid + 1, r);
    }
    return cur;
}

Node *addLine(toAdd) {
    return add(toAdd, -maxN + 1, maxN - 1);
}

```

```

ll query(ll x, int l, int r) {
    int mid = l + (r - l) / 2; // NOT (l+r)/2: that truncates
    ↪ toward ZERO, so on a negative // interval the left child equals
    ↪ the parent -> infinite recursion
    if (l == r || left == NULL)
        return sub(x, line);
    extend(l, r);
    if (x <= mid)
        return min(sub(x, line), left->query(x, l, mid));
    else
        return min(sub(x, line), right->query(x, mid + 1, r));
}

ll query(ll x) {
    return query(x, -maxN + 1, maxN - 1);
}

void clear() {
    if (left != NULL) {
        left->clear();
        right->clear();
    }
    delete this;
}
};

```

```

const int N = 200005; // #versions; also in
    ↪ 01 - Template.cpp
Node *tree[N];

```

8.1.6 KnuthDP

KNUTH OPTIMIZATION - interval DP $dp[i][j] = \min_{i \leq k < j} (dp[i][k] + dp[k+1][j]) + \text{cost}(i, j)$

```

// from  $O(n^3)$  to  $O(n^2)$ , by restricting k to  $[\text{opt}[i][j-1],$ 
    ↪  $\text{opt}[i+1][j]]$ .
// THE HYPOTHESIS IS ON cost, NOT ON opt. 'opt[i][j-1] <= opt[i][j] <=
    ↪  $\text{opt}[i+1][j]$ ' is the
// CONCLUSION - you cannot test it before running. What you must check
    ↪ is BOTH of:
// (1) QUADRANGLE INEQUALITY (Monge):  $\text{cost}(a, c) + \text{cost}(b, d) \leq$ 
    ↪  $\text{cost}(a, d) + \text{cost}(b, c)$ 
// for all  $a \leq b \leq c \leq d$ . Equivalent local form, and the only
    ↪ one checkable by hand:
//  $\text{cost}(a, c) + \text{cost}(a+1, c+1) \leq \text{cost}(a+1, c) + \text{cost}(a, c+1)$ .
// (2) MONOTONE ON INTERVALS:  $a \leq b \leq c \leq d \Rightarrow \text{cost}(b, c) \leq$ 

```

```

    ↪ cost(a,d).
// Both hold for: sum of a non-negative array over [i,j]; (S_j -
    ↪ S_i)^2; f(S_j - S_i) with f convex
// and a[] >= 0; "number of equal pairs inside [i,j]". They FAIL for
    ↪ anything built from max, and
// monotonicity fails as soon as a[] can be negative (QI can still hold
    ↪ - D&C needs only QI,
// Knuth needs both). Applied without them there is no crash and no
    ↪ TLE, just a wrong answer.
// Classic uses: optimal BST, merging n piles of stones, CF 1101F-style
    ↪ interval costs.
struct KnuthDP {
    int n;
    vector<vector<ll>> dp;
    vector<vector<int>> opt;

    KnuthDP(int n = 0) : n(n), dp(n + 2, vector<ll>(n + 2, 0)), opt(n +
    ↪ 2, vector<int>(n + 2, 0)) {}

    // User-defined cost function cost(i, j)
    ll solve(function<ll(int, int)> cost) {
        for (int i = 1; i <= n; i++) {
            opt[i][i] = i;
        }

        for (int len = 2; len <= n; len++) {
            for (int i = 1; i + len - 1 <= n; i++) {
                int j = i + len - 1;
                dp[i][j] = llinf;
                int L = opt[i][j - 1], R = opt[i + 1][j];
                for (int k = L; k <= min(R, j - 1); k++) {
                    ll cur = dp[i][k] + dp[k + 1][j] + cost(i, j);
                    if (cur < dp[i][j]) {
                        dp[i][j] = cur;
                        opt[i][j] = k;
                    }
                }
            }
        }
        return dp[1][n];
    }
};

```

8.1.7 AliensTrick

ALIENS TRICK (Lagrangian relaxation / wqs binary search) - $O(n \log(\text{range}))$.

```

// Drops an "exactly k pieces" constraint: charge lambda per piece,
    ↪ solve the UNCONSTRAINED DP,
// and binary search lambda until the optimum uses k pieces. Answer =
    ↪ cost(lambda) - k*lambda.
// PRECONDITION: f(k), the optimum with exactly k pieces, must be
    ↪ CONVEX in k (for a minimum;
// concave for a maximum). NOT "strictly" - strictness is never needed
    ↪ and rarely true.
// THREE THINGS THAT MAKE IT CORRECT, all easy to get wrong:
// (1) dp_solver MUST return the LARGEST optimal count for that
    ↪ lambda. With f convex, the
// difference d(k) = f(k) - f(k-1) is non-decreasing; the search
    ↪ below finds the largest
// lambda with count >= k, and then d(k) <= -lambda <= d(k+1), so
    ↪ k really is optimal at
// that lambda and cost - k*lambda = f(k). Any other tie-break
    ↪ makes the last line wrong.
// (2) An integer lambda suffices ONLY if all costs are integers.
// (3) [min_penalty, max_penalty] must straddle every slope of f;
    ↪ assert it.
// CONVEX families: "k pairwise non-adjacent items", "exactly k
    ↪ segments with a Monge cost",
// "k disjoint intervals", "min-cost flow of value k" (flow cost is
    ↪ always convex in the value),
// matroid-rank-constrained optima. TRAPS: two coupled constraints, or
    ↪ a cost rewarding a parity
// of k. CHECK IT: brute-force f(k) for n <= 60 and assert f(k+1)-f(k)
    ↪ >= f(k)-f(k-1).

```

```

struct AliensTrick {
    // Binary searches optimal penalty lambda to choose exactly K items
    // Returns minimum cost to choose exactly K items
    static ll solve(int n, int k, ll min_penalty, ll max_penalty,
    ↪ function<pair<ll, int>>(ll)> dp_solver) {
        ll l = min_penalty, r = max_penalty, best_lambda = 0;

        while (l <= r) {
            ll mid = l + (r - l) / 2;
            auto [cost, count] = dp_solver(mid);
            if (count >= k) {
                best_lambda = mid;
                l = mid + 1; // Increase penalty if we used too many
    ↪ items

```

```

        } else {
            r = mid - 1; // Decrease penalty if we used too few
    ↪ items
        }

        // True cost = cost_with_penalty - (k * best_lambda)
        auto [cost, count] = dp_solver(best_lambda);
        return cost - (ll)k * best_lambda;
    }
};

```

8.1.8 AliensTrickExample

ALIENS TRICK (Lagrangian / wqs binary search) - drops the "exactly k pieces" constraint by charging

```

// a penalty lambda per piece, solving the unconstrained DP, and binary
    ↪ searching lambda until the
// optimal solution uses k. REQUIRES the optimum as a function of k to
    ↪ be CONVEX; check that first.
// 1. DP State Structure
struct DPState {
    long long cost;
    long long count; // Number of items/segments picked

    // TIE-BREAKER LOGIC:
    // If costs are equal, we maximize the count to handle collinear
    ↪ points on the convex hull.
    // If you are writing a MINIMIZATION problem with min(), change
    ↪ this to:
    // if (cost != other.cost) return cost > other.cost; (since
    ↪ priority queue / max naturally takes the "biggest")
    // Actually, when using standard >, < operators for min/max DP:
    bool operator<(const DPState& other) const {
        if (cost != other.cost) return cost < other.cost;
        return count < other.count; // Always prefer taking MORE items
    ↪ if costs are tied
    }
};

```

```

void solve_alien() {
    int n, k;
    cin >> n >> k;
    // Read ABC 218 H input: A array is of size N-1
    vector<long long> A(n);
    for(int i = 1; i < n; i++) cin >> A[i];
    // If K > N / 2, swap colors so K <= N / 2 (the score is symmetric)
    if (k > n / 2) k = n - k;
    // Construct 1-indexed values array 'a' where a[i] = A[i-1] + A[i]
    vector<long long> a(n + 1);
    for (int i = 1; i <= n; i++) {
        a[i] = A[i - 1] + (i < n ? A[i] : 0);
    }
    // 2. The Unconstrained DP
    // Returns the optimal {cost, count} for a given penalty.
    auto check_dp = [&](long long penalty) -> DPState {
        // dp[i][0]: Max score in prefix i, ending completely outside a
    ↪ segment
        // dp[i][1]: Max score in prefix i, ending inside an active
    ↪ segment
        vector<vector<DPState>> dp(n + 1, vector<DPState>(2, {0, 0}));

        for (int i = 1; i <= n; i++) {
            // Transition 0: We don't pick item i
            dp[i][0] = max(dp[i-1][0], dp[i-1][1]);

            // Transition 1: We pick item i
            // Rule: ONLY apply penalty when STARTING a new
    ↪ pick/segment!
            // Type B logic: To ensure non-adjacency, we must
    ↪ transition from dp[i-1][0].

            // --- IF MAXIMIZING PROFIT ---
            DPState start_new = {dp[i-1][0].cost + a[i] - penalty,
    ↪ dp[i-1][0].count + 1};

            // Replaced max(start_new, continue_old) with just start_new
            dp[i][1] = start_new;
        }
        return max(dp[n][0], dp[n][1]);
    };
}

```

```

// 3. WQS Binary Search
// The bounds must be large enough to cover the maximum possible
    ↪ penalty.
// FIXED: Dropped bounds to 2e9 to prevent k * mid_penalty from
    ↪ overflowing long long
    long long low = -2e9 - 7, high = 2e9 + 7;

```

```

long long best_ans = 0;

while (low <= high) {
    long long mid_penalty = low + (high - low) / 2;

    DPState res = check_dp(mid_penalty);

    // We want EXACTLY k items.
    // If we picked >= k items, it means the penalty wasn't harsh
    ↪ enough to stop us.
    // Even if res.count > k, we save the answer because of
    ↪ collinear points on the hull.
    if (res.count >= k) {

        // --- IF MAXIMIZING PROFIT ---
        // We subtracted the penalty K times, so we add it back.
        best_ans = res.cost + (1LL * k * mid_penalty);

        // --- IF MINIMIZING COST ---
        // best_ans = res.cost - (1LL * k * mid_penalty);

        // Penalty was too weak, increase it to force picking fewer
        ↪ items next time
        low = mid_penalty + 1;
    } else {
        // We picked too few items. The penalty was too harsh.
        high = mid_penalty - 1;
    }
}

// Output the final answer
cout << best_ans << "\n";
}

```

8.1.9 MaxAverageSubarray

UPPER hull (maximum) and this one is the LOWER hull (minimum) - they are NOT

```

// NOTE: 02 - CHT.cpp also declares 'struct Line' / 'struct
    ↪ HullDynamic', but that one is the
// UPPER hull (maximum) and this one is the LOWER hull (minimum)
    ↪ - they are NOT
// interchangeable. If you need both, rename one pair to
    ↪ UpperHull / LowerHull.
/*
    ↪ =====
* DYNAMIC CONVEX HULL TRICK (CHT) - AVERAGE SUBARRAY TEMPLATE
*
    ↪ =====
*
* HOW TO USE THE 4 VARIATIONS:
*
* 1. MAX Average, LONGEST length (Default)
*   - Code: Use 'lhs < rhs' in Query::operator<
*   - Call: hull.insert_line(i, S[i]); hull.query_tangent_index(i,
    ↪ S[i]);
*
* 2. MAX Average, SHORTEST length
*   - Code: Use 'lhs <= rhs' in Query::operator<
*   - Call: hull.insert_line(i, S[i]); hull.query_tangent_index(i,
    ↪ S[i]);
*
* 3. MIN Average, LONGEST length
*   - Code: Use 'lhs < rhs' in Query::operator<
*   - Call: hull.insert_line(i, -S[i]); hull.query_tangent_index(i,
    ↪ -S[i]); // Negated
*
* 4. MIN Average, SHORTEST length
*   - Code: Use 'lhs <= rhs' in Query::operator<
*   - Call: hull.insert_line(i, -S[i]); hull.query_tangent_index(i,
    ↪ -S[i]); // Negated
*
    ↪ =====
    ↪ */

```

```

struct Query {
    ll i, Si;
};

struct Line {
    ll m, b;
    mutable function<const Line *(> succ;

    bool operator<(const Line &other) const {
        return m < other.m;
    }

    bool operator<(const Query &q) const {
        const Line *s = succ();

```

```

        if (!s) return false;

        // CORRECTED ALGEBRA: Direct cross-multiplication of slopes
        __int128_t lhs = (__int128_t)(q.Si - b) * (q.i - s->m);
        __int128_t rhs = (__int128_t)(q.Si - s->b) * (q.i - m);

        // --- TIE-BREAKING CONTROL ---
        // Change this return statement based on length requirements:
        // return lhs < rhs; // <-- Use for LONGEST subarray
        // return lhs <= rhs; // <-- Use for SHORTEST subarray
        return lhs < rhs;
    }
};

struct HullDynamic : public multiset<Line, less<>> {
    bool bad(iterator y) {
        auto z = next(y);
        if (y == begin()) {
            if (z == end()) return false;
            // CORRECTED: Lower hull keeps smallest y
            return y->m == z->m && y->b >= z->b;
        }
        auto x = prev(y);
        if (z == end()) return y->m == x->m && y->b >= x->b;

        return (__int128_t)(x->b - y->b) * (z->m - y->m) <=
            ↪ (__int128_t)(y->b - z->b) * (y->m - x->m);
    }

    void insert_line(ll m, ll b) {
        auto y = insert({m, b});
        y->succ = [=] { return next(y) == end() ? 0 : &*next(y); };
        if (bad(y)) {
            erase(y);
            return;
        }
        while (next(y) != end() && bad(next(y))) erase(next(y));
        while (y != begin() && bad(prev(y))) erase(prev(y));
    }

    int query_tangent_index(ll i, ll Si) {
        auto l = *lower_bound(Query{i, Si});
        return l.m;
    }
};

```

```

// USAGE (longest subarray with average >= 0, after subtracting the
    ↪ target from every element):
// S[0..n] = prefix sums; the answer for each i is i - (tangent index
    ↪ from the hull of (j, S[j])).
// HullDynamic hull; hull.insert_line(0, 0);
// for (int i = 1; i <= n; i++) {
//     int k = hull.query_tangent_index(i, S[i]); // best j
    ↪ < i
//     ans[i] = i - k;
//     hull.insert_line(i, S[i]);
// }
// WHY IT WORKS: avg(j+1..i) >= 0 <=> S[i] - S[j] >= 0, and among all
    ↪ valid j you want the
// SMALLEST, which is the point of tangency from (i, S[i]) to the lower
    ↪ hull of the points (j, S[j]).
// Binary search on the answer + prefix sums is the shorter O(n log n)
    ↪ alternative; this is O(n).
// THE FOUR VARIATIONS, all from the same hull - only the tie-break and
    ↪ the sign change:
// MAX average, LONGEST segment: insert(i, S[i]), tangent(i, S[i]),
    ↪ strict '<' in the compare
// MAX average, SHORTEST segment: insert(i, S[i]), tangent(i, S[i]),
    ↪ non-strict '<='
// MIN average, LONGEST segment: insert(i, -S[i]), tangent(i,
    ↪ -S[i]), strict '<'
// MIN average, SHORTEST segment: insert(i, -S[i]), tangent(i,
    ↪ -S[i]), non-strict '<='
// (Minimising is just maximising the negated prefix sums; the
    ↪ tie-break decides which of two
// equally-good segments you keep, which is what "longest" versus
    ↪ "shortest" means here.)

```

8.1.10 SlopeTrick

SLOPE TRICK - maintain a CONVEX piecewise-linear $f(x)$ as two heaps of its breakpoints.

```

// L = left slopes (max-heap), R = right slopes (min-heap), minf =
    ↪ current minimum value.
// Every operation below is O(log n). Classic use: "minimum total
    ↪ |change| to make an array
// non-decreasing", scheduling with earliness/tardiness penalties, and
    ↪ any DP where
// dp_i(x) = min over y<=x of dp_{i-1}(y) + |x - a_i|.

```

```

struct SlopeTrick {
    priority_queue<ll> L; // left
    ↪ breakpoints (max-heap)
    priority_queue<ll, vector<ll>, greater<ll>> R; // right
    ↪ breakpoints (min-heap)
    ll minf = 0, addL = 0, addR = 0; // lazy
    ↪ shifts of each side

    ll topL() { return L.top() + addL; }
    ll topR() { return R.top() + addR; }
    void pushL(ll x) { L.push(x - addL); }
    void pushR(ll x) { R.push(x - addR); }

    void addConst(ll c) { minf += c; }
    void addRampR(ll a) { // f(x) +=
    ↪ max(0, x - a)
        if (!L.empty()) minf += max(0LL, topL() - a);
        pushL(a);
        pushR(topL()); L.pop();
    }
    void addRampL(ll a) { // f(x) +=
    ↪ max(0, a - x)
        if (!R.empty()) minf += max(0LL, a - topR());
        pushR(a);
        pushL(topR()); R.pop();
    }
    void addAbs(ll a) { addRampL(a), addRampR(a); } // f(x) +=
    ↪ |x - a|
    void prefixMin() { while (!R.empty()) R.pop(); } // f(x) =
    ↪ min_{y<=x} f(y)
    void suffixMin() { while (!L.empty()) L.pop(); } // f(x) =
    ↪ min_{y>=x} f(y)
    void shift(ll dl, ll dr) { addL += dl, addR += dr; } // f(x) ->
    ↪ min over [x-dr, x-dl]
    ll minVal() const { return minf; }
};
// RECIPE for "make a[] non-decreasing with min sum |b_i - a_i|":
// for each a_i: st.prefixMin(); st.addAbs(a_i); answer =
    ↪ st.minVal()
// For "strictly increasing", first replace a_i by a_i - i.

```

8.1.11 MinPlusConvolution

MIN-PLUS / MAX-PLUS CONVOLUTION: $c[k] = \min_i (a[i] + b[k-i])$.

// In general this is quadratic and CONJECTURED to stay that way (the
 ↪ MinConv hypothesis:
 // MinConv, MaxConv, unbounded knapsack and 0/1 knapsack are all
 ↪ equivalent under subquadratic
 // reductions). So the ONLY fast versions rely on convexity.

// BOTH CONVEX -> $O(n + m)$. The difference sequence of the answer is
 ↪ the sorted MERGE of the two
 // difference sequences: greedily always take the smaller next slope.
 vector<ll> minPlusConvexConvex(const vector<ll>& a, const vector<ll>& b) {

```

    ↪ b) {
        int n = a.size(), m = b.size();
        vector<ll> c(n + m - 1);
        c[0] = a[0] + b[0];
        int i = 0, j = 0;
        for (int k = 1; k < n + m - 1; k++) {
            ll da = (i + 1 < n) ? a[i + 1] - a[i] : LLONG_MAX;
            ll db = (j + 1 < m) ? b[j + 1] - b[j] : LLONG_MAX;
            if (da <= db) c[k] = c[k - 1] + da, i++;
            else c[k] = c[k - 1] + db, j++;
        }
        return c;
    }
}
// MAX-PLUS with both CONCAVE: same code, take the LARGER slope first
    ↪ (or negate and reuse).
vector<ll> maxPlusConcaveConcave(vector<ll> a, vector<ll> b) {
    for (ll& x : a) x = -x;
    for (ll& x : b) x = -x;
    auto c = minPlusConvexConvex(a, b);
    for (ll& x : c) x = -x;
    return c;
}

```

// ONLY ONE CONVEX -> the matrix $X[k][i] = a[i] + b[k-i]$ is totally
 ↪ monotone (Monge), so the row
 // minima can be found with SMAWK in $O(n + m)$, or with divide & conquer
 ↪ in $O(n \log n)$:
 // solve(kl, kr, il, ir): take k = mid, scan i in [il, ir] for the
 ↪ best, recurse with the
 // split at the optimum. Same skeleton as DivideAndConquerDP.
 // 'valid(k)' must return the legal range of i for output index k - for
 ↪ the convolution above
 // that is $[\max(0, k-m+1), \min(n-1, k)]$. Intersecting with it is what
 ↪ keeps 'best' well defined;
 // without it an empty candidate set leaves best = -1 and poisons both

↪ recursive calls.

```

template <class F, class V>
void dcOpt(int kl, int kr, int il, int ir, vector<ll>& c, F cost, V
    ↪ valid) {
    if (kl > kr) return;
    int km = (kl + kr) / 2, best = -1;
    ll bv = LLONG_MAX;
    auto [lo, hi] = valid(km);
    for (int i = max(il, lo); i <= min(ir, hi); i++) {
        ll v = cost(km, i);
        if (v < bv) bv = v, best = i;
    }
    c[km] = bv;
    if (best < 0) best = max(il, lo); // no candidate:
    ↪ keep the bounds sane
    dcOpt(kl, km - 1, il, best, c, cost, valid);
    dcOpt(km + 1, kr, best, ir, c, cost, valid);
}
// NEITHER CONVEX ->  $O(n*m)$  is the best you should plan for, and that
    ↪ is not laziness: min-plus
// convolution, 0/1 knapsack, unbounded knapsack and tree knapsack are
    ↪ subquadratic-EQUIVALENT,
// so a general fast merge would break all four. Look for convexity, a
    ↪ small value range, or a
// bitset instead of a cleverer convolution.
// WHERE THIS APPEARS: merging two "cost as a function of how many you
    ↪ take" curves - group
// knapsack, tree DP where each child contributes a convex cost curve,
    ↪ and combining the
// per-part curves produced by Alien's trick.

```

8.1.12 OneDOneD

1D/1D DP OPTIMISATION - the self-referential case that divide & conquer structurally cannot do.

```

// dp[i] = min over j < i of ( dp[j] + w(j, i) )
// D&C optimisation needs the LAYERED form dp[k][i] = min(dp[k-1][j] +
    ↪ w) so that the whole
// previous layer is already known. Here dp[j] is produced by the very
    ↪ loop that consumes it, so
// you must decide dp[i] before you may use it. That is what this file
    ↪ solves.
// PRECONDITION: w satisfies the QUADRANGLE INEQUALITY (Monge)
// w(a,c) + w(b,d) <= w(a,d) + w(b,c) for all a <= b <= c <= d
// checkable in the local form w(a,c) + w(a+1,c+1) <= w(a+1,c) +
    ↪ w(a,c+1).
// Then the optimal j for i is NON-DECREASING in i, so the array of
    ↪ "who is currently best for
// each future i" is a sequence of INTERVALS, each owned by one j - and
    ↪ finalising dp[i] can only
// append one new interval and overwrite a suffix of the old ones. Keep
    ↪ them on a stack.
//  $O(n \log n)$  (the binary search for the crossover);  $O(n)$  if the
    ↪ crossover is  $O(1)$  to find.

```

```

template <class F>
vector<ll> onedOned(int n, F w) { // w(j,
    ↪ i) for j < i
    vector<ll> dp(n + 1, llinf);
    dp[0] = 0;
    vector<array<int, 3>> st; //
    ↪ {owner j, from, to} intervals
    st.push_back({0, 1, n});
    int head = 0;
    for (int i = 1; i <= n; i++) {
        while (head < (int)st.size() && st[head][2] < i) head++;
        dp[i] = dp[st[head][0]] + w(st[head][0], i); // the
        ↪ owner of position i
        st[head][1] = i;
        // now insert i as a candidate owner for the suffix it wins
        auto better = [&](int a, int b, int x) { // is a
        ↪ better than b at x?
            return dp[a] + w(a, x) <= dp[b] + w(b, x);
        };
        while (!st.empty() && st.back()[1] > i && better(i,
        ↪ st.back()[0], st.back()[1])) st.pop_back();
        if (st.empty() || head >= (int)st.size()) st.push_back({i, i +
        ↪ 1, n});
        else {
            auto [j, l, r] = st.back();
            if (better(i, j, r)) { // i
            ↪ wins part of this interval
                int lo = max(l, i + 1), hi = r, pos = r + 1;
                while (lo <= hi) { //
                ↪ first x where i beats j
                    int m = (lo + hi) / 2;
                    better(i, j, m) ? (pos = m, hi = m - 1) : lo = m +
                    ↪ 1;
                }
            }
        }
    }
}

```

```

        st.back()[2] = pos - 1;
        if (pos <= n) st.push_back({i, pos, n});
    } else if (r < n) st.push_back({i, r + 1, n});
}
}
return dp;
}
/* WHEN YOU NEED WHICH OPTIMISATION - the whole family on one card
dp[i] = min_j (dp[j] + w(j,i)), w Monge, SELF-REFERENTIAL ->
-> THIS FILE, O(n log n)
dp[k][i] = min_j (dp[k-1][j] + w(j,i)), w Monge, LAYERED
-> -> D&C opt (01), O(n k log n)
dp[i][j] = min_k (dp[i][k] + dp[k+1][j]) + w(i,j), w Monge AND
monotone on intervals
-> -> Knuth (06), O(n^2)
dp[i] = min_j (m_j * x_i + b_j)
-> -> CHT (02) / Li Chao (04)
"exactly k pieces", optimum convex in k
-> -> Aliens (07)
min-plus convolution of a CONVEX sequence with anything
-> -> SMAWK / D&C (11)
the query point moves monotonically and lines are range-updated
-> -> kinetic segment tree
VERIFY the quadrangle inequality numerically on random quadruples
-> before trusting any of them:
a DP optimisation applied without its precondition is a silent wrong
-> answer, never a crash.
COMMON MONGE COSTS: (S_j - S_i)^2; f(S_j - S_i) with f convex and
-> a[] >= 0; c * (j - i);
"number of equal pairs inside [i,j]"; a_i * b_j with a
-> non-increasing and b non-decreasing.
NOT MONGE: anything built from max, and anything with negative a[]
-> where you also need the
interval-monotonicity half (D&C only needs QI; Knuth needs both). */

```

8.1.13 KineticSegmentTree

KINETIC SEGMENT TREE ("segment tree beats" for lines). Each position i owns a LINE

```

// f_i(x) = a_i * x + b_i and a current parameter x; the tree supports
// addX(l, r, d) : x += d for every position in [l, r] (d >= 0,
-> MONOTONE)
// queryMax(l, r) : max over the range of a_i * x_i + b_i
// setLine(i, a, b): replace one line
// in AMORTISED O(log^2 n) per operation (O(n log^2 n) total), and that
-> amortisation is the point:
// a Li Chao tree answers "many lines, one query point" while THIS
-> answers "one line per position,
// query points that drift". 12 - OneDOneD.cpp lists it as the tool for
-> "the query point moves
// monotonically and the lines are range-updated" - this is that tool.
// PRECONDITION FOR THE COMPLEXITY (not for correctness): x only ever
-> INCREASES. Each node keeps
// 'melt' = how much further x may advance before the argmax inside it
-> changes; a decrease would
// invalidate every melt and destroy the potential-function argument
-> (the total number of argmax
// changes over the whole run is O(n log^2 n) only for monotone x).
// INDEX CONVENTION: 0-based positions, inclusive ranges, x starts at 0
-> for every position.

```

```

struct KineticSegTree {
    struct Node { ll val, slope, melt; }; // melt
    -> = INF when nothing can flip
    int n;
    vector<Node> t;
    vector<ll> lz; //
    -> pending x-advance for a subtree
    KineticSegTree(const vector<ll>& a, const vector<ll>& b) {
        n = a.size(), t.assign(4 * n, {0, 0, 0}), lz.assign(4 * n, 0);
        build(1, 0, n - 1, a, b);
    }
    Node merge(const Node& L, const Node& R) {
        const Node &w = L.val >= R.val ? L : R, &o = L.val >= R.val ? R : L;
        -> : L;
        ll cross = llinf;
        if (o.slope > w.slope) cross = (w.val - o.val) / (o.slope -
        -> w.slope) + 1;
        return {w.val, w.slope, min({L.melt, R.melt, cross})};
    }
    void build(int v, int l, int r, const vector<ll>& a, const
    -> vector<ll>& b) {
        if (l == r) { t[v] = {b[l], a[l], llinf}; return; }
        int m = (l + r) / 2;
        build(2 * v, l, m, a, b), build(2 * v + 1, m + 1, r, a, b);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }
    void apply(int v, ll d) { // legal
        -> only when d < t[v].melt
        t[v].val += t[v].slope * d;

```

```

        if (t[v].melt != llinf) t[v].melt -= d;
        lz[v] += d;
    }
    void push(int v) {
        if (!lz[v]) return;
        apply(2 * v, lz[v]), apply(2 * v + 1, lz[v]), lz[v] = 0;
    }
    void addX(int v, int l, int r, int ql, int qr, ll d) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr && d < t[v].melt) { apply(v, d); return;
        -> }
        if (l == r) { t[v].val += t[v].slope * d; return; } // a
        -> leaf never "melts"
        push(v);
        int m = (l + r) / 2;
        addX(2 * v, l, m, ql, qr, d), addX(2 * v + 1, m + 1, r, ql, qr,
        -> d);
        t[v] = merge(t[2 * v], t[2 * v + 1]); // the
        -> RECURSION is the beats step
    }
    ll query(int v, int l, int r, int ql, int qr) {
        if (qr < l || r < ql) return LLONG_MIN;
        if (ql <= l && r <= qr) return t[v].val;
        push(v);
        int m = (l + r) / 2;
        return max(query(2 * v, l, m, ql, qr), query(2 * v + 1, m + 1,
        -> r, ql, qr));
    }
    void setLine(int v, int l, int r, int i, ll a, ll b, ll x) {
        if (l == r) { t[v] = {a * x + b, a, llinf}; return; }
        push(v);
        int m = (l + r) / 2;
        i <= m ? setLine(2 * v, l, m, i, a, b, x) : setLine(2 * v + 1,
        -> m + 1, r, i, a, b, x);
        t[v] = merge(t[2 * v], t[2 * v + 1]);
    }
    void addX(int l, int r, ll d) { addX(1, 0, n - 1, l, r, d); }
    ll queryMax(int l, int r) { return query(1, 0, n - 1, l, r); }
    void setLine(int i, ll a, ll b, ll x) { setLine(1, 0, n - 1, i, a,
    -> b, x); }
}

```

/* USING IT

THE MODEL PROBLEM: process items left to right; item i contributes a_i
-> * (how long it has been
open) + b_i , and you repeatedly ask for the best open item. Every
-> "time step" is one range
addX, every answer one queryMax.
SLOPE TRICK / APERIODIC DPs of the form $dp[i] = \max_j (dp[j] + w_j * (i - j))$ become: insert a
line at position j when it becomes available, addX(0, $n-1$, 1) each
-> step, query the max. That
is the self-referential case D&C optimisation cannot do (see 12 -
-> OneDOneD.cpp).
THE MELT FORMULA is integer division; with real-valued slopes use long
-> double and a small
epsilon, or scale to integers. 'cross' must be the FIRST x at which
-> the loser strictly wins,
which is why it is $(diff / dslope) + 1$ - dropping the +1 makes the
-> tree do nothing and the
answers silently stale.
DECREASING x , or arbitrary jumps, still give CORRECT answers (the
-> recursion just triggers more
often) but the complexity degrades to $O(n)$ per operation in the
-> worst case.
SIMPLER ALTERNATIVES FIRST: if all lines are inserted once and queries
-> are offline-sortable, use
CHT (02) or Li Chao (04). Reach for this only when positions are
-> updated in RANGES. */

8.2 Bitmask DP

8.2.1 SOS_DP

Computes the sum of all subsets for every bitmask in $O(N \cdot 2^N)$

```
vector<long long> SOS_DP(int n_bits, const vector<long long>& a) {
    int max_mask = 1 << n_bits;
    vector<long long> dp = a; // dp[mask] initially holds exactly the
    ↪ value for mask

    for (int i = 0; i < n_bits; ++i) {
        for (int mask = 0; mask < max_mask; ++mask) {
            if (mask & (1 << i)) {
                dp[mask] += dp[mask ^ (1 << i)];
            }
        }
    }
    return dp; // dp[mask] now contains sum of all submasks of mask
}
```

8.2.2 BitmaskDP

BITMASK DP - the $n \leq 20$ toolkit.

// MEMORY: vector<vector<ll>> dp(1<<20, vector<ll>(20)) is ~193 MB (24
 ↪ B of vector header per row).
 // Flatten to ONE vector<ll> of size $(1 << n) \cdot n$ (168 MB) or vector<int>
 ↪ (84 MB), or iterate masks in
 // increasing popcount and keep only two layers. At $n = 20$, $2^n \cdot n^2 =$
 ↪ 4e8 - usually not intended.

```
// TSP: shortest Hamiltonian cycle from 0 back to 0.  $O(2^n \cdot n^2)$ .
ll tsp(const vector<vector<ll>>& d) {
    int n = d.size(), N = 1 << n;
    vector<vector<ll>> dp(N, vector<ll>(n, llinf));
    dp[1][0] = 0;
    for (int m = 1; m < N; m++)
        for (int u = 0; u < n; u++) if ((m >> u & 1) && dp[m][u] <
    ↪ llinf)
            for (int v = 0; v < n; v++) if (!(m >> v & 1) && d[u][v] <
    ↪ llinf)
                dp[m | 1 << v][v] = min(dp[m | 1 << v][v], dp[m][u] +
    ↪ d[u][v]);
    ll best = llinf;
    for (int u = 1; u < n; u++) if (dp[N - 1][u] < llinf && d[u][0] <
    ↪ llinf)
        best = min(best, dp[N - 1][u] + d[u][0]);
    return n == 1 ? 0 : best; // drop the
    ↪ +d[u][0] for a Hamiltonian PATH
}
```

```
// MINIMUM SET COVER / partition into the fewest valid groups.  $O(3^n)$ 
    ↪ via submask enumeration.
// ok[m] = true if the set m is a legal single group.
int minPartition(int n, const vector<char>& ok) {
    int N = 1 << n;
    vector<int> dp(N, inf);
    dp[0] = 0;
    for (int m = 1; m < N; m++) {
        int low = m & -m; // fix the lowest
    ↪ element: kills the n! factor
        for (int s = m; s < N; s = (s - 1) & m) if ((s & low) && ok[s])
            dp[m] = min(dp[m], dp[m ^ s] + 1);
    }
    return dp[N - 1];
}
```

// COUNTING the partitions instead: same loop with $dp[m] += dp[m \wedge s]$.
 // If groups are UNORDERED, fixing the lowest element (above) already
 ↪ prevents double counting.

```
// ASSIGNMENT (n tasks to n workers, minimum cost),  $O(2^n \cdot n)$ .  $dp[m] =$   

    ↪ best cost using the  

    // first popcount(m) workers on exactly the task set m.
ll assignment(const vector<vector<ll>>& c) {
    int n = c.size(), N = 1 << n;
    vector<ll> dp(N, llinf);
    dp[0] = 0;
    for (int m = 0; m < N - 1; m++) if (dp[m] < llinf) {
        int i = __builtin_popcount(m); // next worker
        for (int j = 0; j < n; j++) if (!(m >> j & 1))
            dp[m | 1 << j] = min(dp[m | 1 << j], dp[m] + c[i][j]);
    }
    return dp[N - 1]; // n > 20 -> use
    ↪ the Hungarian algorithm
}
```

// COUNTING HAMILTONIAN PATHS / permutations with adjacency
 ↪ constraints: same shape as tsp but
 // $dp[m][u] += dp[m \wedge (1 << u)][v]$ over allowed v.
 // SUBSET-SUM STYLE: when the state is "which items used" AND the order
 ↪ does not matter, prefer

// dp over masks; when only a total matters, use a knapsack - 2^n is
 ↪ not always necessary.
 // MEET IN THE MIDDLE: $n \leq 40$ -> split in half, enumerate 2^{20} each
 ↪ side, sort + two pointers.

8.2.3 SubsetConvolution

ZETA / MOBIUS OVER THE SUBSET LATTICE, and SUBSET CONVOLUTION. 01 -
 SOS_DP.cpp has the plain

```
// "sum over submasks"; this file adds the INVERSE (which is what turns
    ↪ a sum-over-subsets answer
// back into an exact-set answer) and the convolution that requires the
    ↪ parts to be DISJOINT.
// OR-convolution c[k] = sum over i | j = k of a[i] b[j] ->
    ↪ zeta, multiply, mobius
// AND-convolution c[k] = sum over i & j = k of a[i] b[j] ->
    ↪ superZeta, mul, superMobius
// XOR-convolution c[k] = sum over i ^ j = k of a[i] b[j] ->
    ↪ FWHT, see 06 - Math/05/03
// SUBSET conv c[k] = sum over i | j = k AND i & j = 0 ->
    ↪ THIS FILE,  $O(2^n \cdot n^2)$ 
// WHEN: "partition a set into groups" DPs (count ways to split into k
    ↪ connected pieces, minimum
// cost exact cover, chromatic number), where the naive "for each mask,
    ↪ for each submask" is
//  $O(3^n)$  - subset convolution replaces that with  $O(2^n \cdot n^2)$ , which at
    ↪  $n = 20$  is 4e8 vs 3.5e9.
void zeta(vector<ll>& f, int n) { // g[s]
    ↪ = sum over t subset of s
    for (int i = 0; i < n; i++)
        for (int s = 0; s < (1 << n); s++)
            if (s >> i & 1) f[s] = (f[s] + f[s ^ (1 << i)]) % MOD;
}
void mobius(vector<ll>& f, int n) { // exact
    ↪ inverse of zeta
    for (int i = 0; i < n; i++)
        for (int s = 0; s < (1 << n); s++)
            if (s >> i & 1) f[s] = (f[s] - f[s ^ (1 << i)]) % MOD + MOD;
    ↪ % MOD;
}
void superZeta(vector<ll>& f, int n) { // g[s]
    ↪ = sum over t superset of s
    for (int i = 0; i < n; i++)
        for (int s = 0; s < (1 << n); s++)
            if (!(s >> i & 1)) f[s] = (f[s] + f[s | (1 << i)]) % MOD;
}
void superMobius(vector<ll>& f, int n) {
    for (int i = 0; i < n; i++)
        for (int s = 0; s < (1 << n); s++)
            if (!(s >> i & 1)) f[s] = (f[s] - f[s | (1 << i)]) % MOD +
    ↪ MOD) % MOD;
}
// SUBSET CONVOLUTION. Rank by POPCOUNT: after the zeta transform, a
    ↪ product of the rank-i and
// rank-j layers counts pairs with i | j SUBSET of s and  $|i| + |j| = k$ ;
    ↪ reading the answer at
//  $k = \text{popcount}(s)$  forces  $i | j = s$  and  $|i| + |j| = |s|$ , which for a
    ↪ union is exactly  $i \& j = 0$ .
// That ranking is the entire idea - without it, disjointness is not
    ↪ expressible as a transform.
vector<ll> subsetConvolution(const vector<ll>& A, const vector<ll>& B,
    ↪ int n) {
    int N = 1 << n;
    vector<vector<ll>> f(n + 1, vector<ll>(N, 0)), g = f, h = f;
    for (int s = 0; s < N; s++) {
        int p = __builtin_popcount(s);
        f[p][s] = A[s], g[p][s] = B[s];
    }
    for (int k = 0; k <= n; k++) zeta(f[k], n), zeta(g[k], n);
    for (int k = 0; k <= n; k++)
        for (int i = 0; i <= k; i++)
            for (int s = 0; s < N; s++) h[k][s] = (h[k][s] + f[i][s] *
    ↪ g[k - i][s]) % MOD;
    for (int k = 0; k <= n; k++) mobius(h[k], n);
    vector<ll> c(N);
    for (int s = 0; s < N; s++) c[s] = h[__builtin_popcount(s)][s];
    return c;
}
```

/* USING IT

SET PARTITION / EXACT COVER: let $f[s] = 1$ if s is a legal piece. The
 ↪ number of ways to cover the
 full set with exactly k pieces is the k-th subset-convolution POWER
 ↪ of f at s = full; dividing
 by k! makes the pieces unordered. Chromatic number = smallest k with
 ↪ (independent sets) $^k > 0$.
 SUBSET EXP / LOG exist too ($O(2^n \cdot n^2)$) and turn "connected pieces"
 ↪ into "all pieces" and back -
 the standard route for counting CONNECTED labelled structures on

↪ subsets.
 MIN-PLUS version (minimum cost exact cover) does NOT work: min-plus
 ↪ has no inverse, so the
 mobius step is invalid. Use the $O(3^n)$ submask enumeration for
 ↪ min/max objectives.
 $O(3^n)$ SUBMASK ENUMERATION is 'for (int t = s; t; t = (t - 1) & s)'
 ↪ and remains the right answer
 for $n \leq 15$ or when the operation is not a ring - it is shorter and
 ↪ has no modulus.
 XOR needs FWHT, not this; AND/OR need only the transform pair with no
 ↪ ranking.
 OVERFLOW: everything here is mod MOD. Without a modulus use `__int128`
 ↪ or accept unsigned wraparound
 for OR/AND (the mobius step is exact in $\mathbb{Z}$, so wraparound cancels). */

8.2.4 DynamicSubmaskCount

DYNAMIC SUBMASK COUNTING - a multiset of masks with INSERT, ERASE and the query

```

// "how many stored masks a satisfy a & s == a (a is a submask of
// s)"
// in  $O(2^{B/2})$  per operation, B = number of bits. SOS DP answers this
// in  $O(1)$  but only for a
// STATIC multiset (any insert costs a full  $O(B \cdot 2^B)$  rebuild);
// enumerating submasks of s per query
// is  $O(2^B)$ . This is the meet-in-the-middle between the two, and the
// technique generalises to any
// "precompute forwards for heavy items, backwards for light items"
// split.
// THE SPLIT: call a mask HEAVY if popcount > B/2, LIGHT otherwise.
// LIGHT masks are stored twice: once at their own index, and once at
// EVERY SUBMASK of theirs
// (<=  $2^{B/2}$  of them) in sup[], so sup[t] = #light masks that
// CONTAIN t.
// HEAVY masks are stored at every SUPERMASK of theirs (<=  $2^{B/2}$  of
// them) in sub[],
// so sub[t] = #heavy masks that are CONTAINED in t.
// A query with a light s only needs light answers (a submask of s has
// popcount <= popcount(s)),
// and is a submask enumeration over s, <=  $2^{B/2}$  steps. A query with
// a heavy s reads sub[s] for
// the heavy part and inclusion-exclusion over the ZERO bits of s
// (there are < B/2 of them) for
// the light part: #light masks meeting ~s = sum over non-empty T in ~s
// of  $(-1)^{|T|+1}$  sup[T].
template <int B>
struct SubmaskCounter {
    static const int SZ = 1 << B, H = B / 2;
    vector<int> own, sup, sub;
    // own[m] = #light copies of m
    int lightTotal = 0;
    SubmaskCounter() : own(SZ, 0), sup(SZ, 0), sub(SZ, 0) {}
    void modify(int m, int d) {
        // d =
        // +1 to insert, -1 to erase
        if (__builtin_popcount(m) <= H) {
            own[m] += d, lightTotal += d;
            for (int t = m; t; t = (t - 1) & m) { sup[t] += d; if (!t)
            ↪ break; }
        } else {
            int zeros = ((SZ - 1) ^ m);
            for (int t = zeros; t; t = (t - 1) & zeros) { sub[m | t] +=
            ↪ d; if (!t) break; }
        }
        void insert(int m) { modify(m, 1); }
        void erase(int m) { modify(m, -1); }
        int count(int s) {
            //
            ↪ #stored masks that are submasks
            if (__builtin_popcount(s) <= H) {
                int r = 0;
                for (int t = s; t; t = (t - 1) & s) { r += own[t]; if (!t)
                ↪ break; }
                return r;
            }
            int zeros = ((SZ - 1) ^ s), bad = 0;
            for (int t = zeros; t; t = (t - 1) & zeros)
            ↪ inclusion-exclusion, T non-empty
                bad += (__builtin_popcount(t) & 1) ? sup[t] : -sup[t];
            return sub[s] + lightTotal - bad;
        }
    };
};
/* NOTES
THE THRESHOLD H = B/2 balances  $2^H$  against  $2^{B-H}$ ; with skewed
↪ workloads (many more queries
than inserts) move it - all four loops stay correct for any H.
MEMORY is  $3 \cdot 2^B$  ints: B = 20 is 12 MB, B = 22 is 50 MB. Above that,
↪ drop 'own' (it equals the
light part of a separate map) or switch to a hash map for the sparse

```

↪ arrays.
 SUPERMASK COUNTING ("how many stored a with a & s == s") is the same
 ↪ code with the roles of sup
 and sub swapped, or just complement everything: a superset of s is a
 ↪ submask of ~s complemented.
 EXACTLY-EQUAL, OR "how many a with a & s == 0" (disjoint) = count(~s)
 ↪ - both fall out for free.
 STATIC ALTERNATIVE: if all inserts precede all queries, run SOS DP
 ↪ once and answer in $O(1)$. Check
 that first; this structure is only for genuinely interleaved updates
 ↪ and queries. */

8.3 Knapsack

8.3.1 BoundedKnapsackBitset

BOUNDED KNAPSACK, FEASIBILITY ONLY, with a bitset: $O(n \cdot W / 64)$. Binary-group the counts

// (1,2,4,...) so "cnt copies of w" becomes $O(\log \text{cnt})$ shift-ors. Use
 ↪ when you only need which sums
 // are reachable, not the best value.

```

const int MAX_W = 100005;
bitset<MAX_W> dp;

```

```

void multiset_bitset_dp() {
    // Example: We have 13 items, each with weight 5
    int count = 13;
    int weight = 5;

    vector<int> bundled_weights;

    // Binary grouping logic
    int current_power = 1;
    while (count >= current_power) {
        bundled_weights.push_back(current_power * weight);
        count -= current_power;
        current_power *= 2;
    }
    // Add whatever is left over
    if (count > 0) {
        bundled_weights.push_back(count * weight);
    }

    // Now run the ultra-fast Bitset DP on the bundles
    dp[0] = 1;
    for (int bundle_w : bundled_weights) {
        dp |= (dp << bundle_w);
    }
}

```

8.3.2 KnapsackFamily

KNAPSACK FAMILY. W = capacity. All $O(n \cdot W)$ unless noted.

```

// 0/1: iterate capacity DOWNWARD so each item is used once.
ll knap01(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    for (size_t i = 0; i < w.size(); i++)
        for (int c = W; c >= w[i]; c--) dp[c] = max(dp[c], dp[c - w[i]]
        ↪ + v[i]);
    return dp[W];
}
// UNBOUNDED: iterate capacity UPWARD.
ll knapInf(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    for (size_t i = 0; i < w.size(); i++)
        for (int c = w[i]; c <= W; c++) dp[c] = max(dp[c], dp[c - w[i]]
        ↪ + v[i]);
    return dp[W];
}
// BOUNDED (item i available cnt[i] times) via BINARY GROUPING: split
↪ cnt into 1,2,4,...,rest
// and run 0/1 on the groups.  $O(W \cdot \sum \log \text{cnt})$ .
ll knapBounded(const vector<int>& w, const vector<ll>& v, const
↪ vector<int>& cnt, int W) {
    vector<int> W2; vector<ll> V2;
    for (size_t i = 0; i < w.size(); i++) {
        int c = cnt[i];
        for (int k = 1; c > 0; k <= 1) {
            int t = min(k, c); c -= t;
            W2.push_back(w[i] * t), V2.push_back(v[i] * t);
        }
    }
    return knap01(W2, V2, W);
}
// FEASIBILITY ONLY (which sums are reachable) -> bitset,  $O(n \cdot W / 64)$ .
↪ Massively faster.
bitset<100001> reachableSums(const vector<int>& w) {
    bitset<100001> dp;

```

```

dp[0] = 1;
for (int x : w) dp |= dp << x;
return dp;
}
// COUNTING ways (mod p): same loops, dp[c] += dp[c - w[i]].
// Order matters: item-outer/capacity-inner counts COMBINATIONS,
// capacity-outer/item-inner counts PERMUTATIONS
// → (ordered sequences).

```

8.3.3 BoundedKnapsackDeque

BOUNDED KNAPSACK IN $O(n * W)$ - item i may be used at most $\text{cnt}[i]$ times. The binary-grouping

```

// version in 02 - KnapsackFamily.cpp is  $O(W * \sum \log \text{cnt})$ , which is
// → the one to write by default;
// reach for this when sum log cnt is too big (huge counts, or W
// → already at the limit).
// THE REWRITE: fix item i and split the capacities by RESIDUE mod
// → w[i]. Along one residue class,
// dp'[t] = max over k in [0, cnt] of dp[t - k] + k * val
// = ( max over j in [t - cnt, t] of dp[j] - j * val ) + t *
// → val
// which is a SLIDING WINDOW MAXIMUM of width cnt + 1 - a monotone
// → deque,  $O(1)$  amortised per cell.
// PRECONDITION: none beyond w[i] >= 1; this is an exact rewrite, not
// → an approximation. (The
// subtracted term j * val is what makes the window comparable - do not
// → forget to add it back.)
ll boundedKnapsackDeque(const vector<int>& w, const vector<ll>& val,
                        const vector<int>& cnt, int W) {
    int n = w.size();
    vector<ll> dp(W + 1, 0), nd(W + 1, 0);
    for (int i = 0; i < n; i++) {
        int wi = w[i], m = cnt[i];
        ll v = val[i];
        if (wi <= 0 || m <= 0) continue;
        // → weight 0 items: add m * v to all
        if (m > W / wi) m = W / wi;
        // → copies than fit
        for (int s = 0; s < wi && s <= W; s++) {
            deque<pair<int, ll>> dq;
            // {t,
            // → dp[s + t*wi] - t*v}
            for (int t = 0; s + (ll)t * wi <= W; t++) {
                ll cur = dp[s + t * wi] - (ll)t * v;
                while (!dq.empty() && dq.back().second <= cur)
                    // → dq.pop_back();
                dq.push_back({t, cur});
                while (dq.front().first < t - m) dq.pop_front();
                nd[s + t * wi] = dq.front().second + (ll)t * v;
            }
            dp.swap(nd);
        }
        return dp[W];
    }
    // → = best with weight <= c
}

```

/* THE FAMILY, so you pick the right one under time pressure

cnt[i] = 1	→ 0/1 knapsack, $O(n W)$
→ (02 - KnapsackFamily)	
cnt[i] = infinity	→ unbounded, $O(n W)$, forward loop
→ (02 - KnapsackFamily)	
general cnt[i], small sum log	→ binary grouping, $O(W \sum \log \text{cnt})$
→ (02 - KnapsackFamily)	
general cnt[i], W large	→ THIS FILE, $O(n W)$
FEASIBILITY only (no values)	→ bitset, $O(n W / 64)$
→ (02 - KnapsackFamily)	
feasibility, many equal weights	→ $O(S \sqrt{S})$ by distinct value
→ (04 - SubsetSumSqrt)	

COUNTING the number of ways with bounded multiplicity: the deque does

- NOT apply (sums, not maxima) - use prefix sums along the same residue classes instead,
- $\text{dp}'[t] = \text{P}[t] - \text{P}[t - \text{cnt} - 1]$,
- which is the same $O(n W)$ with the same residue decomposition.

RECONSTRUCTION: store which t the deque front came from, or re-run the

- DP layer by layer keeping all n rows ($O(n W)$ memory) and walk back.

IF THE VALUES ARE THE WEIGHTS (subset sum), the reachability bitset

- beats everything by a factor of 64 - check that first, it is one line. */

8.3.4 SubsetSumSqrt

SUBSET SUM IN $O(S \sqrt{S})$ WITH RECONSTRUCTION, where S is the sum of all values (or the target,

```

// whichever is smaller). The bitset DP is  $O(n S / 64)$  and usually
// → wins, but it does NOT
// reconstruct the subset and it degrades when n is large; this one is
// →  $O(S \sqrt{S})$  REGARDLESS of
// n and hands back the item indices.
// THE COUNTING ARGUMENT that gives sqrt: only DISTINCT values matter,
// → and if the values sum to S
// there are at most  $O(\sqrt{S})$  distinct ones. For a fixed value v with
// → multiplicity c, "reachable
// with at most c copies of v" is computed in ONE  $O(S)$  pass: walk each
// → residue class mod v and
// carry a counter of how many copies are still available - a reachable
// → cell refills the counter
// to c, an unreachable cell consumes one copy if any remain. That is
// → the bounded-knapsack
// feasibility trick without a deque.
// PRECONDITION: all values > 0 (a zero value makes the residue walk
// → spin).
// RETURNS the indices of a subset summing to exactly 'target', or an
// → empty vector if impossible.
vector<int> subsetSumSqrt(const vector<int>& a, int target) {
    if (target < 0) return {};
    vector<char> dp(target + 1, 0);
    vector<int> last(target + 1, -1), cnt(target + 1, 0);
    vector<vector<int>> id(target + 1);
    for (int i = 0; i < (int)a.size(); i++)
        if (a[i] > 0 && a[i] <= target) id[a[i]].push_back(i),
        // → cnt[a[i]]++;
        dp[0] = 1;
    for (int v = 1; v <= target; v++) {
        if (!cnt[v]) continue;
        for (int r = 0; r < v; r++) {
            int left = 0;
            // → copies of v still usable
            for (int k = r; k <= target; k += v) {
                if (dp[k]) left = cnt[v];
                // → reachable without any copy here
                else if (left) dp[k] = 1, left--, last[k] = id[v][left];
            }
        }
        if (!dp[target]) return {};
        vector<int> res;
        for (int s = target; s > 0; s -= a[last[s]]) res.push_back(last[s]);
        return res;
    }
}

```

/* WHEN TO USE WHAT

n small, S small	→ plain $O(n S)$ DP, easiest to modify.
feasibility only	→ bitset $\text{dp} = \text{dp} << \text{a}[i]$, $O(n S / 64)$.
→ Fastest by far.	
need the SUBSET back	→ THIS FILE, or the bitset plus $O(n)$ "peel
→ one item at a time" with	
	a per-item bitset snapshot ($O(n S / 64)$)
→ memory - usually too much).	
many equal values	→ THIS FILE is optimal: it is $O(S)$ per
→ DISTINCT value.	
counting the number of subsets, not feasibility	→ the residue walk
→ does not apply; use prefix	
	sums per residue class (bounded) or
→ plain $O(n S)$ (unbounded).	
CLASSIC USES: "split the array into two halves with a given sum",	
→ partitioning tree levels	
(CF 1481F), and any "choose a subset of the multiset of subtree	
→ sizes" argument - those all	
have sum of values = n, hence $O(n \sqrt{n})$.	

THE SAME sqrt ARGUMENT gives the $O(n \sqrt{n} / 64)$ bound for "subset

- sums of an array whose
- values sum to n" with bitsets - group equal values, binary-group
- each group, and the total
- number of groups is $O(\sqrt{n} \log n)$. */

8.4 Classics

8.4.1 DigitDP

DIGIT DP - count numbers in $[L, R]$ with a digit property. $f(R) - f(L-1)$.

```
// State: (position, tight, started, problem state). 'started' handles
// leading zeros.
// THIS CODE IS BASE 10 (to_string, digit cap 9, memo[20] positions).
// For another base, write the
// digits yourself, change the 9, and size memo by the digit count
// (base 2 needs 64 positions).
// L - 1 IS NOT ALWAYS AVAILABLE: if the bounds are given as huge
// decimal STRINGS you cannot
// subtract one, so carry two tight flags (tightLo, tightHi) and count
// [L, R] in a single pass.
string DIG;
ll memo[20][2][2][200];
bool vis[20][2][2][200];

ll go(int p, int tight, int started, int st) {
    if (p == (int)DIG.size()) return started ? /* ACCEPT? */ (st == 0)
    : 0; // <- edit
    if (vis[p][tight][started][st]) return memo[p][tight][started][st];
    vis[p][tight][started][st] = true;
    ll res = 0;
    int hi = tight ? DIG[p] - '0' : 9;
    for (int d = 0; d <= hi; d++) {
        int nStarted = started || d > 0;
        int nSt = nStarted ? (st + d) % 10 : 0;
        // <- edit
        res += go(p + 1, tight && d == hi, nStarted, nSt);
    }
    return memo[p][tight][started][st] = res;
}

bool qualifiesZero = 1; // does the number 0
// satisfy the predicate?
ll countUpTo(ll x) {
    if (x < 0) return 0;
    DIG = to_string(x);
    memset(vis, 0, sizeof vis);
    return go(0, 1, 0, 0) + qualifiesZero; // set qualifiesZero =
    // 1 only if 0 itself counts
}

ll countRange(ll L, ll R) { return countUpTo(R) - countUpTo(L - 1); }
// VARIANTS: track (sum of digits), (value mod k), (last digit), (a
// bitmask of used digits),
// (whether we already saw "13"), (count so far) - anything with a
// small state.
// For "k-th number with the property", binary search on countUpTo.
```

8.4.2 LIS

LIS in $O(n \log n)$, with reconstruction. STRICT increasing by default.

```
// Non-decreasing: change lower_bound -> upper_bound.
// Longest DEcreasing: negate the array. Longest non-increasing: negate
// + upper_bound.
vector<int> lis(const vector<ll>& a) {
    int n = a.size();
    vector<ll> tail; // tail[k] = smallest
    // possible tail of an LIS of length k+1
    vector<int> pos(n), prv(n, -1), idx;
    for (int i = 0; i < n; i++) {
        int k = lower_bound(all(tail), a[i]) - tail.begin();
        if (k == (int)tail.size()) tail.push_back(a[i]),
        // idx.push_back(i);
        else tail[k] = a[i], idx[k] = i;
        pos[i] = k;
        prv[i] = k ? idx[k - 1] : -1;
    }
    vector<int> res;
    for (int i = idx.empty() ? -1 : idx.back(); i >= 0; i = prv[i])
    // res.push_back(i);
    reverse(all(res));
    return res; // indices of one LIS;
    // length = tail.size()
}

// SAME SHAPE SOLVES: longest chain of pairs (sort by x, LIS on y);
// maximum non-overlapping
// intervals; "minimum number of non-increasing subsequences covering
// the array" = LIS length
// (Dilworth). For LIS with an extra weight, replace 'tail' with a
// max-BIT over compressed values.
```

8.4.3 IntervalDP

INTERVAL DP - dp over $[l, r]$, processed by increasing length. $O(n^3)$ ($O(n^2)$ with Knuth).

```
// This is the base pattern that Knuth's optimisation speeds up; write
// this first, optimise later.

// 1) SPLIT form: dp[l][r] = min over k of dp[l][k] + dp[k+1][r] +
// cost(l, r)
// matrix chain multiplication, merging stones, optimal BST, "cost
// to combine a range"
template <class Cost> ll intervalSplit(int n, Cost cost) {
    vector<vector<ll>> dp(n, vector<ll>(n, 0));
    for (int len = 2; len <= n; len++)
        for (int l = 0, r = len - 1; r < n; l++, r++) {
            dp[l][r] = llinf;
            for (int k = l; k < r; k++) dp[l][r] = min(dp[l][r],
            // dp[l][k] + dp[k + 1][r]);
            dp[l][r] += cost(l, r);
        }
    return dp[0][n - 1];
}

// 2) ENDPOINT form: decide about a[l] or a[r] and shrink. Palindromes,
// "remove from either end".
// Minimum insertions to make s a palindrome:
int minInsertPalindrome(const string& s) {
    int n = s.size();
    vector<vector<int>> dp(n, vector<int>(n, 0));
    for (int len = 2; len <= n; len++)
        for (int l = 0, r = len - 1; r < n; l++, r++)
            dp[l][r] = s[l] == s[r] ? dp[l + 1][r - 1] : 1 + min(dp[l +
            // 1][r], dp[l][r - 1]);
            return n ? dp[0][n - 1] : 0;
}

// 3) "BURST BALLOONS" form: choose the LAST element removed in the
// range, so the neighbours are
// the (still intact) boundaries l-1 and r+1.
ll burst(vector<ll> a) {
    int n = a.size();
    a.insert(a.begin(), 1, a.push_back(1));
    vector<vector<ll>> dp(n + 2, vector<ll>(n + 2, 0));
    for (int len = 1; len <= n; len++)
        for (int l = 1, r = len; r <= n; l++, r++)
            for (int k = l; k <= r; k++)
                dp[l][r] = max(dp[l][r], dp[l][k - 1] + a[l - 1] * a[k]
                // * a[r + 1] + dp[k + 1][r]);
            return dp[1][n];
}

// WHEN KNUTH APPLIES (form 1): cost is monotone on intervals and
// satisfies the quadrangle
// inequality; then opt[l][r-1] <= opt[l][r] <= opt[l+1][r] and the
// k-loop becomes amortised  $O(1)$ .
// COUNTING variants (e.g. #ways to parenthesise) replace min with +
// and cost with 1.
```

8.4.4 BrokenProfileDP

BROKEN PROFILE DP - sweep the grid cell by cell carrying a bitmask of the "broken" frontier.

```
//  $O(n * m * 2^m)$ ; always make m the SMALLER dimension.
// mask bit j = "cell (i, j) is already covered by something placed
// earlier".

// Count tilings of an  $n \times m$  grid by  $1 \times 2$  dominoes.
ll dominoTilings(int n, int m) {
    if (m > n) swap(n, m);
    if (((ll)n * m) & 1) return 0;
    int M = 1 << m;
    vector<ll> cur(M, 0), nxt;
    cur[0] = 1;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) {
            nxt.assign(M, 0);
            for (int mask = 0; mask < M; mask++) if (cur[mask]) {
                if (mask >> j & 1) nxt[mask ^ (1 << j)] += cur[mask];
                // already covered
            } else {
                if (i + 1 < n) nxt[mask | (1 << j)] += cur[mask];
                // vertical domino
                if (j + 1 < m && !(mask >> (j + 1) & 1))
                    nxt[mask | (1 << (j + 1))] += cur[mask];
                // horizontal domino
            }
        }
    cur = nxt;
}
```

```

    return cur[0];
}
// BLOCKED CELLS: keep a grid 'bad[i][j]'; a blocked cell must arrive
// → already "covered"
// (skip it by clearing bit j and forbidding any placement on it).
// OTHER PIECES: add one transition per shape; the mask must record
// → every cell the shape
// occupies in a LATER position of the sweep.
// COUNTING INDEPENDENT SETS / no-two-adjacent colourings: mask = which
// → cells of the previous
// row are used; transition over masks with (a & b) == 0 and no two
// → adjacent bits in b.
// AS WRITTEN this is O(n * 4^m) (nested mask loops), not O(n * m *
// → 2^m): at m = 14 that is 2.7e8
// per row. To get O(n * 2^m), note sum over a with (a & b) == 0 of
// → cur[a] = zetaSub(cur)[~b],
// so one subset-zeta pass per row replaces the inner loop (see 05 -
// → Combinatorics/07).
ll independentSets(int n, int m) { // #ways to pick
    // → cells, none orthogonally adjacent
    if (m > n) swap(n, m);
    int M = 1 << m;
    vector<ll> cur(M, 0), nxt;
    for (int b = 0; b < M; b++) if (!(b & (b << 1))) cur[b] = 1;
    for (int i = 1; i < n; i++) {
        nxt.assign(M, 0);
        for (int a = 0; a < M; a++) if (cur[a])
            for (int b = 0; b < M; b++) if (!(b & (b << 1)) && !(a &
// → b)) nxt[b] += cur[a];
        cur = nxt;
    }
    ll r = 0;
    for (ll x : cur) r += x;
    return r;
}
// CONNECTED-COMPONENT DP ("plug DP") is the heavy version: the state
// → stores, for each frontier
// cell, WHICH connected component its plug belongs to (a bracket-like
// → labelling). Use it for
// "count Hamiltonian cycles / single-closed-loop tilings"; it is long
// → - only write it if the
// problem clearly needs connectivity, not just coverage.

```

8.4.5 MaximalRectangle

LARGEST ALL-ZERO (or all-one) RECTANGLE in a binary grid, $O(n \cdot m)$.

```

// Row by row, keep the histogram of consecutive zeros ending at this
// → row, then run the
// largest-rectangle-in-histogram monotonic stack.
ll maximalRectangle(const vector<vector<int>>& g, int target = 0) {
    int n = g.size(), m = n ? g[0].size() : 0;
    vector<int> h(m, 0);
    ll best = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) h[j] = (g[i][j] == target) ? h[j] +
// → 1 : 0;
        vector<int> st; // indices,
// → increasing height
        for (int j = 0; j <= m; j++) {
            int cur = (j == m) ? -1 : h[j];
            while (!st.empty() && h[st.back()] >= cur) {
                int ht = h[st.back()]; st.pop_back();
                int left = st.empty() ? -1 : st.back();
                best = max(best, (ll)ht * (j - left - 1));
            }
            st.push_back(j);
        }
        st.clear();
    }
    return best;
}
// LARGEST SQUARE instead: dp[i][j] = min(dp[i-1][j], dp[i][j-1],
// → dp[i-1][j-1]) + 1 when free.
// COUNTING all all-zero submatrices: for each row, for each column j
// → keep 'run[j]' = the number
// of rectangles ending at (i, j); with a monotonic stack, run[j] =
// → run[prevSmaller] + h[j] *
// (j - prevSmaller), and the total is the sum of run[j] over all
// → cells.  $O(n \cdot m)$ .
ll countAllZeroSubmatrices(const vector<vector<int>>& g) {
    int n = g.size(), m = n ? g[0].size() : 0;
    vector<ll> h(m, 0), run(m, 0);
    ll total = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) h[j] = g[i][j] == 0 ? h[j] + 1 : 0;
        vector<int> st;
        for (int j = 0; j < m; j++) {
            while (!st.empty() && h[st.back()] >= h[j]) st.pop_back();

```

```

            int p = st.empty() ? -1 : st.back();
            run[j] = (p < 0 ? 0 : run[p]) + h[j] * (j - p);
            total += run[j];
            st.push_back(j);
        }
        return total;
    }
}

```

8.4.6 Hirschberg

HIRSCHBERG - reconstruct the ANSWER of a two-sequence DP (LCS, alignment, edit script) in

```

//  $O(n \cdot m)$  time but only  $O(\min(n, m))$  memory instead of the  $O(n \cdot m)$ 
// → table. WHEN:  $n = m = 1e5$  and you
// need the actual subsequence, not its length - the table would be
// →  $1e10$  cells.
// THE IDEA (divide & conquer on the ROW, not on the value): the last
// → row of the DP for the top
// half of a, and the last row of the REVERSED DP for the bottom half,
// → meet at the middle row.
// The column k maximising  $L[k] + R[m - k]$  is a point the optimal path
// → passes through, so the
// problem splits into (top half of a vs  $b[0..k]$ ) and (bottom half of a
// → vs  $b[k..m]$ ).
// COST:  $T(n, m) = O(n \cdot m) + T(n/2, k) + T(n/2, m - k) = O(n \cdot m)$  total -
// → the halves of the second
// level together cost half of the first level, so the series is  $2 \cdot n \cdot m$ .
// PRECONDITION: the DP must be reversible - dp computed left-to-right
// → on the reversed strings
// must equal the suffix DP. True for LCS, edit distance and affine-gap
// → alignment; NOT true for
// DPs whose transition depends on the direction (e.g. "no two chosen
// → positions adjacent" is fine,
// but anything reading a prefix STATE such as a running remainder is
// → not).
vector<int> lcsLastRow(const string& a, const string& b) { //
    // →  $O(|b|)$  memory
    int m = b.size();
    vector<int> prv(m + 1, 0), cur(m + 1, 0);
    for (char ca : a) {
        for (int j = 1; j <= m; j++)
            cur[j] = ca == b[j - 1] ? prv[j - 1] + 1 : max(prv[j],
// → cur[j - 1]);
        swap(prv, cur);
    }
    return prv;
}
string hirschbergLCS(const string& a, const string& b) {
    int n = a.size(), m = b.size();
    if (n == 0 || m == 0) return "";
    if (n == 1) return b.find(a[0]) != string::npos ? string(1, a[0]) :
// → "";
    int i = n / 2;
    string a1 = a.substr(0, i), a2 = a.substr(i);
    string ra(a2.rbegin(), a2.rend()), rb(b.rbegin(), b.rend());
    vector<int> L = lcsLastRow(a1, b), R = lcsLastRow(ra, rb);
    int best = -1, k = 0;
    for (int j = 0; j <= m; j++) if (L[j] + R[m - j] > best) best =
// → L[j] + R[m - j], k = j;
    return hirschbergLCS(a1, b.substr(0, k)) + hirschbergLCS(a2,
// → b.substr(k));
}
/* ADAPTING IT
EDIT DISTANCE / ALIGNMENT: replace lcsLastRow with the edit-distance
// → row (initialised to
0,1,2,... instead of zeros) and take the MINIMUM of  $L[j] + R[m-j]$ 
// → instead of the maximum. The
split column is then a cell the optimal alignment passes through;
// → emit the operations.
AFFINE GAP COSTS need three rows (match / gap-in-a / gap-in-b) and the
// → split must record WHICH
of the three states the path is in at the middle row - otherwise the
// → two halves disagree.
MEMORY:  $O(m)$  for the rows plus  $O(m \log n)$  for the substrings taken at
// → each recursion level; pass
index ranges instead of substrings if that matters.
RECURSION DEPTH is  $O(\log n)$  - safe.
IF YOU ONLY NEED THE LENGTH, do not use this: one rolling row is
// → enough (and for LCS, the
bit-parallel version in 07 - Strings/07 - Sequences/03 is  $O(n \cdot m /$ 
// →  $64)$ ).
THE SAME TRICK reconstructs paths in any layered DAG DP where a middle
// → layer is cheap to
evaluate from both ends - it is the DP analogue of "meet in the
// → middle". */

```

8.4.7 MaximumSubarray

MAXIMUM SUBARRAY (Kadane) and its dimensional relatives. All exact, all one pass per dimension.

```
// KADANE: the best subarray ending at i either extends the best one
//   ending at i-1 or restarts.
// O(n). Returns {sum, l, r} with the segment NON-EMPTY (if every
//   element is negative the answer
// is the largest single element - the empty-segment convention gives 0
//   instead, so decide which
// one the problem wants BEFORE writing the loop; that is the only bug
//   this ever has).
array<ll, 3> maxSubarray(const vector<ll>& a) {
    int n = a.size();
    ll best = LLONG_MIN, cur = 0;
    int bl = 0, br = 0, start = 0;
    for (int i = 0; i < n; i++) {
        if (cur <= 0) cur = a[i], start = i; //
        // restart: the prefix only hurts
        else cur += a[i];
        if (cur > best) best = cur, bl = start, br = i;
    }
    return {best, bl, br};
}

// CIRCULAR (the array wraps): the answer is either an ordinary
//   subarray, or the complement of the
// MINIMUM subarray. Guard the all-negative case - "total - min" would
//   return the empty segment.
ll maxCircularSubarray(const vector<ll>& a) {
    ll total = 0, mx = LLONG_MIN, mn = LLONG_MAX, cx = 0, cn = 0;
    for (ll x : a) {
        total += x;
        cx = max(x, cx + x), mx = max(mx, cx);
        cn = min(x, cn + x), mn = min(mn, cn);
    }
    return mx < 0 ? mx : max(mx, total - mn);
}

// MAXIMUM SUM SUBMATRIX, O(rows^2 * cols): fix the top and bottom
//   rows, compress the columns
// between them into a 1D array by prefix sums, and run Kadane on it.
//   Make ROWS the smaller side.
ll maxSubmatrix(const vector<vector<ll>>& g) {
    int n = g.size(), m = n ? g[0].size() : 0;
    ll best = LLONG_MIN;
    for (int top = 0; top < n; top++) {
        vector<ll> col(m, 0);
        for (int bot = top; bot < n; bot++) {
            for (int j = 0; j < m; j++) col[j] += g[bot][j];
            best = max(best, maxSubarray(col)[0]);
        }
    }
    return best;
}

/* THE VARIANTS, and which structure each one needs
LENGTH AT LEAST k: prefix sums S, answer = max over i >= k of S[i] -
    min(S[0..i-k]) - one running
    minimum, O(n).
LENGTH AT MOST k: S[i] - min over a SLIDING WINDOW of S, so a monotone
    deque, O(n).
AT MOST k NEGATIVE ELEMENTS / at most k "skips": dp[i][j] = best
    ending at i having used j skips.
MAXIMUM AVERAGE subarray (with a length constraint): binary search the
    average and run the
    "length at least k" version on a[i] - mid, or the tangent trick in
    01 - DP Opt/09.
MAXIMUM PRODUCT subarray: carry BOTH the running max and the running
    min (a negative times a
    negative flips them), O(n).
K-TH LARGEST subarray sum: binary search the value plus a
    two-pointer/BIT count of pairs.
MAXIMUM SUBARRAY WITH UPDATES / on a range: segment tree storing
    (total, best prefix, best
    suffix, best inside) - see 02 - Data Structures; this file is the
    static case.
ALL-NEGATIVE ARRAYS are the standard trap in every one of these. */
```

8.4.8 ConnectedComponentDP

CONNECTED-COMPONENT DP (a.k.a. the "broken interval" or profile-free connectivity DP).

```
// SHAPE OF THE PROBLEM: you must build one object out of n items
//   processed in SORTED order, the
// cost depends on how far apart consecutive items end up, and the
//   final object must be CONNECTED
// (one block, one path, one permutation with a cyclic condition). The
//   state is NOT "which items
// are used" (2^n) but "how many disjoint fragments exist so far",
//   because the fragments are
// interchangeable - only their COUNT matters, and every fragment can
//   be glued later.
// STATE: dp[i][c][cost][ends] = first i items placed, c open
//   fragments, accumulated cost, and
// 'ends' = how many of the two outer boundaries have been fixed (0, 1
//   or 2) when the object is a
// PATH rather than a cycle. Drop 'ends' for objects with no
//   distinguished endpoints.
// THE COST TRICK, which is the whole point: when moving from item i to
//   item i+1, EVERY fragment
// boundary that is still open must eventually cross the gap (a[i+1] -
//   a[i]). There are
// (2c - ends) such open boundaries, so add (a[i+1] - a[i]) * (2c -
//   ends) to the cost NOW, once,
// instead of trying to know later which items are adjacent. That is
//   what removes the 2^n.
// COMPLEXITY: O(n^2 * K) states with O(1) transitions, K = cost
//   budget. The classic instance is
// JOI Skyscraper: count permutations of a[] with sum |a[i+1] - a[i]|
//   <= K.
// PRECONDITION: sort a[] first - the cost accounting above is only
//   valid for sorted items.
ll skyscraper(vector<ll> a, int K) {
    int n = a.size();
    sort(a.begin(), a.end());
    if (n == 1) return 1;
    // dp[c][cost][ends]
    vector<vector<array<ll, 3>>> dp(n + 2, vector<array<ll, 3>>>(K + 1,
    // {0, 0, 0}));
    dp[0][0][0] = 1;
    for (int i = 0; i < n; i++) {
        vector<vector<array<ll, 3>>> nd(n + 2, vector<array<ll, 3>>>(K +
    // 1, {0, 0, 0}));
        ll gap = i ? a[i] - a[i - 1] : 0;
        for (int c = 0; c <= i; c++)
            for (int cost = 0; cost <= K; cost++)
                for (int e = 0; e < 3; e++) {
                    ll v = dp[c][cost][e];
                    if (!v) continue;
                    ll nc = cost + gap * (2 * c - e); // pay
                    // for every open boundary
                    if (nc > K) continue;
                    auto add = [&](int c2, int e2, ll ways) {
                        // after placing an item at least one fragment
                        // MUST exist: without this
                        // guard the state (c = 0, ends = 1) survives
                        // and inflates the count
                        if (c2 < 1 || c2 > n || ways <= 0) return;
                        nd[c2][nc][e2] = (nd[c2][nc][e2] + v % MOD *
                        // (ways % MOD)) % MOD;
                    };
                    add(c + 1, e, c + 1 - e); // new
                    // fragment, interior
                    add(c + 1, e + 1, e < 2 ? 2 - e : 0); // new
                    // fragment, at an end
                    add(c, e, 2 * c - e); //
                    // append to one fragment's side
                    add(c, e + 1, e < 2 ? 2 - e : 0); //
                    // append and make it an end
                    if (c >= 2) add(c - 1, e, c - 1); //
                    // merge two fragments
                }
        dp = nd;
    }
    ll res = 0;
    for (int cost = 0; cost <= K; cost++) res = (res + dp[1][cost][2])
    // % MOD;
    return res; // one
    // fragment, both ends fixed
}

/* READING THE TRANSITIONS (they are always these five, in every
//   problem of this family)
NEW FRAGMENT: the item starts a block of its own. Interior blocks can
//   be created in
// (c + 1 - ends) ways (the gaps between existing blocks); if it is to
//   become an outer end of the
```

final object, that is one of the (2 - ends) remaining end slots.
 EXTEND: glue the item to one of the (2c - ends) open sides of an
 → existing block, or spend it as
 an outer end of that block (again 2 - ends ways).
 MERGE: join two adjacent blocks with this item, (c - 1) ordered
 → choices of which pair.
 ACCEPT at the end only c == 1 (connected!) and ends == 2 (both
 → boundaries closed).
 OTHER PROBLEMS IN THIS FAMILY: "arrange n numbers in a row minimising
 → the sum of adjacent
 differences with constraints", counting linear extensions with gap
 → costs, "cut a rod into
 pieces so that ..." and any permutation problem whose cost
 → telescopes over sorted values.
 IF THE OBJECT IS A CYCLE, delete the 'ends' dimension entirely and
 → accept c == 1.
 WATCH THE OVERCOUNTING: 'add(c, e + 1, 2 - e)' and 'add(c + 1, e + 1,
 → 2 - e)' both consume an end
 slot - forgetting one of them, or letting e exceed 2, silently
 → changes the answer. */

8.4.9 PlugDP

PLUG DP WITH CONNECTIVITY (the bracket / minimum-representation profile DP).
 This is the

```
// heavyweight cousin of 04 - BrokenProfileDP.cpp: broken profile only
→ remembers WHICH frontier
// cells are occupied, which is enough for tilings; here the frontier
→ must also remember WHICH
// PARTIAL PATHS ARE CONNECTED TO WHICH, so that closing a loop early
→ can be forbidden.
// ENCODING: the frontier has m + 1 "plugs". Each plug is 0 (nothing),
→ 1 (a LEFT bracket) or
→ 2 (a RIGHT bracket), packed two bits per plug. Because the paths
→ live in a planar grid they
// can never cross, so the connection pattern is a balanced bracket
→ sequence - matching brackets
// = "these two plug ends belong to the same partial path". THAT is why
→ 3 states per plug suffice
// instead of a general set partition of the frontier.
// COMPLEXITY: O(n * m * S) where S = number of reachable states,
→ empirically a few thousand for
// m <= 12 (the Motzkin-like count, far below 3^(m+1)); keep the states
→ in a hash map.
// INDEX CONVENTION (get this wrong and nothing works): cells are
→ 0-based and scanned row by row.
// BEFORE processing cell (i, j) the state holds, in plug index order
→ 0..m:
// indices < j : the BOTTOM edges of the cells (i, 0..j-1) already
→ placed in this row
// index j : the LEFT edge of (i, j) <- p below
// index j+1 : the TOP edge of (i, j) <- q below
// indices > j+1: the TOP edges of (i, j+1..m-1), inherited from the
→ previous row
// AFTER processing, plug j becomes the BOTTOM edge of (i, j) and plug
→ j+1 its RIGHT edge, which
// is exactly the invariant for cell (i, j+1). At the end of a row plug
→ m must be 0 (no path may
// leave the grid) and the state is SHIFTED LEFT by one plug. Plug k
→ uses bits 2k and 2k+1.
// THE PROBLEM SOLVED HERE: count HAMILTONIAN CYCLES in a grid with
→ blocked cells (URAL 1519
// "Formula 1"). Every other plug-DP problem is this loop with
→ different accept/reject rules.
```

```
struct PlugDP {
    int n, m;
    vector<string> g; // '.'
    → free, '*' blocked
    PlugDP(const vector<string>& grid) : g(grid) { n = g.size(), m = n
    → ? g[0].size() : 0; }
    static int get(ll s, int i) { return (s >> (2 * i)) & 3; }
    static ll set(ll s, int i, int v) { return (s & ~(3LL << (2 * i)))
    → | ((ll)v << (2 * i)); }
    int matchRight(ll s, int i) { // find
    → the ')' matching '(' at i
        int cnt = 0;
        for (int j = i; j <= m; j++) {
            int v = get(s, j);
            if (v == 1) cnt++;
            else if (v == 2 && --cnt == 0) return j;
        }
        return -1;
    }
    int matchLeft(ll s, int i) { // find
    → the '(' matching ')' at i
        int cnt = 0;
        for (int j = i; j >= 0; j--) {
            int v = get(s, j);
```

```
        if (v == 2) cnt++;
        else if (v == 1 && --cnt == 0) return j;
    }
    return -1;
}
ll solve() {
    int li = -1, lj = -1; // the
    → LAST free cell
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) if (g[i][j] == '.') li = i, lj
    → = j;
    if (li < 0) return 0;
    ll ans = 0;
    map<ll, ll> cur, nxt;
    cur[0] = 1;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            nxt.clear();
            for (auto& [s, w] : cur) {
                int p = get(s, j), q = get(s, j + 1);
                if (g[i][j] == '*' && g[i][j + 1] == '*') { //
                → blocked: no plug may touch it
                    if (!p && !q) nxt[s] += w;
                    continue;
                }
                if (!p && !q) { // open
                → a new path: down and right
                    if (i + 1 < n && j + 1 < m) nxt[set(set(s, j,
                → 1), j + 1, 2)] += w;
                } else if (!p || !q) { // one
                → plug: continue down or right
                    int v = p ? p : q;
                    if (i + 1 < n) nxt[set(set(s, j, v), j + 1, 0)]
                → += w; // go down
                    if (j + 1 < m) nxt[set(set(s, j, 0), j + 1, v)]
                → += w; // go right
                } else if (p == 1 && q == 1) { // two
                → '(' : merge, retag the match
                    int k = matchRight(s, j + 1);
                    nxt[set(set(set(s, k, 1), j, 0), j + 1, 0)] +=
                → w;
                } else if (p == 2 && q == 2) { // two
                → ')' : merge, retag the match
                    int k = matchLeft(s, j);
                    nxt[set(set(set(s, k, 2), j, 0), j + 1, 0)] +=
                → w;
                } else if (p == 2 && q == 1) { //
                → different paths: just join them
                    nxt[set(set(s, j, 0), j + 1, 0)] += w;
                } else if (i == li && j == lj) { // p ==
                → 1 && q == 2: closes a loop
                    ll t = set(set(s, j, 0), j + 1, 0); // legal
                → ONLY at the last free cell
                    if (t == 0) ans += w;
                }
            }
            cur.swap(nxt);
        }
        nxt.clear(); // end
    }
    → of row: shift the profile
    for (auto& [s, w] : cur) if (!get(s, m)) nxt[s << 2] += w;
    cur.swap(nxt);
    return ans;
}
```

```
};
/* ADAPTING IT - the five cases above never change, only what you do
→ with them
COUNT HAMILTONIAN PATHS (not cycles): allow the "close" case at any
→ cell but require exactly one
open path at the end, or add a virtual start/end cell.
SIMPLE CYCLE THAT NEED NOT COVER EVERY CELL (max weight cycle): let a
→ free cell be SKIPPED (the
'!p && !q' case also allows keeping the state unchanged), and accept
→ the closure at any cell,
taking the max instead of the sum.
PARTITION INTO CLOSED LOOPS: accept every closure and multiply, no
→ "last cell" restriction.
THE THREE-STATE ENCODING ONLY WORKS FOR NON-CROSSING (planar)
→ connections. For general
connectivity (Steiner-tree profile DPs, spanning forests) you need
→ MINIMUM REPRESENTATION:
store, per frontier cell, the id of its component normalised to
→ 0,1,2,... left to right - same
loop, bigger state, and a canonicalisation step after every move.
ALWAYS make m the SMALLER dimension (transpose if needed) - the state
→ count is exponential in m.
```

DEBUGGING: a 4x4 empty grid has 6 Hamiltonian cycles, 6x6 has 1072,
 ↳ 8x8 has 425; a grid with an
 odd number of free cells has none. Check those before running
 ↳ anything bigger. */

8.4.10 DPOverDivisors

DP OVER THE DIVISOR LATTICE OF ONE NUMBER - the exact analogue of SOS DP, with "submask"

```
// replaced by "divisor". The divisors of n form a lattice isomorphic
↳ to a product of chains (one
// per prime, of length e_p), so the same "one dimension at a time"
↳ trick works:
// for each prime p: for divisors in increasing order: g[d] +=
↳ g[d / p]
// gives g[d] = sum over e | d of f[e] in O(D * omega(n)) instead of
↳ O(D^2) pair testing, where
// D = number of divisors (<= 1344 below 1e9) and omega = number of
↳ distinct primes (<= 9).
// WHEN: "for every divisor d of n, count the array elements whose gcd
↳ with n is exactly d" (the
// mobius direction), "sum over all divisors of a divisor",
↳ multiplicative-function DPs restricted
// to one n, and the CF-style "count subsets whose gcd is exactly d".
// INDEX CONVENTION: divs is SORTED ASCENDING and f is indexed by
↳ POSITION in that array; pos maps
// a divisor value to its index. Sorted order guarantees d / p comes
↳ before d, which is what makes
// the in-place update legal - do not shuffle divs.
```

```
struct DivisorLattice {
    vector<ll> divs;
    vector<ll> primes;
    map<ll, int> pos;
    DivisorLattice(ll n, const vector<ll>& primeFactors) :
    ↳ primes(primeFactors) {
        for (ll i = 1; i * i <= n; i++)
            if (n % i == 0) { divs.push_back(i); if (i != n / i)
    ↳ divs.push_back(n / i); }
        sort(divs.begin(), divs.end());
        for (int i = 0; i < (int)divs.size(); i++) pos[divs[i]] = i;
    }
    void zeta(vector<ll>& f) { // g[d]
    ↳ = sum over e | d of f[e]
        for (ll p : primes)
            for (int i = 0; i < (int)divs.size(); i++)
                if (divs[i] % p == 0) f[i] += f[pos[divs[i] / p]];
    }
    void mobius(vector<ll>& f) { // exact
    ↳ inverse of zeta
        for (ll p : primes)
            for (int i = (int)divs.size() - 1; i >= 0; i--)
                if (divs[i] % p == 0) f[i] -= f[pos[divs[i] / p]];
    }
    void superZeta(vector<ll>& f) { // g[d]
    ↳ = sum over d | e of f[e]
        for (ll p : primes)
            for (int i = (int)divs.size() - 1; i >= 0; i--)
                if (divs[i] % p == 0) f[pos[divs[i] / p]] += f[i];
    }
    void superMobius(vector<ll>& f) { // exact
    ↳ inverse of superZeta
        for (ll p : primes)
            for (int i = 0; i < (int)divs.size(); i++)
                if (divs[i] % p == 0) f[pos[divs[i] / p]] -= f[i];
    }
};
```

/* THE CANONICAL USE

cnt[d] = number of array elements divisible by d (easy: for each
 ↳ element, add 1 at gcd(a, n));
 then superZeta turns "exactly d" counts into "divisible by d" counts
 ↳ and superMobius turns them
 back. Counting subsets with gcd exactly d is then: ways[d] =
 ↳ 2^(divisible by d) - 1, followed by
 superMobius over the divisor lattice.
 OVER 1..N INSTEAD OF THE DIVISORS OF ONE n: the same transforms are
 ↳ the harmonic loops
 for (d = 1..N) for (m = 2d, 3d, ...) f[m] += f[d] (divisor zeta,
 ↳ O(N log N))
 for (d = N..1) for (m = 2d, 3d, ...) f[d] += f[m] (multiple zeta,
 ↳ O(N log N))

and their mobius inverses reverse the inner direction. See 04 - Number
 ↳ Theory/04/02.

GCD CONVOLUTION c[k] = sum over gcd(i,j) = k of a[i] b[j]:

↳ multipleZeta both, multiply pointwise,
 multipleMobius the result. LCM convolution is the same with the
 ↳ divisor transforms.

WHY NOT JUST TEST ALL PAIRS: D^2 is 1.8e6 for D = 1344, fine once -
 ↳ but inside a loop over

queries or over n it is not, and the lattice version is 12000

↳ operations.

THE PRIME LIST must be the DISTINCT primes of n (factorise with 04 -
 ↳ NT/02/06 Pollard). Passing a
 prime twice double-counts and silently corrupts every value. */

8.4.11 XorEquationCount

COUNTING SOLUTIONS OF $x_1 \text{ xor } x_2 \text{ xor } \dots \text{ xor } x_n = X$ with $0 \leq x_i \leq a_i$, in $O(n * B)$.

```
// The bounds are DIFFERENT per variable, which is what makes this
↳ hard: with a common bound
// 2^B - 1 the answer is just 2^(B(n-1)) [X < 2^B]. Digit DP over the
↳ bits with an "is this
// variable still tight" flag per variable would need 2^n states - the
↳ trick below avoids that.
// THE TRICK: classify a solution by the HIGHEST bit k at which some
↳ x_i first drops below a_i.
// Above bit k every variable equals a_i exactly, so that prefix is
↳ forced. At bit k, at least one
// variable with a_i's bit set chooses 0 and becomes FREE below k;
↳ every other variable is either
// still tight (its low bits are then bounded by a_i's low bits) or
↳ already free (2^k choices).
// One designated free variable absorbs the xor constraint on the low
↳ bits - it has 2^k choices
// and exactly ONE of them fixes the remaining xor - which is why the
↳ first freed variable
// contributes a factor of 1 and every later one a factor of 2^k.
// If NOBODY drops at bit k, the parity of the k-th bits is forced to
↳ popcount-parity, so the
// recursion continues to bit k-1 only when that parity matches X's
↳ k-th bit.
// PRECONDITION: a_i >= 0 and X >= 0; B must cover every a_i and X (bit
↳ B-1 is the top).
ll countXorSolutions(const vector<ll>& a, ll X, int B = 60) {
    int n = a.size();
    ll res = 0;
    for (int k = B - 1; k >= 0; k--) {
        ll low = (1LL << k) % MOD;
        // dp[freed][parity of the k-th bits chosen so far]
        ll dp[2][2] = {{1, 0}, {0, 0}};
        for (int i = 0; i < n; i++) {
            ll lo = a[i] & ((1LL << (k + 1)) - 1); // bits
    ↳ k..0 of a_i
            ll nd[2][2] = {{0, 0}, {0, 0}};
            if (a[i] >> k & 1) {
                ll keep = (lo - (1LL << k) + 1) % MOD; // x_i
    ↳ bit k = 1, low bits <= a_i
                for (int f = 0; f < 2; f++)
                    for (int p = 0; p < 2; p++) {
                        if (!dp[f][p]) continue;
                        nd[f][p ^ 1] = (nd[f][p ^ 1] + dp[f][p] * keep)
    ↳ % MOD; // stay tight
                        ll fac = f ? low : 1; // the
    ↳ FIRST freed one absorbs xor
                        nd[1][p] = (nd[1][p] + dp[f][p] * fac) % MOD;
    ↳ // drop to 0 here
                    }
            } else {
                ll ch = (lo + 1) % MOD; // x_i
    ↳ bit k = 0, low bits <= a_i
                for (int f = 0; f < 2; f++)
                    for (int p = 0; p < 2; p++) nd[f][p] = dp[f][p] *
    ↳ ch % MOD;
            }
            memcpy(dp, nd, sizeof dp);
        }
        res = (res + dp[1][X >> k & 1]) % MOD;
        int par = 0;
    ↳ nobody dropped at bit k
        for (int i = 0; i < n; i++) par ^= (a[i] >> k) & 1;
        if (par != ((X >> k) & 1)) return res; // the
    ↳ forced prefix already fails
    }
    return (res + 1) % MOD; // the
    ↳ all-tight tuple x_i = a_i
}

/* VARIANTS
SUM OF THE SOLUTIONS, or counting with x_i in [lo_i, hi_i]:
↳ inclusion-exclusion over the lower
  bounds (subtract the count with x_i <= lo_i - 1) - 2^n terms, so
  ↳ only for tiny n; otherwise
  carry a second "already below lo_i" flag, which doubles the state,
  ↳ not the exponent.
AND / OR CONSTRAINTS instead of xor: the same "first variable that
  ↳ drops" decomposition works,
  with the parity condition replaced by the AND/OR of the k-th bits.
```

XOR EQUAL TO AT MOST X: sum the answers over the prefixes of X, i.e.
 ↳ for each bit where X has a
 1, count solutions matching X above that bit and 0 there - one extra
 ↳ outer loop.

LINEAR ALGEBRA ALTERNATIVE: if the bounds are all $2^B - 1$ (or the
 ↳ variables range over a
 subspace), this is a rank computation over $\text{GF}(2)$, not a DP - see 06
 ↳ - Math/07 XOR basis.

SANITY: $n = 1$ gives $[X \leq a_1]$; all $a_i = 2^B - 1$ gives $2^{(B(n-1))}$. */

8.4.12 TransferMatrixDP

TRANSFER-MATRIX DP - when the DP has a HUGE time dimension (n up to $1e18$) but few states.

```
// Needs: 06 - Math/03 - Matrices/01 - MatrixExponentiation.cpp (Mat,
↳ operator*, mpow)
// RECOGNITION TEST: dp[t] is a vector over S states, and dp[t+1] = M *
↳ dp[t] with M INDEPENDENT
// of t. Then dp[n] = M^n * dp[0], computed in  $O(S^3 \log n)$ . If M
↳ depends on t only through a
// small period p, multiply the p matrices once and exponentiate the
↳ product.
// WHEN: count strings of length  $1e18$  avoiding a pattern, count walks
↳ of a given length in a
// graph, count tilings of a  $4 \times 1e18$  strip, any linear recurrence with
↳ n astronomically large.
// PRECONDITIONS, all checkable: (1) the transition is LINEAR in the dp
↳ values (sums of products
// with CONSTANTS - "max" is not linear, see min-plus below); (2) the
↳ state set does not grow with
// t; (3) the same matrix applies at every step (or with a fixed
↳ period).
// Build the matrix by RUNNING ONE STEP of the DP from each unit vector
↳ - that is mechanical and
// far less error-prone than deriving the entries by hand.
template <class Step>
Mat transferMatrix(int S, Step step) { //
↳ step(from, add) enumerates edges
  Mat M(S, vector<Mint>(S, 0));
  for (int i = 0; i < S; i++) step(i, [&](int j, Mint w) { M[j][i] +=
↳ w; });
  return M; //
↳ COLUMN convention: dp' = M * dp
}
// EXAMPLE - count strings of length n over an alphabet of size A that
↳ avoid a set of patterns:
// states = the nodes of the Aho-Corasick automaton with terminal nodes
↳ deleted, and the step is
// for each node u, for each character c: v = advance(u, c); if v is
↳ not terminal, add 1 edge.
// dp[0] = e_root, and the answer is the sum of the entries of M^n *
↳ dp[0].  $O(S^3 \log n)$ 
/* THE VARIANTS
MIN-PLUS ("shortest walk with exactly n edges", "minimum cost after n
↳ steps"): replace (+, *)
with (min, +) and exponentiate the same way - see 06 - Math/03/07 -
↳ MinPlusMatrixPower.cpp.
The matrix product is still associative, which is ALL that
↳ exponentiation needs; only the
 $O(S^3)$  per multiply survives, there is no Strassen and no inverse.
KITAMASA / BOSTAN-MORI (06 - Math/06/03) computes the n-th term of a
↳ linear recurrence of order
k in  $O(k \log k \log n)$  instead of  $O(k^3 \log n)$ . If your matrix is a
↳ companion matrix - i.e. the
DP is a plain linear recurrence - use that instead; k = 1000 is
↳ hopeless for matrices and easy
for Bostan-Mori.
BERLEKAMP-MASSEY finds the recurrence for you: generate 2k terms with
↳ the slow DP, recover the
order-k recurrence, then jump to n. This is the practical route when
↳ the state space is
structured but you cannot describe the matrix cleanly.
PERIODIC / TIME-DEPENDENT transitions: if M_t has period p,
↳ exponentiate  $(M_{\{p-1\}} \dots M_0)$ .
PROFILE DPs on a  $k \times n$  grid become transfer matrices over the  $2^k$ 
↳ profiles - that is how "count
tilings of a  $5 \times 1e18$  board" is solved (and  $2^k$  must stay under  $\sim 200$ 
↳ for the cube to fit).
EIGENVALUES / CLOSED FORMS are almost never worth it in a contest: the
↳ matrix power is  $O(S^3 \log n)$  with no numerical risk, and a closed form needs distinct
↳ eigenvalues you cannot verify.
SANITY: check  $M^1$  and  $M^2$  against the slow DP for small n before
↳ trusting  $M^{1e18}$ . */
```

8.5 Probability

8.5.1 ExpectedValueDP

EXPECTED VALUE / PROBABILITY DP.

```
// Needs: Mint (06 - Math/01) and solveLinear (06 - Math/04 - Linear
↳ Algebra/02 - GaussianElimination.cpp).
// LINEARITY OF EXPECTATION FIRST: E[sum of indicators] = sum of
↳ P(indicator). Most "expected
// number of X" problems collapse to summing per-element probabilities
↳ with no DP at all.
// Only when states depend on each other CYCLICALLY do you need a
↳ linear system.

// E[i] = 1 + sum_j P[i][j] * E[j], with E[absorbing] = 0.
// Move the unknowns left: E[i] - sum_j P[i][j] E[j] = 1 -> (I - P)
↳ E = 1.  $O(n^3)$ .
// Needs solveLinear from GaussianElimination.cpp (values are Mint, so
↳ probabilities must be
// given as modular fractions p * inv(q)).
vector<Mint> hittingTimes(const vector<vector<Mint>>& P, const
↳ vector<char>& absorbing) {
  int n = P.size();
  vector<vector<Mint>> A(n, vector<Mint>(n, 0));
  vector<Mint> b(n, 0);
  for (int i = 0; i < n; i++) {
    if (absorbing[i]) { A[i][i] = 1, b[i] = 0; continue; }
    A[i][i] = 1;
    for (int j = 0; j < n; j++) A[i][j] -= P[i][j];
    b[i] = 1;
  }
  return solveLinear(A, b);
}
/* ABSORBING MARKOV CHAIN, canonical form P = [Q R ; 0 I]:
fundamental matrix N = (I - Q)^{-1}; expected steps before
↳ absorption = N * 1
absorption probabilities B = N * R (B[i][j] = P(absorbed in j |
↳ started in i))

TRIDIAGONAL SYSTEMS (grid random walks that only move left/right
↳ within a row) -  $O(n)$  with
the Thomas algorithm instead of  $O(n^3)$ :
a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i, a_1 = c_n = 0
forward: c'_i = c_i / (b_i - a_i c'_{i-1}), d'_i = (d_i - a_i
↳ d'_{i-1}) / (b_i - a_i c'_{i-1})
back: x_n = d'_n, x_i = d'_i - c'_i x_{i+1}
Diagonally dominant (always true for hitting times), so it is stable.

TREES: substitute E_v = A_v * E_parent + B_v, push A and B up in one
↳ DFS, then resolve at the
root and push back down.  $O(n)$ . (Folklore - no canonical reference;
↳ derive it, do not trust it.)

COMMON TRAPS
- "expected value of a maximum" is NOT the maximum of expected
↳ values; sum over thresholds:
E[max] = sum_{t=1} P(max >= t).
- Probabilities mod p: store p*inv(q); never mix doubles and Mint.
- Infinite processes: E exists only if absorption has probability
↳ 1; check reachability first.
- Conditional expectation: E[X] = sum_y P(Y=y) * E[X | Y=y] -
↳ condition on the FIRST step. */
```

9 Trees

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9.1 Lowest Common Ancestor (LCA)

9.1.1 LCA

LCA by Euler tour + sparse table: $O(n \log n)$ build, $O(1)$ query. 0-based, rooted at 'root'.

// For k-th ancestor / "jump while predicate" use BinaryLifting instead
 ↳ (02 - BinaryLifting.cpp).

```
struct LCA {
  static const int B = 21;
  vector<array<int, 2>> tour;
  vector<array<array<int, 2>, B>> T;
  vector<int> d, in, lg;

  template <class G>
  LCA(int n, const G& adj, int root = 0) : d(n), in(n), lg(2 * n + 1)
  ↳ {
    tour.reserve(2 * n);
    dfs(root, -1, adj);
    int m = tour.size(); // == 2n - 1
```

```

T.resize(m);
for (int i = 2; i <= 2 * n; i++) lg[i] = lg[i / 2] + 1;
for (int i = 0; i < m; i++) T[i][0] = tour[i];
for (int j = 1; j < B; j++)
    for (int i = 0; i + (1 << j) - 1 < m; i++)
        T[i][j] = min(T[i][j - 1], T[i + (1 << (j - 1))][j - 1]);
}
template <class G>
void dfs(int u, int p, const G& adj) {
    in[u] = tour.size();
    tour.push_back({d[u], u});
    for (int v : adj[u]) if (v != p) {
        d[v] = d[u] + 1;
        dfs(v, u, adj);
        tour.push_back({d[u], u});
    }
}
int get(int u, int v) const {
    int l = min(in[u], in[v]), r = max(in[u], in[v]), j = lg[r - l] + 1;
    return min(T[l][j], T[r - (1 << j) + 1][j])[1];
}
int dist(int u, int v) const { return d[u] + d[v] - 2 * d[get(u, v)]; }
};

```

9.1.2 BinaryLifting

BINARY LIFTING - k-th ancestor, LCA, and "jump while a predicate holds", all $O(\log n)$.

```

// This is NOT only a tree tool: the same table jumps  $2^k$  steps in ANY
// functional graph
// (i -> f(i)), which is how you do "apply the operation 1e18 times",
// permutation powers,
// and "min lamps to cover [L,R]" (jump by farthest-reach).
struct Lift {
    int n, B;
    vector<vector<int>> up; // up[k][v] =  $2^k$ -th
    // ancestor (-1 above the root)
    vector<int> d;

    template <class G>
    Lift(int n, const G& adj, int root = 0) : n(n), B(1), d(n, 0) {
        while ((1 << B) < n) B++;
        B++;
        up.assign(B, vector<int>(n, -1));
        dfs(root, -1, adj);
        for (int k = 1; k < B; k++)
            for (int v = 0; v < n; v++)
                up[k][v] = up[k - 1][v] < 0 ? -1 : up[k - 1][up[k - 1][v]];
    }
    template <class G>
    void dfs(int u, int p, const G& adj) {
        up[0][u] = p;
        for (int v : adj[u]) if (v != p) d[v] = d[u] + 1, dfs(v, u, adj);
    }
    int kth(int v, ll k) const { // k-th ancestor, -1 if
    // it does not exist
        for (int i = 0; i < B && v >= 0; i++) if (k >> i & 1) v =
            up[i][v];
        return k >> B ? -1 : v;
    }
    int lca(int u, int v) const {
        if (d[u] < d[v]) swap(u, v);
        u = kth(u, d[u] - d[v]);
        if (u == v) return u;
        for (int i = B - 1; i >= 0; i--) if (up[i][u] != up[i][v]) u =
            up[i][u], v = up[i][v];
        return up[0][u];
    }
    int dist(int u, int v) const { return d[u] + d[v] - 2 * d[lca(u, v)]; }
    int onPath(int u, int v, int k) const { // k-th vertex on the
    // path u -> v (k = 0 gives u)
        int a = lca(u, v), du = d[u] - d[a];
        return k <= du ? kth(u, k) : kth(v, d[u] + d[v] - 2 * d[a] - k);
    }
    // highest ancestor of v still satisfying pred (monotone: true near
    // v, false near the root)
    template <class F>
    int jumpWhile(int v, F pred) const {
        for (int i = B - 1; i >= 0; i--) if (up[i][v] >= 0 &&
            pred(up[i][v])) v = up[i][v];
        return v;
    }
};

```

```

// STANDALONE version for a functional graph f (no tree needed):
// up[0][v] = f(v); up[k][v] = up[k-1][up[k-1][v]]; then apply k
// steps by the bits of k.

```

9.1.3 LCAConstantTime

LCA IN $O(1)$ after $O(n \log n)$ preprocessing - Euler tour + sparse table on DEPTH.

```

// Binary lifting (02) is  $O(\log n)$  per query and needs  $O(n \log n)$ 
// memory too, so reach for this
// when the query count dominates:  $O(1)$  LCA turns "distance between
// every pair in a list" and
// "is u an ancestor of v" from  $n \log n$  into  $n$ . It is also the
// primitive behind  $O(1)$  LCE on a
// suffix tree and behind virtual trees built in bulk.
// The tour has  $2n-1$  entries; the LCA of u and v is the SHALLOWEST
// vertex between their first
// occurrences, and since consecutive tour depths differ by exactly +-1
// a plain sparse table over
// (depth, vertex) pairs is enough - min over an overlapping range is
// idempotent.
struct LCA {
    int n, LOG;
    vector<int> first, depth, tour;
    vector<vector<pair<int, int>>> sp; //
    // (depth, vertex)
    LCA(int n, const vector<vector<int>>& adj, int root = 0)
        : n(n), first(n, -1), depth(n, 0) {
        tour.reserve(2 * n);
        vector<pair<int, int>> st({root, -1}); //
        // iterative: no stack overflow
        vector<int> it(n, 0);
        depth[root] = 0;
        while (!st.empty()) {
            auto& [u, p] = st.back();
            if (first[u] < 0) first[u] = tour.size();
            if (it[u] < (int)adj[u].size()) {
                int v = adj[u][it[u]++];
                if (v == p) continue;
                tour.push_back(u), depth[v] = depth[u] + 1;
                st.push_back({v, u});
            } else {
                tour.push_back(u);
                st.pop_back();
            }
        }
        int m = tour.size();
        LOG = 1;
        while ((1 << LOG) <= m) LOG++;
        sp.assign(LOG, vector<pair<int, int>>(m));
        for (int i = 0; i < m; i++) sp[0][i] = {depth[tour[i]],
            // tour[i]};
            for (int k = 1; k < LOG; k++)
                for (int i = 0; i + (1 << k) <= m; i++)
                    sp[k][i] = min(sp[k - 1][i], sp[k - 1][i + (1 << (k - 1))]);
        }
        int lca(int u, int v) const {
            int a = first[u], b = first[v];
            if (a > b) swap(a, b);
            int k = 31 - __builtin_clz(b - a + 1);
            return min(sp[k][a], sp[k][b - (1 << k) + 1]).second;
        }
        int dist(int u, int v) const { return depth[u] + depth[v] - 2 *
            // depth[lca(u, v)]; }
        bool isAncestor(int u, int v) const { return lca(u, v) == u; }
    };
    // TRUE  $O(n)$  PREPROCESSING is possible (Farach-Colton-Bender): split
    // the +-1 depth array into
    // blocks of size  $\log(n)/2$ , precompute every possible block pattern
    // (there are only  $\sqrt{n}$  of
    // them), and put a sparse table over the block minima. It is ~60 lines
    // and only worth it at
    //  $n \geq 5e5$ , where the  $O(n \log n)$  table's memory is the real problem.
    // --- SUBTREE HASHING: are these two subtrees the same shape?  $O(n \log
    // n)$ . ---
    // Canonical form: sort the children's ids and look the sorted vector
    // up in a map. Two subtrees
    // get the same id iff they are isomorphic as ROOTED trees.
    // Shorter than AHU (09 - Isomorphism) and merges naturally into other
    // tree DPs; AHU is still the
    // one to use when you need the UNROOTED comparison (root at the
    // centres and take the minimum).
    map<vector<int>, int> subtreeId;
    vector<int> subHash;
    int hashSubtree(int u, int p, const vector<vector<int>>& adj) {
        vector<int> ch;
        for (int v : adj[u]) if (v != p) ch.push_back(hashSubtree(v, u,

```

```

    ↪ adj));
    sort(all(ch));
    auto it = subtreeId.find(ch);
    if (it == subtreeId.end()) it = subtreeId.emplace(ch,
    ↪ subtreeId.size()).first;
    return subHash[u] = it->second;
}

// RANDOMISED VARIANT, O(n) and mergeable: h(u) = XOR over children of
    ↪ f(h(child)) with f a fixed
// random 64-bit mixing function (splitmix64). No map, no sorting, and
    ↪ it composes under rerooting
// - which is what you want when you need the hash of every subtree AND
    ↪ of every "rest of tree".

// --- MERGING TWO TREE PATHS. A segment tree node stores the MINIMAL
    ↪ PATH CONTAINING all the
// vertices in its range, as that path's two endpoints ({-1,-1} =
    ↪ empty, {-2,-2} = they do not fit
// on any path). Merging is O(1) given an O(1) LCA: the covering path
    ↪ of two paths, if it exists,
// has its endpoints among the four given ones, so test each of the
    ↪ four candidate pairs (u, v)
// with dist(u,v) == dist(u,w) + dist(w,v) for every endpoint w.
// NOTE THE SEMANTIC: this is the smallest path CONTAINING both, not
    ↪ their union - two disjoint
// pieces of one long path merge into that long path, which is what a
    ↪ covering query wants.
// USES: "is this set of vertices contained in one path", "the smallest
    ↪ path covering a range of
// queries", and segment-tree-over-queries problems on trees.
struct PathNode { int x = -1, y = -1; };
PathNode mergePaths(PathNode a, PathNode b, const LCA& L) {
    if (a.x < 0) return b;
    if (b.x < 0) return a;
    int e[4] = {a.x, a.y, b.x, b.y}; // ALL
    ↪ SIX pairs, not just four:
    for (int i = 0; i < 4; i++) for (int j = i; j < 4; j++) { //
    ↪ (a.x,b.x) and (a.y,b.y) are
        int u = e[i], v = e[j]; //
    ↪ spanning pairs too
        bool ok = true;
        for (int w : e) if (L.dist(u, v) != L.dist(u, w) + L.dist(w,
    ↪ v)) ok = false;
        if (ok) return {u, v};
    }
    return {-2, -2}; // not a
    ↪ path: mark it invalid
}

```

9.2 DSU on Tree

9.2.1 DSUOnTree

DSU ON TREE (small to large) - answers offline subtree queries in $O(n \log n)$ by keeping the heavy

```

// child's data and re-adding only the light subtrees. Each vertex is
    ↪ added  $O(\log n)$  times.
// Use it for "most frequent colour in each subtree" style problems
    ↪ where merging maps would be  $O(n^2)$ .

```

```

class DSUOnTree {
private:
    struct Query {
        int idx, x;
    };

    int n;
    vector<vector<int>> adj;
    vector<int> sz, depth;
    vector<bool> big;
    vector<vector<Query>> queries;
    vector<int> ans;

    // =====
    // PROBLEM SPECIFIC STATE VARIABLES
    // =====
    vector<char> c; // Example: Array of characters per node

    // Example state variables
    // int distinct_count = 0;
    // vector<int> freq;

    void add(int u, int p) {
        // --- ADD LOGIC HERE ---
        // Example: freq[c[u] - 'a']++;

        for (int v : adj[u]) {
            if (v == p || big[v]) continue; // Skip parent and heavy
            ↪ child

```

```

        add(v, u);
    }
}

void remove(int u, int p) {
    // --- REMOVE LOGIC HERE ---
    // Example: freq[c[u] - 'a']--;

    for (int v : adj[u]) {
        if (v == p || big[v]) continue; // Skip parent and heavy
    ↪ child
        remove(v, u);
    }
}

int get_answer(int u, int x = 0) {
    // --- QUERY ANSWER LOGIC HERE ---
    return 0;
}
// =====

void pre(int u, int p = 0) {
    sz[u] = 1;
    depth[u] = depth[p] + 1;
    for (int v : adj[u]) {
        if (v == p) continue;
        pre(v, u);
        sz[u] += sz[v]; // Compute subtree size
    }
}

void dfs(int u, int p = 0, bool keep = false) {
    int mx = -1, bigChild = -1;

    // Find the heavy child
    for (int v : adj[u]) {
        if (v == p) continue;
        if (sz[v] > mx) {
            mx = sz[v];
            bigChild = v;
        }
    }

    // Process light children and clear them
    for (int v : adj[u]) {
        if (v == p || v == bigChild) continue;
        dfs(v, u, false);
    }

    // Process heavy child and keep it
    if (bigChild != -1) {
        dfs(bigChild, u, true);
        big[bigChild] = true;
    }

    // Add current node and its light children
    add(u, p);

    // Answer all queries for the current node
    for (const auto& q : queries[u]) {
        ans[q.idx] = get_answer(u, q.x);
    }

    // Reset the big child flag
    if (bigChild != -1) {
        big[bigChild] = false;
    }

    // Clear state if this node is not meant to be kept
    if (!keep) {
        remove(u, p);
    }
}

public:
    // Initialize with 1-based indexing in mind
    DSUOnTree(int nodes, const vector<char>& chars) : n(nodes),
    ↪ c(chars) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        depth.assign(n + 1, 0);
        big.assign(n + 1, false);
        queries.assign(n + 1, vector<Query>());
        // Initialize problem-specific arrays here (e.g.,
    ↪ freq.assign(26, 0);
    }
}

```

```

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

void add_query(int node, int query_idx, int x = 0) {
    queries[node].push_back({query_idx, x});
}

vector<int> solve(int root, int num_queries) {
    ans.assign(num_queries, 0);
    pre(root, 0); // Precalculate sizes and depths
    dfs(root, 0, false); // Run the sack algorithm
    return ans;
}
};

```

9.3 Heavy Light Decomposition (HLD)

9.3.1 HLD

HEAVY-LIGHT DECOMPOSITION - splits the tree into chains so any root-to-node path crosses $O(\log n)$

// of them; combined with a segment tree it gives path update/query in $\hookrightarrow O(\log^2 n)$.
 // Subtrees are contiguous in tin[], so subtree queries are a single $\hookrightarrow$ range on the same tree.

```

class HLD {
public:
    vector<int> par, sz, head, tin, tout, who, depth;

    int dfs1(int u, vector<vector<int>> &adj) {
        for (int &v: adj[u]) {
            if (v == par[u]) continue;
            depth[v] = depth[u] + 1;
            par[v] = u;
            sz[u] += dfs1(v, adj);
            if (sz[v] > sz[adj[u][0]] || adj[u][0] == par[u]) swap(v,
             $\hookrightarrow$  adj[u][0]);
        }
        return sz[u];
    }

    void dfs2(int u, int &timer, const vector<vector<int>> &adj) {
        tin[u] = timer++;
        for (int v: adj[u]) {
            if (v == par[u]) continue;
            head[v] = (timer == tin[u] + 1 ? head[u] : v);
            dfs2(v, timer, adj);
        }
        tout[u] = timer - 1;
    }

    HLD(vector<vector<int>> adj, int r = 0)
        : par(adj.size(), -1), sz(adj.size(), 1), head(adj.size(), r),
         $\hookrightarrow$  tin(adj.size()), tout(adj.size()),
        who(adj.size()),
        depth(adj.size()) {
        dfs1(r, adj);
        int x = 0;
        dfs2(r, x, adj);
        for (int i = 0; i < adj.size(); ++i) who[tin[i]] = i;
    }

    vector<pair<int, int>> path(int u, int v) {
        vector<pair<int, int>> res;
        for (;;) v = par[head[v]] {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            if (head[u] != head[v]) {
                res.emplace_back(tin[head[v]], tin[v]);
            } else {
                if (depth[u] > depth[v]) swap(u, v);
                res.emplace_back(tin[u], tin[v]);
                return res;
            }
        }
    }

    pair<int, int> subtree(int u) {
        return {tin[u], tout[u]};
    }

    int dist(int u, int v) {
        return depth[u] + depth[v] - 2 * depth[lca(u, v)];
    }

    int lca(int u, int v) {
        for (;;) v = par[head[v]] {

```

```

            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            if (head[u] == head[v]) {
                if (depth[u] > depth[v]) swap(u, v);
                return u;
            }
        }
    }

    bool isAncestor(int u, int v) {
        return tin[u] <= tin[v] && tout[u] >= tout[v];
    }
};

```

9.4 Centroid Decomposition

9.4.1 Centroid

CENTROID DECOMPOSITION - recursively remove the centroid (the vertex whose largest remaining

// component is $\leq n/2$), giving a decomposition tree of depth $O(\log n)$.
 $\hookrightarrow$ Every path in the original
 // tree passes through the centroid of some level, which turns
 $\hookrightarrow$ path-counting problems into
 // "for each centroid, combine the distance lists of its components" in
 $\hookrightarrow O(n \log n)$.

```

class CentroidDecomposition {
private:
    int n;
    vector<vector<int>> adj;
    vector<int> sz, par, nodes;
    vector<bool> vis;

    int dfsSz(int u, int p) {
        sz[u] = 1;
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            sz[u] += dfsSz(v, u);
        }
        return sz[u];
    }

    int getCentroid(int u, int p, int total_nodes) {
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            if (sz[v] * 2 > total_nodes) {
                return getCentroid(v, u, total_nodes);
            }
        }
        return u;
    }

    void collect(int u, int p) {
        nodes.push_back(u);
        for (int v : adj[u]) {
            if (vis[v] || v == p) continue;
            collect(v, u);
        }
    }

    void build(int u, int p) {
        int total_nodes = dfsSz(u, p);
        int centroid = getCentroid(u, p, total_nodes);

        if (p != -1) {
            par[centroid] = p;
        } else {
            par[centroid] = centroid;
        }

        vis[centroid] = true;

        // =====
        // PROBLEM SPECIFIC PROCESSING
        // =====
        // Process paths going through the 'centroid' here
        for (int v : adj[centroid]) {
            if (vis[v]) continue;
            nodes.clear();
            collect(v, centroid);

            // e.g., Update answers using the nodes collected from this
             $\hookrightarrow$  subtree branch
        }
        // =====

        // Recursively decompose the remaining components
        for (int v : adj[centroid]) {
            if (vis[v]) continue;

```

```

        build(v, centroid);
    }
}

public:
    // Initialize with 1-based indexing
    CentroidDecomposition(int nodes_count) : n(nodes_count) {
        adj.assign(n + 1, vector<int>());
        sz.assign(n + 1, 0);
        par.assign(n + 1, 0);
        vis.assign(n + 1, false);
    }

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }

    // Start decomposition, usually from node 1
    void decompose(int root = 1) {
        build(root, -1);
    }

    // Returns the parent of a node in the centroid tree
    int get_centroid_parent(int u) const {
        return par[u];
    }
};

```

9.4.2 PathLengthCounts

NUMBER OF PATHS OF EVERY LENGTH IN A TREE - cnt[L] = how many unordered pairs {u, v} have

```

// Needs: conv (06 - Math/05 - Convolution/01 - FFT.cpp).
// dist(u, v) = L, for ALL L at once, in O(n log^2 n). 0-indexed,
//   ↳ unweighted tree.
// The naive "for each length, count" is O(n^2); a plain centroid
//   ↳ decomposition answers "paths of
// length <= K" in O(n log n) but not all lengths simultaneously. The
//   ↳ fix is to convolve the depth
// HISTOGRAMS instead of merging them pairwise: at each centroid,
//   ↳ square the histogram of all
// distances and subtract the square of each child component's
//   ↳ histogram (inclusion-exclusion for
// the pairs that never actually cross the centroid).
// Reach for it for "for every k, how many pairs are at distance k",
//   ↳ "sum over pairs of f(dist)"
// when f is arbitrary, and distance-distribution / diameter-profile
//   ↳ problems.
vector<ll> pathLengthCounts(int n, const vector<vector<int>>& adj) {
    vector<ll> cnt(n + 1, 0);
    vector<int> sub_(n, 0), dep;
    vector<char> done(n, 0);
    int tot = 0;
    function<void(int, int)> calcSz = [&](int u, int p) {
        tot++, sub_[u] = 1;
        for (int v : adj[u]) if (v != p && !done[v]) calcSz(v, u),
        ↳ sub_[u] += sub_[v];
    };
    function<int(int, int)> centroid = [&](int u, int p) {
        for (int v : adj[u]) if (v != p && !done[v] && sub_[v] * 2 >
        ↳ tot) return centroid(v, u);
        return u;
    };
    function<void(int, int, int)> gather = [&](int u, int p, int d) {
        dep.push_back(d);
        for (int v : adj[u]) if (v != p && !done[v]) gather(v, u, d +
        ↳ 1);
    };
    function<void(int, int, int, int)> add = [&](int u, int p, int d0,
    ↳ int sign) {
        dep.clear(), gather(u, p, d0);
        vector<ll> h(*max_element(all(dep)) + 1, 0);
        for (int d : dep) h[d]++;
        vector<ll> r = conv(h, h);
        ↳ ordered pairs, diagonal included
        for (int i = 0; i < (int)r.size() && i <= n; i++) cnt[i] +=
        ↳ sign * r[i];
    };
    function<void(int, int)> decomp = [&](int u, int p) {
        tot = 0, calcSz(u, p);
        int c = centroid(u, p);
        done[c] = 1;
        add(c, p, 0, 1);
        ↳ all
        ↳ pairs inside this component
        for (int v : adj[c]) if (!done[v]) add(v, c, 1, -1);
        ↳ minus the ones inside one child
        for (int v : adj[c]) if (!done[v]) decomp(v, c);
    };
}

```

```

};
if (n) decomp(0, -1);
cnt[0] = 0;
for (ll& x : cnt) x /= 2;
↳ ordered -> unordered
return cnt;
↳ cnt[L] for L = 1..n-1
}

// WEIGHTED trees: the same decomposition works, but the histogram is
↳ indexed by weight, so it is
// only practical when the total weight is small. Otherwise sort the
↳ distance lists and use two
// pointers - that answers "pairs with dist <= K" but not all lengths
↳ at once.
// SUM OVER PAIRS OF f(dist): convolve as above and dot with f. Any f
↳ works, which is the whole
// point of getting the full distribution.
// ONE LENGTH ONLY, or "count paths with dist <= K": plain centroid
↳ decomposition with sorting,
// O(n log^2 n) and no FFT at all - see 01 - Centroid.

```

9.5 Tree Diameter

9.5.1 TreeDiameter

Tree Diameter via 2-BFS - Time: O(V + E), Space: O(V + E)

```

// Computes diameter length, endpoints (u, v), and the full diameter
↳ path
struct TreeDiameter {
    int n;
    vector<vector<int>> adj;
    vector<int> dist, par;

    TreeDiameter(int n = 0) : n(n), adj(n + 1), dist(n + 1), par(n + 1)
    ↳ {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        adj[v].pb(u);
    }

    int bfs(int src) {
        fill(all(dist), -1);
        queue<int> q;
        q.push(src);
        dist[src] = 0;
        par[src] = -1;

        int farthest = src;
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            if (dist[u] > dist[farthest]) farthest = u;

            for (int v : adj[u]) {
                if (dist[v] == -1) {
                    dist[v] = dist[u] + 1;
                    par[v] = u;
                    q.push(v);
                }
            }
        }
        return farthest;
    }

    // Returns {diameter_length, {endpoint_u, endpoint_v}}
    pair<int, pair<int, int>> get_diameter() {
        int u = bfs(1);
        int v = bfs(u);
        return {dist[v], {u, v}};
    }

    // Returns the sequence of vertices on the diameter path from u to v
    vector<int> get_path() {
        auto [len, endpoints] = get_diameter();
        auto [u, v] = endpoints;
        vector<int> path;
        for (int curr = v; curr != -1; curr = par[curr]) {
            path.pb(curr);
        }
        reverse(all(path));
        return path;
    }
};

```

9.6 Euler Tour

9.6.1 EulerTour

Euler Tour Technique (Tree Flattening into 1D Array Intervals) - $O(V + E)$

```
// Subtree of node u corresponds to contiguous range [in_time[u],
// out_time[u]] in flat_tree
struct EulerTour {
    int n, timer = 0;
    vector<vector<int>> adj;
    vector<int> in_time, out_time, flat_tree;

    EulerTour(int n = 0) : n(n), adj(n + 1), in_time(n + 1), out_time(n
    + 1) {}

    void add_edge(int u, int v) {
        adj[u].pb(v);
        adj[v].pb(u);
    }

    void dfs(int u, int p = -1) {
        in_time[u] = ++timer;
        flat_tree.pb(u);
        for (int v : adj[u]) {
            if (v != p) dfs(v, u);
        }
        out_time[u] = timer;
    }

    void build(int root = 1) {
        timer = 0;
        flat_tree.clear();
        dfs(root);
    }

    // Returns [L, R] 1-based index range for subtree queries on node u
    pair<int, int> get_subtree_range(int u) const {
        return {in_time[u], out_time[u]};
    }
};
```

/* THE EULER-TOUR TOOLBOX - four different tours, each answering a
 ↪ different query family.
 CONVENTION HERE: in_time is 1-based (in_time[u] = ++timer) but
 ↪ flat_tree is a 0-based vector,
 so flat_tree[in_time[u] - 1] == u. Keep that straight when you index a
 ↪ BIT.

- 1) SUBTREE = a contiguous range [in[v], out[v]].
 point update, subtree query → BIT over in[]: add at in[u],
 ↪ query [in[v], out[v]]
 subtree update, point query → DIFFERENCE BIT: +x at in[v], -x
 ↪ at out[v]+1,
 then the value at u is the prefix
 ↪ sum up to in[u]
- 2) PATH TO THE ROOT, using the SAME difference trick in the other
 ↪ direction:
 subtree update, path-to-root query ↔ path update, point query
 ↪ (they are duals).
 PATH UPDATE u..v by +x, point query: +x at in[u], +x at in[v],
 ↪ -x at in[lca],
 -x at in[parent(lca)]
 PATH QUERY u..v, point update: query(in[u]) + query(in[v])
 ↪ - 2*query(in[lca])
 (+ the LCA's own value if
 ↪ vertices carry weights)
- 3) THE 2n TOUR (+1 on entry, -1 on exit) turns ancestor tests and LCA
 ↪ into RMQ:
 u is an ancestor of v iff in[u] ≤ in[v] && out[v] ≤ out[u]
 LCA(u,v) = the vertex of minimum DEPTH in the tour range [in[u],
 ↪ in[v]] - and consecutive
 depths differ by exactly 1, which is what the $O(n)/O(1)$ +-1 RMQ
 ↪ exploits.
- 4) THE EDGE TOUR (each edge appears twice) is what Mo's-on-trees uses:
 ↪ a vertex appearing TWICE
 inside the range cancels, so a path becomes one contiguous range
 ↪ plus the LCA by hand.
 See 02 - Data Structures/06 - Square Root Decomposition/03 -
 ↪ MoOnPaths.cpp.

EDGE WEIGHTS instead of vertex weights: push each edge's weight onto
 ↪ its DEEPER endpoint, then
 every path formula above drops the LCA term (the LCA's incoming edge
 ↪ is not on the path). */

9.7 Rerooting

9.7.1 Rerooting

REROOTING DP - compute $f(v)$ as root for EVERY v in $O(n)$.

```
// Four hooks: E() identity of merge * self(v) contribution of v
// ↪ itself *
// merge(a,b) associative+commutative * lift(a, child)
// ↪ push a child's value
// across one edge into the parent's frame.
// Shipped example: ans[v].first = sum of distances from v to every
// ↪ other node.
struct Reroot {
    typedef pair<ll, ll> T; // {sum of distances,
    // ↪ #nodes}
    static T E() { return {0, 0}; }
    static T self(int v) { return {0, 1}; }
    static T merge(T a, T b) { return {a.first + b.first, a.second +
    // ↪ b.second}; }
    static T lift(T a, int child) { return {a.first + a.second,
    // ↪ a.second}; }

    int n;
    vector<vector<int>> adj;
    vector<T> down, up, ans;
    vector<int> par, ord;

    Reroot(int n) : n(n), adj(n), down(n, E()), up(n, E()), ans(n,
    // ↪ E()), par(n, -1) {}
    void add_edge(int u, int v) { adj[u].push_back(v),
    // ↪ adj[v].push_back(u); }

    void build(int root = 0) {
        ord.clear(), ord.reserve(n);
        vector<int> st = {root};
        par[root] = -1;
        while (!st.empty()) { // iterative order: no
        // ↪ recursion depth limit
            int u = st.back(); st.pop_back(); ord.push_back(u);
            for (int v : adj[u]) if (v != par[u]) par[v] = u,
            // ↪ st.push_back(v);
            }
            for (int i = (int)ord.size() - 1; i >= 0; i--) { // pass 1:
            // ↪ leaves → root
                int u = ord[i];
                down[u] = self(u);
                for (int v : adj[u]) if (v != par[u]) down[v] =
                // ↪ merge(down[u], lift(down[v], v));
                }
                up[root] = E();
                for (int u : ord) { // pass 2: root →
                // ↪ leaves
                    vector<int> ch;
                    for (int v : adj[u]) if (v != par[u]) ch.push_back(v);
                    int k = ch.size();
                    vector<T> pre(k + 1, E()), suf(k + 1, E());
                    for (int i = 0; i < k; i++) pre[i + 1] = merge(pre[i],
                    // ↪ lift(down[ch[i]], ch[i]));
                    for (int i = k - 1; i >= 0; i--) suf[i] = merge(suf[i + 1],
                    // ↪ lift(down[ch[i]], ch[i]));
                    ans[u] = merge(down[u], up[u]);
                    for (int i = 0; i < k; i++)
                        up[ch[i]] = lift(merge(self(u), merge(up[u],
                    // ↪ merge(pre[i], suf[i + 1])), u);
                    }
                }
            };
            // The prefix/suffix trick ("aggregate over all children EXCEPT one")
            ↪ is reusable on its own:
            // use it whenever a DP needs "everything but this branch" and merge
            ↪ has no inverse.
            // OTHER INSTANTIATIONS: max depth from each node (T = ll, lift = a+1,
            ↪ self = 0);
            // count of nodes at even distance; product of subtree answers; tree
            ↪ diameter through each node.
```

9.8 Virtual Tree

9.8.1 VirtualTree

VIRTUAL (AUXILIARY) TREE - compress k marked vertices plus their pairwise LCAs into a tree

```
// with at most  $2k-1$  nodes, in  $O(k \log k)$ . Run the real DP on that
// ↪ instead of on all  $n$  nodes.
// THE pattern for "q queries, each with  $k_i$  marked vertices, sum of
// ↪  $k_i$  bounded".
// Needs tin/tout from a DFS and an LCA (binary lifting or Euler+sparse
// ↪ table).
struct VirtualTree {
    int n;
```

```

vector<int> tin, tout, dep;
vector<vector<int>> adj, vadj;          // vadj = the
↳ compressed tree
int timer = 0;

VirtualTree(int n) : n(n), tin(n), tout(n), dep(n, 0), adj(n),
↳ vadj(n) {}
void add_edge(int u, int v) { adj[u].push_back(v),
↳ adj[v].push_back(u); }
void dfs(int u, int p) {
    tin[u] = timer++;
    for (int v : adj[u]) if (v != p) dep[v] = dep[u] + 1, dfs(v, u);
    tout[u] = timer++;
}
bool isAnc(int u, int v) const { return tin[u] <= tin[v] && tout[v]
↳ <= tout[u]; }

// 'lca' must be a working LCA callable. Returns the roots' node
↳ list; edges land in vadj.
template <class LCA>
vector<int> build(vector<int> key, LCA lca) {
    if (key.empty()) return {};
    sort(all(key), [&](int a, int b) { return tin[a] < tin[b]; });
    key.erase(unique(all(key)), key.end());
    int k = key.size();
    for (int i = 0; i + 1 < k; i++) key.push_back(lca(key[i], key[i
↳ + 1]));
    sort(all(key), [&](int a, int b) { return tin[a] < tin[b]; });
    key.erase(unique(all(key)), key.end());
    for (int u : key) vadj[u].clear();
    vector<int> st = {key[0]};
    for (size_t i = 1; i < key.size(); i++) {
        int u = key[i];
        while (!isAnc(st.back(), u)) st.pop_back();
        vadj[st.back()].push_back(u);          // parent ->
↳ child (directed is enough)
        st.push_back(u);
    }
    return key;          // key[0] is the
↳ root of the virtual tree
}
};
// ALWAYS clear vadj for exactly the nodes you used (done above) -
↳ never clear all n per query,
// or the total cost becomes O(q*n) and defeats the whole point.
// The compressed edge u -> v represents the real path between them;
↳ its "weight" is usually
// dep[v] - dep[u], or a path aggregate you can query in O(1) with
↳ prefix sums / HLD.

```

9.9 Isomorphism

9.9.1 Treelsomorphism

TREE ISOMORPHISM (AHU canonical form). $O(n \log n)$ with a map, $O(n)$ with per-level bucketing.

```

// Gives every rooted subtree a canonical ID: two subtrees are
↳ isomorphic iff their IDs match.
// Exact - unlike random subtree hashing, which can collide.
struct TreeIso {
    map<vector<int>, int> id;          // canonical
↳ child-multiset -> id
    // canonical id of the subtree rooted at u
    int rootedId(int u, int p, const vector<vector<int>>& adj) {
        vector<int> ch;
        for (int v : adj[u]) if (v != p) ch.push_back(rootedId(v, u,
↳ adj));
        sort(all(ch));
        auto it = id.find(ch);
        if (it != id.end()) return it->second;
        int nid = id.size();
        return id[ch] = nid;
    }
    // CENTERS of a tree: 1 or 2 vertices (Jordan). Root there to
↳ canonicalise an UNROOTED tree.
    vector<int> centers(int n, const vector<vector<int>>& adj) {
        if (n == 1) return {0};
        vector<int> deg(n), leaves;
        for (int i = 0; i < n; i++) if ((deg[i] = adj[i].size()) <= 1)
↳ leaves.push_back(i);
        int removed = leaves.size();
        while (removed < n) {
            if (leaves.empty()) break;          // guard: not a
↳ tree / disconnected
            vector<int> nxt;
            for (int u : leaves)
                for (int v : adj[u]) if (--deg[v] == 1)
↳ nxt.push_back(v);

```

```

        leaves = nxt, removed += leaves.size();
    }
    return leaves;          // 1 or 2
}
↳ vertices, adjacent if 2
// Canonical id of an UNROOTED tree: min over its centers.
int unrootedId(int n, const vector<vector<int>>& adj) {
    auto c = centers(n, adj);
    int best = rootedId(c[0], -1, adj);
    if (c.size() == 2) best = min(best, rootedId(c[1], -1, adj));
    return best;
}
};
// Two unrooted trees are isomorphic iff unrootedId is equal (use ONE
↳ shared TreeIso object so
// the id space is common). Two rooted trees: compare rootedId at their
↳ roots.
// USES: "how many distinct tree shapes", dedupe trees, count
↳ isomorphism classes, group
// identical subtrees for a DP over shapes.
// JORDAN: a tree has exactly one or two centers; if two, they are
↳ adjacent.

```

9.10 Tree DP

9.10.1 TreeDP

TREE DP - the most common tree shape in a contest. Root the tree, compute a value per subtree

```

// bottom-up, and combine children. If the answer must be "for every
↳ root", see 07 - Rerooting.

// --- MAXIMUM WEIGHT INDEPENDENT SET / MINIMUM VERTEX COVER on a tree,
↳ 0(n). ---
// dp[v][0] = best for v's subtree with v NOT taken, dp[v][1] = with v
↳ taken.
// A tree is bipartite, so alpha = n - nu and (Konig) min vertex cover
↳ = maximum matching - but
// this DP is shorter than building a matching, and it handles weights,
↳ which Konig does not.
pair<ll, ll> misTree(int v, int p, const vector<vector<int>>& adj,
↳ const vector<ll>& w) {
    ll take = w[v], skip = 0;
    for (int u : adj[v]) if (u != p) {
        auto [s, t] = misTree(u, v, adj, w);
        skip += max(s, t), take += s;          // if v
↳ is taken, no child may be
    }
    return {skip, take};          //
↳ answer = max(skip, take)
}
// MINIMUM VERTEX COVER (unweighted) = n - MIS. MAXIMUM MATCHING on a
↳ tree = the greedy "match
// every leaf to its parent, delete both, repeat" - and that greedy is
↳ optimal (exchange argument).

// --- TREE KNAPSACK, O(n^2) - and the one character that decides n^2
↳ vs n^3. ---
// dp[v][j] = best value using exactly j vertices of v's subtree. Merge
↳ child by child, capping
// BOTH loops by the ALREADY ACCUMULATED size, not by n. Each unordered
↳ pair (a, b) is then
// charged exactly once, at their LCA, giving O(n^2) total. Writing 'u
↳ <= n' instead of
// 'u <= acc' is O(n^3) and looks identical.
vector<ll> knapTree(int v, int p, const vector<vector<int>>& adj, const
↳ vector<ll>& w) {
    vector<ll> dp = {0, w[v]};          // dp[0]
↳ = take nothing at all
    for (int u : adj[v]) if (u != p) {
        vector<ll> c = knapTree(u, v, adj, w);          // c[0]
↳ = skip this child entirely
        vector<ll> nd(dp.size() + c.size() - 1, LLONG_MIN);
        nd[0] = 0;
        for (size_t i = 1; i < dp.size(); i++)          // i >=
↳ 1: v IS taken, so a child
            for (size_t j = 0; j < c.size(); j++)          // may
↳ attach and stay connected
                if (dp[i] != LLONG_MIN && c[j] != LLONG_MIN) nd[i + j]
↳ = max(nd[i + j], dp[i] + c[j]);
                dp = nd;
    }
    return dp;          // dp[j]
↳ = best CONNECTED set of j
}
↳ vertices containing v (dp[0] = 0)
// Dropping the 'i >= 1' restriction gives the UNCONSTRAINED version:
↳ best j vertices anywhere in

```

```
// the subtree, no connectivity - that is the group-knapsack-on-a-tree
↳ variant.
// WITH A CAP k: allocate min(size, k) + 1 slots instead, giving O(n *
↳ k).

// --- DIAMETER / LONGEST PATH by DP, O(n). The version you reuse
↳ inside other tree DPs. ---
// down1[v], down2[v] = the two longest downward paths from v through
↳ DIFFERENT children.
// The answer is max over v of down1[v] + down2[v]; the two-BFS trick
↳ gives the same on an
// unweighted or POSITIVELY weighted tree, but only this DP survives
↳ negative weights.

/* MORE TREE-DP SHAPES, all O(n) unless noted
COUNT SUBTREES / connected subgraphs containing v: prod over
↳ children of (1 + f(child)).
COUNT MATCHINGS / independent sets: same [0/1] split, sum instead of
↳ max.
SUM OF DISTANCES from every vertex: rerooting with (subtree size,
↳ subtree distance sum).
K-COLOURINGS with adjacent-different: k * (k-1)^(n-1). With a colour
↳ list per vertex, DP
dp[v][c] and take the product over children of (sum over c' != c
↳ of dp[u][c']) - keep the
total per child and subtract the forbidden term to stay O(n * k).
LONGEST PATH WITH AT MOST K EDGES / paths of length exactly L:
↳ dp[v][d] merged small-to-large
over the depth arrays, or long-path decomposition for O(n).
BINARY LIFTING ON A DP: when the transition is associative you can
↳ jump 2^k ancestors at once
(max edge on a path, "the first ancestor with property P").
TREE + BITMASK: dp[v][mask] for k <= 20 marked vertices (Steiner
↳ tree on a tree).
REROOTING is the answer to "compute f(v) for EVERY root" - do not
↳ run the DP n times.
AUXILIARY / VIRTUAL TREE (09 - Trees/08) collapses q marked vertices
↳ into an O(q) tree, so a
per-query tree DP costs O(q log q) instead of O(n).
DSU ON TREE (09 - Trees/02) is the answer when the per-subtree state
↳ is a MULTISSET you cannot
merge algebraically (most frequent colour, k-th value), O(n log n).
SEGMENT TREE MERGING (02 - DS/16) does the same in O(n log n) while
↳ keeping the state
queryable, at the cost of memory.
RECURSION DEPTH: a path-shaped tree with n = 2e5 overflows the
↳ default stack once frames get
fat. Move big locals out of the recursive function or convert to
↳ the iterative order used
in 07 - Rerooting.cpp. */
```

9.11 Dynamic Trees

9.11.1 LinkCutTree

LINK-CUT TREE - a FOREST that changes shape, with path queries. O(log n) amortised per op.

```
// This is dynamic HLD: link, cut, re-root, and aggregate over the path
↳ u..v, all online.
// If the tree is STATIC use HLD (09 - Trees/03) - it is faster and far
↳ shorter. Reach for an LCT
// only when edges are added/removed, or when you need makeRoot.
// HOW IT WORKS: each vertex has at most one "preferred" child;
↳ preferred paths are stored as
// splay trees keyed by DEPTH; a non-preferred edge is a path-parent
↳ pointer the splay does not
// know about. access(v) makes root..v one preferred path, and the same
↳ heavy/light potential that
// bounds HLD bounds the number of preferred-child changes, hence the
↳ O(log n) amortised.
```

```
struct LCT {
    struct Node { int ch[2] = {0, 0}, p = 0; ll val = 0, sum = 0; bool
↳ rev = 0; };
    vector<Node> t;
    LCT(int n) : t(n + 1) {} //
↳ 1-indexed; node 0 is the null
    bool isRoot(int x) { return t[x].p.ch[0] != x && t[x].p.ch[1]
↳ != x; }
    void pull(int x) { t[x].sum = t[t[x].ch[0]].sum ^ t[x].val ^
↳ t[t[x].ch[1]].sum; }
    void flip(int x) { if (x) swap(t[x].ch[0], t[x].ch[1]), t[x].rev ^=
↳ 1; }
    void push(int x) { if (t[x].rev) flip(t[x].ch[0]),
↳ flip(t[x].ch[1]), t[x].rev = 0; }
    void rot(int x) {
        int p = t[x].p, g = t[p].p, d = t[p].ch[1] == x;
        if (!isRoot(p)) t[g].ch[t[g].ch[1] == p] = x;
        t[x].p = g;
```

```
t[p].ch[d] = t[x].ch[!d], t[t[x].ch[!d]].p = p;
t[x].ch[!d] = p, t[p].p = x;
pull(p), pull(x);
}
void splay(int x) {
    vector<int> st{x};
    for (int y = x; !isRoot(y); y = t[y].p) st.push_back(t[y].p);
    for (int i = st.size() - 1; i >= 0; i--) push(st[i]);
    while (!isRoot(x)) {
        int p = t[x].p, g = t[p].p;
        if (!isRoot(p)) rot((t[g].ch[1] == p) == (t[p].ch[1] == x)
↳ ? p : x);
        rot(x);
    }
}
int access(int x) { //
↳ root..x becomes one splay tree
    int last = 0;
    for (int y = x; y; y = t[y].p) splay(y), t[y].ch[1] = last,
↳ pull(y), last = y;
    splay(x);
    return last; // = the
↳ LCA on the previous access
}
void makeRoot(int x) { access(x), flip(x), push(x); }
int findRoot(int x) { access(x); while (t[x].ch[0]) push(x), x =
↳ t[x].ch[0]; splay(x); return x; }
bool connected(int x, int y) { return findRoot(x) == findRoot(y); }
void link(int x, int y) { makeRoot(x); if (findRoot(y) != x) t[x].p
↳ = y; }
void cut(int x, int y) { // no-op
↳ if the edge is absent
    makeRoot(x);
    if (findRoot(y) == x && t[y].p == x && !t[y].ch[0]) t[y].p =
↳ t[x].ch[1] = 0, pull(x);
}
ll query(int x, int y) { makeRoot(x), access(y); return t[y].sum; }
↳ // path aggregate
void update(int x, ll v) { access(x), t[x].val = v, pull(x); }
int lca(int r, int x, int y) { makeRoot(r), access(x); return
↳ access(y); }
};
// THE AGGREGATE is XOR here purely because it is self-inverse and
↳ needs no reversal handling.
// For SUM/MIN/MAX just change pull(); for a NON-COMMUTATIVE aggregate
↳ (matrix product, max
// prefix sum) you must store BOTH directions in the node and swap them
↳ inside flip().
// EDGE WEIGHTS: give each edge its own vertex (n + edgeId) with the
↳ weight, and link both ends
// to it. Vertex values then stay 0 and every path query still works.
// SUBTREE aggregates need "virtual subtree" bookkeeping: keep a second
↳ sum over the non-preferred
// children, updated in access() when you swap t[y].ch[1]. That doubles
↳ the code - if the tree is
// static, an Euler tour (09 - Trees/06) is O(log n) and ten lines.
/* WHAT LCT UNLOCKS
ONLINE DYNAMIC CONNECTIVITY ON A FOREST: connected() in O(log n).
↳ For general graphs (cycles)
you need the offline segment-tree-on-time trick (02 - Data
↳ Structures/14) or Holm-de
Lichtenberg-Thorup at O(log^2 n).
DYNAMIC MST: to add edge (u,v,w), if u and v are connected find the
↳ MAXIMUM edge on the path;
if it exceeds w, cut it and link the new edge. O(log n) per update.
MAXIMUM FLOW (Dinic with link-cut) improves the blocking-flow step
↳ to O(VE log V) - almost
never needed at contest sizes.
OFFLINE "PAINT THE ROOT PATH": access() itself is the operation
↳ "recolor root..v with a new
colour"; the number of colour changes is O(n log n) total, so if
↳ each change costs a segment
tree update you get O(n log^2 n). This is the standard trick for
↳ "count distinct substrings
over all substrings of s" and for tree-path-colouring problems. */
```

9.12 Long Path Decomposition

9.12.1 LongPathDecomposition

LONG PATH DECOMPOSITION - split the tree into vertical paths by always following the child with

```
// the DEEPEST subtree (HLD follows the child with the BIGGEST subtree
// instead). 0-indexed, rooted.
// TWO THINGS IT BUYS YOU THAT HLD DOES NOT.
// 1. O(1) K-TH ANCESTOR after O(n log n) preprocessing. Binary
//    lifting is O(log n) per query;
//    this is the standard constant-time answer, and it is only ~30
//    lines.
// 2. DEPTH-INDEXED TREE DP IN O(n) TOTAL ("dp[v][d] = something about
//    the vertices at depth d
//    below v"). Give each path head an array of its length and let
//    every vertex on the path SHARE
//    that array with an offset - the heavy child inherits the
//    parent's array for free, and only
//    the O(sqrt n)-ish path heads allocate. The classic use is "for
//    each v, the most common depth
//    in its subtree" and "count vertices at distance k below v" for
//    all v.
// The number of paths equals the number of LEAVES, and the crucial
// property is: the path containing
// v has length at least (height of v), so jumping to the head of v's
// path already covers a lot.
// K-TH ANCESTOR IN O(1). For every path head h of length L, store the
// L ancestors above h as well
// as the L vertices of the path itself ("up" and "down" tables of size
// 2L, total O(n)). To answer
// kth(v, k): let j = highest power of two <= k, jump v to its 2^j-th
// ancestor with a lifting table
// (one step), and the remaining k - 2^j < 2^j <= (height of the new
// vertex), so the rest of the
// jump lands inside the new vertex's own path table - one array lookup.
struct LongPath {
    int n, B;
    vector<int> par, dep, hgt, heavy, head;
    vector<vector<int>> jmp, tab;
    // tab[h] = ladder of head h
    vector<int> posInTab, headOf, logs;
    LongPath(int n, const vector<vector<int>>& adj, int root = 0)
        : n(n), B(1), par(n, -1), dep(n, 0), hgt(n, 0), heavy(n, -1),
        head(n, -1),
        posInTab(n, 0), headOf(n, -1), logs(n + 1, 0) {
        while ((1 <= B) <= n) B++;
        for (int i = 2; i <= n; i++) logs[i] = logs[i / 2] + 1;
        vector<int> ord;
        // iterative DFS -> preorder
        vector<int> st{root};
        par[root] = -1;
        while (!st.empty()) {
            int u = st.back(); st.pop_back();
            ord.push_back(u);
            for (int v : adj[u]) if (v != par[u]) par[v] = u, dep[v] =
            dep[u] + 1, st.push_back(v);
        }
        for (int i = n - 1; i >= 0; i--) {
            // heights, bottom up
            int u = ord[i];
            for (int v : adj[u]) if (v != par[u] && hgt[v] + 1 > hgt[u])
                hgt[u] = hgt[v] + 1, heavy[u] = v;
        }
        jmp.assign(B, vector<int>(n, -1));
        for (int u = 0; u < n; u++) jmp[0][u] = par[u];
        for (int k = 1; k < B; k++) for (int u = 0; u < n; u++)
            jmp[k][u] = jmp[k - 1][u] < 0 ? -1 : jmp[k - 1][jmp[k -
            1][u]];
        for (int u : ord) head[u] = (par[u] >= 0 && heavy[par[u]] == u)
        ? head[par[u]] : u;
        tab.assign(n, {});
        for (int h = 0; h < n; h++) if (head[h] == h) { // build
        the ladder of this path
            int L = hgt[h] + 1;
            vector<int> down;
            for (int v = h; v >= 0; v = heavy[v]) down.push_back(v);
            vector<int> upv; // L
            // ancestors above h, if they exist
            for (int x = par[h], c = 0; x >= 0 && c < L; x = par[x],
            c++) upv.push_back(x);
            reverse(all(upv));
            for (int v : down) upv.push_back(v); // [far
            // ancestors ... h ... deepest]
            tab[h] = upv;
            int off = (int)tab[h].size() - (int)down.size();
            for (int i = 0; i < (int)down.size(); i++)
                posInTab[down[i]] = off + i, headOf[down[i]] = h;
        }
    }
}
```

```
int kth(int v, int k) const { // k-th
    // ancestor, -1 if too high
    if (k == 0) return v;
    if (k > dep[v]) return -1;
    int j = logs[k];
    v = jmp[j][v], k -= 1 << j; // one
    // lifting step
    int h = headOf[v];
    return tab[h][posInTab[v] - k]; // the
    // rest is inside the ladder
}
};
// WHY THE LADDER IS LONG ENOUGH: after the single 2^j jump, the
// remainder k' < 2^j and the vertex
// we are on has height >= 2^j - 1, so its path (and hence its ladder's
// upper half) covers k'.
// MEMORY is O(n log n) for the lifting table and O(n) for the ladders
// (each path of length L keeps
// 2L entries, and the path lengths sum to n).
// LEVEL ANCESTOR can also be done in O(n) preprocessing (Euler tour +
// the ladder trick without the
// lifting table), but the version above is the one you can write from
// memory.
// SMALL-TO-LARGE ALTERNATIVE for depth-indexed DP: merge
// maps/multisets child by child, O(n log^2)
// - shorter but slower; use it when the DP state is not a plain array
// indexed by depth.
```

9.13 Path Operations

9.13.1 PathIntersectionAndUnion

INTERSECTION AND UNION OF TWO TREE PATHS, O(log n) each. 0-indexed, rooted tree.

```
// Needs: Lift (09 - Trees/01 - Lowest Common Ancestor (LCA)/02 -
// BinaryLifting.cpp).
// A path is the pair of its endpoints {a, b} (unordered); {-1, -1} is
// the EMPTY path and
// {-2, -2} is "these two paths do not union into a path".
// Reach for it for "how many of these q paths pass through a common
// vertex", "the longest prefix
// of the queries whose paths all intersect" (the intersection is
// associative, so put it in a
// segment tree or binary-search it), "merge the constraints u..v must
// be inside one path", and
// road/route-overlap problems.
// KEY DECOMPOSITION: a path a..b splits at l = lca(a, b) into two
// VERTICAL chains l..a and l..b.
// The intersection of two paths is the union of the (at most four)
// pairwise chain intersections,
// and each of those is again a vertical chain that is trivial to
// compute from depths and lca.
struct PathOps {
    const Lift& L;
    typedef pair<int, int> P;
    PathOps(const Lift& L) : L(L) {}
    bool isAnc(int u, int a) const { // is a
        // an ancestor of u (or == u)?
        return L.d[u] >= L.d[a] && L.kth(u, L.d[u] - L.d[a]) == a;
    }
    int len(P p) const { return p.first < 0 ? 0 : L.dist(p.first,
    p.second) + 1; } // #vertices
    // The common part of the vertical chains u..a and v..c, where u is
    // an ancestor of a and v of c.
    P chainMeet(int u, int a, int v, int c) const {
        if (!isAnc(a, v)) return {-1, -1}; // v
        // must lie on the chain u..a
        int x = L.lca(a, c);
        if (L.d[v] < L.d[u]) { if (isAnc(x, u)) return {u, x}; }
        else if (isAnc(x, v)) return {v, x};
        return {-1, -1};
    }
    P joinChains(P x, P y) const { //
        // union, ASSUMED to be a path
        if (x.first < 0) return y;
        if (y.first < 0) return x;
        int can[4] = {x.first, x.second, y.first, y.second}, a =
        can[0], b = -1;
        for (int u : can) if (L.d[u] > L.d[a]) a = u; //
        // deepest endpoint
        for (int u : can) if (!isAnc(a, u) && (b < 0 || L.d[b] <
        L.d[u])) b = u;
        if (b < 0) { b = can[0]; for (int u : can) if (L.d[u] < L.d[b])
        b = u; }
        return {a, b};
    }
    P intersect(P p, P q) const { //
        // {-1, -1} when they are disjoint
```

```

    if (p.first < 0 || q.first < 0) return {-1, -1};
    int u = L.lca(p.first, p.second), v = L.lca(q.first, q.second);
    P r1 = joinChains(chainMeet(u, p.first, v, q.first),
    ↪ chainMeet(u, p.first, v, q.second));
    P r2 = joinChains(chainMeet(u, p.second, v, q.first),
    ↪ chainMeet(u, p.second, v, q.second));
    return joinChains(r1, r2);
}
P unite(P x, P y) const {
    ↪ {-2,-2} if the union is not a path
    if (x.first == -2 || y.first == -2) return {-2, -2};
    if (x.first < 0) return y;
    if (y.first < 0) return x;
    int e[4] = {x.first, x.second, y.first, y.second};
    int need = len(x) + len(y) - len(intersect(x, y)); //
    ↪ [union] by inclusion-exclusion
    for (int i = 0; i < 4; i++) for (int j = i; j < 4; j++) {
        P r[e[i], e[j]];
        ↪ candidate covering path
        bool ok = len(r) == need; // no
        ↪ branch and no gap
        for (int w : e)
            if (L.dist(r.first, w) + L.dist(w, r.second) !=
    ↪ L.dist(r.first, r.second)) ok = 0;
            if (ok) return r;
    }
    return {-2, -2};
}
};
// BOTH OPERATIONS ARE ASSOCIATIVE, so they drop straight into a
    ↪ segment tree over an array of
// paths: "the intersection of a range of queries" or "the longest
    ↪ prefix whose union is a path"
// (descend the tree greedily, which is exactly the classic "sort a
    ↪ permutation along a tree" task).
// SMALLEST PATH CONTAINING two given paths is a THIRD operation, and
    ↪ it is not the union - see
// mergePaths in 01 - LCA/03 - LCAConstantTime, which returns the
    ↪ minimal COVERING path.
// DOES VERTEX w LIE ON PATH a..b? dist(a,w) + dist(w,b) == dist(a,b).
    ↪ That one-liner replaces this
// whole file when the query is that simple.
// PATH INTERSECTION LENGTH is len(intersect(p, q)) in VERTICES;
    ↪ subtract one for edges.

```

10 | Game Theory

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10.1 GrundyAndNim

SPRAGUE-GRUNDY. Every FINITE, ACYCLIC (progressively bounded) IMPARTIAL game under normal

```

/* SPRAGUE-GRUNDY. Every FINITE, ACYCLIC (progressively bounded)
    ↪ IMPARTIAL game under normal
    play (last to move wins) is equivalent to a nim pile of size
    ↪ Grundy(position).
    ACYCLIC IS A HYPOTHESIS, NOT A DETAIL: on a graph with cycles the
    ↪ mex recursion is not
    well-founded and Grundy() below returns silently wrong values - use
    ↪ retrograde BFS instead.
    Grundy(p) = mex{ Grundy(q) : q reachable from p }.
    A position is LOSING for the player to move iff Grundy = 0.
    SUM of independent games: Grundy = XOR of the parts.
    ↪ */
int mex(const vector<int>& v) {
    vector<bool> seen(v.size() + 1, false);
    for (int x : v) if (x >= 0 && x <= (int)v.size()) seen[x] = true;
    int m = 0;
    while (m < (int)seen.size() && seen[m]) m++;
    return m;
}
// Grundy over a DAG of at most n positions, memoised.
vector<int> g_;
int Grundy(int u, const vector<vector<int>>& nxt) {
    if (g_[u] != -1) return g_[u];
    g_[u] = 0;
    ↪ => no real cycles
    vector<int> s;
    for (int v : nxt[u]) s.push_back(Grundy(v, nxt));
    return g_[u] = mex(s);
}
/* NIM AND FRIENDS
    NIM (take any number from one pile). Losing iff XOR of pile sizes ==
    ↪ 0.

```

Winning move: find a pile with $(x \oplus X) < x$ where X is the total
 ↪ xor; reduce it to $x \oplus X$.
 MISERE NIM (last to move LOSES). If every pile is size 1: first
 ↪ player wins iff #piles is EVEN.
 Otherwise identical to normal nim (win iff xor != 0).
 NIM_k / MOORE'S NIM (may take from up to k piles). Losing iff, for
 ↪ every bit, the number of
 piles with that bit set is divisible by $(k+1)$.
 STAIRCASE NIM. INDEX 0 IS THE GROUND - coins there are DEAD and have
 ↪ no move.
 Losing iff XOR of the counts at ODD indices == 0. (If instead a
 ↪ coin on index 0 may still
 slide OFF the board, the ground is a virtual index -1 and you must
 ↪ XOR the EVEN indices.
 Getting this backwards is wrong on 96 of the 256 four-position
 ↪ states.)
 Recognise it as "move something toward a sink, and moving it to an
 ↪ even slot is reversible".
 SUBTRACTION GAME, allowed set S: Grundy is eventually PERIODIC.
 ↪ Compute a prefix, find the
 period, extrapolate - the standard route for n up to 1e9.
 S = {1..k}: Grundy(n) = n mod (k+1).
 WYTHOFF (two piles, take from one or equally from both). Losing
 ↪ positions are
 (floor(k*phi), floor(k*phi^2)), phi = (1+sqrt(5))/2, k >= 0.
 FIBONACCI NIM (first move any amount < pile; then at most 2x the
 ↪ previous move).
 First player loses iff the pile size is a Fibonacci number.
 TURNING / COIN GAMES: Grundy of a set of coins = XOR of the Grundy
 ↪ of each single coin.
 GREEN HACKENBUSH (all edges neutral): Grundy of a path of length n
 ↪ is n; FUSION principle -
 a cycle of odd length fuses to a single edge, an even cycle to
 ↪ nothing; sum branches by XOR.
 COLON PRINCIPLE (trees): Grundy of a node = XOR over children of
 ↪ (Grundy(child) + 1). */

// RETROGRADE ANALYSIS - when the game graph HAS CYCLES (draws
 ↪ possible) or the game is partisan,
 // Grundy does not apply. BFS backwards from terminal positions
 ↪ counting out-degrees.

```

// win[u] = 1 first player to move wins, 0 loses, -1 draw.
vector<int> retrograde(int n, const vector<vector<int>>& radj,
    ↪ vector<int> deg,
    const vector<int>& terminalLose) {
    vector<int> res(n, -1);
    queue<int> q;
    for (int u : terminalLose) res[u] = 0, q.push(u);
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int u : radj[v]) {
            if (res[u] != -1) continue;
            if (res[v] == 0) res[u] = 1, q.push(u); // can move to
    ↪ a losing position
            else if (--deg[u] == 0) res[u] = 0, q.push(u); // all
    ↪ moves lead to wins
        }
    }
    return res; // remaining
    ↪ -1 are draws
}

```

10.2 GameAlgorithms

Concrete game solvers. All impartial games under NORMAL play (last move wins) unless stated.

```

// SUBTRACTION GAME with allowed move set S: Grundy is eventually
    ↪ PERIODIC. Compute a prefix,
// detect the period, extrapolate to n up to 1e18. This is the standard
    ↪ route for huge n.
struct Periodic {
    vector<int> g;
    int start = -1, per = -1;
    Periodic(const vector<int>& S, int prefix = 2000) {
        g.assign(prefix, 0);
        for (int i = 1; i < prefix; i++) {
            vector<bool> seen(S.size() + 2, false);
            for (int s : S) if (s <= i && g[i - s] < (int)seen.size())
    ↪ seen[g[i - s]] = true;
            while (g[i] < (int)seen.size() && seen[g[i]]) g[i]++;
        }
        detect();
    }
    void detect() { // smallest
    ↪ (start, per) that fits
        int n = g.size();

```

```

    for (int p = 1; p <= n / 3; p++) {
        bool ok = true;
        for (int i = n / 3; i + p < n; i++) if (g[i] != g[i + p]) {
            ok = false; break; }
        if (!ok) continue;
        int st = 0; // the LAST
        break, not the first match:
        for (int i = n - p - 1; i >= 0; i--) //
        periodicity must hold from start ON
            if (g[i] != g[i + p]) { st = i + 1; break; }
        if (n - p - st < 2 * p + 20) continue; // demand
        real evidence before believing
        start = st, per = p;
        return;
    }
}
int at(ll n) const {
    if (n < (ll)g.size()) return g[n];
    assert(per > 0); // no period
    found: do NOT extrapolate
    return g[start + (n - start) % per];
}
};
// S = {1..k} => Grundy(n) = n mod (k+1) (no need to compute
// anything)

// WYTHOFF'S GAME (two piles; take any amount from one pile, or the
// SAME amount from both).
// Losing (P-)positions: (floor(k*phi), floor(k*phi^2)) for k >= 0, phi
// = (1+sqrt(5))/2.
// Since phi^2 - phi = 1, the two coordinates differ by EXACTLY k, so k
// = b - a.
// Tested exactly, no floating point: a == floor(k*phi) <=> d <=
// k*sqrt(5) < d+2, d = 2a-k.
bool wythoffLosing(ll a, ll b) {
    if (a > b) swap(a, b);
    ll k = b - a, d = 2 * a - k;
    return d >= 0 && (__int128)d * d <= (__int128)5 * k * k
        && (__int128)5 * k * k < (__int128)(d + 2) * (d + 2);
}
// P-positions: (0,0) (1,2) (3,5) (4,7) (6,10) (8,13) (9,15) (11,18)
// (12,20) (14,23) ...

// MOORE'S NIM_k: you may take from up to k piles in one move.
// Losing iff, for EVERY bit, the number of piles having that bit set
// is divisible by (k+1).
bool mooreLosing(const vector<ll>& piles, int k) {
    for (int b = 0; b < 63; b++) {
        int c = 0;
        for (ll x : piles) c += (x >> b) & 1;
        if (c % (k + 1)) return false;
    }
    return true;
}

// STAIRCASE NIM: coins on positions 0..n-1, a move slides some coins
// from i to i-1.
// INDEX 0 IS THE GROUND: coins there are DEAD and cannot move. Then
// only ODD indices matter
// (a move from an even index is reversible, so it is worth nothing).
// If your problem instead lets a coin at index 0 slide OFF the board,
// the ground is a virtual
// index -1 and you must XOR the EVEN indices. Brute force over 4
// positions: the wrong convention
// is wrong on 96 of 256 states, e.g. (0,0,0,1).
bool staircaseLosing(const vector<ll>& cnt) {
    ll x = 0;
    for (size_t i = 1; i < cnt.size(); i += 2) x ^= cnt[i];
    return x == 0;
}

// GREEN HACKENBUSH (all edges neutral), Grundy of a rooted tree =
// COLON PRINCIPLE:
// g(v) = XOR over children c of (g(c) + 1)
int hackenbushTree(int u, int p, const vector<vector<int>>& adj) {
    int r = 0;
    for (int v : adj[u]) if (v != p) r ^= hackenbushTree(v, u, adj) + 1;
    return r;
}

// FUSION PRINCIPLE for graphs: all vertices of a cycle can be fused
// into one; an ODD cycle
// leaves a single loop (= one edge, Grundy 1), an EVEN cycle leaves
// nothing (Grundy 0).

// COIN TURNING GAMES: a position is a set of "heads". The Grundy value
// of the whole position is
// the XOR of the Grundy values of each single head considered alone

```

→ (Mock Turtles, Ruler, ...).

// MISERE NIM (last to move LOSES): if EVERY pile is 1, the first
 → player wins iff #piles is even;
 // otherwise the normal-play rule (xor != 0 => win) applies unchanged.

10.3 MinimaxAndMatching

Games that Sprague-Grundy does NOT cover: partisan games, scored games, draws.

// Needs: Blossom (03 - Graph Theory/07 - Matching/04 - Blossom.cpp)
 → for the matching part.

```

// MINIMAX with ALPHA-BETA pruning. Returns the value in PLAYER 0's
// units (a fixed sign), not
// "the value for the player to move" - with maxing = false the result
// is still player-0 scored.
// 'moves(state)' lists successors; 'eval(state)' scores terminal/leaf
// states from player 0's view.
template <class S, class Moves, class Eval>
ll alphabeta(const S& s, int depth, ll alpha, ll beta, bool maxing,
    Moves moves, Eval eval) {
    auto nxt = moves(s);
    if (!depth || nxt.empty()) return eval(s);
    if (maxing) {
        ll best = LLONG_MIN;
        for (auto& t : nxt) {
            best = max(best, alphabeta(t, depth - 1, alpha, beta,
            false, moves, eval));
            alpha = max(alpha, best);
            if (beta <= alpha) break; // beta cutoff
        }
        return best;
    }
    ll best = LLONG_MAX;
    for (auto& t : nxt) {
        best = min(best, alphabeta(t, depth - 1, alpha, beta, true,
        moves, eval));
        beta = min(beta, best);
        if (beta <= alpha) break; // alpha
    }
    return best;
}

// Order moves best-first: pruning is only good if you try strong moves
// early.
// Memoise on (state) when states repeat; use iterative deepening when
// depth is unbounded.

// MATCHING GAME ON A GRAPH: a token sits on vertex s; players
// alternately move it along an
// edge to an unvisited vertex; a player who cannot move loses.
// THEOREM: the FIRST player wins iff s is in EVERY maximum matching,
// i.e. iff removing s
// decreases the size of the maximum matching.
// This works on ANY graph, and you get EVERY start vertex at once for
// the price of one matching.
// After a maximum matching is fixed, the vertices left exposed by SOME
// maximum matching are
// exactly those reachable from an already-exposed vertex by an
// EVEN-length alternating path -
// which is precisely the set an unsuccessful blossom search paints as
// OUTER. So run the matching,
// then one failing search from each exposed vertex, and the unpainted
// vertices are the wins.
// O(V^3) for the whole table, versus O(V^4) for "recompute the
// matching without s" per vertex.
// Needs Blossom (03 - Graph Theory/07 - Matching/04 - Blossom.cpp) -
// see the wrapper below.
vector<char> matchingGameWins(Blossom& B) { // B already
    has all edges added
    B.solve();
    vector<char> miss(B.n, 0);
    for (int v = 0; v < B.n; v++) if (B.match[v] < 0) {
        B.augment(v); // always
        fails; it is here for used[]
        for (int u = 0; u < B.n; u++) miss[u] |= B.used[u];
    }
    for (int v = 0; v < B.n; v++) miss[v] = !miss[v]; // in every
    maximum matching -> P1 wins
    return miss;
}

// A vertex with NO edges is exposed in every matching, so the first
// player loses there - which is
// right, they cannot move. ISOLATED vertices therefore need no special
// case.
// GAMES WITH SCORES (not just win/lose): plain minimax DP over states,
// dp[state] = best over moves of (gain + (-dp[next])) for a

```

```

    ↪ zero-sum symmetric game,
// or keep two arrays (dp0, dp1) when the players' objectives differ.
// GAMES WITH DRAWS / CYCLES: retrograde BFS (see 01 -
    ↪ GrundyAndNim.cpp), never Grundy.

```

10.4 NamedGames

NAMED GAMES: the values you cannot re-derive during a contest

```

/* ===== NAMED GAMES: the values you cannot re-derive during a contest
    ↪ =====
Two different families live here: TAKE-AND-BREAK (octal) games, and
    ↪ COIN-TURNING games.
They have different theory - octal games are the splitting rule below,
    ↪ coin-turning games are
XOR of single-head values and, in 2-D, nim MULTIPLICATION. Do not mix
    ↪ them up.
Every value on this card was brute-force verified.
All are impartial, NORMAL play. Grundy of a sum = XOR of parts.

```

SPLITTING RULE (the general take-and-break move):

```
g(v) = mex{ g(a1) XOR g(a2) XOR ... : v -> (a1, a2, ...) }
```

LASKER'S NIM (take any amount from a pile, OR split a pile of size >= 2 into two non-empty)

```

g(4k+1) = 4k+1      g(4k+2) = 4k+2      g(4k+3) = 4k+4      g(4k+4) =
    ↪ 4k+3

```

KAYLES (remove 1 or 2 adjacent pins, possibly splitting the row)

Periodic with period 12 from n = 71 onward (Winning Ways says 72; 71 ↪ is tight).

```

    block for n >= 71:  7 4 1 2 8 1 4 7 2 1 8 2
    14 exceptions (n, G):
    ↪ (0,0)(3,3)(6,3)(9,4)(11,6)(15,7)(18,3)(21,4)(22,6)(28,5)(34,6)
                                (39,3)(57,4)(70,6)      last exception n
    ↪ = 70

```

DAWSON'S CHESS (octal game .137)

```

    Period 34 from n = 52 onward.
    block for n >= 52: 3 3 0 1 1 3 0 2 1 1 0 4 5 3 7 4 8 1 1 2 0 3 1 1
    ↪ 0 3 3 2 2 4 4 5 5 9
    7 exceptions (n, G): (0,0)(14,0)(16,2)(17,2)(31,2)(34,0)(51,2)
    ↪ last exception n = 51

```

DAWSON'S KAYLES (.07) period 34 from n = 53, same block, 8 ↪ exceptions, last at n = 52.

GRUNDY'S GAME (split a pile into two UNEQUAL non-empty piles)

```

    Periodicity is an OPEN PROBLEM - more than 2*10^10 values computed
    ↪ and no period found.
    Do NOT assume a period here; compute the prefix you need.

```

GUY-SMITH PERIODICITY THEOREM (how to PROVE a period, not just guess ↪ it)

```

    For an octal game whose largest non-zero code digit has index k: if
    G(n + p) = G(n) for all n with n0 <= n < 2*n0 + p + k
    then the period holds for ALL n >= n0. So checking up to 2*n0 + p +
    ↪ k is a proof.

```

TURNING GAMES (a position is a set of heads; the whole position's ↪ Grundy value is the XOR of

the values of each single head considered alone)

```

    Twins (turn exactly 2 coins), 0-indexed: g(x) = x
    ↪ [plain nim]

```

```

    Mock Turtles(turn 1, 2 or 3 coins), 0-indexed: g(x) = the x-th
    ↪ ODDIOUS number

```

```

    (odious = odd popcount): 1, 2, 4, 7, 8, 11, 13, 14, 16,
    ↪ ...

```

```

    g(x) = 2x if 2x is odious, else 2x + 1
    Ruler (turn any number of CONSECUTIVE coins), 1-indexed:
    g(x) = largest power of two dividing x = x & (-x) ->
    ↪ 1,2,1,4,1,2,1,8,...

```

```

    TRAP: Ruler is 1-INDEXED while Twins and Mock Turtles are 0-INDEXED.
    ↪ Mixing them up gives a
    silently wrong answer.

```

NIM WITH ONE PASS (Guy's model: the pass may be used once, at any ↪ NON-terminal position)

```

    1 heap: G(0) = 0, G(n) = n+1 for n odd, G(n) = n-1 for n even >=
    ↪ 2.

```

```

    2 heaps: P-positions are exactly (0,0) and (a, a+1) with a ODD.
    3+ heaps: OPEN. No pattern; 34 P-positions with a<=b<=c<=13 and
    ↪ neither xor==0 nor xor==1

```

comes close. Never guess.

```

    If the pass is ALSO legal at a terminal position (a weaker rule),
    ↪ the game collapses:

```

```

    G = (XOR of piles) XOR 1, so P-positions are exactly XOR == 1.

```

```

    ↪ */

```

```

bool odious(ll x) { return __builtin_parityll(x); }
ll mockTurtles(ll x) { return odious(2 * x) ? 2 * x : 2 * x + 1; } //
    ↪ 0-indexed
ll ruler(ll x) { return x & -x; } //
    ↪ 1-indexed
ll lasker(ll n) { // g(0) = 0, NOT -1
    ll r = n % 4;
    if (r == 0) return n ? n - 1 : 0;
    return r == 3 ? n + 1 : n;
}
// Grundy of a sum of turning-game heads = XOR of the single-head
    ↪ values.
ll turningGameValue(const vector<ll>& heads, ll (*g)(ll)) {
    ll r = 0;
    for (ll h : heads) r ^= g(h);
    return r;
}

```

10.5 NimbersAndSums

NIMBERS, OCTAL GAMES, AND THE SUM VARIANTS. Everything on this page was brute-force verified.

```

// --- NIM PRODUCT. [0, 2^(2^k)] is a FIELD under (XOR, nimMul). ---
// Definition: a (*) b = mex{ a'(*)b ^ a(*)b' ^ a'(*)b' : a' < a, b' <
    ↪ b }.
// Computed with the two identities: for a Fermat 2-power F = 2^(2^k),
// x (*) F = x * F for x < F, and F (*) F = 3F/2.
// USES: Turning Corners (turn the 4 corners of a rectangle) has g(x,y)
    ↪ = x (*) y, and the TARTAN
// THEOREM says the 2-D product of two coin-turning games has g(x,y) =
    ↪ g1(x) (*) g2(y). Also
// CF 1310F, which asks you to solve a^(*)x = b in the nimber field of
    ↪ size 2^64.

```

```

typedef unsigned long long u64; // nimbers
    ↪ are UNSIGNED, always
map<pair<u64, u64>, u64> nmMemo;
u64 nimMul(u64 a, u64 b) {
    if (a < b) swap(a, b);
    if (b == 0) return 0;
    if (b == 1) return a;
    auto it = nmMemo.find({a, b});
    if (it != nmMemo.end()) return it->second;
    int L = 0;
    while (L < 6 && (a >> (1u << L))) L++;
    u64 F = 1ULL << (1u << (L - 1)); //
    ↪ largest Fermat 2-power <= a
    u64 ah = a / F, al = a % F, bh = b / F, bl = b % F;
    u64 A = nimMul(ah, bh), B = nimMul(ah, bl), C = nimMul(al, bh), D =
    ↪ nimMul(al, bl);
    return nmMemo[{a, b}] = ((A ^ B ^ C) * F) ^ D ^ nimMul(A, F / 2);
}

```

```

// --- OCTAL (TAKE-AND-BREAK) GAME GRUNDY GENERATOR. ---
// The code is ".d1d2d3...": digit d_k governs "remove exactly k tokens
    ↪ from one heap".
// bit 1 (value 1): you may take the WHOLE heap (i.e. leave
    ↪ zero heaps)
// bit 2 (value 2): you may leave ONE non-empty heap
// bit 4 (value 4): you may SPLIT the remainder into TWO non-empty
    ↪ heaps
// So .333... = Nim, .77 = Kayles, .07 = Dawson's Kayles, .137 =
    ↪ Dawson's chess.
// Being able to read a statement into a code string turns "invent a
    ↪ Grundy recurrence" into a
// function call. O(N^2 * digits) because of the split term; O(N *
    ↪ digits) without it.
vector<int> octal(const string& code, int N) { // code
    ↪ = the digits AFTER the dot
    vector<int> d;
    for (char c : code) d.push_back(isdigit(c) ? c - '0' : c - 'a' +
    ↪ 10);
    vector<int> g(N + 1, 0);
    for (int n = 1; n <= N; n++) {
        set<int> s;
        for (int i = 0; i < (int)d.size(); i++) {
            int k = i + 1;
            if (k > n) break;
            int rem = n - k;
            if ((d[i] & 1) && rem == 0) s.insert(0);
            if ((d[i] & 2) && rem >= 1) s.insert(g[rem]);
            if (d[i] & 4) for (int a = 1; a < rem; a++) s.insert(g[a] ^
    ↪ g[rem - a]);
        }
        int m = 0;
        while (s.count(m)) m++;
        g[n] = m;
    }
}

```

```

    }
    return g;
}
// GUY-SMITH PERIODICITY THEOREM turns a guess into a PROOF: with k =
    ↪ the index of the last
// non-zero code digit, if  $G(n + p) = G(n)$  for all  $n_0 \leq n < 2n_0 + p +$ 
    ↪ k, the period holds for
// ALL  $n \geq n_0$ . See 02 - GameAlgorithms.cpp for the (fixed) period
    ↪ detector.

// --- EUCLID'S GAME. (a, b); subtract a positive multiple of the
    ↪ smaller from the larger. ---
// FIRST PLAYER WINS iff  $a = b$  OR  $\max/\min > \phi$ . The usual phrasing
    ↪ " $\geq \phi$ " is WRONG on the
// whole diagonal: (a, a) is always an immediate win.
bool euclidWin(ll a, ll b) {
    if (a < b) swap(a, b);
    if (b == 0) return false;
    if (a == b || a >= 2 * b) return true;
    return !euclidWin(b, a - b);
}

// --- ANTI-SG / the SJ theorem: MISERE play on a SUM of impartial
    ↪ games. ---
// PRECONDITION, and it is not optional: in every component,  $SG(u) == 0$ 
    ↪ must imply u is TERMINAL.
// (Without it this rule is wrong on ~1.3% of random impartial games;
    ↪ with it, exact.)
// Misere nim satisfies it, which is why the classic misere-nim rule
    ↪ looks so clean.
bool antiSGFirstWins(const vector<int>& g) {
    int x = 0, mx = 0;
    for (int v : g) x ^= v, mx = max(mx, v);
    return (x && mx > 1) || (!x && mx <= 1);
}

// --- EVERY-SG: you must move in EVERY unfinished component. Not a
    ↪ function of the XOR at all. ---
// step(v) = 0 if v is terminal
// step(v) = 1 + max{ step(u) :  $SG(u) == 0$  } if  $SG(v) > 0$  (drag
    ↪ it out - you will win it)
// step(v) = 1 + min{ step(u) } if  $SG(v) == 0$  (end
    ↪ it fast - you will lose it)
// First player wins iff max over components of step(root) is ODD.
// The mirrored max/min assignment is WRONG; this orientation is the
    ↪ verified one.
void everySGStep(const vector<vector<int>>& succ, const vector<int>&
    ↪ sg, vector<int>& step) {
    int n = succ.size();
    step.assign(n, 0);
    for (int u = 0; u < n; u++) { // u in
    ↪ REVERSE topological order
        if (succ[u].empty()) { step[u] = 0; continue; }
        if (sg[u] > 0) {
            int b = 0;
            for (int v : succ[u]) if (!sg[v]) b = max(b, step[v] + 1);
            step[u] = b;
        } else {
            int b = INT_MAX;
            for (int v : succ[u]) b = min(b, step[v] + 1);
            step[u] = b;
        }
    }
}

/* ===== VERIFIED CLOSED FORMS AND FACTS =====
COIN-TURNING FAMILY (a position is a set of heads; the whole position
    ↪ = XOR of single-head
values). THE INDEXING IS LOAD-BEARING - mixing 0- and 1-indexed values
    ↪ is silently wrong.
Twins          turn exactly 2          0-idx  g(x) = x
TurningTurtles turn 1 or 2             1-idx  g(x) = x
MockTurtles    turn 1, 2 or 3          0-idx  g(x) = 2x or 2x+1,
    ↪ whichever is ODDIOUS
( = the x-th odious
    ↪ number: 1 2 4 7 8 11 13 )
Ruler          any CONSECUTIVE run     1-idx  g(x) = x & -x
Triplets       turn exactly 3          0-idx  g(x) = 0 for x < 2,
    ↪ MockTurtles(x-2) otherwise
Mogul          turn 1..5               NO closed form (1 2
    ↪ 4 8 16 31 32 64 103 ...)
TurningCorners turn 4 rectangle corners g(x, y) = x (*) y
    ↪ [nim product]
TARTAN THEOREM: for the 2-D product of two coin-turning games,
    ↪  $g(x, y) = g_1(x) (*) g_2(y)$ .

k-WYTHOFF (Fraenkel): take from one pile, or  $k_1, k_2 > 0$  from both with
    ↪  $|k_1 - k_2| < k$ .

```

P-positions: $a_n = \text{mex}\{a_i, b_i : i < n\}$, $b_n = a_n + k \cdot n$.
 Closed form: $(\text{floor}(n \cdot \alpha), \text{floor}(n \cdot \beta))$ with $\alpha = (2 - k + \sqrt{k \cdot k + 4}) / 2$,
 $\beta = \alpha + k$. $k = 1$ is ordinary Wythoff.

SILVER DOLLAR (coins at $p_1 < \dots < p_k$ on a strip, slide one left
 ↪ onto an empty square, no
 jumping): with gaps $g_i = p_i - p_{i-1} - 1$ and $p_0 = -1$, the GRUNDY
 ↪ VALUE is
 $g_k \text{ XOR } g_{k-2} \text{ XOR } g_{k-4} \text{ XOR } \dots$ - staircase nim on alternate
 ↪ gaps counted FROM THE RIGHT.

CHOMP: every $m \times n$ board is a FIRST-PLAYER WIN except 1×1 , by strategy
 ↪ stealing - and there is
 NO known constructive winning move. Two-row Chomp: P-positions are
 ↪ exactly $(b+1, b)$.
 Strategy stealing is a pure EXISTENCE proof; if the problem wants
 ↪ the move, it is useless.

BLUE-RED HACKENBUSH (partisan!): a string of edges from the ground has
 ↪ a NUMBER value, not a
 number. The first run of m equal colours contributes $+m$ (Blue
 ↪ positive), and every later edge
 contributes $+1/2, +1/4, \dots$ halving. This is the cleanest concrete
 ↪ model of surreal numbers.
 Partisan values in general: 0 = second player wins, $*$ = $\{0|0\}$ =
 ↪ first player wins,
 $up = \{0|*\}$ is positive but smaller than every positive number, a
 ↪ SWITCH $\{a|b\}$ with $a > b$ is
 "hot" (both players want to move there), and the TEMPERATURE is what
 ↪ it costs the opponent to
 let you move first. SIMPLICITY RULE: if some number lies strictly
 ↪ between the Left and Right
 options, the game IS the simplest such number. NUMBER AVOIDANCE:
 ↪ never move in a number
 component while a non-number component exists.

MISERE THEORY, honestly: misere nim's clean rule is an ACCIDENT. In
 ↪ general the misere outcome
 of a sum is NOT a function of the Grundy values; the correct
 ↪ invariant is the GENUS, and only
 "tame" games (genus matching nim's) obey nim-like rules. Anti-SG
 ↪ above is exactly the tame
 case, which is why it carries a precondition.

COMPLEXITY LANDSCAPE - when to stop looking for a formula:
 UNDIRECTED vertex geography is in P (the maximum-matching theorem in
 ↪ 03 - MinimaxAndMatching).
 DIRECTED vertex geography, Node Kayles, generalized geography and
 ↪ generalized Hex are all
 PSPACE-complete; generalized chess and Go are EXPTIME-complete. So a
 ↪ $1e5$ -sized instance of
 directed geography MUST have extra structure - find it instead of
 ↪ searching.

BOGUS NIM / REVERSIBLE MOVES: adding a move that the opponent can
 ↪ immediately undo does not
 change the Grundy value. That is precisely why the even indices in
 ↪ staircase nim are worth
 nothing, and it is the impartial case of the "bypass a reversible
 ↪ option" simplification. */

10.6 Hackenbush

BLUE-RED HACKENBUSH ON A TREE - the one PARTISAN game you can actually
 compute. The board is a

```

// tree rooted at the ground; every edge is Blue (only Left may delete
    ↪ it) or Red (only Right may
// delete it), and deleting an edge also deletes everything no longer
    ↪ connected to the root. The
// value of such a position is a DYADIC RATIONAL (a surreal NUMBER,
    ↪ never a number), and the usual
// number rules then decide everything: value > 0 means Left wins
    ↪ whoever moves, value < 0 means
// Right wins whoever moves, value == 0 means the SECOND player wins.
// THE RECURSION. For a stalk (a path from the ground) the value is
    ↪ easy and worth remembering:
// the first run of m equal colours is worth  $+m$ , and every edge after
    ↪ it is worth  $+1/2, +1/4,$ 
// halving each time. For a tree, evaluate each subtree, then push its
    ↪ value v through the edge
// that joins it to its parent: Blue edge  $\rightarrow (v + p) / 2^{(p-1)}$  with p
    ↪ the smallest positive
// integer making  $v + p > 1$ ; Red edge  $\rightarrow (v - p) / 2^{(p-1)}$  with p
    ↪ smallest making  $v - p < -1$ .
// Sum the results over the children. That is the "one edge above a
    ↪ game" simplification, and it

```

```
// is what makes the whole tree computable in near-linear time.
// WHEN: exactly the CodeChef GERALD08 shape - a tree of black/white
// → edges, two players delete
// their own colour, print who wins moving first and who wins moving
// → second. More generally, any
// partisan game whose components are trees of forced-order moves.
// COMPLEXITY:  $O(n \log^2 n)$  with small-to-large merging of the fraction
// → maps. The value needs
// Theta(n) bits in the worst case, which is why it is stored as an
// → integer part plus a map of
// set fraction bits rather than as a long double - a double loses the
// → answer at depth 53.
// DEGENERATE: a single vertex (no edges) has value 0, so the second
// → player wins. All-Blue trees
// have value equal to the number of edges' contribution and are a Left
// → win; the routine handles
// them with no special case. EXACT - all arithmetic is integer bit
// → shuffling.
struct Surreal { // value = val
    // → + sum of frac bits, in [0,1)
    int val = 0, w = 0; // absolute bit
    // → position = key + w
    map<int, int> frac; // only
    // → positions < 0 survive normalisation
    void clear() { frac.clear(), val = w = 0; }
    void add(int x) { // set bit at
        // → key x, then carry upward
        frac[x]++;
        auto it = frac.find(x);
        while (it != frac.end() && it->se != 1) {
            if (it->se > 1) frac[it->fi + 1] += it->se >> 1, it->se &=
        }
        // → 1;
        auto d = it++;
        if (!d->se) frac.erase(d);
    }
    void divide(int x) { // value /= 2^x
        for (int i = 0; i < x; i++) {
            if (!val) break;
            if (val & 1) add(i - w); // bit i of val
        }
        // → lands at position i - x
        val >>= 1;
        w -= x;
    }
    void operator+=(Surreal& o) {
        val += o.val;
        for (auto& [k, b] : o.frac) if (b) add(k + o.w - w);
        while (!frac.empty() && frac.rbegin()->fi + w >= 0) {
            // → // promote carries into val
            if (frac.rbegin()->se) val += 1 << (frac.rbegin()->fi + w);
            frac.erase(prev(frac.end()));
        }
    }
    void blueEdge() { // v -> (v + p)
        // → / 2^(p-1)
        int p = max(1, 1 - val);
        if (frac.empty() && p + val == 1) p++; // v + p > 1
        // → is STRICT
        val += p, divide(p - 1);
    }
    void redEdge() { // v -> (v - p)
        // → / 2^(p-1)
        int p = max(1, 1 + val);
        if (val - p == -1) p++;
        val -= p, divide(p - 1);
    }
    int sign() { return val > 0 ? 1 : val < 0 ? -1 : !frac.empty(); }
};
// g[u] = {v, colour}, colour 0 = Blue (Left / first player's edges), 1
// → = Red. Tree rooted at 0.
// Returns the sign of the value: +1 Left wins always, -1 Right wins
// → always, 0 second player wins.
int hackenbush(int n, const vector<vector<pii>>& g) {
    vector<Surreal> a(n);
    vector<int> id(n), vis(n, 0);
    iota(all(id), 0);
    function<void(int)> dfs = [&](int u) {
        vis[u] = 1;
        for (auto& [v, c] : g[u]) {
            if (vis[v]) continue;
            dfs(v);
            c ? a[id[v]].redEdge() : a[id[v]].blueEdge();
            if (sz(a[id[v]].frac) > sz(a[id[u]].frac)) swap(id[v],
        }
        // → id[u]); // small to large
        a[id[u]] += a[id[v]], a[id[v]].clear();
    }
}
```

```
};
dfs(0);
return a[id[0]].sign();
}
// READING THE ANSWER: sign > 0 -> Left wins moving first AND second;
// → sign < 0 -> Right wins both;
// sign == 0 -> whoever moves second wins. There is no "first player
// → wins whoever they are" case,
// because a Hackenbush tree value is always a number and numbers are
// → never fuzzy with 0.
// GREEN HACKENBUSH (every edge deletable by both) is impartial and is
// → a NIMBER instead: the
// colon principle, Grundy(u) = XOR over children of (Grundy(child) +
// → 1) - see 02 - GameAlgorithms.
// A tree with edges of all three colours has no closed form; you are
// → back to computing surreal
// values of general games, which is exponential.
// THE PARTISAN TOOLKIT this file is the concrete instance of (05 -
// → NimbersAndSums has the rest):
// 0 = second player wins, * = {0|0} = first player wins, up = {0|*}
// → beats every 0 but loses to
// every positive number, a SWITCH {a|b} with a > b is hot, TEMPERATURE
// → measures how much the move
// is worth, SIMPLICITY RULE picks the simplest number strictly between
// → the Left and Right options,
// and NUMBER AVOIDANCE says never move in a number component while a
// → non-number one exists.
```

11 Geometry

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11.1 Geometry 2D

11.1.1 Core

Point / vector core. $P < l l >$ = exact (use for all predicates), $P < l d >$ = real answers.

```
// cross(a,b) > 0 <=> b is counter-clockwise from a.
// a.cross(b,c) > 0 <=> c lies strictly LEFT of the directed line a->b.
// → [= orient(a,b,c)]
int sgn(ll x) { return (x > 0) - (x < 0); }
int sgn(ld x) { return (x > eps) - (x < -eps); }
int sgn(int x) { return (x > 0) - (x < 0); } // without these
// → two overloads a bare int or
int sgn(double x) { return sgn((ld)x); } // double argument
// → is AMBIGUOUS: hard error
ll gcdll(ll a, ll b) { return b ? gcdll(b, a % b) : a; } //
// → std::__gcd is libstdc++-only
```

```
template <class T>
struct P {
    typedef P Pt; T x = 0, y = 0;
    P(T x = 0, T y = 0) : x(x), y(y) {}
    Pt operator+(Pt p) const { return {x + p.x, y + p.y}; }
    Pt operator-(Pt p) const { return {x - p.x, y - p.y}; }
    Pt operator*(T d) const { return {x * d, y * d}; }
    Pt operator/(T d) const { return {x / d, y / d}; }
    T dot(Pt p) const { return x * p.x + y * p.y; }
    T cross(Pt p) const { return x * p.y - y * p.x; }
    T cross(Pt a, Pt b) const { return (a - *this).cross(b - *this); }
    // → // orient
    T dist2() const { return x * x + y * y; }
    ld dist() const { return sqrtl((ld)dist2()); }
    ld angle() const { return atan2l(y, x); } // (-pi, pi]
    Pt perp() const { return {-y, x}; } // rotate +90
    // → deg
    Pt unit() const { return *this / dist(); }
    Pt rot(ld a) const { return {x * cosl(a) - y * sinl(a), x * sinl(a)
    // → + y * cosl(a)}; }
    bool operator<(Pt p) const { return tie(x, y) < tie(p.x, p.y); }
    bool operator==(Pt p) const { return tie(x, y) == tie(p.x, p.y); }
    friend istream& operator>>(istream& i, Pt& p) { return i >> p.x >>
    // → p.y; }
    friend ostream& operator<<(ostream& o, Pt p) { return o << p.x << '
    // → ' << p.y; }
};
typedef P<ll> Pi;
typedef P<ld> Pd;
```

```
// Signed angle from a to b in (-pi, pi]. Sign tells you the rotation
// → direction.
template <class T> ld angleTo(P<T> a, P<T> b) { return
    // → atan2l(a.cross(b), a.dot(b)); }
// Angle a-0-b in [0, 2pi).
template <class T> ld angleAt(P<T> a, P<T> 0, P<T> b) {
    ld r = angleTo(a - 0, b - 0);
    return r < 0 ? r + 2 * PI : r;
```

```

}

// EXACT polar sort, no atan2. Orders by angle in [0, 2pi) starting at
// ↪ +x axis.
// sort(all(v), angCmp<ll>); ties (same direction) keep input order.
template <class T> bool half(P<T> p) { return p.y < 0 || (p.y == 0 &&
    ↪ p.x < 0); }
template <class T> bool angCmp(P<T> a, P<T> b) {
    return half(a) != half(b) ? half(a) < half(b) : a.cross(b) > 0;
}

// OVERFLOW: |coord| <= 1e9 => cross fits in ll (max ~4e18, ll max
// ↪ 9.2e18).
// dist2 of a difference fits too. Anything squared twice
// ↪ needs __int128.

```

11.1.2 Lines

Lines, segments, rays. "line ab" = infinite line through a and b.

```

// Predicates are exact on P<ll>; anything returning a point needs
// ↪ P<ld>.

template <class P> ld lineDist(P p, P a, P b) { // distance
    ↪ point -> line
    return fabsl((ld)(b - a).cross(p - a)) / (b - a).dist();
}
template <class P> P proj(P p, P a, P b) { // foot of
    ↪ perpendicular
    return a + (b - a) * ((ld)(p - a).dot(b - a) / (b - a).dist2());
}
template <class P> P refl(P p, P a, P b) { return proj(p, a, b) * 2 -
    ↪ p; }

template <class P> bool onSeg(P p, P a, P b) { // inclusive
    ↪ of endpoints
    return a.cross(b, p) == 0 && (a - p).dot(b - p) <= 0;
}
template <class P> ld segDist(P p, P a, P b) { // distance
    ↪ point -> segment (exact-safe)
    if (a == b) return (p - a).dist();
    ld d = (ld)(b - a).dist2(), t = (ld)(p - a).dot(b - a);
    if (t <= 0) return (p - a).dist();
    if (t >= d) return (p - b).dist();
    return fabsl((ld)(b - a).cross(p - a)) / sqrtl(d);
}
template <class P> P closestOnSeg(P p, P a, P b) {
    if ((p - a).dot(b - a) <= 0) return a;
    if ((p - b).dot(b - a) >= 0) return b;
    return proj(p, a, b);
}

// LINE x LINE. first = 1 unique, 0 parallel-disjoint, -1 same line.
template <class P> pair<int, P> lineInter(P a, P b, P c, P d) {
    auto x = (b - a).cross(d - c);
    if (sgn(x) == 0) return {-(a.cross(b, c) == 0), P()};
    auto p = c.cross(b, d), q = c.cross(d, a); // p + q == x
    ↪ (affine weights)
    return {1, (a * p + b * q) / x}; // P must be
    ↪ floating
}

// SEGMENT x SEGMENT. Returns 0, 1, or 2 points (2 = collinear overlap
// ↪ endpoints).
// Exact on P<ll> when the answer has 0 or 2 points. The 1-point case
// ↪ DIVIDES and the numerator
// is ~2*C^3, so it is only exact for |coordinate| <= ~1.6e6. Above
// ↪ that use segCross for the
// boolean, or Frac (01 - Miscellaneous/13) for an exact rational point.
template <class P> vector<P> segInter(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b), oc = a.cross(b, c), od
    ↪ = a.cross(b, d);
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0)
        return {(a * ob - b * oa) / (ob - oa)};
    set<P> s;
    if (onSeg(c, a, b)) s.insert(c);
    if (onSeg(d, a, b)) s.insert(d);
    if (onSeg(a, c, d)) s.insert(a);
    if (onSeg(b, c, d)) s.insert(b);
    return {all(s)};
}

// Just "do they touch?" - fully exact, NO division and no allocation,
// ↪ so it is both safe at
// |c| = 1e9 and ~30x faster than asking segInter. Use this whenever
// ↪ you only need a bool.
template <class P> bool segCross(P a, P b, P c, P d) {
    auto oa = c.cross(d, a), ob = c.cross(d, b), oc = a.cross(b, c), od
    ↪ = a.cross(b, d);
    if (sgn(oa) * sgn(ob) < 0 && sgn(oc) * sgn(od) < 0) return true;
}

```

```

    return onSeg(c, a, b) || onSeg(d, a, b) || onSeg(a, c, d) ||
    ↪ onSeg(b, c, d);
}

template <class P> ld segSegDist(P a, P b, P c, P d) {
    if (segCross(a, b, c, d)) return 0;
    return min({segDist(a, c, d), segDist(b, c, d), segDist(c, a, b),
    ↪ segDist(d, a, b)});
}

// RAY from s through e.
template <class P> bool onRay(P p, P s, P e) {
    return s.cross(e, p) == 0 && (p - s).dot(e - s) >= 0;
}
template <class P> ld rayDist(P p, P s, P e) {
    return (p - s).dot(e - s) <= 0 ? (p - s).dist() : lineDist(p, s, e);
}

// Perpendicular bisector of ab, returned as two points on it.
// FLOATING P ONLY: (a+b)/2 truncates on P<ll> and returns a DIFFERENT
// ↪ line (for a=(0,0),
// b=(1,1) it gives y = -x instead of y = -x + 1). For an exact integer
// ↪ version work on the
// doubled lattice: the line through (a+b) and (a+b) + (b-a).perp().
template <class P> pair<P, P> perpBisector(P a, P b) {
    P m = (a + b) / 2;
    return {m, m + (b - a).perp()};
}
// Interior angle bisector direction at B for angle A-B-C (unit vectors
// ↪ summed).
template <class P> P bisectorDir(P a, P b, P c) { return (a - b).unit()
    ↪ + (c - b).unit(); }

// Line as a*x + b*y = c -> two points on it (needs floating P).
template <class P> pair<P, P> fromABC(ld A, ld B, ld C) {
    P n(A, B), p = n * (C / n.dist2());
    return {p, p + n.perp()};
}

```

11.1.3 Polygon

Polygons. Vertices in order (CW or CCW); convex routines want CCW, no collinear points.

```

template <class P> auto area2(const vector<P>& v) { // SIGNED
    ↪ 2*area. >0 <=> CCW.
    decltype(v[0].cross(v[0])) a = 0; // exact on
    ↪ P<ll>; safe on empty
    for (int i = 0, n = v.size(); i < n; i++) a += v[i].cross(v[(i + 1) %
    ↪ n]);
    return a;
}
template <class P> ld area(const vector<P>& v) { return
    ↪ fabsl((ld)area2(v)) / 2; }
template <class P> ld perimeter(const vector<P>& v) {
    ld r = 0;
    for (int i = 0, n = v.size(); i < n; i++) r += (v[i] - v[(i + 1) %
    ↪ n]).dist();
    return r;
}
template <class P> P centroid(const vector<P>& v) { // area
    ↪ centroid (needs floating P)
    P r; ld A = 0;
    for (int i = 0, j = v.size() - 1, n = v.size(); i < n; j = i++) {
        auto c = v[j].cross(v[i]);
        r = r + (v[i] + v[j]) * c; A += c;
    }
    return r / A / 3;
}

// A completely collinear (zero-area) polygon reports TRUE here - guard
// ↪ with area2 != 0 if that
// matters. Bow-ties and spikes are correctly rejected.
template <class P> bool isConvex(const vector<P>& v) {
    int n = v.size(), pos = 0, neg = 0;
    for (int i = 0; i < n; i++) {
        int s = sgn(v[i].cross(v[(i + 1) % n], v[(i + 2) % n]));
        s > 0 ? pos++ : s < 0 ? neg++ : 0;
    }
    return !pos || !neg;
}

// Point in ANY simple polygon, O(n), exact. strict=true -> boundary
// ↪ counts as outside.
template <class P> bool inPoly(const vector<P>& v, P a, bool strict =
    ↪ true) {
    int cnt = 0, n = v.size();
    for (int i = 0; i < n; i++) {
        P q = v[(i + 1) % n];
        if (onSeg(a, v[i], q)) return !strict;
    }
}

```

```

    cnt ^= ((a.y < v[i].y) - (a.y < q.y)) * a.cross(v[i], q) > 0;
}
return cnt;
}

// Convex hull, CCW, strictly convex (collinear points dropped). O(n
↳ log n).
// Change <= 0 to < 0 in the pop test to KEEP collinear points on the
↳ hull.
template <class P> vector<P> hull(vector<P> p) {
    if (p.size() <= 1) return p;
    sort(all(p));
    vector<P> h(p.size() + 1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(p)))
        for (P q : p) {
            while (t >= s + 2 && h[t - 2].cross(h[t - 1], q) <= 0) t--;
            h[t++] = q;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}

// Point in CONVEX polygon (CCW, strict), O(log n). 1 inside, 0 on
↳ boundary, -1 outside.
// h must be a strictly convex CCW hull with n >= 1. Returns 1 inside,
↳ 0 on the boundary,
// -1 outside.
template <class P> int inConvex(const vector<P>& h, P p) {
    if (h.empty()) return -1;
    int n = h.size();
    if (n == 1) return p == h[0] ? 0 : -1;
    if (n == 2) return onSeg(p, h[0], h[1]) ? 0 : -1;
    if (h[0].cross(h[1], p) < 0 || h[0].cross(h[n - 1], p) > 0) return
↳ -1;
    int l = 1, r = n - 1;
    while (r - l > 1) { int m = (l + r) / 2; (h[0].cross(h[m], p) >= 0
↳ ? l : r) = m; }
    int s = sgn(h[l].cross(h[l + 1], p));
    if (s < 0) return -1;
    if (s == 0) return 0;
    return (h[0].cross(h[1], p) == 0 || h[0].cross(h[n - 1], p) == 0) ?
↳ 0 : 1;
}

// Index of the hull vertex furthest in direction 'dir', O(log n). CCW
↳ hull, no collinear.
// Use to get tangents from a far-away direction, or max/min dot
↳ product.
#define _cmp(i, j) sgn(dir.perp().cross(h[(i) % n] - h[(j) % n]))
#define _ext(i) (_cmp(i + 1, i) >= 0 && _cmp(i, i - 1 + n) < 0)
template <class P> int extreme(const vector<P>& h, P dir) {
    int n = h.size(), lo = 0, hi = n;
    if (_ext(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (_ext(m)) return m;
        int ls = _cmp(lo + 1, lo), ms = _cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == _cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}

// Cut polygon by the line s->e, KEEP the part on the LEFT (boundary
↳ included). O(n).
// Needs floating P. Boundary-inclusive matters: with a strict test,
↳ vertices lying exactly on
// the line are dropped from BOTH halves and you lose area. Repeat over
↳ many lines = half-plane
// intersection. cut(v,e,s) gives the other half; the two areas sum to
↳ area(v).
// Keeps the part LEFT of the directed line s->e, boundary INCLUSIVE.
↳ Vertices exactly on the
// line are kept, so a line through a vertex produces a DUPLICATE point
↳ - fine for area, but
// dedup before feeding the result to hull / isConvex / minkowski /
↳ diameter2.
template <class P> vector<P> cut(const vector<P>& v, P s, P e) {
    vector<P> r;
    for (int i = 0, n = v.size(); i < n; i++) {
        P cur = v[i], prv = v[(i + n - 1) % n];
        bool side = sgn(s.cross(e, cur)) >= 0;
        if (side != (sgn(s.cross(e, prv)) >= 0))
↳ r.push_back(lineInter(s, e, cur, prv).second);
        if (side) r.push_back(cur);
    }
    return r;
}

```

```

// Minkowski sum of two CONVEX CCW polygons. O(n+m). Result is convex
↳ CCW.
// BOTH polygons must be CONVEX and COUNTER-CLOCKWISE. On clockwise
↳ input the merge below makes
// no progress and loops forever (an unexplained TLE/MLE, not a wrong
↳ answer) - so normalise.
template <class P> vector<P> minkowski(vector<P> A, vector<P> B) {
    if (area2(A) < 0) reverse(all(A));
    if (area2(B) < 0) reverse(all(B));
    auto low = [](vector<P>& v) {
        auto lo = [](P a, P b) { return tie(a.y, a.x) < tie(b.y, b.x);
↳ };
        rotate(v.begin(), min_element(all(v), lo), v.end());
    };
    low(A); low(B);
    A.push_back(A[0]); A.push_back(A[1]);
    B.push_back(B[0]); B.push_back(B[1]);
    vector<P> r; size_t i = 0, j = 0;
    while (i + 2 < A.size() || j + 2 < B.size()) {
        r.push_back(A[i] + B[j]);
        auto c = (A[i + 1] - A[i]).cross(B[j + 1] - B[j]);
        if (c >= 0 && i + 2 < A.size()) i++;
        if (c <= 0 && j + 2 < B.size()) j++;
    }
    return r;
}

// Rotating calipers. Hull must be CCW strictly convex.
template <class P> auto diameter2(const vector<P>& h) { // max
↳ squared distance
    int n = h.size(), j = n < 2 ? 0 : 1;
    decltype(h[0].dist2()) r = 0;
    for (int i = 0; i < j; i++)
        for (; j = (j + 1) % n; ) {
            r = max(r, (h[i] - h[j]).dist2());
            if ((h[(j + 1) % n] - h[j]).cross(h[i + 1] - h[i]) >= 0)
↳ break;
        }
    return r;
}

template <class P> ld width(const vector<P>& h) { // min
↳ distance between parallel supports
    int n = h.size(); if (n <= 2) return 0;
    ld r = 1e18; int j = 1;
    for (int i = 0; i < n; i++) {
        while ((h[(i + 1) % n] - h[i]).cross(h[(j + 1) % n] - h[j]) >=
↳ 0) j = (j + 1) % n;
        r = min(r, lineDist(h[j], h[i], h[(i + 1) % n]));
    }
    return r;
}

// Minimum-area enclosing rectangle. One side always lies on a hull
↳ edge (that's the theorem),
// so try every edge. O(n^2) on the HULL only - fine, and immune to
↳ caliper tie bugs.
template <class P> ld minRectArea(const vector<P>& h) {
    int n = h.size(); if (n < 3) return 0;
    ld best = 1e18;
    for (int i = 0; i < n; i++) {
        P d = h[(i + 1) % n] - h[i];
        ld L = d.dist(), lo = 1e18, hi = -1e18, ht = 0;
        for (P q : h) {
            ld u = (ld)d.dot(q - h[i]) / L, v = (ld)d.cross(q - h[i]) /
↳ L;
            lo = min(lo, u); hi = max(hi, u); ht = max(ht, fabs(v));
        }
        best = min(best, (hi - lo) * ht);
    }
    return best;
}

// PICK'S THEOREM. Lattice polygon: A = I + B/2 - 1.
template <class P> ll boundaryPts(const vector<P>& v) { // B
    ll b = 0;
    for (int i = 0, n = v.size(); i < n; i++) {
        P d = v[(i + 1) % n] - v[i];
        b += gcdll(llabs(d.x), llabs(d.y));
    }
    return b;
}

// PICK'S THEOREM, and it needs a SIMPLE polygon with integer vertices.
↳ On a self-intersecting
// or degenerate (zero-area) polygon the result is meaningless, not
↳ merely approximate.
template <class P> ll interiorPts(const vector<P>& v) { // I = A -
↳ B/2 + 1

```

```
    return (llabs(area2(v)) - boundaryPts(v)) / 2 + 1;
}
```

11.1.4 Circles

Circles. All of these want a floating P (Pd).

```
// --- circle x line: 0, 1 (tangent) or 2 points on segment/line ab ---
template <class P> vector<P> circleLine(P o, ld r, P a, P b) {
    P p = proj(o, a, b);
    ld h2 = r * r - (p - o).dist2();
    if (h2 < -eps) return {};
    if (h2 < eps) return {p};
    P h = (b - a).unit() * sqrt(h2);
    return {p - h, p + h};
}

template <class P> vector<P> circleSeg(P o, ld r, P a, P b) {
    vector<P> out;
    for (P p : circleLine(o, r, a, b)) if (segDist(p, a, b) < eps)
        out.push_back(p);
    return out;
}

// --- circle x circle: the 0/1/2 intersection points ---
template <class P> vector<P> circleCircle(P a, ld r1, P b, ld r2) {
    P d = b - a; ld d2 = d.dist2();
    if (d2 < eps) return {}; // concentric
    (equal => infinite)
    ld p = (d2 + r1 * r1 - r2 * r2) / (2 * d2), h2 = r1 * r1 - p * p *
    d2;
    if (r1 + r2 < sqrt(d2) - eps || fabs(d2 - r1 * r1) > sqrt(d2) + eps)
        return {};
    P mid = a + d * p, per = d.perp() * sqrt(max((ld)0, h2) / d2);
    if (h2 < eps) return {mid};
    return {mid + per, mid - per};
}

// --- AREA of the lens (intersection region) of two circles ---
template <class P> ld circleInterArea(P a, ld r1, P b, ld r2) {
    ld d = (a - b).dist();
    if (d >= r1 + r2) return 0;
    if (d + min(r1, r2) <= max(r1, r2)) return PI * min(r1, r2) *
    min(r1, r2);
    ld A = acos((d * d + r1 * r1 - r2 * r2) / (2 * d * r1));
    ld B = acos((d * d + r2 * r2 - r1 * r1) / (2 * d * r2));
    return r1 * r1 * (A - sinl(2 * A) / 2) + r2 * r2 * (B - sinl(2 * B)
    / 2);
}

// --- TANGENTS. Each result is a pair (touch point on c1, touch point
    on c2).
// inner=false -> the 2 external tangents; inner=true -> the 2 internal
    (crossing) ones.
// Tangent lines from a POINT p to circle (c,r): tangents(p, 0, c, r,
    false).
template <class P> vector<pair<P, P>> tangents(P c1, ld r1, P c2, ld
    r2, bool inner) {
    if (inner) r2 = -r2;
    P d = c2 - c1;
    ld dr = r1 - r2, d2 = d.dist2(), h2 = d2 - dr * dr;
    if (d2 < eps || h2 < -eps) return {};
    vector<pair<P, P>> out;
    for (ld s : {-1.0L, 1.0L}) {
        P v = (d * dr + d.perp() * sqrt(max((ld)0, h2)) * s) / d2;
        out.push_back({c1 + v * r1, c2 + v * r2});
    }
    if (h2 < eps) out.pop_back();
    return out;
}

// --- circumcircle / incircle of a triangle ---
template <class P> P circumcenter(P a, P b, P c) {
    P B = c - a, C = b - a;
    return a + (B * C.dist2() - C * B.dist2()).perp() / B.cross(C) / 2;
}

template <class P> ld circumradius(P a, P b, P c) {
    return (b - a).dist() * (c - b).dist() * (a - c).dist() / (2 *
    fabs((ld)(b - a).cross(c - a)));
}

template <class P> P incenter(P a, P b, P c) {
    ld A = (b - c).dist(), B = (c - a).dist(), C = (a - b).dist();
    return (a * A + b * B + c * C) / (A + B + C);
}

template <class P> ld inradius(P a, P b, P c) {
    ld s = ((b - a).dist() + (c - b).dist() + (a - c).dist()) / 2;
    return fabs((ld)(b - a).cross(c - a)) / 2 / s;
}

template <class P> P orthocenter(P a, P b, P c) { return a + b + c -
```

```
    circumcenter(a, b, c) * 2; }
```

```
// --- MINIMUM ENCLOSING CIRCLE, expected O(n) (Welzl, incremental) ---
template <class P> pair<P, ld> mec(vector<P> p) {
    if (p.empty()) return {P(), 0};
    shuffle(all(p),
    mt19937(chrono::steady_clock::now().time_since_epoch().count()));
    P o = p[0]; ld r = 0, E = 1 + 1e-9;
    for (int i = 0; i < (int)p.size(); i++) if ((o - p[i]).dist() > r *
    E) {
        o = p[i], r = 0;
        for (int j = 0; j < i; j++) if ((o - p[j]).dist() > r * E) {
            o = (p[i] + p[j]) / 2, r = (o - p[i]).dist();
            for (int k = 0; k < j; k++) if ((o - p[k]).dist() > r * E)
                o = circumcenter(p[i], p[j], p[k]), r = (o -
                p[i]).dist();
        }
        return {o, r};
    }
}
```

```
// --- AREA of (circle centered at c, radius r) INTERSECT polygon ps ---
template <class P> ld circlePolyArea(P c, ld r, vector<P> ps) {
    auto arg = [] (P p, P q) { return atan2l(p.cross(q), p.dot(q)); };
    auto tri = [&] (P p, P q) -> ld {
        ld r2 = r * r / 2;
        P d = q - p;
        ld a = d.dot(p) / d.dist2(), b = (p.dist2() - r * r) /
        d.dist2(), det = a * a - b;
        if (det <= 0) return arg(p, q) * r2;
        ld s = max((ld)0, -a - sqrt(det)), t = min((ld)1, -a +
        sqrt(det));
        if (t < 0 || 1 <= s) return arg(p, q) * r2;
        P u = p + d * s, v = p + d * t;
        return arg(p, u) * r2 + u.cross(v) / 2 + arg(v, q) * r2;
    };
    ld sum = 0;
    for (int i = 0, n = ps.size(); i < n; i++) sum += tri(ps[i] - c,
    ps[(i + 1) % n] - c);
    return fabs(sum);
}
```

```
// --- power of a point / radical axis ---
// pow(p) = |p-c|^2 - r^2 : <0 inside, 0 on circle, >0 outside; equals
    (tangent length)^2.
template <class P> ld power(P p, P c, ld r) { return (p - c).dist2() -
    r * r; }
```

```
// Radical axis of two non-concentric circles, as two points on it.
template <class P> pair<P, P> radicalAxis(P a, ld r1, P b, ld r2) {
    P d = b - a; ld t = ((r1 * r1 - r2 * r2) / d.dist2() + 1) / 2;
    P m = a + d * t;
    return {m, m + d.perp()};
}
```

11.1.5 Sweep

Global / sweep-line geometry: problems about a whole SET of objects.

```
// --- CLOSEST PAIR of points, O(n log n). Returns the two points.
    Exact on P<ll>. ---
template <class P> pair<P, P> closestPair(vector<P> v) {
    assert(v.size() > 1); // n < 2 has
    no pair to return
    set<P> S; // ordered by
    (x, y)
    sort(all(v), [](P a, P b) { return a.y < b.y; });
    pair<ll, pair<P, P>> best{LLONG_MAX, {P(), P()}};
    int j = 0;
    for (P p : v) {
        P d(1 + (ll)sqrt((ld)best.first), 0);
        while (v[j].y <= p.y - d.x) S.erase(v[j++]);
        auto lo = S.lower_bound(p - d), hi = S.upper_bound(p + d);
        for (; lo != hi; ++lo) best = min(best, {(lo - p).dist2(),
        (*lo, p)});
        S.insert(p);
    }
    return best.second;
}

// --- AREA OF UNION OF AXIS-ALIGNED RECTANGLES, O(n log n). ---
// rects: {x1, y1, x2, y2} half-open [x1,x2) x [y1,y2). Sweep x,
    segment tree counts covered y.
struct CoverTree { // "count">0
    length" segment tree
    int n; vector<int> cnt; vector<ll> len; vector<ll> ys;
    CoverTree(vector<ll> y) : ys(y) {
        n = ys.size() - 1, cnt.assign(4 * n, 0), len.assign(4 * n, 0);
    }
}
```

```

void upd(int v, int l, int r, int ql, int qr, int d) {
    if (qr <= l || r <= ql) return;
    if (ql <= l && r <= qr) cnt[v] += d;
    else {
        int m = (l + r) / 2;
        upd(2 * v, l, m, ql, qr, d); upd(2 * v + 1, m, r, ql, qr,
        ↪ d);
        len[v] = cnt[v] ? ys[r] - ys[l] : (r - l == 1 ? 0 : len[2 * v]
        ↪ + len[2 * v + 1]);
    }
    void upd(int l, int r, int d) { upd(1, 0, n, l, r, d); }
    ll get() { return len[1]; }
};
ll rectUnionArea(vector<array<ll, 4>> r) {
    vector<ll> ys;
    for (auto& q : r) ys.push_back(q[1]), ys.push_back(q[3]);
    sort(all(ys)); ys.erase(unique(all(ys)), ys.end());
    if (ys.size() < 2) return 0;
    auto id = [&](ll y) { return (int)(lower_bound(all(ys), y) -
    ↪ ys.begin()); };
    vector<array<ll, 4>> ev; // {x, +-1,
    ↪ y1, y2}
    for (auto& q : r) ev.push_back({q[0], 1, q[1], q[3]}),
    ↪ ev.push_back({q[2], -1, q[1], q[3]});
    sort(all(ev));
    CoverTree T(ys);
    ll area = 0, px = ev.empty() ? 0 : ev[0][0];
    for (auto& e : ev) {
        area += T.get(e) * (e[0] - px); px = e[0];
        T.upd(id(e[2]), id(e[3]), (int)e[1]);
    }
    return area;
}
// PERIMETER of the union: run the same sweep twice (once per axis)
↪ accumulating
// |len(after) - len(before)| at each event x, and add the vertical
↪ contributions.

// --- MAX POINTS ON A LINE, O(n^2 log n) ---
template <class P> int maxOnLine(vector<P> p) {
    int n = p.size(), best = min(n, 1);
    for (int i = 0; i < n; i++) {
        map<pair<ll, ll>, int> c; int dup = 0;
        for (int j = 0; j < n; j++) if (j != i) {
            ll dx = p[j].x - p[i].x, dy = p[j].y - p[i].y;
            if (!dx && !dy) { dup++; continue; } //
            ↪ duplicate points
            ll g = gcdll(llabs(dx), llabs(dy)); dx /= g; dy /= g;
            if (dx < 0 || (dx == 0 && dy < 0)) dx = -dx, dy = -dy;
            c[{dx, dy}]++;
        }
        best = max(best, dup + 1);
        for (auto& [k, v] : c) best = max(best, v + dup + 1);
    }
    return best;
}

// --- COUNT PAIRS OF INTERSECTING SEGMENTS, O(n^2) exact. ---
// For "do ANY two of them cross", use Shamos-Hoej (17), which is O(n
↪ log n) and returns the pair.
// For O((n+k) log n) reporting of ALL of them use Bentley-Ottmann;
↪ almost never needed.
template <class P> ll countSegInter(vector<pair<P, P>> s) {
    ll c = 0;
    for (int i = 0; i < (int)s.size(); i++)
        for (int j = i + 1; j < (int)s.size(); j++)
            c += segCross(s[i].first, s[i].second, s[j].first,
            ↪ s[j].second);
    return c;
}

// --- CONVEX LAYERS (onion peeling): repeatedly strip the hull. O(n^2
↪ log n) naive. ---
// Fine to n ~ 3000; past that use the decremental hull in 22 -
↪ OnionDecomposition.cpp,
// O(n log^2 n). Note this version DROPS points lying inside a hull
↪ edge, 22 keeps them.
template <class P> vector<vector<P>> onion(vector<P> p) {
    vector<vector<P>> L;
    while (!p.empty()) {
        auto h = hull(p);
        L.push_back(h);
        for (P q : h) p.erase(find(all(p), q));
    }
    return L;
}

```

11.1.6 HalfPlaneIntersection

HALF-PLANE INTERSECTION - O(n log n). A half-plane is "everything LEFT of the directed

```

// line a -> b". Returns the CCW vertices of the intersection, or {} if
↪ it is empty.
// A big bounding box is added, so the result is always bounded.
// 2D LINEAR PROGRAMMING is exactly this: constraints -> half-planes,
↪ then optimise over the
// resulting convex polygon (ternary search on the boundary, or check
↪ all vertices).
struct HP {
    Pd p, d; // keep the side LEFT
    ↪ of p -> p + d
    HP() {}
    HP(Pd a, Pd b) : p(a), d(b - a) {}
    bool out(Pd r) const { return d.cross(r - p) < -eps; }
};
Pd hpInter(const HP& a, const HP& b) { // assumes a.d and b.d
    ↪ are not parallel
    return a.p + a.d * ((b.p - a.p).cross(b.d) / a.d.cross(b.d));
}
vector<Pd> halfPlaneInter(vector<HP> h, ld BOX = 1e9) {
    Pd box[4] = {Pd(-BOX, -BOX), Pd(BOX, -BOX), Pd(BOX, BOX), Pd(-BOX,
    ↪ BOX)};
    for (int i = 0; i < 4; i++) h.push_back(HP(box[i], box[(i + 1) %
    ↪ 4]));
    sort(all(h), [](const HP& a, const HP& b) {
        if (half(a.d) != half(b.d)) return half(a.d) < half(b.d);
        ld c = a.d.cross(b.d);
        if (fabs(c) > eps) return c > 0;
        return a.out(b.p); // same direction: most
        ↪ restrictive first
    });
    deque<HP> dq;
    for (auto& x : h) {
        if (!dq.empty() && fabs(dq.back().d.cross(x.d)) <= eps &&
            dq.back().d.dot(x.d) > 0) continue;
        while (dq.size() > 1 && x.out(hpInter(dq[dq.size() - 1],
        ↪ dq[dq.size() - 2]))) dq.pop_back();
        while (dq.size() > 1 && x.out(hpInter(dq[0], dq[1])))
        ↪ dq.pop_front();
        dq.push_back(x);
    }
    while (dq.size() > 2 && dq[0].out(hpInter(dq[dq.size() - 1],
    ↪ dq[dq.size() - 2]))) dq.pop_back();
    while (dq.size() > 2 && dq.back().out(hpInter(dq[0], dq[1])))
    ↪ dq.pop_front();
    if (dq.size() < 3) return {};
    vector<Pd> r;
    for (size_t i = 0; i < dq.size(); i++) r.push_back(hpInter(dq[i],
    ↪ dq[(i + 1) % dq.size()]));
    return r;
}
// If the constraints arrive ONLINE, or you must report after each one,
↪ use 21 - DynamicHalfPlane.
// If you only have a handful of half-planes, repeated cut() (03 -
↪ Polygon.cpp) on a big box is
// shorter to retype and just as correct - O(n^2) but n is small.

```

11.1.7 Unions

AREA OF UNIONS. The three shapes that actually show up.

```

// --- UNION OF POLYGONS (any simple polygons, may overlap), O(n^2 log
↪ n). ---
// Works by, for every edge, measuring how much of it is "on the
↪ boundary of the union" and
// accumulating the shoelace contribution with the right multiplicity.
ld edgeRatio(Pd a, Pd b) { return sgn(b.x) ? a.x / b.x : a.y / b.y; }
ld polyUnion(vector<vector<Pd>>& ps) { // every
    ↪ polygon must be CCW
    ld ret = 0;
    for (size_t i = 0; i < ps.size(); i++)
        for (size_t v = 0; v < ps[i].size(); v++) {
            Pd A = ps[i][v], B = ps[i][(v + 1) % ps[i].size()];
            vector<pair<ld, int>> seg = {{0, 0}, {1, 0}};
            for (size_t j = 0; j < ps.size(); j++) {
                if (i == j) continue;
                for (size_t u = 0; u < ps[j].size(); u++) {
                    Pd C = ps[j][u], D = ps[j][(u + 1) % ps[j].size()];
                    int sc = sgn(A.cross(B, C)), sd = sgn(A.cross(B,
                    ↪ D));
                    if (sc != sd) {
                        ld sa = C.cross(D, A), sb = C.cross(D, B);
                        if (min(sc, sd) < 0) seg.push_back({sa / (sa -
                        ↪ sb), sc > sd ? 1 : -1});
                        } else if (!sc && !sd && j < i && sgn((B - A).dot(D

```

```

    <- C)) > 0) {
        seg.push_back({edgeRatio(C - A, B - A), 1});
        seg.push_back({edgeRatio(D - A, B - A), -1});
    }
    sort(all(seg));
    for (auto& s : seg) s.first = min(max(s.first, (ld)0),
    <- (ld)1);
    ld sum = 0;
    int cnt = seg[0].second;
    for (size_t j = 1; j < seg.size(); j++) { // fraction
    <- of AB on the union boundary
        if (!cnt) sum += seg[j].first - seg[j - 1].first;
        cnt += seg[j].second;
    }
    ret += A.cross(B) * sum;
    }
    return ret / 2;
}

// --- UNION OF CIRCLES, O(n^2 log n) by integrating each arc that is
    <- on the outer boundary. ---
ld circleUnion(vector<Pd> c, vector<ld> r) {
    int n = c.size();
    ld total = 0;
    for (int i = 0; i < n; i++) {
        vector<pair<ld, int>> ev;
        int cover = 0;
        for (int j = 0; j < n; j++) {
            if (i == j) continue;
            ld d = (c[i] - c[j]).dist();
            if (d + r[i] <= r[j] + eps &&
            <- (r[i] < r[j] || (fabsl(r[i] - r[j]) < eps && j < i))) {
            <- cover = -1; break; }
            if (d >= r[i] + r[j] - eps || d + r[j] <= r[i] + eps)
            <- continue;
            ld a = (c[j] - c[i]).angle();
            ld w = acosl((d * d + r[i] * r[i] - r[j] * r[j]) / (2 * d *
            <- r[i]));
            ld L = a - w, R = a + w;
            if (L < -PI) L += 2 * PI;
            if (R > PI) R -= 2 * PI;
            if (L > R) ev.push_back({L, 1}), ev.push_back({PI, -1}),
            ev.push_back({-PI, 1}), ev.push_back({R, -1});
            else ev.push_back({L, 1}), ev.push_back({R, -1});
        }
        if (cover < 0) continue;
        ev.push_back({-PI, 0}), ev.push_back({PI, 0});
        sort(all(ev));
        int cnt = 0; ld last = -PI;
        for (auto [a, t] : ev) {
            if (!cnt && a > last) { // arc
            <- [last, a] is on the boundary
                Pd p1 = c[i] + Pd(cosl(last), sinl(last)) * r[i];
                Pd p2 = c[i] + Pd(cosl(a), sinl(a)) * r[i];
                total += p1.cross(p2) / 2 + r[i] * r[i] * ((a - last) -
            <- sinl(a - last)) / 2;
            }
            cnt += t;
            last = max(last, a);
        }
    }
    return total;
}

// --- PERIMETER OF A UNION OF AXIS-ALIGNED RECTANGLES ---
// Run the rectUnionArea sweep TWICE (once per axis). At each event x,
    <- the vertical contribution
// is |coveredLen(after) - coveredLen(before)|; sum those, then repeat
    <- with x/y swapped.

```

11.1.8 Transforms

COORDINATE TRANSFORMS - the tricks that turn a hard metric into an easy one.

```

// CHEBYSHEV <-> MANHATTAN. rot45(p) = (x+y, x-y).
// Manhattan distance in the original = Chebyshev distance after
    <- rot45 (and vice versa).
// Use it for: "max Manhattan distance over a set" (becomes max over 4
    <- coordinate extremes),
// "count pairs within Manhattan distance k", chessboard-king problems.
template <class P> P rot45(P p) { return P(p.x + p.y, p.x - p.y); }
template <class P> P unrot45(P p) { return P((p.x + p.y) / 2, (p.x -
    <- p.y) / 2); }
// Max Manhattan distance between any two of the points, O(n).
ll maxManhattan(const vector<Pi>& v) {
    ll best = 0;

```

```

    for (int s = 0; s < 2; s++) { // the 2^1 sign
    <- patterns for (x +- y)
        ll mn = LLONG_MAX, mx = LLONG_MIN;
        for (auto& p : v) { ll t = s ? p.x - p.y : p.x + p.y; mn =
    <- min(mn, t), mx = max(mx, t); }
        best = max(best, mx - mn);
    }
    return best;
}
// In d dimensions the same idea uses all 2^(d-1) sign patterns.

// MANHATTAN MST - the MST under L1 distance, O(n log n) instead of
    <- O(n^2).
// Key fact: every point only needs edges to its nearest neighbour in
    <- each of 8 octants; by
// symmetry 4 sweeps suffice (rotate/reflect the plane between sweeps).
vector<array<ll, 3>> manhattanEdges(vector<Pi> ps) { // {w, i, j};
    <- indices stay ORIGINAL
    int n = ps.size();
    vector<int> id(n);
    iota(all(id), 0);
    vector<array<ll, 3>> e;
    for (int s = 0; s < 4; s++) {
        sort(all(id), [&](int a, int b) { return ps[a].x + ps[a].y <
    <- ps[b].x + ps[b].y; });
        map<ll, int> act; // key = -y,
    <- value = index
        for (int i : id) {
            for (auto it = act.lower_bound(-ps[i].y); it != act.end();
    <- act.erase(it++)) {
                int j = it->second;
                Pi d = ps[i] - ps[j];
                if (d.y > d.x) break;
                e.push_back({d.x + d.y, (ll)i, (ll)j});
            }
            act[-ps[i].y] = i;
        }
        for (auto& p : ps) { if (s & 1) p.x = -p.x; else swap(p.x,
    <- p.y); } // next octant pair
    }
    return e; // feed these
    <- to Kruskal
}

// LINEAR TRANSFORMATION mapping p0->q0 and p1->q1 (rotation + scaling
    <- + translation).
Pd linTrans(Pd p0, Pd p1, Pd q0, Pd q1, Pd r) {
    Pd dp = p1 - p0, dq = q1 - q0, num(dp.cross(dq), dp.dot(dq));
    return q0 + Pd((r - p0).cross(num), (r - p0).dot(num)) / dp.dist2();
}

// POINT <-> LINE DUALITY: point (a, b) <-> line y = a*x - b.
// "max points on a common line" becomes "max lines through a common
    <- point"
// convex hull of points <-> upper/lower envelope of lines (this IS
    <- the CHT correspondence)
// incidences are preserved, above/below is preserved.

```

11.1.9 KDTree

KD-TREE over 2D points: nearest-neighbour and "count points in a rectangle".

```

// Build O(n log n); NN query O(log n) expected, O(n) worst. The O(sqrt
    <- n) bound is for
// RECTANGLE queries; a rectangle count costs O(sqrt n + k).
// Use when coordinates are large/sparse and a grid or sort-sweep will
    <- not do.
struct KD {
    struct Node {
        Pi p; // the
    <- splitting point
        ll x0 = LLONG_MAX, x1 = LLONG_MIN, y0 = LLONG_MAX, y1 =
    <- LLONG_MIN; // bounding box
        int l = -1, r = -1, cnt = 0;
    };
    vector<Node> t;
    vector<Pi> pts;

    KD(vector<Pi> v) : pts(v) { if (!v.empty()) build(0, v.size()),
    <- false; }

    int build(int lo, int hi, bool divX) {
        if (lo >= hi) return -1;
        int id = t.size();
        t.push_back({});
        int mid = (lo + hi) / 2;
        nth_element(pts.begin() + lo, pts.begin() + mid, pts.begin() +
    <- hi,
            [&](Pi a, Pi b) { return divX ? a.x < b.x : a.y <

```

```

    b.y; });
    Node nd;
    nd.p = pts[mid], nd.cnt = hi - lo;
    for (int i = lo; i < hi; i++)
        nd.x0 = min(nd.x0, pts[i].x), nd.x1 = max(nd.x1, pts[i].x),
        nd.y0 = min(nd.y0, pts[i].y), nd.y1 = max(nd.y1, pts[i].y);
    t[id] = nd;
    int L = build(lo, mid, !divX), R = build(mid + 1, hi, !divX);
    t[id].l = L, t[id].r = R;
    return id;
}
ll boxDist(int i, Pi q) const { // squared
    distance from q to the box
    const Node& n = t[i];
    ll dx = max({(ll)0, n.x0 - q.x, q.x - n.x1}), dy = max({(ll)0,
    n.y0 - q.y, q.y - n.y1});
    return dx * dx + dy * dy;
}
void nn(int i, Pi q, ll& best, Pi& bp, bool skipSelf) const {
    if (i < 0 || boxDist(i, q) >= best) return;
    ll d = (t[i].p - q).dist2();
    if (d < best && !(skipSelf && d == 0)) best = d, bp = t[i].p;
    int a = t[i].l, b = t[i].r;
    if (a >= 0 && b >= 0 && boxDist(b, q) < boxDist(a, q)) swap(a,
    b);
    nn(a, q, best, bp, skipSelf), nn(b, q, best, bp, skipSelf);
}
// skipSelf skips EVERY point at distance 0, not just q itself -
// with duplicate input points
// that is wrong for "nearest OTHER point". If duplicates are
// possible, count multiplicity
// separately and treat a repeated point as its own nearest
// neighbour at distance 0.
pair<ll, Pi> nearest(Pi q, bool skipSelf = false) const {
    if (t.empty()) return {llinf, Pi{}};
    ll best = LLONG_MAX; Pi bp;
    nn(0, q, best, bp, skipSelf);
    return {best, bp}; // {squared
    distance, point}
}
int count(int i, ll x0, ll y0, ll x1, ll y1) const { // #points in
    [x0,x1] x [y0,y1]
    if (i < 0 || t[i].x1 < x0 || t[i].x0 > x1 || t[i].y1 < y0 ||
    t[i].y0 > y1) return 0;
    if (x0 <= t[i].x0 && t[i].x1 <= x1 && y0 <= t[i].y0 && t[i].y1
    <= y1) return t[i].cnt;
    int r = (x0 <= t[i].p.x && t[i].p.x <= x1 && y0 <= t[i].p.y &&
    t[i].p.y <= y1);
    return r + count(t[i].l, x0, y0, x1, y1) + count(t[i].r, x0,
    y0, x1, y1);
}
int count(ll x0, ll y0, ll x1, ll y1) const { return t.empty() ? 0
    : count(0, x0, y0, x1, y1); }
};

```

11.1.10 Triangulation

EAR CLIPPING - triangulates any SIMPLE polygon (convex or not, no self-intersections, no holes)

```

// Needs: Core.cpp, Polygon.cpp (area2).
// into n-2 triangles, O(n^2). Exact with P<ll>. Returns index triples
// into the ORIGINAL vector.
// An "ear" is a convex vertex whose triangle contains no other vertex
// of the polygon.
// USES: area/integral of anything over a polygon, rendering, splitting
// a region for a sweep,
// point-location preprocessing, computing a polygon's centre of mass
// with a weight function.
template <class P>
vector<array<int, 3>> triangulate(const vector<P>& poly) {
    int n = poly.size();
    vector<array<int, 3>> tri;
    if (n < 3) return tri;
    vector<int> id(n);
    iota(all(id), 0);
    if (area2(poly) < 0) reverse(all(id)); // work
    counter-clockwise
    auto inTri = [&](int a, int b, int c, int q) { //
    closed triangle
        return poly[a].cross(poly[b], poly[q]) >= 0 &&
        poly[b].cross(poly[c], poly[q]) >= 0 &&
        poly[c].cross(poly[a], poly[q]) >= 0;
    };
    while (id.size() > 2) {
        int m = id.size(), cut = -1, flat = -1;
        for (int i = 0; i < m && cut < 0; i++) {
            int a = id[(i + m - 1) % m], b = id[i], c = id[(i + 1) % m];
            auto o = poly[a].cross(poly[b], poly[c]);

```

```

            if (o == 0) { flat = i; continue; } //
            collinear vertex: just drop it
            if (o < 0) continue; //
        }
        reflex
        bool ok = true;
        for (int j = 0; j < m && ok; j++)
            if (id[j] != a && id[j] != b && id[j] != c && inTri(a,
            b, c, id[j])) ok = false;
        if (ok) cut = i, tri.push_back({a, b, c});
    }
    if (cut < 0) cut = flat;
    if (cut < 0) break; // not
    a simple polygon
    id.erase(id.begin() + cut);
}
return tri;
}
// FASTER: monotone decomposition (sweep, O(n log n)) then triangulate
// each monotone piece in O(n).
// Only worth writing for n beyond ~2000; ear clipping handles
// everything a contest throws at you.
// DELAUNAY TRIANGULATION (13 - Delaunay.cpp, O(n log n)) maximises the
// minimum angle instead, and
// its dual is the VORONOI DIAGRAM (14 - Voronoi.cpp). Cheap derivation
// to remember: lift each
// point to the paraboloid (x, y, x^2+y^2) and take the LOWER hull of
// the 3D convex hull - its
// faces project to the Delaunay triangles.
// Delaunay contains: the Euclidean minimum spanning tree, the
// nearest-neighbour graph, and the
// closest pair. So "EMST of points in the plane" = Delaunay (O(n log
// n) edges) + Kruskal.
// In-circle test (a,b,c counter-clockwise): d is strictly inside the
// circumcircle of a,b,c iff
// the 3x3 determinant of rows (a-d, |a-d|^2), (b-d, |b-d|^2), (c-d,
// |c-d|^2) is > 0. [__int128]
// ART GALLERY THEOREM: floor(n/3) guards always suffice for a simple
// n-gon, and are sometimes
// necessary (comb polygon). Proof = triangulate, 3-colour the
// triangulation graph, take the
// smallest colour class. The triangulation above is exactly what that
// construction needs.
// POLYGON WITH HOLES: connect each hole to the outer boundary with a
// "bridge" edge (pick the hole's
// rightmost vertex, shoot a ray right, split the hit edge) to get one
// simple polygon, then clip.

```

11.1.11 PolygonOps

POLYGON AGAINST POLYGON: clipping, intersection, containment, distance, and the O(log n)

```

// Needs: Core.cpp, Lines.cpp, Polygon.cpp.
// queries you get once one side is a convex hull.

// --- IS THIS POLYGON SIMPLE? O(n^2), exact on P<ll>. ---
// Every polygon routine (inPoly, area, triangulate, Pick, polyUnion)
// silently returns nonsense on
// a self-intersecting polygon, so this is also the oracle you
// stress-test them with.
template <class P> bool isSimple(const vector<P>& v) {
    int n = v.size();
    if (n < 3) return false;
    for (int i = 0; i < n; i++) {
        if (v[i] == v[(i + 1) % n]) return false; //
        zero-length edge
        for (int j = i + 1; j < n; j++) {
            int a = i, b = (i + 1) % n, c = j, d = (j + 1) % n;
            if (a == c || a == d || b == c || b == d) { //
                adjacent edges: they may only
                if (a == d || b == c) { //
                    share the one common vertex
                    P sh = (a == d) ? v[a] : v[b];
                    for (P q : {v[a], v[b], v[c], v[d]})
                        if (!(q == sh) && onSeg(q, v[a], v[b]) &&
                        onSeg(q, v[c], v[d])) return false;
                }
                continue;
            }
            if (segCross(v[a], v[b], v[c], v[d])) return false;
        }
    }
    return true;
}
// --- WINDING NUMBER. inPoly uses the EVEN-ODD rule, which is wrong
// for a self-intersecting
// outline and for "polygon with holes given as one vertex list". The
// nonzero rule uses this. ---
template <class P> int windingNumber(const vector<P>& v, P p) { // 0

```

```

    ↪ = outside (nonzero rule)
    int n = v.size(), w = 0;
    for (int i = 0; i < n; i++) {
        P a = v[i], b = v[(i + 1) % n];
        if (onSeg(p, a, b)) return 0; // on the
    }
    ↪ boundary: caller decides
    if (a.y <= p.y) { if (b.y > p.y && a.cross(b, p) > 0) w++; }
    else if (b.y <= p.y && a.cross(b, p) < 0) w--;
    }
    return w;
}
// --- SUTHERLAND-HODGMAN: clip any polygon by a CONVEX window, O(n *
    ↪ m). ---
// The window must be convex and counter-clockwise; the subject may be
    ↪ concave.
// This is the general "intersect with a convex region" and it is just
    ↪ 'cut' in a loop.
template <class P> vector<P> clipByConvex(vector<P> subject, const
    ↪ vector<P>& win) {
    int m = win.size();
    for (int i = 0; i < m && !subject.empty(); i++)
        subject = cut(subject, win[i], win[(i + 1) % m]);
    return subject;
}
// --- CONVEX INTERSECT CONVEX in O(n + m) is the classic O'Rourke
    ↪ walk, but clipByConvex is
// O(nm) and fits on one line - at contest sizes (n, m <= a few
    ↪ thousand) prefer it. Use
// halfPlaneInter (06) when the polygons come as half-plane
    ↪ constraints, not vertex lists.

// --- MINIMUM DISTANCE BETWEEN TWO CONVEX POLYGONS, O(n * m) as
    ↪ written. ---
// 0 if they touch or overlap. The O(n+m) rotating-calipers version
    ↪ exists but this is 6 lines and
// the distance is a floating answer anyway. Equivalent trick:
    ↪ dist(origin, minkowski(A, -B)).
template <class P> ld convexDist(const vector<P>& A, const vector<P>&
    ↪ B) {
    for (P p : A) if (inPoly(B, p, false)) return 0;
    for (P p : B) if (inPoly(A, p, false)) return 0;
    ld best = 1e18;
    int n = A.size(), m = B.size();
    for (int i = 0; i < n; i++) for (int j = 0; j < m; j++) {
        P a = A[i], b = A[(i + 1) % n], c = B[j], d = B[(j + 1) % m];
        if (segCross(a, b, c, d)) return 0;
        best = min(best, segSegDist(a, b, c, d));
    }
    return best;
}
// --- TANGENTS FROM A POINT TO A CONVEX POLYGON, O(n) here (O(log n)
    ↪ with a binary search). ---
// Returns the two vertex indices l, r such that the whole polygon lies
    ↪ to one side of p->h[l]
// and of p->h[r]: the visible arc is h[r], h[r+1], ..., h[l]. p must
    ↪ be strictly OUTSIDE.
// USES: visibility ("how much of the wall can I see"),
    ↪ shadow/silhouette, and "add p to the hull".
template <class P> pair<int, int> pointTangents(const vector<P>& h, P
    ↪ p) {
    int n = h.size(), l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (p.cross(h[i], h[l]) > 0) l = i; //
    }
    ↪ h[l] is the CCW-most
    if (p.cross(h[i], h[r]) < 0) r = i; //
    ↪ h[r] is the CW-most
    }
    return {l, r};
}
// --- LINE x CONVEX POLYGON in O(log n): which two edges does the line
    ↪ a->b cross? ---
// Returns {-1,-1} for no hit; {i,-1} for a single touched corner i;
    ↪ {i,j} when the line crosses
// the edges (i,i+1) and (j,j+1). h must be a strictly convex CCW hull
    ↪ (no collinear vertices).
// USES: "length of the segment inside the polygon" per query, in O(log
    ↪ n) instead of O(n).
template <class P> pair<int, int> lineHull(const vector<P>& h, P a, P
    ↪ b) {
    int n = h.size();
    int endA = extreme(h, (a - b).perp()), endB = extreme(h, (b -
    ↪ a).perp());
    auto cmpL = [&](int i) { return sgn(a.cross(h[i], b)); };
    if (cmpL(endA) < 0 || cmpL(endB) > 0) return {-1, -1}; //
    ↪ whole hull on one side
    auto bs = [&](int lo, int hi) { //
    ↪ last index still >= 0

```

```

        int step = (hi - lo + n) % n;
        for (int s = step; s > 0; s /= 2)
            while (s && cmpL((lo + s) % n) >= 0) lo = (lo + s) % n,
    ↪ step -= s;
        return lo;
    };
    int i = bs(endB, endA), j = bs(endA, endB);
    if (i == j) return {i, -1};
    return {i, j};
}
/* MORE POLYGON FACTS AND RECIPES
AREA OF INTERSECTION of two CONVEX polygons: clipByConvex then area().
    ↪ For two ARBITRARY simple
    polygons, triangulate one (10 - Triangulation.cpp) and clip each
    ↪ triangle by the other.
AREA OF THE UNION: |A| + |B| - |A n B|, or polyUnion (07 - Unions.cpp)
    ↪ for many polygons at once.
SYMMETRIC DIFFERENCE: |A| + |B| - 2|A n B|.
POINT IN A POLYGON WITH HOLES: outer boundary CCW, every hole CW, then
    ↪ the sign of the total
    winding number is the answer (this is exactly why windingNumber
    ↪ exists).
CONVEX POLYGON CONTAINS CONVEX POLYGON: every vertex of B is in A
    ↪ (inConvex, O(m log n)).
CENTROID OF THE PERIMETER (not of the area): sum of edge midpoints
    ↪ weighted by edge LENGTH.
    The two differ - "balance the wire frame" vs "balance the plate".
POLYGON DIAMETER / WIDTH / MIN RECTANGLE: 03 - Polygon.cpp (rotating
    ↪ calipers).
REGULAR n-GON of circumradius R centred at c: c + R*(cos(2*pi*k/n +
    ↪ t), sin(2*pi*k/n + t)).
    Its area is (n/2) R^2 sin(2 pi/n); with INRADIUS r it is n r^2
    ↪ tan(pi/n).
SPLIT A POLYGON BY A LINE into both halves: cut(v, s, e) and cut(v, e,
    ↪ s).
LARGEST INSCRIBED / SMALLEST ENCLOSING: smallest enclosing circle is
    ↪ mec (04 - Circles.cpp);
    smallest enclosing rectangle is minRectArea; largest inscribed
    ↪ CIRCLE in a convex polygon is
    the Chebyshev centre, 18 - InscribedCircle.cpp (a nested ternary
    ↪ search, no LP needed);
    largest inscribed AXIS-ALIGNED rectangle needs a sweep and is
    ↪ genuinely hard - do not improvise.
OFFSET / BUFFER a convex polygon by d: push every edge outward by d
    ↪ along its normal and
    intersect the half-planes (06 - HalfPlaneIntersection.cpp). The
    ↪ exact offset of a NON-convex
    polygon has circular arcs at the reflex corners - approximate them.
TRIANGLE CONTAINS POINT, exact: the three orientations all have the
    ↪ same sign (or one is 0 for
    the boundary). Faster than inPoly for n = 3 and it is the inner loop
    ↪ of triangulate. */

```

11.1.12 Primitives

THE SMALL-FUNCTION LAYER. Individually trivial, collectively the thing you actually spend

```

// Needs: Core.cpp, Lines.cpp, Circles.cpp.
// contest minutes re-deriving. Everything here is one to six lines.

// --- ANGLE BISECTORS OF TWO LINES. Two lines have TWO bisectors,
    ↪ perpendicular to each other.
// Direction of the internal one = the sum of the unit direction
    ↪ vectors; external = difference.
// (Take the pair whose sign matches the side you want; if the lines
    ↪ are parallel the bisector is
// the parallel line halfway between them.)
template <class P> pair<P, P> lineBisectors(P d1, P d2) { //
    ↪ returns the two DIRECTIONS
    P u = d1.unit(), v = d2.unit();
    return {u + v, u - v};
}
// Internal bisector of the angle at b in triangle a-b-c, as a
    ↪ direction from b:
template <class P> P angleBisector(P a, P b, P c) { return (a -
    ↪ b).unit() + (c - b).unit(); }
// The internal bisector from A meets BC at the point dividing it in
    ↪ ratio AB : AC.
template <class P> P bisectorFoot(P a, P b, P c) {
    ld x = (a - b).dist(), y = (a - c).dist();
    return (b * y + c * x) / (x + y);
}
// --- ROTATE / SCALE ABOUT AN ARBITRARY CENTRE (rot() in Core is about
    ↪ the origin). ---
template <class P> P rotAbout(P p, P c, ld a) { return c + (p -
    ↪ c).rot(a); }
template <class P> P scaleAbout(P p, P c, ld k) { return c + (p - c) *
    ↪ k; }

```

```
// Reflect p across the POINT c (a 180-degree rotation):
template <class P> P reflPoint(P p, P c) { return c * 2 - p; }
// --- SIGNED AREA OF A TRIANGLE, and the barycentric coordinates of p
// in a-b-c. ---
template <class P> auto triArea2(P a, P b, P c) { return a.cross(b, c); }
// 2 * signed
template <class P> array<ld, 3> barycentric(P a, P b, P c, P p) {
    ld s = a.cross(b, c);
    return {(ld)p.cross(b, c) / s, (ld)a.cross(p, c) / s,
            (ld)a.cross(b, p) / s};
}
// p is inside the triangle iff all three coordinates are >= 0 (they
// always sum to 1).
template <class P> bool inTriangle(P a, P b, P c, P p) {
    // exact, boundary included
    int s1 = sgn(a.cross(b, p)), s2 = sgn(b.cross(c, p)), s3 =
    sgn(c.cross(a, p));
    return (s1 >= 0 && s2 >= 0 && s3 >= 0) || (s1 <= 0 && s2 <= 0 && s3
    <= 0);
}
// --- SEGMENT x CIRCLE: clip the segment a-b to the disc, returning
// the surviving piece. ---
template <class P> vector<P> segCircle(P a, P b, P c, ld r) {
    auto v = circleLine(c, r, a, b);
    // 0, 1 or 2 points
    vector<P> res;
    for (P p : v) if (onSeg(p, a, b)) res.push_back(p);
    return res;
}
// --- CIRCLE THROUGH THREE POINTS = circumcenter (04 - Circles.cpp);
// its radius = circumradius.
// CIRCLE THROUGH TWO POINTS WITH A GIVEN RADIUS: the two centres are m
// +- h * (b-a).perp().unit()
// where m is the midpoint and h = sqrt(r^2 - |b-a|^2/4). None if 2r <
// |b-a|.
template <class P> vector<P> circlesThrough(P a, P b, ld r) {
    ld d2 = (b - a).dist2(), h2 = r * r - d2 / 4;
    if (d2 < eps || h2 < 0) return {};
    // a
    // == b: infinitely many
    P m = (a + b) / 2, dir = (b - a).perp().unit() * sqrt(h2);
    return h2 < eps ? vector<P>{m} : vector<P>{m - dir, m + dir};
}
// --- CIRCLE OF RADIUS r THROUGH p AND TANGENT TO THE LINE a-b. Its
// centre is at distance r from
// the line, so on one of the two PARALLELS at offset r, and at
// distance r from p, so on the
// circle (p, r). Intersect the two: 0, 1 or 2 centres. With d =
// distance from p to the line,
// there are 2 for d < 2r (both, tangentially, when d == 0), 1 when d
// == 2r, and none beyond.
template <class P> vector<P> circleTangentLine(P a, P b, P p, ld r) {
    vector<P> res;
    P n = (b - a).perp().unit() * r;
    for (int s = -1; s <= 1; s += 2)
        for (P c : circleLine(p, r, a + n * s, b + n * s))
            res.push_back(c);
    return res;
}
// --- APOLLONIUS CIRCLE: the locus of p with |pa| / |pb| = k (k != 1)
// is a circle whose diameter
// endpoints are the two points dividing ab internally and externally
// in ratio k : 1.
template <class P> pair<P, ld> apollonius(P a, P b, ld k) {
    P u = (b * k + a) / (k + 1), v = (b * k - a) / (k - 1);
    // internal, external
    return {(u + v) / 2, (u - v).dist() / 2};
}
// --- IS THE QUADRILATERAL a,b,c,d CONCYCLIC? Exact on P<ll> with
// __int128 (in-circle test). ---
// > 0: d strictly inside the circumcircle of a CCW triangle a,b,c; ==
// 0: concyclic.
template <class P> int inCircle(P a, P b, P c, P d) {
    auto f = [&](P p) { return (__int128)(p - d).dist2(); };
    __int128 det = (__int128)(a.x - d.x) * ((b.y - d.y) * f(c) - (c.y -
    d.y) * f(b))
        - (__int128)(a.y - d.y) * ((b.x - d.x) * f(c) - (c.x -
    d.x) * f(b))
        + f(a) * ((__int128)(b.x - d.x) * (c.y - d.y)
        - (__int128)(c.x - d.x) * (b.y - d.y));
    return (det > 0) - (det < 0);
}
// --- MAXIMUM POINTS COVERED BY A CIRCLE OF RADIUS r, 0(n^2 log n). ---
// For each i, every j within 2r gives an ARC of centre-angles that
// cover both; sweep the arcs.
template <class P> int maxCovered(const vector<P>& p, ld r) {
    int n = p.size(), best = 1;
    for (int i = 0; i < n; i++) {
```

```
        vector<pair<ld, int>> ev;
        int base = 1;
        // p[i], plus its duplicates
        for (int j = 0; j < n; j++) if (j != i) {
            ld d = (p[j] - p[i]).dist();
            if (d > 2 * r + eps) continue;
            if (d < eps) { base++; continue; }
            // DUPLICATE of p[i]: the
            // division below would be 0/0
            // and it is always covered
            ld th = (p[j] - p[i]).angle(), ph = acosl(min((ld)1,
            max((ld)-1, d / (2 * r))));
            ld s = th - ph, e = th + ph;
            // the arc of valid centres
            while (s < -PI) s += 2 * PI;
            // normalise, then SPLIT the
            e = s + 2 * ph;
            // arcs that wrap past +pi -
            // 0 = arc opens, 1 = arc closes. Pair-sorting then puts
            // the OPEN first at an equal
            // angle, which is what makes a TANGENTIAL arc (d == 2r, so
            // the arc is a single point)
            // still count - with {+1}/{-1} the close sorts first and
            // the point is lost.
            if (e <= PI) ev.push_back({s, 0}), ev.push_back({e, 1});
            else {
                ev.push_back({s, 0}), ev.push_back({PI, 1});
                ev.push_back({-PI, 0}), ev.push_back({e - 2 * PI, 1});
            }
            sort(all(ev));
            int cur = base;
            best = max(best, cur);
            for (auto& [a, t] : ev) best = max(best, cur += t ? -1 : 1);
        }
        return best;
    }
    // the circle passes through p[i]
}
// --- GEOMETRIC MEDIAN (minimise the SUM of distances). Nested ternary
// search - the objective is
// convex, so this is exact to any eps and cannot get stuck. Weiszfeld
// iteration is faster but
// breaks when the iterate lands exactly on a sample point.
template <class P> P geoMedian(const vector<P>& p) {
    auto f = [&](ld x, ld y) { ld s = 0; for (auto& q : p) s += (P{x,
    y} - q).dist(); return s; };
    auto fy = [&](ld x) { ld lo = -2e9, hi = 2e9;
        for (int it = 0; it < 100; it++) { ld a = lo + (hi - lo) / 3, b
        = hi - (hi - lo) / 3;
            f(x, a) < f(x, b) ? hi = b : lo = a; } return (lo + hi) /
        2; };
    ld lo = -2e9, hi = 2e9;
    for (int it = 0; it < 100; it++) {
        ld a = lo + (hi - lo) / 3, b = hi - (hi - lo) / 3;
        f(a, fy(a)) < f(b, fy(b)) ? hi = b : lo = a;
    }
    ld x = (lo + hi) / 2;
    return {x, fy(x)};
}
/* THE REST OF THE SMALL LAYER, as one-liners you can reconstruct
// instantly
COLLINEAR a, b, c          a.cross(b, c) == 0
PARALLEL / PERPENDICULAR  d1.cross(d2) == 0 / d1.dot(d2) == 0
SAME SIDE OF A LINE        sgn(a.cross(b, p)) == sgn(a.cross(b, q))
POINT ON A RAY / SEGMENT   onRay, onSeg (02 - Lines.cpp)
DISTANCE point-line/segment lineDist, segDist ray: rayDist
// seg-seg: segSegDist
PROJECT / REFLECT over a line proj, refl (02 - Lines.cpp)
ANGLE between two vectors atan2(cross, dot) -> signed, in (-pi,
// pi]. NEVER acos(dot/|a||b|):
// it loses all precision near 0 and pi and
// cannot give you a sign.
ROTATE 90 DEGREES          p.perp() = (-y, x). Twice = negation.
// This is the whole of "turn
// left"; you almost never need a general
// rot() for a right angle.
SORT BY ANGLE              angCmp (01 - Core.cpp), NOT atan2 - exact
// and 5x faster.
AREA OF A TRIANGLE         |cross| / 2. Heron only if you are given
// the three SIDES, and then
// use the sorted stable form: sort a >= b
// >= c and
// A =
// sqrt((a+(b+c))(c-(a-b))(c+(a-b))(a+(b-c))) / 4.
TRIANGLE CENTRES           centroid (a+b+c)/3, circumcenter,
```

↳ incenter, orthocenter (04).
 ↳ OG:GN:NH = 2:1:3, and
 ↳ midpoint of OH, radius $R/2$.
 TRIANGLE IDENTITIES
 ↳ A/s ; $r_a = A/(s-a)$
 ↳ $R \geq 2r$
 ↳ $m + c^2 n = a(d^2 + mn)$
 CYCLIC QUADRILATERAL
 ↳ order). For ANY four
 ↳ iff concyclic in that order.
 ↳ $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.
 ↳ angle-free):
 ↳ with p, q the diagonals.
 POWER OF A POINT
 ↳ through p meeting the circle at
 ↳ it. Outside: = $\tan^2$.
 ↳ perpendicular to the centre line.
 ↳ non-collinear centres: the
 DESCARTES CIRCLE THEOREM
 ↳ $k_i = +1/r_i$ (negative
 ↳ $= 2(k_1^2 + k_2^2 + k_3^2 + k_4^2)$,
 ↳ $k_3 k_1$.
 LATTICE POINTS
 ↳ endpoints included.
 ↳ (simple polygons only).
 ↳ Number Theory/01/03).
 HELLY (2D)
 ↳ meet, all of them meet.
 RADON (2D)
 ↳ hulls intersect.
 CARATHEODORY (2D)
 ↳ of at most THREE points of S
 ↳ enclosing circle needs only 3.
 ART GALLERY
 ↳ simple n-gon (triangulate,
 ↳ 3-colour, take the smallest class).
 ↳ Orthogonal polygon: $\text{floor}(n/4)$.
 EULER, PLANAR
 ↳ position cut the plane into
 ↳ bounded; n circles in
 DUALITY
 ↳ Incidence and above/below are
 ↳ becomes "max concurrent lines"
 ↳ envelope. */

EULER LINE: O, G, N, H are collinear with
 $H = A + B + C - 2O$. Nine-point centre =
 $A = rs = abc/(4R)$; $R = abc/(4A)$; $r =$
 law of cosines $c^2 = a^2 + b^2 - 2ab \cos C$
 law of sines $a/\sin A = 2R$
 $OI^2 = R(R - 2r) \Rightarrow$ EULER'S INEQUALITY

Stewart (cevian d to BC split m + n): b^2
 median $m_a = \sqrt{2b^2 + 2c^2 - a^2}/2$
 PTOLEMY: $AC \cdot BD = AB \cdot CD + BC \cdot AD$ (in cyclic
 points $AC \cdot BD \leq AB \cdot CD + BC \cdot AD$, equality
 BRAHMAGUPTA area =
 General quadrilateral (BRETSCHNEIDER,
 $A = \sqrt{4p^2 q^2 - (b^2 + d^2 - a^2 - c^2)^2}/4$

$\text{pow}(p) = |pc|^2 - r^2$; for ANY line
 A and B the signed product $PA \cdot PB$ equals
 RADICAL AXIS = equal power, a line
 RADICAL CENTRE of three circles with
 three pairwise axes are concurrent.
 four mutually tangent circles, curvatures
 for the enclosing one): $(k_1 + k_2 + k_3 + k_4)^2$
 so $k_4 = k_1 + k_2 + k_3 \pm 2 \sqrt{k_1 k_2 + k_2 k_3 +$

on a segment: $\gcd(|dx|, |dy|) + 1$
 inside a polygon: PICK, $I = A - B/2 + 1$
 under a line, in $O(\log)$: floorSum (04 -
 if every THREE of a family of convex sets
 any FOUR points split into two sets whose
 a point in the hull of S is in the hull
 - which is exactly why the minimum
 $\text{floor}(n/3)$ guards always suffice for a
 3-colour, take the smallest class).
 $V - E + F = 1 + C$. n lines in general
 $1 + n + C(n, 2)$ regions, $C(n-1, 2)$ of them

general position give $n^2 - n + 2$ regions.
 point (a,b) $\leftrightarrow$ line $y = ax - b$.
 preserved, so "max collinear points"
 and a convex hull becomes an upper

11.1.13 Delaunay

DELAUNAY TRIANGULATION, $O(n \log n)$, divide and conquer over the quad-edge structure.

// Needs: Core.cpp.
 // Returns index triples into the input, each one COUNTER-CLOCKWISE.
 ↳ The defining property, and
 ↳ the only one you ever use: the circumcircle of every returned
 ↳ triangle is EMPTY of input points.
 // WHEN: EUCLIDEAN MST of points in the plane - the EMST is a subgraph
 ↳ of the Delaunay graph, so
 // Delaunay ($O(n)$ edges) + Kruskal replaces the $O(n^2)$ complete graph.
 ↳ Also the nearest-neighbour
 // graph, the closest pair, the largest empty circle, "maximise the
 ↳ minimum angle" meshing, and
 // the VORONOI DIAGRAM, which is exactly the dual (14 - Voronoi.cpp).
 // COMPLEXITY: $O(n \log n)$ time. At most $2n - 2 - h$ triangles and $3n - 3$
 ↳ - h edges, h = hull size.
 // Allocates quad-edges with new and never frees them; budget ~200

↳ bytes per point.
 // DEGENERATE: DUPLICATE POINTS ARE NOT ALLOWED - dedupe before
 ↳ calling. Fewer than three points,
 // or ALL POINTS COLLINEAR, returns {} - there is genuinely no
 ↳ triangle, handle it upstream.
 // Four or more CONCYCLIC points make the triangulation non-unique; you
 ↳ get one legal answer.
 // EXACT on P<ll> for |coord| <= 1e9: orient fits in ll (<= 8e18) and
 ↳ inCircle is __int128.
 typedef struct QuadEdge* QE;
 Pi qArb(LLONG_MAX, LLONG_MAX); // a sentinel
 ↳ unequal to any real point
 struct QuadEdge {
 QE rot, o; Pi p = qArb; bool mark = 0;
 Pi& F() { return r()->p; } // the far endpoint
 ↳ of this directed edge
 QE& r() { return rot->rot; } // the same edge
 ↳ reversed
 QE prev() { return rot->o->rot; }
 QE next() { return r()->prev(); }
 };
 bool inCirc(Pi p, Pi a, Pi b, Pi c) { // is p strictly
 ↳ inside circumcircle(a, b, c)?
 __int128 p2 = p.dist2(), A = a.dist2() - p2, B = b.dist2() - p2, C
 ↳ = c.dist2() - p2;
 return p.cross(a, b) * C + p.cross(b, c) * A + p.cross(c, a) * B >
 ↳ 0;
 }
 QE qeEdge(Pi orig, Pi dest) {
 QE q[] = {new QuadEdge{0, 0, orig}, new QuadEdge{0, 0, qArb}, new
 ↳ QuadEdge{0, 0, dest},
 ↳ new QuadEdge{0, 0, qArb}};
 for (int i = 0; i < 4; i++) q[i]->o = q[-i & 3], q[i]->rot = q[(i +
 ↳ 1) & 3];
 return *q;
 }
 void qeSplice(QE a, QE b) { swap(a->o->rot->o, b->o->rot->o),
 ↳ swap(a->o, b->o); }
 QE qeConnect(QE a, QE b) {
 QE q = qeEdge(a->F(), b->p);
 qeSplice(q, a->next()), qeSplice(q->r(), b);
 return q;
 }
 QE dNext(QE e, bool lf) { return lf ? e->o : e->prev(); } //
 ↳ walk left / right of the base
 bool dValid(QE e, QE base) { return e->F().cross(base->F(), base->p) >
 ↳ 0; }
 QE dDel(QE init, QE base, bool lf) { // drop edges
 ↳ that the new base invalidates
 QE e = dNext(init, lf);
 if (dValid(e, base))
 while (inCirc(dNext(e, lf)->F(), base->F(), base->p, e->F())) {
 QE t = dNext(e, lf);
 qeSplice(e, e->prev()), qeSplice(e->r(), e->r()->prev());
 e = t;
 }
 return e;
 }
 pair<QE, QE> dRec(const vector<Pi>& s) { // {ccw hull edge at
 ↳ left end, cw at right}
 if (s.size() <= 3) {
 QE a = qeEdge(s[0], s[1]), b = qeEdge(s[1], s.back());
 if (s.size() == 2) return {a, a->r()};
 qeSplice(a->r(), b);
 auto side = s[0].cross(s[1], s[2]);
 QE c = side ? qeConnect(b, a) : 0; // side == 0:
 ↳ three collinear points, no face
 return {side < 0 ? c->r() : a, side < 0 ? c : b->r()};
 }
 QE A, B, ra, rb;
 int half = s.size() / 2;
 tie(ra, A) = dRec({s.begin(), s.end() - half});
 tie(B, rb) = dRec({s.begin() + (s.size() - half), s.end()});
 while ((B->p.cross(A->F(), A->p) < 0 && (A = A->next())) || //
 ↳ find the lower common tangent
 (A->p.cross(B->F(), B->p) > 0 && (B = B->r()->o));
 QE base = qeConnect(B->r(), A);
 if (A->p == ra->p) ra = base->r();
 if (B->p == rb->p) rb = base;
 for (;) { // zip the two
 ↳ hulls together, bottom to top
 QE lc = dDel(base->r(), base, 1), rc = dDel(base, base, 0);
 if (!dValid(lc, base) && !dValid(rc, base)) break;
 if (!dValid(lc, base) || (dValid(rc, base) && inCirc(rc->F(),
 ↳ rc->p, lc->F(), lc->p)))
 base = qeConnect(rc, base->r());
 else base = qeConnect(base->r(), lc->r());

```

    }
    return {ra, rb};
}
vector<array<int, 3>> delaunay(const vector<Pi>& pts) {
    vector<Pi> p = pts;
    sort(all(p));
    vector<array<int, 3>> res;
    if (p.size() < 3) return res;
    bool line = true; // guard collinear
    ↪ input: dRec spins on it
    for (int i = 2; i < sz(p) && line; i++) line = p[0].cross(p[1],
    ↪ p[i]) == 0;
    if (line) return res;
    QE e = dRec(p).first;
    vector<QE> q = {e};
    while (e->o->F().cross(e->F(), e->p) < 0) e = e->o;
    vector<Pi> out;
    auto walk = [&](QE e) { // emit one face,
    ↪ queue its neighbours
        QE c = e;
        do { c->mark = 1, out.pb(c->p), q.pb(c->r()), c = c->next(); }
    ↪ while (c != e);
    };
    walk(e), out.clear(); // the first face
    ↪ walked is the OUTER one
    for (int i = 0; i < sz(q); i++) if (!(e = q[i])->mark) walk(e);
    map<pair<ll, ll>, int> id;
    for (int i = 0; i < sz(pts); i++) id[{pts[i].x, pts[i].y}] = i;
    for (int i = 0; i + 2 < sz(out); i += 3)
        res.pb({id[out[i].x, out[i].y], id[out[i + 1].x, out[i +
    ↪ 1].y]],
            id[out[i + 2].x, out[i + 2].y]}]);
    return res;
}
// EUCLIDEAN MST: collect the 3 edges of every triangle, dedupe, sort
    ↪ by squared length, Kruskal.
// vector<array<ll,3>> E; for (auto& t : delaunay(p)) for (int k = 0;
    ↪ k < 3; k++)
//     E.pb({p[t[k]] - p[t[(k+1)%3]].dist2(), t[k], t[(k+1)%3]}]);
// LARGEST EMPTY CIRCLE whose centre is inside the hull: its centre is
    ↪ a Delaunay circumcentre or
// a point on the hull boundary - check all O(n) circumcentres, then
    ↪ the hull edges.
// The DELAUNAY GRAPH also contains the nearest-neighbour graph and the
    ↪ Gabriel graph, so any
// "for each point, its closest other point" question is answered by
    ↪ scanning its Delaunay edges.
// FLIP ALGORITHM (any triangulation -> Delaunay by flipping illegal
    ↪ edges) is O(n^2) and simple,
// but you already have this; the paraboloid lift (x, y, x^2+y^2) +
    ↪ LOWER 3D hull is the other
// derivation and is the one to remember when you need the connection
    ↪ to convexity.

```

11.1.14 Voronoi

VORONOI DIAGRAM. The cell of site i is the set of points closer to site i than to any other

```

// Needs: Core.cpp, HalfPlaneIntersection.cpp (Delaunay.cpp for the
    ↪ fast variant).
// site; it is a convex polygon (unbounded for sites on the convex
    ↪ hull, so everything here is
// clipped to a box of half-width BOX - make BOX comfortably larger
    ↪ than any coordinate you care
// about, and remember that a vertex of the clipped cell may be an
    ↪ artefact of the box).
// The cell is just the intersection of the half-planes "closer to i
    ↪ than to j" over all  $j \neq i$ ,
// and each of those is the side of the perpendicular bisector of  $s_i$ 
    ↪  $s_j$  containing  $s_i$ .
// WHEN: nearest-site queries with a planar point location structure on
    ↪ top; largest empty circle;
// "which post office serves this house"; any facility-location or
    ↪ growth/flood simulation where
// regions expand at equal speed. If you only need nearest neighbours,
    ↪ a KD-tree (09) is shorter.
// COMPLEXITY:  $O(n^2 \log n)$  for all  $n$  cells as written ( $O(n \log n)$  per
    ↪ cell). That is fine up to
//  $n$  around 2000. For  $O(n \log n)$  total, build the DELAUNAY dual instead
    ↪ - see the tail.
// DEGENERATE: duplicate sites give two identical cells that are both
    ↪ empty ({} is returned);
// dedupe first.  $n == 1$  gives the whole box. Collinear sites give slab
    ↪ cells, handled correctly.
// FLOATING POINT: the bisectors are exact only if you keep sites
    ↪ integral, but halfPlaneInter is
// an eps algorithm, so the cell vertices carry the usual  $1e-9$  relative
    ↪ error. Four or more
// cocircular sites make a Voronoi vertex of degree  $> 3$  and the cell

```

```

    ↪ may pick up a zero-length
// edge; strip edges shorter than eps if you are counting them.
vector<Pd> voronoiCell(const vector<Pd>& s, int i, ld BOX = 1e9) {
    vector<HP> h;
    for (int j = 0; j < sz(s); j++) if (j != i) {
        Pd m = (s[i] + s[j]) / 2; // the
        ↪ perpendicular bisector, directed so
        h.pb(HP(m, m + (s[j] - s[i]).perp())); // that the
        ↪ LEFT side contains s[i]
    }
    return halfPlaneInter(h, BOX);
}
vector<vector<Pd>> voronoi(const vector<Pd>& s, ld BOX = 1e9) {
    vector<vector<Pd>> c;
    for (int i = 0; i < sz(s); i++) c.pb(voronoiCell(s, i, BOX));
    return c;
}
// O(n log n) VIA THE DELAUNAY DUAL (13 - Delaunay.cpp). One Voronoi
    ↪ vertex per Delaunay triangle,
// namely its CIRCUMCENTRE; one Voronoi edge per Delaunay edge, joining
    ↪ the circumcentres of the
// two triangles that share it. Recipe: run delaunay(p); for every
    ↪ triangle t store
// circumcenter(p[t[0]], p[t[1]], p[t[2]]); for each site collect the
    ↪ circumcentres of the
// triangles incident to it and sort them by angle around the site -
    ↪ that is its cell, in CCW
// order. A site on the CONVEX HULL has an unbounded cell: its two hull
    ↪ edges contribute rays
// along their outward perpendicular bisectors, so clip them to the box
    ↪ by hand.
// NEAREST SITE for a query point = the cell containing it: build the
    ↪ diagram, then run planar
// point location (15). For a handful of queries just scan all sites.
// LARGEST EMPTY CIRCLE with centre inside the hull: its centre is a
    ↪ Voronoi vertex (= a Delaunay
// circumcentre) or a point where a Voronoi edge crosses the hull
    ↪ boundary. Check both families.
// DELAUNAY EDGE <-> VORONOI EDGE, and the MST / nearest-neighbour
    ↪ consequences, are all in 13.
// WEIGHTED (POWER / LAGUERRE) DIAGRAM: replace the bisector by the
    ↪ RADICAL AXIS of the two
// circles (04 - Circles.cpp) - the cells stay convex polygons, so this
    ↪ exact code works with the
// bisector swapped out. The MULTIPLICATIVELY weighted (Apollonius)
    ↪ diagram has circular arcs for
// boundaries and none of this applies.

```

11.1.15 PointLocation

PLANAR POINT LOCATION, offline, by a left-to-right sweep. Input: a set of segments that are

```

// Needs: Core.cpp.
// pairwise NON-CROSSING (they may share endpoints) - i.e. the edges of
    ↪ a planar subdivision - and
// a list of query points. Output, for each query, the index of the
    ↪ segment DIRECTLY BELOW it, or
// -1 if the query is below everything. That single answer is the whole
    ↪ primitive: label every
// segment with the id of the face lying immediately ABOVE it and the
    ↪ query's face falls out.
// WHEN: "n disjoint polygons, q points, which polygon contains each"
    ↪ with n and q both  $1e5$ , so
// the  $O(nq)$  scan is dead. Also "which cell of an arrangement", and
    ↪ nearest-site queries once you
// have built the Voronoi diagram (14). For ONE convex polygon use
    ↪ inConvex (03) instead; for a
// handful of polygons just call inPoly (03) on each.
// COMPLEXITY:  $O((n + q) \log n)$  time,  $O(n + q)$  memory. Everything is
    ↪ offline - all queries must be
// known up front. (Online needs a persistent segment tree over the
    ↪ vertical slabs, which is
//  $O(n \log n)$  memory and rarely worth typing.)
// DEGENERATE: NO VERTICAL SEGMENTS - the comparator below assumes
    ↪ every segment points strictly
// rightwards. If you have them, shear the plane with (x, y) -> (x *
    ↪ (2C + 1) + y, y), which is
// exact on integers and makes all x distinct, but squares your
    ↪ coordinates (use __int128 cross).
// A query exactly ON a segment reports that segment (endpoints
    ↪ included), because segments
// starting at x are inserted before the queries at x and segments
    ↪ ending at x are removed after.
// Zero-length segments must be filtered out first.
// EXACT on P<ll> with |coord| <= 1e9; the comparator is pure
    ↪ orientation signs, no division.
vector<int> locateBelow(vector<pair<Pi, Pi>> s, const vector<Pi>& qs) {
    int n = sz(s), m = sz(qs);

```

```

for (auto& e : s) if (e.second < e.first) swap(e.first, e.second);
s.pb({Pi(), Pi()}); // slot n: the
↳ current query, as a point
// Two non-crossing segments are comparable wherever their x-ranges
↳ overlap: ask which side of
// a's line BOTH endpoints of b fall on, and if that is
↳ inconclusive (shared endpoint) ask the
// mirror question. Only ever consulted while both are in the sweep
↳ set, so this is a valid
// strict weak ordering at every instant even though it changes as
↳ the sweep advances.
auto below = [&](int i, int j) {
    auto& [a, b] = s[i]; auto& [c, d] = s[j];
    int v = sgn(a.cross(b, c)) + sgn(a.cross(b, d));
    if (v) return v > 0;
    return sgn(c.cross(d, a)) + sgn(c.cross(d, b)) < 0;
};
set<int, decltype(below)> t(below);
vector<int> add(n), del(n), qi(m), res(m, -1);
iota(all(add), 0), iota(all(del), 0), iota(all(qi), 0);
sort(all(add), [&](int i, int j) { return s[i].first.x <
↳ s[j].first.x; });
sort(all(del), [&](int i, int j) { return s[i].second.x <
↳ s[j].second.x; });
sort(all(qi), [&](int i, int j) { return qs[i].x < qs[j].x; });
int ia = 0, id = 0, iq = 0;
while (ia < n || id < n || iq < m) { // sweep EVERY
↳ event x, not just query x:
    ll x = LLONG_MAX; // the set
↳ order is only valid while all
    if (ia < n) x = min(x, s[add[ia]].first.x); // members
↳ still span the sweep line
    if (id < n) x = min(x, s[del[id]].second.x);
    if (iq < m) x = min(x, qs[qi[iq]].x);
    while (ia < n && s[add[ia]].first.x == x) t.insert(add[ia++]);
↳ // open,
    while (iq < m && qs[qi[iq]].x == x) {
↳ // then answer,
        int k = qi[iq++];
        s[n] = {qs[k], qs[k]}; // a
↳ zero-length segment = the query point
        auto it = t.lower_bound(n);
        if (it != t.begin()) res[k] = *prev(it);
    }
    while (id < n && s[del[id]].second.x == x) t.erase(del[id++]);
↳ // then close
}
return res;
}
// ATTACHING FACES. Build the subdivision as half-edges: every segment
↳ carries the face id above
// it and the face id below it (the unbounded face is -1). Then:
// f = locateBelow(...)[k]; answer = f < 0 ? -1 : faceAbove[f];
// and the query is ON the boundary iff qs[k].cross(s[f].first,
↳ s[f].second) == 0.
// For DISJOINT SIMPLE POLYGONS the labelling is free: walk each
↳ polygon in CCW order and give
// every edge whose direction points LEFT (a.x > b.x) the polygon's id
↳ as its "above" face, and
// every rightward edge the id of whatever the polygon sits in (-1 for
↳ a top-level polygon).
// ARRANGEMENT OF LINES: n lines cut the plane into 1 + n + C(n,2)
↳ faces; build the arrangement by
// sorting the intersections along each line, then run this on the
↳ resulting O(n^2) segments.
// A different, shorter answer to "point in one of many disjoint
↳ polygons": if the polygons are
// convex and you only need containment, sort by x-range and binary
↳ search, or build the union
// hull. Reach for point location only when the subdivision is
↳ genuinely a subdivision.

```

11.1.16 DynamicHull

DYNAMIC CONVEX HULL - insert points ONLINE, ask "is this point inside the hull so far" in

```

// Needs: Core.cpp.
// O(log n), amortised O(log n) per insert. The structure below is the
↳ UPPER hull only, stored as
// a map x -> max y; run a second copy fed the negated points for the
↳ lower hull and you have the
// full hull. Storing one y per x is what makes this short: the upper
↳ chain has strictly
// increasing x by definition, and if two points share an x only the
↳ higher one can be on it.
// WHEN: points arrive one at a time and you must answer containment /
↳ support / area queries
// between insertions; "add point, print hull area"; maximising a

```

```

↳ linear function over a growing
// set (that special case is CHT / Li Chao, 08 - DP Optimizations, and
↳ is simpler - use it if the
// query is always a dot product). Deletions are NOT supported; for
↳ those, use offline divide and
// conquer over time, or Kinetic / segment-tree-on-time rebuilding.
// COMPLEXITY: O(log n) amortised insert (each point is erased at most
↳ once), O(log n) query,
// O(n) memory.
// DEGENERATE: COLLINEAR points are dropped (the chain keeps only
↳ strict turns), so a set of
// collinear points collapses to its two extremes and under() still
↳ answers correctly. Inserting a
// point already inside is a no-op. An empty structure answers false to
↳ everything.
// EXACT on integers: the only arithmetic is a cross product, so
↳ |coord| <= 1e9 keeps it in ll.
struct UpHull { // the upper
↳ chain, left to right
    map<ll, ll> h;
    typedef map<ll, ll>::iterator it_t;
    Pi at(it_t i) { return Pi(i->fi, i->se); }
    bool under(ll x, ll y) { // is (x, y) on
↳ or below the chain?
        auto it = h.lower_bound(x);
        if (it == h.end() || (it == h.begin() && it->fi != x)) return
↳ false; // outside the span
        if (it->fi == x) return y <= it->se;
        return at(prev(it)).cross(at(it), Pi(x, y)) <= 0;
↳ // not strictly left
    }
    bool bad(it_t it) { // does *it sit
↳ on or below its neighbours?
        if (it == h.begin() || next(it) == h.end()) return false;
        return at(prev(it)).cross(at(next(it)), at(it)) <= 0;
    }
    void add(ll x, ll y) {
        if (under(x, y)) return;
        h[x] = y;
        auto it = h.find(x);
        while (next(it) != h.end() && bad(next(it))) h.erase(next(it));
        while (it != h.begin() && bad(prev(it))) h.erase(prev(it));
    }
};
// FULL HULL: UpHull up, dn; insert p as up.add(p.x, p.y) and
↳ dn.add(-p.x, -p.y);
// inside(p) = up.under(p.x, p.y) && dn.under(-p.x, -p.y).
// EXTREME POINT in direction d, O(log n): binary search the upper
↳ chain for the first edge whose
// direction turns from d-positive to d-negative, i.e. the last vertex
↳ with
// dot(next - cur, d) > 0. With a map you have to walk, so keep the
↳ chain in a
// vector-backed balanced BST (or an ordered set with order_of_key) if
↳ you need this a lot.
// AREA UNDER INSERTIONS, the usual companion query: keep ll s2 = sum
↳ of cross(p_i, p_{i+1}) over
// consecutive chain vertices. Every erase of a middle vertex b between
↳ a and c does
// s2 -= cross(a,b) + cross(b,c) - cross(a,c);
// and every insert of b between a and c does the same with a plus
↳ sign; erases and inserts at an
// end touch only the single adjacent term. The hull's area is |s2_up +
↳ s2_down| / 2 with the
// lower chain's points negated, since negating both coordinates leaves
↳ every cross product alone.
// OFFLINE ALTERNATIVE, almost always easier: if you know all the
↳ points up front and the queries
// are interleaved, sort by time and rebuild the hull in O(n log n)
↳ once per sqrt-block, or run
// divide and conquer on the time axis. Only type this file when the
↳ input is genuinely online.

```

11.1.17 ShamosHoeoy

SHAMOS-HOEY: do ANY TWO of n segments intersect? O(n log n), and it returns the witness pair.

```

// Needs: Core.cpp, Lines.cpp (segCross).
// The sweep keeps the active segments in a set ordered by their y at
↳ the sweep line. Two segments
// can only be the FIRST crossing pair if they are adjacent in that
↳ order immediately before they
// cross, so it is enough to test a segment against its two neighbours
↳ when it is inserted, and to
// test the two segments that become neighbours when one is removed.
// WHEN: "is this polygon simple", "do these n roads/walls cross", "is
↳ this a planar graph
// drawing" - anything where you only need YES/NO or one example, with

```

```

    ↪ n up to 1e5 so the  $O(n^2)$ 
// double loop (05 - Sweep.cpp, countSegInter) is out. isSimple (11 -
    ↪ PolygonOps.cpp) is the
// polygon-shaped special case and is  $O(n^2)$ ; replace it with this when
    ↪ n is large.
// COMPLEXITY:  $O(n \log n)$  time,  $O(n)$  memory.
// DEGENERATE: touching at a shared endpoint COUNTS as an intersection
    ↪ (segCross is inclusive), so
// for "is this polygon simple" you must skip the adjacent pairs
    ↪ yourself - easiest is to shrink
// every polygon edge by a hair, or to special-case the pair (i, i+1
    ↪ mod n) in the answer.
// Overlapping collinear segments are reported. Zero-length segments
    ↪ are reported against anything
// they touch. Vertical segments are fine here (unlike point location).
// The ANSWER (which pair) is exact - segCross is pure integer
    ↪ orientation. Only the SET ORDER
// uses floating point, and only to decide adjacency, so an eps slip
    ↪ can at worst make the sweep
// miss a crossing whose two segments are within  $1e-9$  of each other in
    ↪ y; with |coord|  $\leq 1e9$  and
// ld that does not happen for integer input.
struct SSeg {
    Pi a, b; int id; // normalised
    ↪ so that a.x  $\leq$  b.x
    ld yAt(ld x) const {
        return a.x == b.x ? (ld)a.y : a.y + (ld)(b.y - a.y) * (x - a.x)
    ↪ / (b.x - a.x);
    }
    bool operator<(const SSeg& o) const { // compare
    ↪ where both are already alive
        ld x = max((ld)a.x, (ld)o.a.x);
        return yAt(x) < o.yAt(x) - eps;
    }
};
pair<int, int> anyIntersect(const vector<pair<Pi, Pi>>& in) {
    int n = sz(in);
    vector<SSeg> s(n);
    vector<array<ll, 3>> ev; // {x, +1 open
    ↪ / -1 close, id}
    for (int i = 0; i < n; i++) {
        s[i] = {in[i].fi, in[i].se, i};
        if (s[i].b < s[i].a) swap(s[i].a, s[i].b);
        ev.pb({s[i].a.x, 1, i}); ev.pb({s[i].b.x, -1, i});
    }
    sort(all(ev), [](const array<ll, 3>& p, const array<ll, 3>& q) {
        return p[0] != q[0] ? p[0] < q[0] : p[1] > q[1];
    ↪ // open before close
    });
    set<SSeg> t;
    vector<set<SSeg>::iterator> pos(n);
    typedef set<SSeg>::iterator sit;
    auto hit = [&](sit x, sit y) {
        return x != t.end() && y != t.end() && segCross(x->a, x->b,
    ↪ y->a, y->b);
    };
    for (auto& e : ev) {
        int i = (int)e[2];
        if (e[1] > 0) {
            sit nx = t.lower_bound(s[i]), pv = nx == t.begin() ?
    ↪ t.end() : prev(nx);
            sit me = t.insert(nx, s[i]); // insert
    ↪ first, then test both neighbours
            if (hit(nx, me)) return {nx->id, i};
            if (hit(pv, me)) return {pv->id, i};
            pos[i] = me;
        } else {
            sit nx = next(pos[i]), pv = pos[i] == t.begin() ? t.end() :
    ↪ prev(pos[i]);
            if (hit(nx, pv)) return {pv->id, nx->id}; // they become
    ↪ adjacent once i leaves
            t.erase(pos[i]);
        }
    }
    return {-1, -1};
}
// COUNT or REPORT ALL k intersections: BENTLEY-OTTMANN,  $O((n + k) \log$ 
    ↪ n). Same sweep, but the
// event queue is a priority queue that also receives the intersection
    ↪ points themselves; when two
// segments swap order at a crossing you re-test each against its new
    ↪ neighbour. It is long,
// fiddly with degeneracies (three segments through one point), and
    ↪ almost never what a problem
// wants - if you need a COUNT of crossing pairs of ARBITRARY segments
    ↪ and n is 1e5, look for
// extra structure (all horizontal/vertical -> BIT sweep; all chords of

```

```

    ↪ a circle -> inversions).
// IS A POLYGON SIMPLE, in  $O(n \log n)$ : shrink every edge towards its
    ↪ midpoint by a factor
//  $(1 - 1e-9)$  so consecutive edges no longer share their endpoint, then
    ↪ this returns {-1, -1}
// exactly for a simple polygon. Shrinking is the honest fix; filtering
    ↪ "adjacent indices" out of
// the answer is not, because the sweep stops at the FIRST adjacent
    ↪ pair it meets.

```

11.1.18 InscribedCircle

LARGEST INSCRIBED CIRCLE IN A CONVEX POLYGON - the CHEBYSHEV CENTRE.
Returns {centre, radius}.

```

// Needs: Core.cpp, Polygon.cpp (area2).
// The trick is that  $f(p) = \min$  over edges of the signed distance from
    ↪ p to that edge's LINE is a
// minimum of affine functions, hence CONCAVE on the whole plane, and
    ↪ it is exactly the distance
// to the boundary for p inside. So a nested ternary search on (x, y)
    ↪ finds its maximum with no
// case analysis at all: no medial axis, no half-plane shrinking, no
    ↪ binary search on r.
// WHEN: "place a disc of maximum radius inside this region", "the
    ↪ point of a convex region
// farthest from its boundary", robot clearance, and 2D LP with an
    ↪ objective that is a min of
// linear functions (this IS that LP - maximise r subject to n linear
    ↪ constraints).
// COMPLEXITY:  $O(n * \text{it}^2)$  with it = 100 iterations per axis; about 1e4
    ↪ * n operations, so keep n
// below a few thousand. If n is large, run the same search on the  $O(n)$ 
    ↪ half-plane form or binary
// search r and test emptiness with halfPlaneInter (06) at  $O(n \log n)$ 
    ↪  $\log(1/\text{eps})$ .
// DEGENERATE: the polygon must be CONVEX and given in order; the code
    ↪ reverses a CW input for
// you. A degenerate (zero-area) polygon returns radius 0 at some point
    ↪ of it. Fewer than 3
// vertices returns radius 0. For a NON-CONVEX polygon f is not concave
    ↪ and this silently returns
// a local optimum - the correct answer there lives on the medial axis
    ↪ and is genuinely hard.
// FLOATING POINT, always: the answer is irrational. 100 ternary
    ↪ iterations over a  $2e9$ -wide box
// leave the centre accurate to  $\sim 1e-9$  * box, which is the best ld can
    ↪ do anyway.
pair<Pd, ld> inscribedCircle(vector<Pd> v) {
    int n = sz(v);
    if (n < 3) return {n ? v[0] : Pd(), 0};
    if (area2(v) < 0) reverse(all(v)); // f must be
    ↪ POSITIVE inside
    auto f = [&](ld x, ld y) { // distance to
    ↪ the nearest edge LINE
        ld r = 1e18;
        for (int i = 0; i < n; i++)
            r = min(r, v[i].cross(v[(i + 1) % n], Pd(x, y)) / (v[(i +
    ↪ 1) % n] - v[i]).dist());
        return r;
    };
    ld lo = 1e18, hi = -1e18, bo = 1e18, bi = -1e18;
    for (Pd& p : v) lo = min(lo, p.x), hi = max(hi, p.x), bo = min(bo,
    ↪ p.y), bi = max(bi, p.y);
    auto best = [&](ld x) { // max over y
    ↪ for a fixed x, still concave
        ld a = bo, b = bi;
        for (int it = 0; it < 100; it++) {
            ld m1 = a + (b - a) / 3, m2 = b - (b - a) / 3;
            f(x, m1) < f(x, m2) ? a = m1 : b = m2;
        }
        return (a + b) / 2;
    };
    for (int it = 0; it < 100; it++) {
        ld m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        f(m1, best(m1)) < f(m2, best(m2)) ? lo = m1 : hi = m2;
    }
    ld x = (lo + hi) / 2, y = best(x);
    return {Pd(x, y), max((ld)0, f(x, y))};
}
// WHY THE NESTED SEARCH IS LEGAL:  $g(x) = \max$  over y of  $f(x, y)$  is the
    ↪ partial maximisation of a
// concave function, and that is concave in x. The same argument
    ↪ licenses geoMedian (12) and every
// other nested ternary search - check concavity of the PARTIAL max,
    ↪ not just of f.
// LP FORM, if you prefer it or need n large: maximise r subject to
// cross(b_i - a_i, p - a_i) / |b_i - a_i|  $\geq$  r for every edge,
// which is n constraints in three variables (x, y, r) - Seidel's

```

```

    ↪ randomised LP solves it in O(n)
// expected. Equivalently, binary search r, push every edge line inward
    ↪ by r, and ask whether the
// half-plane intersection is non-empty.
// LARGEST INSCRIBED CIRCLE CENTRED ON A GIVEN LINE: one ternary search
    ↪ along the line.
// LARGEST INSCRIBED AXIS-ALIGNED RECTANGLE is a different and much
    ↪ harder problem - it needs a
// sweep with a stack and is not a two-line variation of this. Do not
    ↪ improvise it under time.
// MINIMUM ENCLOSING CIRCLE is mec (04 - Circles.cpp); minimum
    ↪ enclosing rectangle is minRectArea
// (03 - Polygon.cpp) by rotating calipers.

```

11.1.19 LineClip

TOTAL LENGTH OF A LINE INSIDE A POLYGON, including NON-CONVEX and self-intersecting ones, in

```

// Needs: Core.cpp.
// O(n log n). The line a -> b generally enters and leaves the region
    ↪ many times; this sums all of
// the pieces. Works by collecting the parameters t at which the line
    ↪ crosses each edge, tagging
// each crossing with the direction the boundary sweeps past it,
    ↪ sorting, and integrating the
// stretches where the running WINDING NUMBER is non-zero. cut() (03 -
    ↪ Polygon.cpp) only handles
// the convex case and only gives you one side; this gives you the
    ↪ measure.
// WHEN: "how much of this laser / road / sight line is inside the
    ↪ region", area of a polygon
// swept by a moving line, and any integral over a region done by
    ↪ integrating over parallel lines
// (fix a direction, sweep the offset, and this is the inner integrand).
// COMPLEXITY: O(n log n) for the sort, O(n) memory.
// DEGENERATE: a vertex lying exactly ON the line is handled by
    ↪ treating a zero orientation as
// POSITIVE, which is the same as nudging the line infinitesimally
    ↪ toward its right; that is
// consistent, so a vertex merely touching the line contributes nothing
    ↪ and a vertex the boundary
// genuinely passes through contributes once. An EDGE lying along the
    ↪ line is skipped for the same
// reason - if you need boundary-along-the-line to count, add its
    ↪ length yourself.
// SELF-INTERSECTING input is fine and uses the non-zero winding rule
    ↪ (windingNumber, 11), so a
// doubly-wound region counts once, not twice. Orientation (CW or CCW)
    ↪ does not matter.
// The crossing TEST is exact on P<ll> (|coord| <= 1e9); only the
    ↪ parameter t is floating point,
// so nearly-tangent edges can shift a boundary by ~1e-9 relative.
template <class P> ld lineInPoly(const vector<P>& v, P a, P b) {
    P d = b - a;
    int n = sz(v);
    vector<pair<ld, int>> ev;
    for (int i = 0; i < n; i++) {
        P p = v[i], q = v[(i + 1) % n];
        int s1 = sgn(d.cross(p - a)), s2 = sgn(d.cross(q - a));
        if (!s1) s1 = 1; // a point ON
        ↪ the line counts as LEFT
        if (!s2) s2 = 1;
        if (s1 == s2) continue;
        ev.pb({(ld)(p - a).cross(q - p) / d.cross(q - p), s1 > s2 ? 1 :
        ↪ -1});
    }
    sort(all(ev));
    ld len = 0;
    int c = 0;
    for (int i = 0; i + 1 < sz(ev); i++)
        if ((c += ev[i].second)) len += ev[i + 1].fi - ev[i].fi;
    ↪ // inside <=> winding != 0
    return len * d.dist();
}
// SEGMENT instead of a line: same events, but clamp the accumulated
    ↪ interval to [0, 1] before
// adding, i.e. len += max((ld)0, min((ld)1, ev[i+1].fi) - max((ld)0,
    ↪ ev[i].fi)).
// RAY: clamp to [0, infinity) the same way.
// THE PIECES THEMSELVES, not just their total length: emit the
    ↪ interval [ev[i].fi, ev[i+1].fi]
// whenever the running count is non-zero; a + d * t gives the
    ↪ endpoints.
// AREA OF THE POLYGON, as a sanity check on this file: integrate
    ↪ lineInPoly over a family of
// parallel lines. That is also how you get "the area of the part of a
    ↪ polygon on one side of an
// arbitrary curve" when the curve is not a line and cut() does not

```

```

    ↪ apply.
// LINE x CONVEX POLYGON in O(log n): lineHull (11 - PolygonOps.cpp)
    ↪ gives the two crossed edges
// directly; there is exactly one interval and you do not need this
    ↪ file.
// LINE x CIRCLE is circleLine (04); LINE x DISC intersected with a
    ↪ polygon: clip against both.

```

11.1.20 MaxAreaPolygon

MAXIMUM-AREA POLYGON WITH GIVEN SIDE LENGTHS. Given n side lengths in any order, the polygon

```

// of largest area that has exactly those sides is CYCLIC - all its
    ↪ vertices lie on one circle
// (and the order of the sides does not matter, which is the surprising
    ↪ half of the theorem). So
// the whole problem is "find the circumradius R", and each side a_i
    ↪ subtends a central angle
// 2 * asin(a_i / (2R)). Two cases: if the centre is INSIDE, the angles
    ↪ sum to 2 * pi; if the
// longest side is long enough to push the centre OUTSIDE, its angle is
    ↪ subtracted instead and the
// signed sum is 0. Both are monotone in R, so binary search.
// WHEN: "arrange these n sticks into the polygon of largest area"; the
    ↪ n = 3 case is Heron and
// the n = 4 case is Brahmagupta, and this is the general answer for
    ↪ both. Also the standing proof
// that a regular polygon maximises area among equilateral ones.
// COMPLEXITY: O(n * 100) - one binary search with 100 iterations, each
    ↪ O(n).
// DEGENERATE: no polygon exists at all unless the longest side is
    ↪ strictly shorter than the sum
// of the others (the polygon inequality); the routine returns -1 for
    ↪ that, and for n < 3. Sides
// of length 0 are harmless. All sides equal gives the regular n-gon,
    ↪ area n r^2 sin(2 pi / n) / 2
// with r = a / (2 sin(pi / n)) - use that to check the code.
// FLOATING POINT, necessarily: R is irrational. asin loses precision
    ↪ when a_i / (2R) is near 1,
// which is exactly the degenerate "longest side is a diameter" case,
    ↪ so clamp the argument.
ld maxAreaPolygon(vector<ld> a) {
    int n = sz(a);
    if (n < 3) return -1;
    int m = max_element(all(a)) - a.begin();
    ld s = 0;
    for (int i = 0; i < n; i++) if (i != m) s += a[i];
    if (a[m] >= s) return -1; // the polygon
    ↪ inequality fails
    auto ang = [&](ld R, int sg) { // sum of
        ↪ central angles, a[m] signed
        ld t = 0;
        for (int i = 0; i < n; i++) {
            ld u = 2 * asinl(min((ld)1, a[i] / (2 * R)));
            t += (i == m ? sg * u : u);
        }
        return t;
    };
    ld lo = a[m] / 2, hi = 1e18; // R >= a[m] /
    ↪ 2 always
    int sg = ang(lo, 1) > 2 * PI ? 1 : -1; // centre
    ↪ inside (+1) or outside (-1)
    ld tgt = sg > 0 ? 2 * PI : 0; // ang(.) is
    ↪ DECREASING in R for sg = +1,
    for (int it = 0; it < 200; it++) { // INCREASING
        ↪ for sg = -1
        ld mid = (lo + hi) / 2;
        ((ang(mid, sg) > tgt) == (sg > 0) ? lo : hi) = mid;
    }
    ld R = (lo + hi) / 2, area = 0;
    for (int i = 0; i < n; i++) {
        ld u = 2 * asinl(min((ld)1, a[i] / (2 * R)));
        area += (i == m ? sg : 1) * R * R * sinl(u) / 2;
    }
    return area;
}
// THE VERTICES, if the problem wants them and not just the area: walk
    ↪ the angles. Start at
// (R, 0), and after each side rotate by its central angle (negated for
    ↪ the long side when the
// centre is outside). p_0 = R * (cos 0, sin 0), theta_{k+1} = theta_k
    ↪ + u_k, p_k = R * (cos, sin).
// TRIANGLE (n = 3), closed form, no search: HERON with the stable
    ↪ sorted form,
// sort a >= b >= c, A = sqrt((a+(b+c))(c-(a-b))(c+(a-b))(a+(b-c))) /
    ↪ 4.
// CYCLIC QUADRILATERAL (n = 4): BRAHMGUPTA, A =
    ↪ sqrt((s-a)(s-b)(s-c)(s-d)) with s the

```

```
// semiperimeter - and by this theorem that IS the maximum over all
//   ↳ quadrilaterals with those
// sides. General quadrilateral with diagonals p, q (BRETSCHNEIDER):
//   A = sqrt(4 p^2 q^2 - (b^2 + d^2 - a^2 - c^2)^2) / 4.
// MAXIMUM AREA WITH GIVEN PERIMETER, no side constraints: the circle.
//   ↳ Among n-gons: the regular
// one. This file is the constrained version of the same isoperimetric
//   ↳ fact.
// A DIFFERENT PROBLEM often confused with this one: pick the triangle
//   ↳ of maximum area from a set
// of given points - that is rotating calipers on the hull, O(n^2), and
//   ↳ is not here.
```

11.1.21 DynamicHalfPlane

DYNAMIC HALF-PLANE INTERSECTION - a convex region you can CUT online, in O(log n) amortised per

```
// Needs: Core.cpp (angCmp).
// cut, while asking for its area or its extreme point in any direction
//   ↳ between cuts. The static
// routine (06 - HalfPlaneIntersection.cpp) needs every constraint up
//   ↳ front and re-sorts; this one
// does not, so it is what you want when the constraints arrive one at
//   ↳ a time, or when you must
// report an answer after each one.
// REPRESENTATION: the region's boundary, as a multimap from EDGE
//   ↳ DIRECTION to the vertex where
// that edge begins, in counter-clockwise angular order. That single
//   ↳ choice is the whole trick -
// the edge a new cut attacks is found by one lower_bound on its
//   ↳ direction, and every vertex is
// erased at most once, so the amortised cost is a lookup plus the
//   ↳ erases you paid to insert.
// It starts as a huge box, so the region is always bounded; make B
//   ↳ larger than any coordinate you
// care about and remember that a surviving box corner is an artefact,
//   ↳ not an answer.
// WHEN: "after each of q constraints, print the area of the feasible
//   ↳ region"; online 2D LP;
// incremental visibility / lit-region problems; kernel of a polygon
//   ↳ built edge by edge.
// COMPLEXITY: O(log n) amortised per cut, O(log n) per extreme-point
//   ↳ query, O(n) per area query
// (keep a running sum of cross products if you need area after every
//   ↳ cut - update it in the same
// places the code erases and inserts).
// DEGENERATE: cutting with a line that misses the region entirely is a
//   ↳ no-op; cutting everything
// away leaves the map EMPTY and every later cut returns immediately,
//   ↳ so check empty() for
// infeasible. Duplicate directions are allowed (that is why it is a
//   ↳ multimap). A cut exactly
// along an existing edge removes the now-redundant vertex.
// FLOATING POINT: directions and vertices are ld and compared with the
//   ↳ usual eps through sgn.
// Feed it integer-valued Pd if you can; a cut that is within eps of
//   ↳ tangent may keep or drop a
// zero-length edge, so filter edges shorter than eps before counting
//   ↳ them.
```

```
struct DirLess { bool operator()(Pd a, Pd b) const { return
//   ↳ angCmp<ld>(a, b); } };
struct HPSet {
    typedef multimap<Pd, Pd, DirLess> M;
    typedef M::iterator it_t;
    M m; // edge
    // direction -> start vertex of edge
    HPSet(ld B = 1e9) {
        m.insert({Pd(1, 0), Pd(-B, -B)}), m.insert({Pd(0, 1), Pd(B,
//   ↳ -B)});
        m.insert({Pd(-1, 0), Pd(B, B)}), m.insert({Pd(0, -1), Pd(-B,
//   ↳ B)});
    }
    it_t nx(it_t i) { return next(i) == m.end() ? m.begin() : next(i); }
    it_t pv(it_t i) { return i == m.begin() ? prev(m.end()) : prev(i); }
    it_t fix(it_t i) { return i == m.end() ? m.begin() : i; }
    void cut(Pd a, Pd b) { // keep only
//   ↳ the part LEFT of a -> b
        if (m.empty()) return;
        int old = sz(m);
        Pd d = b - a;
        auto ev = [&](it_t i) { return sgn(a.cross(b, i->second)); };
        auto ix = [&](it_t i) { // where i's
//   ↳ edge meets the line a -> b
            return i->second + i->first * (d.cross(a - i->second) /
//   ↳ d.cross(i->first));
        };
        it_t it = fix(m.lower_bound(d)); // the vertex
//   ↳ farthest RIGHT of d
```

```
        if (ev(it) >= 0) return; // it survives,
//   ↳ so everything does
        while (sz(m) && ev(pv(it)) < 0) m.erase(pv(it));
        while (sz(m) && ev(nx(it)) < 0) it = fix(m.erase(it));
        if (m.empty()) return;
        if (ev(nx(it)) > 0) it->second = ix(it); // pull this
//   ↳ vertex onto the cutting line
        else it = fix(m.erase(it));
        if (old <= 2) return;
        it = pv(it);
        m.insert(it, {d, ix(it)}); // the new edge
//   ↳ starts where we re-entered
        if (ev(it) == 0) m.erase(it); // the old
//   ↳ vertex is now redundant
    }
    Pd extreme(Pd dir) { // vertex
//   ↳ maximising dot(v, dir), O(log n)
        return fix(m.lower_bound(dir.perp()))->second;
    }
    ld area() {
        if (sz(m) <= 2) return 0;
        ld s = 0;
        for (it_t i = m.begin(); i != m.end(); ++i) s +=
//   ↳ i->second.cross(nx(i->second);
        return fabs(s) / 2;
    }
    vector<Pd> poly() { // the region
//   ↳ as a CCW vertex list
        vector<Pd> v;
        for (auto& e : m) v.pb(e.second);
        return v;
    }
};
// USAGE: HPSet h; then h.cut(p, q) for each constraint "left of the
//   ↳ directed line p -> q", which
// is the same convention as HP in 06. To cut with the half-plane {x :
//   ↳ n.dot(x) <= c}, take a
// point p on the boundary line and cut(p, p + n.perp()).
// AREA AFTER EVERY CUT in O(log n): keep ld s = sum of cross(v_i,
//   ↳ v_{i+1}). Every erase of a
// vertex b between a and c does s -= cross(a,b) + cross(b,c) -
//   ↳ cross(a,c), every insert adds it,
// and moving a vertex is an erase plus an insert. Same bookkeeping as
//   ↳ 16 - DynamicHull.cpp.
// 2D LP ONLINE: after each constraint the optimum of a linear
//   ↳ objective c is extreme(c) - one
// lower_bound, no scan. That is the reason to prefer this over
//   ↳ rebuilding with 06.
// DELETIONS are not supported. If constraints are added AND removed,
//   ↳ go offline: segment tree on
// time, and rebuild with the static routine in each node.
```

11.1.22 OnionDecomposition

ONION DECOMPOSITION (convex layers) in O(n log^2 n): strip the convex hull, then the hull of

```
// Needs: Core.cpp.
// what remains, and so on, and report which layer each point lands in.
//   ↳ The naive loop in
// 05 - Sweep.cpp rebuilds a hull per layer and is O(n^2 log n), which
//   ↳ dies at n = 1e5 on the
// worst input (points on nested circles give Theta(n^(2/3)) layers,
//   ↳ and a spiral gives Theta(n)).
// THE STRUCTURE: a DECREMENTAL hull - it only supports DELETE, which
//   ↳ is exactly what peeling
// needs. A segment tree over the points sorted by y stores, in every
//   ↳ node, the two endpoints of
// the BRIDGE joining the hulls of its two children; pull() finds that
//   ↳ bridge by the classic
// simultaneous descent in O(log n), so a delete costs O(log^2 n) and
//   ↳ reading off the current hull
// costs O(hull size). Two copies handle the two sides: the LEFT hull
//   ↳ of the points, and the left
// hull of the negated-and-reversed points, which is the right hull.
// WHEN: "how many peels until the set is empty", "which layer is point
//   ↳ i on", k-th layer queries,
// and the classic "repeatedly remove the hull" simulation. Also the
//   ↳ fastest route to the DEPTH of
// a point set (the number of layers), which some
//   ↳ rotating-calipers-free arguments need.
// COMPLEXITY: O(n log^2 n) time, O(n) memory. Beats the naive version
//   ↳ from about n = 3000.
// DEGENERATE: ALL POINTS MUST BE DISTINCT - dedupe first. COLLINEAR
//   ↳ points on a hull edge are
// KEPT on that layer (the bridge search uses strict turns, so a point
//   ↳ in the middle of an edge is
// still reported), which differs from hull() in 03 - Polygon.cpp; if
//   ↳ you need them dropped,
```

```
// filter each layer afterwards. n <= 2 puts everything on layer 1.
// EXACT on P<ll>: cross fits in ll for |coord| <= 1e9, and the one
  ↳ product of a cross with a
// coordinate inside pull() is widened to __int128, so nothing here
  ↳ overflows at that budget.
struct LeftHull {                                // points must
  ↳ be sorted by (y, x)
  struct Nd { int bl, br, L, R, lc, rc; };
  vector<Pi> ps;
  vector<Nd> t;
  int root = 0;
  bool leaf(int w) { return t[w].lc < 0 && t[w].rc < 0; }
  void pull(int w) {                               // find the
  ↳ bridge between the two children
    int l = t[w].lc, r = t[w].rc;
    ll sy = ps[t[r].L].y;
    while (!leaf(l) || !leaf(r)) {
      int a = t[l].bl, b = t[l].br, c = t[r].bl, d = t[r].br;
      if (a != b && ps[a].cross(ps[b], ps[c]) > 0) l = t[l].lc;
      else if (c != d && ps[b].cross(ps[c], ps[d]) > 0) r =
  ↳ t[r].rc;
      else if (a == b) r = t[r].lc;
      else if (c == d) l = t[l].rc;
      else {
        ll s1 = ps[a].cross(ps[b], ps[c]), s2 =
  ↳ ps[b].cross(ps[a], ps[d]);
        if (s1 + s2 == 0 || ((__int128)s1 * ps[d].y +
  ↳ ((__int128)s2 * ps[c].y <
        ↳ ((__int128)sy * (s1 + s2)) l =
  ↳ t[l].rc;
        else r = t[r].lc;
      }
    }
    t[w].bl = t[l].L, t[w].br = t[r].L;
  }
  void build(int w, int L, int R) {
    t[w].L = L, t[w].R = R;
    if (R - L == 1) { t[w].lc = t[w].rc = -1, t[w].bl = t[w].br =
  ↳ L; return; }
    int M = (L + R) / 2;
    t[w].lc = w + 1, t[w].rc = w + 2 * (M - L);
    build(t[w].lc, L, M), build(t[w].rc, M, R), pull(w);
  }
  int erase(int w, int L, int R) {
    if (R <= t[w].L || L >= t[w].R) return w;
    if (L <= t[w].L && R >= t[w].R) return -1;
    t[w].lc = erase(t[w].lc, L, R), t[w].rc = erase(t[w].rc, L, R);
    if (t[w].lc < 0) return t[w].rc;
    if (t[w].rc < 0) return t[w].lc;
    return pull(w), w;
  }
  void walk(int w, int l, int r, vector<int>& res) {
    if (leaf(w)) res.pb(t[w].L);
    else if (r <= t[w].bl) walk(t[w].lc, l, r, res);
    else if (l >= t[w].br) walk(t[w].rc, l, r, res);
    else walk(t[w].lc, l, t[w].bl, res), walk(t[w].rc, t[w].br, r,
  ↳ res);
  }
  LeftHull(const vector<Pi>& p) : ps(p), t(max<size_t>(2, p.size() *
  ↳ 2)) { build(0, 0, sz(p)); }
  void erase(int i) { root = erase(root, i, i + 1); }
  vector<int> hull() {
    vector<int> res;
    if (root >= 0) walk(root, 0, sz(ps) - 1, res);
    return res;
  }
};
// layer[i] = 1-based convex-layer index of the ORIGINAL point i. All
  ↳ points must be distinct.
vector<int> onionLayers(const vector<Pi>& p) {
  int n = sz(p);
  if (!n) return {};
  vector<int> ord(n), lay(n), res(n);
  iota(all(ord), 0);
  sort(all(ord), [&](int i, int j) { return tie(p[i].y, p[i].x) <
  ↳ tie(p[j].y, p[j].x); });
  vector<Pi> a(n), b(n);
  for (int i = 0; i < n; i++) a[i] = p[ord[i]];
  for (int i = 0; i < n; i++) b[i] = Pi(-a[n - 1 - i].x, -a[n - 1 -
  ↳ i].y); // the right hull
  LeftHull L(a), R(b);
  for (int id = 1, cnt = 0; cnt < n; id++) {
    set<int> h;
    for (int i : L.hull()) h.insert(i);
    for (int i : R.hull()) h.insert(n - 1 - i);
    for (int i : h) cnt++, lay[i] = id, L.erase(i), R.erase(n - 1 -
  ↳ i);
  }
};
```

```
}
for (int i = 0; i < n; i++) res[ord[i]] = lay[i];
return res;
}
// The same structure answers "hull of the set after these deletions"
  ↳ at any moment, so it also
// does OFFLINE DELETIONS on a hull: process the deletes in order and
  ↳ read hull() in between.
// INSERTIONS instead of deletions are 16 - DynamicHull.cpp; you cannot
  ↳ have both here.
// LAYER COUNT bounds worth knowing: n points in convex position give 1
  ↳ layer, a spiral gives n/3,
// and a uniform random square gives Theta(n^(2/3)) - so "peel until
  ↳ empty" is NOT O(sqrt n).
```

11.2 Geometry 3D

11.2.1 Geometry3D

3D point / vector, planes, and the few 3D things that actually appear.

```
template <class T>
struct P3 {
  typedef P3 Pt; T x = 0, y = 0, z = 0;
  P3(T x = 0, T y = 0, T z = 0) : x(x), y(y), z(z) {}
  Pt operator+(Pt p) const { return {x + p.x, y + p.y, z + p.z}; }
  Pt operator-(Pt p) const { return {x - p.x, y - p.y, z - p.z}; }
  Pt operator*(T d) const { return {x * d, y * d, z * d}; }
  Pt operator/(T d) const { return {x / d, y / d, z / d}; }
  T dot(Pt p) const { return x * p.x + y * p.y + z * p.z; }
  Pt cross(Pt p) const { return {y * p.z - z * p.y, z * p.x - x *
  ↳ p.z, x * p.y - y * p.x}; }
  T dist2() const { return x * x + y * y + z * z; }
  ld dist() const { return sqrtl((ld)dist2()); }
  Pt unit() const { return *this / dist(); }
  ld phi() const { return atan2l(y, x); } //
  ↳ longitude, (-pi, pi]
  ld theta() const { return atan2l(sqrtl((ld)(x * x + y * y)), z); }
  ↳ // colatitude, [0, pi]
  Pt rot(ld a, Pt ax) const //
  ↳ Rodrigues, ccw around ax
    Pt u = ax.unit(); ld s = sinl(a), c = cosl(a);
    return u * dot(u) * (1 - c) + (*this) * c + u.cross(*this) * s;
  }
  bool operator<(Pt p) const { return tie(x, y, z) < tie(p.x, p.y,
  ↳ p.z); }
  bool operator==(Pt p) const { return tie(x, y, z) == tie(p.x, p.y,
  ↳ p.z); }
  friend istream& operator>>(istream& i, Pt& p) { return i >> p.x >>
  ↳ p.y >> p.z; }
  friend ostream& operator<<(ostream& o, Pt p) { return o << p.x <<
  ↳ p.y << p.z; }
};
typedef P3<ll> Pi3;
typedef P3<ld> Pd3;

// PLANE n.dot(X) = d. Build from 3 points, or from a normal and a
  ↳ point.
template <class P> struct Plane {
  P n; ld d;
  Plane(P n = P(), ld d = 0) : n(n), d(d) {}
  Plane(P a, P b, P c) : n((b - a).cross(c - a)), d(n.dot(a)) {}
  ld side(P p) const { return n.dot(p) - d; } //
  ↳ sign = which side
  ld dist(P p) const { return fabsl(side(p)) / n.dist(); }
  P proj(P p) const { return p - n * (side(p) / n.dist2()); }
  P refl(P p) const { return p - n * (2 * side(p) / n.dist2()); }
};
// LINE a->b vs plane. {0 parallel-disjoint, 1 unique, -1 line lies in
  ↳ plane}
template <class P> pair<int, P> planeLine(const Plane<P>& pl, P a, P b)
  ↳ {
  ld da = pl.side(a), db = pl.side(b);
  if (fabsl(da - db) < eps) return {fabsl(da) < eps ? -1 : 0, P()};
  return {1, (b * da - a * db) / (da - db)};
}
// PLANE x PLANE -> a line, given as {point on it, direction}. dir==0
  ↳ means parallel.
template <class P> pair<P, P> planePlane(const Plane<P>& A, const
  ↳ Plane<P>& B) {
  P dir = A.n.cross(B.n);
  if (dir.dist2() < eps) return {P(), P()};
  P p = (B.n * A.d - A.n * B.d).cross(dir) / dir.dist2();
  return {p, dir};
}
// VOLUME of the tetrahedron abcd, signed. The x6 form is exact on
```

```

    ↪ P3<ll> only up to
// |coordinate| ~ 5.7e5, since |vol6| <= 48 * C^3; above that make the
    ↪ return type __int128.
template <class P> auto vol6(P a, P b, P c, P d) { return (b -
    ↪ a).cross(c - a).dot(d - a); }
template <class P> ld tetraVolume(P a, P b, P c, P d) { return
    ↪ fabsl((ld)vol6(a, b, c, d)) / 6; }

// SPHERE x LINE: points where line a->b meets the sphere (o, r).
template <class P> vector<P> sphereLine(P o, ld r, P a, P b) {
    P d = b - a, f = a - o;
    ld A = d.dist2(), B = 2 * f.dot(d), C = f.dist2() - r * r, D = B *
    ↪ B - 4 * A * C;
    if (D < -eps) return {};
    D = sqrtl(max((ld)0, D));
    if (D < eps) return {a + d * (-B / (2 * A))};
    return {a + d * ((-B - D) / (2 * A)), a + d * ((-B + D) / (2 * A))};
}

// GREAT-CIRCLE distance on a sphere of radius R, inputs in DEGREES
    ↪ (haversine).
ld greatCircle(ld lat1, ld lon1, ld lat2, ld lon2, ld R) {
    auto rad = [] (ld d) { return d * PI / 180; };
    ld dL = rad(lat2 - lat1) / 2, dg = rad(lon2 - lon1) / 2;
    ld h = sinl(dL) * sinl(dL) + cosl(rad(lat1)) * cosl(rad(lat2)) *
    ↪ sinl(dg) * sinl(dg);
    return 2 * R * asinl(sqrtl(min((ld)1, h)));
}
}

11.2.2 Lines3D
LINES, SEGMENTS AND TRIANGLES IN 3D. L3 is the line {o + t*d} through two
points, carrying
// Needs: Geometry3D.cpp, Core.cpp (sgn).
// distance / projection / reflection of a point. On top of it the two
    ↪ things everyone re-derives
// under pressure: the SMALLEST DISTANCE BETWEEN TWO LINES (project the
    ↪ gap onto the common normal
// d1 x d2) and the PAIR OF CLOSEST POINTS realising it, one on each
    ↪ line. Then point-segment,
// segment-segment and point-triangle distance, and SEGMENT x TRIANGLE
    ↪ intersection (does segment
// d-e cross triangle abc, and where). Everything is O(1) time and
    ↪ memory.
// You reach for this the moment the input stops being planar: two 3D
    ↪ lines generically MISS, so
// lineInter is meaningless and "distance between the lines" replaces
    ↪ it; and mesh problems
// (visibility, is this ray blocked, closest approach of two moving
    ↪ points) are exactly segTri and
// segSegDist3.
// DEGENERATE: parallel lines fall back to point-line, and lineClosest
    ↪ then returns {a.o, its foot
// on b} - one of infinitely many optimal pairs. A zero-length segment
    ↪ degrades to its point. A
// collinear triangle makes triDist3 return the min over its three
    ↪ edges and makes segTri report
// -1. segTri with d == e reports 0 (or -1 if that point is in the
    ↪ plane of abc), never 1.
// EXACTNESS: the predicates are sgn(vol6), so segTri is EXACT on
    ↪ P3<ll> up to |coord| <= 5.7e5
// (the vol6 budget of Geometry3D.cpp; wrap vol6 in __int128 for 1e9).
// triDist3's inside test and
// L3::dist2 are DEGREE-4 forms (((v-u) x (p-u)).n and |d x (p-o)|^2),
    ↪ so they hold in ll only for
// |coord| <= 1.5e4 - again __int128 lifts that to 9e8. Anything
    ↪ returning a distance or a point
// divides, so it comes back in ld whatever you passed in; and note
// proj / refl / lineClosest silently TRUNCATE on P3<ll>. The eps tests
    ↪ below are absolute, so
// scale your input to O(1)..O(1e6) if you care about near-degenerate
    ↪ cases.
template <class P> struct L3 {
    P o, d; // {o +
    ↪ t*d}; d must be nonzero
    L3(P a = P(), P b = P()) : o(a), d(b - a) {} // point +
    ↪ DIRECTION: L3(o, o + dir)
    ld dist2(P p) const { return (ld)d.cross(p - o).dist2() /
    ↪ d.dist2(); }
    ld dist(P p) const { return sqrtl(dist2(p)); } // ratio
    ↪ of two exact integer forms
    P proj(P p) const { return o + d * ((ld)d.dot(p - o) / d.dist2()); }
    P refl(P p) const { return proj(p) * 2 - p; }
    bool cmpProj(P p, P q) const { return d.dot(p) < d.dot(q); } //
    ↪ sort points ALONG the line
};
// SMALLEST DISTANCE BETWEEN TWO LINES: the gap vector measured along
    ↪ the common normal.

```

```

template <class P> ld lineLineDist(L3<P> a, L3<P> b) {
    P n = a.d.cross(b.d);
    if (n == P()) return a.dist(b.o); //
    ↪ parallel: any point of b will do
    return fabsl((ld)(b.o - a.o).dot(n)) / n.dist();
}
// The pair of points realising it, {on a, on b}. The point on a is
    ↪ where a pierces the plane
// spanned by b and the common normal, so the "closest point on a"
    ↪ needs only one division.
template <class P> pair<P, P> lineClosest(L3<P> a, L3<P> b) {
    P n = a.d.cross(b.d);
    if (n == P()) return {a.o, b.proj(a.o)}; //
    ↪ parallel: infinitely many pairs
    P na = b.d.cross(n), nb = a.d.cross(n);
    return {a.o + a.d * ((ld)(b.o - a.o).dot(na) / a.d.dot(na)),
        ↪ b.o + b.d * ((ld)(a.o - b.o).dot(nb) / b.d.dot(nb))};
}
// POINT -> SEGMENT. Clamp the line parameter to [0, 1] and expand the
    ↪ norm, so no point of type
// P is ever constructed: this stays correct when you hand it P3<ll>.
template <class P> ld segDist3(P p, P a, P b) {
    P u = b - a, w = p - a;
    ld L = (ld)u.dist2(), uw = (ld)u.dot(w);
    if (L < eps) return w.dist(); // a == b
    ld t = min((ld)1, max((ld)0, uw / L));
    return sqrtl(max((ld)0, (ld)w.dist2() - t * (2 * uw - t * L)));
}
// SEGMENT -> SEGMENT. The minimum is either at an endpoint or at the
    ↪ unconstrained closest pair
// of the two LINES, and that one only counts when both parameters land
    ↪ strictly inside (0, 1).
// den = |u|^2|v|^2 - (u.v)^2 loses absolute precision at huge
    ↪ coordinates, but a garbage (s, t)
// still names two REAL points of the segments, so it can only fail to
    ↪ improve the answer.
template <class P> ld segSegDist3(P a, P b, P c, P d) {
    P u = b - a, v = d - c, w = a - c;
    ld A = (ld)u.dist2(), B = (ld)u.dot(v), C = (ld)v.dist2();
    ld D = (ld)u.dot(w), E = (ld)v.dot(w), W = (ld)w.dist2(), den = A *
    ↪ C - B * B;
    ld r = min({segDist3(a, c, d), segDist3(b, c, d), segDist3(c, a,
    ↪ b), segDist3(d, a, b)});
    if (den > eps) {
        ld s = (B * E - C * D) / den, t = (A * E - B * D) / den;
        if (s > 0 && s < 1 && t > 0 && t < 1) // |w +
        ↪ s*u - t*v|^2, expanded
            r = min(r, sqrtl(max((ld)0, W + s * s * A + t * t * C
                ↪ + 2 * s * D - 2 * t * E - 2 * s *
                ↪ t * B)));
    }
    return r; //
    ↪ parallel / collinear: endpoints win
}
// POINT -> TRIANGLE (the solid triangle). Project onto the plane; if
    ↪ the projection is inside,
// the plane distance is the answer, else the min over the three edges.
    ↪ The inside test uses p
// itself, NOT the projection: p and its projection differ by a
    ↪ multiple of n, and ((v-u) x n).n
// is 0, so the three orientations are unchanged - which keeps the test
    ↪ exact on integers.
template <class P> ld triDist3(P p, P a, P b, P c) {
    P n = (b - a).cross(c - a);
    ld e = min({segDist3(p, a, b), segDist3(p, b, c), segDist3(p, c,
    ↪ a)});
    if (n.dist2() < eps) return e; //
    ↪ collinear abc: no plane to hit
    auto f = [&](P u, P v) { return (v - u).cross(p - u).dot(n); };
    auto s1 = f(a, b), s2 = f(b, c), s3 = f(c, a);
    bool in = (s1 >= 0 && s2 >= 0 && s3 >= 0) || (s1 <= 0 && s2 <= 0 &&
    ↪ s3 <= 0);
    return in ? fabsl((ld)n.dot(p - a)) / n.dist() : e; // no eps:
    ↪ the two branches AGREE on
}
    ↪ boundary, so either is right
// SEGMENT d-e x TRIANGLE a-b-c, by four orientation signs and nothing
    ↪ else.
// {1, p} they meet in the single point p (grazing a vertex or an edge
    ↪ counts); {0, _} disjoint;
// {-1, _} d-e is COPLANAR with abc, where the meet can be empty, a
    ↪ point or a whole segment -
// map both to 2D with 03 - CoplanarFrame.cpp and clip there.
template <class P> pair<int, Pd3> segTri(P a, P b, P c, P d, P e) {
    auto va = vol6(a, b, c, d), vb = vol6(a, b, c, e);
    int sa = sgn(va), sb = sgn(vb);
    if (!sa && !sb) return {-1, Pd3()}; // (also

```

```

    ↪ fires when abc is collinear)
    if (sa == sb) return {0, Pd3()}; // segment
    ↪ strictly on one side
    int t1 = sgn(vol6(d, e, a, b)), t2 = sgn(vol6(d, e, b, c)), t3 =
    ↪ sgn(vol6(d, e, c, a));
    // The LINE d-e pierces the triangle iff it turns the same way
    ↪ around all three edges.
    if (!(t1 >= 0 && t2 >= 0 && t3 >= 0) || (t1 <= 0 && t2 <= 0 && t3
    ↪ <= 0)) return {0, Pd3()};
    Pd3 D((ld)d.x, (ld)d.y, (ld)d.z), E((ld)e.x, (ld)e.y, (ld)e.z);
    return {1, (E * (ld)va - D * (ld)vb) / ((ld)va - (ld)vb)}; // =
    ↪ planeLine, in vol6 terms
}
/* RELATED
SEGMENT x PLANE      planeLine (01) then check the parameter lies
    ↪ in [0, 1].
TRIANGLE -> TRIANGLE min over the 6 (edge, triangle) pairs of
    ↪ segment-to-triangle distance;
    ↪ that one is a ternary search over the segment
    ↪ parameter - the distance
    ↪ from a point to a convex set is convex, so it
    ↪ is safe.
SEGMENT -> TRIANGLE same ternary search; 0 exactly when segTri
    ↪ says 1.
RAY x TRIANGLE      same four signs, but replace the sa != sb
    ↪ test with sa != 0 and sa
    ↪ opposite to the side the ray points at
    ↪ (Moller-Trumbore is this test
    ↪ with the divisions folded in - no reason to
    ↪ type it).
TWO SEGMENTS MEET   in 3D only if coplanar: vol6(a, b, c, d) ==
    ↪ 0, then use the 2D
    ↪ segInter inside a coplanar frame. segSegDist3
    ↪ == 0 is the cheap test.
CLOSEST APPROACH of two points moving linearly = segSegDist3 of the
    ↪ two trajectories with a
    ↪ shared parameter; if the speeds differ,
    ↪ minimise |w + t(v1 - v2)|^2
    ↪ directly, t* = -w.(v1-v2) / |v1-v2|^2 clamped
    ↪ to the time window. */

```

11.2.3 CoplanarFrame

THE COPLANAR FRAME - the single most valuable file in 3D geometry, because it deletes 3D.

```

// Needs: Geometry3D.cpp, Core.cpp, Polygon.cpp.
// Given points known to lie on one plane, it builds a frame (o, dx,
    ↪ dy, dz) with dx, dy in the
// plane and dz along the normal, and maps each point to a 2D point:
    ↪ pos2d. From there EVERY 2D
// routine in the notebook (hull, area2, inPoly, inConvex, rotating
    ↪ calipers, cut, minkowski,
// Pick, ...) applies UNCHANGED, and to3d maps the answer back. pos3d
    ↪ keeps the third coordinate,
// which is the signed offset off the plane - 0 exactly for the
    ↪ coplanar points.
// You reach for it whenever a statement says "a polygon in space",
    ↪ "the points lie on a common
// plane", or you have just cut a polyhedron by a plane and hold the
    ↪ cross-section as a bag of
// unordered 3D points. Building it is O(1); mapping is O(1) per point.
//
// TWO FRAMES, AND THE CHOICE IS THE WHOLE POINT.
// ORTHONORMAL (ortho = true, the default): dx, dy, dz are unit and
    ↪ mutually perpendicular, so the
// map is a rigid motion - DISTANCES, ANGLES and AREAS are preserved
    ↪ exactly as they are in 3D.
// It normalises, so it needs P3<ld> and inherits float error.
// AFFINE (ortho = false): dx = q - p and dy = r - p as they came in,
    ↪ no normalisation, so it
// stays on P3<ll> and every 2D predicate downstream stays EXACT. It
    ↪ does NOT preserve distances
// or angles - a square can come out as a parallelogram - but it is
    ↪ an invertible linear map of
// the plane, so it preserves COLLINEARITY, CONVEXITY, ORIENTATION,
    ↪ ratios along a line and
// inside/outside. That is exactly enough for hull, inPoly, inConvex,
    ↪ isConvex, orientation
// tests and point location. Areas are multiplied by the constant |dx
    ↪ x dy| (the 2D area2 comes
// out |dx x dy|^2 times the area in plane coordinates, and the true
    ↪ 3D area is the 2D area
// divided by |dx x dy|), so ratios of areas survive too. Never feed
    ↪ the affine map to anything
// metric: closest pair, minimum enclosing circle, width, perimeter,
    ↪ angle sorting.
// DEGENERATE: p, q, r must not be collinear (both constructors produce
    ↪ dz = 0 otherwise, and the

```

```

// orthonormal one divides by 0). pos2d does not check that its
    ↪ argument is on the plane - an
// off-plane point is silently flattened onto it, so test (p -
    ↪ o).dot(dz) == 0 first if you are
// not sure. to3d is the inverse of pos2d for the ORTHONORMAL frame
    ↪ only; for the affine frame
// carry the input INDEX through the 2D routine instead of mapping back
    ↪ (see hullCoplanar), or
// solve the 2x2 Gram system: with A = dx.dx, B = dx.dy, C = dy.dy, det
    ↪ = AC - B^2,
// a = (C*X - B*Y)/det, b = (A*Y - B*X)/det and the point is o + dx*a +
    ↪ dy*b.
// EXACTNESS: affine frame on P3<ll> is exact; pos2d is a dot product,
    ↪ so |coord| <= 1e9 keeps it
// in ll, and a downstream area2 squares that again - drop to |coord|
    ↪ <= 3e4 or use __int128
// there. Orthonormal frame is floating throughout.
template <class Q> struct Frame {
    typedef decltype(Q().dot(Q())) T; // ll for
    ↪ P3<ll>, ld for P3<ld>
    Q o, dx, dy, dz; // dz is
    ↪ the plane NORMAL
    Frame(Q p, Q q, Q r, bool ortho = true) : o(p), dx(q - p), dy(r -
    ↪ p) {
        dz = dx.cross(dy); // exact
    ↪ normal, before any scaling
        if (ortho) { dx = dx.unit(); dz = dz.unit(); dy = dz.cross(dx);
    ↪ }
    }
    P<T> pos2d(Q p) const { return {(p - o).dot(dx), (p - o).dot(dy)}; }
    Q pos3d(Q p) const { return {(p - o).dot(dx), (p - o).dot(dy), (p -
    ↪ o).dot(dz)}; }
    Q to3d(P<T> p) const { return o + dx * p.x + dy * p.y; } //
    ↪ ORTHONORMAL frames only
};
// CONVEX HULL OF COPLANAR 3D POINTS, returned as 3D points, ccw when
    ↪ seen from the dz side.
// Exact on P3<ll>: it uses the affine frame and carries the point
    ↪ itself through a map, so
// nothing is ever mapped back. O(n log n).
template <class Q> vector<Q> hullCoplanar(vector<Q> p) {
    sort(all(p)); p.erase(unique(all(p)), p.end());
    int n = sz(p), j = 2;
    if (n < 3) return p;
    while (j < n && (p[1] - p[0]).cross(p[j] - p[0]) == Q()) j++;
    if (j == n) return {p[0], p[n - 1]}; // all
    ↪ collinear: the two extremes
    Frame<Q> F(p[0], p[1], p[j], false);
    vector<decltype(F.pos2d(p[0]))> q;
    map<decltype(F.pos2d(p[0])), Q> back; // pos2d
    ↪ is injective on the plane
    for (Q x : p) q.push_back(F.pos2d(x)), back[q.back()] = x;
    vector<Q> res;
    for (auto& y : hull(q)) res.push_back(back[y]);
    return res;
}
// IS A 3D POINT INSIDE A COPLANAR POLYGON? Exact on P3<ll>. The plane
    ↪ test comes FIRST: without
// it a point floating above the polygon projects straight into it and
    ↪ reports true.
template <class Q> bool inCoplanar(const vector<Q>& v, Q p, bool strict
    ↪ = true) {
    int n = sz(v), j = 2;
    while (j < n && (v[1] - v[0]).cross(v[j] - v[0]) == Q()) j++;
    if (j == n) return false; //
    ↪ degenerate polygon: no plane
    Frame<Q> F(v[0], v[1], v[j], false);
    if (sgn((p - F.o).dot(F.dz)) != 0) return false; // off the
    ↪ plane
    vector<decltype(F.pos2d(p))> q;
    for (Q x : v) q.push_back(F.pos2d(x));
    return inPoly(q, F.pos2d(p), strict);
}
/* RECIPES
AREA OF A PLANAR 3D POLYGON no frame needed: |sum p_i x p_{i+1}| /
    ↪ 2, the vector area
    ↪ (faceArea in 06 - Polyhedron.cpp).
    ↪ Through the frame you get the
    ↪ same number as area(pos2d(..)) for an
    ↪ ORTHONORMAL frame, and
    ↪ area(pos2d(..)) / |dx x dy| for the
    ↪ affine one - use that identity
    ↪ as your sanity check that the frame is
    ↪ built right.
IS A SET OF 3D POINTS COPLANAR - fix the first three non-collinear
    ↪ ones, then vol6 == 0 for the
    ↪ rest. O(n), exact.

```

CUT A CONVEX POLYHEDRON BY A PLANE - clip each face, collect every new
 → point on the cutting
 plane, and close the hole with
 → hullCoplanar of those points.
 SORT COPLANAR POINTS AROUND THEIR CENTROID - map with the affine frame
 → and use angCmp (01 -
 Core.cpp) - exact, and gives the ccw
 → face order a mesh wants.
 POLYGON x POLYGON IN SPACE only meaningful if they are coplanar
 → (planePlane first); then map
 both with ONE frame and run the 2D clip.
 A 3D ROTATION MATRIX from the frame: the rows dx, dy, dz of an
 → orthonormal frame ARE the matrix
 that takes the plane to $z = 0$, and its
 → transpose takes it back. */

11.2.4 ConvexHull3D

3D CONVEX HULL, incremental. Input: n points (duplicates allowed, any order). Output: the hull

```
// Needs: Geometry3D.cpp.
// surface as TRIANGULAR faces, each an index triple {i, j, k} into the
// input vector, ordered so
// that (p[j]-p[i]) x (p[k]-p[i]) points OUTWARD - equivalently
// vol6(p[i], p[j], p[k], x) <= 0 for
// every input point x. Coplanar facets come back cut into triangles by
// an arbitrary
// triangulation, and points strictly inside a facet are simply absent
// from the output.
// Reach for it for "volume/surface of the hull", "is this point inside
// the hull", "diameter of a
// 3D point set", "smallest enclosing box", or any time a 2D hull idea
// has to survive one more
// coordinate. O(n^2) faces touched (O(n^2 log n) as written, the log
// being the sort of the
// horizon edges - swap in an n x n matrix to drop it; it has never
// mattered).
// Memory O(n) for the face list, O(n) for the horizon.
// DEGENERATE, and this is the part people get wrong: n < 4, ALL POINTS
// EQUAL, ALL POINTS
// COLLINEAR and ALL POINTS COPLANAR each return an EMPTY face list - a
// flat hull has no volume
// and no consistently outward orientation, so it cannot be expressed
// in this format. Detect it
// (the return is empty) and rerun the problem in 2D through 03 -
// CoplanarFrame.cpp, which is
// where a coplanar input belongs. Duplicate points are harmless: a
// repeat never strictly sees a
// face, so it is skipped and only one copy can appear in the output.
// EXACTNESS: no divisions anywhere - only vol6 signs - so on P3<ll>
// the hull is EXACT with
// |coord| <= 3.6e5 (the orientation test sums four vol6, 192*C^3 <=
// 9.2e18; make vol6 return
// _int128 to reach 1e9). The three seeding tests convert to ld to
// compare against eps, which is
// what makes the same code safe on P3<ld>; there eps is absolute, so
// scale your input to O(1).
template <class P> vector<array<int, 3>> hull3d(const vector<P>& p) {
    int n = sz(p), k;
    if (n < 4) return {}; // no
    // solid hull exists
    vector<array<int, 3>> F;
    vector<int> id(n);
    iota(all(id), 0);
    auto nz = [](P v) { return max({fabsl((ld)v.x), fabsl((ld)v.y),
    // fabsl((ld)v.z)} > eps; };
    for (k = 1; k < n && !nz(p[id[k]] - p[id[0]]); k++); // a
    // second DISTINCT point
    if (k == n) return {};
    swap(id[1], id[k]);
    for (k = 2; k < n && !nz((p[id[1]] - p[id[0]]).cross(p[id[k]] -
    // p[id[0]])); k++);
    if (k == n) return {}; // every
    // point on one line
    swap(id[2], id[k]);
    for (k = 3; k < n && fabsl((ld)vol6(p[id[0]], p[id[1]], p[id[2]],
    // p[id[k]])) < eps; k++);
    if (k == n) return {}; // every
    // point on one plane
    swap(id[3], id[k]);
    // Orient against the centroid of the seed tetrahedron, which is
    // strictly inside it and so
    // stays strictly inside every later hull. Summing the four vol6 IS
    // testing that centroid,
    // scaled by 4, so the test never ties and never divides.
    auto face = [&](int a, int b, int c) {
        auto s = vol6(p[a], p[b], p[c], p[id[0]]) + vol6(p[a], p[b],
        // p[c], p[id[1]])
```

```
        + vol6(p[a], p[b], p[c], p[id[2]]) + vol6(p[a], p[b],
        // p[c], p[id[3]]);
        F.push_back(s > 0 ? array<int, 3>{a, c, b} : array<int, 3>{a,
        // b, c});
        for (int i = 0; i < 4; i++) face(id[(i + 1) % 4], id[(i + 2) % 4],
        // id[(i + 3) % 4]);
        for (int t = 4; t < n; t++) {
            int u = id[t];
            vector<array<int, 3>> keep;
            vector<pii> E; //
            // directed edges of the dying faces
            for (auto& f : F)
                if (vol6(p[f[0]], p[f[1]], p[f[2]], p[u]) > eps) {
                    // u strictly SEES f
                    for (int j = 0; j < 3; j++) E.push_back({f[j], f[(j +
                    // 1) % 3]});
                } else keep.push_back(f); // u is
            if (E.empty()) continue; // inside the hull: nothing to do
            F = keep;
            sort(all(E));
            // The HORIZON is the directed edges whose reverse did not also
            // die: those are the ones
            // whose other face survives. Cone them to u.
            for (auto& [a, b] : E) if (!binary_search(all(E), pii(b, a)))
                face(a, b, u);
            return F;
        }
    }
    /* RELATED
    VOLUME / AREA / CENTROID of the result: build the face list and call
    // 06 - Polyhedron.cpp -
    vector<vector<P>> fs; for (auto& f : hull3d(p))
    // fs.push_back({p[f[0]], p[f[1]], p[f[2]]});
    // the faces are already consistently outward, so polyVol6 comes out
    // POSITIVE.
    IS A POINT INSIDE THE HULL vol6(p[f[0]], p[f[1]], p[f[2]], q) <= 0
    // for every face, O(F).
    EXTREME POINT in a direction d: max dot over the hull VERTICES (the
    // indices that appear in F).
    DIAMETER of a 3D point set: no rotating calipers in 3D worth typing -
    // take the hull vertices
    // and brute force the pairs, it is O(h^2) with a tiny h.
    O(n log n) HULL exists (divide and conquer, or randomised incremental
    // with a conflict graph).
    // It is 150 lines of merge bookkeeping for a constant factor you do
    // not need: n = 2000 here is
    // 4e6 orientation tests, a few milliseconds.
    DELAUNAY TRIANGULATION of 2D points: lift each (x, y) to (x, y, x^2 +
    // y^2) on the paraboloid,
    // take the 3D hull, and keep the faces whose outward normal has
    // NEGATIVE z (the LOWER hull) -
    // those triangles, projected back, are exactly the Delaunay
    // triangulation. The upper hull is
    // the farthest-point Delaunay. Lifting squares the coordinates, so
    // |coord| <= 1e4 to stay in
    // ll here; the Voronoi diagram is the dual of what comes out.
    HALFSpace INTERSECTION in 3D is the dual problem: map each plane n.x =
    // d (d > 0, so that the
    // origin is interior) to the point n/d, take the hull of those, and
    // dualise the faces back. */
```

11.2.5 Spherical

SPHERICAL GEOMETRY: solid angles, spherical triangles and polygons, caps, the lens of two

```
// Needs: Geometry3D.cpp.
// spheres, the circumsphere of four points and the smallest enclosing
// sphere of a point set.
// Points "on the sphere" are always passed as vectors FROM ITS CENTRE
// and only their DIRECTION is
// read, so you may hand in unnormalised vectors, and a sphere centred
// elsewhere is handled by
// subtracting the centre first. Angles are radians, areas are on a
// sphere of radius R.
// You reach for it for radar/antenna coverage, "what fraction of the
// sky", area of a country on
// a globe, overlapping-ball volumes, and the two enclosing-object
// questions. All O(1) except
// sphPolyArea (O(n)) and minSphere (O(n) per iteration).
// DEGENERATE: solidAngleTet returns 0 when a, b, c are coplanar with
// the origin (a flat corner)
// and is unaffected by their lengths. sphAngle is undefined when a is
// parallel to b or c (a
// zero-length spherical segment) and returns 0 there. sphPolyArea
// needs a SIMPLE polygon, ccw
// seen from outside, with no two consecutive vertices antipodal (the
```

```

    ↪ arc between them is not
// unique); it returns the area of the complement if you feed it cw.
    ↪ circumsphere needs four
// points in general position - vol6 == 0 makes it divide by zero.
    ↪ minSphere needs a non-empty
// input and returns {p[0], 0} for a single point.
// EXACTNESS: this whole file is floating point - every routine either
    ↪ normalises, calls a trig
// function, or divides. The predicates it rests on (vol6, dot, cross)
    ↪ are still exact on P3<ll>,
// so use integer input where you can and only the final value carries
    ↪ error. atan2 is used
// everywhere in place of acos: acos(dot/|a||b|) loses half its digits
    ↪ near 0 and pi, which is
// exactly the regime these formulas live in. eps is absolute here, so
    ↪ normalise your radii to
// 0(1)..0(1e6).
// SOLID ANGLE of a cone of half-angle th (steradians): the full sphere
    ↪ is 4*pi.
ld solidAngleCone(ld th) { return 2 * PI * (1 - cosl(th)); }
// SOLID ANGLE of the tetrahedron corner at the ORIGIN spanned by a, b,
    ↪ c (VAN OOSTEROM-STRACKEE):
// tan(Ωm/2) = |a.(b x c)| / (|a||b||c| + (a.b)|c| + (a.c)|b| +
    ↪ (b.c)|a|).
// atan2 takes the correct branch by itself, so this is right on the
    ↪ whole range [0, 2*pi] with no
// sign patching, and it stays accurate for the needle-thin corners
    ↪ that Girard's formula ruins.
template <class P> ld solidAngleTet(P a, P b, P c) {
    ld A = a.dist(), B = b.dist(), C = c.dist();
    ld num = fabsl((ld)a.dot(b.cross(c)));
    ld den = A * B * C + (ld)a.dot(b) * C + (ld)a.dot(c) * B +
    ↪ (ld)b.dot(c) * A;
    return 2 * atan2l(num, den);
}
// ANGLE at a of the spherical triangle abc = the dihedral angle
    ↪ between the planes Oab and Oac,
// in [0, pi]. The ORIENTED version measures how far ccw you turn from
    ↪ arc a->b to arc a->c, in
// [0, 2*pi], which is what a signed area needs.
template <class P> ld sphAngle(P a, P b, P c) {
    P u = a.cross(b), v = a.cross(c);
    return atan2l(u.cross(v).dist(), (ld)u.dot(v));
}
template <class P> ld sphAngleOr(P a, P b, P c) {
    ld t = sphAngle(a, b, c);
    return a.cross(b).dot(c) >= 0 ? t : 2 * PI - t;
}
// SPHERICAL TRIANGLE AREA on a sphere of radius R. GIRARD: R^2 * (A +
    ↪ B + C - pi), the spherical
// EXCESS - a spherical triangle's angles overshoot pi by exactly its
    ↪ area. Computed through the
// solid angle instead, which is the same number without the
    ↪ catastrophic cancellation of summing
// three angles that are each close to pi/2 and subtracting pi.
template <class P> ld sphTriArea(P a, P b, P c, ld R) { return R * R *
    ↪ solidAngleTet(a, b, c); }
// SPHERICAL POLYGON AREA, vertices in ccw order seen from OUTSIDE: R^2
    ↪ * (sum of the interior
// angles - (n-2)*pi), the same excess with n corners. Handles
    ↪ non-convex polygons.
template <class P> ld sphPolyArea(const vector<P>& p, ld R) {
    int n = sz(p);
    ld s = -(n - 2) * PI;
    for (int i = 0; i < n; i++) s += sphAngleOr(p[(i + 1) % n], p[(i +
    ↪ 2) % n], p[i]);
    return R * R * s;
}
// SPHERICAL CAP of height h on a sphere of radius R (a plane at
    ↪ distance R - h from the centre;
// from the polar half-angle th instead, h = R * (1 - cos th) and the
    ↪ base radius is R * sin th).
// capArea is the CURVED part only - add pi*a^2 for the flat disc, a^2
    ↪ = h * (2R - h).
ld capArea(ld R, ld h) { return 2 * PI * R * h; }
ld capVolume(ld R, ld h) { return PI * h * h * (3 * R - h) / 3; }
// SPHERICAL SEGMENT / zone: the slab between two parallel planes, base
    ↪ radii a and b, height h.
ld zoneArea(ld R, ld h) { return 2 * PI * R * h; } // curved
    ↪ part; independent of WHERE
ld zoneVolume(ld a, ld b, ld h) { return PI * h * (3 * a * a + 3 * b *
    ↪ b + h * h) / 6; }
// SPHERE x SPHERE, centres d apart: {surface area, volume} of the lens
    ↪ they share. The lens is
// two caps glued on the plane of the intersection circle. CONTAINMENT
    ↪ is checked FIRST - there
// the cap formula silently returns nonsense, and the answer is just

```

```

    ↪ the smaller ball.
pair<ld, ld> sphereSphere(ld R, ld r, ld d) {
    if (R < r) swap(R, r);
    if (d >= R + r) return {0, 0}; //
    ↪ disjoint (tangent counts)
    if (d + r <= R) return {4 * PI * r * r, 4 * PI * r * r * r / 3};
    ↪ // small one swallowed
    ld x = (R * R - r * r + d * d) / (2 * d); // plane
    ↪ of the circle, from centre R
    ld h1 = R - x, h2 = r - (d - x); // the two
    ↪ cap heights
    return {capArea(R, h1) + capArea(r, h2), capVolume(R, h1) +
    ↪ capVolume(r, h2)};
}
// The intersection CIRCLE itself has radius sqrt(R^2 - x^2) and sits
    ↪ at x along the centre line.
// CIRCUMSPHERE of four points, {centre, radius}. Needs P3<ld>; the
    ↪ denominator IS 2*vol6, so
// coplanar input divides by zero - test vol6 first.
template <class P> pair<P, ld> circumsphere(P a, P b, P c, P d) {
    P B = b - a, C = c - a, D = d - a;
    ld den = 2 * (ld)B.dot(C.cross(D));
    P o = a + (C.cross(D) * (ld)B.dist2() + D.cross(B) * (ld)C.dist2()
    ↪ + B.cross(C) * (ld)D.dist2()) / den;
    return {o, (o - a).dist()};
}
// SMALLEST ENCLOSED SPHERE, {centre, radius}, by shrinking steps
    ↪ toward the current farthest
// point. Start at the centroid and step a fraction q toward the
    ↪ farthest point, with q decaying
// geometrically: the centre converges to the true one and the step
    ↪ budget bounds the error by
// (initial spread) * 0.999^30000 ~ 1e-13 of the diameter - fine for
    ↪ 1e-6 answers, NOT a proof.
// O(n) per iteration. Welzl with a 4-point basis is the exact O(n)
    ↪ expected version and is 40
// lines; this is 8 and has never lost a contest problem.
template <class P> pair<P, ld> minSphere(const vector<P>& p) {
    P c;
    for (P q : p) c = c + q;
    c = c / sz(p);
    for (ld q = 0.1; q > 1e-13; q *= 0.999) {
        int f = 0;
        for (int i = 1; i < sz(p); i++) if ((c - p[i]).dist2() > (c -
    ↪ p[f]).dist2()) f = i;
        c = c + (p[f] - c) * q;
    }
    ld r = 0;
    for (P x : p) r = max(r, (c - x).dist());
    return {c, r};
}
// GREAT CIRCLE, the part 01 - Geometry3D.cpp does not cover: its
    ↪ greatCircle takes lat/lon in
// DEGREES, these take 3D vectors. Distance along the surface between
    ↪ two points of the sphere,
// and the lat/lon -> vector conversion (latitude from the equator,
    ↪ longitude east, degrees).
template <class P> ld sphDist(P a, P b, ld R) {
    return R * atan2l(a.cross(b).dist(), (ld)a.dot(b));
}
Pd3 fromLatLon(ld R, ld lat, ld lon) {
    lat *= PI / 180, lon *= PI / 180;
    return {R * cosl(lat) * cosl(lon), R * cosl(lat) * sinl(lon), R *
    ↪ sinl(lat)};
}
/* RELATED
TANGENT CONE from an external point at distance d: half-angle acos(R /
    ↪ d), and the cap it cuts
    ↪ off the sphere has height R * (1 - R / d). d < R means the point
    ↪ is inside: no tangent.
SPHERE x PLANE a circle of radius sqrt(R^2 - dist(centre, plane)^2),
    ↪ centred at the projection.
SPHERE x LINE / SEGMENT sphereLine (01 - Geometry3D.cpp), then clamp
    ↪ the parameters.
VOLUME OF A UNION OF BALLS - no closed form beyond two; for three or
    ↪ more integrate over one axis
    ↪ numerically, or use inclusion-exclusion only when the statement
    ↪ promises few overlaps.
SPHERICAL SEGMENT x SEGMENT - two great-circle arcs ab and cd cross
    ↪ where the plane normals a x b
    ↪ and c x d cross: the candidate directions are +((a x b) x (c x
    ↪ d)); accept the one lying
    ↪ inside both arcs, which is the exact 3D test sgn((a x b).c) !=
    ↪ sgn((a x b).d) and vice versa.
SPHERICAL CONVEX HULL of directions = the 3D convex hull of the unit
    ↪ vectors {04}, as long as
    ↪ they fit in a hemisphere.

```

SOLID ANGLE OF A POLYGONAL CONE - triangulate it from one vertex and
 ↳ sum solidAngleTet - that is
 exactly what sphPolyArea does with $R = 1$. */

11.2.6 Polyhedron

POLYHEDRA GIVEN AS A FACE LIST: $fs[i]$ is one face, a vector of its vertices in order around it,

```
// Needs: Geometry3D.cpp, Spherical.cpp (winding3 only).
// coplanar, no repeat of the first vertex at the end. Faces may be any
// ↳ polygon, not just
// ↳ triangles. Everything here wants the list CLOSED (every edge shared
// ↳ by exactly two faces) and
// CONSISTENTLY oriented (all normals outward, or all inward); reorient
// ↳ produces the consistency
// from a face list that has it in neither direction, and the sign of
// ↳ polyVol6 tells you which of
// the two you ended up with. Computes: signed 6*volume, surface area,
// ↳ volume centroid, the
// orientation fixup, and the 3D winding number that answers "is the
// ↳ origin inside this thing".
// You reach for this file after a 3D hull, after cutting a solid by
// ↳ planes, or when the statement
// hands you a mesh (an .obj-shaped input) and asks for volume, mass
// ↳ centre, or containment.
// COMPLEXITY: volume, area, centroid  $O(\text{total vertices})$ . reorient  $O(E)$ 
// ↳ log  $E$  for the edge map,
//  $O(E)$  memory. winding3  $O(\text{total vertices})$  with a few trig calls each.
// DEGENERATE: an empty face list gives volume 0, area 0 and centroid
// ↳ 0/0 - guard it. A face with
// fewer than 3 vertices contributes nothing (its fan is empty). A
// ↳ polyhedron that is NOT closed
// still returns a number from polyVol6, and that number is meaningless
// ↳ - there is no cheap check,
// so verify that every directed edge appears exactly once (which is
// ↳ what reorient's map does).
// reorient BFSes the face adjacency graph, so a DISCONNECTED surface
// ↳ (two separate shells) comes
// back with only the component of face 0 fixed; run it per component.
// ↳ Non-manifold edges (three
// faces meeting) break the map, which keeps only the first face per
// ↳ directed edge.
// EXACTNESS: polyVol6 and faceArea's vector sum are exact on  $P3<ld>$ 
// ↳ ( $|coord| \leq 5.7e5$  for vol6);
// faceArea then takes one sqrt, and centroid divides, so those return
// ↳ ld / need  $P3<ld>$ . reorient
// compares vertices for EQUALITY, so on  $P3<ld>$  two faces only share an
// ↳ edge if the coordinates
// are bit-identical - with float input, or any input where the same
// ↳ vertex was computed twice,
// index your vertices and run reorient on vector<vector<int>> with the
// ↳ map keyed by pii instead.
// SIGNED 6*VOLUME, by the divergence theorem: fan every face from the
// ↳ ORIGIN and sum the
// tetrahedra. The parts outside the solid cancel exactly. Positive iff
// ↳ the faces are OUTWARD.
template <class P> auto polyVol6(const vector<vector<P>>& fs) {
    decltype(fs[0][0].dot(fs[0][0])) s = 0;
    for (auto& f : fs) for (int i = 1; i + 1 < sz(f); i++) s +=
        ↳ vol6(P(), f[0], f[i], f[i + 1]);
    return s;
}
template <class P> ld polyVolume(const vector<vector<P>>& fs) {
    return fabs(s(polyVol6(fs))) / 6;
}
// AREA of one planar face, from its VECTOR AREA sum  $p_i \times p_{(i+1)}$ 
// ↳ (which is 2A times the unit
// normal, and is independent of where the origin sits because the face
// ↳ is closed and planar).
template <class P> ld faceArea(const vector<P>& f) {
    P s;
    for (int i = 0, m = sz(f); i < m; i++) s = s + f[i].cross(f[(i + 1)
        ↳ % m]);
    return s.dist() / 2;
}
template <class P> ld surfaceArea(const vector<vector<P>>& fs) {
    ld a = 0;
    for (auto& f : fs) a += faceArea(f);
    return a;
}
// CENTROID of the SOLID (not of the vertices, not of the surface): the
// ↳ volume-weighted average of
// the tetrahedron centroids of the same fan. Needs  $P3<ld>$ . Signs
// ↳ cancel exactly as in polyVol6.
template <class P> P polyCentroid(const vector<vector<P>>& fs) {
    P c; ld V = 0;
    for (auto& f : fs) for (int i = 1; i + 1 < sz(f); i++) {
        ld v = (ld)vol6(P(), f[0], f[i], f[i + 1]); // 6 *
```

```
↳ signed volume of  $O-f_0-f_i-f_{i+1}$ 
    c = c + (f[0] + f[i] + f[i + 1]) * (v / 4); // its
↳ centroid is  $(O+a+b+c)/4$ 
    V += v;
}
return c / V;
}
// FACE-ORIENTATION FIXUP. Flip a subset of the faces so that all
// ↳ normals agree (either all out or
// all in - which one you get depends on face 0; check sgn(polyVol6)
// ↳ afterwards and reverse every
// face if you wanted the other). THE RULE: two faces sharing an edge
// ↳ agree exactly when they
// traverse that edge in OPPOSITE directions, so seeing the SAME
// ↳ directed edge twice means exactly
// one of the two faces must flip. BFS that parity over the face
// ↳ adjacency graph.
template <class P> vector<vector<P>> reorient(vector<vector<P>> fs) {
    int n = sz(fs);
    if (!n) return fs;
    vector<vector<pii>> g(n); //
    ↳ {neighbour, same direction?}
    map<pair<P, P>, int> es;
    for (int u = 0; u < n; u++)
        for (int i = 0, m = sz(fs[u]); i < m; i++) {
            P a = fs[u][i], b = fs[u][(i + 1) % m];
            if (es.count({b, a})) { int v = es[{b, a}]; g[u].pb({v,
                ↳ 0}); g[v].pb({u, 0}); }
            else if (es.count({a, b})) { int v = es[{a, b}];
                ↳ g[u].pb({v, 1}); g[v].pb({u, 1}); }
            else es[{a, b}] = u;
        }
    vector<int> vis(n, 0), flip(n, 0), q = {0};
    vis[0] = 1; // face 0
    ↳ defines the common direction
    for (int i = 0; i < sz(q); i++)
        for (auto& [v, same] : g[q[i]])
            if (!vis[v]) vis[v] = 1, flip[v] = flip[q[i]] ^ same,
            ↳ q.push_back(v);
        for (int u = 0; u < n; u++) if (flip[u]) reverse(all(fs[u]));
        return fs;
}
// 3D WINDING NUMBER about the ORIGIN: how many times the surface wraps
// ↳ around it. 0 = the origin
// is OUTSIDE; +1 = inside with the faces oriented outward; -1 = inside
// ↳ with them oriented inward.
// Each face is projected onto the unit sphere, where it covers a
// ↳ signed solid angle; the total is
// 4*pi per wrap. remainder folds each face's contribution into  $(-2\pi,$ 
// ↳  $2\pi]$  so that faces covering
// more than a hemisphere still add up right. To test a point other
// ↳ than the origin, subtract it
// from every vertex first. The origin exactly ON the surface makes a
// ↳ face's solid angle
// undefined - handle that case before calling (segTri / inCoplanar in
// ↳ 02 and 03).
template <class P> int winding3(const vector<vector<P>>& fs) {
    ld s = 0;
    for (auto& f : fs) s += remainderl(sphPolyArea(f, 1), 4 * PI);
    return (int)llroundl(s / (4 * PI));
}
/* RELATED
IS A POINT INSIDE A CONVEX POLYHEDRON - do not call winding3: test sgn
↳ of the side of every face
plane, 0(F), exact on integers. Convex + outward faces => inside
↳ iff all sides <= 0.
EULER  $V - E + F = 2$  for any closed surface of genus 0; a
↳ triangulated one has  $E = 3F/2$ , so
 $F = 2V - 4$  and  $E = 3V - 6$ . Use it as a free assertion on the
↳ output of a 3D hull.
SURFACE OF A NON-PLANAR "face" - faceArea silently returns the area of
↳ its projection onto the
best-fit plane. Triangulate first if the input can be skew.
MOMENTS / INERTIA - the same fan works: sum the tetrahedron inertia
↳ tensors weighted by vol6.
CUT BY A PLANE - clip every face (keep the vertices with side <= 0
↳ plus the crossing points), then
close the hole with hullCoplanar of the new points (03 -
↳ CoplanarFrame.cpp).
MINKOWSKI SUM of two convex polyhedra: hull of the pairwise vertex
↳ sums,  $O((nm)^2)$  but n and m
are tiny in practice.
VOLUME OF A UNION of polyhedra has no cheap formula - sweep a plane
↳ and integrate the
cross-section area, or inclusion-exclusion on convex pieces. */
```

12 | Problem Solving

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12.1 Reframing

REFRAMING: CHANGE THE PROBLEM, NOT THE EFFORT

```
/* ===== REFRAMING: CHANGE THE PROBLEM, NOT THE EFFORT
   ↳ =====
```

This chapter has no algorithms - it has the moves you make BEFORE you
 ↳ know which template to
 paste. Stuck for 10 minutes? Do not re-read the statement a fourth
 ↳ time. Walk the six questions,
 then walk the TRIGGERS below and try each against your problem.

THE SIX QUESTIONS

1. What object am I counting / optimising over? Say it in one
 ↳ sentence with no input format.
2. What is INVARIANT under the allowed operation?
3. What does the ANSWER look like (a prefix? one of the inputs? a
 ↳ threshold? always $\leq$??)
4. Can I CHECK a candidate faster than I can find one? →
 ↳ binary search the answer
5. What do the constraints forbid, and what does the leftover budget
 ↳ exactly equal?
6. What would the brute force print for $n = 1..8$?

--- REVERSE THE PROCESS ---

Run time backwards: deletions become insertions, splits become
 ↳ merges. Every structure is
 add-friendly and delete-hostile (DSU above all), so reversing turns
 ↳ impossible into trivial.
 TRIGGER: things are removed one at a time and you must answer after
 ↳ each removal.
 EXAMPLE: CF 722C Destroying Array - process destructions in reverse,
 ↳ union with restored
 neighbours. Also CSES 1163 Traffic Lights.
 TRAP: emit answers in reverse too, and check the off-by-one at both
 ↳ ends with $n = 1$.

--- WORK BACKWARDS FROM THE GOAL ---

Define the answer at terminal states and propagate backwards.
 ↳ Standard for games and for
 "what must have been true one step earlier".
 TRIGGER: few terminal states, huge forward branching, small backward
 ↳ branching.
 TRAP: on a CYCLIC game graph this is NOT a plain DP - you need
 ↳ retrograde BFS with out-degree
 counters (a position is losing only once ALL successors are known
 ↳ winning). See 10 - Game
 Theory. Writing $\text{win}[v] = \text{OR over children on a cyclic graph}$
 ↳ silently mislabels draws.

--- COMPLEMENTARY COUNTING ---

Count the bad ones and subtract; count $P(X \leq x)$ instead of $P(X = x)$.
 TRIGGER: "at least one", "there exists", "not all equal", "some
 ↳ pair".
 EXAMPLE: $E[\max] = \text{sum over } x \text{ of } P(\max \geq x)$, with $P(\max \leq x) =$
 ↳ $(x/n)^k$.
 TRAP: the complement is easy for AT MOST / AT LEAST, not for
 ↳ EXACTLY. Getting from one to the
 other is binomial inversion: $\text{exactly}_k = \text{sum}_{\{j \geq k\}} (-1)^{(j-k)}$
 ↳ $C(j,k)$ atleast_j.

--- THE CONTRIBUTION TECHNIQUE ---

Instead of "for each subarray, compute f", ask "for each element, in
 ↳ how many objects does it
 contribute, and how much". Turns $O(n^2)$ enumeration into $O(n)$.
 TRIGGER: "sum over all subarrays / pairs / subsets of $[\min | \max |$
 ↳ $\text{gcd} | \# \text{distinct} | \text{bottleneck}]$ ".
 EXAMPLE: sum over subarrays of the minimum → monotonic stack gives
 ↳ each element's span.
 Sum over pairs of the bottleneck → Kruskal, each merge
 ↳ contributes $w * |A| * |B|$ pairs.
 TRAP: ties. With equal values the span must be strict on ONE side
 ↳ and non-strict on the other,
 or every duplicated minimum is counted twice. Decide which side
 ↳ BEFORE coding.

--- SWAP THE ORDER OF SUMMATION ---

$\text{sum}_i \text{sum}_d \{ i \} f(d) = \text{sum}_d f(d) * \text{floor}(n/d)$. Move the outer
 ↳ loop to the variable whose
 inner count has a closed form.
 TRIGGER: a double sum whose inner range depends on the outer index;
 ↳ divisors, multiples,
 ancestors, intervals containing a point.

EXAMPLE: sum of $\text{sigma}(i)$ for $i \leq n = \text{sum}_d d * \text{floor}(n/d)$, then
 ↳ divisor blocks → $O(\sqrt{n})$.

--- LINEARITY OF EXPECTATION ---

$E[\text{sum } X_i] = \text{sum } E[X_i]$, with NO independence needed. Decompose into
 ↳ indicators.
 TRIGGER: "expected number of ...", or an expectation over a process
 ↳ you cannot untangle.
 TRAP: linearity does not survive max, min, or division. $E[\max] \neq$
 ↳ $\max E$, $E[1/X] \neq 1/E[X]$.
 For an expected maximum use the complementary-counting tail sum
 ↳ above.

--- FIX THE EXTREMUM ---

Add an artificial outer loop "for each choice of the maximum / the
 ↳ leftmost / the last move".
 Everything else becomes bounded relative to the fixed thing.
 TRIGGER: the statement mentions max or min over a range; a
 ↳ Cartesian-tree recursion is available.
 EXAMPLE: CF 1156E - fix the position of the maximum, then enumerate
 ↳ the SMALLER side.
 TRAP: always iterate the smaller side (small-to-large, $O(n \log n)$
 ↳ total). Iterating the left
 side unconditionally is $O(n^2)$ on a sorted permutation - and it
 ↳ passes the samples.

--- DUALIZE ---

max-flow = min-cut; max bipartite matching = min vertex cover
 ↳ (Konig); min path cover of a DAG
 $= n - \text{matching}$; longest antichain = min chain cover (Dilworth); "a
 ↳ perfect matching exists" =
 "no Hall violator".
 TRIGGER: the statement asks for a MINIMUM removal / blocking set /
 ↳ number of chains - that is
 almost always the dual of a maximum you can compute.
 TRAP: Konig gives the SIZE for free, but recovering the cover needs
 ↳ the alternating-BFS
 construction $(L \setminus Z) \cup (R \cap Z)$. And Konig is BIPARTITE ONLY.

--- MODEL IT AS A GRAPH ---

Objects → nodes, relations → edges. States can be tuples: (cell,
 ↳ fuel), (vertex, parity),
 (position, mask).
 TRIGGER: "can I get from A to B", "minimum number of operations",
 ↳ "are these constraints
 consistent", "each x points at exactly one y" (functional graph).
 TRAP: compute $|V| * |E|$ explicitly before coding. In a layered graph
 ↳ the answer is usually
 "any layer", not layer 0.

--- MODEL IT AS INTERVALS / A TIMELINE ---

Recast objects as [start, end] and use a sweep, greedy-by-endpoint,
 ↳ or a difference array.
 Tree subtrees become intervals via the Euler tour.
 TRIGGER: "from time a to time b", "covers l..r", "is available
 ↳ during", subtree aggregates.
 TRAP: decide ONCE whether [1,3] and [3,5] overlap, and encode it in
 ↳ exactly one place.

--- MODEL IT AS A POLYNOMIAL ---

"For every possible sum, count the ways" is a convolution.
 ↳ Multiplying GFs = combining
 independent choices; x^k = "used k of the budget".
 TRIGGER: modulus 998244353 (NTT is intended); independent parts
 ↳ whose sizes add.
 TRAP: convolving an array with itself double-counts $i \neq j$ and
 ↳ mis-counts $i = j$.

--- MODEL IT AS LINEAR ALGEBRA OVER GF(2) ---

Each element is a bit-vector; "can I make X" = "is X in the span";
 ↳ the number of subsets giving
 a representable Y is $2^{(n - \text{rank})}$.
 TRIGGER: XOR anywhere; "product is a perfect square" (exponent
 ↳ parity vector over primes);
 switch-toggling puzzles; "each constraint satisfied an even number
 ↳ of times".
 TRAP: $2^{(n - \text{rank})}$ counts solutions only if the system is CONSISTENT -
 ↳ check first. And subtract
 the empty subset when the problem wants a non-empty one.

--- MODEL IT AS THE SUBSET LATTICE ---

Bitmask DP plus zeta/Mobius (SOS) to aggregate over submasks in
 ↳ $O(2^n n)$ instead of $O(3^n)$.
 TRIGGER: $n \leq 20$ with a "set of chosen items" state, or a ≤ 22 -bit
 ↳ value domain with queries
 "over all y contained in x".
 TRAP: the bit loop goes OUTSIDE the mask loop. Swapping them gives a

```

↳ wrong-but-plausible array.
  Submask enumeration must be for (s = m; ; s = (s-1) & m) { ...;
↳ if (!s) break; };

--- THINK ABOUT WHAT THE ANSWER LOOKS LIKE ---
Characterise the optimum's SHAPE first: it is a prefix; it is one of
↳ the inputs; it is at most
2; it has a threshold structure. Then search only that shape.
TRIGGER: the output is one small number regardless of n; the problem
↳ screams "constructive".
TRAP: "the answer is at most k" needs a matching lower-bound
↳ example, or you print k-1.

--- CHANGE OF VARIABLES ---
Difference array (range add -> two point updates; "make all equal"
↳ -> zero the differences),
prefix sums (subarray -> pair of points), Chebyshev <-> Manhattan
↳ via (x+y, x-y), logs
(products -> sums).
TRIGGER: operations act on ranges; the objective is max(|dx|, |dy|)
↳ or |dx|+|dy|.
TRAP: C++ % is negative for negative operands - use ((x % n) + n) %
↳ n. Seed the empty prefix.

--- PER-BIT / PER-PRIME INDEPENDENCE ---
If the objective decomposes across bits (XOR/AND/OR sums, popcount
↳ weights) or across primes
(gcd/lcm exponent-wise), solve 30 tiny problems and recombine.
EXAMPLE: sum over pairs of (a_i XOR a_j) = sum_b 2^b * cnt0_b *
↳ cnt1_b.
TRAP: bit-independence holds for SUMS, not for COMPARISONS. "Count
↳ pairs with a_i^a_j < k" is
not separable - that needs a binary trie walked high bit first.

--- REDUCE TO SOMETHING YOU ALREADY HAVE ---
"minimum adjacent swaps to sort" = inversions; "maximum
↳ pairwise-compatible set" = matching;
"minimise the maximum edge on a path" = MST bottleneck / Kruskal
↳ reconstruction tree; "longest
chain of pairs" = LIS; "pairwise gap constraints" = difference
↳ constraints + Bellman-Ford;
"each person points at one other" = functional graph.
TRIGGER: you can state the problem without the story. Do that, then
↳ match it.
TRAP: the disguise is usually ALMOST the classic - one extra
↳ constraint breaks the reduction.
  Check the reduction on the samples by hand before trusting it.
=====
↳ */

```

12.2 SearchAndProof

SEARCH: TURN "FIND" INTO "CHECK"

```
/* ===== SEARCH: TURN "FIND" INTO "CHECK" =====
```

```

BINARY SEARCH ON THE ANSWER. If feasible(x) is monotone, stop
↳ optimising and find the boundary.
TRIGGER: "minimise the maximum", "maximise the minimum", "minimum
↳ time such that", "possible
within k". Also: the answer is an integer in a huge range and no
↳ formula presents itself.
TRAPS: (1) the check itself overflows - break early once a running
↳ count passes the target;
(2) 'hi' must be PROVABLY feasible - seed it with the trivial
↳ answer, not 1e9;
(3) if the predicate is not monotone you get a plausible wrong
↳ answer and no crash.

BINARY SEARCH THE VALUE, COUNT WITH A SECOND SEARCH. For "k-th
↳ smallest X" over a set you cannot
materialise (pairs, products, distances, subarray sums): binary
↳ search x, count how many are
<= x. The counting routine is the real problem.
TRAP: use firstTrue(count(x) >= k) so the returned x actually occurs.

PARAMETRIC SEARCH / LAGRANGIAN (Aliens, wqs). Replace "exactly k" by a
↳ penalty per item and
binary search the penalty. Requires the optimum to be CONVEX in k.
↳ See 08 - DP/01/07.
TRAP: verify convexity by brute-forcing f(k) for small n; and the
↳ inner DP must break ties
toward the largest count or the final subtraction is off by an
↳ arbitrary amount.

TERNARY SEARCH. Unimodal f: cut a third each step. Reals: fixed ~200
↳ iterations (never an eps
loop). Integers: while (hi - lo > 2), then scan the survivors.

```

```

TRAP: PLATEAU KILL IT. f(m1) == f(m2) on a flat region can discard
↳ the true optimum. For any
discretely convex f use binary search on the difference instead:
  argmin = firstTrue(lo, hi-1, [&](ll x){ return f(x+1) >= f(x);
↳ });

```

```

TWO POINTERS - and why it is legal. Legal precisely when validity is
↳ MONOTONE IN l for fixed r:
if [l, r] is valid then so is [l', r] for every l' >= l.
TRAP: "subarray with sum <= S" is two-pointerable with positive
↳ values and WRONG with negative
ones. Every time you reach for two pointers, say out loud
↳ "shrinking can only help". If it
cannot, stop.

```

```

SWEEP LINE. Sort events by one coordinate, maintain a structure over
↳ the other.
TRAP: event ordering at TIES is the whole problem - for closed
↳ intervals process opens first;
for touching rectangles process closes first. Getting it wrong is
↳ off-by-exactly-the-ties.

```

```

COORDINATE COMPRESSION. Only the order matters; map <= 2n values to
↳ 0..m-1.
TRAP: if the problem cares about GAPS (lengths, areas, "how many
↳ integers strictly between"),
keep the original values to weight each compressed cell. Insert
↳ both l and r+1.

```

```

DIVIDE AND CONQUER ON TIME. Put each object's lifetime on a segment
↳ tree over the timeline, DFS
it with a ROLLBACK structure, answer at the leaves. This is how you
↳ delete from a DSU.
TRAP: rollback DSU must use union by size only - path compression
↳ makes undo impossible, so
find is O(log n), not alpha. Budget accordingly.

```

```

MEET IN THE MIDDLE. Split in half, enumerate 2^(n/2) per side,
↳ recombine by sorting + binary
search / two pointers / hash map.
TRIGGER: n <= 40..44 with a subset-flavoured question.
TRAP: memory (2^20 ll = 8 MB per array); and in "mod m" variants you
↳ must also consider the
wrap-around pair. Put the extra element of an odd n in the CHEAPER
↳ half.

```

```

SQRT DECOMPOSITION OF THE PROBLEM (not the data structure). Two
↳ complementary bad algorithms,
O(x) and O(n/x): threshold at sqrt and use each where it wins. Heavy
↳ vs light vertices, small
vs large divisors, big vs small queries, "sum a_i <= 1e5 so at most
↳ sqrt distinct values".
TRAP: tune the threshold to the ACTUAL costs; if updates are 10x
↳ cheaper the optimum is far
from sqrt(n). Measure.

```

```

OFFLINE REORDERING. If nothing forces you online, sort the queries: by
↳ right endpoint (+ BIT),
by threshold (+ DSU), by Mo's ordering, by time. The highest-yield
↳ contest move for "q queries
on a static array".
TRIGGER: no "xor with the previous answer" marker; all queries
↳ readable up front.
TRAP: store answers by the ORIGINAL query id and print in input
↳ order.

```

```

SMALL TO LARGE. Always insert the smaller container into the larger:
↳ each element moves
O(log n) times. SWAP THE HANDLES, never copy - copying the big one
↳ in is O(n^2) and looks
identical.

```

```

AMORTIZATION / POTENTIAL. An occasionally-expensive operation is cheap
↳ on average if each
expensive step DESTROYS something expensive to create. Justifies
↳ monotonic stacks, ODT, beats.
TRAP: ODT's amortization needs assign-heavy input; it is O(n) per op
↳ in the worst case.
Monotonic-stack amortization dies if you ever re-push an index.

```

```

ADD ONE ELEMENT AT A TIME. Solve for prefixes and maintain the answer
↳ incrementally. Turns "for
every prefix, output X" from O(n^2) into O(n log n) and usually
↳ reveals the structure.
TRAP: LIS - lower_bound gives strictly increasing, upper_bound gives
↳ non-decreasing. The tails
array is NOT the subsequence; reconstruct with parent pointers.

```

PARALLEL BINARY SEARCH. q independent binary searches over one shared
 ↳ timeline: bucket queries
 by their current mid, sweep the timeline once, resolve one bit for
 ↳ every query.
 TRAP: reset or rebuild the shared structure between LEVELS.
 ↳ Forgetting produces monotone-looking garbage.

===== PROOF TOOLS: THE TEN MINUTES THAT SAVE THE SUBMIT
 ↳ =====

EXCHANGE ARGUMENT (for greedy). Take any optimum; show that swapping
 ↳ two ADJACENT elements
 toward your greedy order never makes it worse. For a sorting greedy
 ↳ this DERIVES the
 comparator: compare $\text{cost}(a \text{ then } b)$ against $\text{cost}(b \text{ then } a)$.
 TRAP: the derived comparator must be a STRICT WEAK ORDERING. Ties
 ↳ returning true, or an
 overflowing cross-multiplication, makes `std::sort` run off the
 ↳ array - an RE that looks like
 a logic bug. Test with all-equal input, and cross-multiply in
 ↳ `__int128`.

INVARIANTS. A quantity unchanged by every legal operation. If start
 ↳ and target differ on it the
 answer is "impossible" - and you have a proof instead of a guess.
 TRAP: an invariant proves IMPOSSIBILITY only. You still owe a
 ↳ construction for the other side.

MONOINVARIANTS. A quantity that strictly decreases and is bounded:
 ↳ proves termination and bounds
 the number of operations. (Euclid: $a+b$ strictly decreases.)
 TRAP: on reals a monovariant proves nothing - you need integrality
 ↳ or a floor.

PARITY AND COLOURING. Colour the board so every legal move changes the
 ↳ colour counts in a fixed
 way, then count. Try in order: checkerboard, columns mod k , $(i+j)$
 ↳ mod 3, a weight function.
 EXAMPLE: the mutilated chessboard has $32/30$ of the two colours, so
 ↳ no domino tiling exists.
 TRAP: if no colouring produces an imbalance in five minutes, the
 ↳ obstruction is not parity.

THE EXTREMAL PRINCIPLE. Argue about the largest / smallest / deepest
 ↳ object - extremes have
 neighbours on one side only, which collapses the cases.
 EXAMPLE: two-BFS tree diameter is correct because the farthest
 ↳ vertex from ANY start is an
 endpoint of SOME diameter.
 TRAP: that theorem is TREE-ONLY. It is false on general graphs and
 ↳ on negative weights.

PIGEONHOLE. $n+1$ objects in n boxes. In practice: $n+1$ prefix sums but
 ↳ only n residues, so some
 subarray sum is divisible by n .
 TRAP: the constructive version needs the FIRST repeat - store the
 ↳ earliest index per residue.

SYMMETRY AND WLOG. If the problem is invariant under swapping two
 ↳ roles, assume an order and
 divide by the symmetry factor at the end.
 TRAP: handle the diagonal $i = j$ separately: $(\text{total} - \text{diagonal})/2 +$
 ↳ diagonal. Under a modulus,
 "divide by 2" means multiply by inv2 .

STRATEGY STEALING. In a symmetric game where an extra move never
 ↳ hurts, the first player wins or
 draws: if the second player had a winning strategy the first could
 ↳ steal it.
 TRAP: pure EXISTENCE. It gives you no winning move, and it fails
 ↳ under misere play.

ADVERSARY ARGUMENTS / QUERY LOWER BOUNDS. q binary queries distinguish
 ↳ at most 2^q inputs, so
 $q \geq \log_2(\# \text{inputs})$; comparison sorting needs $\log_2(n!) \sim n \log n$.
 TRAP: assume the interactor is ADAPTIVE unless told otherwise - it
 ↳ fixes nothing in advance,
 only stays consistent. Strategies relying on "the answer was fixed
 ↳ before I started" lose.

===== CONSTRAINTS ARE A MESSAGE FROM THE SETTER =====
 $n \leq 10$ permutations $n!$, or backtracking with pruning
 $n \leq 20$ 2^n bitmask DP, SOS, 3^n submask partition
 $n \leq 24..28$ 2^n with a small factor
 $n \leq 40..44$ MEET IN THE MIDDLE, $2^{(n/2)}$

$n \leq 100$ $O(n^3)/O(n^4)$: Floyd, Gauss, interval DP, min-cost
 ↳ flow, Hungarian
 $n \leq 500$ $O(n^3)$ or $O(n^2 \log n)$; flows on n^2 edges
 $n \leq 5000$ $O(n^2)$: LCS, tree knapsack, or $O(n^2/64)$ with a bitset
 $n \leq 1e5$, $\log^2$ slack segtree of segtrees, merge-sort tree,
 ↳ parallel binary search, HLD
 $n \leq 2e5$ $O(n \log n)$. This is the default and it means "find
 ↳ the $n \log n$ "
 $n \leq 1e6$ $O(n)$ or $O(n \log n)$ with a SMALL constant: no map, no
 ↳ set, fast IO mandatory
 $n, q \leq 1e5$, offline Mo's ($n \sqrt{q}$) is in budget, $\sim 3e7$ pointer
 ↳ moves
 $\text{value} \leq 1e9$, n small coordinate compression, or binary search the
 ↳ value
 $a_i \leq 1e18$ Miller-Rabin + Pollard rho, `__int128`, no sieve
 $\text{sum } n \leq 2e5$ over tests per-test $O(n \log n)$; NEVER `memset` a MAXN
 ↳ global per test
 $t \leq 1e4$, no sum bound the intended solution is $O(1)$ or $O(\log)$ per
 ↳ test
 mod 998244353 NTT / polynomial methods intended
 mod $1e9+7$ counting DP / combinatorics, NOT NTT
 real output, $1e-6$ geometry, ternary/binary search on reals, or
 ↳ expectation
 $\leq 20-30$ oracle queries binary search ($2^{20} > 1e6$) or bit-by-bit
 ↳ determination
 TL 3-5 s at $n \leq 1e5$ the intended solution really is $n \sqrt{n}$ or
 ↳ $n \log^2 n$
 ML 32-64 MB offline/streaming, rolling DP arrays, no persistence

THE /64 ARGUMENT. An "impossible" $O(n^2)$ or $O(n^3)$ over BOOLEANS
 ↳ becomes $n^2/64$ - fine up to
 $\sim 1e10$ raw ops. Reachability, subset sum, LCS, string matching, set
 ↳ intersection.
 TRAP: memory. n bitsets of n bits is $n^2/8$ bytes: at $n = 5e4$ that is
 ↳ 312 MB. Process in
 blocks of ~ 1000 and reuse. bitset needs a compile-time size; a
 ↳ vector<u64> hand-roll is
 $\sim 30\%$ slower and always available.

THE "ANSWER IS SMALL" ARGUMENT. Prove the answer is $\leq$ a tiny
 ↳ constant, then brute force over
 it. Each operation halves something / clears a bit / merges two
 ↳ components; "OR of a range
 equals its SUM" means the values are bit-disjoint so at most 30 are
 ↳ nonzero; $\text{sum } a_i \leq 1e5$
 means at most ~ 632 distinct values; $a_i \leq 1e6$ means at most 7
 ↳ distinct primes, 240 divisors;
 gcd over an expanding range takes only $O(\log V)$ distinct values.
 TRAP: prove the bound, do not observe it on samples, and handle
 ↳ "impossible" explicitly.

↳ */

12.3 ContestCraft

CONSTRUCTIVE, INTERACTIVE, RANDOMISED, EMPIRICAL

/* ===== CONSTRUCTIVE, INTERACTIVE, RANDOMISED, EMPIRICAL
 ↳ =====

BUILD GREEDILY AND PROVE AS YOU GO. Emit the answer one decision at a
 ↳ time, always taking the
 best local choice that KEEPS THE REMAINDER FEASIBLE - that check is
 ↳ the whole problem.
 TRAP: for LEXICOGRAPHICALLY SMALLEST, greedy alone is wrong. You
 ↳ need "pick the smallest
 candidate such that the rest is still completable", and the
 ↳ completability test must be
 exact. If you cannot write that test, you cannot do lexicographic
 ↳ greedy.

BUILD RECURSIVELY. Solve n from $n/2$ (or $n-1$, or four quadrants) and
 ↳ glue with a fixed pattern.
 TRIGGER: n is a power of two, or the object is self-similar (Gray
 ↳ code, Hanoi, de Bruijn).
 TRAP: base cases are usually $n = 1$ AND $n = 2$. Recursive output of
 ↳ 2^n lines needs buffered
 output - endl will TLE.

EXTEND FROM A KNOWN SMALL CASE. Brute force $n = 1..8$, find the pattern
 ↳ of CONSTRUCTION (not just
 the count), then prove the inductive step.
 TRAP: the small cases are exactly where the pattern fails. Find the
 ↳ smallest n where it first
 works and special-case everything below. Nearly every constructive
 ↳ WA is a small- n exception.

CERTIFICATE CONSTRUCTION (prove the NO). Output the witness: the

→ negative cycle, the odd cycle, the Hall violator, the parity mismatch. Producing it also validates
 → your YES logic.
 TRAP: after n Bellman-Ford rounds you have a vertex AFFECTED BY a
 → negative cycle, not one ON
 it. Walk n parent pointers back first, then follow parents until a
 → vertex repeats.

INTERACTIVE: BUDGET FIRST, ALGORITHM SECOND. Compare 2^Q against the
 → number of possibilities.
 $Q \sim \log n \rightarrow$ binary search; $Q \sim n \rightarrow$ one query per element; $Q \sim n$
 → $\log n \rightarrow$ sorting/merging.
 TRAPS: (1) FLUSH after every query (endl, or fflush) or you get
 → "idleness limit exceeded",
 which looks nothing like the real bug; (2) count your queries
 → locally and assert the limit;
 (3) assume the interactor is ADAPTIVE - it only has to stay
 → consistent with past answers.

RANDOM SHUFFLE to destroy adversarial order: Kuhn's matching,
 → quickselect, incremental convex
 hull, minimum enclosing circle (which is $O(n)$ only after a shuffle),
 → KD-tree splits.
 TRAP: seed from the clock (mt19937_64 with steady_clock), never
 → srand(time(0)) + rand().
 unordered_map needs the splitmix64 custom hash or it is $O(n^2)$ on
 → a crafted test.

RANDOM SAMPLING. If a target occupies more than a $1/c$ fraction, k
 → samples miss it with
 probability $(1 - 1/c)^k$.
 TRIGGER: "majority element of a range", "a value shared by at least
 → half".
 TRAP: sampling gives a CANDIDATE - you must verify it exactly. And
 → the failure probability
 compounds over q queries: with $q = 1e5$ you need per-query failure
 → below $1e-12$, so ~ 40
 samples, not 20.

RANDOM WEIGHTS FOR UNIQUENESS (Zobrist / XOR hashing). Give each value
 → a random 64-bit label;
 then multiset equality, "every value appears a multiple of k times",
 → and "this subtree shape
 occurred before" become single-number comparisons.
 TRAP: one random assignment has a real false-positive rate - repeat
 → ~ 30 times with FRESH
 randoms. A fixed small set of randoms is breakable by Gaussian
 → elimination.

HASHING AS $O(1)$ EQUALITY. Polynomial hashing for substrings, 2D for
 → submatrices.
 TRAP: RANDOM base, and either double-mod or one 61-bit Mersenne mod.
 → Fixed base 31 with mod
 $1e9+7$ is anti-hashed by a Thue-Morse construction. Birthday bound:
 → comparing m hashes
 pairwise collides with probability $\sim m^2/(2*MOD)$, so $m = 1e6$
 → against a single $1e9+7$ mod is a
 coin flip.

THE REPETITION MATH. k independent trials fail with p^k . Budget PER
 → QUERY, not per test, then
 multiply by 100 for safety. For primality use the DETERMINISTIC
 → witness set
 $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$ (valid below $3.3e24$) - zero
 → failure probability, same cost.

STRESS TESTING AS A THINKING TOOL. gen + brute + a diff loop; then
 → SHRINK the counterexample
 until it fits on one line and stare at it.
 TRAPS: (1) the generator must produce DEGENERATE cases - all equal,
 → $n = 1$, all zero, maximum
 values, duplicate coordinates. A uniform generator over $[1, 1e9]$
 → never produces a tie and
 will never find your tie bug. (2) If the problem says "print any",
 → diffing is useless -
 write a CHECKER. (3) Stress the brute force against the samples
 → first.

BRUTE FORCE SMALL CASES, THEN LOOK IT UP. Compute $n = 1..12$ exactly,
 → then search OEIS, or feed
 the sequence to Berlekamp-Massey, or fit a polynomial by Lagrange
 → interpolation.
 TRAP: an OEIS match on 4 terms is noise - get 8, then confirm the
 → 9th. BM needs terms computed
 under the SAME prime you will use, and at least 2x the order.

PRINT THE DP TABLE AND LOOK FOR STRUCTURE. Dump the $n \times n$ table (and

→ the argmin table) for
 $n = 20$ and look for monotone rows, monotone $opt[i][j]$, convex
 → differences, sparsity.
 That is how you decide between Knuth, divide & conquer, and Aliens.
 TRAP: monotonicity that holds at $n = 20$ can fail at $n = 200$. Also
 → verify the quadrangle
 inequality numerically on random quadruples.

INSTRUMENT, DO NOT GUESS (for TLE). Print a counter of inner-loop
 → iterations on your largest
 local test and compare against $1e8-1e9$. That distinguishes "wrong
 → complexity" from "wrong
 constant", which need completely different fixes. Compile locally
 → with $-O2$.

===== CONTEST CRAFT =====

READING THE SET

Minute 0-15: all three read DIFFERENT problems, front to back.
 → Nobody codes. On the board, per
 problem: a one-line restatement, n, and a guess at the family.
 Order by (confidence $\times$ 1/effort), not by letter. Setters mis-order
 → $\sim 30\%$ of the time.
 Use the scoreboard from minute 30 - it is the best difficulty signal
 → you have.
 ONE coder at the keyboard, always. A problem is not "started" until
 → it has a written plan with
 its complexity and its main data structure.

WHEN TO ABANDON - when any TWO hold:

- * 25+ minutes with no NEW idea (the idea count stopped growing, not
 → the code).
- * The complexity is 10x over budget with no plan to close it.
- * Two teammates failed to find the bug and 5 minutes of stress
 → testing found nothing
 → you are testing the wrong thing or misread the statement.
- Re-read it OUT LOUD.
- * Another unattempted problem has more solves.
 Write the state on paper before switching, and set a hard return
 → time.

DEBUGGING UNDER TIME PRESSURE - in this order:

1. RE-READ THE STATEMENT: output format, the modulus, 1-based vs
 → 0-based, multi-test.
- About a quarter of "impossible" bugs are misreads.
2. Re-read your input parsing (n and m swapped; looping to n over an
 → array sized m).
3. Run ALL the samples, including the one you skipped.
4. Run $n = 1$, $n = 2$, and the all-equal case.
5. Stress test. This beats reading code whenever you have a brute
 → force.
6. Only then read the code - out loud, to a teammate, for INTENT.
 → Silent re-reading just
 re-hallucinates the same thing.
7. Nothing after 15 minutes: REWRITE the suspicious function from
 → scratch.

PRE-SUBMIT CHECKLIST (40 seconds, every submit)

- [] Overflow: every product of two inputs; inside comparators;
 → inside prefix sums.
- [] Modulus: right constant, +MOD after subtraction, inverse instead
 → of division, non-negative.
- [] $n = 1$, $m = 1$, $k = 0$, single-element array.
- [] Degenerate input: empty string, zero edges, one-node tree, all
 → equal, all zero, negatives.
- [] STATE RESET between test cases: every global, vector, counter,
 → visited array, the answer
 variable, and any function-local static. Clear only what you
 → touched if sum n is bounded.
- [] Right bound read: MAXN vs the actual limit; the sum over queries
 → can exceed the per-test n.
- [] Uninitialised: dp arrays that need -INF not 0; DSU parent[i] = i.
- [] Recursion depth: DFS on a path with $n = 2e5$ is $2e5$ frames - move
 → big locals out.
- [] Output format: newline vs space, YES vs Yes, -1 vs IMPOSSIBLE,
 → setprecision.
- [] endl removed from any loop running more than $\sim 1e4$ times; debug
 → output removed.
- [] The 'cin >> tc' line matches whether the problem is multi-test.

VERDICT TRIAGE

WA → which test? Test 1 means a misread or a format error; a late
 → test means an edge case or
 overflow. Then: modulus, overflow, ties/duplicates (strict vs
 → non-strict), stress test,
 unreset state, floating-point comparison.
 TLE → is the complexity really what you think (count the loops on

↪ paper)? endl in a loop /
 missing sync_with_stdio. map/set in a hot loop. Reallocation
 ↪ and pass-by-value. memset
 of a huge global per test. Infinite loop: a binary search
 ↪ with l = mid.
 Only then change algorithm. A /64 win never fixes a wrong
 ↪ complexity class.
 RE -> array bounds first: rebuild with -fsanitize=address,undefined
 ↪ -D_GLIBCXX_DEBUG and run
 the samples; it finds it in seconds. Then stack overflow,
 ↪ division by zero, a comparator
 that is not a strict weak ordering (crashes INSIDE
 ↪ std::sort), .back()/.top() on an
 empty container, 0- vs 1-indexed access.
 MLE -> n x n arrays (5000*2 ints = 100 MB); vector<vector<int>> for
 ↪ a graph (5-8x the raw
 data - use CSR); persistent structures (count the nodes
 ↪ before allocating); recursion
 frames; go offline and stream instead of storing versions.
 Idleness on an interactive -> a missing flush, or debug output on
 ↪ stdout.

TEAM PROTOCOL

Never submit without the checklist: a rejection costs 20 penalty
 ↪ minutes plus morale.
 One person owns the printed notebook and fetches pages; the coder
 ↪ does not browse.
 Paste, adapt, then read once - the #1 template bug is an
 ↪ index-convention mismatch.
 Two solutions disagree? Trust the brute force. Always.
 Last 30 minutes: stop starting new problems; finish the nearest one
 ↪ and re-check the submits.

THE TEN MOST EXPENSIVE RECURRING BUGS (by contest-time cost)

1. Overflow in an intermediate product, invisible because the final
 ↪ answer is small.
2. State not reset between test cases - passes locally, fails on
 ↪ test 2.
3. Wrong strictness (< vs <=) in a comparator, a binary-search
 ↪ predicate, or a tie-break.
4. Two pointers on a non-monotone predicate: plausible output,
 ↪ silently wrong.
5. Binary search bounds: hi not provably feasible, or l = mid
 ↪ causing an infinite loop.
6. 0- vs 1-based off-by-one when pasting a template.
7. endl in an output loop, or a missing sync_with_stdio, at n = 1e6.
8. Negative modulo in C++ (-3 % 5 == -3).
9. A greedy that was never proved and passes the samples.
10. Recursion depth on a path-shaped tree with n = 2e5.

=====
 ↪ */

Index of every routine (alphabetical)

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